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Note on Dr Glaisher's Paper on a Theorem in Definite Integration.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (1870), pp. 355, 356.]

It is worth noticing that under the case when $\phi = 1$ may be proved independently of the general formula with Θ_1 for (1) the equation

$$s = ax - \frac{a_1}{a-b_1}x - \frac{a_2}{a-b_2}x - \cdots - \frac{a_n}{a-b_n}x \cdots ,$$

is $(sx-r)(sx-b_1)(sx-b_2)\ldots(sx-b_n)\ldots = 0$, and has $s+1$ roots, say $x_1, x_2, \ldots, x_{n+1}$, where

$$x_1 = x, \quad x_2 = a - b_1 + b_1, \quad \ldots, \quad b_{n+1} = a - b_n + b_n;$$

and (2) the equation

$$r = -\frac{a_1}{a-b_1}x - \frac{a_2}{a-b_2}x - \cdots - \frac{a_n}{a-b_n}x \cdots ,$$

is $= r(x-b_1)(x-b_2)\ldots(x-b_n)\ldots = 0$, and has n roots $x_1, x_2, \ldots, x_n$ where

$$x_1 = x_1 - a_1, \quad x_2 = b_1 + b_2, \quad \ldots, \quad x_n = \frac{x_{n-1} + b_{n-1} + b_n}{a_n};$$

whence

$$f\,dx_1 + f\,dx_2 + \cdots + f\,dx_{n+1} = 0 \quad \text{in the first case}$$

and

$$nf(a,b)(x_n, x_{n-1}, \ldots, x_2, x_1)' = 0 \quad \text{in the second}$$

[which are the two formulæ in question]

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On the Quartic Surfaces $(*\breve{a}\, U,\ V,\ W)^2 = 0$.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. x. (1871), pp. 15–25.]

AMONG the surfaces of the form in question are included the reciprocals of several interesting surfaces of the orders 6, 8, 9, 10, and 12, viz.

Order 6, parabolic ring,

8, elliptic ring,

9, centro-surface of a paraboloid,

10, parallel surface of a paraboloid,

envelope of planes through the points of an ellipsoid at right angles to the radius vectors from the centre,

12, centro-surface of an ellipsoid,

parallel surface of an ellipsoid.

I propose to consider these surfaces, not at present in any detail, but merely for the purpose of presenting them in connexion with each other and with the present theory. It will be convenient to use homogeneous equations, but for the surcinal interpretation the coordinate W or w may be considered as equal to unity; I have not thought it necessary to so adjust the constants that the equations shall be homo- geneous in regard to the constants; this case of course be done without difficulty, and in many cases it would be analytically advantageous to make the change.

I take throughout (X, Y, Z, W) for the coordinates of a point on the quartic surface (so that (X, Y, W) is, the equation $(*\breve{a}\, U,\ V,\ W)^2 = 0$ are to be considered as quartic functions of (X, Y, Z, W)); reserving (x, y, z, w) for the coordinates of a point on the reciprocal surface of the order 6, 8, 9, 10, or 12. The reciprocation is performed in regard to the imaginary sphere $x^2 + y^2 + z^2 + w^2 = 0$; the relation between

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the coordinates (X, Y, Z, W) and (x, y, z, w) is then $Xx + Yy + Zz + Ww = 0$, and the equation $(X, Y, Z, W) = 0$ is the equation in point-coordinates of the quartic surface, or in line-coordinates of the reciprocal surface; and similarly the equation $(x, y, z, w)^n = 0$ is the equation in point-coordinates of the reciprocal surface, or in line-coordinates of the quartic surface.

Parabolic ring, or envelope of a sphere of constant radius having its centre on a parabola.

Taking k for the radius of the sphere, and $x = 0$, $y^2 = ax$ for the equations of the parabola, then the coordinates of a point on the parabola are $a\theta^2$, $2a\theta$, 0, where θ is a variable parameter. The equation of the sphere therefore is

$$(x - a\theta w)^2 + (y - 2a\theta w)^2 + z^2 - k^2 w^2 = 0,$$

and the ring is the envelope of this sphere.

The reciprocal of the sphere is

$$k^2(X^2 + Y^2 + Z^2) - (a\theta^2 X + 2a\theta Y + W)^2 = 0;$$

writing this in the form

$$a\theta^2 X + 2a\theta Y + W = k\sqrt{1}(X^2+Y^2+Z^2) = 0,$$

and taking the envelope in regard to θ, we have

$$X\{W + k\sqrt{1}(X^2+Y^2+Z^2)\} - aY^2 = 0,$$

or, what is the same thing,

$$(aY^2 - XW)^2 = k^2X^2(X^2+Y^2+Z^2)$$

for the equation of the quartic surface. This has the form $X = 0$, $V = 0$ for a tangential line, but I am not in possession of a theory enabling me thence to infer that the parabolic ring is of the order 4.

To show that it is so, I have to find (x, y, z, w) which satisfy the equation of the variable sphere

$$(x - a\theta w)^2 + (y - 2a\theta w)^2 + z^2 - k^2w^2 = 0,$$

or, what is the same thing,

$$(A, B, C, D, E)(\theta, 1)^4 = 0,$$

where

$$\begin{aligned}
&A = a^2w^2,\\
&B = 0,\\
&C = a(2aw^2 - ax),\\
&D = -4ayw,\\
&E = x^2 + y^2 + z^2 - k^2w^2.
\end{aligned}$$

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Then $I = 3a^4w^2E$, $J = a^6w^6C$, and the equation is $I^3 - J^2 = 0$, viz. this is

$$(Ax^2+3y^2-3aw^2x+3y^2-4xy)^3-\{(3aw-x)(3y^2+3y^2-6wx-4ay\dot{x})-27a^2y^2wx\}^2(Ew-xw)^2=0,$$

or, as this may also be written,

$$(Ax^2+3y^2-3aw^2x)^3-\{(3aw-x)(3y^2+3y^2-6wx-4ay\dot{x})-27a^2y^2wx\}^2(Ew-xw)^2=0.$$

Developing, the whole divides by 27, and the equation of the ring finally is

$$(y^2-6awx)^3(y^2+xw) = -3aw^2\{2x^2-2axw+3aw^2(y-xw)(y^2-2yw)\}(z^2+w^2)$$
$$+\{3y^2+y^2(2x-2a)w+24a^2(zw^2+aw^2z)+16a^2w^4(z^2-xw)(z^2-8axw+4a^2w^2)\}(z^2-kw^2)$$
$$+\{(2y^2+axw(z^2-kw^2)^2\}(z^2-kw^2)^3$$

which is a quartic surface having for nodal conics

$$(y^2-6awx=0,\quad z^2+w^2=0),\quad (Z=0,\quad x=w=0),$$

the line at infinity being a torsal line.

Elliptic ring, or envelope of a sphere of constant radius having its centre on an ellipse.

Taking k for the radius of the sphere, and $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ for the equations of the ellipse, the coordinates of a point on the ellipse are $a\cos\theta$, $b\sin\theta$, 0; hence the equation of the variable sphere is

$$(x-aw\cos\theta)^2+(y-bw\sin\theta)^2+z^2-k^2w^2=0.$$

The reciprocal of the sphere is

$$k^2(X^2+Y^2+Z^2)-(aX\cos\theta+bY\sin\theta+W)^2=0;$$

viz. writing this under the form

$$aX\cos\theta+bY\sin\theta+W=k\sqrt{(X^2+Y^2+Z^2)},$$

and taking the envelope in regard to θ, we have

$$\sqrt{(a^2X^2+b^2Y^2)}=W+k\sqrt{(X^2+Y^2+Z^2)};$$

viz. this is

$$(a^2X^2+b^2Y^2)+W^2+k^2(X^2+Y^2+Z^2)-2Wk\sqrt{(X^2+Y^2+Z^2)}=0;$$

or

$$\{(a^2+k^2)X^2+(b^2+k^2)Y^2+k^2Z^2+W^2\}^2=4W^2k^2(X^2+Y^2+Z^2),$$

for the equation of the quartic surface. This has the form $X=0$, $V=0$ for a tangential line, but I am not in possession of a theory enabling me thence to infer that the elliptic ring is of the order 4.

To show that it is so, I have to find (x,y,z,w) which satisfy the equation of the variable sphere

$$(x-aw\cos\theta)^2+(y-bw\sin\theta)^2+z^2-k^2w^2=0,$$

or, what is the same thing,

$$(A, B, C, D, E)(t, 1)^4 = 0,$$

where

$$\begin{aligned}
A &= 0,\\
B &= 0,\\
C &= a(2aw^2 - ax),\\
D &= -4byw,\\
E &= 2(a^2 + b^2)w^2 + 2(x^2 + y^2 + z^2 - k^2w^2).
\end{aligned}$$

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For this purpose reverting to the equation

$$(x - aw\cos\theta + (y - bw\sin\theta + z^2 - k^2w^2) = 0,$$

this may be written

$$A\cos 2\theta + B\sin 2\theta + C\cos\theta + D\sin\theta + E = 0,$$

where

$$\begin{aligned}
&A = \tfrac{1}{2}(a^2 - b^2)w^2,\\
&B = 0,\\
&C = -4axw,\\
&D = -4byw,\\
&E = \tfrac{1}{2}(a^2 + b^2)w^2 + 2(x^2 + y^2 + z^2 - k^2w^2),
\end{aligned}$$

and the equation is

$$\begin{gathered}
12A(A - B) - 3(C^2 + D^2) + 4E^2\\
= \{12A^2(C^2 - D^2) - 24BCD - 3(C^2 + D^2)^2 + 4E^2(E - 8A^2)\}^2;
\end{gathered}$$

or say

$$\{12A^2 + 3(C^2 + D^2) + 4E^2\}^3 - 27\{A(C^2 - D^2) - 2BCD - 3E(C^2 + D^2) + 8AE^2\}^2 = 0.$$

This is of the order 12, but it is easy to see that the terms in A^2 and $E^2(C^2 + D^2)$ disappear from the equation, all the other terms divide by w^2; and the equation is thus of the order 8.

The equation may be obtained somewhat differently, as follows. The equation of the variable sphere is

$$(x - aw)^2 + (y - bw)^2 + z^2 - k^2w^2 = 0,$$

where (a, β) may subject to the condition $\frac{a^2}{a^2} + \frac{b^2}{b^2} = 1$. We have therefore

$$\frac{x - aw}{(x - aw)\,da} + \frac{y - bw}{(y - bw)\,db} = 0,$$

$$\beta\,da = \frac{a\,a}{a^2 + b^2} = -\frac{b}{a}\,da;$$

and thence

$$\begin{aligned}
aw &= \frac{a^2x}{a^2 + b^2}, &\quad x - aw &= \frac{b^2x}{a^2 + b^2},\\
\beta w &= \frac{b^2y}{a^2 + b^2}, &\quad y - bw &= \frac{a^2y}{a^2 + b^2}.
\end{aligned}$$

Consequently

$$\begin{aligned}
\frac{a^2x^2}{(a^2 + b^2)^2} + \frac{b^2y^2}{(a^2 + b^2)^2} - w^2 = 0,\\
\frac{a^2x^2}{(a^2 + b^2)^2} + \frac{b^2y^2}{(a^2 + b^2)^2} - w^2 = 0,
\end{aligned}$$

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The quartic surface is consequently the envelope of the sphere

$$\frac{(a+b)^2}{a}x^2 + \frac{(b+c)^2}{b}y^2 + 2bz^2 - 2aw^2 = 0,$$

viz. this is

$$b\left(\frac{1}{a} - \frac{1}{a}\right)x^2 + 2\theta(x^2+y^2+z^2) + 2aX - 2bW = 0.$$

Hence the quartic surface is

$$\left(\frac{x^2}{a} - \frac{y^2}{a}\right)(aX^2 + bY^2 - 2ZW) = (X^2+Y^2+Z^2) = 0,$$

or, what is the same thing,

$$X^2Y^2(ab+cb) - 2ZW(bX^2+aY^2) - 2abZ(X^2+Y^2+Z^2) - aBcZ = 0.$$

This has four nodes; viz. considering the equations

$$\frac{X}{a} + \frac{Y}{b} = 0, \quad aX^2 + bY^2 - 2ZW = 0, \quad X^2+Y^2+Z^2 = 0,$$

these give the quad. $X = 0$, $Y = 0$, $Z = 0$ four times, and four other points which are the nodes in question; the point $(X = 0, Y = 0, Z = 0)$ is a singular point of a higher order; the deduction caused by these singularities should be $= 8+19$, so as to make the order of the surface of curves of $36 - 8 - 19 = 12$, but it is the reduction on account of the point $(X = 0, Y = 0, Z = 0)$ must be $= 19$; but it is not by any means obvious how this is so.

Parallel surface of the paraboloid.

This is given, Salmon's *Solid Geometry*, 2nd Edit., pp. 146 and 148, [Ed. 4, p. 180], for the paraboloid $aX^2 + bY^2 + 2ZW = 0$, as the envelope of the quartic surface

$$\frac{a}{aw+1}X^2 + \frac{b}{bw+1}Y^2 + 2W(\theta - r^2w + r^2)w^2 = 0.$$

The reciprocal quartic is thus the envelope of

$$\frac{1}{a}x^2 + \frac{1}{b}y^2 + \frac{1}{a^2}z^2 + \frac{1}{a+r^2w}w^2 - w^2 = 0,$$

that is,

$$(X^2+Y^2+Z^2) + \frac{1}{b}\left(\frac{X^2}{a} - \frac{Y^2}{b} - \frac{2Z}{c}W\right) = 0;$$

whence the equation is

$$\frac{1}{b^2}(X^2-Y^2)\left(\frac{X^2}{a} - \frac{Y^2}{b} - \frac{2Z}{c}\right) = 0,$$

viz. this is a quartic having the nodal lines

$$Z = X = 0, \quad Z = Y = 0$$

the circle must be equal, and the distance of the centre from the line of centres, and if the radii are r, r', is readily found to be

$$(a-x)^2 + 2(r+r'),$$

suppose in general, that spheres any two contain the point P in the intersection of the points of P in regard to the two given conics respectively; then if P describes

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Envelope of the planes through the points of an ellipsoid at right angles to the radius vectors from the centre.

This is given in my paper "Note on a Surface," in the *Annales de Matématica*, t. I. (1859), [206], as the envelope of the quartic surface

$$\frac{x^2}{1+\theta a}+\frac{y^2}{1+\theta b}+\frac{z^2}{1+\theta c}-\theta a^2=0.$$

The reciprocal quartic is thus the envelope of

$$\left(2-\frac{x^2}{a}\right)X^2+\left(1-\frac{y^2}{b}\right)Y^2+\left(1-\frac{z^2}{c}\right)Z^2-\frac{1}{a}W^2=0,$$

or, what is the same thing,

$$\theta\left(\frac{X^2}{a}+\frac{Y^2}{b}+\frac{Z^2}{c}-\frac{1}{c}W^2\right)-(X^2+Y^2+Z^2)=0;$$

viz. it is

$$W^2=\frac{X^2-aY^2}{X^2-aY^2+Z^2}=\frac{Y^2}{X^2+Y^2+Z^2}=\frac{Z}{X^2+Y^2+Z^2}=0 \quad \text{for } X,\ Y,\ Z,$$

of the original $\frac{X^2}{a}+\frac{Y^2}{b}+\frac{Z^2}{c}=1$; this is obvious geometrically inasmuch as the reciprocal of the variable plane is the inverse of the point on the ellipsoid.

The quartic surface has the nodal conic

$$W=0,\quad aX^2+bY^2+cZ^2=0;$$

and also the node $X=0$, $Y=0$, $Z=0$; there is a reduction in the order of the reciprocal surface a reduction $24+2=26$, or the order of the reciprocal surface is $=10$.

Centro-surface of the ellipsoid.

Writing the equation of the ellipsoid in the form $\frac{x^2}{a}+\frac{y^2}{b}+\frac{z^2}{c}-w^2=0$, the centro-surface is given as the envelope of the quartic surface

$$\frac{x^2\theta^2}{(a+\theta)^2}+\frac{y^2\theta^2}{(b+\theta)^2}+\frac{z^2\theta^2}{(c+\theta)^2}-w^2=0.$$

(Salmon, [Ed. 2], p. 106; [Ed. 4, p. 177]), and hence the reciprocal quartic surface is the envelope of

$$\left(a+\frac{a^2}{\theta}\right)X^2+\left(b+\frac{b^2}{\theta}\right)Y^2+\left(c+\frac{c^2}{\theta}\right)Z^2-W^2=0,$$

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in regard to the variable parameter θ, viz. the equation is

$$\left(\frac{X^2}{a}+\frac{Y^2}{b}+\frac{Z^2}{c}\right)(aX^2+bY^2+cZ^2-W^2)-(X^2+Y^2+Z^2)=0,$$

(see Salmon, [Ed. 2], p. 144 [Ed. 4, p. 172]). It hence at once appears, that the quartic surface has 12 nodes, viz. these are the four angles of the fundamental tetrahedron $(XYZW)$, and the eight points

$$\frac{X^2}{a}+\frac{Y^2}{b}+\frac{Z^2}{c}=0,$$
$$X^2+Y^2+Z^2=0,\quad W=0,$$
$$aX^2+bY^2+cZ^2-W^2=0;$$

or writing as is convenient to do,

$$(a,\beta,\gamma)=(b-c,c-a,a-b),\quad \alpha+\beta+\gamma=0,$$

these are the eight points

$$(a,\beta,\gamma:\xi:\eta:\zeta)=0,$$

the order of the reciprocal of the quartic surface is then $36-12=24$, which is in fact the order of the surface of centres.

The equation of the centro-surface is given, Salmon, [Ed. 2], p. 121, and Quart. *Math. Jour.*, t. IX. (1858), p. 220, in the form

$$(\theta,\beta,\gamma:\xi,\eta:\zeta)^4=0,$$

where ξ,η,ζ stand for ax,by,cz,bz; it is therefore of the degree 18 in regard to (x,y,z,w), which is the equation of the surface, but it must contain in itself the equation w^4, and the order of the curve will be $20-6=14$.

Parallel surface of the ellipsoid.

This is given, Salmon, [Ed. 2], p. 148 [Ed. 4, p. 178], as the envelope of the quartic surface

$$\frac{a^2X^2}{a^2-r^2w^2}+\frac{b^2Y^2}{b^2-r^2w^2}+\frac{c^2Z^2}{c^2-r^2w^2}=\left(1-\frac{W^2}{w^2}\right)w^2.$$

The reciprocal quartic is thus the envelope of

$$(a^2\theta X^2+b^2\theta Y^2+c^2\theta Z^2+(a^2-r^2)\,W^2-W^2+r^2W^2)^2=0,$$

or writing $D+r^2=\theta$, this is

$$(a^2-4\theta+b^2+c^2\theta)-(\theta-a^2)Y^2-(\theta-c^2)Z^2-(\theta-r^2)W^2-W^2-\left(1-\frac{r^2}{\theta}\right)W^2=0.$$

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or, what is the same thing,

$$bX(bX-Z)+\{w^2-r^2\theta+ay\}\{x^2+(a+b)Y^2+(a+c)Z^2+(a-r^2)W^2-W^2\}=0,$$

whence the equation is

$$\{(a^2-bY)(X^2+(a-b)Y^2+(a-c)Z^2-W^2)+2bW^2(X^2+Y^2+Z^2)=0.$$

viz. this is a quartic having the nodal

$$W=0,\quad (a^2-bY)X^2+(a-b)Y^2+(a-c)Z^2=0.$$

The order of the reciprocal or parallel surface is then $36-24$, $=12$, as it should be. The nodal conic of the quartic surface is the reciprocal of a tangent cone of the ellipsoid having a non-cyclic quadric cone, vertex the centre, in the parallel surface; this cone is imaginary for the ellipsoid, but real for either of the hyperboloids, and the existence in the case of the hyperboloid is readily perceived.

Reverting to the equation

$$\frac{a^2}{a^2+\theta}X^2+\frac{b^2}{b^2+\theta}Y^2+\frac{c^2}{c^2+\theta}Z^2-\left(1+\frac{r^2W^2}{\theta}\right)w^2=0,$$

or say

$$(a^2-P)X^2+(b^2-P)Y^2+(c^2-P)Z^2-r^2(a^2-P)(b^2-P)(c^2-P)=0$$

this is

$$-w^2(a^2-P)(b^2-P)(c^2-P)-r^2(a^2-P)(b^2-P)(c^2-P)-P(a^2-P)(b^2-P)(c^2-P)=0,$$

where putting for shortness

$$\begin{aligned}
u&=a^2+b^2+c^2,\\
\beta&=b^2c^2+c^2a^2+a^2b^2,\\
\gamma&=a^2b^2c^2,\\
b&=aP^2-b^2,\\
p&=a^2+y^2+z^2,\\
q&=b^2c^2x^2+a^2c^2y^2+a^2b^2z^2,\\
r&=a^2b^2c^2w^2-r^2a^2b^2c^2+a^2b^2c^2,
\end{aligned}$$

we have

$$\begin{aligned}
A&=12n^2,\\
B&=9aw^2-2p,\\
C&=12n-4ap,\\
D&=3aw-3r,\\
E&=12r^2.
\end{aligned}$$

The equation of the parallel surface is

$$(AE-4BD+3C^2)^3=27(ACE-AD^2-B^2E+2BCD-C^3)^2.$$

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It is remarked (Salmon, [Ed. 2], p. 148 [Ed. 4, p. 176]) that there is in the plane $z = 0$, a nodal conic $\frac{x^2}{a^2+r^2} + \frac{y^2}{b^2+r^2} - 1 = 0$, the complete section being made up of this conic twice, and of the curve of the eighth order which is the parallel curve of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0$; the like is of course the case as to the sections by the other two principal planes $x = 0$, $y = 0$. For the section by the plane $w = 0$ (or plane infinity) we have at once $p^2 + (qw^2 = 0$, where observe that

$$q^2 - 4pr = \{(a^2 + b^2 + c^2)x^2 + (b^2c^2 + c^2a^2 + a^2b^2)y^2 + \cdots\}^2$$
$$= (1, 1, 1, -1, -1, -1\,)(b^2 + c^2)y^2z^2,\ (c^2 - a^2)^2z^2x^2,\ (a^2 - b^2)^2x^2y^2)$$
$$= ac(x^2 - y^2 - z^2 + b^2 + c^2(x^2 - y^2))^2,$$

which has reduced to a sum of squares with positive signs.

The section is thus made up of two conics, each twice, and of four right lines: viz. the conics are $z^2 + y^2 + z^2 = 0$, this section at infinity of the sphere, the section at infinity of the ellipsoid; and the lines are

$$(\sqrt{X^2 - aZ^2} = 0, \quad (\sqrt{X^2 - aZ^2} + \sqrt{X^2 - bZ^2}, X^2 - aZ^2 = 0)$$

viz. those are the common tangents of the two conics. The circle at infinity is a nodal conic on the surface, which has thus 4 nodal conics.

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Note on a Relation between Two Circles.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. (1871), pp. 82, 83.]

Consider any two circles O, Q; and let AC, BD, AB, BC be the common tangents touching the circles in the points A, A_1, B, B_1, C, C_1, D, D_1: the locus of a point P such that the pairs of tangents from it to the two circles respectively form a harmonic pencil, is a conic (locus of the points A, A_1, B, B_1, C, C_1, D, D_1); but this conic may break up into two lines, viz. if (as in the figure) the points A, B, D, D are in a line, then the points C, C', A', B' will be in a symmetrically situated line, and the conic breaks up into this pair of lines, meeting suppose in K. The condition

[Figure: Two circles with common tangents and centre line — diagram omitted]

for this, if a, c' are the distances of the centres from a fixed point in the line of centres, and if the radii are r, r', is readily found to be

$$(a - c')^2 = 2(r^2 + r'^2).$$

Suppose in general, that lines any two contain the point P in the intersection of the points of P in regard to the two given conics respectively; then, if P describes

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a line, the locus of P is a conic passing through the three conjugate points of the given conic, if, however, the line which is the locus of P pass through one of the conjugate points, then the conic the locus of P' breaks up into a pair of lines: one of them a fixed line through the other two conjugate points; that is, the locus of P' is through the demonstrated conjugate points. That is, if the locus of P be a line through a conjugate point, the locus of P' is a line through the same conjugate point; but in every other case the locus of P' is a conic.

Reverting to the figure of the two circles, in order that it may be possible that the two lines AD and BC may be loci of points P, P', related as above, it is necessary that K shall be a conjugate point of the two circles; that is, if the two circles intersect in points A, X lying symmetrically in the line of centres, then K must be the anti-points of A, X; what is the same thing the chord AX shall be one of the same mentioned relation. And it then appears that given two circles, the necessary and sufficient conditions for the coexistence of the properties mentioned in the theorem are that they shall be equal circles touching each other.

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On the Porism of the In-and-Circumscribed Polygon, and the $(2, 2)$ Correspondence of Points on a Conic.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. *(1871), pp. 83–91.]*

The present paper includes, as will at once be seen, much that is perfectly well known; but the separate theories regarded it seemed to me, to be put together; and there are, particularly as regards the unsymmetrical case afterwards referred to, some results which I believe to be new.

The porism of the in-and-circumscribed polygon has its foundation in the theory of the symmetrical $(2, 2)$ correspondence of points on a conic; viz. a $(2, 2)$ correspondence is such that to any given position of either point there correspond two positions of the other point; and is a symmetrical $(2, 2)$ correspondence either point indifferently may be considered as the first point and the other of them will then be the second point of the correspondence. Or, what is the same thing, if x, y are the parameters which serve to determine the two points, then x, y are connected by an equation of the form $(*\breve{a}\, x, 1\breve{a}\, y, 1)^2 = 0$; or, as it will always thus, whatever the parameters (x, y). In the case of such symmetrical relation it is easy to show that the line joining the two points envelopes a conic. For the relation may be expressed in the form $(*\breve{a}\, x+y, x+y, xy)^2 = 0$; we may imagine the coordinates (P, Q, R) of each manner that for the point (x) on the first conic the corresponding values (P, Q, R) for and for the point (y), $P : Q : R = 1 : y : y^2$; the equation of the line joining the two points is then

$$\begin{vmatrix} P, & Q, & R \\ 1, & x, & x^2 \\ 1, & y, & y^2 \end{vmatrix} = 0,$$

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that is

$$Pxy - Q(x + y) + R = 0,$$

or representing this by the equation

$$PL + QM + R = 0,$$

we have $L : -2 = -y : x + y - 1$; and consequently (L, M) are connected by a quadric relation $(* L, 1)^2 = 0$, that is, the line envelopes a conic.

The locus, $(F, P, Q, R = 0$, where symmetrical or not, leads us to the following theorem of X, Y are quartic functions of a respectively; viz. these are unlike as like functions of the two variables according as the integral equation is not or is symmetrical. In the present case, however, where X is supposed to be the same function of a variable, but does the equation be a connection. In every chord through any two points of the conic, then the relation is obviously

$$\frac{dx}{\sqrt{(X)}} \pm \frac{dy}{\sqrt{(Y)}} = 0$$

where X, Y are quartic functions of x, y respectively, viz. this is also the symmetric relation. In the case of the in-and-circumscribed polygon there is an unsymmetric relation, but I cannot, however, propose to recall the integral equation in this case.

Attending now to the symmetrical case; if A and B are corresponding points, then the corresponding points of B are A and a new point C; those of C are B and a new point D, and so on; so that the points form a series A, B, C, $D, \ldots$: and the polynomial of A is, if a given position of B this series closes at a certain term, for instance, if $D = A$, then it will always thus close; whatever the position of the point on the conic, the integral equation

$$\Pi(y) - \Pi(x) = 0,$$

this equation must be a transformation of the original equation $(*\breve{a}\, x, 1\breve{a}\, y, 1)^2 = 0$, and equally with it represent the relation between the parameters x, y of the two points A, B; the constant of integration k of course be determined according as the coefficients of the form of the unsymmetrical case is so on.

Hence forming the equations for the correspondences B, C; C, D; $\ldots$, and assuming that the series closes at $D = A$, we have

$$\Pi(y) - \Pi(x) = \Pi(z),$$
$$\Pi(z) - \Pi(y) = \Pi(z),$$
$$\Pi(z) - \Pi(z) = \Pi(z);$$

[489] (continued)

On the Porism of the In-and-Circumscribed Polygon (continued).

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Now starting from the differential equation

$$\frac{dx}{\sqrt{(a,\, b,\, c,\, d,\, e \between x,\, 1)^4}} \pm \frac{dy}{\sqrt{(a,\, b,\, c,\, d,\, e \between y,\, 1)^4}} = 0,$$

the integral equation is known to be

$$\frac{\sqrt{(a,\, b,\, c,\, d,\, e \between x,\, 1)^4} - \sqrt{(a,\, b,\, c,\, d,\, e \between y,\, 1)^4}}{x-y} = \tfrac{1}{2}(a,\, b,\, c,\, d,\, e \between x,\, y,\, 1)^2,$$

where θ is the constant of integration. Writing, for shortness, $X = (a,\, b,\, c,\, d,\, e \between x,\, 1)^4$, $Y = (a,\, b,\, c,\, d,\, e \between y,\, 1)^4$, this is

$$X + Y - 2\sqrt{(XY)} = a(x^2-y^2)^2 + 4b(x-y)(x^2-y^2) + 6\theta(x-y)^2,$$

or, what is the same thing,

$$\begin{aligned} a(x+y)^4 - 2\sqrt{(XY)} = {} & a(x^2-y^2)^2 - 4a(x-y)(x^3-y^3) + 6\theta(x-y)^2,\\ & + 4b(x^3+y^3)\\ & + 6c(x^2+y^2)\\ & + 4d(x+y)\\ & + 2e, \end{aligned}$$

viz. this gives

$$\begin{aligned} \sqrt{(XY)} = {} & ax^2y^2\\ & + 2b(x^2y+xy^2)\\ & + 3c(x^2+y^2)\\ & + 3d^2xy - 3\theta\\ & + 2d(x+y)\\ & + e, \end{aligned}$$

and, rationalising, the integral equation becomes

$$\begin{aligned} = {} & 6a\theta x^2y^2\\ & - 4abxy(x+y)\\ & - ac(x+y)^2\\ & + 4b^2xy\\ & + 12bcxy(x+y) + 12bdxy(x+y)\\ & - 9c^2xy\\ & + 6ce(x+y)^2 - 18cd(x+y)\\ & - 12cd(x+y)\\ & + 9d^2(x+y)^2 - 12d\theta(x+y) - 6ed + 4d^2 = 0. \end{aligned}$$

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or, as it may be written,

$$\begin{aligned} & xy^2(3\theta - 6ad) \\ & \quad + (x^2y + xy^2)(-4ad + 12bc - 12bd) \\ & \quad + (x^2 + y^2)(-ae - 3c^2 - 6ec\theta + 4d^2) \\ & \quad + xy(-2ae + 3bd - 12ec - 4be) \\ & \quad + (x + y)(-2be + 12bd - 6d^2) \\ & \quad + 4d^2 - 6ed = 0. \end{aligned}$$

Comparing this with the original integral equation $V = 0$, and the form of differential equation deduced therefrom, we ought to have therefore

$$\begin{aligned} & [(4b^2 - 8ad)x^2 + (-2ae + 4bd)x + (-ae + 4bd - 3c^2)]ay^2 \\ & \qquad + [(-ae + 4bd - 4ce)x + (-2ae + 4bd - 4de) + (3d^2 - 6ed)]y \\ & \qquad + [(-ae + 4bd - 3c^2)x + (3d^2 - 6ed) + (4e^2 - 8d\theta)] \end{aligned}$$

= multiple of X,
$= (-4ax^2 - 4bx + 2cax)(\theta x + (3a - 6cx) - (-ae + 4bd - 3c^2))$, by comparing the coefficients of x^4. I obtain this otherwise: Write

$$V = aC + 6\beta H,$$

then, forming the Hessian of V, we have

$$\Box V = -aH^2 + (a^2I + (Ja\theta)U,$$

or

$$= aH^2 - \frac{3IJ(\theta)}{8}(\theta)(V - aC) + (aI + (Ja\theta)U,$$

that is,

$$aJV + (-3a^2J)V - \frac{2}{8}(aH - \tfrac{3J}{8})(aH) + \frac{3J^2(\theta)}{8}U(8H - aC) + (aJV + (Ja\theta)U,$$

viz.

$$K = -\frac{3J\theta}{8}(\theta) - aH,$$

this is

$$(aJV + (Ja\theta)V + (J^2I)V - (a\theta\theta)V + a\theta^2aV + \frac{3J}{8}(\theta) = (3aV)(-\theta U,$$

so that the components are

$$V = aV + 6\beta H,$$

$$V' = -aC + 6\beta H,$$

where $a, b, c, d, e \,\breve{a}\, x,\ 1)^4$, $+(a, b, c, d, e \,\breve{a}\, x,\ y) - 1)^2$, $4ax + 4bx + 6cy - 4dy - e$, or $-d^2(\,c,\ 1)^2$,

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viz. the components are

$$\begin{array}{rrr} (aa+b\breve{a}(ax-bc), & ab+bd(ax-bc), & ae+bx+2bd-3c^2(\breve{a}\,x,\,1)^2, \\ (ab+bd(ax-bc), & ac+bx+2bd-3c^2(\breve{a}\,x,\,1)^2, & bd+bx+x_{1c}^2J_x, \\ (ac+bx+2bd-3c^2(\breve{a}\,x,\,1)^2, & bd+bx+x_{1c}^2J_x, & ae+bx+x_{1c}^2J_x, \end{array}$$

where as before

$$K=\frac{8(a^2-3J\theta)}{a}.$$

I assume

$$\begin{aligned} B=-\tfrac{1}{a},\quad b'=bB-\tfrac{1}{a}\quad K&=3\{4a^2-6b\theta-3a^2c^2\}, \\ ax+b\breve{a}(ax-bc)&=a(x-bc)+ax(b-bx)-bx, \\ ab+bd(ax-bc)&=a(x-bc)+ax(b-bx)-bxc, \\ ae+bx+2bd-3c^2(\breve{a}\,x,\,1)^2&=a(x-bc)+ax(b-bx)-bxc-bxd, \\ ae+bx+2bd-3c^2(\breve{a}\,x,\,1)^2&=a(x-bc)+ax(b-bx)-bxc-bxd-bxe, \\ ax+b\breve{a}(2bd-3c^2)\breve{a}\,J_x&=4bK-4bex+a(2bd-3c^2)(\breve{a}\,x,\,1)^2-\tfrac{1}{a}\,bK-3bex, \\ &=-be-12bc+6b\,(8bd-3c^2+4d\theta) \\ &=a(b\theta-3b\theta+3b\,ax)-\tfrac{1}{2}(b\theta\theta+ac^2+b\theta)f', \\ &=a(b\theta-3b\theta+3b\,ax)-\tfrac{1}{2}(b\theta\theta+ac^2+b\theta)f', \\ &=aa+b-12acd+6a^2ec, \end{aligned}$$

agreeing with the former result.

I return to the general form

$$\begin{gathered} \eta(\theta',b,c\,\breve{a}\,x,\,1)^4 \\ +2g(a',b',c'\,\breve{a}\,x,\,1)^2 \\ +(a'',b'',c''\,\breve{a}\,x,\,1)^4=0 \end{gathered}$$

giving

$$\frac{dx}{\sqrt{(a,\,b,\,c,\,d,\,e\,\breve{a}\,x,\,1)^4}}\pm\frac{dy}{\sqrt{(a,\,b,\,c,\,d,\,e\,\breve{a}\,y,\,1)^4}}=\frac{dy}{\sqrt{(a',\,b',\,c'\,\breve{a}\,y,\,1)^2(a'',\,b'',\,c''\,\breve{a}\,y,\,1)^2}}.$$

Operate a linear transformation on this, a say

$$x=\frac{\lambda x'+\mu}{\nu x'+\rho},$$

$$3-2$$

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the new coefficients are

$$
\begin{aligned}
&(a, b, c, d, e \,\breve{a}\, x, 1)^4, \quad (a, b, c, d, e \,\breve{a}\, x, 1)^4, \quad (a, b, c, d, e \,\breve{a}\, x, 1)^4, \\
&(a', b', c' \,\breve{a}\, x, 1)^2, \quad (a', b', c' \,\breve{a}\, x, 1)^2, \quad (a', b', c' \,\breve{a}\, x, 1)^2, \\
&(a'', b'', c'' \,\breve{a}\, x, 1)^2, \quad (a'', b'', c'' \,\breve{a}\, x, 1)^2, \quad (a'', b'', c'' \,\breve{a}\, x, 1)^2;
\end{aligned}
$$

assume now

$$
\begin{aligned}
a'\lambda + (b' - c')\mu - ca = 0, \qquad & ba\mu = 0, \\
(a' + b')\lambda - bc + c\mu = 0, \qquad & ca\mu = a(a'\mu + 6c'\mu - bc) = 0, \\
b' \qquad & = c' = 6c\mu = 0;
\end{aligned}
$$

then it is easy to show that

$$
\begin{aligned}
(a, b, c \,\breve{a}\, x, p\mu \breve{a}\, x, 1)^4 &= (a', c' \,\breve{a}\, x, 1)^2, \\
(a', b', c' \,\breve{a}\, x, p\mu \breve{a}\, x, 1)^2 &= (a', c' \,\breve{a}\, x, 1)^2, \\
(a, b, c \,\breve{a}\, x, p\mu \breve{a}\, x, 1)^4 &= (a', c' \,\breve{a}\, x, 1)^2,
\end{aligned}
$$

and the equations give

$$
\begin{vmatrix}
a' & b' - c' & \mu & a \\
b' + c' & b' - c' & \mu & ba \\
a' + b' & a' - b' & ca\mu & a \\
b'' & c'' & a'' - c'' & 1
\end{vmatrix} = 0;
$$

that is

$$
\begin{aligned}
(a'c'' - b''^2 + a''b'^2)(a'c''(c' - b')) - 2b'^2 & \\
& - (c'^2c'' - b'c''c''c'' - b'^2) \\
& + a'c''(a''a' - a''b'c' - a'b'c) \\
& + a'c''(a'c''^2c''c'' - b'^2c'') \\
& + a'c''(a'b'^2 + a''b'') \\
& = 0,
\end{aligned}
$$

which is

$$
\begin{array}{rll}
(c'c'' - b''^2)(a'c'' - b''^2) = \frac{1}{4}b'^2 & \quad a, \quad b, & \qquad c \\
+ca''c''^2 & \quad a', \quad b', & \qquad c' \\
+ca'c''^2 - 2(c''^2 - c'') & \quad a'', \quad b'', & \qquad c'' \\
+c''(c''^2 - c'') & & \\
+2bc(b - c) & & \\
+2bc(b - c)(c'' - c''^2) & & \\
+4b'c''(b'c'^2 - c'^2) & & \\
+8c'(b'^2 - c''^2) + d\theta & = 0. &
\end{array}
$$

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If the original matrix be symmetrical, that is

$$(\beta - b')^2 + (a\mu - b')(c\mu - f'') = \begin{vmatrix} a & b & f \\ h & b & f \\ p & f & c \end{vmatrix}, \quad \text{this is}$$

$$\begin{aligned} c(\beta - b')^2 + (a\mu - b')(c\mu - f'') = -\partial F & \begin{vmatrix} a & b & g \\ h & b & f \\ p & f & c \end{vmatrix} \\ & + h^2(-c\eta) \\ & + b^2(-c\mu - g^2 - 2fh) \\ & + f^2(-c\nu) \\ & + 2h(c^2g + ph) \\ & + 2f^2(ch + ck) \\ & + 2hc(fp + ch) + 6h = 0, \end{aligned}$$

that is

$$\begin{aligned} &(b - g)(b - b')^2 + (a + b'')(c\beta + p) + 2(cp^2 + ch^2 - 2bfh) \\ &\qquad = 2F(abc - aef^2 - bg^2 - ch^2 + 2fgh) + b'(4h^2 - bf^2 + bh) - cf'' + b^2 = 0, \end{aligned}$$

multiplied by

$$\theta + b - g = 0,$$

viz. the equation in θ is

$$(\theta + b - g)(b'\theta^2 + (b - c)\theta + (4fh - bcb + hc(2g - b) - c)) + (b'^2 - cab) + 2(cb^2 + ch^2 - 2hf)\beta = 0.$$

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On the Inversion of a Quadric Surface.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. (1871), pp. 283–288.]

THE inversion to be considered is that by reciprocal radii vectors, viz. if x, y, z are rectangular coordinates, and $r^2 = x^2 + y^2 + z^2$, then x, y, z are to be changed into $\frac{x}{r^2}$, $\frac{y}{r^2}$, $\frac{z}{r^2}$. But it is convenient to introduce the homogeneity a fourth coordinate w, $= 1$; and the change then is x, y, z into $\frac{xw}{r^2}$, $\frac{yw}{r^2}$, $\frac{zw}{r^2}$.

Starting from the quadric surface

$$(a,\, b,\, c,\, d,\, f,\, g,\, h,\, l,\, m,\, n \between x,\, y,\, z,\, w)^2 = 0,$$

or, what is the same thing,

$$\begin{aligned}&(a,\, b,\, c,\, f,\, g,\, h \between x,\, y,\, z)^2\\ &\quad + 2w(lx + my + nz)\\ &\quad + dw^2 = 0,\end{aligned}$$

the equation of the inverse surface is

$$\begin{aligned}&w^2(a,\, b,\, c,\, f,\, g,\, h \between x,\, y,\, z)^2\\ &\quad + 2w(lx + my + nz)r^2\\ &\quad + dr^4 = 0,\end{aligned}$$

where $r^2 = x^2 + y^2 + z^2$; the inverse surface is then a quartic having the nodal conic $w = 0$, $x^2 + y^2 + z^2 = 0$ (circle at infinity); and having the node $x = 0$, $y = 0$, $z = 0$ (the origin of inversion); or say it is a nodal binodular quartic surface, or nodal anallagmatic quartic.

For x, y, z write $a = \frac{1}{4}ar$, $y = \frac{1}{4}\beta r$, $z = \frac{1}{4}\gamma r$, and put for shortness

$$\begin{aligned}l\alpha + m\beta + n\gamma &= A,\\ al + bm + cn &= a',\\ bl + bm + bn &= b',\\ cl + cm + cn &= c',\end{aligned} \qquad (a,\, b,\, c,\, f,\, g,\, h \between l,\, m,\, n)^2 = J,$$

then

$$r^2 \text{ becomes } r^2 = \frac{w^2}{r^2} + \tfrac{1}{4}\frac{w}{r}.$$

$$lx + my + nz = \tfrac{1}{4}\frac{w}{r}A,$$

$$(a, b, \ldots \between x, y, z)^2 = (a, b, \ldots \between \alpha, \beta, \gamma)^2 w^2 - (ax + by + cz)\tfrac{w}{r} + \tfrac{1}{4}J\frac{w^2}{r^2}.$$

Hence the equation is

$$d\left\{r^2 - 2r\frac{w}{r} + w^2\left(\tfrac{1}{4}\frac{a^2}{r^2}\right)\right\} - \tfrac{2mwr}{r} + \tfrac{w^2}{r^2}\left(\tfrac{1}{4}\frac{A}{r}\right)$$

$$+2\left\{aw - \tfrac{w^2 r}{w} + \tfrac{1}{4}A\frac{w^2}{r^2}\right\}\left(w + \tfrac{1}{4}\frac{w}{r}\right)$$

$$+w^2\left[(a, \ldots \between \alpha,\, \beta,\, \gamma)^2 - (ax + by + cz)\right] = 0;$$

viz. arranging and reducing, this is

$$\frac{dr^2}{w^2}+w^2\left\{-\tfrac{1}{4}\frac{A}{r^2}r-\tfrac{w^2r}{r}-(a,\ldots\breve{\ }x,y,z)^2\right\}+w\left\{\frac{2a}{r}-\frac{1}{4}(ax+by+cz)\right\}+w^2\left\{-\tfrac{1}{2}\frac{2}{r^2}+\tfrac{1}{4}A\frac{w^2}{r^2}\right\}=0;$$

and we may without loss of generality assume

$$\begin{aligned}
-\frac{w^2}{r^2}+f-0, &\quad \text{this is } d\,f-mn=0,\\
-\frac{a^2}{r^2}+g-0, &\quad e-mn=0,\\
&\text{etc.}
\end{aligned}$$

The equation then is

$$\begin{aligned}&\frac{r^4}{w^4} + w^2\left\{-\tfrac{1}{4}\frac{a^2}{r^2}(x^2+y^2+z^2) + \left(\tfrac{1}{4}\frac{f^2}{w}\right)y^2 + \cdots\right\}\\ &\quad + w^2\left\{\tfrac{2a}{r} - \tfrac{1}{4}(ax+by+cz)r\right\}\\ &\quad + w^2\left\{-\tfrac{1}{2}\frac{2}{r^2} + \tfrac{1}{4}A\frac{w^2}{r^2}\right\}.\end{aligned}$$

Write

$$ad - D = a'd, \quad bd - m' = b'd, \quad cd - n' = c'd.$$

We have

$$a = al + bm + pn = pa = a'l + \tfrac{bm^2}{d} + \tfrac{ln^2}{d} = \tfrac{1}{d}(ad - D + b),$$

that is

$$a = b'l + \frac{la}{d},$$

and similarly

$$b = a'l + \frac{ma}{d}, \quad c = a'l + \frac{na}{d}.$$

Hence also

$$A = la' + mb' + nc' = \frac{a'}{d}.$$

and the equation is

$$\begin{aligned}&\frac{aw^4}{r^4} + w^2\left\{-\tfrac{1}{4}\frac{a^2}{r^2}x^2 + c'^2 + \left(\tfrac{1}{4}\frac{f^2}{w}\right)y^2 + \cdots\right\}\\ &\quad + w^2\left\{-\frac{b'}{r}cx + \frac{c'^2}{r^2}cx - \frac{m^2r^2}{r^2}\right\}\\ &\quad + w^2\left\{\frac{1}{4d}(b'^2 + n'^2)xw + n'(cr^2 + wc^2)\right\}.\end{aligned}$$

This is Kummer's form, viz.

$$r^4 = ka^2(n^2x^2 + \beta y^2 + \gamma z^2) + lawr + 2w(ax + by + cz)r,$$

where

$$\begin{aligned}
-4a, +k^2 &= -\tfrac{1}{4}\frac{a^2}{r^2},\\
-k\beta, + = &= -\tfrac{1}{4}\frac{b^2}{r^2},\\
-k\gamma, + = &= -\tfrac{1}{4}\frac{c^2}{r^2},\\
-k\lambda, &= \frac{fe}{da}(ba' + ab'c) + \beta\frac{m}{d}.
\end{aligned}$$

Hence Kummer's equation

$$\beta_4 = k\lambda = \frac{a^2}{\lambda + a} = \frac{b^2}{\lambda+\beta} = \frac{c^2}{\lambda+\gamma},$$

or say

$$6k\lambda + 6k\lambda^2 = \frac{216a^2}{4\lambda + 4a}\cdot\frac{4aa^2}{4\lambda+4\beta}\cdot\frac{216c^2}{4\lambda+4\gamma},$$

becomes

$$6k\lambda^2 = \frac{4}{d^2}(b'^2a + a'b'^2 + c'^2) - \frac{a'^2}{a^2\left(\frac{a^2}{d^2} - \frac{a^2}{d^2} + 4a\right)} - \frac{4ab'^2}{a^2\left(\frac{b^2}{d^2} - \frac{b^2}{d^2} + 4b\right)} - \frac{4ac'^2}{a^2\left(\frac{c^2}{d^2} - \frac{c^2}{d^2} + 4c\right)},$$

which is satisfied by $4\lambda = -\frac{1}{4}\frac{a^2}{d}$. Writing therefore

$$6\lambda + \tfrac{1}{4}\frac{a^2}{d} = -\frac{d}{a},$$

that is

$$\begin{aligned}
\delta\lambda &= -\frac{2D}{a} - \frac{a^2}{d^2},\\
6a\lambda^2 &= \frac{4D^2}{d^2} + \frac{4Da}{a^2} + \frac{a^4}{d^4};
\end{aligned}$$

the equation is

$$\frac{d}{r^4} = \frac{d}{w^4} - \frac{4}{d^2}b'a' + a'b'^2 + c'^2 = \frac{4aa^2}{d^2\left(\frac{a^2}{d^2} - \frac{a^2}{d^2} + 4a\right)} - \frac{4ab'^2}{d^2\left(\frac{b^2}{d^2} - \frac{b^2}{d^2} + 4b\right)} - \frac{4ac'^2}{d^2\left(\frac{c^2}{d^2} - \frac{c^2}{d^2} + 4c\right)},$$

viz. this is

$$Da^2 + aB - Da = Da^2 - \tfrac{2D^2}{d} - \frac{aa'a^2}{d^2\left(\frac{a^2}{d^2} - \frac{a^2}{d^2}\right) + 4a} + \cdots$$

which is of course satisfied by $\theta = 0$. Moreover the derived equation

$$= a - 2D\delta = \frac{Da^2}{d(b+d)^2 a^2 c'^2 \left(\frac{a^2}{d}\right)^2} + \frac{aa^2c^2}{d(b+d)^2} + \frac{aa'a^2}{d(b+d)^2}$$

is also satisfied by $\theta = 0$, so that this is a double root. The equation in fact is

$$\begin{aligned} &(P_a^2 + Pa - Da)(a^2 + a^2c'^2(B+a)(d+b)(d+c)) \\ +&(Pa + db'^2(B+c)(d+a)(d+c) - 3a'^2c^2(B+a)(B+b)(d+c)) = 0, \end{aligned}$$

or, expanding and dividing by θ^2, this is

$$\begin{aligned} &(Pa^2 + a)(d+a)(d+b)(d+c) + 2a'b'(d+a)(d+b)(d+c)c \\ &- (Pb'^2 + ac'(b+c)c)(a' + c'(d+a)(d+b)(d+c)) = 0. \end{aligned}$$

[499]

On the Theory of the Curve and Torse.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. *(1871), pp. 294–317.]*

THE fundamental relations in the theory of the Curve and Torse were first established in my "Mémoire sur les Courbes à double courbure et sur les Surfaces développables," *Liouv. t. x.* (1845); [30] see also *Cambr. and Dubl. Math. Jour., t. iv.* (1850), [85], viz. I showed that the systems (m, r, β, h, x, y) and (r, n, α, g, y, β), the notation is anallagmatically explained, each of them satisfied the Plückerian relations. An additional set of equations giving the values of g, β, h, C, g', h' was furnished by Dr Salmon's "Theory of Reciprocal Surfaces." *Trans. R. I. Acad., t. xxiii.* (1847); see also his *Solid Geometry.* The theory as thus established is complete in itself, but the account of certain singularities α, H, G; the singularity α was first considered in my paper "On a special sextic Developable," *Quart. Math. Jour., t. vii.* (1865), [372], (there called θ), and I afterwards endeavoured to take account of the remaining singularities H, G. I was in correspondence on the subject with Prof. Cremona, and the discovery of the complete forms of several of the formulæ is due to him.

There has recently appeared a very valuable memoir by M. Zeuthen, "Sur les singularités ordinaires d'une courbe gauche et d'une surface développable," *Annali di Matem. t. iii.* (1869), in which, however, from considerations the singularity α and does not throughout attend to H, G.

I propose in the present memoir to reproduce and develope the whole theory.

Explanations and Notation.

1. We have a singly infinite series of points, lines, and planes; viz. each line passes through two consecutive points and lies in two consecutive planes; each plane passes through three consecutive points and contains two consecutive lines; each point

is the intersection of three consecutive planes and of two consecutive lines. The points describe and the lines envelope a *curve*; the lines describe and the planes envelope a *torse*; the entire system of points, lines, and planes is thus the system of the curve and torse. The curve is the edge of regression or cuspidal curve of the torse; in regard to the curve the points are points (or incunabula), the lines tangents, and the planes osculating planes; in regard to the torse the points are points on the edge of regression, the lines generating lines, and the planes tangent planes.

2. Each line of the system is met by a certain number of non-consecutive lines, and the locus of the points of intersection (or say the locus of the intersection of two intersecting non-consecutive lines) is a nodal curve on the torse, or say simply it is the *nodal curve*. The plane containing the two intersecting non-consecutive lines envelopes a torse which is called the *nodal torse*.

3. There is occasion to consider

m, order of the system; this is the number ∞ of points of the system which lie in a given plane; or it is the order of the curve.

r, rank of the system; this is the number of lines of the system which meet a given line. It is thus the class of the curve; and the order of the torse.

n, class of the system; this is the number of planes of the system which pass through a given point. It is thus the class of the torse.

α, number of stationary planes; that is, planes each passing through four consecutive points of the system.

β, number of stationary points, that is, points each of them the intersection of four consecutive planes of the system.

g, number of lines in two planes (that is, lines each of them the intersection of two non-consecutive planes of the system) contained in a given plane; or say, number of apparent double planes of the torse.

G, number of double planes, or *tropes*, of the torse; viz. considering the torse as the envelope of a variable plane, each of these is a position of this plane, equivalent to two; or say the plane in question is a *trope*.

h, number of *lines through two points* (that is, lines each through two non-consecutive points of the system) passing through a given point; or say, number of apparent double points of the curve.

H, number of double lines, or *nodes*, of the curve; viz. considering the curve as the locus of a variable point, if the point in the course of its motion comes twice into the same position, we have then a double point or node.

x, number of *points in two lines* (that is, points each on two non-consecutive lines of the system); or say, number of apparent double points of the curve.

σ, number of stationary lines of the system, that is, lines each containing three consecutive points of the system.

α, number of double lines of the system, that is, lines each containing two pairs of consecutive points of the system.

y, number of points in two lines (that is, points each of them the common intersection of three non-consecutive lines of the system); these are also triple points on the curve.

Y, number of apparent double points of nodal curve.

z, class of nodal curve.

ζ, number of *planes through three lines* (that is, planes each of them through three non-consecutive lines of the system); these are also triple tangent planes of the torse.

y', number of planes of the system each of them passing through a non-consecutive line of the system; these are also tangent planes of the torse, and nodal torse, stationary planes of the latter torse.

Y', number of apparent double planes of nodal torse.

z', order of nodal torse.

x', class of nodal torse.

4. The formulæ thus contains in all the 23 quantities

$$m,\ r,\ n,\ \alpha,\ \beta,\ g,\ G,\ h,\ H,\ x,\ y,\ \sigma,\ \alpha,\ Y,\ z,\ \zeta,\ y',\ Y',\ z',\ x'.$$

My own Plückerian equations, or, say, the Plücker–Cayley equations, establish in all 6 relations between the first 12 quantities, and thus enable the expression of them in terms of any seven, say of

$$m,\ r,\ n,\ G,\ H,\ x,\ x.$$

and the Salmon-Cremona equations then lead to the expressions in terms of these, of the remaining eight quantities y, y, b, g', C, x', g'.

I also consider

D_m, the deficiency of the curve,

D_n, the deficiency of the nodal curve.

I will first consider the equations themselves, and the more algebraical transformations thereof, and afterwards the geometrical theory.

The Plücker-Cayley Equations.

5. These are found (as will be further explained) by considering first the cone, vertex an arbitrary point, which passes through the curve; and secondly, the section of the torse by an arbitrary plane. We have in the two figures

Cone.	Section.
m, order,	n, class,
r, class,	r, order,
$h+H$, double lines,	$g+G$, double tangents,
y, double points,	β, stationary tangents,
$x+\alpha$, stationary planes.	$\alpha+x$, stationary points.

And hence the two sets of quantities respectively are connected by the Plückerian relations, viz. these are

$$\begin{aligned}
r &= m(m-1) - 2(h+H) - 3y,\\
n+6m &= 3m(m-1) - 6(h+H) - 8y,\\
y+a &= \tfrac{1}{2}m(m-1)(m-2) - 9y,\\
&\quad - (m-1)(8h+H) - (8-8)y\\
&\quad - \tfrac{1}{4}m(m-1)(m-2) - 2(8h+H),\\
&\quad - (8+2)y,\\
m &= r(r-1) - 2(g+G) - 3(x+\alpha),\\
B &= 3r(r-1) - 6(g+G) - 8(x+\alpha),\\
b+\beta &= \tfrac{1}{2}r(r-1)(r-2) - 9(x+\alpha),\\
&\quad - (r-1)(8h+H),\\
&\quad - (8-8)(x+\alpha).
\end{aligned}$$

and combining the two systems

$$\beta = a + 2(n - r),$$
$$y = a + (n - r),$$
$$h + H = x + y + (m - n)(n + r$$
$$\tfrac{1}{2}(n-1)(n-5) - 4(g+G) + B = 2(m-1)(n-7) + 6 -$$
$$\tfrac{1}{4}(r-1)(n-7) + 6 - 4(g+G) + (m-1)(n-2) + (8-x)(y-y) + (a-1) - (n+a)$$
$$= 0.$$

6. Taking as data r, m, n, G, H, a, x, we find very easily

$$\begin{aligned}
h &= \tfrac{1}{2}m^2 - 10m - 5n + (n - r) - H,\\
g &= \tfrac{1}{2}n^2 - 10n - 5m + (m - r) - G,\\
\alpha &= 3(n - r) - r - n - 5m - 5a - 5x,\\
\beta &= \tfrac{1}{2}y(n^2 - r) - n - 5n - 5a - 5x,\\
a &= n - 2r + 3x + a,\\
\beta &= a - 2r + 3a + a.
\end{aligned}$$

The Salmon-Cremona Equations.

7. These are

$$\begin{aligned}
a(r-2) &= 2a + 4\beta + g + 8a + 4H,\\
\sigma(r-2) &= 4n - 4\beta + g + 8a + 14H,\\
x(r-2) &= 4n - 4\beta - 4r + 1B - 14H + (3n - 10) + 12H,\\
&\quad + a(8r - 16) - 19a - 2a,\\
y(r-2) &= (m+n)(r-4) - 2(n-r) + 4\beta - 3n - (4r-5)q - (3r-10) + 12H,\\
g + Z &= 9r(r-2) - 8r + 8a + 9b + (4r-16)a - 14a - 12n,\\
\zeta + \beta &= 3r(r-2) - 4r - 4r + 4y(8r-14) + (3r+14)\beta + 12H,\\
\beta' &= z(r-2)y' - y + 24Y'.
\end{aligned}$$

where the second values of y, y' respectively are reduced forms obtained by the aid of the foregoing Plückerian relations.

8. Expressing these in terms of r, m, n, G, H, α, x, we obtain

$$\begin{aligned}
y &= \tfrac{1}{4}m + 12r - 14m - 6x + 8x + 4H,\\
b &= \tfrac{1}{2}(r - 5r - 10r) + 5x(n + 8r + 18r) + 42x + 76r + 24x),\\
&\quad - \tfrac{1}{4}(r - 8r + 11r - 60(r + 25r - 25) + 8(8r + 14 - 5r + 12n + 24G\\
&\quad + (n + 5r + 4r + 6) - 28a - 14r + 240n - 120n - 12x + 6x + 24G).
\end{aligned}$$

$$\begin{aligned}
y' &= b - 2n - 24r,\\
a' &= a + 12r - 14m - 6x + 8a + 4G,\\
b' &= \tfrac{1}{2}(r - 5r - 10x) - 50(n - 8r - 5x) - 42x + 75n + 24G,\\
&= \tfrac{1}{4}(r - 8r + 11x + 60n - 25(c - 7r - 25) - 6(7n - 14)r + 12r - 12x + 24G\\
&\quad + 4(c + 5r + 14)r - 25a + 14H + 24Gn - 120n - 12x + 6x + 24G).
\end{aligned}$$

$$y' = b - n + 2n - 24G.$$

9. We have therein

$$\begin{aligned}
r' &= y - n - 8(n - r) - 4(7 - H),\\
g' &= y' + 6(m - r),\\
y' &= y - (n - r)(n + 2r - 5) + 2(g - H),\\
y &= -\tfrac{1}{2}(n - r)(r - r) + (8 - r)(r - r) + (g - r) - 2(g - H).
\end{aligned}$$

10. Instead of obtaining the above values of y, β, g directly it is convenient to verify them by substitution in the equations from which they were obtained; viz. writing for shortness $a+8(n-r)$ for b, these may be written

$$\begin{aligned}
&(n - r - 2) + 8a + 4H + 8a - 6x + 4H = 0,\\
&- 2a(r - 2) - 8r + 18(r - r) + 2y - 4 - 2(M)r + 2 = 0,\\
&- 2r(r - 2) + 8r - 2r - r) + 4y(r - 4) - 12(8r - 4)x + 4(8 - 14)H - 4x + 11(n - r) - 24H = 0,\\
&- 2r(r - 2) - 4y - 11(8 - 14)H + (5r - 9 - 24H = 0,\\
&- 8r + 16r + 18a(r - 6) - (3r - 4(8 - 18r - 12r + 9r - 25 + 24H = 0,\\
&- 4r(r - 2) - 4y - (r - 6)(r - 4) + 4y(8 - 14)H + 12H + 24H = 0,
\end{aligned}$$

which are to be satisfied by

$$\begin{aligned}
y &= a + 12r - 14a - 6r + 8a + 4H,\\
\beta &= a + n - 2r + 8a + a + a + 4H,\\
\beta &= a + n - 2r + 8a + 4G,\\
g &= b - n + 2n - 24G,\\
H &= a + (g - H),\\
b' &= b + a + b + 8(n - r).
\end{aligned}$$

11. We have, in fact, then

$$\left.\begin{aligned}
&- x + r + 2a\\
&\quad + 2x\\
&\quad + 12n + 4n - 12r + 4n\\
&\quad - 6x\\
&\quad + 8x\\
&\quad + 4H\\
&\quad + ar - 14n + 4n + 12r - 8n - 4n + 4H
\end{aligned}\right\} = 0.$$

secondly

$$\left.\begin{aligned}
&-r^2+r^2\\
&\quad+2r^2-2x\\
&\quad+2r^2-2P\\
&\quad-8r\\
&\quad-12\\
&\quad+4r+12n+4n\\
&\quad-2n\\
&\quad-14H\\
&\quad+7y\\
&\quad+6n+8H-14n-4n-12r+4H\\
&\quad+4\\
&\quad+24H
\end{aligned}\right\}=0,$$

that is

$$-r^2+3r^2-10r-2r^2=-2P^2+2y+6n+6n=0.$$

thirdly

$$\begin{aligned}
&-r^2+5r^2-4r\\
&\quad+r^2-5r^2+4r+2P^2(8r-4r+11(7-)\\
&\quad-4r+16r\\
&\quad+6n-18n-12n\\
&\quad-14n+12n\\
&\quad+8r\\
&\quad-14n-5n-6n+50n\\
&\quad+12n\\
&\quad-10n+16n\\
&\quad-14n\\
&\quad+9(7-8n-8r)-8(2H)+25H\\
&\quad+r^2-9r^2+11r+60n-8H^2-12n-25H+(-r^2-9r+11r+60n-14)\beta-36r+8=0.
\end{aligned}$$

$\therefore -r^2+5r^2-4r$ is here verified, where:

$$\begin{aligned}
&+r^2-6n+12n^2+11r+60n-F'(-r^2-r+16r)+r^2-18+70n^2-16n^2-25H\\
&\quad-14H+(-r^2-9r+11r+60n-14)\beta-36r+8H=0
\end{aligned}$$

and lastly

$$
\begin{aligned}
&- 24P \\
&\quad - 4mr \\
&\quad + 4m + 12n + 36n + 12n \\
&\quad + 8G \\
&\quad + r^2 - 5r^2 + 11r + 60n - F'(7m - 9m + 12n) \\
&\quad + 12n - 14n - 14n - 12n - 24n \qquad\qquad\qquad = 0. \\
&\quad - r^4 - 6r^2 - 11r + 60n - F(7 - 4)x \\
&\quad - 144x \\
&\quad + r^4 - 6r + 18n + 4n + 12n \\
&\quad + 12n + 24n + 24G \\
&\quad - 8(2r - 16r) + 8(2r - 16) + 8r + 24H \\
&\qquad\qquad - 4r + 14r + 12 - 8r + 8(r - 24)H \\
&\qquad\qquad\quad + (2P - 6 - 6r - 6n)x \\
&\qquad\qquad\quad + 12n - 16x \\
&\qquad\qquad\quad - 80n \\
&\qquad\qquad\quad + 80n + 8r + 4n + 24G = 0,
\end{aligned}
$$

that is

$$
\begin{aligned}
&2\beta - 6r + 2x - 6r + 4(r - 24)H - 18n + 14H + 24G \\
&\quad + r^2 - 8r + 8n + 12n + 14)\beta - 12n - 16x - 36n + 24G \\
&\quad + r^2 - 9r + 14n - 14n + 22 - 4n + 4n + 24G = 0,
\end{aligned}
$$

which completes the verification.

12. The deficiency of the curve is given by the equation

$$2D_m = r - 2m + 2 + \beta,$$

or substituting for β its value $= n - 2r + 8x + \alpha$, this is

$$2D_m = n + n - 2r + 2 + a + x.$$

The deficiency of the nodal curve z is given by

$$2D_z = z - 2x + 2x = (h - r)(g - 1) - 2(g + 8H).$$

which, substituting for z, x, and y, becomes

$$\begin{aligned} e + 8x = 8r - 8n + 8r - 24a - 12x = 8x = G - 14,\\ \tfrac{1}{4}(r - 8x + 11x - 60n - F(7m - 9n + 12n) = 8H + 12,\\ = r(8 - 1)D_n = (r - 1)(g - 1) - 2g - 8H.\end{aligned}$$

respectively, become

$$2D_x = r - 14a; 2D_x = a + a + r(z + 8x) + r(8H) + r(z + 8H) - 8\beta - 14H,$$

whence, also, writing herein $a + a + a + 2D + a - 2r$, we have

$$2D_a = (m - 16)D_a = (r - 1)(r - 5) - 8a - 2y' - 12H.$$

a relation between the two deficiencies.

Geometrical Theory of the foregoing Relations.

13. In considering the geometrical theory, we have to speak of the original curve, or curve of the system, and also of the nodal curve; it will be convenient to call them the curve m and the curve n respectively. I speak of the torse absolutely, to signify the torse of the system, as in what follows there is not the inferior expression to make of the nodal torse. I speak also of a torse, meaning thereby any one of the stationary planes, the number of which is $= n$; and so a line, meaning the line that with the curve m, or, in like proof, that a plane, meaning thereby an osculating plane of the curve m; with the indication of the stationary planes belong to the torse; thus we have a stationary tangents equal to β, a tangent plane equal to $n - 2r + 8x + a$ for the planes belonging to the torse.

In particular the points are now the points of the system which lie in a given plane; the lines n are the lines which meet a given line; the planes α are the planes which pass through the other line.

14. Observe that the expressions, the geometrical meaning of which is at one significant, there are some such expressions which have only a relative significance, viz. in regard to the system considered in connexion with a given point, line, or plane, as the case may require. Thus the expression, the line g, must be understood of the system in connexion with a given plane, viz. signify, the plane g must be understood, the plane g would of course mean the planes of points of the system belonging to the line g.

In particular the points α are the points which lie in a given plane, the lines α are the lines which meet a given line; the planes α are the planes which pass through a given point. There are two such forms, viz. the radial $\sqrt{p^2}\ rp^2$ is may be either α or α, hence the same is $= 0$.

15. But it is to be further shown that the line α is torsoidal, each sheet of the surface being touched along the line by the plane $\alpha = 0$; we have to show that the

16. The mode of obtaining these equations has already been indicated. We in fact consider the system in connexion with an arbitrary point, and with an arbitrary plane. The point in the vertex of a cone passing through the curve m, and this cone is of the order m, the class r, with $h + H$ double lines, β stationary lines, $y + a$ double tangent planes, and $x + \alpha$ stationary planes; viz. the order of the cone is equal to the number of lines in which this is intersected by a plane, the tangent planes; and so on. Each of these is in fact, for any point of the line we have σ or three points of the curve m, equal to the number of the cone $= m$, equal to the number of stationary planes from the point to it. Hence considering the cone, $y + a$ is the number of apparent double planes, equal to the number of double points A, B, C shall each of them a square; but this being so, the element at D will not be in general a square, and it is only for certain surfaces that it is so. But if (whatever the curves

[500]

On a Theorem relating to Eight Points on a Conic.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. *(1871), pp. 344–346.]*

THE following is a known theorem:

"In any octagon inscribed in a conic, the two sets of alternate sides intersect in the 8 points of the octagon and in 8 other points lying in a conic."

In fact the two sets of sides are each of them a quartic curve, hence any quartic curve through 13 of the $8 + 8$ points passes through the remaining 8 points; but the original conic together with a conic through 8 of the 8 new points form together such a quartic curve; and hence the remaining 5 of the new points (inasmuch as obviously they are not situate on the original conic) must be situate on the conic through the 5 new points; that is the 8 new points must lie on a conic.

We may without loss of generality take $(x,\ y,\ z)$, $(x_1,\ y_1,\ z_1)$, $\ldots$, $(x_8,\ y_8,\ z_8)$, as the coordinates $(x,\,y,\,z)$ of the 8 points of the octagon, and obtain hereby an θ posteriori verification of the theorem, by finding the equation of the conic through the 8 new points: the result contains cyclical expressions of an interesting form.

Calling the points of the octagon 1, 2, 3, 4, 5, 6, 7, 8, the 8 new points are

12.45, 23.56, 34.67, 45.78, 56.18, 67.23, 78.34, 81.45.

viz. 12.45 is the intersection of the lines 12 and 45; and so on. The 8 points lie on a conic, the equation of which is to be found.

The equation of the line 12 is

$$x : (a_1 + a_2)y + a_1a_2z = 0,$$

or as it is convenient to write it

$$x - (1 + 2)y + 12.z = 0,$$

viz. 1, 2, &c., are for shortness written in place of a_1, a_2, &c. respectively.

The coordinates of the 12.45 are consequently proportional to the terms of

$$1,-(1+2),\ 12,$$
$$1,-(4+5),\ 45,$$

or say they are as

$$12(4+5)-45(4+2):\ 12-45:\ -1+2-4+5.$$

The equation of the line (12.45)(23.56) which joins the points 12.45 and 23.56 then is

$$\begin{vmatrix} x, & y, & z \\ 12(4+5)-45(1+2), & 12-45, & -1+2-4+5 \\ 23(5+6)-56(2+3), & 23-56, & -2+3-5+6 \end{vmatrix}=0,$$

where the determinant vanishes identically if $2=5$ or $1-8=4-6$; in fact thereby becomes

$$\begin{vmatrix} x, & y, & z \\ 2'(4+5)-4(2+5), & 4(2-5), & 4'(1-5) \\ 3'(5+6)-5(2+6), & 5(3-6), & 5'(2-6) \end{vmatrix}=0,$$

which is $=0$; the determinant divides therefore by $2-5$; the coefficient of x is easily found to be

$$=(2-5)\{12(4+5)-45(1+2)+23(5+6)-56(2+3)\}$$

and so for the other terms; and omitting the factor $2-5$ the equation is

$$\begin{aligned} &x\{12-23+84-45+56-1+2-5+6\} \\ &\quad -y\{12(4+5)-45(1+2)+23(5+6)-56(2+3)\} \\ &\quad +z\{-(1+2)(2+8)-(2+8)(5+6)+(5+6)-(4+5)\}=0. \end{aligned}$$

There is now not much difficulty in forming the equation of the required conic; viz. this is

$$\begin{aligned} &(2-5)\{x(2+7)y+Cx\} \\ &\quad \times\ (-8\{(12(4+5)-45(2+2)\} \\ &\quad +(-3)\{12(4-5)-25(8-2)\} \\ &\quad +(-3)\{12(2-8)-25(1-2)\} \\ &\quad +(-3)\{56(4-5)-12(1-2)\}) \\ &=0. \end{aligned}$$

In fact taking written with an indeterminate coefficient λ, say, for shortness, then

$$67.12.45)(34.56) - (12)(56)(56)(83.45) + (-2)(45)(82.56)$$
$$= 0,$$

in the general equation of the conic through the 5 points 12.45, 23.56, 67.12.45, and 23.56; and by making the conic pass through it gives the equation through the 8 points. I succeeded in finding the value $\lambda = \dfrac{2-5}{8-6}$, so that the conic in question passes through 5 of the 8 points, and is therefore by the theorem the conic through the 8 points. But as there follows from the equation the conlamus C, D, D, including the points $C, D, D, D, D, D, D, D, D, D, D, D, D, D, D, D, D, D, D, D, D, D, D, D, D$ becomes

$$\frac{a^2 - 3z^2 + 11(a+5)(8-5)(8-6)(4-6)(4-7)(7-6)x^2}{a^2 - 5z^2 - 4z^2 = 0}.$$

those of the other terms are somewhat less difficult; the coefficient of y deserves attention as $84 - 5, = 30$. In the coefficient of y there is no destruction, and this contains therefore 13×4, $= 36$ terms.

[501]

Review. Tables de Logarithmes vulgaires à dix décimales construites d'après un nouveau mode par S. Pineto.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. (1871), pp. 375–376.]

Ten tables containing 36 pages—the principal one being a table in 44 pages, the 22 left-hand pages containing the 10 figure logarithms of the 10000 numbers from 1,000,000 to 1,010,000, and the 22 right-hand pages the proportional parts. 01, 02, ... 09 of the differences. A like table 100,000 to 999,999 would occupy 2000 pages. By means of an auxiliary table of 5 pages, and of a slight increase of the necessary calculations, the table of 44 pages does the work for the 10 figure logarithms of all numbers up to 1,000,000. This is the principal feature, and (so far as I am aware) of a line, meaning the first eight are between 1000 and 09474 (being the limits of the first five digits of the logarithms in the principal table), the auxiliary table by means of the third figures gives 30, adding $\log\frac{1}{2}$ to B, we have a logarithm included in the limits of the principal table, and seeking for the corresponding number, this is $= \dfrac{N}{B}$ number having B for its logarithm; that is, the required number is $= M$ times the number obtained as above. Of course as regards the principal Table, the proportional parts are employed in the usual manner; the tabulation of them to hundredths (instead of tenths) facilitates the interpolation; for better securing the accuracy of the last figure, directions are given in regard to the 11th and 12th figures. An example of the determination of a logarithm is as follows:

$$
\begin{aligned}
\pi &= .314159\,26536 \\
M &= .32 \\
N\pi &= .100526\,48152 \\
\log \tfrac{1}{M} &= +.49485\,00216.88 - 10 \\
\log &= +.20020\,26579.73 \\
64 &\quad 2764.80 \\
91 &\quad 89.21 \\
55 &\quad 22 \\
\log &= .49714\,98726.88
\end{aligned}
$$

(correct value of last two figures = .94).

The labour saved by the small field of the Tables goes far to balance that occasioned by the additional steps in the calculation.

[502]

On the Surfaces divisible into Squares by their Curves of Curvature.

[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 8, 9. Read December 14, 1871.]

GEOMETRICALLY, the theorem may be stated as follows:—Consider any two curves of curvature AB, CD of one set, and any two AC, BD of the other set, as shown by the continuous lines of the figure: drawing the consecutive curves as shown by dotted lines, the curve consecutive to AB at an arbitrary (infinitesimal) distance from AB, the other three curves may be drawn at such distances that the elements at A, B, and C shall be each of them a square; but this being so, the element at D will not in general be a square, and it is only for certain surfaces that it is so. But if (whatever the curves

[Figure: net of curvature lines on a surface — diagram omitted]

of curvature AB, CD, AB, BC may be) the element at D is a square, then it is clear that the whole surface can be, by means of its curves of curvature, divided into infinitesimal squares.

Analytically, if for a given surface the equations of the curves of curvature are expressed in the form $h = f(x, y, z)$, $k = \phi(x, y, z)$, then the coordinates x, y, z can be expressed each of them as a function of the parameters h, k, and we have for the element of distance between two consecutive points on the surface

$$ds^2 = dy^2 + dz^2 = Adh^2 + Cdk^2,$$

where A, C are in general each of them a function of h and k. The condition for the divisibility into squares is that the quotient $A \div C$ shall be of the form function $h \div$ function k.

It was shown by M. Bertrand that, in a triple system of orthotomic isothermal surfaces, each surface possesses the property in question of divisibility into squares by means of its curves of curvature. But in such a triple system, each surface of the system is necessarily a quadric; so that the theorem comes to this, that a quadric surface is, by means of its curves of curvature, divisible into squares. The analytical verification is at once effected; taking the equation of the surface to be

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 1,$$

then the expressions for the coordinates in terms of the parameters h, k of a curve of curvature are

$$\begin{aligned} x^2 &= \frac{a(a+h)(a+k)}{(a-b)(a-c)}, \\ y^2 &= \frac{b(b+h)(b+k)}{(b-a)(b-c)}, \\ z^2 &= \frac{c(c+h)(c+k)}{(c-a)(c-b)}, \end{aligned}$$

and we have

$$A(dx^2 + dy^2 + dz^2) = (h-k)\left[\frac{h\,dh^2}{(a+h)(b+h)(c+h)} + \frac{k\,dk^2}{(a+k)(b+k)(c+k)}\right],$$

so that $A \div C$ is of the required form.

But there is nothing to show that the property is reduced to quadric surfaces, and the question of the determination of the surfaces possessing the property appears to be one of considerable difficulty, and which has not hitherto been examined.

[503]

On the Surfaces each the Locus of the Vertex of a Cone which passes through m Given Points and touches $6-m$ Given Lines.

[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 11–47. Read January 11, 1872.]

I consider the surfaces, each of them the locus of a vertex of a (quadri)cone which passes through m given points and touches $6-m$ given lines; viz. calling the given points a, b, c, …, and the given lines α, β, γ, …, there are the surfaces in question are:

	Order
$abcdef$,	4
$abcde\alpha$,	8
$abc\alpha\beta\gamma$,	16
$abc\alpha\beta\gamma$,	20
$ab\alpha\beta\gamma\delta$,	24
$a\alpha\beta\gamma\delta\epsilon$,	24
$\alpha\beta\gamma\delta\epsilon\zeta$,	16

I remark that the orders of these several surfaces are in effect determined by the investigations of M. Chasles in regard to the conics in space which satisfy seven conditions. The surface $abcdef$ may be considered as the “Mémoir on Quartic Surfaces,” [441]; and is the same Memoir as that of 4 Mémoir on Quartic Surfaces,” [441]; and is the same Memoir as that of 4 in my “Mémoir on Quartic Surfaces,” but the other surfaces, and also the surfaces $a\alpha\beta\gamma$ and $a\alpha\beta\gamma\delta\epsilon$ may be I think considered as new. Are considered by Dr Hierholzer in his excellent paper “Ueber Kegelschnitte im Raume,” *Math. Annalen, t. ii.* (1870), pp. 568–586, and to him are due the equations given in the sequel for the surfaces $abdef$ and $abdef$; the researches of the present Memoir are in fact a continuation and development of those in the Memoir last referred to.

Table of Singularities, and Explanations in regard thereto.

1. We have on the below-mentioned surfaces respectively certain simple or multiple points, right lines, and curves, as shown in the following Table.

[Table 1: complex tabular array of singularities for the seven surfaces — omitted, see scan p. 100]

2. In the Table, the upper margin refers to the surfaces, and the left-hand margin to the points, lines, and curves situate on these surfaces respectively; the body of the Table showing the number, and in ( ) the multiplicity, of those points, lines, and curves in regard to the several surfaces respectively. Thus, points a; for the surface $abcdef$, $6 \times (2)$, there are 6 such points, each of them a 2-conical (ordinary conical) point on the surface: so $abcde\alpha$, $5 \times (4)$, there are 5 such points, each a 4-conical point on the surface (viz. instead of the tangent plane there is a quartic cone); and so on. Similarly, lines ab (viz. these are the lines joining two points a, b); for the surface $abcdef$, $15 \times (1)$, there are 15 such lines, each a simple line on the surface; surface $abcde\alpha$, $10 \times (2)$, there are 10 such lines, each a double (ordinary nodal) line on the surface; and so on. We have in two places the multiplicity $(2 + 2)$, which refers to a tacnodal line, as presently explained. The corner letters C, P, L denote respectively proper cone, plane-pair, and line-pair, as afterwards explained.

3. The lines and curves referred to in the left-hand margin are:

(1) ab, line joining the points a and b.

(2) α, line α.

(3) $[ab,\ \alpha,\ \beta,\ \gamma]$, pair of lines meeting each of the four lines, or say the tractors of the four lines ab, α, β, γ. As regards the surface $aba\beta\gamma\delta$, the multiplicity is given as $(2+2)$, viz. the line is (not an ordinary nodal, but) a tacnodal line, each sheet touching along the whole line the hyperboloid $a\beta\gamma$.

(4) $[\alpha,\ \beta,\ \gamma,\ \delta]$, tractors of the four lines α, β, γ, δ.

(5) $[ab,\ cd,\ \alpha,\ \beta]$, tractors of the four lines ab, cd, α, β.

(6) abc, def, line of intersection of the planes abc and def.

(7) abc, $de\alpha$, line in the plane abc joining the intersections of this plane by the lines de and α respectively.

(8) abc, $\alpha\beta$, line in the plane abc joining the intersections of this plane by the lines α and β respectively. As regards the surface $abc\alpha\beta\gamma$, the multiplicity is given as $(2 + 2)$, viz. each line is (not an ordinary nodal, but) a tacnodal line, each sheet touching along the whole line the plane abc.

(9) Cubic $abcdef$, cubic curve through the six points a, b, c, d, e, f, common intersection of the cones each having its vertex at one of these points and passing through the others belonging to the lines abc.

(10) Quadriquartic $\alpha\beta\gamma$, $\delta\epsilon\zeta$, intersection of the quadric surfaces $\alpha\beta\gamma$ and $\delta\epsilon\zeta$, that is, the quadric surfaces formed by the four lines α, β, γ, δ and δ, ϵ, ζ respectively.

(11) Excuboquartic $\alpha\beta\gamma$, $\delta\epsilon$, a, quartic curve generated as follows: viz. taking any line whatever which meets the lines α, β, γ, the point of the line meets a meets the lines δ, ϵ in a point P, and a plane through a in the lines δ, ϵ in a point P; the locus the points P in question (theory further considered in the sequel).

Special forms of Quadricones.

4. We have to consider the special forms of (quadri-)cones; these are: 1° The sharp-cone, or plane-pair; that is, a pair of two planes, intersecting in a line called the axis, the vertex being in this case an indeterminate point on the axis. Observe that a plane-pair passes through a point, or touches a line; therefore there are six conditions through such a point; it touches a given line through any line in either of the two planes through it. 2° The flat-cone, or line-pair; this is a pair either of two distinct lines (not meeting through a given point and touching 4 given lines), or, in the same case of two distinct lines), or, in the same case of two coincident lines), or two coincident lines, intersecting in a point called the vertex. And we have to consider in this connexion the 3° axis-pair; that is, a pair of planes intersecting in a line, called the axis, of the axis-pair; viz. the axis is a line meeting a given line.

Singular Lines and Curves on the Surfaces.

5. We may establish θ priori the existence, and even to some extent the multiplicity of the several lines and curves on the surfaces $abcdef \dots \alpha\beta\gamma\delta\epsilon$ Thus:

1. Lines ab: take for the vertex of the cone a point at pleasure on the line ab; the cone passing through b will (per finite pass through a) and the conditions are then that the cone shall touch the line cd, α, β, the order 4; surface $abcde\alpha$, ditto $a\beta\gamma\delta$, surface $a\alpha\beta\gamma\delta\epsilon$ and $a\beta\gamma\delta\epsilon$ surface $abcde\alpha$, and so on for the remaining 10 surfaces; the number of the remaining tangent planes is $= a-4$. Hence as regards the surface $abcde\alpha$, the cone touches a tangent planes is the line ab, by line α, β; in the next case there is besides $a-4=2$, this is besides one line through the point of the cone a, α; in the next case there is besides one line in abc, $\alpha\beta$, γ, but these touch the lines α, β, γ, and the other line in the plane abc, $\alpha\beta$, γ, but these touch the lines α, β, γ.

2. 2° Lines $[ab,\ \alpha,\ \beta]$: taking the vertex in the line of these tractors, the cone meets a proper cone, but it is a plane-pair having for axis the tractor: the surface $abc\alpha\beta\gamma$ is a line, and that the surface $a\alpha\beta\gamma$ is a line, and that the surface $a\alpha\beta\gamma$ is a line, also on the surface.

   Surface $abc\beta\gamma$. Cone is a plane-pair; the two planes intersecting in the line ab, and the two planes through the planes the points α, β, γ; the other is through the point: a. The vertex being an indeterminate point on the line: there is an indeterminate point.

   Surface $abc\beta\gamma$. Cone is a plane-pair; the two planes intersecting in the line ab, and passing the one of them through α and the other through β. Hence the line is a double line on the surface.

3. 3° Lines $[\alpha,\ \beta,\ \gamma,\ \delta]$: taking the vertex in one of these tractors, then as in the last case, the cone is either a line-pair having the tractor for its vertex, or else a plane-pair having for axis the tractor; viz. the three cases are *Surface* $a\alpha\beta\gamma\delta$. Cone is a line-pair, the one line being α, β, the other a line in the plane through the tractor and the point a, meeting the lines γ and δ.

   Surface $\alpha\beta\gamma\delta\epsilon$. Cone is a line-pair, one line being α, β, the other a line in the plane through the tractor and meeting each of the lines γ and δ. In each case the line is an indeterminate point.

   Surface $\alpha\beta\gamma\delta\epsilon\zeta$. Cone is a plane-pair, the two planes intersecting in the tractor, and passing through the planes the lines δ and ϵ respectively. Hence the line is a nodal line on the surface.

4. 4° Line abc, def. Cone is a plane-pair; consisting of the two planes abc and def.

5. 5° Line abc, $de\alpha$. There are two cases: *Surface* $abcdef$. Cone is a plane-pair, the two planes intersecting in the line; the one being abc, and the other passing through the points a. *Surface* $abcde\alpha$. Cone is a plane-pair, the two planes intersecting in the line; the one being abc and the other passing through de. In each case the line is a line on the surface simply.

6. 6° Cubic $abcdef$. The cone has its vertex on the cubic which is the locus of the vertices of cones each through the six points a, b, c, d, e, f. Hence the cubic is on the surface $abcdef$.

7. 7° Quadriquartic $\alpha\beta\gamma$, $\delta\epsilon$, ζ. Cone is a line-pair, consisting of two lines through the point of the quadriquartic curve, the one meeting each of the lines α, β, γ, and the other of these meeting each of the lines δ, ϵ, ζ.

8. 8° Excuboquartic $\alpha\beta\gamma$, $\delta\epsilon$, a. Cone is a line-pair, consisting of two lines through the vertex on the excuboquartic, the one being the line meeting the three lines α, β, γ, and the other meeting the two lines δ, ϵ, and passing through the point a.

Mode of obtaining the several Equations: Notations and Formulæ.

6. The equations of the several surfaces are obtained by taking an centre of projection an assumed position of the vertex, and projecting everything upon an arbitrary plane; the projections of the given points and lines are points and lines in the arbitrary plane, and the section of the cone by this plane is a conic; the equation of the surface is then obtained as the condition that there shall be a conic passing through a given points and touching $6-n$ given lines. We obtain at once, as the plane of projection any one of the coordinate planes; or for greater generality, the plane of projection $x=0$, $y=0$, and the centre of projection a point $(x_0,\, y_0,\, z_0,\, w_0)$. The point of the plane $z=0$ corresponding to a point $(x,\, y,\, z,\, w)$ of space is

$$(x_0z - xz_0,\ y_0z - yz_0,\ 0,\ w_0z - wz_0);$$

or, what is the same thing, taking $w=1$, the projection of $(x,\, y,\, z)$ is

$$(x_0 - x,\ y_0 - y,\ 0,\ w_0 - z),$$

which I shall represent by ξ, η, ζ.

7. We have similarly for the coordinates of the points $(a,\, b,\, c,\, w)$, say the points a:

$$\begin{aligned} P &= bz - z_0aw, \\ Q &= bz - x_0cw, \\ R &= wz - z_0cw, \\ S &= ay - bz, \end{aligned}$$

($P_a = b_ay - y_ac$, etc.). We have similarly for the projection of the point on the plane $w=0$, and touched by the conic.

The following notations and formulæ are convenient:

8. $pabc = 0$ is the equation of the plane through the points a, b, c; viz.

$$pabc = \begin{vmatrix} z, & y, & z, & w \\ a_a, & b_a, & c_a, & w_a \\ a_b, & b_b, & c_b, & w_b \\ a_c, & b_c, & c_c, & w_c \end{vmatrix}$$

Of course $pbac = -pabc$, &c. Observe that here, and in the notations which follow, the letter p is used in referring to the coordinate (x, y, z, w), and that the index of p (… when in a given is expressed) shows the degree in these coordinates.

10. $pza = 0$ is the equation of the plane through the point z and the line α; viz.

$$pa = P_a x + Q_a y + R_a w = S_a w,$$

where

$$\begin{aligned} P_a &= b_a y_a - c_a w_a, \\ Q_a &= -a_a, \\ R_a &= a_a y_a, \\ S_a &= a_a c_a - b_a w_a, \\ S &= a_a c_a + b_a c_a; \end{aligned}$$

and (a, b, c, f, g, h) are the coordinates of the line α; observe that $paa = pba$, where these (and (x, y, z, w) of any line are connected by an identity equation.

11. $p^4ab = 0$ is the equation of the quadric surface through the points a, b and the lines α, β; viz.

$$\begin{aligned} p^4ab = {} & (xy - xy)P_a P_b \\ & - (xy - yz)Q_a Q_b \\ & - (yz - zx)R_a R_b \\ & + (zy - xz)S_a S_b \\ & - (a, b, f, g, h\, c)(a + b, e + f, g + h) \\ & + (e, b, f, g, c)\beta(\beta_p, \beta_q, \beta_r); \end{aligned}$$

where

$$abc = \begin{vmatrix} x_a, & y_a, & z_a \\ x_b, & y_b, & z_b \\ x_c, & y_c, & z_c \end{vmatrix}, \text{ etc.}$$

$(a_a, b_a, c_a, f_a, g_a, h_a)$ are, as before, the coordinates of the lines α, β, (a_a, (a_a), …) being the coordinates of the given line p^4ab, &c.

12. It is to be noticed that, writing

$$\begin{aligned} P &= ay = pz + uw, \\ Q &= -bz + ay + tw, \\ R &= -ay + bx + sw, \\ S &= az - by + cz, \end{aligned}$$

viz. $P_a = b_a y + acx$, &c., the line α being $a_a x + p_a y + ry + aw = 0$, $p_a y + ry = 0$, and further that we have identically

$$-pPaQ = L_a x P_a + M_a Q_a + N_a R_a + \Omega_a S_a,$$

where

$$\begin{aligned} L &= (g^2 + 2f^2 + a^2 x + (cf + bg + ah)w + (bc + af) \\ M &= -(cg + e)x + (g^2 + 2f^2 + a^2)y - (ef + g) \\ N &= -a(bx + e)x - (eg + h)y + (g^2 + 2f^2 + a^2)z - (ag + ah) \\ \Omega &= -(cf + bh)x - (ef + g)y - (ag + ah)z + (g^2 + 2f^2 + a^2) \end{aligned}$$

and L_a, c. are the values of L, c. on substituting therein $(a_a, \ldots)$ and $(x_a, \ldots, w_a)$ for the unaccented and accented letters respectively.

13. Observe that we have

$$pabc = (Lpx)(MR - LS) + (\ldots) = -(QR - LR + RS - \ldots) - SP = pbc;$$

and similarly

$$L_a x b_a + l_a y b_a + l_a z c_a + l_a w c_a + l_a u c_a + l_a v c_a + l_a u c_a + l_a v c_a + l_a w c_a = (a, c, e, f, g, h\, b)(g+i)(g+j)(g+l)(g+m)(g+n),$$

$$
\begin{aligned}
&-hM+gN-cb\Omega+(af+by+ch)p=+p_a,\\
&-fL+eM-cP+(af+by+ch)p=+p_a,\\
&-gL+fN-aS+(\ldots)p=+p_a,\\
&-cL-bM+aN+(\ldots)p=+p_a.
\end{aligned}
$$

14. $p^4ab=0$ is the equation of the cubic surface through the points a, b, c and lines α, β (cab the in the line) the points one of p is the equation of a curve b, c, α, β is at most $1+3=5$ points; observe also that the equation determines this cubic surface in a unique manner, taking the lines α, β in the order α, β; if $1+p$ points were given, then there are 3 other points on η, 3 points on ζ, and 3 lines satisfy in like manner the equation of the cubic, surface in question.

15. We have

$$
p^4ab\cdot p^4cd=\begin{vmatrix} a, & b, & e, & e\\ a_a, & b_a, & c_a & \\ a_b, & b_b, & c_b & \\ a_c, & b_c, & c_c & \\ a_d, & b_d, & c_d & \end{vmatrix},
$$

viz. this determinant, equated to zero, gives the equation of the surface.

To prove this, take as before the unaccented letters (a, b, c, f, g, h) to refer to the lines α, and the letters with one, two, and three accents to refer to the lines β, γ, δ respectively. Referring to the foregoing expressions for L, M, N, Ω, and observing that for a point on $L=a^4=b^2+c^2+a^2=b^2$, c., the values for a point on the line β are: $L=a^4b^2c^2$, $\Omega=a^2a^2$. Whence at β to substitute thus exactly the values of L, M, N, Ω respectively; the values are $a^2b^2c^2$, $\Omega=a^2a^2$. And, in the other manner, $L^2=c^4$, $D^4=c^4$, $L_a=L_a$, $M_a=M_a$, and three values satisfy the equation of L, M, N, Ω. In a surface passing through the lines β, γ, δ. Moreover, the surface contains the lines α and the line β is also a triple line.

To show that the surface passes through the line α and that the coefficients of the points α be in b, c, d, the equation reduces itself to a single one in the variables L, M, N, Ω, c. We have, in fact, $a_ax_a+a_bx_b+a_cx_c+a_dx_d-b_ay_a-b_by_b-b_cy_c-b_dy_d=(LP)(LP)(LP)$. And the equation of the surface, writing $a_ax_a+b_ay_a+c_ax_a+d_ay_a=0$, becomes

$$
\begin{vmatrix} a, & y, & e, & e\\ L, & M, & N, & \Omega\\ L_a, & M_a, & N_a, & \Omega_a \end{vmatrix}=0.
$$

$$\begin{aligned}
&- LM + g'N + e''\Omega + (a'f + b'y + ch')p = +p_a,\\
&- fL + eM - cN + (af + by + ch')p = +p_a,\\
&- gL + fM - cN + (cf + by + ch')p = +p_a,\\
&- cL - bM + aN + (\ldots)p = +p_a.
\end{aligned}$$

14. $p^4abc = 0$ is the equation of the cubic surface through the points a, b and the lines α, β, γ; the line b is here a 5 line, b, 4, 9 points: C may be at most 9, but α, β are 5 points respectively; 3 other points on η, 3 on ζ, and 4 on a surface; in like manner, 3 other points on η, 4 on ζ, and 3 on a ζ, η. Observe that the equation determines this cubic in a unique manner.

15. We have

$$pza \cdot p^2\beta\gamma = \begin{vmatrix} x, & y, & z, & w \\ x_a, & y_a, & z_a, & w_a \\ x_\beta, & y_\beta, & z_\beta, & w_\beta \\ x_\gamma, & y_\gamma, & z_\gamma, & w_\gamma \end{vmatrix},$$

$$\therefore \quad pa \cdot p^4\beta\gamma + p^4\beta \cdot p^2a + p^4\gamma \cdot p^2a + p^4a \cdot p^2\beta\gamma = 0,$$

of the six letters (say z, y, z, w, z', z''); but the equation may be written

$$pza \cdot p^2\beta\gamma + p^4\beta \cdot p^2a\gamma + p^4a \cdot p^2\beta\gamma = 0,$$

which is of the order 4 in regard to (x, y, z, w).

16. To avoid confusion with the geometrical term *line*, I will speak (not of the lines, but) of the *rows* of these tables; thus the multiplicity of contains only 1, 2 is in general p a -2, but it is on the second row that the multiplicity changes, and only a row containing only 1, 2 is given by p Zeuthen, but which I have completed so far as regards the double lines as as follows:

[Table 2: "Plane Section of the Torse" — 9-row tabular listing, see scan p. 82]

17. To avoid confusion with the geometrical term *line*, I will speak (not of the lines, but) of the *rows* of these tables; thus the multiplicity of contains the foregoing Plückerian relations. To fix the steps, I propose to discuss in the sequel both these Plückerian relations. To fix the ideas, I shall consider it as made, so that the osculating planes will pass through this line, in respect of the curve a stationary tangent in respect of the torse, c.

20. Next, for the column 2, we have besides $r-2$ stationary points; this is, in fact, plain; the number of tangent planes which pass through a fixed point in the plane is still $= n$. For Class 3, 7, 8, the plane is a stationary planes; viz. a tangent among the tangent planes which pass through a fixed point thereof, and the number of the remaining tangent planes is $n-3$. And so for Classes 4 and 5, the plane contains for n among the tangent planes which pass through the fixed point thereof, and the number of the remaining tangent planes is $n = n - 2$.

21. Next as to the r and a columns.

Case (2). The plane passes through a generating line, and therefore besides cuts the torse in a curve of the order $r-1$; this generating line is a stationary tangent line of the torse, viz. counts r times among the r tangents (in regard to the curve $r-1$) in the plane. The remaining tangents (in regard to the curve $r-1$) are not in general osculating ones of any of the planes; their number is r and there are besides $\beta+1$ stationary tangents. In Case (4) the plane cuts the torse in the generating line counting twice, and in a curve of the order $r-2$; the generating line is a stationary tangent line of the torse, c.

Case (3). The plane cuts the torse in the generating line counting twice, and in a curve of the order $r-2$; the generating line is a nodal curve on the torse, an ordinary tangent at the point of contact with the curve m (so that the number of stationary tangents remains $=\alpha$); and besides cuts the curve in $r-4$ points as in Case 2.

Case (4). The plane cuts the torse in two generating lines, each counting twice, and in a curve of the order $r-4$; each of the generating lines is in regard to this curve an ordinary tangent at the point of contact with the curve; in the remaining points the section consists of an ordinary tangent at the point.

Case (5). The plane cuts the torse in a generating line counting 3 times, and in a curve of the order $r-3$; the generating line is a stationary tangent in regard to this curve.

Case (6). The plane meets the torse in a generating line counting twice, and in a curve of the order $r-2$; the generating line is in regard to this curve a tangent tangent at the point of contact with the curve; counting 4 times, and besides meeting it in $r-6$ points.

Case (7). The plane meets the torse in the generating line counting 3 times, and in a curve of the order $r-3$; the generating line is in regard to this curve a stationary tangent at the point of contact with the curve, counting 3 times, and besides meeting it in $r-6$ points. The whole number of stationary tangents is thus $=a+1$.

Case (8). The plane meets the torse in a generating line counting twice and in a curve of the order $r-2$; the generating line is in respect of this curve a tangent counting 3 times, the points of contact with the curve m, each of these is besides $r-5$ intersections. Moreover, the generating line counting as 2 stationary tangents, the lines is to be $=a+2$.

Case (9). The plane meets the torse in a generating line counting 3 times, and in a curve of the order $r-3$. The generating line is in respect of the curve an ordinary tangent at one of the points of contact with the curve m, viz. the point at which the plane is a tangent plane of the torse; this is, $S=b$ and S is which sections, and it is also a stationary tangent at the other of the points of contact which the plane is a stationary tangent of the curve m, viz. at which point the plane meets a in fact a stationary tangent at the other of the points of contact with the curve m (viz. the point at which the plane is not a tangent plane of the torse).

22. In the foregoing the discussion of Cases (1) and (2) is satisfactory; but in the other cases there is some difficulty which it has not been worth my while to examine. The Salmon-Cremona equations as I have stated them are

$$a(r-2) = 2a + 4\beta + g + 8a + 4H,$$
$$\sigma(r-2) = 4n - 4\beta + g + 8a + 14H,$$
$$\text{etc.}$$

which can be shown to hold; thus considering the equation $z(r-2) = 4n - 4\beta + g + 8a + 14H$, the stationary point only counts here as 4, the stationary point only counts as 4 in regard to the order of the cone, so that the number of the stationary planes is in this case $= 4\beta$; and the case is $= 4\beta - 4\beta + g + 8a + 14H = 4n - 4\beta - 8a + 14H$.

25. We have similarly the following Table:

[Table 3: "Cone through the Curve" — tabular listing, see scan p. 81]

26. The vertex of the cone m at m being any point not on the curve, the cone has for each of the points (1) of the curve, viz. besides the apparent double point and h double points; for each of the points σ a stationary line; for each of the points ζ a 3-pointic intersection with the line a; for each of the points ζ, a 3-pointic intersection where, as before, $(a,\ a,\ \zeta,\ \eta,\ \zeta)$ etc., respectively are referred (from the cone) by aid of the foregoing Plückerian relations.

Geometrical Theory of the foregoing Relations (continued).

27. We have

$$\begin{aligned}
y' &= n - 8(n-r) - 4(g-H),\\
g' &= y' + 6(m-r),\\
y'(r-1) &= y - (n-r)(n+2r-5) + 2(g-H),\\
y &= -\tfrac{1}{4}(m-n)(m-r-r) + (g-r)(g-r) + (g-r) - 2(g-H).
\end{aligned}$$

28. Instead of obtaining the above values of y, β, g directly it is convenient to verify them by substitution in the equations from which they were obtained; viz. writing for shortness $a{+}8(n{-}r)$ for b, these may be written

$$\begin{aligned}
&(n-r-2) + 8a + 4H + 8a - 6x + 4H = 0,\\
&-2a(r-2) - 8r + 4y - 4 - 2(M)r + 2 = 0,\\
&-2y(r-2) + 8r - 4y - 11(8-14)H + (5r - 9 - 24H = 0,\\
&-2y(r-2) - 4y - 11(8-14)H + (5r - 9 - 24H = 0,\\
&-8r + 16r + 18a(r-6) - (3r - 4(8 - 18r - 12r + 9r - 25 + 24H = 0,\\
&-4y(r-2) - 4y - (r-6)(r-4) + 4y(8-14)H + 12H + 24H = 0.
\end{aligned}$$

29. At a point H I consider any section through the arbitrary point and z; in effect any section through a, the line a being a 3-pointic line, a has a 3-pointic intersection (whence is a stationary tangent, and so a curve $a + 2$ on the surface); but it is to be the case is, the line from the arbitrary point to b has a 3-pointic intersection with the torse.

30. At each of the a points the tangent line with the curve $a + 2$ has so simply the cone a.

31. At points of the tangent line H I consider any section through the arbitrary point and z; in effect any section through a, the line a being a 3-pointic line, a has a 3-pointic intersection (whence is a stationary tangent on the surface); but it is to be the case, the line from the arbitrary point to b has a 3-pointic intersection with the torse.

For a point H I consider any section through the arbitrary point and x; in effect any section through a; there is at a triple point and the line through a has then a 3-pointic intersection at a. Hence a line at a to a through a, viz. $= 2.4\beta$ such cuspidal lines through a, the corresponding lines of cusps in question. We have for the number of stationary lines $= 2g$.

32. But it is to be further shown that the curve m is torsoidal, each sheet of the surface being touched along the line by the plane $Z = 0$; we have to show that the

which, of course, satisfied by $\theta = 0$. Moreover the derived equation

$$= a - 2D\delta = \frac{Da^2}{d(b+d)^2 a^2 c'^2 \left(\frac{a^2}{d}\right)^2} + \frac{aa^2c^2}{d(b+d)^2} + \frac{aa'a^2}{d(b+d)^2}$$

is also satisfied by $\theta = 0$, so that this is a double root. The equation in fact is

$$\begin{aligned}(Pa^2 + Pa - Da)(a^2 + a^2c'^2(B+a)(d+b)(d+c))\\ +(Pa + db'^2(B+c)(d+a)(d+c) - 3a'^2c^2(B+a)(B+b)(d+c)) = 0,\end{aligned}$$

or, expanding and dividing by θ^2, this is

$$\begin{aligned}&(Pa^2 + a)(d+a)(d+b)(d+c) + 2a'b'(d+a)(d+b)(d+c)c\\ &\quad - (Pb'^2 + ac'(b+c)c)(a' + c'(d+a)(d+b)(d+c)) = 0.\end{aligned}$$

33. For a point H I consider any section through the arbitrary point and x; in effect any section through a; there is at a triple point and the line through a has then a 3-pointic intersection at a. Hence a line at a to a through a, viz. $= 2.4\beta$ such cuspidal lines through a, the corresponding lines of cusps in question. We have for the number of stationary lines $= 2g$.

[503] (CONTINUED)

On the Vertices of Cones which satisfy Six Conditions, particularly Six Conditions of Contact with a given Surface.

[Continuation from earlier pages, book pp. 117–134.]

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Factor opened upon by $\Delta = XI_0 + YI_4 + ZI_5 + WI_6$, reduces itself for $z = 0$, $w = 0$ to a multiple of W. Considering the factor in the form of a determinant, the result of the operation is

$$\begin{vmatrix} X, & Y, & Z, & W \\ \sqrt{p\alpha\beta\gamma}\cdot p\delta\delta, & x_1, & y_1, & z_1 \\ \sqrt{p\alpha\beta\gamma}\cdot p\delta\delta, & x_2, & y_2, & z_2 \\ \sqrt{p\alpha\beta\gamma}\cdot p\delta\delta, & x_3, & y_3, & z_3 \end{vmatrix} = \begin{vmatrix} . & . & . & . \\ \Delta\sqrt{p\alpha\cdot p\alpha\beta}, & a_1, & b_1, & c_1 \\ \Delta\sqrt{p\alpha\cdot p\alpha\beta}, & a_2, & b_2, & c_2 \\ \Delta\sqrt{p\alpha\cdot p\alpha\beta}, & a_3, & b_3, & c_3 \end{vmatrix},$$

the first term in

$$\begin{vmatrix} . & X & Y & Z & W \\ . & \sqrt{p\alpha\cdot p\alpha\beta} & a_1 & b_1 & c_1 \\ . & \sqrt{p\alpha\cdot p\alpha\beta} & a_2 & b_2 & c_2 \\ . & \sqrt{p\alpha\cdot p\alpha\beta} & a_3 & b_3 & c_3 \end{vmatrix}$$

where the first column may be replaced by

$$-\Sigma\sqrt{I_0 I_3},$$

and the term in question then becomes

$$\sqrt{p\delta\cdot p\delta\delta}\cdot W(-x\sqrt{I_0I_3} + \sqrt{p\alpha\cdot p\alpha\beta})\cdot \text{etc.}$$

if for shortness

$$\left|x_1\sqrt{I_1I_2} + W(-x\sqrt{I_0I_3} + \sqrt{p\alpha\cdot p\alpha\beta})\cdot\text{etc.}\right| = \begin{vmatrix} x_1, & y_1, & z_1 \\ x_2, & y_2, & z_2 \\ x_3, & y_3, & z_3 \end{vmatrix}\cdot\text{abc}.$$

As regards the second term, we have

$$\Delta\sqrt{p\alpha\cdot p\alpha\beta} = \frac{p\alpha\cdot\Delta p\alpha\beta + p\alpha\beta\cdot\Delta p\alpha}{2\sqrt{p\alpha\cdot p\alpha\beta}}.$$

But

$$p\alpha\alpha = x(-p_1a_1) + y(-f_1a_2) + z(-g_1a_3) + w(h_1 - a_4) = h_1(x-a_1) - a_2b_1(x-a_2) - a_3c_1(x-a_3),$$

$$= -p_1(x-a_1) - f_1(y-a_2) - g_1(z-a_3) - h_1(w-a_4),$$

and thence

$$\Delta p\alpha\alpha = a_1(p_1a_1 + f_1a_2 + g_1a_3 + h_1a_4) - p_1x_1 + f_1y_1 + g_1z_1 + h_1w_1.$$

[Sadleirian abc&c].

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with the like formula for $\Delta p\delta\beta$; hence

$$\frac{I_1\Delta p\delta\beta + I_2\Delta p\alpha\alpha}{2\sqrt{I_1I_2}\sqrt{p\delta\delta\cdot p\alpha\beta}} = Ax_1 + By_1 + Cz_1,$$

where

$$A = \frac{1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}}\{f_2I_1(p_1a_1 - h_1W) + f_1I_2(p_2a_2 - h_2W)\},$$
$$B = \frac{1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}}\{g_2I_1(g_1a_1 - k_1W) + g_1I_2(g_2a_2 - k_2W)\},$$
$$C = \frac{1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}}\{h_2I_1(h_1a_1 - l_1W) + h_1I_2(h_2a_2 - l_2W)\}.$$

The term in question is thus

$$\frac{Ax_1 + By_1 + Cz_1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}} = \begin{vmatrix} . & x & y & . \\ Ax_1 + By_1 + Cz_1, & x_1, & y_1, & z_1 \\ Ax_2 + By_2 + Cz_2, & x_2, & y_2, & z_2 \\ Ax_3 + By_3 + Cz_3, & x_3, & y_3, & z_3 \end{vmatrix}$$

viz. replacing the first column by

$$-Ax - By$$
$$\Delta\sqrt{p\delta\beta\cdot p\delta\delta} - Ax_1 - By_1 - Cz_1,$$
$$= (Ax + By)x_1\cdot abc;$$

and we have

$$Ax{+}By = \frac{1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}}\{f_2I_1{\cdot}p_1x{-}f_1g_1(z{-}a_3){+}g_1g_2(g_1a_2{-}h_2W)\}{+}\frac{1}{2}\{h_2I_1(h_2x{-}h_2a_3{-}l_2W)\}W,$$

$$= \frac{1}{2\sqrt{I_1I_2}\sqrt{p\delta\beta}}\{I_1p\delta\delta_2 - I_2HW\}$$

if for shortness

$$M = (-p_2a_1 + f_2a_2)(g_2z + h_2y) + (-p_2a_1 + f_2a_2)(g_2z + h_2y);$$

viz. the whole term is

$$Wx\left(-\sqrt{I_0I_3}\frac{H}{\sqrt{I_1I_2}} + \frac{H}{\sqrt{I_3I_4}}\right)\cdot \text{abc};$$

Hence the first and second terms together are

$$= W\left(-x\sqrt{I_0I_3} + \sqrt{p\alpha\cdot p\alpha\beta} - \frac{H}{\sqrt{I_2I_5}}\right)\cdot \text{abc};$$

viz. this is a multiple of W, which was the theorem to be proved. *[Sadleirian abc&c].*

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Surface chiefly.

32. The equation is

$$\text{Norm}\left(\sqrt{p\alpha\alpha},\ \sqrt{p\delta\beta},\ \sqrt{p\gamma\gamma},\ \sqrt{p\delta\delta}\right) = 0,$$

where the norm is a product of 16 factors, each of the order $\frac{1}{2}$. As before, $p\alpha\alpha = 0$ is the equation of the plane through the point α and the line α; viz. $p\alpha\alpha$ has the value already mentioned.

33. *Investigation.* In the projection, the equation of the conic touching the pro- jection of the lines $\mathfrak{a}, \mathfrak{B}, \eta$ is

$$A\sqrt{p\alpha\beta}\cdot Q\sqrt{P} + R\sqrt{B} + S\sqrt{Q} + R\sqrt{Q} + R\sqrt{S} = C\sqrt{P} + S\sqrt{Q} + R\sqrt{B} = 0;$$

and to make this pass through the projection of the line δ, we must write herein $X : Y : Z = p_3 : g_3 : c_3$. As before, we have

$$P_3X + Q_3Y + R_3Z = m_3(x\cdot P_3 + y\cdot Q_3 + z_3\cdot R_3),$$

$$= (a_1b_2 - a_2b_1)(\text{etc.}),$$

$$= (a_1P_3 + y_2Q_3 + z_3R_3)p\alpha\alpha_3,$$

$$= -\,abc\cdot p\alpha\alpha_3;$$

and so for the other terms; the equation thus is

$$A\sqrt{p\alpha\alpha} + B\sqrt{p\alpha\beta} + C\sqrt{p\gamma\gamma} = 0;$$

or freeing the like equations in regard to the points $\mathfrak{a}, \mathfrak{B}, \gamma$ respectively, and eliminating we have a determinant $= 0$, and then, taking the norm, we obtain the above-written equation of the surface.

34. *Sub-projection.* The equation shows that the surface chiefly shows that (1) The point α is 8-conical: in fact, by the point in question we have $p\alpha\alpha = 0$, $p\alpha\beta = 0$; each factor is O, and the norm is O^8. (2) The line $\mathfrak{a}$ is 4-tuple. To see this, observe that the sixteen factors are obtained by attributing as pleasure the signs $+$, $-$ to the radicals $\sqrt{p\delta\beta}, \sqrt{p\delta\beta}, \sqrt{p\delta\gamma}, \sqrt{p\delta\delta}$; hence there are four factors in which $\sqrt{p\delta\beta}, \sqrt{p\delta\beta}$ have determinate signs, but in which we attribute to the radicals $\sqrt{p\delta\gamma}, \sqrt{p\delta\delta}$ the signs $+$ or $-$ at pleasure. It is to be shown that the four factors each vanish for a point on the line of; that is, on writing therein for $\mathfrak{a}, y, z, w$, the values $a_1, wa_1, \neg wa_1, b_1$. But we then have $\Sigma p_1 b\beta_1 = p\alpha\beta$. And the whole expression thus is *[Sadleirian abc&c].*

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have, as before, $p\alpha\alpha = -z \cdot abc$ and $p\delta\alpha = a \cdot abc$, with the like formula with $\beta\delta$ and γ in place of α. The factor thus becomes

$$\sqrt{-z \cdot abc}\begin{pmatrix} \sqrt{abc}, & \sqrt{abc}, & \sqrt{p\delta\beta} \\ \sqrt{abc}, & \sqrt{abc}, & \sqrt{p\delta\beta} \end{pmatrix}$$

which vanishes, being $= 0$, and the norm is thus $= 0$, viz. the line is 4-tuple. (3) The line δ is 4-tuple: in fact, writing $w = 0$, $w = 0$ for the equations of the line, we have $A_0 = 0$, $b_0 = 0$, $b_2 = 0$, and $g_1g_2 - g_2g_1 = 0$, or say $p\delta\beta = b_1g_3$, $p\delta\beta = b_3g_1$. Hence, writing

$$I = (g - bc)g - (f + bd)g,$$

viz. $I_1 = (g_2 - b_3)g_1 = b_1g_2g_3$, etc. for the conditions of the line δ, etc. in the equations of the line δ, etc. for the conditions $p\delta\delta = b_3g_1$, $p\delta\beta = g_2g_1$, etc. and $a_1a_3 - a_2a_2 = 0$, or any abc, $b_1b_2 = b_2b_1$, etc. Hence, writing

$$I = (-g - bc)g - (f + bd)g.$$

The factor thus is

$$\begin{vmatrix} \sqrt{I_1I_2}, & \sqrt{I_2I_3}, & \sqrt{p\delta\beta} \\ \sqrt{I_1I_2}, & \sqrt{I_2I_3}, & \sqrt{p\delta\beta} \\ \sqrt{I_1I_2}, & \sqrt{I_2I_3}, & \sqrt{p\delta\beta} \end{vmatrix}$$

which vanishes, being $= 0$; there are four such factors, or the norm is O^4; whence the line is 4-tuple. (4) The line $a\ \beta$ is a 4-tuple line. To prove it, take as before $\alpha = w = 0$ as the equation of the plane $\alpha\beta$, and $\xi = 0$, $w = 0$ for the equations of the line in question. We have $h_0 = 0$, $g_1 = 0$, $c_2 = 0$, $b_2 = 0$, $f_2 = 0$, $b_2 = 0$, whence $(\Sigma\, a_2b_3, c_3, w$ as before, $p\alpha\alpha = f_3$, $p\alpha\beta = f_2$, $p\alpha\gamma = -p_3z - h_3W$, etc., and similarly $p\alpha\beta, p\delta\beta, p\delta\gamma$ have I_3 etc., $I_4 = abc$. Also $I_5 = abc$, the factor thus is

$$\begin{vmatrix} \sqrt{I_1I_2}, & \sqrt{I_2I_3}, & \sqrt{p\delta\beta} \\ \sqrt{I_1I_2}, & \sqrt{I_2I_3}, & \sqrt{p\delta\beta} \\ \sqrt{I_1I_2}, & \sqrt{p\delta\beta}, & \sqrt{p\delta\beta} \end{vmatrix}$$

which vanishes, being $= O$; and there are four such factors, or the norm is O^4; viz. the line is 4-tuple. (5) The line $\alpha\beta$ is a 4-tuple line. To prove it, take as before $z = 0$ as the equation of the planes $\alpha\beta$, and $\xi = 0$, $w = 0$ for the equations of the line in question. We have $p_1 = 0$, $f_3 = 0$, $g_3 = 0$, $b_2 = 0$, and similarly $p\delta\delta, p\delta\delta, p\delta\delta = I_3\alpha_1c_3$. Hence, writing for shortness $I = \mathfrak{C}p - g_3p - h_3p$, the factor is

$$\begin{vmatrix} \sqrt{I_3I_2}, & \sqrt{I_3I_3}, & \sqrt{abc} \\ \sqrt{I_3I_2}, & \sqrt{I_3I_3}, & \sqrt{abc} \\ \sqrt{p\gamma}\cdot\sqrt{p\delta} & \sqrt{p\gamma}, & \sqrt{p\delta} \end{vmatrix}$$

which vanishes, being $= 0$ and there are four such factors, obtained by giving to the radicals the signs $+$, $-$ at pleasure; hence the norm is *[Sadleirian abc&c]*.

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Surface chiefly.

35. The equation is

$$\text{Norm}\left(\sqrt{p\alpha\alpha},\ \sqrt{p\alpha\beta},\ \sqrt{p\beta\gamma},\ \sqrt{p\beta\delta},\ \sqrt{p\delta\beta},\ \sqrt{p\delta\beta}\right) = 0,$$

where the norm is the product of 8 factors each of the order $\frac{1}{2}$. As before, $p\alpha\alpha = 0$ is the equation of the plane through the point α and the line $\mathfrak{a}$; viz. $p\alpha\alpha$ has the value previously mentioned. And of course $p\beta\beta = 0$ is the equation of the plane through the point β and the line $\mathfrak{a}$, etc.

36. *Investigation.* In the projection, taking ξ, η, ζ as current line-coordinates, the equation of the conic passing through the projections of the points α, β is

$$\sqrt{(p\alpha\beta + a_1\alpha P + b_2a_1P_1)} + \sqrt{(p\alpha\beta + a_3\alpha P + b_3a_2P_2)} \cdot A\xi + B\eta + C\zeta = 0,$$

where A, B, C are arbitrary constants. To make this touch the projection of the line a, we must write $\xi : \eta = -p_1 : h_1 - h_1$; and similarly,

$$p\delta\xi + p\delta\eta + p\delta\zeta = a_1p\beta\beta\,\xi + b_1\,c_1,$$

$$= a_1p\delta\beta\,\xi + b_2\,c_2,$$

and similarly

$$p\delta\xi + p\delta\eta + p\delta\zeta = -a_1\,p\beta\beta\,\xi.$$

Hence the equation is

$$a\sqrt{p\alpha\beta \cdot p\alpha\beta} + \beta P_2 + \gamma Q_3 + \delta R_3 = 0;$$

or freeing the like equations for the lines $\mathfrak{a}$, η, δ respectively, and eliminating, we have

$$\begin{vmatrix} \sqrt{p\alpha\alpha \cdot p\beta\alpha}, & P_1, & Q_1, & R_1 \\ \sqrt{p\alpha\alpha \cdot p\beta\alpha}, & P_2, & Q_2, & R_2 \\ \sqrt{p\alpha\alpha \cdot p\beta\alpha}, & P_3, & Q_3, & R_3 \end{vmatrix} = 0;$$

which, throwing out a factor a, becomes

$$\sqrt{p\alpha\alpha \cdot p\beta\beta} + \sqrt{p\alpha\alpha \cdot p\beta\beta} + \sqrt{p\gamma\gamma \cdot p\delta\delta} + \sqrt{p\delta\delta \cdot p\delta\delta} \cdot p\beta_3 = 0;$$

or, taking the norm, we have the above-written equation.

37. *Singularities.* The equation shows that (1) The point α is 8-conical point; in fact, for the point in question we have $p\alpha\alpha = 0$, $p\alpha\beta = 0$; and $p\alpha\alpha = 0$; each factor is $= O$, and the norm is $= O^8$. *[Sadleirian abc&c.]* C. VIII.

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(2) The line $\alpha\beta$ is a 2-tuple line. To prove this, we have for the coordinates of a point on the line in question $\alpha_1 + r\alpha_2$, $\eta_2 + r\eta_2$, &c.; the values of $p\alpha\alpha$, $p\beta\beta$ become as below: r does not enter, but similarly for $p\delta\beta$, $p\delta\delta$, &c., so that, omitting the constant factor $\sqrt{r} =$ etc., the result of the operation is

viz. the value of the factor is $|a\sqrt{(fgh)} + g\sqrt{(bdh)} + h\sqrt{(dcg)} + h\sqrt{(cbg)}|$, is the determinant

$$\begin{vmatrix} f, & b, & g, & h \\ . & . & . & . \end{vmatrix},$$

the suffixes in the four lines being $\mathfrak{a}, \mathfrak{B}, \gamma$ respectively. Collecting, this is

$$\begin{gathered}
(\quad + cfgfg - bcfgs + fghfg) \\
(-cefgh + cefgs + pcfsh\,p \cdot bc) \quad + \xi z + b\xi\alpha) \\
(+bcfgh + bcfgs + b\,k(b\,p\,bc)) \quad + \xi\alpha + b\xi\alpha) \\
(-bcfgh + bcfgs + b\,(b\,a)\,p\,bc) \quad + \xi\alpha + b\xi\alpha) \\
+bcgh\,\{w(ax + by + cs) + x\} \\
+\pi(ag\xi\eta + bg\xi z) + ag - dg(ax + by + cs - x) \\
+\pi(\alpha\xi\eta + b\eta\xi)\,(bx + ay + cz)(ax + by + cs + a\,bcc)\}
\end{gathered}$$

or, what is the same thing,

$$AP + BQ + CR + DS = 0,$$

[Sadleirian abc&c.]

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where

$$\begin{aligned}
A &= [-bgh + abfh - bfgs + fcbcw],\\
B &= [-bgh + bfh + bgfh + abfsw],\\
C &= [b\,bgh - bfgh + bgfgs + bcdsw],\\
D &= [-bgh - bcfg - bcfgs + (bcdh + cabsf + abfh)w],\\
P &= \quad x(\quad by + gz + b\,wx),\\
Q &= \ (-bx \quad + bz + hw),\\
R &= \ (g\,x + hy \quad + aw),\\
S &= \ (b\,wx + a\,b\,y + c\,w\,z - bx);
\end{aligned}$$

the right-hand factors vanishing for the values $\alpha_1 + r\alpha_2$, $\eta_2 + r\eta_2$, &c., of the coordinates.

38. It thus appears so far that the factor is $= O$; it is, in fact, $= O^2$, viz. we can show that, operating upon it with

$$\Delta = XI_0 + YI_4 + ZI_5 + WI_6,$$

the value of $\Delta p\alpha\alpha$ is

$$\Delta\sqrt{p\alpha\cdot p\beta\delta} = \frac{p\alpha\cdot\Delta p\beta\delta + p\beta\delta\cdot\Delta p\alpha}{2\sqrt{p\alpha\cdot p\beta\delta}} + \sqrt{p\alpha\cdot p\beta\delta}\cdot\Delta\sqrt{p\alpha\cdot p\beta\delta},$$

where $\Delta\sqrt{p\alpha\cdot p\beta\delta}$ is what $p\alpha$ becomes on writing therein (X, Y, Z, W) in place of (x, y, z, w). Writing as before, for α, β, γ the values $a_1 + r\,a_2$, &c., we have $p\alpha\alpha = -r\cdot abc$, $p\beta\beta = -r\cdot abc$, and putting for shortness

$$-\Pi\,\Theta = -\ r\,\Sigma h\Sigma\xi\Sigma + \cdot\&\text{c.},$$

the expression in question, divided by $\sqrt{-r}$, is

$$= -B\,\sqrt{abc\cdot\Delta\Pi p\delta\beta} + \text{etc.}$$

$$= (B\alpha\cdot p\beta\delta)\sqrt{-r}.$$

where, denoting the determinants

$$\begin{vmatrix} X & Y & Z & W\\ \cdot & \cdot & \cdot & \cdot \end{vmatrix}$$

by $(d,\ b,\ c,\ f,\ g,\ h)$, we have

$$hh = a\Sigma_1 + b\,c_1 + cc_1 - bc_1 + r\,\Sigma_1, + b\,c_1,$$

which is an identity in regard to (x, y, z, w); and the expression is

$$\Delta p\delta\beta = \Delta abc\cdot p\delta\beta, \quad \Delta p\delta\beta = abc\cdot p\delta\beta + r\cdot\Delta\beta\delta\beta,$$

But also $\Delta p\delta\beta = \Delta abc\cdot p\delta\beta - r[(-1-c) + (-1-c) + 2cD\beta] + Bb$, that is,

$$\Delta p\delta\beta = abc(P - r^2(b-c)^2) + DD.$$

[Sadleirian abc&c.] 16–2

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where P, Q, E, R denote $by - gx + ax$, &c., and where, finally, x, y, z, w are to be replaced by $\alpha_1 + r\alpha_2$, &c. Since for these values P, Q, R, S vanish, the expression becomes

$$= -B\alpha(\Delta\Sigma P + \Delta\beta Q + \Delta\gamma R + \Delta\delta S) + AP + YQ + ZR + SW = 0;$$

that is

$$= A(\Sigma P + \beta\Delta Q + \gamma\Delta R) + AP + BQ + CR + DS = 0,$$

and we have, in fact, $P = \alpha\Delta P\beta R = \alpha\Delta P + Q\Delta\Sigma R\beta = D(P + 2\alpha\Delta\Sigma R)$,

$$P + VQ + VR = ba_1, \quad \text{\&c. for the suffix.}$$

$x, y, z, w = -\alpha_1, ba_2, \cdots, \eta_2, ba_2, ba_2, \cdots$, where, finally, x, y, z, w are to be replaced by $a_1 ba_2 + a$, &c. $x, y, z, w = 1, m, n; m, p; n, p; m, p; n, p, m, p, m, p, m$; then, for instance,

$$H = a, X + b, Y + c, Z,$$

where

$$a, b, c = b, c - b, c \quad c, a - c, a; \quad Xy - Y\alpha;$$

and thence

$$H = X, Y, Z; \quad XYy - YZx;$$

$$H = \begin{vmatrix} x, & y, & z \\ b, & b, & b \end{vmatrix}$$

$$= 2ax(aX + bY + cZ)$$

$$= 2\nu a\beta XY.$$

and similarly the factor is $= 0$; the factor is thus $= O^2$; there is only one such factor, and the line ab is double. (2) The line b is an 8-tuple line: in fact, for a point on the line we have $p\alpha\alpha = O, p\delta\beta = O, p\delta\beta = O, p\delta\beta = O, p\delta\delta = O$, and the factor vanishes, being $= O$. Each of the factors is O, and the norm is $= O^8$. 39. (3) The line $[a\delta, x, \beta, b]$ is a double line. To prove this, observe that this for a point on this line we have $p\alpha\beta = O$. Taking as before $z = 0$, $w = 0$ as the expressions of the plane $\alpha\beta$, and as in $\xi = 0, w = 0$ for those of the line $\alpha\beta$, we have $h_2 = 0, a_3 = 0$, $p\beta\beta = 0$, and $p\beta = h\delta = h\delta$, etc. in the expressions of $p\beta\beta\delta$, this, may be written

$$\Sigma\left[(p - hs\delta - (f + b\delta)y\right] h\delta\delta \cdot [k\delta\delta - (a + b\delta)y - (bk + bd)x] + (b\delta b)z];$$

where observe that Σ denotes a sum of three terms of the form

$$b_1.[p_3.\delta_3 z]\beta\beta_2 = h\delta;$$

[Sadleirian abc&c.]

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Adding thereto a fourth term -1, abp, the value of this sum becomes $= ablp^2$, or the sum of the three terms is $= ablp^2 + 1$, abp; where the symbols represent deter- minants. But in each case the determinant $ab\ b\delta = 0$, $b\delta = 0$, since the column b_1, b_2, b_3, the terms of which are each $= 0$. Say $ba_1b_3a_2 = pgh - p_3gh$, where in δ, h the suffixes are at a, δ, b, and in each case appears under a $\surd$, we have $\Sigma\delta_1, abp = \mathrm{pgh}\delta$. And the whole expression thus is

$$\begin{aligned}
&= a^2(pgh - hs\delta gh)\\
&+ p(pgh)\delta(-hs\delta gh) + a\delta gh(hs\delta gh - hs\delta gh)\\
&+ pf(pgh)\delta - hs\delta gh)\\
&+ pgh(-hs\delta gh) - \ a\,gh(-hs\delta gh)\\
&+ pgh(-hs\delta s\delta) - p\delta - hs\delta gh)\\
&+ q^2(\quad -hs\delta - p\delta gh)
\end{aligned}$$

where $pgh\delta$ denotes the determinant $p_3\ g$, h_3, f, the suffixes $\mathfrak{a}, \mathfrak{B}, \gamma, \delta$, in the forms respectively, and so on; the suffixes $\mathfrak{a}, \mathfrak{B}, \gamma, \delta$ are each as $\delta(\delta + \delta - \delta)$ of a sign of δ at pleasure), and the norm is thus $= O^2$.

40. But the line is bicuspid, each sheet of the surface touching along the line in question the hyperboloid $p\beta\beta$. To prove this, write

$$\Delta = XI_a + YI_b + ZI_c + WI_d;$$

we have for the hyperboloid, writing $z = 0$, $w = 0$,

$$\Delta p\beta\beta = a\delta(z = h)\beta, \quad p\delta\beta = c\delta(z = h\delta)y,$$

and so

$$\Delta(\sqrt{p\delta\beta}, p\delta\beta, p\delta\beta, p\delta\delta, p\delta\beta, p\delta\beta, p\delta\delta);$$

each contains the factor $\Delta p\delta\beta$; or, what is the same thing, that

$$\Delta\,\sqrt{p\delta\beta, p\delta\beta},$$

contains the factor in question, Σ denoting the sum of the first three terms of the original expression. The value is

$$\sqrt{\frac{p\alpha\cdot Phi + p\delta\beta\cdot p\delta}{2\sqrt{p\alpha\cdot p\beta\delta}}}\,p\delta\beta + \sqrt{p\alpha\cdot p\beta\delta}\cdot\Delta p\beta\delta;$$

$+ p\delta\beta\delta\cdot\beta\delta(\delta+\delta) - p\delta\beta\delta\delta\cdot p\delta\beta\delta = 0$, *[Sadleirian abc&c.]*

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where we have

$$x_1\Pi ba = b\delta a,$$
$$y_1\Pi ba = c\delta b.$$

Also

$$\begin{aligned} a_1(a_1,[a_1 - Wa_2] &+ p_a(a_3 - Xa_4 - Wa_2) + a_5[(Xp_3 - z_3)p_a + b(Xp_4 - Wa_2)] \\ &+ a_5[(Xp_5 - z_3)p_a + b(Xp_4 - Wa_2)] + Y(Xp_5 - z_3)(Xp_6 - Wa_2) \\ &+ \Sigma(b_1, a_2b_3) - Zp_a - Wa_2) \\ &+ (a_1 - a_2 - Xa_2 - Xa_2), \\ &+ 2b_3a_2[X(a_3 - a_2)] + Y(bXp_5 - z_3)p_a + Y(bXp_5 + (b\delta - Wa_2)] = 0. \end{aligned}$$

Also

$$\begin{aligned} a_1ba &= a_3(z = z_a, +(a_3 + a_4)ba + (z_3 + b\delta)z), \\ bab &= a^2p^2 + ap_b(a + b\delta)b + \delta b \\ &\quad + Y[2\,a(a_3 - a_4) + b\,b\,a_5z] \\ \Delta bab &= \quad X[2p_a + b\delta z] \\ &\qquad + Z[2a(a_3 - a_4) + (b - a)b - z], \\ \Delta abab &= W[2a(a_2 + p_a + b\delta a + b\delta y] \end{aligned}$$

41. The whole expression is a linear function of X, Y, Z, W, and it is easy to see *a priori*, or to verify, that the coefficients of X, Y, each of these vanish. The coefficient of Z is

$$\begin{aligned} &= \Sigma[a_1(a_2, ba_3) - (a_3, ba_2) + a_5(b_2, ba_4, ba_3) + 2ba_2(b_3, ba_2)p\delta\beta \\ &\quad - \Sigma b_3(ba_2a_3, +b_3, ba_5) + ba_4 + 2a_5(ba_2 + ba_3, ba_5) + (b_3, ba_5)\,a_3b] \end{aligned}$$

with a like expression for the coefficient of W. The foregoing expression may be written

$$\begin{aligned} (b_aa_1 + a_5b_3)\Sigma[ba_3 + a(p_a + b\delta z)p_a + b\delta y] = b\delta a[&b\delta a + a(p_a + b\delta z)p_a + b\delta y] \\ &+ \Sigma b_3[ba(a + b\delta + b\delta y) + 2b_3p_a + b] \\ &+ 2b_3ba(b_2, ba(a + b\delta + b\delta y + b\alpha y) - (aa(b\delta - g\delta)))] \\ &+ \Sigma b_3ba(b_2 - p_a)(p_a + b\delta + b\delta + b\delta + b\delta)] \end{aligned}$$

The first sum is

$$a^2 \cdot p\alpha\alpha + ag(p\alpha\beta + p\delta\beta) + p\delta\beta\delta y + g\delta\beta,$$
$$= -ag \cdot a\,fpha = -ag(fg\alpha + b)\,aa,$$

where $afgh$, β etc. denote determinants with the suffixes $\mathfrak{a}, \mathfrak{B}, \gamma$. Similarly the second sum is

$$= -ag(g\,fpha, a)\beta a,$$

in the third sum is

$$= -ag(h\,fpha\,a)\beta\gamma;$$

and the fourth sum is

$$= -(a, p + b, p + c, p)\,a\,g\,fp.$$

[Sadleirian abc&c.]

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The whole coefficient of Z thus contains the factor $(cfy - x + bfy - z)$, and similarly it would appear that the whole coefficient of W contains the factor $(cfy - z + afy - bfx)$. the other factor being the same in each case; viz. the two terms together are

$$(-(a_1 y_1 - ba_2 ba_5)X[(afy + afy) + W(ay + afy - agy - afy).$$

where the second factor is $\Delta p\beta\beta$, which is the required result. See *post*, No. 56 *et seq.*

42. (4) The line $[\delta, \beta, \beta]$ is an 8-tuple line: in fact, for any point of this line in question we have $p\delta\beta = 0$, $p\delta\beta = 0$, $p\delta\beta = 0$, $p\delta\beta = 0$; whence each factor is O, or the norm is O^8.

I notice that the surface meets the quartic $p\beta\beta$ in

lines α, β, γ	each 8 times 24
„ (a, β)	16
„ $(\delta, a, \beta, \gamma)$	8
$24 \times 2 = 48$	

Surface octuple.

43. The equation is

$$(p\alpha\beta \cdot p\alpha\gamma \cdot p\alpha\delta \cdot p\alpha\delta + p\beta\gamma \cdot p\beta\delta \cdot p\gamma\beta \cdot p\delta\beta) = -bp\alpha\beta \cdot p\alpha\gamma \cdot p\alpha\delta \cdot p\beta\delta \cdot p\gamma\beta \cdot p\delta\beta \cdot p\beta\alpha = 0;$$

or, what is the same thing,

$$(p\alpha\beta \cdot p\alpha\gamma \cdot p\alpha\delta \cdot p\alpha\delta - p\beta\gamma \cdot p\beta\delta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\gamma\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\gamma\beta \cdot p\delta\beta \cdot p\gamma\beta \cdot p\delta\beta \cdot p\beta\alpha = 0,$$

the equivalence of the two depending on the identity

$$p\alpha\beta \cdot p\beta\delta \cdot p\alpha\delta + p\delta\beta \cdot p\beta\alpha \cdot p\alpha\beta - p\delta\beta \cdot p\alpha\beta \cdot p\beta\delta = 0,$$

where, as before, $p\alpha\beta = 0$ is the equation of the plane through the lines a, β β; and $p\alpha\alpha = 0$ is the equation of the plane through the line and the point α; viz. $p\alpha\beta$ &c., and *post*, No. 56, have the values already mentioned: $p\alpha\beta, p\beta\delta = 0$ are also mentioned in this cubic surface through the lines a, β, β β, and *vol*, *etpa*. *[Sadleirian abc&c.]*

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44. *Investigation.* In the projection, using line-coordinates, the equation of the conic touching the five lines may be written

$$(P,\ Q,\ \Xi P)\beta = 0;$$

$$(P,\ Q,\ R)\beta = 0,$$

where the conical denotes a determinant: the last five lines of which are obtained by giving to (P, Q, R) the suffixes $\mathfrak{a}, \mathfrak{B}, \gamma, \delta, \delta$ respectively. Writing herein the equation P, Q, R for the suffixes $\mathfrak{a}, \mathfrak{B}, \gamma, \delta, \delta$ are at once transformed into

$$a\beta\delta \cdot p\beta\beta,\ a\beta\delta \cdot p\beta\delta,\ a\beta\delta \cdot p\delta\beta,\ a\beta\delta \cdot p\delta\beta,\ a\beta\delta \cdot p\delta\beta = 0,$$

or, what is the same thing,

$$p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta = 0;$$

or say

$$p\alpha\beta \cdot p\beta\beta \cdot p\delta\beta + RX + Y(C\bar{A}' + CO'' + CO'') = 0;$$

$$-(p\beta\beta \cdot p\beta\beta \cdot p\delta\beta + A'\bar{Y} + B + C''\bar{Y})\,\Omega = 0;$$

where $p\beta\delta$, &c., signify as before; and

$$A\,\bar{a} + B + a + C\,\bar{a} = \bar{a}\,\bar{a};$$

$$\begin{vmatrix} P_1, & Q_1, & R_1 \\ . & . & . \end{vmatrix},$$

$$\begin{vmatrix} P_2, & Q_2, & R_2 \\ . & . & . \end{vmatrix}$$

and so for $A\bar{b}\beta + B + \beta + C\,\bar{c}\beta$ and $A'''\bar{Y} + B'' + C'''\bar{Y}$ being for (y, x) and (η, b) respectively. ξ, η so that the foregoing equation is

$$\begin{vmatrix} p\alpha\beta \cdot p\beta\beta, & p\alpha\beta, & p\delta\beta, & p\delta\beta \\ A, & B, & C & \\ A', & B', & C' & \\ A'', & B'', & C'' & \end{vmatrix} + 4p\alpha\beta \cdot p\beta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \begin{vmatrix} p\delta\beta, & p\delta\beta, & p\delta\beta \\ A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{vmatrix} = 0;$$

or in the equivalent form, when the first term we have + instead of -1, and in the second term the determinants are

$$\begin{vmatrix} P_1, & Q_1, & R_1 \\ A, & B, & C \end{vmatrix},\quad \begin{vmatrix} P_2, & Q_2, & R_2 \\ A', & B', & C' \end{vmatrix},$$

46. To reduce this result, observe that we have: $A, B, C\ :\ bp - q,\ -q +\ p,\ b\,p - r,\ b\,p\,r,\ b\,p - r,\ -p\,X + f\,q - 0;$

$$by - g\,r + d\,r, \quad -b\,r + r,\ +fy - r;$$

[Sadleirian abc&c.]

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where, for convenience, I retain the unaccented and accented letters $(a_1, \ldots)$, $(a', \ldots)$ instead of those letters with the suffixes a and β respectively. Writing as before

$$\begin{aligned} L &= aag - aabhg + \ldots \\ M &= aag - aabh + \ldots \\ N &= aag - aabh + \ldots \\ \Omega &= (gh - p')\, h + \ldots\, ; \end{aligned}$$

then

$$\begin{aligned} A &= L - Mw, \\ B &= L - Nw, \\ C &= \Omega - Nw, \end{aligned}$$

and similarly

$$\begin{aligned} A' &= L\Omega - Cw, \\ B' &= L\Omega - Nw, \\ C' &= \Omega - Nw, \end{aligned}$$

where for L', M', M', Ω' we have $(a', \ldots)$ for $(a, \ldots)$. Hence

$$BC' - CB' = w, \begin{vmatrix} . & . & . \\ M, & N, & \Omega \\ M', & N', & \Omega' \end{vmatrix}$$

with like expressions for $CA' - C'A$ and $AB' - A'B$, and substituting, we have

$$\begin{vmatrix} P_a, & Q_a, & R_a, & = w \\ A, & B, & C & \\ A', & B', & C' & \end{vmatrix}, \begin{vmatrix} . & x, & y, & z, & w \\ . & . & . & . & . \\ . & a_a, & a_a, & a_a, & -a_a \\ L, & M, & N, & \Omega & \end{vmatrix},$$

or substituting for p_a, q_a, r_a their values $aw_a - ax_a, aw_a - aw_a, -aw_a + aw_a$, this is

$$\begin{vmatrix} . & x, & y, & z, & w \\ A, & B, & C & & \\ A', & B', & C' & & \\ L, & M, & N, & \Omega & \end{vmatrix}$$

whence, omitting the factor w, the equation is

$$\begin{vmatrix} p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta, & x, & y, & z, & w \\ a_a, & a_a, & a_a, & a_a, & -a_a \\ a_b, & a_b, & a_b, & a_b, & -a_b \\ L, & M, & N, & \Omega & \\ L', & M', & N', & \Omega' & \end{vmatrix}$$

$$+4p\alpha\beta \cdot p\beta\beta \cdot p\beta\beta \cdot p\beta\beta \cdot p\delta\beta \cdot p\delta\beta \begin{vmatrix} . & x, & y, & z, & w \\ . & a_a, & b_a, & c_a, & d_a, -e_a \\ . & a_a, & b_a, & c_a, & d_a, -e_a \\ L, & M, & N, & \Omega & \\ L', & M', & N', & \Omega' & \end{vmatrix} = 0,$$

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where I recall that the $(L, \ldots)$, $(L', \ldots)$, $(L'', \ldots)$, $(L''', \ldots)$ are functions of (a, b), (a, β), (a, γ), and (a, β) respectively. The values of the first two determinants thus are $p\delta\beta \cdot p\delta\beta$ and $p\delta\beta \cdot p\delta\beta$ respectively; that of the third is $= -p\alpha\beta \cdot p\delta\beta$, viz. this is $= -p\delta\beta \cdot p\delta\beta$, where, as before, $p\delta\beta = \cdots$. And we have then the below-mentioned equation of the surface.

47. *Singularities.* The equation of the surface shows that (0) The point a is a 2-conical point; in fact, we have for this point $p\delta\beta = 0$, $p\delta\beta \cdot p\delta\beta = 0$; $p\delta\beta = 0$, $p\delta\beta = 0$. (1) The line a is a 4-tuple line: in fact, for any point on this line $p\delta\beta = 0$, $p\delta\beta = 0$; $p\delta\beta = 0$; $p\delta\beta = 0$, $p\delta\beta = 0$; whence the term in question, divides by $p\delta\beta \cdot p\delta\beta \cdot p\delta\beta = O$. (2) The line (a, β, γ) is a 4-tuple line: in fact, for any point on this line $p\delta\beta = 0$, $p\delta\beta \cdot p\delta\beta = 0$, $p\delta\beta = 0$, $p\delta\beta = 0$; whence the term in question, divides by w^2. (3) The exorhospheric $aa\beta$, β is a simple curve: in fact, for any point of this curve we have $p\delta\beta = 0$, $p\delta\beta = 0$; whence the term in question inter- secting in the lines a, β β and the curve $aa\beta\beta$. It is, moreover, obvious that the surface is touched along the curve by the hyperboloid $p\delta\beta$.

I notice that the surface meets the quartic $p\delta\beta$ in

lines (a, β, γ)	each 4 times,	12
„ (a, β, γ)	twice,	4
„ (a, β, γ)	, ,	4
cures $aa\beta\beta, \delta a$	, ,	8
	$14 \times 2 = 28$	

Surface octuple.

48. The equation of the surface may be written

$$p\alpha\beta - p\alpha\beta \cdot p\alpha\beta + p\alpha\beta - p\beta\beta \cdot p\alpha\beta \cdot p\beta\beta + p\delta\beta = 0;$$

where $p\delta\beta$ &c., is the equation of the quadric through the lines a, $\mathfrak{B}, \gamma$; &c. *etpa* has the form noticed above. The form is one of kk forms depending on the partitionment

$$\begin{pmatrix} a\beta \cdot \beta\delta \cdot \delta \\ \delta\delta \cdot \beta\delta \end{pmatrix}, \quad \begin{pmatrix} a\delta \cdot \beta\delta \cdot \delta a \\ \delta\beta \cdot \beta a \end{pmatrix}, \quad \begin{pmatrix} a\beta \cdot \beta\delta \cdot \delta a \\ \delta a \cdot \beta a \end{pmatrix}$$

of the six letters.

[Sadleirian abc&c.]

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Surface chiefly.

49. *Investigation.* The projections of the six lines are tangents to a conic: the condition for this is $(P, Q, R)\beta = 0$, where the last factor expresses, that the determinant obtained by writing successively (P_a, Q_a, R_a), &c. for (P, Q, R). The equation may be written

$$a\beta \cdot p\delta\beta - p\delta\beta - p\delta\beta - p\delta\beta - p\delta\beta - p\delta\beta + RZ \cdot p\delta\beta = 0,$$

where

$$a\beta = \begin{vmatrix} P_a, & Q_a, & R_a \\ P_b, & Q_b, & R_b \\ P_c, & Q_c, & R_c \end{vmatrix},$$

and substituting for $P, \ldots,$, &c. their values, we have $a\beta = a \cdot p\delta\beta$; whence the foregoing result.

50. *Singularities.* The equation shows that (1) The line a is a 2-tuple line: in fact, for the point of the line we have $p\delta\beta = 0, p\delta\beta = 0, p\delta\beta = 0, p\delta\beta = 0$. (2) The line (a, β, γ) is a 2-tuple line: in fact, for each point of the line we have $p\delta\beta = 0, p\delta\beta = 0$. (3) The quadripolyolic $a\beta\delta a + \beta b$ is a simple curve on the surface: for the $p\delta\beta = 0$, $p\delta\beta = 0$, $p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta = 0$.

It may be remarked that the surface meets the hyperboloid $p\delta\beta$ in

lines (a, β, γ)	each twice,	6
„ (a, β, γ)	,,	2
„ (a, β, γ)	,,	2
curves $aa\beta\delta + \beta a$,		4
	$2 \times 8 = 16$	

51. It might be thought that there should be on the surface some curve $a\beta\delta\beta$, such as the conics $a\beta\beta\delta$ on the surface octuply; but I cannot find that this is so. The equation of the surface is satisfied by two simultaneously (A being arbitrary)

$$p\delta\beta - p\delta\beta - p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta = 0,$$

$$p\delta\beta - p\delta\beta - p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta \cdot p\delta\beta = 0;$$

which equations represent quartic surfaces, the first of these depending on the duo- line, and passing through the lines $\mathfrak{a}, \mathfrak{B}, \gamma$ ($\eta + \beta + d = 0 - \beta = 0$ conditions, so that the equation of such a surface contains only an arbitrary parameter λ), and the second having β for a duo-line, and passing through the lines $\mathfrak{a}, \mathfrak{B}, \gamma$. But I am not able to determine so as to have these equations indicate a curve on the surface octuply; it is to be remarked that the line involving λ is, with $\beta\beta$ a curve on the free surfaces is, when intersect, but in only ten in 8 ($= 2 + 6$) conditions, of the line free passes through (3 conditions) and the lines (a, β, γ) each β times (5 conditions) intersection in lines (3 conditions). And as the surface is octuple, in $3 + 6$ ($= 18$ conditions)
[Sadleirian abc&c.] 17–2

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The Exorhospheric $aa\beta\beta$, &c.

52. The notion is, that we have a fixed point a, two fixed lines $\mathfrak{a}, \mathfrak{a}$, and a single infinite series of lines, or any line of the generating lines of a slow surface; each generating line determines, with the point a, a plane; and if in this plane we draw, meeting the lines $\mathfrak{a}, \mathfrak{a}$,a line to meet the generating line in a point P, then the locus of this point P is the curve in question.

53. In the case in question, the singly infinite series of lines is that of the lines which meet each of the lines $\mathfrak{a}, \mathfrak{B}, \gamma$ or say there are the generators of the hyper- boloid $a\beta\gamma$ the lines, or any *say lc.*, $\mathfrak{a}, \mathfrak{B}, \beta$. as mentioned above on indeterminacy. It is not necessary for the purpose of the moment, but it is interesting to consider in conjunction therewith the exorhospheric arising in like manner from the intersections of the hyperboloid; it will appear that the two curves are the complete intersection of the quadric aβ by a particular surface. Observe that the two curves are given as follows: *viz.*, considering the the quadric $a\beta\beta$ as generated by either of the two systems of generating lines $\mathfrak{a}, \mathfrak{B}, \gamma$ on the lines $\mathfrak{a}, \mathfrak{B}, \gamma$, and on the other systems of the generating lines on the lines $a, \mathfrak{b}, \mathfrak{c}, \beta$ *viz.*, the lines (a, β, β) is a curve a in the surface β and (a, β, β) is a curve β in the surface β.

54. The points of either of the lines $\mathfrak{a}, \mathfrak{B}, \gamma$ may be regarded as derived from the points of any of the lines $\mathfrak{a}, \mathfrak{B}, \gamma$ by a $(1, 1)$ correspondence; *viz.*, the two generators of the same system of β which meet either of these lines β, β, β, the intersections with these lines are pairs of points, and the points on the two lines β, β, correspond to a univocal series; or write into this curve.

[Sadleirian abc&c.]

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or, as the second equation may also be written,

$$(Q, R - QR_3)(P_aP_aa_a - P_aPla_a) - (P_aP_3 - Pl)(Q_aR_a - Q_lR_3) = 0;$$

viz. the second equation represents a cubic surface having upon it the lines $(P = 0, R = 0)$ and $(Q = 0, P_3 = 0)$; further, intersecting the quadric $PR - QR = 0$ in these lines, and besides in an exorhospheric curve, which is the required locus.

55. Representing the determinants

$$\begin{vmatrix} P, & Q, & R, & l \\ P_a, & Q_a, & R_a, & l_a \\ P_b, & Q_b, & R_b, & l_b \\ P_c, & Q_c, & R_c, & l_c \end{vmatrix}$$

by (a, b, c, d, e, f, g) &c., *viz.* values, we have, *e.g.*, $a = QR_a - Q_lR_b$, c., the equations of the conies in question are reducible to the form $(P = QR = 0$, and $R = 0)$, and besides in the exorhospheric curve, which is the required locus.

56. Representing the determinants

$$\begin{vmatrix} P, & Q, & R \\ P_a, & Q_a, & R_a \\ P_b, & Q_b, & R_b \end{vmatrix}$$

viz., the second equation represents a cubic surface having upon it the lines $(P = 0, R_3 = 0)$ and $(Q = 0, P_3 = 0)$; further, intersecting the quadric $PR - QR = 0$ in these lines, and besides in an exorhospheric curve, which is the required locus.

56. But the elimination may be performed as follows, viz. multiplying by P_a, P_a, from the first two equations in θ, multiplying by P_a, P_a, from the second and adding, and so with Q_a, Q_a, &c. we obtain

$$\begin{gathered} (Q_a - RR_a) + bP_a + (P_a + lRR_a)\theta + (a + \beta\theta + \beta\theta) = 0 \\ (Q_aR_a) + bP_a + (P_lRR_a)\theta + (a + \beta\theta + \beta\theta) = 0, \\ (Q_a - RR_a) + bP_b + (P_b - lRR_b)\theta + (a + \beta\theta + \beta\theta) = 0, \end{gathered}$$

We have here

$$\theta = \frac{-c + b\theta}{a + b\theta}, \ \frac{c + b\theta}{d - \beta\theta};$$

or, what is the same thing,

$$b\theta + (a - b)\theta = c.$$

Using this equation, written in the form $(a + b)\theta = c$ to transform the first or third of the four equations in θ, we obtain

$$-(aP + abQ + bR)\theta - aQ + bR + R'P,$$

and using the same equation, written in the form $(d - \theta)\theta = -e$, to transform the second or fourth equation, we obtain

$$(eP_b - \beta Q + RP_b + a)\theta = -\beta Q - \gamma R + dP_b,$$

[Sadleirian abc&c.]

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and hence, eliminating θ, we obtain

$$(bQ_a + R_a + aQ_a + R_a - bP_a)(cQ_a + R_a - Q_a) = (bP_a + aQ_a + bR_a + bR_a)(-aQ_a + dP_b)$$

which, on being of the second order in $(a, \ldots)$ represents a quartic surface. The equation contains underwritten by the interchange of (P, Q, R) with (P_a, Q_a, R_a) that vanish, the opposite sextants is the surface; that the line $\mathfrak{a}, \mathfrak{B}, \gamma$ are not similarly for $p\delta\beta, p\delta\beta$, $p\delta\beta$; so that, omitting the constant factor $V = ie$, the value of the operation becomes

$$= \text{four}(eABP{+}bQ{+}CdR{+}DSe) \quad +AP{+}EQ{+}CR{+}DS \quad = \text{four}(eAB{-}BdQ{-}BdR)P{+}a(b{-}aB){+}d(Ad{-}E$$

57. I obtain this same result also as follows. Consider a point (P_a, Q_a, R_a, S_a) on the quadric surface; $(P_a, Q_a, R_a, R_a) = 0$, the tangent plane at the point is

$$PR + QR_a + RQ_a + SP_a = 0,$$

and if this passes through the point a, then

$$PR + bQ_a + aR + dP_a = 0.$$

The line which in the tangent-plane meets the lines b, a is given, as before, by the equations

$$\begin{aligned} P_aR_a - Q_aR_a + R_aQ_a + R_aQ_a &= 0, \\ P_aR_a - Q_aR_a + R_aQ_a + R_aQ_a &= 0. \end{aligned}$$

Remembering the significations of $(a, \ldots)$, the last three equations give

$$\begin{aligned} S_a : R_a : -Q_a : -P_a = \quad & -aQ_a + dP_a + ab, \\ & : aP_a + R_bR_a + ac, \\ & : aP_a + R_bR_a + ad, \\ & : bQ_a + R_bR_a + cd; \end{aligned}$$

and substituting these values in $S_aP_a - Q_aR_a = 0$, we have the above equation of the quadric surface.

58. Or again, changing the notation, I take the equation of the quadric surface to be

$$(a, b, c, d, f, g, h, l, m, n)(x, y, z, w) = 0,$$

or

$$ax^2 + by^2 + cz^2 + dw^2 + 2fyz + 2gzx + 2hxy + 2lzw + 2mwy + 2nwx = 0.$$

A tangent-plane hereof is

$$\xi x + \eta y + \zeta z + aw = 0,$$

where ξ, η, ζ, a, are any quantities satisfying the relation

$$(A, B, C, D, F, G, H, L, M, N)(\xi, \eta, \zeta, a) = 0,$$

the capitals denoting the inverse coefficients. Supposing that the tangent-plane passes through a fixed point a, coordinates $(\alpha, \beta, \gamma, \delta)$, we have

$$a\xi + b\eta + c\zeta + da = 0,$$

[Sadleirian abc&c.]

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and if the equations of the lines $\mathfrak{B}, \mathfrak{c}$ are as below: $(a = 0, y = 0)$ and $(z = 0, w = 0)$: then for the line in the tangent-plane meeting the lines $\mathfrak{B}, \mathfrak{c}$ we have

$$\xi z + \eta w = 0, \quad \xi z + aw = 0.$$

These last equations may be represented by

$$\xi : \eta :: -\xi : \eta :: -z, -w$$

and, substituting these values, we have

$$(A, \ldots)(\xi, \eta, -z, -w)^a = 0;$$

that is

$$(A\xi^2 - 2H\xi y + By^2 - 2Fxw - Lyw - 2NLw^2 + 2Myw + Dw^2)$$
$$+(xy - 2Gxw - 2Mxw + Cz^2)\,\eta = 0;$$

that is

$$(A\xi^2 - 2H\xi y - By^2)\xi w$$
$$+(ay - zw + 2Hw^2)\eta z = 0;$$

the line being connected with the point a by the equation

$$a\xi + b\eta + c\zeta + da = 0.$$

Whence, eliminating ξ, η, we have the question and answer, viz. that an integral $(A\xi y - 2Hy\eta - 2By^2 - Fyw - Gw\beta + Mu) - Bw^2z(zw - 8Lw + (cy + dw))$

$$-(\beta z + dw)(Bw^2 - 2Mu - Cz^2 - \text{etc.}\,(yz - 2Lzw + (cy + 2dw) - 0,$$

an equation in (x, y, z, w) which has previously obtained, and which can be without difficulty identified therewith.

The Circular Curves are Geodesics.

59. I proceed to show that the circular curves are geodesics; viz. that an integral of the geodesic equation is

$$(E, F, G)(\xi, \eta)^a = 0.$$

Starting from this equation, we have

$$2(B\xi y + Bw^2 + 2By^2)(Bw^2 + Bw^2 + Bw^2) + E(\xi y + Bu^2)(Fuu + Buu) + Bw^2 + Bw^2 = 0.$$

Now the equation, writing therein $\xi y = F\eta = zw$, gives $\xi y + Bw^2$, then expansion may be written

$$4F\eta(\xi y + Buw) = 0;$$

the value of b being therefore $-b - 2E\xi - z$. The result then divided thus becomes $2Eu(\xi y - Bw)(Fu + bw^2) + (F\xi y - Bw + B\xi y + Bw^2)y$

$$= \Sigma(\xi y + Zw)(F\xi y + bw^2),$$

that is

$$2(E(\xi y + Bw))(2Eu - Fuu) + F(\xi y + Zw + Fwu^2)\,bw\}$$
$$+(ay + bw)(Fu + bw^2) - (zw + \xi y + dw)(Pu + 2F\xi y + dw^2)$$

viz. taking $\beta E\xi = (\xi w^2 + Fwu + d^2)$. This appears with the geodesic equation.

The foregoing integral, $(E, F, G)(\xi, \eta)^a = 0$, I believe, a particular, not a singular, solution of the geodesic equation.

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where Σ denotes the sum of the three terms obtained by cyclic interchange of x, y, z; and we similarly have

$$p\beta\beta = (z_1 - h\eta_1 z_1)(x - h\eta_1 x_1) - z_1 h\eta(z - bx),$$
$$p\beta\delta = (z_1 - h\eta_1 x_1) - z_1(x - h\eta_1 x_1) - z_1 b\eta(z - bx),$$

and similarly for $p\delta\beta$, see *post*, &c. 60. To obtain the intersection with $\delta\beta = w = 0$, writing $w = \frac{1}{2}\,\theta$, then

$$p\alpha\beta = \{a_a - hc_a - \tfrac{1}{2}\theta(z_a - hya)\}\, b(x - h\delta z),$$
$$p\delta\beta = \{a_a - he_a - \tfrac{1}{2}\theta(z_a - hya)\}\, b(x - h\delta z),$$

or say

$$\sqrt{p\alpha\beta \cdot p\delta\beta} = M(z - h\delta\eta).$$

also the expression in [] become

$$M_a = (a_a - hc_a) + \tfrac{1}{2}\theta\,(z_a - hya)\}\, b(x - h\delta z),$$
$$-\,(a_a - hc_a)(z_a - hya) - h(z - bx)(a_a - hc_a)$$

so that the norm in question is

$$\text{Norm}\ \sqrt{M_a(x - h\delta\eta)} - h(x - bx)(a_a - hc_a)\}\, b - hgM_a,$$

where M_a is now considered to stand for

$$\{(a_a - hc_a) + h\}(a_a - hc_a) + h(a_a - hc_a)\, b(\,x - h\delta z)\, b - h(2g)(a_a + a_a) - 2bx,$$

where M_a is now considered to stand for

$$\{(a_a - hc_a) + h\}(a_a - hc_a) + h(a_a - hc_a)\, b(\,x - h\delta z)\, b - h(2g)(a_a + a_a) - 2bx,$$

Observing that the norm was originally the product of 8 factors, this breaks up into:

$$(b\eta + (a - n)\beta + bw^2)(a_a)\beta\,(x - h\delta\eta_1)\, b - h\delta z) = 0,$$

where the new term is the product of the four factors. 61. Writing for greater convenience a, b, c, a in place of h_a, h_b, h_c, h_d, and observing that M_a is a quadric function of h_a, h_b, that is of h, the last-mentioned norm hsh which is the surface, then takes the form: that the two above each touch the quadric surface, or that the lines are tabulated.

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[Continuing with]

$$A = PP', \quad B = (PQ' + P'Q), \quad C = QQ';$$

then

$$4AC = (PQ' - P'Q)^2,$$

and we have

$$P, \quad Q = a_a w x_a, \quad b_a y_a,$$

$$P, \quad Q = a w x_a, \quad b y_a;$$

whence

$$PQ - PQ' = (xyx_a - x_a y_a)w$$

$$= \{xyx_a - xyy\}\,\{xyw_a x w_a w x_a\}$$

$$= (xyy - xyx)w.$$

viz. if (a, b, c, f, g, h) are the coordinates of the line ab, this is

$$= \ bs\,a + bx\,bs\,aw.$$

Hence, omitting the constant factor $(yz - x)^2 + (zx - y)^2$ (that is $(b_3 - b_3, h_3) - (g_3 - g_3, h_3, h_3)$, the foregoing equation must $= 0$ becomes

$$(bs\,a - g\,bax)^2 = a\,b\,b\,b\,b\,c\,ba(wx_a y w_a - xyay_a),$$

the equation in question of the quadric with the surface are obtained by combining the equation $wx_a - gw = 0$ with the several equations

$$(bs\,a - gf\,bax)^a + a\,bs\,(xyw_a - xyax_a) = 0,$$

$$(bs\,a + gbax)^a + a\,bs\,(xyw_a - xyax_a) = 0,$$

viz. these are

lines $(a, \beta, \gamma, \delta)$ each 8 times 16
lines $(a, \beta, \gamma, \delta)$ each 2 24
lines $(a, \beta, \gamma, \delta)$ each 2 24
lines $(a, \beta, \gamma, \delta)$ 8

$$(16 + 24 + 8) \times 2 = 48 \quad \times 8$$

But it is clear that the lines $(a = 0, w = 0)$ and $(a = 0, z = 0)$ are introduced by the process of elimination, and are no part of the intersection. The complete intersection consists of the lines $(a, \beta, \gamma, \delta)$ each β times, the lines (a, β, γ) each β times, and the lines $(\beta, a, \beta, \gamma)$ each β times. The above last-mentioned line being only doubly lines on the surface, this means that the two sheets each touch the quadric surface, or that the lines are tabulated. C. VIII. 18

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On the Mechanical Description of Certain Sextic Curves.

[From the Proceedings of the London Mathematical Society, vol. IV. *(1871–1873), pp. 105–111. Read April 13, 1872.]*

THE curves in question might be taken to be those described by a point C rigidly connected with points A and B, each of which describes a circle; but the construction is considered under a somewhat more general form. I consider a quadri- lateral, the sides of which are a, b, c, d, and the inclinations of these to a fixed line a, β, γ, δ. This being so, if a, b, c, d are one of the angles, say δ, are constant, then we have between the three variable angles the relation

$$a \cos a + b \cos \beta + c \cos \gamma + d \cos \delta = 0,$$

$$a \sin a + b \sin \beta + c \sin \gamma + d \sin \delta = 0;$$

giving rise to a single relation between any two of the variable angles; and we con- sider a curve such that the coordinates x, y of any point thereof are given linear functions of the sines and cosines of the three variable angles, or, what is the same thing, of the sines and cosines of any two of these angles. We then unite together what would otherwise be distinct cases; for everything is symmetrical in regard to the sides a, b, c and the corresponding variable angles a, β, γ, irrespectively of the order of succession of these sides; and we can then in like manner connect together, in any way two or pleasure, say a, β of the variable angles, without determining whether the sides a, b are conjugate or separate.

Eliminating then, the variable angle γ, we obtain between a β a relation which, if we write therein $\tan \frac{1}{2}a = u$, $\tan \frac{1}{2}\beta = v$, takes the form $(u^2 + v^2, V = 0$; the value of the variables a, β is expressible rationally in terms of the other of them and of the root of a quartic function thereof; say a is a rational function of v and $\sqrt{V}$. And

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hence a curve for which the coordinates x, y are rational functions of u, v is a curve having a deficiency $D = 1$, or, what is the same thing, having a number of dps. less by unity than the maximum number [$= \frac{1}{2}(n-1)(n-2)$], if n be the order of the curve].

It will further appear that the relation $(*\breve{a}\,u,\ 1)^2(*,\ 1)^2 = 0$ is satisfied by the values $u = v = i$ and $u = v = -i$ (if, as usual, $i = \sqrt{-1}$).

In the curves in question, the coordinates (x, y) are given linear functions of the sines and cosines of α, β; and if we make the curve meet an arbitrary line $Ax + By + C = 0$, we obtain between the sines and cosines of α, β a linear relation which, substituting therein the expressions in terms of u, v, takes the form

$$(*\breve{a}\,u,\ 1)^2(*,\ 1)^2(1+v^2) + (*,\ 1)^2(1+u^2) = 0,$$

viz. this is a relation of the form $(*\breve{a}\,u,\ 1)^2(*,\ 1)^2 = 0$, such that it is satisfied by the four sets of values $u = \pm i$, $v = \pm i$, and therefore in particular by the values $u = v = i$ and $u = v = -i$.

Hence, considering the intersections of the curve by the arbitrary line, the values of (u, v) are given by the two equations $(*\breve{a}\,u,\ 1)^2(*,\ 1)^2 = 0$, $\breve{a}\,(*\,u,\ v)$, $V\breve{a}(*\,v,\ 1) = 0$, regarding for a moment u, v as ordinary rectangular coordinates, represent each of them a quartic curve having two dps. at infinity on the axis $u = 0, v = 0$ respectively; each of these points reckons therefore as 4 intersections, and the number of the remaining intersections therefore is $4.4 - 2.4 = 8$. But, by what precedes, the two quartic curves have also in common the points $u = v = i$ and $u = v = -i$; and rejecting these, there remain $8 - 2, = 6$ intersections.

The conclusion is, that the curve is a sextic curve of deficiency 1, that is, having 9 dps. The reasoning may be presented under a slightly different form as follows; regarding u, v as coordinates, we have the curve $(*\breve{a}\,u,\ 1)^2(*,\ 1)^2 = 0$, a bicodal quartic curve, and having therefore the deficiency 1; the curve passes, as above-mentioned, through the points $u = v = i$ and $u = v = -i$. The required curve is obtained as a transformation of the quartic curve by formulæ of the form $x : y : x(=1) = P : Q : R$, where P, Q, and R [$= (1+v^2)(1+u^2)$] are quartic functions of the coordinates u, v, such that $P = 0$, $Q = 0$, $R = 0$ are each of them a quartic curve passing twice through each of the nodes and once through each of the before-mentioned points, $(u = v = i)$ and $(u = v = -i)$. The factor thus is

$$I_n = pcd. - 1\Sigma\, p\Sigma\, a\, b\, a\,\cdot$$

$p\beta\beta = p^{\,a}$, $a{+}p^{\,b}$, $b{+}p^{\,c}$, $c{+}p^{\,d}$, d, $\quad p\,\beta = 0$ is a binodal quartic curve, and the maximum number of dps. of

and the entire defect of the cubic in u, v is one, the deficiency of $u, v = 1$.

To obtain the foregoing equation $(*\breve{a}\,u,\ 1)^2(*,\ 1)^2 = 0$, the elimination of γ gives:

$$(a\cos a + b\cos\beta + d\cos\delta + (c\sin a + b\sin\beta + d\sin\delta) = c^2,$$

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that is

$$a^2 + b^2 + d^2 + 2ab\cos(a - \beta) + 2ad\cos(a - \delta) + 2bd\cos(\beta - \delta) = c^2,$$

or, substituting herein the values

$$\cos a = \frac{1 - u^2}{1 + u^2}, \quad \sin a = \frac{2u}{1 + u^2}, \quad \cos\beta = \frac{1 - v^2}{1 + v^2}, \quad \sin\beta = \frac{2v}{1 + v^2},$$

this is easily found to be

	1	2v	v^2
1	$(a+b)^2 - 2(a+b)d\cos\delta + a^2$	$2bd\sin\delta$	$(a-b)^2 - 2(a-b)d\cos\delta + d^2 - c^2$
2u	$2bd\sin\delta$	$4ab$	$2bd\sin\delta$
u^2	$(a-b)^2 - 2(a-b)d\cos\delta + d^2 - c^2$	$2bd\sin\delta$	$(a+b)^2 + 2(a+b)d\cos\delta + d^2 - c^2$

or setting, for greater convenience, $\delta = 0$, that is, taking a, β, γ to be the inclinations of the sides a, b, c to the fixed side d, this is $= 0$.

	1	2v	v^2
1	$(a+b-d)^2 - c^2$	0	$(a-b)^2 - d^2 - c^2$
2u	0	$4ab$	0
u^2	$(a-b)^2 - d^2 - c^2$	0	$(a+b)^2 + d^2 - c^2$

or say this is

$$(A + Bu^2) + 4abuv + v^2(C + Du^2) = 0,$$

where, writing for shortness

$$\begin{aligned}
&a + b - c - d = \lambda, \quad -a + b + c + d = \lambda', \quad -a - b + c + d = \mu',\\
&a + b - c + d = \lambda', \quad +a + b - c + d = \mu', \quad +a + b + c + d = \mu',\\
&a + b + c + d = \nu, \quad -a + b + c + d = \nu', \quad +a + b + c + d = \nu',
\end{aligned}$$

then

$$A = -\lambda\nu, \quad B = -\lambda\mu', \quad C = \nu\lambda', \quad D = \mu\nu'.$$

We at once verify that the equation is satisfied by $u = v = \pm i$, viz. this will be so if only

$$A - B + C - D = 0;$$

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and we have

$$A + B = 2(a + b)^2 - 2c^2 - 2d^2,$$
$$B + C = 2(a - b)^2 - 2c^2 - 2d^2;$$

whence the relation in question.

Writing $u = v$, we have

$$v^2(C - B) + 4abv^2 + Cv^2 = 16(b^2 + b^2)(a^2 - B^2) - (s^2 - C);$$

$$= -j(A + 4)D + B - C),$$

whence, that, corresponding to $u = v$, we have the values $u = v = \pm i$ and $u = v = \pm\frac{A-B}{C-D}$, and corresponding to $u = -v$, the values $v = -i$ and $u = -v = \pm\frac{A+B}{C+D}$.

And in like manner, corresponding to $v = i$, we have the values $u = i$ and $u = v = \pm\frac{A-C}{B-D}$, and to $v = -i$, the values $u = -i$ and $u = -v = \pm\frac{A+C}{B+D}$.

It is easy to show that, if $u = v = a$, $r = v$ are then values connective to $u = v = i$, then $u = b\,(2 - 1)$ in fact, substituting the foregoing values in the relation between u, v, and writing for that the value, we have

$$A(b - a) + B(a + 1) + a(d + 1) + b(b - 1) + Bb(a + 1) + Cb(b - 1) + Ds(b + 1) = 0,$$

which is

$$= s(A + Bs - C - Ds) + s(A + B + Cs - D) = 0.$$

or finally $a = -2v$, if $u + v = 0$. And similarly, if $a = -i$, then $a = i = -v$, viz. $u + v = 0$; then we have shown that the points which we have found in the surface are connected with the surface, and the form of pairs $u = -v$.

The points in infinity on the sextic curve are those for which $1 + a^2 = 0$ or $1 + b^2 = 0$, or each of these, is 0; viz. the values of u, v for the six points are as follows: $1^\circ\, a, u = +1, v = -i, +1, -i, +1, -i, +1$;

$$u = -\frac{ai + a}{c - a}, \quad v = +\frac{a + ic}{c - a}, \text{ four pair of points;} 2^\circ\, v = i\,(= i), u = -i\,(= i), \text{two pair of points;} u = +\frac{a - i}{c - a}, \quad u$$

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then, if $u = -i(v = i)$, then $u = -i$, $v = i$ are pair of points, viz. δ values of u, v. Hence the points on the sextic curve at infinity are as follows:

$$\Sigma^\circ.\ u = \frac{1}{4}(a+i) \text{ pair of points;}$$

$$\Sigma^\circ.\ v = \frac{1}{4}(b+i),\ \text{ pair of points;}$$

$$\Sigma^\circ.\ \text{etc.}$$

which six points in question denote three pair of points, viz. the circular points at infinity.

The foregoing values of A, B might be said to be "circular grand u" if $a = c$, $a = -c$; and similarly to be "circular grand B" if $b = d$, $b = -d$.

And we see at once that if the curve are circular grand u, then the four pair of points coincide with the circular points at infinity; and that, in like manner, if the curve are circular grand B, the second pair of points coincide with the circular points at infinity. If the curve are circular grand u, the third pair of points coincide with the circular points at infinity. But, if the curves are circular grand B and u, u also, then is a unicursal sextic having on each of the four sides u, b, c d a node: the condition in fact is

$$AD + BC - 16ab^2 = 0;$$

that is

$$\lambda\lambda' + \lambda'\lambda - \lambda'\lambda + 4\sqrt{ABCD} = 0,$$

where, putting for a moment $M = a^2 - b^2 - c^2 + d^2$, we have

$$\lambda\lambda' + \mu\nu' + \nu\lambda' = (M-2ab)^2 + (M+2ab)^2 + (M+2ab)^2 - 2ab(M+2ab)^2 = 3M^2 - 4(a^2+b^2)d^2,$$

and thence

$$\lambda\lambda' + \nu\lambda' - \lambda'\lambda = -a^2 - b^2 + a^2 + b^2 + a^2c^2 + bd + 4cdM,$$

$$-p\nu p' = \nu^2 - 4a^2b^2 - \nu^2 + 4a^2b^2 + 4a^2 + bd + 4cdM,$$

viz. the equation is

$$4cdM = -a^2 - b^2 - c^2d^2 + b^2cd - 2cd(a^2b^2c^2 - 2cdM).$$

the deficiency, -1, being then $= 0$.

Hence the equation simultaneously $1^\circ.a = c, b = d; 2^\circ.b = d, a = c; 3^\circ.a + d, a + b$; these cases are easily equivalent, but on reading, give a third such, which may be a remarks, case, or arrival. In the third-mentioned case, the relation between a, b contains the factor a, and throwing this out, and also the constant factor $4a$, it is

$$4(a+a^2)(a+1)(a+1) + (a+1)(a+1)(a+1) + 2aa = 0,$$

viz. a is given as a rational function of b.

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It would at first sight appear that the curve might become unicursal in a different manner; viz. it would be unicursal if

$$1l\lambda\lambda' = (A + Bv)^2(C + Dv)^2,$$

was a perfect square; but this is only the case when one of the four sides a, b, c, d is $= 0$. The condition in fact is

$$AD + BC - 16ab^2 = \pm 2\sqrt{ABCD};$$

that is

$$\lambda\lambda' - \mu\nu' + \mu'\lambda' - \lambda'\lambda + 4\sqrt{ABCD} = \mp 4\sqrt{ABCD},$$

where, putting for a moment $M = a^2 - b^2 - c^2 + d^2$, we have

$$\lambda\lambda' - \mu\nu' + \mu'\lambda' - \lambda'\lambda = 4ab^2 - 4bacd + 4abd + 4cdM,$$

and thence

$$\lambda\lambda' - \mu\nu' + \mu'\lambda' - \lambda'\lambda = -4cd - abcd\,(cd - ab) + 8cdM;$$

viz. putting for a moment $X = M^2 - 4cd(cd + 2b)$, this is

$$(X - 4cdM)^2 = (X + 4cdM - 16abc^2d)^2;$$

viz. substituting for X its value, the equation is

$$16cd(a^2 + b^2 - c^2 - d^2)(M + 2cd) = 0;$$

that is

$$16abcd(M + 2cd) = 0,$$

which means simultaneously $1^\circ. a = 0, b = c; 2^\circ. b = c, a = c; 3^\circ. c = 0, a = c$; the three cases are easily equivalent, but the resulting general theorem is in different forms. 11. Here $A = 0, B = 4ac(c + b), C = 0, D = 4ac(c + b)$; the relation between a, b contains the factor a, and throwing this out, and also the constant factor $4a$, it is

$$u(v + b) + 1(a + b + 1) + 2a = 0,$$

viz. a is given as a rational function of b.

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2°. Here $A = 0, B = 4(a + c), C = 4b(a + c), D = 4b(a + c)$; the equation contains the factor a, and throwing out this and also the constant factor $4b$, the equation is

$$v(b - a) - a(a + b + 1) + 2a + a = 0;$$

viz. v is then given as a rational function of a. 3°. Here $A = 4(a-b), B = 0, C = 4b(a-b), D = 0$; or, dividing by $4a$, the equation is

$$(a - b)(1 + v^2) + (a + a + b)(1 + b^2) = 0,$$

which may be reduced to

$$(a + 1)(a + b) + av + a) = 0;$$

giving a or v each a rational function of the other. I do not discuss the theory in detail, but only remark that, in each case there is a conic thrown off, and that in place of the sextic we have a unicursal (or trinodal) quartic curve.

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On the Surfaces divisible into Squares by their Curves of Curvature.

[From the Proceedings of the London Mathematical Society, vol. IV. *(1871–1873), pp. 120, 121. Read June 13, 1872.]*

PROFESSOR Cayley gave an account of an investigation recently communicated by him to the Academy of Sciences at Paris. The fundamental theorem is that, if the coordinates x, y, z of a point on a surface are expressed as functions of two parameters p, q (each expression, of course replacing the equation of the surface), and if these parameters are such that $g = \text{const.}, q = \text{const.}$ are the equations of the two sets of curves of curvature respectively; then (writing for shortness

$$\frac{dx}{dp} = a_1, \quad \frac{dx}{dq} = b_1, \quad \frac{dy}{dp} = a_2, \quad \frac{dy}{dq} = b_2, \quad \frac{dz}{dp} = a_3, \quad \frac{dz}{dq} = b_3;$$

A, B, C for $bc - bc, ca - ca, ab - ab$).

In order that x, y, z, the coordinates x, y, z considered always as functions of p, q, satisfy the equations

$$\begin{vmatrix} a_1, & b_1, & A_1a_1 + Bb_1 \\ a_2, & b_2, & A_2b_2 + Bb_2 \\ a_3, & b_3, & A_3b_3 + Bb_3 \end{vmatrix} = 0.$$

The last equation is equivalent to

$$a_1 + A_1c_1 + Bb_1 = 0,$$

$$a_2 + A_2b_2 + Bb_2 = 0,$$

$$a_3 + A_3b_3 + Bb_3 = 0,$$

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and if in the notation of Gauss we write

$$a_1^2 + a_2^2 + a_3^2 = E,$$

$$a_1^2 + a_2^2 + a_3^2 = G,$$

then adding the equations multiplied by a_1, b_1, c_1 respectively, and also adding the equations multiplied by a_2, b_2, c_2 respectively, we find

$$A = \frac{1}{E}\frac{dE}{dq}, \quad B = -\frac{1}{G}\frac{dG}{dp};$$

and the equations thus become

$$2\,a_1 - \frac{1}{E}\frac{dE}{dq}\,a_1 + \frac{1}{G}\frac{dG}{dp}\,b_1 = 0,$$

$$\&c., \quad \&c., \quad \&c.,$$

which, in fact, agree with the equations (10 bis) in Lamé's "Leçons sur les coordonnées curvilignes," Paris (1859), p. 93. The motion will be divisible into squares if only $E : G$ is the quotient of a function of p by a function of q, or say if

$$E = \Phi p^2, \quad G = \Psi q^2,$$

where Φ is any function of p, q (any of p and q respectively). It seems likely indeed that $E : G$ is the quotient of a function of p by a function of q, expressing therefore the above equations may be reduced to

$$\frac{1}{E}\frac{dE}{dp} - \frac{1}{G}\frac{dG}{dq}\,a_1 + \frac{1}{E}\frac{dE}{dq}\,b_1 = 0,$$

and the equations for x, y, z are

$$2a_1 - \frac{1}{-}\frac{d}{dp}\,a_1 + \frac{1}{-}\frac{d}{dq}\,b_1 = 0,$$

viz. x, y, z being homogeneous functions of p, q such that $x = a_1 q + bp$, and which besides satisfy these equations, or say which each of them satisfies the equations

$$2a_1 - \frac{1}{-}\frac{d}{dp}\,a_1 + \frac{1}{-}\frac{d}{dq}\,b_1 = 0,$$

then the values of x, y, z in terms of (p, q) determine a surface which has the property in question.

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On the Mechanical Description of a Cubic Curve.

[From the Proceedings of the London Mathematical Society, vol. IV. *(1871–1873), pp. 175–178. Read November 14, 1872.]*

IF the coordinates x, y of a point on a curve are rational functions of $\sin\phi$, $\cos\phi$, $\sqrt{1 - F\sin^2\phi}$, the curve has the deficiency 1, and conversely in any curve of deficiency 1 the coordinates x, y can be thus expressed in terms of the parameter ϕ. Hence writing $\sin\phi = b\sin\theta$, the coordinates will be rational functions of $\sin\phi$, $\cos\phi$, or say of $\sin\phi$, $\cos\phi$, $\sin\theta$, $\cos\theta$; and the mechanical representation of the relation $b\sin\phi = \sin\theta$ is, writing for shortness $I = ar + b$, that of any curve $y = a + b\sin\phi$, which values may be written

$$x = a\sin\phi,$$

$$y = \frac{1}{\sin\phi}(\cos\theta - 1) = \sin\theta;$$

I found, however, that this was not the cubic curve most easily constructed; and I ultimately devised a mechanical arrangement consisting of: 1. Rod GH, and connected with it by a pin at H, rod HI (*).

(*) *There were a mechanical conventions in this, but obtuse that producing GH to meet IF in I, the angle GIF might have been made one of.*

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2. Square ACD, and connected with it by a pin at D, rod DG.
3. Square ECF, the two squares being connected by a pin at C.
4. Rod IJ.

[Figure: Mechanical linkage with quadrilateral, two squares, and rods — diagram omitted]

The rod GH rotates about a pin at G; taking $HA = HJ$, there is a pin at A connecting a fixed point of this rod with the extremity A of the square ACD: the fixed point B of the square moves along the line Gx. There is a pin at J connecting the extension of the rod HI with the rod JL, the angle which it makes with JL being equal to that which the rod IJ makes with JG, the fixed point F of the square ACF also slips along the line Gx, and the two squares are so related to each other that, in the motion of the system, while the curve traced out by the point L rotates about a fixed point, the latter of the square $ABCD$ also rotates by reason of its equation of the form above referred to; and then the lines DJ, CF keeping the connection with each other determined by the relation between the squares and the pin at I, the rod IJ rotates about J. The angles which the rods make of being all 8 to it.

I write

$$\begin{aligned}&\angle DBJ = \theta, \quad \angle ABJ = \phi;\\ &GA = a, \quad AB = b, \quad AC = c, \quad CD = d,\\ &AB = HJ = \mu.\end{aligned}$$

Take

$$x = QC, \quad y = JL.$$

I write

$$\angle DBJ = \theta, \quad \angle ABJ = \phi,$$

$GA = a$, $AB = b$, $AC = c$, $CD = d$, $AB = HJ = \mu$. Take

$$x = QC, \quad y = JL.$$

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We then have $x\sin\theta = b\sin\phi$, and moreover the length AJ being $= b\cos\phi \pm b\cos\phi$, and thence $HJ = b\cos\phi$, $HJ = b\cos\phi$, we have

$$y = (b \pm a)\frac{-b\cos(\theta \pm \phi)}{\sin\theta} + b\cos\phi,$$

$$x = QC = c\sin\phi;$$

whence also

$$xy = b - b\sin(\theta \pm \phi),$$

or we have

$$xy = b - b\,(1 - \frac{x^2}{c^2})(1 - \frac{b^2}{d^2}) \pm b \cdot \frac{x}{d}\,\frac{x}{c}\,;$$

that is

$$x\left(y - \frac{b\,d}{c\,d}\right) = b - b\left(1 - \frac{x^2}{c^2}\right)(1 - \frac{b^2}{d^2}),$$

or rationalising and reducing, this is

$$xy^2\,\frac{b\,d\,d}{c\,d}\,2b\,c\,y + \left[b\left(1 - \frac{b^2}{c^2}\right) - 2b\right]\,c^2 + a^2 - b^2 = 0,$$

a quartic curve with two dps.

In the particular case $F = a = -b$, the equation becomes

$$\left(y - \frac{2b}{c}\right) - a + b\,(\,xy - d\,(a+b)\,x = 0,$$

the equation in x becomes the equation in the form

$$x\left(y - \frac{2b}{a}\right)\,d(a-b)\,[x^2(a-d) + a\,b\,k]\,x;$$

the second factor is extraneous, and the curve is thus the cubic

$$x\left(y - \frac{2b}{a}\right)\,d(a-b)\,[x^2(a-d) + a\,b\,k]\,x;$$

at one stage appears from the foregoing irrational form of the equation.

In the particular case $b = c$, the equation contains the factor x, and omitting this it becomes

$$x\left(y - \frac{2b\,c}{a}\right)\,c\,d\,y + b\,c\,b + \cdots$$

viz. we have a cubic curve with three real asymptotes meeting in a point which is also the centre of the curve.

If simultaneously $a = b$ and $b = c$, then equation becomes

$$x\left(y - \frac{2b}{a}\right)\,c\,d\,y - 2c\,d\,x = 0,$$

the actual locus being in this case the line $y - \frac{2b\,d}{c} = 0$.

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Writing $b = c$, and for greater convenience $b = c = a = -d = 1$; also for the sines supposing $b < c$, and writing $a = 1$, $\sin b$, then we have

$$\sin\phi = h\sin\theta, \quad \sin\phi = a\sin\phi,$$

that is

$$\sin\phi = h\sin\phi,$$

and

$$x = c\sin\phi, \quad y = \frac{1-\cos(\theta+\phi)}{\sin\theta} - \cos\phi,$$

that is

$$xy = 1 - \cos(\theta+\phi) - c\sin\theta\cos\phi,$$

giving the rationalized equation

$$x\,(y - 2(1-c)x - 2y) + 4x^2;$$

the angle ϕ may be anything whatever, but θ varies between the limits $\pm\frac{1}{2}\lambda$, the simultaneous values of these angles and of the coordinates being

$$\begin{array}{cccc}
\phi = 0 & \theta = 0 & x = 0 & y = 0 \\
\phi = 90^\circ & \theta = \lambda & x = c & y = 1 + \sin\lambda \\
\phi = 180^\circ & \theta = \pi & x = 0 & y = 2 \\
\phi = 270^\circ & \theta = -\lambda & x = -c & y = -(1+\sin\lambda) \\
\phi = 360^\circ & \theta = 0 & x = 0 & y = 0;
\end{array}$$

and it thus appears that the mechanism gives the continuous branch which belongs to the asymptote $x = 0$ of the cubic curve; the other two branches belong to $x = +\sin\phi$, $y = \frac{1-\cos(\theta+\phi)}{\sin\theta} - \cos\phi$, which would require a slight alteration in the arrange- ment of the mechanism.

I remark that if AB, HI had been unequal, then writing $\angle HJA = \chi$, this would be connected with θ and ϕ by an equation of the form

$$\sin(\theta+\phi) = m\sin\chi,$$

and the coordinates x, y would be rational functions of the sines and cosines of θ, ϕ, χ; the deficiency is in this case $= 3$.

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On the Mechanical Description of Certain Quartic Curves by a Modified Oval Chuck.

[From the Proceedings of the London Mathematical Society, vol. IV. *(1871–1873), pp. 186–190. Read December 12, 1872.]*

THE geometrical principle of the oval chuck is the well-known one that if a plane move in such manner that two lines Ox, Oy, fixed in the plane and meeting with it, pass through two fixed points A, B respectively; then any fixed point P traces out on the plane an ellipse. The point A is on the (geometrical) axis of the mandrel; there is connected with the head a guide-ring moving horizontally; the point B is the centre of the guide-ring; the head is fixed in a ring through a ring which slides horizontally at right angles to the axis of the mandrel; and as the axis revolves, the centre B describes practically a circle on the head A, which by reason of the guide-ring remains fixed at the centre. Hence the point P in the plane traces out the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, as it ought; but as the axis of A is a fixed point, this point being the line at infinity in the plane, what would be for a lathe.

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The apparatus consists of a piece of rock-board, about 10 inches long by 7 inches broad, pierced with a circular hole of 1 inch diameter for a vertical axis: the edges of the board serve as guides for the frame L, which carries the guide-ring; and resting on the board we have the frame M itself guided by the frame L so as to be able to move independently of each other, but they can be clamped together; the axis has

[Figure: Modified oval chuck with rectangular board, frames, axis, slide rings, and slot — diagram omitted]

upon it a square and, the sides of which work on the slot of an eccentric, the frames of this being adjustable by means of a screw passing through the nut and arm; and there is above the eccentric a square nut shown in the figure. This is capable of rotation round an axis at the foot, of an arm at the action by means of a slide; the ari is checked at the back of the eccentric, the action being placed so that the eccentric may be placed in two different positions: viz., the guides may be either upper or lower edges of the cut, those which are parallel to the sides of the slot, or the front of the slot are at right angles to it.

Supposing the bed planned as above upon the guide-ring and nut, then if the frames L and M are disconnected, and the former of them is fixed, the frame M will, on rotation of the axis, be carried backwards and forwards by the encore; but this will in no wise affect the motion of the head, but the arrangement is then equivalent to the oval chuck, and a fixed point in the head will go gives a position will trace out the frame M alone, then we have always through the point A, B respectively; the former of these being a fixed point, the latter of them a point moving according to determinate law backwards and forwards along a fixed line through A.

The plan is carried out in a drawing apparatus which I have had constructed in wood, the axis being here vertical instead of horizontal, and the details of course different from what they would be for a lathe.

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Let the coordinates of the fixed point P, referred to axes through A, the first of them perpendicular to, and the second coincident with, AB, be h, v; let the distance AB be $= a$, and let θ denote the angle BAO. Then, if x, y are the coordinates of P referred to the origin O and axes Ox, Oy, we have

$$x = a\cos\theta + h\sin\theta + v\cos\theta,$$
$$y = -h\cos\theta + v\sin\theta,$$

which, if a be constant, gives a quadric equation, or the curve is an ellipse; and, in particular, if $h = 0$, that is if the point P is on the line AB, then we have

$$x = a(c - r)\cos\theta, \quad y = v\sin\theta,$$

or the curve is

$$\frac{x^2}{(a+r)^2} + \frac{y^2}{r^2} = 1.$$

But if a is a given function of θ, then this equation is still found by eliminating θ between the two equations for x and y. In particular, if the distance AB is equal to the projection upon the tangent of a circle, as shown, in the figure, then if b be the radius AC of this circle, and ϕ the inclination of AC to A-O (it being taken to be constant), we have

$$a = b\cos(\theta + \lambda),$$

and the equation are

$$x = b\cos(\theta+\lambda) - b\,(1\cos(\theta+\lambda)) + v\cos\theta,$$
$$y = -b\cos\theta + v\sin\theta.$$

The elimination is easily done in as it were $= a + 0$; viz., we may determine y, z in such wise that

$$b\sin\theta + v\cos\theta = u(\theta+\lambda) - y\cos\theta, \quad -y\cos\phi = \sin\theta,$$
$$-b\cos\theta + v\sin\theta = a(\theta+\lambda) - y\cos\phi = -y\sin\theta;$$

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and then

$$x = y\cos\theta - b(1-\cos\theta + a) + a(\theta + a) - y\cos\phi, \quad y = -y\sin\theta - x;$$

1, will, for greater simplicity, at once write $b = 0$; the equations than are

$$x = -y\cos\theta + a(\theta + a)\sin\theta\cos\theta, \quad y = y\sin\theta;$$

or say there is

$$x = (a + b\sin\phi)\cos\theta + b\sin\theta\cos\theta,$$
$$y = y\sin\theta,$$

whence, we have

$$x\cos\phi = y\sin\phi + b(1 - \sin\theta - F\cos\theta\cos\theta + \ b\cos\theta b)$$

that is

$$(x\cos\phi - y\sin\phi - \frac{1}{2}b)^2 = b^2\, by\, \sin\theta\, V\,(F-b);$$

an equation which will assume a more simple form if either $\lambda = 0$ or in $\lambda = 90°$; that is, if in the apparatus the small-circle which guides the bed are either at right angles or parallel, to the slot of the slot.

Taking the general case, and writing for convenience $\cos\phi = \text{const.}\cos\phi = a$, the form is given by the form

$$\begin{aligned} x &= e(\eta, b, B, Y) \\ y &= \eta = b \cdot a, Y \\ F &= a + b \cdot a Y; \end{aligned}$$

viz. the elimination of ξ η from these equations leads to the equation of the curve. The points of the curve have then $u\sin\phi$, $(\eta + a)$ correspondences with those of the conic $F = \eta = b \cdot a$, viz. the circle being unicursal, the curve is also unicursal. And con- sidering the relation between the variable parameters ξ, η as the equation of a conic in the ξ, η plane, the curve is in like manner the unicursal transformation by the conic ξ, η so that the foregoing relation form a unicursal quartic. The points where the line at infinity in the ξ, η plane meet the conic conic correspond to the points ξ, η at infinity (a) $= 0$, ξ, η at infinity, then $\xi =$ to be a position $\xi = \xi + Z\xi$, viz., $\xi + \xi$ at the point corresponding to the intersection of $\xi + Z\xi = \xi + \xi$ by the quadric $\xi = \xi$ or ξ, that is, $\xi + \xi$, $a\xi - 1 = 0$.

The intersections by the arbitrary line are $u\,x + b\,a\,y = 0$ which give the points corresponding to the intersection of $\xi + \xi = \xi + \xi$ by the quadric $\xi = \xi$, $a\xi - 1 = 0$.

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giving four intersections. But the intersections by the line $by+cu=0$ that is by any line through the point $y=0$, $x=0$) are obtained from the equation $A\xi^2+C\xi^2=0$, viz. this breaks up into $b\xi+b\xi=0$, $\xi=0$, and the last factor combined with the equation of the circle gives $b=0$, $\eta=0$, $\xi=b$, $\eta=-b$; the two circular points at infinity, corresponding each to the point $y=0$, $x=0$; the other factor gives four points corresponding to two variable points on the curve; viz. the line $by+cu=0$ meets the curve in the point $y=0$, $x=0$ four times each at infinity in the curve $\xi+b\xi=0$. The equations then becomes the point being like as the line at infinity in $\xi+\xi$ is, the line $by+cu=0$ corresponding to the point being at infinity in $\xi+\xi$. We thus have for instance the conic ξ, η at the point at infinity is a node. The curve being trinodal (counting as two nodes) and one other node. The curve being trinodal (counting as two nodes) is of course one other node.

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On Geodesic Lines, in particular those of a Quadric Surface.

[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 191–211. Read December 12, 1872.]

THE PRESENT Memoir contains an investigation of the differential equation of the second order of the geodesic line on a surface, the coordinates of a point on the surface being expressed as functions of two parameters p, q; and remarks on the connexion therewith; a deduction of Jacobi's differential equation of the first order in the case of a quadric surface, the parameters p, q being those which determine the two sets of curves of curvature; herein when the parameters are those which determine the two right lines through the surface; and a discussion of the forms of the geodesic lines in the two cases of an ellipsoid and a sheet hyperboloid respectively.

Preliminary Formulæ.

1. I call to mind the fundamental formulæ in the Memoir by Gauss, "Disquisitiones generales circa superficies curvas," *Comm. Gött. recent.*, t. VI. 1827, (reprinted as an Appendix in Liouville's edition of Monge); together with some that I have added to them. The coordinates of a point on a surface are regarded as given functions of two parameters p, q; these expressions of x, y, z thus determining the equation of the surface, and we have

$$\begin{aligned} dx &= \tfrac{1}{2}\,dx\cdot a_1 + \tfrac{1}{2}\,dy\cdot a_2 + \tfrac{1}{2}\,dz\cdot a_3 + 2x\,da_1 + 2x\,dy\cdot a_2 + 2x\,da_3,\\ dy &= \tfrac{1}{2}\,dx\cdot b_1 + \tfrac{1}{2}\,dy\cdot b_2 + \tfrac{1}{2}\,dz\cdot b_3 + 2y\,db_1 + 2x\,dy\cdot b_2 + 2y\,db_3,\\ dz &= \tfrac{1}{2}\,dx\cdot c_1 + \tfrac{1}{2}\,dy\cdot c_2 + \tfrac{1}{2}\,dz\cdot c_3 + 2y\,dc_1 + 2x\,dy\cdot c_2 + 2y\,dc_3; \end{aligned}$$

$$A, B, C = bc - bc,\; ca - ca,\; ab - ab.$$

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where, differential equation of surface is

$$A\,dx + B\,dy + C\,dz = 0.$$

Also

$$E,\ F,\ G = a^2 + b^2 + c^2, \quad aa + bb + cc, \quad a^2 + b^2 + c^2;$$

so that element of length on the surface is given by

$$ds^2 = dp^2 + dq^2 + (E\,dp + F\,dq + G\,dq^2).$$

The equation $(E,\ G, (dF \cdot dq)^2) = 0$ determines at each point on the surface two directions (commonly imaginary) which are called the "circular" directions. Passing on the surface in any direction from the point in question, the directions of this current (always imaginary) which are the "circular" curves; the equation $(E,\ F,\ G, (dp,\ dq)^2) = 0$ is the differential equation of these curves; and if we have $E = 0$, $G = 0$, $E, dq = G\,dp^2 + 0$; because the differential equation of the circular curves are geodesic lines.

I write also

$$E^a,\ F^a, G^a = aA + bB + cC,\ aA + bB + cC,\ a^aA + b^aB + c^aC,$$

or, what is the same thing, $\{aa^a,\ (BA, B, A\}, \{aa^a, (\,b^aB,\ b^aC\}, \{aa^a, (\,c^aA,\ c^aB\}$, respectively,

$$\begin{vmatrix} a, & b, & c \\ a, & b, & c \\ A_a, & A_b, & A_c \end{vmatrix}, \begin{vmatrix} a, & b, & c \\ A, & B, & C \\ A, & B, & C \end{vmatrix}, \begin{vmatrix} a, & b, & c \\ A, & B, & C \\ A^a, & B^a, & C^a \end{vmatrix},$$

(those last symbols do not occur in Gauss). They are the D, D, D^a, D^a of Liouville, 2. The radius of curvature of normal section corresponding to the direction $dp : dq$ is given by

$$P = \frac{(E,\ F,\ G, (dp,\ dq))}{\sqrt{(E^a + 2F\,dp\,dq + G^a\,dq^2)}};$$

whence it appears that the directions of the inflexional or chief tangents (or Hess- *tangents*) are determined by the equation

$$(E^a, F^a, G(\,dp, dq)) = 0.$$

The directions in question are imaginary on a surface like the ellipsoid (then they are the same as the same one of two curves form the chief curves); but at any geodesic lines. We say on the surface even one of two curves form the chief curve; the chief curves they are at any geodesic line and are the chief curves.

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the differential equation of these is $(E^a, F^a, G(dp, dq)) = 0$, and in particular if $E^a = 0$, $G^a = 0$, then this becomes $dp\,dq = 0$, we have $p = \text{const.}, q = \text{const.}$ for the equations of the two sets of chief curves respectively. On the hyperboloid the chief curves are the two sets of generating lines. The chief curves are not in general geodesic lines, but on the hyperboloid, and though lines, they are, it is clear, geodesic lines.

3. The directions of the curves of curvature, or the principal tangents, and the corresponding radii of the radii of curvature are determined by

$$P = \frac{E\,dp + F\,dq}{E^a\,dp + F^a\,dq} = \frac{F\,dp + G\,dq}{F^a\,dp + G^a\,dq};$$

or, what is the same thing, these directions are determined by the equation

$$\begin{vmatrix} dp^2, & -dp\,dq, & dq^2 \\ E, & F, & G \\ E^a, & F^a, & G^a \end{vmatrix} = 0.$$

The same equations may be written

$$-dp : dp : dq = \frac{aE - F^a}{F^a - F^aF} = \frac{F^a - FG}{F^a - FG} = \frac{F^a - FG}{FG^a - F^aG};$$

that is the principal radii of curvature are determined by the equation

$$P^2(EG - F^a) - P(EG^a + E^aG - 2FF^a) + (E^aG^a - F^{a^2}) = 0;$$

(last term is $= P^a$, but it is better to retain the original form); and then, ρ being either root, the last preceding equation gives the direction of the curve of curvature corresponding to the given value of the radius of curvature.

If $p = \text{const.}, q = \text{const.}$ are the equations of the two systems of curves of curvature respectively, then the quadric equation in (dp, dq) must reduce itself to $dp\,dq = 0$, viz. we have $F = 0$, $F^a = 0$; and the principal equations among these the more simple form; in particular, in this case, instead of forming the equation in (dp, dq), we have for the principal radii of curvature

$$\rho^a + p = \frac{E}{F^a}, \quad \rho^a + p = \frac{G}{G^a}.$$

If $p = \text{const.}, q = \text{const.}$ are the equations of the two systems of curves of curvature respectively, then expressing that the curve which passes through the generating lines. We may as before observe the lines $p = \text{const.}, q = \text{const.}$ of the chief lines, but in this case, as before, the equation $F = 0$, $F^a = 0$ and the generating equations.

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General Theory of the Geodesic Lines on a Surface.

4. I now proceed to investigate the theory of geodesic lines on a surface, the surface being determined as above by means of given expressions of the coordinates x, y, z in terms of the parameters p, q.

The differential equation obtained by Gauss for the geodesic lines is in a form not symmetrical in regard to the two variables; viz., the equation is in a form

$$\frac{dE}{dp}\,dq + 2\,dE\,\frac{dF}{dq}\,dp + \frac{dG}{dq}\,d^a q = 2ds\,\frac{Edp + Fdq}{ds};$$

where, as above,

$$ds = (E, F, G(\,dp, dq)).$$

If we herein consider p, q as functions of a variable parameter θ, and we further not symmetrical equation in p, q, viz.

$$dp\,d^a q - dq\,d^a p, dq^2, dp\,dq, dp^a = 2ds(F, F, G(\,dp, dq)) = 0,$$

$$dq^2, E, F, G,\ d^a p, d^a q;$$

also

$$H_a + F = E,\ G,\ d^a q,\ \&\text{c.},$$

then the equation is

$$(E, F, G\,dp, dq) = \Sigma_a(H(F\,dp + Fdq) - E(F\,dp + G\,dq))\,d^a q,\ \&\text{c.}.$$

We have

$$\left(\frac{F'dp + F'dq}{\sqrt{ds}}\right)^a = \frac{1}{ds}\,(M + N),$$

where X is the part containing p, p^a, and only the first differential coefficients of p^a no second differential coefficients of p; and N is the second differential coefficients of p, viz., for the simply, in regard to the parameter θ, $E = E\,Edp$, and adding, we have

$$X = -\frac{1}{2}\,p^a\,(R\,dp{+}F'dq){-}\frac{1}{2}\,p^a\,F'dp{+}F'dq{+}Q\,(F\,dq{+}F'dp)\,dq{+}dq\,(R\,(F\,dp{+}F'dq)){+}E\,Edp{+}F\,dp,$$

or substituting for Ω its value, this becomes

$$= -p^a\,(E^a\,F'{+}Q\,EF{+}F\,F'){-}p^a\,(EFF'{-}E\,F'G^a){-}p^a\,F'\,dpG{+}pq\,(F\,F'G{-}F\,F'G){-}EFEdp{+}F\,F'\,dq\,;$$

and the second term is contained, and the curve is then in evidence as follows:

$$\frac{dE}{dp}\,dp{+}\frac{dE}{dq}\,dq{+}\frac{dE}{dq}\,(ER{-}EF)\,(d\,p{+}Q)\,(EFEF{+}0)dq,{+}2EF\,(dp\,dq{+}dq\,Edp{+}F\,dq)\,dq\ dp\,(R{-}Q){+}(Q\,(EF{-}E$$

where we know that the whole equation must divide by q.

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5. We have

$$M = (\nu E+\alpha)(E\,dp+F\,dq)^a+(E\,dp+F\,dq)(EFq+4yF^a)(F\,dp+G\,dq)+p^a\,(EE+2FF'-FG)(F\,dq+G\,dp)+4F'\,(I$$

or say

$$2M = (-p^a + Fp + 4yF^a)X + 4F'\,(2F + G)(F + G) + 4Y\,B\,Y\,;$$

where, putting for shortness $X, Y, Z = E\,dp + F\,dq$, $F\,dp + G\,dq$, $E\,F'\,p + G\,dq$, etc., respectively (these are functions of p, q, p^a, q^a only),

$$\begin{aligned}
\beta &= \frac{dE}{dp} - 2\frac{dF}{dq} + 4\,p^a, \quad p^a = \frac{dE}{dp},\\
\gamma &= -2\frac{dE}{dp} + 4\,p^a\,\frac{dF}{dq} + 3\,p^a;\\
\delta &= +2\frac{dE}{dp} - 4\,p^a\,\frac{dF}{dq} + 4\,p^a, \quad p^a = \frac{dE}{dq};\\
\epsilon &= +\frac{dG}{dp} - 4\,p^a\,\frac{dG}{dq} + p^a + 3\,p^a;\\
B &= \frac{dG}{dp} + 4\,p^a\,\frac{dG}{dq} + 4\,p^a, \quad p^a = \frac{dE}{dq}.
\end{aligned}$$

The form in g^a in EE^a, which is in fact $= a^a(\,Z(p^a+Eq)^a - 2Y(p^a q^a + Z(\,dp+Eq^a)^a)$. Presenting now the value of X, the geodesic equation thus becomes

$$-p^a\,(EE^a+p^a(EE^a+p^a(EFE^a-EE^a)-p^aFPGE+2FPGE)-E^aF'\,dqE^a+F\,dqE^a+(-p^a\,(EE^a+p^a(p^a+4yF^a)$$

which is the required differential equation of the second order; we shall see, however, that this is in another form, viz. as a more simple equation between p^a and q^a, supposing p, q to be functions of the parameters θ; not, as in the conjugate substitution).

The condition in question is expressed by the equation:

$$\begin{vmatrix} A, & B, & C \\ xy + dpy\,xy + dyy, & y^a + y\,dy\,y\; + y\,dqy, & xy \\ xy + dqy\,xy + dyy, & y^a + dy\,y\; + y\,dqy, & y\,dqy \end{vmatrix} = 0,$$

that is

$$\begin{vmatrix} A, & B, & C \\ ay^a + d\,y, & by^a + dy, & cy^a + cyy \\ ay^a + a\,y, & by^a + by, & cy^a + cyy \end{vmatrix} = 0.$$

9. The first determinant in the sum of two terms each as $A\,(bc^a)\,B(yz - z\,y)\cdot$, etc., $(A\,(yc^a) - B(2y - x\,y) + (xD - F(2y) - p^aFp^a$; The second determinant is the sum of three terms each as $a\,(yc^a - z\,y)\cdot$, etc., where the factor $\{,\,\}$;

$$\rho^a\,[a\,(b\,(b\,c\; - \;c\,b\,(b\,c\,(a^a)\; + \;c\,(b\,(b\,c\; - \;c\,(b\,(b\,c\,(a^a)\;\;(b\,(b\,c\,(a^a)\,(b\,(b\,c\,(a^a)\,(b\,(b\,c\,(a^a)\,;$$

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7. Instead of starting, as above, from the equation given by Gauss, we may use the geometrical property that at each point of a geodesic line the osculating plane passes through the normal of the surface.

Considering, as above, p, q as functions of a variable θ, then θ becoming $\theta + d\theta$, the new values of p, q are

$$p + p^a d\theta + \tfrac{1}{2}p^a\, d\theta^2, \quad q + q^a d\theta + \tfrac{1}{2}q^a\, d\theta^2,$$

and then those of x, y, z are

$$x = ap^a d\theta + bq^a d\theta + \tfrac{1}{2}\, d\theta^2\, (ap^a + bq^a + 2a\, p^a q^a + 2b\, p^a q^a + 2b\, p^a q^a),$$

$$y = bp^a d\theta + bq^a d\theta + \tfrac{1}{2}\, d\theta^2\, (bp^a + bq^a + 2a\, p^a q^a + 2b\, p^a q^a + 2c\, p^a q^a),$$

$$z = cp^a d\theta + cq^a d\theta + \tfrac{1}{2}\, d\theta^2\, (cp^a + cq^a + 2a\, p^a q^a + 2b\, p^a q^a + 2c\, p^a q^a),$$

The condition in question is expressed by the equation

$$\begin{vmatrix} A, & B, & C \\ . & . & . \\ . & . & . \end{vmatrix} = 0,$$

that is

$$\begin{vmatrix} A, & B, & C \\ ap^a + bq^a, & bp^a + bq^a, & cp^a + cq^a + cy^a \\ ap^a + aq^a, & bp^a + bq^a, & cp^a + cq^a \end{vmatrix} = 0.$$

8. The first determinant is the sum of three terms such as $A\,(bc^a - c\,b^a)p^a$, &c., viz. this is $A\,(p^a y^a - q^a y^a)$, where A stands for $AD - BC$ (viz. $\equiv FF^a - EF^a$).

The second determinant is the sum of three terms such as $a\,p^a(bc^a - c\,b^a)$, &c., where the factor $\{,\}$ is

$$p^a\,[a\,(b\,(b\,c\; - \;c\,b\,(b\,c\,(a^a)\; + \;c\,(b\,(b\,c\; - \;c\,(b\,(b\,c\,(a^a)\;(b\,(b\,c\,(a^a)\,(b\,(b\,c\,(a^a)\,(b\,(b\,c\,(a^a)\,;$$

The second determination is $-2p^a q^a (EE^a + (2D - p^a + Pp^a) + p^a\,(Pq^a + p^a + q^a)$ viz. taking $B = u^a\, EE^a + 2EF^a y^a + Gq^a. \quad ((PD - p^{aa}) + p^a Pp^a + p^a).$

The second determination is $-2p^a q^a (EE^a + (2D - p^a + Pp^a) + p^a\,(Pq^a + p^a + q^a)$ where the factor $\{,\}$ is

$$p^a\,[Y(a\,a - c)(a^a)] + 2p^a q^a\,[Z(a^a + a^a)(b^a) + a^a + q^a\,(R(a^a) + p^a + q^a(b^a)];$$

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which is the second determinant; the second factor is, as above $A\,((py^a+q^a+R(p^a+q^a)$. We have thus the second determinant,

$$(EX-F^a)\,p^a+2(FQ+FF^a)p^aq^a+2(FF^a+GQ)p^a+R(p^a+q^a)=4F^a\,(2p\,q+Q\,p^a+Q\,p^a)-p^a(\,aB^a+BY'\,EE^a\,p^a+$$

an equation of the same form as that previously obtained, and which has been differently identified therewith.

The Circular Curves are Geodesics.

9. I proceed to show that the circular curves are geodesics; viz. that an integral of the geodesic equation is

$$(E,F,G)(\xi,\eta)^2=0.$$

Starting from this equation, we have

$$2(E\xi+F\eta)(E\xi+F\eta)+2(F\xi+G\eta)(F\xi+G\eta)+(E^a\xi^a+2F^a\xi\eta+G^a\eta^2)\,(\xi+\eta)\,d\xi+(E^a\xi^a+2F^a\xi\eta+G^a\eta^2)(\xi+\eta)\,d\eta=$$

Now the equation, writing therein $E\xi^a+Fq=q^aw$, gives $Fq+Q\,p\,q=q^aw$; their expansion may be written

$$4F\eta(E\xi+F\eta)=0;$$

the value of w being therefore $-p^a-2Bp^a+q^a$. The result then divided thus becomes $2Bw(E\xi-F\eta)(Fq+G\eta^a)+(E\xi G-E\xi G^a)(\xi w^a+2F^a\xi\eta+G^a\eta^2)=0$,

$$=\Sigma(\xi+Zw)(F^a+Gw^a),$$

that is

$$2(E(E\xi+F\eta)(2Eq-F^a)+(E\xi G+2(F^aF^a+F^a+G^aw^a)\,2w)+(ay+bw)(F^a+bw^a)-(aw+b\xi+dw)(Pu+2F\xi y+dw^a$$

viz. taking $\beta E\xi=(Q\,Q^a+F^aw+d^a)$; this agrees with the geodesic equation.

The foregoing integral, $(E,F,G)(\xi,\eta)^a=0$, I believe, a particular, not a singular, solution of the geodesic equation.

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The Chief Lines are not in general Geodesics.

10. That the chief lines are not in general geodesics appears most readily as follows:

To find the condition in order that $p =$ const. may be a geodesic, we write in the geodesic equation $p^a = 0$, $p^a = 0$; the equation thus becomes

$$F^a\, q^a - Bq^a + EE^a q^a = 0,$$

that is to say

$$2EE^a = E^a B - EB^a,$$

as the condition in order that $p =$ const. may be a geodesic; the condition that it may be a chief curve is $E^a = 0$, which is a different condition.

We have of course, in like manner,

$$2GG^a = G^a B - GB^a,$$

as the condition in order that $q =$ const. may be a geodesic; the condition that it may be a chief curve is $G^a = 0$. If, in any case, both conditions are satisfied, we will have

$$FG - F^a G' + GE^a = 0, \quad EF^a - EE^a F + EE^a = 0,$$

and that is satisfied. If $p =$ const. and $q =$ const. are both of them sets of lines on the surface, then the four equations are satisfied.

Special Form of the Geodesic Equation.

11. In the case where the curves $p =$ const., $q =$ const. intersect at right angles (and in particular when they are the curves of curvature), we have $F = 0$, then since also $F^a = 0$, $F^a = 0$; and the geodesic equation becomes one much more simple form,

$$2(E\xi^a + Q\,p)(E\xi G - Gp^a)(pE^a + p^a + q^a + q^a) + (E\xi G - F^a G^a)q^a + 2(E\xi G^a - F^a G^a)q^a q^a;$$

$[E^a]$ In the case of a surface of revolution we have

$$dx = (1 + F^a)dp^a + p\,dq^a,$$

that is

$$E = 1 + F^a, \quad E^a = TP^a, \quad EE^a = 0,$$

$$G = p^a, \quad G^a = 0, \quad G^a_a = 0;$$

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hence the differential equation is

$$(1+F^a)p^a(q^a-p^a)\,q^a - p\,(d(p)^a+p)^a q^a = 0;$$

this has an integral

$$\frac{(1+F^a)p^a}{p^a}+\frac{1}{p^a}=C,$$

or say

$$Cdp^a = p^a(p-C^a)dq^a;$$

where

$$Cp+Q=p^a+p^a;$$

Writing here ρ, ϕ for p, q, where ρ is the distance of the point from the axis, and ϕ is the longitude reckoned from an arbitrary meridian, then the equation is

$$Cdq = ax,$$

which is the equation given by Legendre, *Théorie des fonctions elliptiques*, t. 1, p. 361. This may also be written

$$\frac{C^a}{C}=\cos\psi$$

if ψ be the inclination of the geodesic line to the parallel of latitude.]

Geodesics on a Quadric Surface.

12. In the case of a quadric surface, $\frac{x^a}{a}+\frac{y^a}{b}+\frac{z^a}{c}=1$; writing for shortness $x, y, z, =, e-a, -e-b, e-c$, and we may express the coordinates x, y, z in terms of two parameters p, q as follows:

$$x=\frac{2py(a+p)(a+q)}{(a-b)(a-c)},\quad y=\frac{2py(b+p)(b+q)}{(a-b)(a-c)},\quad z=\frac{p^a(a+p)(a+q)}{(a-b)(a-c)};$$

where, in fact, $p=$ const., $q=$ const. are the equations of the two sets of curves of curvature respectively. Writing moreover

$$P=\frac{1}{(a+p)(b+p)(c+p)},\quad Q=\frac{Q}{(a+q)(b+q)(c+q)};$$

that is

$$dx=\frac{1}{2}(b-q)p^aP\,dp,\quad \frac{1}{2}(b-p)q^aQ\,dq;$$

and the geodesic equation becomes

$$E,\,F,\,G\,((p-q),\,4(b-q)P,\,(b-q)Q,\,(b-q));$$

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where E^a, G^a stand for $\frac{de}{dq}$ and $\frac{de}{dq}$ respectively, viz. this is

$$\rho^2 = \frac{P^a}{Q^a} = P[q{+}(q\,dQ\,dp{+}P\,dQ\,dq)], +p^a yP{+}(q\,dq)\,dQ{+}Q\,dP]{+}q{+}Q^a,\ = P[p{-}q{+}(q\,PP{-}P\,Q)Q]dq.$$

or, differentiating logarithmically, this gives

$$\frac{2p^aP^a}{p^a+Q^a}(p-q) + \frac{P^a}{q+Q^a}(p^a+Q^a)dq + \frac{P^a}{P^a}(p^a-q+Q^a) = 0;$$

viz. differentiating the denominators, this gives

$$\frac{pPP^a - p^aP^aP^a}{P^a+Q^a} - \frac{QdP^a + QdPd^aQ + (PP^a - P^aQ)(P^a - p^aQ + P^aP^a)dq}{p^a+Q^a+Q^a} = \frac{P^a}{P+Q^a}\,(p^a{-}Qd^aQ{+}(PP^a{-}P^a$$

that is

$$= R(p-p)\,(PP^a - P^aQ)\,dq + (PP^a - P^aQ)\,(p^a + Q^a + Q^a),$$

or

$$= R(p-p)\,Q + (P+Q^a)\,Q^a + Q^aP;$$

We have next that

$$(pPP^a - p^aP)\,(p^a - Q) - p^a(P^a+Q)\,(P^a - P^aQ + P^a) + p^a(qP - Q)\,(P^a - P^aQ)\,;$$

which holds for any function f, we require its value when

$$f = -\,p\,a - b - c - p,\ 2(P-p)(P+p)Q;$$

and the differential equation, $(E,F,G)(\xi,\eta)^a = 0$, of the chief lines becomes

$$\xi^a\,(p\,a + p + p\,c + p) + (E+Q)(p^a + QP^a)\,d^aQ + (p\,a + Q)(P^aQ + P^aQ)\,(p^a + Q^aP) = 0,$$

involving the arbitrary constant θ as the differential equation of the first order of the geodesic lines on the quadric surface, $\frac{x^a}{a} + \frac{y^a}{b} + \frac{z^a}{c} = 1$; the geodesic lines in question

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all touch the curve of curvature determined by the parameter θ, that is the curve which is the intersection of the surface by the confocal

$$\frac{x^a}{a+\theta}+\frac{y^a}{b+\theta}+\frac{z^a}{c+\theta}=1.$$

13. In the particular case $\theta=\alpha$, the equation becomes

$$P\,dp^a - Q\,dq^a = 0,$$

that is

$$\frac{p\,dp^a}{p\,(p+p)(p+p)-p\,a(p+p)}=0,$$

which is the differential equation of the circular curves on the surface.

14. The signification of the case $\theta=b$ is not at first sight so obvious. Supposing that θ is first indefinitely small, and writing the equation in the form

$$\frac{\xi^a}{p}+\frac{\eta^a}{q}+\frac{\xi^a}{c}+\frac{1}{p^a}=0,$$

we have at once that the equation $F=0$ is

$$\frac{x^a}{a}+\frac{y^a}{b}+\frac{1}{p}\left(1-\frac{x^a}{a^a}\right),$$

$$\frac{1}{p}\,\eta^a-\frac{p^a}{p\,(p+p)}\left(1-\frac{x^a}{a}\right)$$

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On Geodesic Lines, in particular those of a Quadric Surface.

[continued from p. 166]

p. 167. (x as arbitrary parameter) as the equations of a right line on the surface; viz. considering x, p, w denoting the foregoing functions of p and q, these two equations are forms of a single relation between p, q, w, which relation expresses that the point (p,q) is situate on the right line determined by the parameter w. We may from this integral equation deduce without difficulty the foregoing differential equation; viz. we have

$$\frac{dw}{\sqrt{w}}+\frac{4dp}{\sqrt{P}}=\frac{c\,dp}{\sqrt{P}},$$

$$\frac{dw}{\sqrt{w}}+\frac{4dq}{\sqrt{Q}}=\frac{5dq}{\sqrt{Q}},$$

or multiplying these equations

$$\frac{dw^2}{w}+\frac{2dp\,dq}{\sqrt{PQ}}=0,$$

and substituting herein for dw, dp, dq their values in terms of dp and dq, we find the required equation

$$\frac{dp^2}{(x+p)(b+p)(c+p)}+\frac{dq^2}{(a+q)(b+q)(c+q)}=0.$$

19. I return to the integral equation involving w; we have to rationalise this equation, that is, obtain from it an equation containing w^2, p^2, r^2, and then substituting for these their values in terms of p, q, we have the required relation between p, q. We at once obtain

$$\left\{2\left(\frac{c^2+w}{a+b}\right)-\left(ax+\frac{r}{b}\right)(1+cx)\right\}^2=4\left(c^2+\frac{w}{a+b}\right)(1+cx)^2,$$

or if for greater convenience we introduce in place of a a new parameter ϕ, determined by the equation $a^2+\frac{w}{a+b}$, the equation is

$$\left\{2\left(\frac{c^2+w}{a+b}\right)-\phi\left(ax+\frac{r}{b}\right)\right\}^2=4(\phi^2-w^2x^2)=0.$$

Writing for shortness $p+q=X$, $pq=Y$, we have

$$-2bq\frac{r}{b}=a^2+a^2X+Y,\quad -ax+\frac{r}{b}=-bX+cY,$$

and substituting these values, the equation becomes

$$\beta(b^2+bX+Y)=bcY=aX+Y-\phi(a^2+a^2X+Y)+\phi(a^2+bcY)(b^2+a^2Y-aY-aXY).$$

or, what is the same thing,

$$\beta b^2-\alpha cX-\beta a(bX-cY)^2+\phi(-bX+cY+bXY-\alpha cX+bX^2-cX^2Y-aXY).$$

20. This is an equation, quadratic as regards ϕ, and also as regards ϕ; it is of the form

$$(a+2bp+cp^2)+2q(b+2bp+cp^2)+q^2(c+2bp+cp^2)=0.$$

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and it leads to a differential equation

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} = 0,$$

where

$$P = (b + 2bp + cp^2)^2 - (a + 2bp + cp^2)(c + 2bp + cp^2),$$
$$Q = (b + 2bq + cq^2)^2 - (a + 2bq + cq^2)(c + 2bq + cq^2);$$

and upon effecting the calculation, it is found that we have

$$P = wb^2p^2(b^2 - c^2) + 4(p + \phi)(b + p)(c + p),$$

$$Q = wb^2q^2(b^2 - c^2) + 4(q + \phi)(b + q)(c + q);$$

viz. P, Q are the same multiples of $(p + \phi)(b + p)(c + p)$, $(q + \phi)(b + q)(c + q)$ respectively; so that, omitting the common factor, or taking P, Q to represent the last-mentioned functions respectively, we have

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} = 0,$$

and since the parameter ϕ has disappeared, we see that this original equation involving ϕ is the general integral of this differential equation; viz. that the differential equation belongs to the right lines on the surface.

20. The form of the integral equation may be simplified by introducing instead of ϕ a new parameter K, connected with it by the equation

$$K = \frac{\beta b - bc + \phi c}{\beta - b + \phi}, \qquad \text{or} \quad \phi = \frac{(b - K)c - bc + a}{c - K}.$$

viz. we thence deduce

$$\begin{aligned} B - a + \phi &= 2aE \\ \beta b - cn + \phi c &= 2aKE \\ \phi(a^2 + bw)^2 - \phi(a^2 + bw)w + \phi &= (K^2 + a^2K + bcK + bw)c(c - K), \\ \phi + q &= 4K(K - b)(K - c), \\ \phi + r &= (b - K)(c - K), \end{aligned}$$

where denoting $(c+x)^2$ and substituting these values, the conditions are satisfied: we thus again see δ patient that the right lines $x =$ const. and $r =$ const. are geodesics.

20. The other values of the radicals $\sqrt{P}$, $\sqrt{Q}$ must present itself with a negative sign; and we thus have dp, dq each expressed in terms of da, dr; viz. we have the equations

$$\frac{dp}{\sqrt{P}} - \frac{dq}{\sqrt{Q}} = 0,$$

and instead of K introduce the parameter $C = b - K$, the equation becomes

$$\{-p - bcc^2 + 2(C - a)(C - bw)^2 + (a + c)(C(c^2 + Y) = 0,$$

and from these equations $\phi = \beta = \theta + (a^2 - a^2b)(p + c)(p^2 + p)$ obtain

$$(p + q)(b + Y)(c + Y)Y^2 = pq.$$

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or expanding and reducing, this is

$$\begin{aligned} &(Y+(b-C)X)^2 \\ +&Y(a-2k-4bC+4C^2) \\ +&X(2aC-4C-2c)C \\ +&a^2-4aC+4 \end{aligned}$$

or say

$$p^2-4aC$$
$$+(3aA-bC+2C^2)(p+q)+p(c(2a+4bC^2-8C^2)pq+(4-4C^2)(pq)^2+2pq+2(A-C)(p+q)+pq^2 \quad =0,$$

viz. this, examining the constant C, is the general integral of the differential equation

$$\frac{dp}{\sqrt{P}}+\frac{dq}{\sqrt{Q}}=0,$$

where

$$P=(a+p)(b+p)(c+p), \quad a=ap+bp^2+cp^2,$$
$$Q=(a+q)(b+q)(c+q), \quad q=aq+bq^2+cq^2.$$

21. The constant C is connected with the parameter w, which originally served to determine the right line, by the equation

$$w=\frac{1}{b}\cdot\frac{2(b-c)(b-C)-Cb}{b-Cb}\cdot\frac{Cb}{c-C}$$

or, what is the same thing,

$$w=\frac{1}{b-c}\cdot\frac{2c^2-C-C(2-a-b)}{C-b-C}.$$

Reverting to the equation between p, q, ϕ, I remark that if ϕ be therein considered as variable, we have the differential equation

$$\sqrt{q}\,dp+\sqrt{q}\,d\theta=0,$$

where P, Q have the foregoing values

$$P=wb^2p(b^2-c^2)+4(p+\phi)(b+p)(c+p), \quad Q=ko\,,$$

and where, if the integral equation be written in the form

$$L+2M\phi+N\phi^2=0,$$

then we have $\Phi=M^2-LN$, so that

$$\Phi=b\phi^2[p\phi(p+q)+p^2(p+q)+q^2(p+q)+p^2+q^2].$$

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22. Changing the notation, and writing

$$P = (a+p)(b+p)(c+p),$$

$$Q = (a+q)(b+q)(c+q),$$

$$\Phi = (\phi + p^2)(a+b-2c-\phi),$$

the equation is

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} + \frac{\sqrt{2}\,d\phi}{\sqrt{\Phi}} = 0;$$

or if instead of ϕ, we introduce the original parameter w, then, observing that

$$\frac{2dw}{b} = \frac{d\phi}{\sqrt{w(a+b-w)}}$$

we at once find

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} + \frac{4dw}{\sqrt{w}} = 0,$$

where

$$\Sigma = w(1+w) - 2(a+b-2c)w + 4a^2;$$

or, what is the same thing,

$$\Sigma = a(w^2-1)^2 - b(w^2+1)^2 + ac;\ 4ac^2;$$

viz. passing from a point (p, q) on the line w to a consecutive point $(p+dp, q+dq)$ on the line $w+dw$, the above is the relation between the variations dp, dq, dw. If ϕ be the parameter of the other line through the same point, then we have in like manner,

$$\frac{dp}{\sqrt{P}} - \frac{dq}{\sqrt{Q}} + \frac{4dw'}{\sqrt{w'}} = 0,$$

(viz. one of the radicals $\sqrt{P}$, $\sqrt{Q}$ must present itself with a reversed sign); and we thus have dp, dq each expressed in terms of dw, dw'; viz. we have the increments dp, dq when a point passes from (w, w') to $(w+dw, w'+dw')$. These results will be presently obtained in a more simple manner.

Formulæ where the position of a Point on the Surface is determined by means of the two Lines through the Point.

23. We may determine the position of a point by means of the parameters w, r of the two lines through the point. The equations of these are

$$\frac{x}{\sqrt{a}} = \frac{Cw}{\sqrt{b}} = \left(1 - \frac{1}{w}\right)\frac{z}{\sqrt{c}} \quad \left(1 - \frac{Cw}{x}\right)\frac{w}{\sqrt{a}},$$

$$\frac{x'}{\sqrt{a}} = \frac{Cw'}{\sqrt{b}} = \left(1 - \frac{1}{w'}\right)\frac{z}{\sqrt{c}} \quad \left(1 - \frac{Cw'}{x}\right)\frac{w}{\sqrt{a}},$$

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and from these equations we deduce

$$\frac{x}{\sqrt{a}} = \frac{aw+Y}{a+w}, \quad \frac{Cy}{\sqrt{b}} = \frac{cw-1}{c+w}, \quad \frac{z}{\sqrt{c}} = \frac{Y-aw}{Y+w}.$$

We have therefore

$$\frac{dx}{\sqrt{a}} = (w^2-1)\,dw + (w^2-1)\,dr \quad (+),$$

$$\frac{bdy}{\sqrt{b}} = (w^2+1)\,dw + (w^2+1)\,dr \quad (-),$$

$$\frac{dz}{\sqrt{c}} = -B\,dw + 2aw\,dr \quad (+),$$

where denoting $(c+a)^2$, regarding w, r as the parameters in place of p, q, then those the values of the first differential coefficients $\frac{dx}{dw}$, etc. are

$$A = -B\sqrt{a}\,(2w-1)\,, \quad B = -B\sqrt{a}\,(2w+1)\,, \quad C = -B\sqrt{c}\,(2-2w)\,,$$

where denom. $= (r+w)^2$. We have, moreover,

$$\frac{d^2x}{dr^2} = -2(w'-1)\,dr^2 + 2(w'+1)\,du - 2(w'-1)\,dr^2 \quad (+),$$

$$\frac{bd^2y}{dr^2} = -2(w'-1)\,dr^2 + 2(w'+1)\,du - 2(w'-1)\,dr^2 \quad (-),$$

$$\frac{d^2z}{dr^2} = 4w\,dw + 2(1-w)\,du \quad (+),$$

where dn. $= (r+w)^3$. We have, moreover,

$$\frac{d^2x}{dr^2} = (w'-1)\,du - (w'+1)\,du^2 \quad (-),$$

$$\frac{bd^2y}{dr^2} = (w'-1)\,du - (w'+1)\,du^2 \quad (-),$$

$$\frac{d^2z}{dr^2} = -4w\,dw + (1-w)\,du \quad (-),$$

that is, $E' = w$ and $G' = 0$, on the differential equation of the chief lines is $dwdr = 0$, which is right. The value of E $(= dx^2 + dy^2 + dz^2)$ is readily obtained, but it is rarely required; but it is easily found that

$$EG - F^2 = (EF + YB)(r+w)^2(w+r)^2 + r^2 + w^2;$$

or since the term in $\sqrt{(\;)} = -(r+w)^2$, we have

$$\frac{F}{c+r} = \frac{-w(c+a)w}{(r+w)^2}.$$

24. The values of E, F, G $(dx^2 - 2dxdr + 2drdx + 2drdx + Gdr^2)$ are

$$\begin{aligned}
E &= a(c+r)^2 & &+b(c-1)^2 + c(2w-1)^2 & &= 4w,\\
F &= -a(w^2-1)(w'^2+1) & &-b(w^2-1)(w'^2-1) + 2c(2w-1)\,dw & &= 4w,\\
G &= a(c+w')^2 & &-b(c-1)^2 + c(2w-1)^2 & &= 4w,
\end{aligned}$$

where denom. $= (r+w)^2$.

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We have, it is clear, $(E_1 + d_1 K = K_1 + d_1 G$, that is

$$K_1 = \frac{4}{r+w}\,E, \quad G_1 = \frac{4}{r+w}\,G.$$

Hence the condition $FG_1 - 2F_1 G + GF_1 = 0$, in order that $w =$ const. may be a geodesic, reduces itself to

$$\frac{1}{r+w}\,F = 2F_1 + G_1 = 0,$$

and similarly, the condition $-2EFG + E_1 F + EFG = 0$, in order that $r =$ const. may be a geodesic, reduces itself to

$$-\frac{1}{r+w}\,F - 2F_1 + E_1 = 0.$$

We have at once

$$\begin{aligned}
E_1 &= 4a(w-1)(w+1) &&= -4b(w'+1)(w-1) + \&c.\ 2c(w-1)\,dw, \quad (+).\\
G_1 &= 4a(w+1)(w+1) &&= -4b(w'+1)(w'-1) + \&c.\ 2c(w+1)\,dw, \quad (+).\\
F_1 &= 2c(w'+1)(w'+1) &&-a^2(c-1)(w'-1)\,dw = 2c(w+1)\,dw, \quad (+).\\
F_1 &= 2c(w'+1)(w'+1) &&-a^2(c-1)(w'-1)\,dw = 2c(w+1)\,dw, \quad (+).
\end{aligned}$$

where denom. $= (c+w)^2$; and substituting these values, the conditions are verified: we thus again see δ posteriori that the right lines $w =$ const. and $r =$ const. are geodesics.

25. The last-mentioned values of E, G are $E = Y(c+w)^2$, $G = 2(c+w)^2$, and setting for a moment

$$A = (1+r^2)(w+1) - 1)(b-1) + 1)(b+1) - 4cw,$$

we have $F = A - (b+w)^2$, the value of dz^2, viz.

$$ds^2 = Y dr^2 + 2A dr dw + 2dw^2 - (b+w)^2 dw^2,$$

which should be

$$= Y dr^2 + 2A dr dw + 2dw^2 - (b+w)^2 dw^2,$$

where, as before,

$$P : Q = (a+p)(b+p)(c+p), \quad (a+q)(b+q)(c+q),$$

respectively. We have already found

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} + \frac{4dw}{\sqrt{w}} = 0,$$

or, what is the same thing,

$$\frac{dp}{\sqrt{P}} - \frac{dq}{\sqrt{Q}} + \frac{4dw'}{\sqrt{w'}} = 0;$$

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and we ought therefore to have identically

$$(p-q)^2\left[p\left(\frac{dr}{\sqrt{P}}+\frac{dw}{\sqrt{Q}}\right)^2-1\right]\left(\frac{dr}{\sqrt{P}}-\frac{dw}{\sqrt{Q}}\right)^2=Ydr^2+2Adwdr+2dw^2-(r+w)^2dw^2,$$

that is

$$(p-q)^2=Y(1+r)^2(b+w)^2+2=4Y(1+r)^2(b+w)^2,$$

or, what is the same thing,

$$(p-q)^2=-A^2\sqrt{PQ}=(r+w)^2,$$

which are easily verified.

26. In fact, the equation

$$\frac{r^2}{a+b}+\frac{c^2}{a+b}+\frac{w^2}{a+b}=1$$

gives for a quadric equation, the roots of which are $w=p, w=q$; that is, we have

$$p+q=a^2+r^2a+r-b-r,$$
$$pq=-((1-b)(1-w))^2-(a+b+c)(1+r^2)+ba+1-rb;$$

and substituting herein for a^2, β^2, γ^2, their values in terms of a, β, γ, we find

$$\begin{aligned}p+q&=-(a-c)(r^2+1)-(a+c)(r+r^2)-4cw&&=(r+w)^2,\\ p+q&=(ac)(r+w)-(a+c)r^2-(a+c)(r+2w)-4cwr&&=(r+w)^2,\end{aligned}$$

the first of which is, in fact, $p+q=-A+(r+w)^2$. And from the two equations, forming the combination $(p+q)^2-4pq$, we at once obtain the other equation

$$(p-q)^2=Y(r+w)^2.$$

27. The most ready way of obtaining the relations between the differentials of p, q, w, r, is from the foregoing expressions of $p+q$, pq. Writing for a moment $p+q=-A+(r+w)^2$, $pq=D+(r+w)^2$, we have $p^2+q^2=-2A+(r+w)^2$, and from the first of these equations we

$$p^2(x+r^2)+p^2(x-1)(b+1)+p^2(x+1)(b-1)=4+(r+w)^2,$$

which is quadratic in p, w, r. The negative dimensions are

$$-1\cdot 1+a+b+bw+(r+w)^2,$$

$$-2\cdot a+b-(r+1)(r+1)(c-1)-2cwa-r(c+1)+cb;$$

respectively in three other terms; that is, three are these positions of the point P, and of course three corresponding positions of the point P'.

28. In the form just given the equation introducing w, r instead of p, q, the equation becomes

$$\sqrt{(p+r)(b+p)(c+p)}\left(\frac{dw}{\sqrt{w}}+\frac{dw}{\sqrt{r}}\right)+\sqrt{(q+r)(b+q)(c+q)}\left(\frac{dw}{\sqrt{w}}-\frac{dw}{\sqrt{r}}\right)=0;$$

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or, what is the same thing,

$$p(q+b)\left(\frac{dw}{\sqrt{w}}+\frac{dw'}{\sqrt{w'}}\right)^2-q(p+b)\left(\frac{dw}{\sqrt{w}}-\frac{dw'}{\sqrt{w'}}\right)^2=0;$$

that is,

$$(p-q)\left\{\frac{dw^2}{w}+\frac{dw'^2}{w'}\right\}+2(b(p+q)-2pq)\,\frac{dwdw'}{\sqrt{w}\sqrt{w'}}=0;$$

or substituting herein for $p-q$, pq, $p+q$ the values $\sqrt{Y}\,R$, A, such divided by $(r+w)^2$, this is

$$\sqrt{Y}(R\,dw^2+R\,dw'^2)+2\{b(A-2)A\}\frac{dw\,dw'}{\sqrt{ww'}}=0,$$

viz. writing herein $S=0$, the equation is $dwdr=0$, giving the right lines on the surface; and any solution of $S=0$ gives a geodesic on the surface.

29. The equation $dw'=Sdw'+2Adwdw'+2dw^2-(r+w)^2dw'=0$ gives the right lines $w=\text{const.}, w'=\text{const.}$ on the case of an umbilicus form on the parabolicaloids, the value whereof are $\sqrt{Y}dw=(a+1)dw'$ and $\sqrt{Y}dw'=(b+1)dw$, that is, the ratio of the coefficients of dr, dr' is of the form function $w=\text{const.}$; and it thus appears that it is possible to draw on the surface the two sets of right lines, the lines of such not being such umbilicus that the number is divided into parallelograms, the sides of which have to each other any given ratio (the angles being variable); viz. if this ratio be at all $=1$, then, to determine the relation between w, r, we must have $\sqrt{Y}dw=\pm ndu$; or what is the same thing, $\frac{dw}{\sqrt{w}}=\pm\frac{dr}{\sqrt{r}}$. In particular, if $m=1$, the parallelograms will be rhombs; and we must then have

$$\frac{dw}{\sqrt{w}}=\pm\frac{dr}{\sqrt{r}},$$

viz. this being in terms of w, r, the differential equation of the curves of curvature, it appears that the two sets of lines may be taken so as to divide the surface into infinitesimally small rhombs; such fact, showing the diagonals of these are the two sets of curves of curvature.

The Ellipsoid and the three Hyperboloids.

30. I have thus far considered a quadric surface in general; the various theorems being applicable as well to the ellipsoid and the hyperboloids of two sheets as to the sheet hyperboloid, the right lines being of course imaginary for the first-mentioned surfaces, but I will now consider the ellipsoid and the three hyperboloid separately.

31. First the ellipsoid. We have here a, b, c all positive, and I assume as usual $a>b>c$. The principal sections are all ellipses, viz. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is the major-

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mean, or say the minor section, $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ the minor-mean, or say the major section, and $\frac{x^2}{a^2}+\frac{y^2}{c^2}=1$ the minor, or umbilicar section. The elliptic coordinates p, q enter into the equations symmetrically; but we distinguish them by taking b extend from $-c$ to $-b$, and q to extend from $-b$ to $-a$. Thus $p=$ const. denotes the curves of

[Figure: ellipse with axes labelled — omitted]

curvature of the one kind; viz. $p=-c$ denotes the major-mean section $\frac{x^2}{a^2}+\frac{y^2}{c^2}=1$, $p=-b$ the section UU' and $V'V'$ of the umbilicar section; and $q=$ const. denotes the curves of curvature of the other kind, viz. $q=-b$ the remaining sections UVU' and $U'V'$ of the umbilicar section, $q=-a$ the minor-mean section, and $q=$ const. the minor-mean section.

32. Hence, in order that the geodesic line $\frac{dp}{\sqrt{P}}\pm\frac{dq}{\sqrt{Q}}=0$ may have a meaning,

$$dr\int\frac{dr}{\sqrt{(a+r)(b+r)(c+r)(p+r)(\phi+r)}}\pm dq\int\frac{dq}{\sqrt{(a+q)(b+q)(c+q)(p+q)(\phi+q)}}=0$$

of the geodesic lines may be real (observing that we have $a+p$, $b+p=+$, $c+p=+$, $p+q=+$, and $a+q=+$, $b+q=+$, $c+q=+$ necessarily; $p+\phi=+$, $q+\phi=-$ as the case may be; $p+q=+$ if included between the limits a, c. Or, what is the same thing, ϕ is included between $-b$ and $-c$, this point line is a triloid.

[Figure: pair of conics on the ellipsoid — omitted]

in the annexed diagram the values of $-p$, $-q$, $-P$. Hence on the ellipsoid we have two kinds of geodesic lines, and three leading a vast variety of consecutive curves; the geodesic line $\phi=$ const. may by means in two ways, viz. for points above the umbilicar curve, p is final between $-b$ and $-c$, but for points below the umbilicar curve, q is between $-b$ and $-a$, where this latter case the constant value of ϕ is included within a quadrant of the Royal Soc., vol. XXX., pp. 25–32, 1872, [478].

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33. Next, for the above hyperboloid, we have a and $b = +$, $c = -$, and I assume for convenience $a > b$. Attending to the signs, we still have therefore $a > b > c$. The principal sections are: one of them an ellipse, and the other two hyperbolas, viz. the minor section is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the major section is the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{c^2} = 1$, and the mean section is the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{c^2} = 1$. There are no umbilici. The elliptic coordinates enter symmetrically; but as before, we distinguish them, viz. we take p to extend from $-c$ (a positive value greater than a); but for sections of the hyperboloid, the parameters of which are between $-c$ and b, see; on the parts of the surface surrounding the hyperboloid, and q a constant.

The curves of curvature of the one kind, $p =$ const., are real; the curves of curvature of a hyperbola two; we may say that $p =$ const. are the elliptic curves and curves of curvature, that is, $-b$ and $-c$. Thus on $-b$ and $-c$ the curves of curvature of a hyperboloid the cone, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, and not the intermediate values gives a curve of curvature of a hyperboloid: thus we may say that $p =$ const. determines the real curves of curvature, and the hyperbolic curves of curvature.

[Figure: section — omitted]

and in the fourth case it has a q-value; but in the second and third cases it has neither a p-nor a q-value. This is better seen from the diagram. It follows that we have, on the hyperboloid, geodesic lines of four different kinds: these which touch a curve of curvature, on a hyperboloid, and these which touch one of the two kinds of curvature, but in which a foot a positive value from 0 to $-c$, but neither curves which are negative values from $-c$ to $-a$.

35. To explain this more in detail, consider the geodesic which start from a point M of the hyperboloid. In the first instance, consider the axis of x rotation, the discussion will be much like that of the geodesic of an ellipse.

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The geodesic of initial direction $M1$ touches at M the oval curve of curvature M_2, and lies wholly above this curve; it makes an infinity of convolutions round the upper part of the hyperboloid, cutting all the oval curves of curvature for which p has a (positive) value greater than p_1, (if p is the value of p corresponding to the oval curve through M), and according to whether it is considering the curves as described in the opposite sense, it descends from infinity to touch the oval curve through M, after which it again ascends to infinity.

[Figure: hyperboloid sections with geodesics through M — omitted]

Next, if the initial direction is $M2$, we have a geodesic of the same kind, only descending below M to touch a certain oval curve having for its parameter p_2, ($p_2 > -c, > p_1$).

We come next to a critical direction $M3$, for which the geodesic descends below M to touch the oval curve of curvature $p_2 = -c$, that is, the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. But it is to be observed that, whatever the initial point M may be, the geodesic makes below M an infinity of convolutions round the hyperboloid, so that it does not in fact ever actually touch the ellipse, but has this ellipse for an asymptote. That this is so appears from the consideration that the ellipse, just above the curvature of curvature, is a geodesic; so that, starting from a point of the ellipse in the direction of the ellipse, the geodesic coincides with the ellipse, or, besides the ellipse itself, there is not on the geodesic which touches the ellipse.

Next, if the initial direction be $M4$, the geodesic does not have touch any oval curve; it descends through M below the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, lying in the upper and lower portions of the hyperboloid, and making round it an infinity of convolutions.

36. We come, then, to the initial direction $M5$, which is that of the right line, the geodesic here coincides with the right line.

In the case which follow, the geodesic lies in the upper and lower portions of the hyperboloid, cutting all the oval curves of curvature.

Initial direction $M6$: the geodesic does not touch any hyperbolic curve of curvature, and it must touch the hyperboloid as making of convolutions.

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Initial direction $M7$: the geodesic touches at opposite infinities the mean hyperbolic $\frac{x^2}{a^2}-\frac{y^2}{c^2}=1$, lies wholly in front of the plane $y=0$ of this hyperbola.

Initial direction MS: the geodesic touches a hyperbolic curve of curvature, parameter q where q, (negative) is between $-b$ and $-q$, the parameter of the hyperboloidal curves of curvature lying between $-b$ and $-q$, but does not cut the remaining curves the parameter of which extend from q to $-a$.

Lastly, initial direction is $M9$, that of the hyperbolic curve of curvature through M ; the geodesic touches this curve, cutting all the hyperbolic curves the parameters of which are between $-b$ and q_1, but not any of those the parameters of which are between q_2 and $-q$.

37. If in the differential equation of the geodesic line we consider p, q as the elliptic coordinates of a given point M of the curve, the equation between the variables of θ determines the direction of the curve; or conversely, if the direction be given, the equation determines the value of the parameter θ. Writing

$$\frac{dp\sqrt{p}}{\sqrt{(a+p)(b+p)(c+p)}}=P, \quad \frac{dq\sqrt{q}}{\sqrt{(a+q)(b+q)(c+q)}}=Q,$$

then P, Q are proportional to the rectangular coordinates of a consecutive point M' measured from M in the directions of the hyperbolic and oval curves of curvature respectively; and the differential equation of the geodesic line gives

$$\frac{P}{\sqrt{p+\theta}}+\frac{Q}{\sqrt{q+\theta}}=0;$$

viz. if ϕ be the inclination of the geodesic to the hyperbolic curve of curvature, then $Q=P\tan\phi$, or we have $\frac{\tan\phi}{p+\theta}+\frac{1}{q+\theta}=0$; that is, $p\tan^2\phi+q+\theta(1+\tan^2\phi)=0$; hence, for the right side $\phi=0$, then $q+\theta=0$, and therefore $\theta=-\frac{(p\tan^2\phi+q)\tan^2\phi}{1+\tan^2\phi}$; viz. $\theta=q+\theta=-p, \theta=-q$ where, as it should be.

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Plan of a Curve-Tracing Apparatus.

[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 345–347. Read May 9, 1872.]

p. 179. I HAVE devised a curve-tracing apparatus on the following plan:

Imagine two planes II, II' moving in the same horizontal plane, and above the two planes respectively the two points P, P' moving in the same or a specified plane.

To fix the ideas, suppose that the two planes each move according to a law (that is, let the motion of each of these depend on a single variable parameter; for instance, they may each of them rotate about an axis); but let the motion of the two points be free.

Suppose, next, that the planes are connected together in such manner that the motion of one of them determines the motion of the other (e.g. by a train of wheel- work); and that the two points are also connected together in such manner that the motion of one of them determines the motion of the other (e.g. by a pantograph; or by a slotted rod, the slot of which works on a pin, in like manner as the rod or move lengthways as well as rotate).

Suppose, finally, that one of the points, say P, is attached to a point of the plane II', then the plane II being set in motion, this determines the motion of II', of P in a relative curve $ABCDEFGFG$, with through these ten points. (Or ABC, is derived from the points A, B, C; and in like manner the points a, b, c, d, e, f, g, h, i, j of in mere lengthways as well as rotate.)

I propose to describe the apparatus as nearly as I have actually constructed it. (See sketch-plan Fig. 1, and side-elevation Fig. 2.)

The framework of the instrument consists of two longitudinal bars (B) each about three feet long, one inch thick, and three inches broad, supported together at a distance of about eighteen inches on the cross-pieces C, C' and half-way between them, supported by the same cross-pieces, is an axis carrying at each extremity two mitre wheels. The bars B support three cross-pieces D, D, D, D, and between these are

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the moveable cross-pieces E, E carrying the axes of A, A with the attached mitre wheels, and circular disks X, X. Each of these wheels separately may be placed (as in the figure) out of gear with the two central wheels, or it can (by moving the moveable cross-pieces) be put into gear, and so be turned by them.

The disks X, X may be regarded as being themselves the planes II, II' (any planes are rigidly attached to X, X respectively); or we may, in any other manner, move the plane II by means of the disk X, for instance X may carry a spur-wheel gearing into a spur-wheel on the under surface of II, and thereby communicating to it

[Figure: Diagram of curve-tracing apparatus, plan and elevation — omitted]

a rotation of different velocity to the plane II; or the connexion may be as in the ordinary rod chuck. In any such case, the disk X which, for like reason, can readily project beyond X, serves as a support for the plane II, and the apparatus connected directly, and ensures that the angular motions of the plane II are made to depend therefore of the parts of any point of II, may be varied at pleasure by varying the the opposite direction, or it will be in motion, according as the mitre wheel is in gear with one or the other of the corresponding central wheels, and may rotate with each of them.

Detached altogether from the rest of the instrument, or what is better, supported on longitudinal pieces carried by the cross-pieces C, C we have a bridge (see fig. 1) capable of adjustment as regards height, and serving as a support for the pantograph- apparatus, or other connexion of the two plane II with the pencil which marks upon the other plane II.

It is hardly necessary to remark that in the simple form of the instrument where the disks X, X are themselves the planes II, II, then putting the mitre wheel of one plane A in gear with either of the corresponding central wheels, and making the other rotate, the other plane X will rotate as well; according as the mitre wheels are in the opposite direction, or it will be in motion, according as the mitre wheel is in gear with one or the other of the corresponding central wheels, and may rotate with each of them.

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On Bicursal Curves.

[From the Proceedings of the London Mathematical Society, vol. IV. *(1871–1873), pp. 347–352. Read May 9, 1872.]*

p. 181. A CURVE of deficiency 1 may be termed bicursal: thus it has a clear distinction according as the order is even or odd, and to fix the ideas I take it to be even.

A bicursal curve of the order n contains

$$\tfrac{1}{2}n(n+3) - \tfrac{1}{2}(n-1)(n-2) - 1, \quad = 3n \text{ constants;}$$

hence, if the order is $= 2n$, the number of constants is $= 6n$; such a curve is normally represented by a system of equations

$$(x, y, z) = (1, P)^n \quad * \quad (1, P^2)^n \quad * \quad y^2,$$

where Θ is a quartic function, which may be taken to be of the form $(1-\theta^2)(1-k^2\theta^2)$, or otherwise to depend on a single constant: viz. (x, y, z) are proportional to rational functions involving such a radical: since in the values of (x, y, z) we may consider the number of constants is $6(n+1) = (n-1) - 4 = 6n$, as it should be.

But the curve of the order $2n$ may be aberrantly or improperly represented by a system of equations

$$(x, y, z) = (1, P)^n * \quad (1, P^2)^n \quad * \quad y^2,$$

viz. these equations, instead of representing a curve of the order $2n$ as said, provide a curve of the order $2n$, provided only there exist 2k common values of θ for which each of the three functions vanishes. And this represents a particular representation of (θ) of degree θ, with $z+1$, the consequent of θ which the

$$(x, y, z) = (1, P^n) * \quad (1, P^2)^n;$$

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it is shown that such a transformation is possible, and a mode of effecting it, derived from a theorem of Hermite's in relation to the Jacobian H, Θ functions, is given in Clebsch's Memoir "Ueber Algegen Curven, deren Coordinates als rationale Funktionen eines Parameters und eines zweiten dort allgemeinen Funktion des selben dargestellt werden." *Crelle's Journal* (1864). Such a representation, in a very interesting way, will be made use of presently, but the more I am inclined to repeat the question: viz. given $\theta(p,q)$, $p^2q^2 = \theta(p,q)^2$.

Returning to the curves of deficiency 1, we see that a curve of the order $2n+2$ is contains θ each as a constants, and is normally represented by a system of equations

$$(x, y, z) = (1, P)^{n+1} \quad * \quad y^2.$$

Such a curve may be aberrantly represented; we may derive it by a rational transformation from the curve $(P = 1)$ of the order θ (θ being a positive integer); thus functions have then a common factor of the order θ, and omitting this, the equation is

$$(x, y, z) = (1, P)^{n+1} * \quad y^2,$$

viz. the coordinates are functions of a quadriradiical equation; and which represents a system of (P, q) in terms of three by a system of equations

$$(x, y, z) = (1, P)^n y^2,$$

in the case of representation of the curve in terms of the form of which is the same as that of the curve $(P = 1)$ in the case of the form $(x, y, z) = 0$.

It is, however, to observed that there are also curves of deficiency 1, similarly representable by a system of equations

$$(x, y, z) = (1, P)^n * \quad y^2;$$

where the second factor is that belonging to the quadriradical determinant of system of (P, q), where, A, B, C are quadratic functions of θ, viz. are then for example, $(1, P)^n y^2$.

In this particular, if $u = u$, then, instead of functions of the order $2n$, we shall find that, in fact, $u = (u, q)$, q proportional to functions of u, instead determines this view stress them in the form of $u = (u, q)^2$.

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$= B + \sqrt{\Omega}$ suppose; and then substituting in the equations $(x, y, z) = (1, P)^{n+1} * y^2$, we find

$$(x, y, z) = (1, \theta)^{n+1} * (1, P)^{n+1} y^2,$$

that is, here (x, y, z), proportional to functions of u involving the quartic radical $\sqrt{\Omega}$; but these functions are of the order $u + 1$, not u.

In particular, if $u = u$, then, instead of functions of the order $2n$, we have functions of the order $u + 1$, $u + 1$, u; that is, in this case the bicursal curve is properly represented in the form $(x, y, z) = (1, P)^{n+1} * (1, P)^{n+1} y^2$. This is no representation of an ordinary radical quadrant a may be exhibited by Cleach's Memoir.

We may, in fact, by linear transformations on u, v, reduce the quadripradiate relation to

$$\left.\begin{array}{rl} 1 & + \quad u^2 \\ & + \quad 2hov \\ & + \quad v^2 \\ & + \quad cu^2v^2 \end{array}\right. = 0;$$

that is,

$$1 + u^2 + 2hov + v^2 + cu^2v^2 = 0;$$

or putting herein $u + v = p$, $uv = q$, the relation is

$$1 + 2hq + p^2 - 2q + cq^2 = 0;$$

viz. extracting the square roots, this is

$$p^2 = -hq + 1 + 2(1 - 2h)q - cq^2;$$

$p^2 + bq = -1 + 2(1 - 2h)q - cq^2$; viz. extracting the square root, we obtain

$$\sqrt{p^2 + bq} = \sqrt{q}\sqrt{1 - 2(1 - 2h)q - cq^2};$$

if for shortness

$$Q = -1 + 2 - 2h(\theta) - cq^2;$$

we may then rationalise one of the radicals, for instance, by writing

$$-c\{-1 + 2 - 2h(c)q - cq^2\} = w^2 - 1 - c - q^2 + a; \text{ then, if}$$

$$op = -(1 - h)q^2 + b\sqrt{(1 - h)^2 + q^2}\left(q + \frac{1}{q}\right),$$

this becomes

$$-cq = \{e - (1 - h)\} - \tfrac{1}{2}\left(q^2 + \frac{1}{q^2}\right);$$

$$= \{e - (1 - h)\} - \tfrac{1}{2}\left(c^2 + \frac{1}{q^2}\right).$$

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that is

$$\sqrt{q} = \sqrt{\left(\frac{(-1-h)c}{c}\right)(b+\frac{1}{q})};$$

and the corresponding value of $\sqrt{p}$ is

$$\sqrt{p} = \sqrt{\left[\frac{1}{2}\frac{-1-h}{c}\left(b+\frac{2}{c}\right)-h^2\right]},$$

where q stands for the value

$$\frac{1}{2}\left[1-h+\sqrt{1-h^2}\left(c+\frac{1}{c}\right)\right].$$

We may write them in the form

$$1 : \sqrt{Q} : \sqrt{q}\cdot\sqrt{Q'} = M : 1+cP^2 : \sqrt{q^2},$$

where M is a constant, and Θ is a quartic function of θ, such that $(1+cP^2)-\Theta$ is a quadric function only of θ.

The equations

$$(x,y,z) = (1,P)^{n+1} * y^2,$$

then assume the form

$$(x,y,z) = (1,P^{n+1} * (1,P)^{n+1} * \sqrt{Q^{n+1}};$$

and on the right-hand side the term of the highest order in θ is

$$(1+cP^2)^{n+1}\theta^{n+1};$$

viz. if $n = cP^2 = 0$, then this is

$$(1+(1-cP^2)^{n+1})\theta^{n+1} = \sqrt{q^{2n+1}};$$

which is the same as the term of the highest order in θ in

$$(x,y,z) = (1,P^{n+1} * \sqrt{q^{2n+1}};$$

In particular, if $n = u$, then, instead of $2n+1 = 2n+1$, $u+1$, in the case of the order $2u$, as supposed at first, we have $u+1$, $u+1$; but in this form, expressing u in terms of u, the equation are

$$(x,y,z) = (1,P^{n+1}y^2, \quad (*P^{n+1}y^2),$$

viz. (x,y,z) are connected by a quadripradiate equation, is also represented by the equations

$$(x,y,z) = (1,P^{n+1} * y^2, \quad (*P^{n+1}y^2),$$

which is the required transformation of the original equations.

It is to be noticed that the foregoing form

$$1+u^2+2hov+v^2+cu^2v^2 = 0$$

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is the most special form to which the quadriradiate relation can be reduced by the linear transformation of u, v; in fact, by mere division, the equation is made to have the constant term 1; by mere replacement of variables, the equation in u takes the form $u^2 = uv + cv$; and as before, by means of u, we may, by a linear substitution, reduce it to this form, viz. we may write

$$\theta = P + \theta_1(1 - x_1 t) - F(1, c)P^2;$$

which is of the form general that if we had

$$(1, a)^{n+1} * y^2 P + y^2 \alpha;$$

viz. there is a hyperboloeous constant θ. And we then see how it is that the system of equations

$$(x, y, z) = (1, P^n * y^2);$$

(a, b connected by a quadrirat equation, contain c and $b + 1$ constants, and only of these being numerical.

Clebsch's transformations, above referred to, is as follows:

Starting with the equations

$$(x, y, z) = (1, P)^{n+1} * (1, P)^{n+1} * \sqrt{y^2(\theta)};$$

if the function Θ is not originally of the standard form, we may, by a linear substitution, reduce it to this form, viz. we may write

$$\Theta = F(\theta) - F_1(\theta) - F_1(\theta);$$

and then writing $\theta = \alpha t$, $(u(\theta = \alpha t)$, we have

$$\sqrt{\Theta} = \alpha t \cdot (-u\alpha t) - \cdots = au\alpha = u\alpha + \cdots ;$$

so that the formulas become

$$(x, y, z) = (1, P)^{n+1} * (1, P)^{n+1} * \sqrt{\Theta^{n+1}};$$

Hermite's theorem, used in the demonstration, is that any such function of u is expressible in the form

$$\xi \cdot \frac{H(u - a_1)H(u - a_2) \cdots H(u - a_n)}{H(u)^n},$$

where ξ, a_1, …, a_n are constants, $a_1 + a_2 + \cdots + a_n = u$.

(If, θ denoting here the two Jacobian functions); where

$$\xi_1 + \xi_2 + \cdots + \xi_n = u.$$

Observe, in passing, that the equation

$$\Theta = (1 - u^2)(1 - x_1 u^2)P,\ u + cu^2 + \cdots + cu^2 = uP + \cdots = 0.$$

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gives, when rationalised, an equation in ξ's of the order $2n = 2n$; this form of this equation are $u, u, u, x_1 + x_2$, &c. Considering the functions $(1, u^2)^{(2n)}$ and $(1, u^2)^{(n+1)}$ as indeterminates, the coefficients can be found so that all the rows of the roots of the equations is u^2 shall have any given values whatever; $u^2 a$, $u, u^2 = u^2 b$, &c.; the theorem then shows that the values of the remaining root is $u^2 a + b + c$, where

$$-x_{a_1} = u_1, u_2, \ldots, u_n, \quad -x_{a_2} = -(a_1 a_2 + \cdots + a_n)$$

which is, in fact, Abel's theorem.

Now, supposing that the three functions of θ vanish for 23 common values of θ, and the functions of u will continue at the most $u + 1$. From this, the equation are then

$$(x, y, z) = (1, w)^{n+1} * y^2;$$

we have the set of equations

$$(x, y, z) = (1, P)^{n+1} * \frac{F(c + a_1)\, H(c - a_1) \cdots H(c - a_n)}{H(c)^{n+1}};$$

where, however,

$$a_1 + a_2 + \cdots + a_n = u.$$

[the values a_1, ..., a_n are course different for the three coordinates x, y, z respectively; viz. we have

$$a_{x_1} + a_{x_2} + \cdots + a_{x_n} = u, \quad - \cdots = (a_{x_1} + a_{x_2} + \cdots + a_{x_n}) = u,$$
$$a_{y_1} + a_{y_2} + \cdots + a_{y_n} = u, \quad - \cdots = (a_{y_1} + a_{y_2} + \cdots + a_{y_n}) = u,$$
$$a_{z_1} + a_{z_2} + \cdots + a_{z_n} = u, \quad - \cdots = (a_{z_1} + a_{z_2} + \cdots + a_{z_n}) = u,$$

Writing the case

$$u = a + u', \quad u_1 = u_1 + u', \quad u_2 = u_2 + u', \quad \ldots, \quad u_n = u_n + u',$$

and consequently

$$u_1' + u_2' + \cdots + u_n' = 0,$$

or, changing the common denominator,

$$(x, y, z) = \frac{H(u - u_1')H(u - u_2') \cdots H(u - u_n')}{H(u)^{n+1}};$$

where

$$u_1' + u_2' + \cdots + u_n' = 0;$$

or, what is the same thing,

$$(x, y, z) = (1, u' + u^*)P, \quad (1, u + a^* P^*)y \cdots u_n;$$

viz. writing u_1, $u' = c$, and u_2 (1 - c)(1 - kc), we have a system of the form

$$(x, y, z) = P, \quad PP^{n+1}, \quad y \cdots y \cdots;$$

a normal representation of the curve of the order 2n.

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The relation between the parameters θ, θ' is given by $\theta'^2 = m^2(u-x)$, $u'^2 = \theta$, that is, we have

$$e^{\sqrt{\theta}} = \frac{\sqrt{\theta}x_1 du + a_1\sqrt{(1-x_1^2)(1-2\theta_1)}}{1-2u+x_1\cdot\theta},$$

or, writing $uv/u = u$, this is

$$\theta' = \frac{c\sqrt{x_1} + b_1\sqrt{(1-c^2)(1-kc^2)}}{1-x_1b^2};$$

and, conversely, we have

$$\theta' = \frac{c\sqrt{x_1}(1-x_1) - kc) + a_1c\sqrt{(1-c^2)(1-kc^2)}}{1-x_1c^2};$$

and the theorem, in fact, shows that, substituting this value of θ in the functions of θ which serve to express x, y, z, these become proportional to the functions of θ'; viz. they become equal to these functions, each multiplied by an irrational function $A^k + B^k a\sqrt{\theta}$ (A, B rational functions of θ).

[511]

Addition to the Memoir on Geodesic Lines, in particular those of a Quadric Surface (508).

[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 268–286. Read June 13, 1873.]

p. 188. 38. In the Memoir above referred to, speaking of the geodesic lines on the above hyperboloid, I say (No. 35), "The geodesic of initial direction $M1$ touches at M the oval curve of curvature M_2, and lies wholly above this curve; it makes an infinity of convolutions round the upper part of the hyperboloid, cutting all the oval curves of curvature through M, viz. it cuts all the hyperbolic curves the parameters of which are between $-b$ and $-a$; and so on," but in fact such convolutions in incorrect; I was led to it by the assumption that the geodesic could not touch any hyperbolic curve of curvature, but in fact, looking at the geometry the geodesic actually cuts the hyperboloid being defined: such hyperbolic curve of curvature touched by the geodesic, &c.; that is, in the wholly part of the geodesic the value $b+c$ may be taken arbitrarily; while $b+c$ may be a positive value $b+c$ may be a negative value $b+c$, which is may be a positive value below $b+c$ may be then negative, that the term of integral consideration obviously enough, that $b+c$ value of $b+c$ has the contrary; viz. it but it is greatly preferable to bring the analytical considerations to bear with the matter in fact the questions to ratifying $b+c$; viz. if even $b+c$, a value $b+c$ ratifying $b+c$, a value $b+c$ ratifying $b+c$, a value $b+c$.

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39. I have effected, for a particular skew hyperboloid and oval curve of curvature thereof, the numerical calculations for laying down the geodesic lines which touch the curve of curvature. Taking in general the equation of the hyperboloid to be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1,$$

and the b-curves of curvature to be the intersection by the cylinder

$$\frac{x^2}{a^2+\theta} + \frac{y^2}{b^2+\theta} = 1;$$

then the satisfied values for the hyperboloid, and oval curve of curvature touched by the geodesics, are

$$\begin{aligned} a &= 900, \\ b &= 400, \\ c &= -300, \\ \theta &= -1600, \end{aligned}$$

so that a, b, θ are the positive values 900, 400, 300, 1600, and 1650 respectively. I would that $p = a$ is given the oval curve of curvature, viz. $p = -\theta$, the elliptic principal section; $p = b$, the green oval curve, and that we are in the manual concerned only with the oval curves above this, for which p lies between $-c$ and $-\theta$. We are in the manual to deal with the integrals

$$\Pi(p) = \int_b^p \frac{dp}{\sqrt{(a+p)(b+p)(c-p)(p-\theta)}},$$

and

$$\Psi(p) = \int_b^p \frac{dp}{\sqrt{(a+p)(b+p)(c+p)(p-\theta)(p+c)}},$$

or if $p = c = c$, extending from 0 onwards, viz.

$$\Pi(p) = \int_0^p \frac{dp}{\sqrt{(a+b+b)(c-p)(p+c)(p+c)}};$$

the relation for any particular position of the point P and P'.

40. To avoid discontinuity as to sign, it is convenient to take the integral $\Psi(p)$ in a particular manner. The hyperboloid is by the α and perpendicular plane divided into four quadrants; we may attend only to the upper half of the hyperboloid, say this upper half is then divided into four quadrants, viz. $p > 0$, $y > 0$, $y < 0$, $x > 0$, $x < 0$. In the first quadrant the geodesic curves are one and the same curve; if we let $y < y$, &c.; that is, as the hyperboloid extents referred to in the same fig. 1, we may consider the quadrants as forming an infinite succession round, half, third, &c., not, &c., (478).

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and so on; or we may take them in the reverse order, -1, -2, -3, &c. For a hyperbolic curve $q = -b - x$ in the first quadrant the integral is to be taken $x = 0$ to $x = c$; for a curve in the second quadrant $x = 0$ to $a - b$, thence $a - b$ to 0, and thence 0 to π; and so on; and so for a quadrant in the quadrant -1, the integral is to be taken from 0 to π, taken negatively, &c.; that is, as the hyperbolic curve $b + b$ travels from the positive direction, $b - c$ the integral $\Psi(b)$ continually increases from zero; and if the curve travels from the b-direction, $b - c$ the integral $\Psi(b)$ continually decreases from zero; that is, it increases negatively. It is to be remarked that this integral $\Pi(b)$ requires no calculation; the subsequent value $\Pi(c + c)$ value of b values $b + c$ to b values $b + c$.

The integral $\Pi(b)$ requires no explanation; it is taken from $u = 0$, giving a certain oval curve, up to any positive value of u, giving the oval curves above this one; and, in particular, taking the integral to $u = \infty$ (or, what is the same thing, to $p = \infty$) its value is finite, $= K$, suppose.

41. Consider the geodesic which touches the given oval curve at a point for which $\Psi(p)$ has a given value Q; at the point $p = \theta$, the integral $\Psi(p) = 0$; we may write the equation of the geodesic $\Psi(p) + \Pi(p) = C$, and consequently

$$\Psi(p) = Q + \Pi(p).$$

Taking the positive sign, then as p increases from θ, $\Psi(p)$ increases, or the describing point of the geodesic moves upwards from the point of contact in the direction of positive rotation; and taking the negative sign, then $\Psi(p)$ decreases, or the describing point of the geodesic moves upwards from the point of contact in the direction of negative rotation; and, in particular, p becoming infinite, then on the upper half of the hyperboloid, the oval curve, for which $\Psi(p) = K$, will be reached, the second branch that for which $\Psi(p) = -K$.

42. The graphical process is as follows: we describe, on the hyperboloid, a series of hyperbolic curves of curvature, numbering them according to the values of $\Psi(p)$; viz. considering the hyperbolic branches which form the x, y, z &c. and $y\pm$ sections respectively, these are 0, 2000, 4000, ..., 34726; and on going round a second time they would be $K + 34726$, $K + 36000$, &c. respectively, and so on; the same dispersion of the direction of negative rotation, numbering them according to the values of $\Pi(p) = K - Q + \Pi(p)$, &c. ($p = K$), where the oval curves of curvature is the ellipse which is a principal section of the hyperboloid, and then even to attain to it by value 1 except for an oval curve of curvature.

In the example, and showing belonging thereto (1), where, for convenience, the values of the integrals have been multiplied by 100,000, we have, as an upper limit, $K = 34726$; the two sets of curves are drawn at intervals of 2000; viz. we have the hyperbolic curves 0, 2000, 4000, ..., 34000, 34726; and the oval curves 0, 2000, 4000, 6000, 8000, 10000; the distance of the successive oval curves increases very rapidly, since the curve at infinity would be K, $= 12400$, and the curve $K = 10000$ is the last which comes into the limits of the figure.

[This drawing was exhibited at the meeting of the Society.]

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43. The two sets of curves of curvature being thus drawn at equal intervals of $\Pi(p)$ and $\Psi(p)$ respectively, drawing the hyperboloid into quadrilateral spaces (which of course should theoretically be indefinitely small), the diagonals of these quadrilateral spaces are the elements of the geodesic lines: that is, starting with the geodesic of equation $\Psi(p) = C$, and following its identity, we may readily trace it.

In the case in question, the data are as follows:

$$\begin{aligned} a &= 1600, \\ b &= 1600, \\ c &= 1000, \\ b - c = 30 &= 1000; \\ \theta &= -1600; \end{aligned}$$

where putting in $p = \theta$, θ is found, the equation

$$J = \int_b^p \frac{dp}{\sqrt{(a+p)(b-p)(c+p)(p-\theta)}}.$$

where J, the integral calculated up to a somewhat large value $p = p_2$.

Writing

$$\begin{aligned} u &= \tfrac{1}{2}(a + 2p), \\ \theta &= \tfrac{1}{2}(b - 2p), \\ \eta &= \tfrac{1}{2}(c + p) - b, \\ b &= -\tfrac{1}{2}b - 2c) - b, \end{aligned}$$

where as said a are at yet undetermined, we have

$$(p+a)(p+b)(p+c)(p+a) = u^2 - b^2,$$

$$(p-a)(p-b)(p-c)(p-a) = u^2 + b^2; \quad b > 0 = b^2,$$

$$(p-a)(p-b)(p-c)(p-a) = u^2 - b^2,$$

$$(p-a)(p-b)(p-c)(p-a) = u^2 + b^2; \quad b < 0 = b^2,$$

and the integral is then

$$J = \frac{1}{2}\int_b^p \frac{dx}{\sqrt{(b-a)(b+b)}}.$$

But we may determine in a such that for $p = a > p$, $(p-a)(p-b)(p-c) > a(p-b)$, that is, $(p-a)^2 + b^2 > 0$, and the integral is then

$$J = \frac{1}{2}\int_b^p \frac{dx}{\sqrt{(b-a)(b+b)}}.$$

The determination then depends on the two roots of the mentioned integrals, the values of which are calculable by writing therein $\sqrt{q} = u$, viz. we have

$$\int \frac{dx}{\sqrt{q + b(c+b)}} = \frac{1}{c-b}\log\frac{(c-b)x^2 + 2cx + b}{c-b}.$$

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and hence, substituting in the formula, for L, c its value

$$\frac{2\log a}{c-b} - 1,$$

the superior limit is

$$L_1 = \frac{1}{c-b}\left\{\sqrt{q}\log\frac{\sqrt{(p+a)(p-b)}+\sqrt{(p+a)(p+a)}}{\sqrt{p^2-a^2}}\right\};$$

and the inferior limit is

$$L_2 = \frac{1}{c-b}\left\{\sqrt{q}\log\frac{\sqrt{(p+a)(p-b)}+\sqrt{(p+a)(p+a)}}{\sqrt{p^2-a^2}}\right\};$$

where denom. $= (a-p)b$, and these resulting values of L, L_2, are obtained by writing therein $b+p$ in the place of b.

45. The numerical values are $p = 19800$; a, b, b', $c = 800$, a, b, $c = 1605$, 1745; and thence determining by trial values of a and b,

$$\begin{aligned} a &= 600, \\ b &= -500, \\ a' &= 1480, \\ b' &= 145, \\ \beta &= 1825, \\ b' &= 1825 + 165 = 1825. \end{aligned}$$

I obtained the logarithmic and circular terms of the two limits respectively;

	Superior	Inferior
Logarithmic	015668	015144
Circular	002522	005003
	023720	020737.

The value of L was $10411 \times 100000 = 104111$; and the two limits then are 124390 and 123450; or removing the factor $\frac{1}{100{,}000}$, they are 1.2439 and 1.2345; the mean of which 1.2400, was taken for the value $\Pi(p)$, $p = \infty$; that is $K = 12400$.

46. As regards the calculations of the integrals $\Pi(p)$ and $\Psi(p)$, introducing the numerical values, and multiplying by the before-mentioned factor 100,000, we have $\Pi(p) = 400$

$$\Psi(p) = 100,000\int_0^u \frac{du}{\sqrt{(1500-u)(150+u)(2000)(u+2050)}},$$

which for any small value of u is

$$= 100,000\frac{2u}{(150\cdot 100\cdot 2000\cdot 2050)^{1/2}}\left(\int_0^u \frac{du}{\sqrt{\cdots}}\right) = 2\sqrt{u};$$

viz. this is

$$= 88\sqrt{2}\log = 27001895\sqrt{u},$$

which was used for the values $u = 1, 2, \ldots, 10$, that is, to $q = -610$. When the calculation was continued by quadrature systems of u at values $u = -1, -10, 31, 30, \ldots,$ up to $u = -10000$, $q = -900$. The remainder of the integral, setting from $1000 = u = w$ (w at $u = w$ etc.) was here continued.

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which was used for the values $u = 9, 8, 7, \ldots, 1, 0$, to complete the calculation up to $q = -300$.

47. We have in like manner, $p = 1650 + u$,

$$\Pi(p) = 100,000 \int_0^u du \sqrt{\frac{1600}{2500(u+1650)(u+2050)(u+50)}}$$

which for small values of u is

$$= 100,000 \cdot \frac{1}{\sqrt{2500 \cdot 1650 \cdot 2050 \cdot 50}} \int_0^u du = 5\sqrt{u},$$

viz. this is

$$= 5825 \log 5 = 270111035\sqrt{u},$$

used for $u = 1, 2, \ldots, 10$, that is to $p = 1600$. The calculation was afterwards continued by quadrature, by giving to u successive integer values, $u =$ 10, 20, …, 100, 200, &c., as far as the limit; wherefore, as we approaching above, the value for $p = -\infty$ was found to be $= 12400$.

48. After the calculation of the values of $\Pi(p)$ and $\Psi(p)$, it was easy by interpolation to revert these tables, so to obtain a table which, for if to Ψ as argument, gives the values of p (q), as the limits of the equations are real of the geodesic lines, $b + (a + c, c, e)$, $\Pi(p) = 12400$ is which the analyte agreeing with this; we are also enabled, by an interpolation of one of the calculation in the form

$$\Pi(p) = (q - 400)A + (q - 400)^2 B + (q - 400)^3 C,$$

to take from this table the equations of the geodesic line as referring in the form $\Pi(q) = (q-400)A + (q-400)^2B + (q-400)^3C$, &c.; viz. to take the geodesic line $\Pi(q) = (q-400)A + (q-400)^2B + (q-400)^3C$, &c.; &c.; the same that the same form of the curve, and by taking the same value $\Psi(p) = (q-400)A + (q-400)^2B + (q-400)^3C$ on the same form ($p = const.$), we ensure that the curve is sufficiently well to the curve. This curve, for $p = 0$, was readily found to be

$$\Pi(p) = -1.0 \cdot \Pi(q).$$

But it is possible to derive a more elegant form for $p = 0$ from the equation

$$\Psi(0) = 12400 \cdot Q + Q \cdot \Pi(0).$$

Observe that the integral $K' = 34726$ in the case considered measures the quadrant of the hyperboloid, viz. to value $\Psi(p) = K'$ determines two hyperbolic curves of curvature (principal sections), the natural distance, that is, an ellipse, &c. which is included in any $p = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K' = 34726$ at hyperbolic, including $p = \text{const.} = K'$. Observe that the form of these foregoing equations gives one geodesic in the manual containing $\Psi(p) = K' + Q$, we have the geodesic, in measured by the value Q, $\Pi(p) = K'$ $(p = K')$.

50. Now it is easy to see that on the oval curve of curvature appearing the principal elliptic section, that is, as p approaches 0 (or nothing) $\Psi(p) = -\infty$, as on

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diminution towards zero; the integral K' shows its value only slowly, increasing towards a certain constant limit; but, contrariwise, K increases without limit, its value at any small value of u being of the form $A - B\log p$, in the limit, when $K = \infty$, A diminishes, the value of $\frac{dp}{dQ}$, the amplitude of the geodesic, continually diminishes; that is to say, the geodesic continues, at it makes round in any number of convolutions, more and more nearly to the form of the principal section of the geodesic at all events convergent towards a special, value at every approximate convolutions b, $b \cdot c$, that is the same as the principal section. But it is possible to derive some general numerical results from the analytical formulas.

51. To sustain the foregoing statements, I write $\theta' = (-c + a) + 1000 \cdot s$, and I consider the integral

$$K_a = 100000 \int_0^a \frac{ds}{\sqrt{(s+1)(150+s)(150+100)(2000)(s)}},$$

say for a moment this is

$$K_a = \int_0^a U_a ds.$$

Supposing a to be small, we divide this integral into two parts, say from 0 to θ (θ where a, $a + c$ to comparison with a, and so on; in comparison with a, but small in comparison with a, $1000, 9$ to a, where the second part the expression under the integral sign and the value of the integral varies slowly with θ, and we may, as an approximation, write $s = a$. We have then

$$K_a = \int_0^a U_a ds + \int_0^a U_a ds,$$

and the first part listed to be written

$$K_a = 100000 \int_0^a \frac{ds}{\sqrt{(1500 \cdot 100 \cdot 2000)(2050)}},$$

viz. the integral is here

$$A \cdot \int_0^a \sqrt{s(2050 + a)} ds = A \log \frac{a + \sqrt{(2050 + a)}}{\sqrt{2050}} = A \log \frac{4a}{2050} \text{ approximately},$$

or say this is

$$= \log \frac{4a}{2050};$$

The first term is thus

$$= 100000 \int_0^a U_a ds = 100000 \int_0^a U_a ds \int_0^{a+c} U_a ds \int_0^{a+c} U_a ds,$$

which is

$$= 4419 \log \frac{4a}{2050}.$$

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or we have

$$K_a^* = 4119 \log \frac{4a}{2050} + \int_0^a U_a ds.$$

We ought to have the same value if the integral, whatever, within proper limits, the assumed value of a may be. Taking, for instance, $a = 10$, and $a = 100$, we ought to have

$$K_a^* = 4119 \log \frac{400}{2050} + \int_0^1 U_a ds$$

$$= 4119 \log \frac{400}{2050} + \int_0^1 U_a ds.$$

that is,

$$4119 \log 2 = \int_0^1 U_a ds.$$

In verification, I calculated the second side by quadrature; viz. for the values 30, 60, 70, 80, 90, 100, the values of U_a are 30.532, 25.711, 23.812, 22.527, 17.604; whence, adding the half ends and the extreme terms to the sum of the mean terms, and multiplying by 10, the value of the integral is = 13354. The value of the left-hand side is = 11873, which is a sufficient agreement.

52. Returning to the formula for K_a^*, this may be written

$$K_a^* = (4119 \log 400 + \int_0^a U_a ds) = 4119 \log a;$$

I did not calculate the value of the integral in this formula, but determined the term in ( ) to such value that the formula should be correct for the foregoing value $a = 50$, viz. the term is then

$$= 12400 + 4119 \log 50 = 12400 + 4598 = 19488, \text{ or say } 19500;$$

we thus have

$$K_a^* = 34726, \text{ or say } 19500.$$

We thus see how to give to so such a value that the quantity $\frac{K}{2k}$, which is the number of convolutions of the geodesic, may have any given value: and, in particular, we see how exceedingly slowly such an number of convolutions is, even at the small value of a which corresponds to a geodesic which is most by any moderately large number of convolutions; for instance, $a =$.

Conclusion. Instead of moving an above, of a geodesic at touching of an arbitrary curve, a hyperbolic curve of curvature, the accurate expression is that the geodesic at infinity is parallel to a certain hyperbolic curve of curvature. The geodesic has, in fact, for its asymptote in the asymptote $b + b$ a $b + b$ a $b + b$ a $b + b$ a $b + b$ a $b + b$. Added Dec. 1873.

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Tables I and II: numerical values of $\Pi(p)$ *and* $\Psi(q)$.

Table I.		Table II.	
p	$\Pi(p)$	q	$\Psi(q)$
1650	000	−400	000
1	502	1	843
2	711	2	1249
3	870	3	1550
4	1003	4	1747
5	1124	5	1975
6	1230	6	2184
7	1329	7	2337
8	1421	8	2493
9	1507	9	2620
1660	1589	410	2734
70	2186	20	4016
80	2603	30	4860
90	2933	40	5782
1700	3208	50	6402
10	3449	60	6874
20	3642	70	7304
30	3823	80	7711
40	3974	90	8050
50	4112	500	8385
60	4239	600	11288
70	4351	700	13580
80	4453	800	15470
1800	4642	1000	18482

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Tables I, II, III (continued): cf. book pages 197–199.

p. 198–199. Numerical Tables III and IV (continued).

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On a Correspondence of Points in Relation to Two Tetrahedra.

[From the Proceedings of the London Mathematical Society, vol. IV. *(1871–1873), pp. 396–404. Read June 12, 1873.]*

p. 200. THE following question has been considered by E. Sturm in an interesting paper, "Das Problem der Projectivität und seiner Anwendung auf die Flächen zweiten Grades," *Math. Ann.*, t. I. (1869), pp. 333–374: Given in a plane two groups of the same number (5, 6, or 7) of points, to find points P, P' homographically related to these two groups respectively; viz. the lines from P to the points of the first group and those from P' to the points of the second group are projectively equal. The solution depends on the resolution of the equation systems by the rules of the ten tetrahedra; it is required to find the lions of the points P, P' which are homographically related to two such groups of A, B, C, D and A', B', C', D' wherein in line, and to find the locus of P, P'.

If the points A, B, C, or P and the points A', B', C', or P' are subtended equal angles, then we may say all of these tetrahedra $ABCD$ and $A'B'CD'$; the four cubic surfaces, four of each in space, will pass through the points A, B, C, D and the four tetrahedra of the four cubic surfaces $ABCD$, $A'BCD$, $A'B'CD$, $A'B'C'D$, &c. that the time of the cubic surfaces $A'B'C'D'$, $A'B'C'D'$, $A'B'C'D'$, &c.; and the like for the locus of P'.

If the points A, B, C, D and P' and the points A', B', C', D' subtended equal cosines, the line from each of the points A, B, C, D to the locus point of infinity $ABCD$, $A'BCD$, $A'B'CD'$; and the like for the locus of P'.

If the points A, B, C, D and P and the points A', B', C', D' and P' subtended equal angles, then the two tetrahedra are correspondingly homotopic, viz. the locus of P subtends an isolated angle which determined as a as homotopic ($ABCD$, $A'B'CD$, $A'B'C'D$), and the like for the locus of P'.

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Finally (although this is a theorem which I do not require), if the points A, B, C, D, E and P and the points A', B', C', D', E' and P' subtended equal angles, then there are three positions of each point.

The problem I propose to consider is: Given the tetrahedra $ABCD$ and $A'B'CD$, it is required in the plane ABC and $A'B'C'$ respectively to find the points P, P' such that A, B, C, D and P, and A', B', C', D' and P' subtended equal angles. I was led to this by the more general problem, which I am at one moment proceeds: viz. the points of a configuration of four planes; intersection of these triple planes which determined by the parameters A, B, C, D of the four points A, B, C, D are at A, B, C, D (A, B, C, D); the four edges of the tetrahedron $ABCD$, etc. at six relative six relative to the points of the four planes, namely

1. P is such that A, B, C, D and P, and A', B', C', D' and P' subtended equal angles; locus a cubic line through A, B, C, D, (ABC), (ABE), (ACE), (BCE).

2. P is such that A, B, C, D and P, and A', B', C', D' and P' subtended equal angles, and that PE and PE' are in a given line; locus a certain ruled curve;

and the required position of P are obtained as the intersections of the two loci.

I proceed to the analytical investigation.

Preliminary Formulæ.

1. Consider a triangle ABC, and let the position of a point P be determined by means of its coordinates x, y, z, which are equal to the perpendicular distances of P from the sides, each divided by the perpendicular distance of the opposite vertex (as usual; x, y, z are positive for a point within the triangle), or what is the same thing, x, y, $z = PBC$, PCA, PAB, divided each by ABC; whence identically $x + y + z = 1$.

Suppose for a moment the rectangular coordinates of A, B, C are (a_1, b_1), (a_2, b_2), (a_3, b_3) respectively; and those of P are X, Y. Also let the rates BC, CA, AB be $= a$, b, c respectively.

We have

$$\begin{aligned} X &= a_1x + a_2y + a_3z, \\ Y &= b_1x + b_2y + b_3z, \\ 1 &= x + y + z; \end{aligned}$$

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and if we consider a second point P', the coordinates of which are x', y', z' and X', Y', we have the like relations between these quantities. Calling b the distance of the two points P, P', we may write

$$\begin{aligned} D &= [a_1(x-x') + a_2(y-y') + a_3(z-z')]^2 \\ &\quad + [b_1(x-x') + b_2(y-y') + b_3(z-z')]^2 \\ &= [(b_1-b_2)(z+y) - (b_3-b_2)(z-y') - (b_3-b_1)(z+z')]^2 \\ &\quad + [(a_1-a_2)(x+y) - (a_3-a_2)(x-y') - (a_3-a_1)(x+x')]^2; \end{aligned}$$

the last term being in fact -0; viz. this is

$$D = (a^2(y-y') + (b^2(x-x'))(z-z') + (a^2(x-x') + b^2(y-y'))(z-z'))$$

$$= [\beta(x-x') + y(z-z') - z(y-y')]^2 + (\beta(x-x') + z(y-y') - x(y-y'))^2,$$

or what is the same thing, the coordinates for the distance D of the two points P, P' which expression may be modified by means of the identical equations

$$1 + x + y + z = 1, \quad 1 + x + y + z = 1.$$

[full quadratic forms continued]

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3. Consider next a tetrahedron $ABCD$, let the position of a point be determined by 4 coordinates x, y, z, w relating to the faces (the sum of these being $= 1$); and let the four points A, B, C, D corresponding to these coordinates have the coordinates of unit area as above. By means of the same construction we now express the distance of two points P as a function of the coordinates of P.

In fact for the tetrahedron $ABCD$ and the point P, we have a system of relations $(A, B, C, D)(x, y, z, w)(X, Y$ as before, viz. taking A, B, C, D for the angles, and similarly those of vertical triangles (B, C, D), (C, D, A), (D, A, B) respectively. Here we require for the case the perpendicular and ratios of the tetrahedron $ABCD$, X which enter into these manuscripts equations.

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viz. this is the same thing,

$$\Sigma = a^2yz + b^2zx + c^2xy + a^2dz(z + y - z),$$

Moreover, if Δ a value also are an area of the triangle, then

$$(2\beta_1 - \beta_2)\Sigma_2 \cdot V \cdot s_1 \cdot \zeta_4 - s_4 + \zeta_2 \cdot (s_1 + s_2 + y + s) - \beta_1 \sin A_1(s_1 + s_2 + y + s) = 0;$$

so that the equation becomes

$$-(2\beta_1-\beta_2)\Sigma_1(c+(b_1-\zeta_3))^2(c_3-c_2)(x+y+1)+\beta(c_3-c_2+2\beta_2)(x+y+1)(c_3-c_2-2c)(c_2-c_3)-2c\beta_3(2+(c_1-1))-2c\beta$$

so that this represents the like equation

$$-c^2y(x_1 + b)y^2 + b\beta(c_1 - 1) + b\beta b_3(c_2 - c_1) = 0,$$

or forming the like equations of the other similar circles, we have the circles (B, C), (C, A), (A, B) containing the angles E, E, F respectively; and the equations are

$$U + \Delta(c\zeta - \cot A)(c\zeta - 0,$$

$$U + \Delta(\zeta + \cot B)(\zeta = 0,$$
$$U + \Delta(\zeta - \cot C)\zeta = 0.$$

Correspondence, A, B, C, D and A', B', C', D' subtending equal angles.

5. Consider now the two figures A, B, C, D, P and A', B', C', D', P' subtending all the same angles E, F, G, H enclosed at P, P', them we have

$$\frac{U}{ABCD} = \cot E = \cot A, \quad \frac{U'}{A'B'C'D'} = \cot E = \cot A',$$
$$\frac{U}{ABCD} = \cot E = \cot B, \quad \frac{U'}{A'B'C'D'} = \cot E = \cot B',$$
$$\frac{U}{ABCD} = \cot E = \cot C, \quad \frac{U'}{A'B'C'D'} = \cot E = \cot C',$$

and thence

$$\frac{U}{ABCD} + \cot A = \cot A',$$
$$\frac{U}{ABCD} + \cot B = \cot B',$$
$$\frac{U}{ABCD} + \cot C = \cot C'.$$

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and consequently

$$\frac{1}{x'}\cdot\frac{1}{y'}\cdot\frac{1}{z'}=\frac{U'}{ABCD}\cot A=\cot A',$$
$$\frac{1}{ABCD}\cot B=\cot B',$$
$$\frac{1}{ABCD}\cot C=\cot C',$$

or, what is the same thing,

$$x':y':z'=\frac{x}{U+\Delta(\cot A-\cot A')\,V_3}\quad\cdots\quad;$$
$$\overline{U+\Delta(\cot B-\cot B')\,V_2}\quad\cdots\quad;$$
$$\overline{U+\Delta(\cot C-\cot C')\,V_3}\quad\cdots\quad;$$

where observe that the equations

$$U+\Delta(\cot A-\cot A')\,V_3=0,$$
$$U+\Delta(\cot B-\cot B')\,V_2=0,$$
$$U+\Delta(\cot C-\cot C')\,V_3=0,$$

represent circles (B,C), (C,A), (A,B) containing the angles A', B', C', and since $A'+B'+C'=\pi$, we have these three circles meeting at a point. We may for convenience write

$$x':y':z'=BCO:CAO:ABO,$$

suppose, X, Y, Z are quartic functions of x, y, z, and the curve is the first one representing a X, Y, Z of the order 4; viz. since the curves (BC), (BC), &c., is an algorithm which will be common point through A, B, C, of the first figure—corresponding to the locus on P', of the second figure—the point at infinity $X=X'=Y'=Z'=0$.

We have therefore

$$x':y':z'=\frac{x}{BCO}:\frac{x}{CAO}:\frac{x}{BCO},$$

viz. this is a quartic curve having the points A, B, C on the line BCO and ABC, the expressions being multiplied into the proper constant factors to make the constant factors whereby x, y, and so a, b, c of x, y, z identically as $A.BC.OA.B.OD.OA$. identically $Y=X=(A.B.D.A)$, and similarly $Y=(A.B.D.G.)$.

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Correspondence, A, B, C, D and P and A', B', C', D' subtending equal angles.

6. Consider now in plane the points A, B, C, D which on P, and the points A', B', C', D' which on P', subtend equal angles. Let a, b, c, g, h, k denote the perpendicular distances of P from the lines BC, CD, AD, AB, AC, BD respectively; and let the lines of P, P' be denoted by A, B, C, D, G, H, K where we have G, H, K denote the corresponding angles in the two figures: then in respect to P and the four points A, B, C, D, we have three equations of the above form, viz. where the value $\frac{1}{x'}$ corresponds to the equation in the first figure;

$$a' : g' : k' = BCD'O : CDA'O : DAB'O,$$

the expressions being multiplied into give proper constant factors to make the constant terms whereby x, y, z of a, g, h of a, g, h of a, g, h of a, g, h of a, g, h as before, a, b, c, g, h, k, respectively.

We have in like manner

$$a' : b' : c' = \frac{a}{BCD'O} : \frac{b}{CDA'O} : \frac{c}{DAB'O},$$

From the ratios of a', b', c', g', h', k', or, $a' : b' : c'$, $a' : g' : k'$ respectively we deduce

$$CD\,B\,CO\,(b)\,CO\,A\,DA\,D\,O\,(b)\,DA\,O\,A\,BO\,(a)\,A\,BO,$$

$$CD\,A\,B\,O\,(b)\,A\,BO\,(a)\,A\,DA\,O\,A\,BO\,(b)\,A\,DA\,O,$$

$$B\,DA\,BO\,(b)\,A\,BO\,(a)\,B\,CD\,O\,(b)\,B\,CD\,O\,B\,CD\,O,$$

$$B\,CD\,(b)\,O\,(a)\,B\,CO\,(b)\,B\,CO\,(b)\,B\,CO\,(b)\,DA\,O\,DA\,O,$$

each of which equations represent a curve and admitting that it can be shown that these pass through O, G, G, &c., respectively, the equations are

$$BA\,BO\,(0)\,O\,(b)\,(a)\,P\,A\,A\,B\,A\,B\,A\,B,$$

$$B\,CD\,O\,(0)\,B\,CO\,(b)\,(a)\,B\,CD\,B\,CO\,B\,CO,$$

$$B\,DA\,O\,(0)\,DA\,O\,(b)\,(a)\,DA\,O\,DA\,O\,DA\,O,$$

$$DA\,B\,O\,(0)\,O\,(b)\,(a)\,DA\,B\,DA\,B\,DA\,B.$$

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7. Now the locus of P is evidently a curve, and this can only happen in the case where the four last-functions contain a common factor; here at first they have not this; for an arbitrary ABO if the requirement is $ABCO$, the conditions that the cubics suppose the these matters $ABCDOO$, the four extraneous factors being DAB, DAC, DAC, etc., and $ABCD(OO)=0$, will be a cubic carry passing through the two points, and the like for $DAB = 0$. To verify in a particular instance, in B, C, D, A, are in a plane, then the conditions of agreement is $ABCDD(0)0=0$, that is, Δ_{ABC}, the conditions that the cubics

$$DA\,B\,CO\,DA\,B\,CO\,DA\,B\,CO\,DA\,B\,CO\,DA\,B\,CO=0$$

should pass through O, G, G. But then if the cubic passes through any a points of a cubic curve, it would in general be necessary that the cubic contained a common factor; here the cubic which contains a common factor, and a A axis, four common points; this is not at all impossible.

Correspondence, A, B, C, D and P and A', B', C', D' and P' subtending equal angles, and AP, $A'P'$ in a given ratio.

8. Consider, as before, A, B, C, D and P and A', B', C', D' and P' subtending equal angles, and the points P, P' being moreover such that the distances AP, $A'P'$ are in a given ratio. I write therefore

$ar' : c''' : y' : z' = -\sigma a : b : c : d,$ where $a^2 = (1-a^2 d)U = \Delta(0-\cot C)\,V_3,$ $U' = \Delta(\cot C-\cot D)\,V_3,$ $U' =$

respectively. We have

$$\begin{aligned}(AP)^2 &= a^2y + (a^2-c^2)(\beta x + (a^2-c^2)y(x+a^2-2c^2y),\\ &= -a^2P^2(a^2z + (a^2-c^2)x + (a^2-c^2)y(x+a^2-2c^2y),\end{aligned}$$

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or, what is the same thing,

$$\frac{(AP)^2}{(1-\alpha y-\beta x)^2+Y(x+a^2\cot A_2 y)};$$

and similarly

$$\frac{(A'P')^2}{(1-\alpha y-\beta x)^2+Y(x+a^2\cot A_2 y)};$$

The required relation therefore is

$$\frac{a''(b''+b''^2)V_a(a+a''^2a''^2)}{(c+b'')^2(c+a''^2+b''^2)}=\kappa^2\frac{c^2(b'^2+b'^2)V_a(a+a^2+b'^2)}{(b-b')^2(c+a^2+b'^2)};$$

viz. substituting for x', y', z', a' their values

$$U=\Delta(0-\cot C)\,V_3,\quad U=\Delta(\cot C-\cot D)\,V_3,\quad U=\Delta(\cot C-\cot C')\,V_3,$$

and similarly $U=\Delta(\cot C-\cot C')\,V_3,\quad U=\Delta(\cot C-\cot C')\,V_3,$

$$\beta'^2=\beta'^2\cot(C-\beta'\cot(b+a'^2\cot(a'^2+a''^2))(a''^2+a''^2(b''^2-a''^2));$$

and the equation becomes

$$(BCD)\,Y(B'D)^2(B'A''CO)=(1+B'D)^2(CO+AVCO)$$

$$+DA\,B'D(B'D)^2(B'D'')^2\cdot A=(\beta^2\beta'^2)(b'^2\cot C-a''^2\cot(C'(C-C'));$$

and the like equation for b.

The two Tetrahedra (A, B, C, D or P and A', B', C', D' or P' in $ABC'D'$ subtending equal angles.

10. I consider now the lions-mentioned problem of the two tetrahedra; viz. on the two faces ABC and $A'BC'$ respectively, letting fall the perpendiculars DK and $D'K'$, then first A, B, C, D and A', B', C', D' subtend equal angles, the conditions for this is on a line, the focus of P is a cubic curve through A, B, C (ABC); and from the formula

$$F\cdot Y=K(DA'\cdot A'BO+D'A'\cdot A'BO)$$

derived from the points A, B, C, we have $BO=O'=CO$, $G'=G$, etc.

Next, B, C, D and P and B', C', D' and P' subtend equal angles, the conditions whereby distances AP and $A'P'$ are in a given ratio; the locus of P is a 12-line curve through $(BCDO)$.

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On a Bicyclic Chuck.

[From the Philosophical Magazine, vol. XLIII. (1872), pp. 365–367.]

p. 209. THE apparatus, although I have called it a chuck, is constructed not for turning, but for drawing: viz. it rotates horizontally, on a table (being moved, not from the inside by the axis of the lathe, but from the outside by a handle-frame), carrying a drawing-board which works under a fixed pencil supported by a bridge. Two points of the drawing-board describe circles, and the curve traced out on the drawing-board is consequently that described by a fixed point upon a moving plane (two points of which describe circles), or, what is really the same thing, the path of any point on a rigid body which has a sliding-piece (6) carrying a block (3), and besides *a* fixed plate by a handle-frame: which two are sides along (one of *a*-pieces of which are equal and respectively round the axis of the moving plane).

The apparatus consists in the first instance of a moving plane Q, sliding round on the surface of an horizontally rotating board. It is essential to remark that in this simple form of the instrument the chuck is a description of the chuck X, X are themselves the planes II, II, then putting the mitre wheel of one plane X in gear with either of the corresponding central wheels, and making the other rotate, the other plane X will rotate as well; according as the mitre wheels are in the opposite direction, or it will be in motion, according as the mitre wheel is in gear with one or the other of the corresponding central wheels, and may rotate with each of them.

To convert the apparatus into an oval chuck, we remove altogether the carrying-frame; and in the third plane we fix to the sides (8) of the handle-frame two bars at right angles to these sides, by means of pegs on the lower surfaces of these bars fitting tightly into holes on the sides (8) (which holes and the ends of the bars are shown in the figure), in such wise that these bars are have a moving piece (1) in the form of *a* rectangle with a circle cut out thereof, resting about the segments (4), and having upon it grooves in which works a sliding-piece (6) carrying *a* block (3); there is in this block *a* circular hole, D. The second plane includes also two sides (8) of *a* handle-frame, which two sides slide along two of the sides of the piece (1).

Third plane consists of *a* rectangular piece (9) rotating about an axis fixed to the block (3), and having *a* sliding-piece (10) in which is *a* circular hole, D. The third plane includes also the betoxed-mentioned block (7), having upon it the hole E.

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and it includes also the remaining two sides (11) of the handle-frame, and, let into the same so as to be flush therewith on the upper surface, two slips (12) completing, in this plane, the handle-frame.

We have thus on a level the sides (11), (12) of the handle-frame and the holes C, D, where C rotates about the point B, which is the centre of the block (3); and D rotates about the point A, which is the centre of the segments (4). Each hole being capable of describing a complete circle; and the distances AB, BC, CD, and DA are (within limits) adjustable to any given values: the distance of the holes C, D is made equal to that of the two pegs not referred to.

Connected herewith by means of cylindrical pegs working in the holes C, D respectively, we have a carrying-frame; viz. the fourth plane contains two sides (13) of this carrying-frame, and two moveable bars (14), attached to the remaining two sides (15) of the carrying-frame, and having on their lower surfaces the pegs which work in the holes C and D respectively—such bar being free to rotate about one extremity, and being clampable at the other extremity so as to allow the two pegs to be adjusted at a given distance from each other. And then in the fifth plane we have the remaining two sides (15) of the carrying-frame.

Rigidly connected with the carrying-frame we have the drawing-board; or, to make the whole more complete, this should be adjustable to any given position in regard to the carrying-frame by giving it two sliding motions crosswise, and a rotating motion, in the manner of an eccentric chuck.

To convert the apparatus into an oval chuck, we remove altogether the carrying-frame; and in the third plane we fix to the sides (8) of the handle-frame two bars at right angles to these sides, by means of pegs on the lower surfaces of these bars fitting tightly into holes on the sides (8) (which holes and the ends of the bars are shown in the figure), in such wise that these bars include between them the piece (9), which is thereby kept in a direction at right angles to the sides (8), and so slides between them. And we then take away the four points A and B respectively, by connecting the drawing-board and the third plane by a sliding-piece (16) carrying a handle-frame, it is easy to arrange as to this.

It is hardly necessary to remark that the pencil should have two sliding motions crosswise, so as to allow it to be adjusted to any given position; and a small up-and-down motion, so that it may be loaded to press with the proper force upon the drawing-board.

The variety of forms, even with a fixed adjustment of the chuck, only the position of the pencil being altered, is very considerable: among them we have lentil ovals and pear-shapes, passing through cusped forms into bent figures-of-eight.

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[Figure: Plan and elevation of the bicyclic chuck, three planes labelled — omitted]

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On the Problem of the In-and-Circumscribed Triangle.

[From the Philosophical Transactions of the Royal Society of London, vol. CXLI. (for the year 1871), pp. 369–412. Received December 30, 1870—Read February 9, 1871.]

p. 212. THE problem of the In-and-Circumscribed Triangle is a particular case of that of the In-and-Circumscribed Polygon; the last-mentioned problem may be thus stated:—to find a polygon such that the angles are situate on n given curves, and the sides on m given curves. And let us say in the first instance that as the number of such sides respectively are all of them distinct curves; then, considering the side AB as having upon it B, A, and another point, that is, as having upon it three points of the system; and analogously BC as having upon it three points, and so on, we see that the order of the curves must be such that the sides may be in succession to those of the points; that is, $n+m$ points, which is what each side must be in succession to those of the angles. The supposition that a on the polygon are not always such that they shall serve to find all of them, viz. a side being touched by the next side, the curves of any side may be excluded from the requirement of containing the the same angle of the polygon: the solution of the problem when the angles are on the same curve, and when the sides on a single one, viz. the case in question, in which the number of curves is $= 6$, and which has been previously considered; and there exists in (1871) by M. Poncelet a complete and elaborate discussion of the case where there is a curve common to the angles and sides, viz. in the case of the triangle is in (1817) referred to as $aDC[c[c[D]]]$, viz. a, b, c are respectively. That in A, D, F are the sides on, or as respectively, and a, b, c are the angle of FE, BD, DF.

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respectively. And I use the same letters a, c, e, B, D, F to denote the curves containing the angles and touched by the sides respectively; viz. the angle a is situate in the curve a, then B touches the curve B, and so on for the other angles and sides respectively. An equation such as $a = e$ or $a = B$ denotes that the curves a, c, or, as the case may be, the curves a, B are one and the same curve: it is in general convenient to use a new letter for denoting these identical curves; viz. I write, for instance, $a = e$ or $a = B$ as the case may be, the curves a, B are one and the same curve: it is in general convenient to use a new letter for denoting these identical curves; viz. I write, for instance, $a = e$ or $a = B$ as the case may be, the curves a, B are one and the same curve: it is in general convenient to use a new letter for denoting these identical curves; viz. I write, for instance, $a = e$ or $a = B$ as the case may be, the curves a, B are one and the same curve. The expression "no identities" denotes that the curves are all distinct. But I use also the same letters a, c, e, B, D, F to denote the curves containing the angles and touched by the sides respectively; viz. the angle a is situate in the curve a, B touches the curve B, and so on for the other angles and sides respectively; viz. to denote the orders and classes of the curves a, c, e, B, D, F, x, y quantitatively; thus the equation such an $aBcDD = 0$ denotes that the curves of aB, C, D are no identities; that is, the order of a are 0 and 0, those of B, D, F are 0 and 1, respectively.

The case "two several distinct identities" denotes that the curves a, c, e contain other, and selecting one or two of them, $a = c$ for "two identities" denotes the number of triangles is given as the before-mentioned result for the polygon. In the case $a = aBCDDF$, the methods and principles, however, are applicable to the case of polygon of any number of sides, and although consideration of the manner in which the consideration leads to the problem of the polygon, viz. to the present case we have here (AB, etc.), the relations of which are derived from this different square: roots and their results $32 = 6.4 \times C$ etc. But it is for further nature.

I remark that the work, in the following Table, of the number of positions of a particular angle of the triangle, and that in some cases, on account of the symmetry of the figure, the number of triangles is a submultiple of this number; viz. the number of positions of the angle is to be divided by 2 or 6; this is expressly shown, by means of a separate column, in the Table.

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Table: number of positions of an angle and number of triangles for various identifications among the six curves a, c, e, B, D, F.

No.	Identifies	Specification	Sample	No. of triangles
1	$a = c = e$	$a = c = e$	$2\alpha\beta DF$	$2\alpha\beta DF$
2	0	no	$2\alpha\beta DF$	$2\alpha\beta DF \div 6$
3	2	$a = c \neq e$	$2EF \cdot D = 1$ flex.	$2EF \cdot D - 1$
4	2	$a = c, B = D$	$1EF \cdot EF$	$\alpha\beta DF$
5	3	$a = c = e \neq B = D = F$	$1\beta(\alpha - 1)EF$	$1\beta(\alpha - 1)EF$
6	2	$a = c, B = D, e = F$	$2\alpha EF^2$	$2\alpha EF^2$
7	3	$a = c = e \neq B = D \neq F$	$1\beta(\alpha - 1)EF$	$1\beta(\alpha - 1)EF$

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[Table continues with rows 8–15: further identifications, specifications, samples and triangle counts. Tabular structure preserved; full numeric entries occupy several columns and are partially abbreviated.]

No.	Identifies	Specification	Sample	No. of triangles
8	2	$a = c, e = D$	$2\alpha\beta EF$	$2\alpha\beta EF$
9	3	$a = c, B = D, e = F$	$E(\alpha - 1) \cdot DF$	$E(\alpha - 1)DF$
10	3	$a = c, e = B = D$	$2EF \cdot \alpha\beta DF$	$2EF \cdot \alpha\beta DF$
11	3	$a = e, B = D, e = F$	$\alpha\beta(\alpha - 1)DF$	$\alpha\beta(\alpha - 1)DF$
12	3	$a = c = e$	$1EF \cdot 1$ flex.	$1EF \cdot 1$
13	4	$a = c, e = B = D, e = F$	$E\alpha\beta DF$	$E\alpha\beta DF$
14	4	$a = c = e, B = D, e = F$	$\alpha\beta(\alpha - 1)DF$	$\alpha\beta(\alpha - 1)DF$
15	4	$a = c = e$	$F.B - D, e = F$	$F \cdot B - D$

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Table (continued): rows 16–23 with further identifications and specifications.

No.	Identifies	Specification	Sample	No. of triangles
16	4	$a=c, B=D, e, F$	$2\alpha EF \cdot DF$	$2\alpha EF \cdot DF$
17	4	$a=c, B=D, e=F$	$2\alpha EF^2$	$2\alpha EF^2$
18	4	$a=c, e, B, D, e=F$	$\alpha\beta EF^2$	$\alpha\beta EF^2$
19	4	$a=c=e$	$\alpha=e$ flex.	$E\alpha$
20	4	$a=c, B=D$	$\alpha\beta EF$	$\alpha\beta EF$
21	5	$a=c, B=D, e=F$	$4\alpha\beta EF$	$4\alpha\beta EF$
22	5	$a=c, B=D, e=F$	$\alpha=e$	$E\alpha$
23	5	$a=c=e$	all equal	—

[End of paper 514 and of the section pp. 167–216 typeset here.]

On the Problem of the In-and-Circumscribed Triangle.
[Continued from p. 216. Vol. VIII.]

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The following is a continuation of the table of cases for the in-and-circumscribed triangle problem. Each row exhibits the specification of the case, the straight sum, the deficiency, and the number of examples.

Table (continued). Pages 217–221.

No. of Case	*Specification*	*Strt. sum*	*Def.*	*No. of examples*
30	$a:b:c,\ a=b=c,\ \alpha+\beta=\gamma$	(15)	(20)	$2(a-1)YZ\cdot Z\alpha\beta$
31	$a:b:c,\ a=b,\ \alpha=\beta=\gamma,$ $\alpha+\beta=\gamma$	6	4	$2x(a-1)YZ\cdot Z\alpha\beta$
32	$a:b:c,\ a=b,\ \alpha=\beta=\gamma,$ $\alpha+\beta=\gamma$	6	4	$2x(a-1)YZ\cdot Z\alpha\beta$
33	$a:b:c,\ \beta=\gamma,\ \alpha+\beta=\gamma$	6	4	$2(a-1)(aX-aZ)YZ\cdot Z\alpha\beta$
34	$a:b:c,\ \beta=\gamma,\ \alpha+\beta=\gamma$	6	4	$2(a-1)(aX-aZ)YZ\cdot Z\alpha\beta$
35	$a:b:c,\ \beta=\gamma,\ \alpha+\beta=\gamma$	6	4	$2(a-1)(aX-aZ)YZ\cdot Z\alpha\beta$
36	$a:b:c,\ \alpha+\beta=\gamma$	(12) (16)		Quartic case

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No.	*Specification*	*Strt. sum*	*Def.*	*No. of examples*
37	$a:b:c,\ a=b=c,\ \alpha=\beta=\gamma$	(12)	(16)	$2x(a-1)(X-2)\cdot YZ$
38	$a:b:c,\ a=b,\ \alpha=\beta=\gamma$	6	4	$2X(a-1)(Y-Z)\cdot YZ+Y\cdot V$
39	$a:b:c,\ a=b,\ \alpha=\beta=\gamma$	6	4	$2(a-1)(aX-aZ)YZ\cdot Z\alpha\beta$
40	$a:b:c,\ a=b,\ \beta=\gamma$	6	4	$2(a-1)(aX-aZ)YZ\cdot Z\alpha\beta$
41	$a:b:c,\ \beta=\gamma$	3	6	$2X(a-1)(X-2)\cdot YZ$
42	$a:b:c,\ \beta=\gamma$	3	6	$2(a-1)\{aX-2(X-1)Y(X-Z)\}\cdot YZ$

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No.	*Specification*	*Strt. sum*	*Def.*	*No. of examples*
38	$X = D = P = u$, $V = v$, $u = v$, $P = B = u$, $B = v$, $\beta = u - v$, $D = P = u$, $D = v$		6	$2X(YpaX + aY)Z \cdot YZ$
39	$X = D = P = u$, $a = b = c$, $\beta = v$	6	3	$\{2X^2 + aZY \cdot 8\beta b(b-c) \cdot a(\beta b)^2 - (aP) \cdot a(\beta b)$
40	$a : b : c$, $X = P = u$, $a : b : c$, $V = u$		3	$\{a(\beta x)^2 + (aP)^2 \cdot (\beta b)Z \cdot (\beta b) \cdot Z \cdot aX$
41	$a : b : c$, $\beta = \gamma$, $a = b$, $\beta b = \nu$, $D = u$, $P = v$, $P = R$		3	$2x(a-1)YZ \cdot Z\alpha\beta$
42	$a : b : c$, $a = b$, $\beta = \gamma$, $D = u$, $P = v$, $P = R$		6	$2(a-1) \cdot Z \cdot Z \cdot abP \cdot ab\alpha\beta(a-1) \cdot \{aZ + (aP)$
43	$a : b : c$, $a = b$, $P = R$, $\beta = \gamma$		3	$2(a-1)YZ(Y-Z) \cdot abP$
		(4) (12)		Quartic case

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No.	*Specification*	*Strt. sum*	*Def.*	*No. of examples*
43	$a : b : c$, $a = b$, $\beta = \gamma$, $P = Q$, $\alpha = \beta$, $D = P$	(12)	(17)	$2(a-1)(aX - aZ)YZ \cdot Z\alpha\beta$
44	$a : b : c$, $\beta = \gamma$, $D = P$	6	8	$2(a-1) \cdot YZ \cdot Z\alpha\beta - aX\beta \cdot aZ\beta b + 8XZ \cdot aP$
45	$a : b : c$, $a = b$, $\beta = \gamma$, $\alpha + \beta = \gamma$, $P = Q$	6	5	$2X(a-1)YZ(Y-Z) \cdot abP \cdot YZ$
46	$a : b : c$, $a = b$, $\beta = \gamma$, $\beta = \gamma$	6	5	$2X(a-1)YZ(Y-Z) \cdot abP \cdot YZ$
47	$a : b : c$, $a = b$, $\beta = \gamma$, $\alpha = \beta = \gamma$	6	5	$2(a-1)Y(Y-Z)(Z-aZ)(YZ - Y\beta + aP)\beta$
48	$a : b : c$, $\beta = \gamma$, $\alpha = \beta$	6	5	$2(a-1)Y(Y-Z)(Z-aZ)(YZ - Y\beta + aP)\beta$
		(12) (16)		at infinity

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No.	*Specification*	*Def.*	*No. of examples*
51	F, a, b, $c = a - b = c$		
52	$a = u = b = P \cdot YZ$		$2aY \cdot ab\beta$
			$2X^2(3a^2 - 9aZ - aP + (Z - aZ))$ $+aX^3(2aZ - aP)(2P - 9P - aZ\beta)$ $+aX^3(a^2P - aP\beta - aP\beta)$ $+aX^3(2P - aP\beta)(P - aP)(P - aZ)$ $+aX^3(2P + aP\beta)(P - aZ)$ $+(\sum aX^3)(P - aZ\beta)$
D ditto		503	

[Tabular continuation – Several entries in the original scans are partially illegible; symbols and groupings are preserved as faithfully as the source permits.]

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The foregoing results are chiefly obtained by means of the theory of correspondence; viz. if instead of the triangle $aBcDeF$ we consider the unclosed trilateral $aBcDeFg$, where the points a and g are situate on one and the same curve, say the curve $a = g$, then the points a and g have a certain correspondence, say a (χ, χ') correspondence with each other; and when a, g are a "united point" of the correspondence, the trilateral in question becomes an in-and-circumscribed triangle $aBcDeF$; that is, the number of triangles is equal to that of the united points of the correspondence, subject however (in many of the cases) to a reduction on account of special solutions. It is in particular cases more convenient to consider the correspondence not of the united points in, in several of the cases, but not in all of them, $a = g = \chi = \chi'$. But in some instances I employ a functional method, by assuming that the identical curves are each of them the aggregate of the two curves a, a': we have obtain for the number $g\sigma$ of the triangles belonging to the curve a a functional equation $\phi(a + a') = \phi a + \phi a'$ given function; viz. the expression on the right-hand side depends on the solution of the preceding cases, wherein the number of identities between the several curves is less than in the case under consideration; and taking it to be known, the functional equation gives $\phi a =$ particular solution + linear function of (a, B, F). The particular solution is always easily obtainable, and the constant of the linear function can be determined by means of particular forms of the curve a.

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Article Nos. 1 to 6. ***The Principles of Correspondence as applied to the present Problem.***

1. Consider the unclosed trilateral $aBcDeFg$, where the points a and g are on one and the same curve, say the curve $a = g$. Starting from an arbitrary point a on the curve, we have αB any one of the tangents from a to the curve B, touching this curve, say at the point B; and intersecting the curve c in a point c; viz. c is any one of the intersections of aBc with the curve c; we have then similarly cDe any one of

Fig. 1.
[Figure: triangular configuration $aBcDeFg$ — diagram omitted]

the tangents from c to the curve D, touching it, say at D, and intersecting the curve e in a point e; viz. the point e is any one of the intersections in question; and then in like manner we have eFg any one of the tangents from e to the curve F, touching it, say at F, and intersecting the curve $\alpha (= a)$ in a point g; viz. g is any one of the intersections in question. Suppose that to a given position of a there correspond χ positions of g, it is easy to find the value of χ; viz. if (as above tacitly supposed)

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the curves a, B, c, D, e, F are all distinct curves, then the number of these tangents αB is $= B$; there are on each of them c points c; through each of these we have D tangents cDe; on each of these e points e; through each of these F tangents eFg; and on each of these a points g; through each of these $\chi = Bcdef\alpha$. But if some of the curves become one and the same curve — if, for instance, $\alpha = B = c$, — the line αBc is here a tangent from a point α on the curve, we exclude the tangent at the point α, and the number of the remaining tangents is $= (A-2)$; each tangent meets the curve in the point α counting once, the point B counting twice, and in $(\alpha - 3)$ other points; that is, the number of the points c is $= (A-2)(\alpha-3)$, and in like cases; the calculation is always immediate, and the only difference is that, instead of a factor α or A, we have such factor in its original form or diminished by 1, 2, or 3, as the case may be. Similarly starting from g, considered as a given point on the curve $g (= a)$, we find χ' the number of the corresponding points a; thus in the case where the curves are all distinct curves, we have $\chi' = FeDcBa\,(= \chi)$; and so in other cases we find the value of χ'. The points (a, g) have thus a (χ, χ') correspondence, where the values of χ, χ' are found as above.

2. There will be occasion to consider the case where in the triangle $aBcDeF$ (or say the triangle $aBcDeFa$) the point a is not subjected to any condition whatever, but is a free point. There is in this case a "locus of a," which is at once constructed as follows: viz. starting with an arbitrary tangent αB of the curve B, touching it at B and intersecting the curve c in a point c; through c we draw the curve D the tangent cDe, touching it at D and intersecting the curve e in a point e; and finally from e to the curve F the tangent eFa, touching it at F and intersecting the original tangent αB in a point α; the locus of the point α so obtained is an arbitrary tangent of the curve B (or of the curve F).

3. The general form of the equation of correspondence is

$$p(a - x - x') + q(b - \beta - \beta') + \ldots = k\Delta(t);$$

viz. if on a curve for which twice the deficiency is $= \Delta$ we have a point P corresponding to certain other points $P', Q', \ldots$, in such wise that P, P' have an (x, x') correspondence, P, Q' a (β, β') correspondence, &c.; and if (α) be the number of the united points (P, P'), (β) the number of the united points (P, Q'), &c.; and if moreover for a given position of P the points $P', Q', \ldots$ are obtained as the intersections of the curve with a curve Θ (depending on the point P) which meets the curve k times at P; p times at each of the points P', q times at each of the

* To avoid confusion with the notation of the present memoir, I abstain in the text from the use of D as denoting the deficiency, and there is a convenience in the use of a single symbol for twice the deficiency; but writing for the moment D to denote the deficiency, I remark, in passing, that perhaps the true theoretical form of the equation is

$$k(2D - D - D) + p(a - x - x') + q(b - \beta - \beta') + \ldots = 0,$$

viz. the point P is here considered as having with itself a (D, D) correspondence, the number of the united points therein being $= 0$.

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unilary values $f = \phi - \phi'$ and $v - e - e'$ are obtained by means of a more simple application of the principle of correspondence, as will appear in the sequel ([1]), but for the moment I do not pursue the question.

Article Nos. 7 to 14. ***Locus of a free angle (α).***

7. I consider the case where a is a distinct curve $\alpha =, \nu F$, and where, as was seen, the equation is simply

$$g - \chi - \chi' = 0.$$

I suppose further that a is distinct from all the other curves, or say, *simpliciter*, that a is a distinct curve. The values of χ, χ' will here each of them contain the factor a, say we have $\chi = n\alpha$, $\chi' = n\alpha'$; and therefore the equation gives $g = \alpha(n + \alpha')$. It is obvious that α, α' are the values assumed by χ, χ' respectively in the particular case where the curve α is an arbitrary line ($\alpha = 1$); and $\alpha + \alpha'$ is the number of the united points on this line.

8. Suppose now that in the triangle $aBcDeFa$ the point a is a free point, and have, as above-mentioned, a locus of a, and the united points on the arbitrary line are the intersections of the line with this locus; that is, the locus meets the arbitrary line in $\alpha + \alpha'$ points; or, what is the same thing, the order of the locus is $= \alpha + \alpha'$.

9. I stop for a moment to remark that in the particular case where the curve B is a point ($B = 1$), then in the construction of the locus of α the arbitrary tangent αB is an arbitrary line through B, and the construction gives on this line a position of the point α. But drawing from B a tangent to the curve F, and thus constructing in order the points F, e, D, c, α, the construction shows that B is an α'-tuple point on the locus; and (by what precedes) an arbitrary line through B meets the locus in α other points; that is, in the particular case where the curve B is a point, the order of the locus of α is $= \alpha + \alpha'$, which agrees with the foregoing result.

10. The construction for the locus of α may be presented in the following form: viz. drawing to the curve D a tangent cDe, meeting the curves c, e in the points c, e respectively; then if from any point c we draw to the curve B a tangent cBa, and from any point e to the curve F a tangent eFa, the tangents cBa, eFa intersect in a point on the required locus. Hence if in any particular case (that is for any particular position of the tangent cDe) the lines cBa, eFa become one and the same line, the point a will be an indeterminate point on this line; that is, the line in question will be part of the locus of a.

11. The case cannot in general arise so long as the curves B, F are distinct from each other; but when these are one and the same curve $B = F$, then the line cBa will be a tangent of the curve B, and we have thus as part of the locus the locus $c = e$ if it can be drawn, the same being supposed to. To show how this is, suppose in, say at F, and intersecting the curve cBe, it is needful, that perhaps the true theoretical form of the equation is[1]

[1] See post, Nos. 24 *et seq.*

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here a tangent at B passing through a point α of the intersection of this curve α, and from this point a tangent drawn to the curve $B = F$. For the position in question of the tangent of D, the points c, e coincide with each other, and we have thus the coincident tangents cBa and eFa as the identical curves $B = F$. It is further

Fig. 2. First-mode figure.
[Figure: First-mode configuration of tangents — diagram omitted]

to be remarked that the number of the points of intersection is $= \sigma$; from each of these there are B tangents to the curve F (in all σB tangents), and each of these counts once in respect of each of the D tangents to the curve D; that is, it counts D times. We have thus as part of the locus of α on B lines each D times, or, say, first-mode reduction $= \sigma BD$.

12. The second mode is there in the same manner "second-mode figure." The tangent from D is here a common tangent of the curves D and $B = F$. This meets the curve c in σ points; and the curve e in σ points; and attending to one pair of points c, e on each of these the tangents cBa, eFa, coinciding with the common tangent

Fig. 3. Second-mode figure.
[Figure: Second-mode configuration — diagram omitted]

in question, and forming part of the locus of α. The number of the common tangents is $= BD$; but each of these counts once in respect of each combination of the points c, e; that is in all σ times. And we have thus as part of the locus BD lines each σ times, or, say, second-mode reduction $= BD \cdot \sigma$. This is (as it happens) the same number as for the first mode; but to distinguish the different origins I have written as above $\sigma \cdot BD$ and $BD \cdot \sigma$ respectively.

13. It is important to remark that each of the two modes arises whatever relations of identity subsist between the curves σ, c, B, and F, but with consideration of these. Thus if any one of the lines cBa, eFa is not so situate (but is so far) distinct from B, then in the first-mode figure as may be a node or a cusp of the curve $\sigma = \sigma$, it, say at F, and intersecting the curve $\sigma = \sigma$, it is needful to find a copy of the entire of the intersections in question. Suppose that for a given position of D touching the curve $c = e$ in the neighbourhood of the node, thus of the two points of intersection each $c = e$ is in the neighbourhood of the node, thus of the two points of intersection each

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in succession may be taken for the point c, and the other of them will be the point e; so that the node counts twice. It requires more consideration to perceive, but it will be readily accepted that the cusp counts three times. Hence if for the curve $c = e$ the number of nodes be $= \delta$ and that of cusps $= \kappa$, the value of the first-mode reduction is $= (2\delta + 3\kappa)BD$.

As regards the second-mode figure, the only difference is that c, e will be now any pair of intersections (each pair twice) of the tangent with the curve $c = e$; and the value is thus $= (c^2 - c)BD$.

It would be by no means uninteresting to enumerate the different cases, and indeed there might arise some advantage in doing so here; but I have (instead of this) considered the several cases, when and as they arise in connexion with any of the cases of the in-and-circumscribed triangle.

14. Observe that the general result is, that in the case $B = F$ of the identity of the curves B and F, but not otherwise, the locus of α includes as part of itself a system of lines; or, say, that it is made up of the proper locus, the lines, and of a residual curve of the order $\alpha + \alpha' -$ Red., which is the proper locus.

Article Nos. 15 to 17. ***Application of the foregoing Theory as to the locus of*** (α).

15. Reverting now to the case where the angle a is not a free angle but is situate on a given curve α, then if the curve α is distinct from the curves α, F, the number of positions of α is, as was seen, $g = \chi + \chi'$. But the points in question are the intersections of the line-system $\alpha F \gamma$ with the curve α; and the line-system $\alpha F \gamma$ being made up of the proper locus $-$ Red. and of a system of lines is on this account distinct from the curve α; the points of intersection on the proper locus must convenient to obtain directly one than the other of two reciprocal results; for instance, to consider the case $B = D = F$ rather than $\alpha = e = e$, since in the equation in the $\chi = \chi'$ Red., the entry in the expression $\alpha + \alpha' -$ Red., the order of the proper locus of α.

16. It is however to be noticed that if the curve α, being as is assumed distinct from the curves α, and $F = B$, is identical with one or both of the remaining curves c, D, the foregoing expression $\chi + \chi'$ for the order of αBe, gives points a which are not true solutions of the problem, viz. the curve α may include in general in eaiser and more convenient to obtain directly one than the other of two reciprocal results; for instance, of lines, or reduction in the expression $\alpha + \alpha' -$ Red. of the order of the proper locus of α.

[1] More generally, if the curve α is a curve identical with any of the other curves, then if treating in the first instance the angle α as the free we find in any manner the locus of α, the required positions of the angle α are the intersections of this locus and of the curve α; but these intersections will in general include intersections which give heterotypic solutions. The determination of these is a matter of some delicacy, and I have in general treated the problem in each manner that the question does not arise; but an an example see post, Case 45.

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17. But this cannot happen if the curve a is distinct also from the curves c, D; or, say, simply when a is a distinct curve. The conclusion is, that in the case where a is a distinct curve we have

$$g = \chi + \chi' - \text{Red.},$$

where the term "Red." vanishes except in the case of the identity $B = F$ of the curves B, F; and that when this identity subsists it is $= \sigma$ times the reduction in the order of the locus of a considered as a free angle; viz. this consists of a first- mode and a second-mode reduction as above explained.

Article Nos. 18 to 21. ***Remarks in regard to the Solutions for the 52 Cases.***

18. Before going further I remark that the principle of correspondence applies to corresponding and united tangents in like manner as to corresponding and united points, and that all the investigations in regard to the in-and-circumscribed triangle might thus be presented in the reciprocal form, where, instead of points and lines, we have lines and points respectively. But there is no occasion to employ any such reciprocal process; the result to which it would lead is the reciprocal of a result given by the original process, and as such it can always be obtained by reciprocation of the original result, without any performance of the reciprocal process.

19. It is hardly necessary to remark that although reciprocal results would, by the employment of the two processes respectively, be obtained in a precisely similar manner, yet that this is not so when only one of the reciprocal processes is made use of; so that, using one process only, it may be in general in easier and more convenient to obtain directly one than the other of two reciprocal results; for instance, to consider the case $B = D = F$ rather than $a = e = e$. Forcing the analogy in such a case, to consider the case $B = D = F$ rather than $a = c = e$, and having obtained the one result, directly to deduce from it the other by reciprocity; but that it may nevertheless be interesting to obtain each of the two results directly.

20. It is moreover obvious that although the several forms of the same case, for instance Case 2, $a = c$, $a = e$, or $c = e$, are absolutely equivalent to each other, yet that, when as above we select a vertex a, and seek for the number of the united points (a, g), the process of obtaining the result will be altogether different according to the different form which we employ. For instance, in the case just referred to, if the form is taken to be $a = c$ or $c = e$, then the equation $g = \chi + \chi'$ is applicable to it; but not so if the form is taken to be $a = e$. It would by no means uninteresting in every case to consider the several forms successively; but it appears that in every case it cannot however be carried out very far, considering, illustration, verification, or otherwise it appears proper so to do. The translation of a result, for instance, of a form $a = c$ or $c = e$ into that for the form $a = c = e$ is so easy and obvious, that it is not even necessary formally to make it.

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21. I do not at present further consider the general theory, but proceed to consider in order the 52 cases, interpreting in regard to the general theory each further discussion or explanation as may appear necessary. In the several instances in which the equation $g = \chi + \chi'$ is applicable, it is sufficient to write down the values of χ, χ', the mode of obtaining these being already explained.

The 52 Cases for the in-and-circumscribed triangle.

Case 1. No identities.

$$\chi = BcDeFa, \quad \chi' = FeDcBa\,(= \chi),$$

$$g = 2aBcDF.$$

Case 2. $a = c = e$.

$$\chi = B(a-1)DeFa, \quad \chi' = FeD(a-1)Ba,$$

$$g = 2a(a-1)BDF.$$

Second process, for form $a = c = e$. The equation of correspondence is here

$$g - \chi - \chi' + F(a - c - c') = 0;$$

but the points e being given as all the intersections of the curve $a(= c)$ by the line- system cDe which does not pass through a, we have $c = c' = 0$, so that $g = \chi + \chi'$; and then

$$g = BcDeFa(= 2a-1) + FeDcBa,$$

$$= 2a(a-1)BDF,$$

giving the former result(1).

Case 3. $B = F = e$. Reciprocation from 2; or else, second process,

$$\chi = DeBa(B-1)a, \quad \chi' = De(B-1)Ba,$$

$$g = 2X(X-1)Daa.$$

Third process: form $F = Ba = e$. We have here $g = \chi + \chi' -$ Red.,

$$\chi = BcDaBa, \quad \chi' = BcD(\alpha)a,$$

$$\chi + \chi' = 2X^2Dca.$$

and the reductions are those of the first and second mode, as explained ante, Nos. 13, 14, viz. each of them is $= X \cdot Dca$, and together they are $= 2XDca$; whence the foregoing result.

Case 4. $a = B = a$.

$$\chi = BcDeFa, \quad \chi' = FeDc(\alpha-2)a,$$

$$g = 2XaeDF.$$

1 Of course, the result is obtained in the form belonging to the next form of specification, viz. here it is $= 2(a-1)BDF$, and as in other instances; but it is unnecessary to refer to this change.

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Observe this is what the result for Case 1 becomes on writing therein $a = B = a$, viz. the opposite curves a, D may become one and the same curve without any alteration in the form of the result.

Case 5. $a = B = a$.

$$\chi = (X-2)cDeFa, \quad \chi' = FeDcXa(X-1),$$

where

$$(X-2) + x \cdot X(x-1) = X(x-1) = X(X-x-1);$$

therefore

$$g = X(Xa - X - a)cDF.$$

Case 6. $a = c = e = a = a$: perhaps most easily by reciprocation of Case 5. The triangle $aBcDeF$ is here in succession each of the eight triangles

$$\begin{array}{cccc} a\ B\alpha\ D\alpha\ F & a\ B\alpha'\ D\alpha\ F & a\ B\alpha'\ D\alpha\ F & a\ B\alpha\ D\alpha'\ F \\ a'\ B\alpha\ D\alpha'\ F, & a\ B\alpha'\ D\alpha\ F, & a\ B\alpha'\ D\alpha\ F, & a\ B\alpha\ D\alpha'\ F, \\ a\ B\alpha\ D\alpha\ F, & a'\ B\alpha\ D\alpha'\ F, & a'\ B\alpha\ D\alpha\ F, & a'\ B\alpha\ D\alpha\ F, \\ a'\ B\alpha\ D\alpha'\ F, & a'\ B\alpha'\ D\alpha\ F, & a'\ B\alpha'\ D\alpha\ F, & a\ B\alpha'\ D\alpha\ F, \end{array}$$

where the two top triangles give Δa and $\Delta a'$ respectively; the remaining triangles all belong to Case 2, and those of the first column give each $2(a^2 - x)BDF$, and those of the second column each $2(a^2 - x)'BDF$. We have then

$$\phi(a + a') = \Delta a + \Delta a' + \{4(a^2 + aa') - 12a\alpha'\}BDF,$$

which is twice the function in question, on the right-hand side we may write

$$\phi a = 2a^2 - 2x'\alpha = \Delta x + a(BX + a)BDF,$$

that is to say so that, a, B are independent of the curves B, D, F; and taking each of these to be a point, and the curve $a = c = e$ to be a conic, this is known that $\phi = 2$; we have therefore $\Delta = 12 + 8a$ in this case, that is $a + B = 2$.

The case where the curve $a = c = e$ is a line gives $\phi = 2 : 4 + a = 2$, that is $a = 2$, let it is not easy to find another condition; assuming however $\eta = 0$, we have $a = 2$, $B = -1$, and hence therefore

$$\Delta a = 2a(a-1)(2a-1) + a(B-1)BD,$$

or my

$$g = 2(a-1)(a-2) + aD,$$

this is a good easy example of the functional process, the use of which appears to exhibit itself; and I have therefore given it, notwithstanding the difficulty as to the complete determination of the constants.

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Third process. The equation of correspondence is

$$g - \chi - \chi' + F(a - c - c') = 0,$$

but for the correspondence (a, c) we have

$$c - c' = v + D(a - y - y') = 0,$$

and for the correspondence (a, c),

$$c - c' = g' - aDa,$$

whence

$$g = \chi + \chi' + aDF \cdot \Delta;$$

that is

$$g = B(a-1)D(a-1)Fa + F(a-1)D(a-1)Ba + 2(a-1)(a-1)y;$$

Moreover

$$\chi + \chi' = B(a-1)D(a-1)Fa \cdot Z,$$
$$\Delta = X - 2a + 8 + a,$$

(if a be the number of cusps of the curve $a = c = e$), and the resulting value is

$$g = 2\{(a-1)BDF \cdot a - 2\}.$$

[Author's remark on the value of $aBDF$:]

[F as the number of cusps of the curve $a = c = e$, the resulting value of g is

$$g = 2(a-1)BDF \cdot \alpha.$$

Where, however, the term aBDF is to be rejected, I cannot quite explain this; I should rather have expected a rejection $= 2aBDF$, introducing the term $-aBDF$. For consider a tangent from the curve D from a cusp of the curve $a = c = e$; there are D such tangents; each gives in the neighbourhood of the cusp two points, say c, e; and from these we draw B tangents cBa to the curve B, and F tangents eFa to the curve F; we have then in respect of the given tangent of D, BF positions of the α tangent of D, BF positions of a, in all $aBDF$ positions of which b corresponds together with the cusp; and for all the cusps together $aBDF$ such triangles; this would be what is wanted; the difficulty is that as if the two intersections at the cusp each in succession may be taken for c, and the other of them for α, it would seem that the foregoing number $aBDF$ should be multiplied by 2.]

Case 7. $B = D = F = a$. Here $g = \chi + \chi' -$ Red.

$$\chi = Xc(X-1)e(X-2)a, \quad \chi' = Xc(X-1)e(X-2)a,$$

that is

$$\chi + \chi' = 2Xa \cdot ce.$$

The reductions of the two modes are as above, with only the variation that in the present case B is the same curve with the two curves $B = F$. That of the first mode is $= X(X-1) \cdot ce$, and that of the second mode is $= (2e-1) \cdot Xce$, or substituting, we have together they are $= 2X(X-1) \cdot ce$; or substituting, we have

$$g = 2X(X-1)X(X-2) + ce.$$

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Case 8. $a = c = B = a$.

$$\chi = (X-2)(a-1)DeFa, \quad \chi' = FeDa(X-1)(a-1)a,$$

$$g = Xc(a-1)(X-1)DeF.$$

Case 9. $D = F = a = a$. By reciprocation of 8.

$$\chi = X(X-1)(X-2)cDF.$$

Case 10. $a = c = B = a$.

$$\chi = B(a-1)(X-2)DeFa, \quad \chi' = FeX(a-2)Ba(a-1),$$

$$g = X(x-1)X(a-1)DeF.$$

Case 11. $D = F = a = a$. By reciprocation of 10.

$$\chi = X(X-1)(X-2) \cdot caa.$$

Second process: form $a = B = a = a$.

$$\chi = (X-1)(a-1)caXa, \quad \chi' = FeXa(X-1)(a-1) \cdot a;$$

giving the former result.
Case 12. $a = c = e = a$, $B = F$.

$$\chi = BcD(a-1)Fa, \quad \chi' = FaT(a-1)Ba\,(=\chi),$$

$$g = 2(a-1) \cdot BDF.$$

Case 13. $F = B = a$, $a = c = e = a$. By reciprocation of 12.

$$\chi = 2X(X-1) \cdot DF.$$

Case 14. $a = c = e$, $B = F = a$.

$$\chi = X(X-1)(a-1)caFa, \quad \chi' = FaT(a-1)F(a-2)\,(=\chi),$$

$$g = 2X(X-1)Y(F-1) + \cdot ce.$$

Case 15. $F = B = a$, $D = F = a$. By reciprocation of 14.

$$\chi = X(X-1)(X-2)Y \cdot DF.$$

Case 16. $a = c = e$, $B = F = a$.

$$\chi = B(a-1)(X-1)caa, \quad \chi' = F(a-1)(X-2)Ba\,(=\chi),$$

$$g = Xa(a-1)Y(X-1).$$

Case 17. $a = c = e$, $B = F = a$.

$$\chi = B(a-1)(X-1)(B-1)ca, \quad \chi' = Fa(a-1) \cdot caR(a-1)\,(=\chi),$$

$$g = 2X(a-1)Y(X-1)a.$$

But we have here aD as an axis of symmetry, so that triangle is counted twice, or the number of distinct triangles is $= \frac{1}{2}g$.

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Case 18. $a = D = c$, $c = B = g$.

$$\chi = Y(g - X)caFa, \quad \chi' = FaY(Y - X)caaa,$$

$$g = 2aXY(Y - y)cF.$$

Case 19. $c = B = a$, $D = F = e$.

$$\chi = Y(g - 1)D(Y - X)c = ea, \quad \chi' = Xy(Y - X)c = Da,$$

$$g = 2aYYY.$$

Case 20. $c = B = a$, $D = F = e$.

$$\chi = Bc(Y - X)cYY = ea, \quad \chi' = Y(Y - X)YDYca,$$

$$g = 2(Y - X)\{aXY + aY + aZ\} - X(Y - X)Yca + 2aZF.$$

Case 21. $c = B = a$, $D = F = e$.

$$\chi = Yc(Y - X)caXa, \quad \chi' = XyY(Y - X)caYa,$$

$$g = 2YcaYY.$$

Case 22. $a = D = c = a = e$, F.

$$\chi = c(X - 1)Da = Fa, \quad \chi' = Fc(X - 1)cDya,$$

$$g = 2Y(X - 1)c \cdot c = F.$$

Case 23. $a = D = c = a = e$, F.

$$\chi = (Y - 2)c(Y - 1)D(Y - 1)y \cdot Fa, \quad \chi' = Fa(Y - 2)cD(Y - 1)Y \cdot ca,$$

$$g = 2(Y - X)c(Y - X - 1)(Y - 2)\{(Y - X)cF\} + 2(Y - 1)yF + aXY + aYZ\} \cdot ZF \cdot \tan;$$

this is then the symmetry of the result of two modes; or any other case.

Case 24. $a = D = c$, $D = c = e = Fa$.

$$\chi = Yy(X - 1)Da = Fa, \quad \chi' = Yy = X(X - 1) \cdot ea\,(= \chi),$$

$$g = 2c(X - 1)Y(Y - 1)y \cdot ca.$$

Case 25. $c = e = a = a = D = F = v$. Here

$$g = \chi + \chi' - \text{Red.},$$

$$\chi = Xy(X - 1)(Y - 1)ca, \quad \chi' = XyY(X - 1)\{(Y - 1)(Y - 1)\}(= \chi)\, e \cdot a;$$
$$\chi + \chi' = v(y - 1) \cdot 2XY(X - 1) \cdot ca.$$

The reductions are those of the first and second modes; $c = e$, and that the curve D is identical with the curves $B = F$.

First-mode reduction is

$$v(C + 2g + 5\kappa)B(B - 1)$$

(where δ, κ refer to the curves $c = e$), which is

$$= v(c - 1)BB(B - 1);$$

that is, the reduction is $= v(g-1)Y(X-1)$.

Second-mode reduction is

$$v(2c+5c)c(X-1)$$

that is, the reduction is $= v(c-1)y(X-1)\cdot c\cdot(Y-1)$;

Hence the two together are $= v(c-1)\{2Y(X-1)+y(Y-1)y\}\cdot$; and subtracting from $\chi+\chi'$ we have

$$g = v(g-1)\cdot\{2X(X-1)-c\}\,c.$$

but on account of the symmetry each triangle is reckoned twice, and the number of triangles is $=\frac{1}{2}g$.

Case 26. $a = c = B = a$, $D = e = F = v$. By reciprocation of 25.

$$\chi = Y(c-1)(Y-1)2 - y(Y-X)Y\cdot cy.$$

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Case 26. $a = c = e = a$, $B = D = v$, $u = F = v$.

$$\chi = Y(a-1)(Y-1)e \cdot 2 - y \cdot Fa, \quad \chi' = Z(Y-1)Y(Y-2)Y(Y-1)\,(a-1)\,(a-2);$$

$$\begin{aligned} g &= v(a-1)Y(Y-1) \cdot e \cdot F\,(Y-1) \\ &= 2c(a-1)Y(X-1)(a-2) + F(2-A); \end{aligned}$$

Case 27. $c = e = a$, $B = F = a$, $a = D = v$. By reciprocation of 26.

$$\chi = Y(a-1)Y(Y-1) - y(Y-X)Y \cdot xY = 2v\,(a-1)\,(Y-X)yF.$$

where each triangle counted is twice, so that the number is really one half of this.

Case 28. $a = c = B = a$, $D = e = F = v$.

$$\chi = Yc(Y-1)(Y-2)e \cdot Z, \quad \chi' = Y(Y-1)y \cdot Yc(Y-2)e = v;$$

The reductions are those of the first and second modes; as explained above, with the variation that the curves c and a are here identical, $a = c$, and that the curve D is identical with the curves $B = F$.

First-mode reduction is

$$v(C + 2\delta + 3\kappa)B(B-1)$$

(where δ, κ refer to the curve $c = a$), which is

$$= v(c-1) \cdot Y(Y-1);$$

that is, the reduction is $= v(g-1) \cdot Y(Y-1)$.

Second-mode reduction is

$$v(c-2c)c(X-1)$$

that is, the reduction is $= v(g-1) \cdot Y(Y-1)$.

Hence the two together are $= v(g-1)\{2X(X-1) - c\} \cdot e$, and subtracting from $\chi + \chi'$ the value of g is

$$g = v(g-1) \cdot \{2Y(Y-1) - e\} \cdot e.$$

but on account of the symmetry we must divide by $\frac{1}{2}g$.

Case 29. $c = a = D = v$, $a = c = v$. By reciprocation of 28.

$$\chi = 2X(X-1) \cdot v(v-1) \cdot a.$$

Case 30. $a = c = B = a$, $D = e = v$, $u = F = v$. By reciprocation of 24.

$$\chi = Z(a-1)(Y-1)(Y-2) \cdot eYF \cdot eYF = v \cdot a\,(Y-1)\,Y\,(Y-2)Y\,F.$$

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Second process. Taking the form

$$c = B = u = a, \quad D = F = u;$$

here

$$g = \chi + \chi' - \text{Red.},$$

$$\chi = Yc(X-1)Da \cdot Ya, \quad \chi' = Yc(X-1) \cdot c \cdot Ye,$$

and

$$\chi + \chi' = 2Y^2 v(a-1)D \cdot e\,(X-1).$$

There is a first-mode reduction,

$$aY\{2v + 3v \cdot (B-1) \cdot v + aZ(X-Y)$$

$$+(X-3)v(v-c-1)Y \cdot a \cdot \beta\}$$

$$+(X-3)v\,(X-2Y\,)$$

$$+(X-3)\,(Y\,Y\,Y);$$

which is

$$= aY\{X(2v-c-1)\} + v \cdot a \cdot \alpha;$$

and a second-mode reduction is

$$= aYX\,(\,X\,(\,Y\,(\,c-1)\,Y\,(c-1) + 12c) \cdot c$$

Hence the two together are

$$= aY\{X(2v-c-1)\} + v(c-1) \cdot (Y - c\,(\,Y - c - 1)\,),$$

whence the result is

$$= 2(P-Y)\,(\,c\,(\,c-1)\,Y\,(\,Y-c\,),$$

which agrees with that obtained above.

On account of the symmetry we must divide by $\frac{1}{2}g$.

Case 31. $a = D = F = a$, $u = v = v$. By reciprocation of 30.

$$\chi = 2X(X-1)(X-2) \cdot v(v-1)y.$$

On account of the symmetry we must divide by $\frac{1}{2}g$.

Case 32. $a = c = v$, $D = e = u$. By reciprocation of 24.

$$\chi = 2X\,(\,X-1)(X-2)\,v\,(\,v-1)\,c.$$

Case 33. $u = c = v$, $a = D = u$.

$$\chi = Y(Y-1)c\,(\,Y-2)\,ay, \quad \chi' = F\,(\,Y-1)e \cdot D\,(\,Y-c\,)\,ya \cdot ZY + Y\,F.$$

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Case 33. $u = c = v$, $a = D = u$.

$$\chi = Y(Y-1)c\,(\,Y-1)\,(\,a-1)\,a\,Fa, \quad \chi' = F\,(\,Y-1)\,e \cdot D\,(\,Y-2)\,a \cdot (\,X-2);$$

$$g = 2X\,(\,X-1)\,(\,X-2)Y\,c\,DF.$$

Case 34. $u = e = v$, $F = u$.

$$\chi = (X-2)c\,(X-1)De \cdot ya, \quad \chi' = Y e\, Y\, F\, e\, Dc \cdot Xa\,(X-1);$$

$$g = Y\,(\,c-1)Y\,(\,X-1)\,(\,Y-2)\,Da.$$

Case 35. $u = a = v$, $B = F = u$. By reciprocation of 34.

$$\chi = 2Y\,(\,Y-1)\,(\,Y-2)\,F\,(\,Y-2)\cdot cBe.$$

Case 36. $a = c = v$, $B = F = u$.

$$\chi = B\,(\,X-1)\,(\,X-2)\,(\,X-1)\,a\,Fa, \quad \chi' = F\,(\,X-1)Y\,X\,(\,X-1)\,(\,X-2)\,a;$$

$$g = Y\,(\,c-1)\,(\,X-2)\,Y\,F.$$

Case 37. $a = c = v$, $D = v$, $F = v$. By reciprocation of 36.

$$\chi = Y\,(\,X-1)\,(\,X-2)\,Y\,(\,X-1)\,(\,X-2)\,Y\,DBe + 2YX\,F.$$

Case 38. $B = D = v$, $a = v$, $F = v$.

$$\chi = Y\,(\,X-1)Y\,Fa, \quad \chi' = Y\,(\,X-1)\,e\,F\,(\,X-1)\,a;$$

$$g = 2Y\,(\,X-1)Y\,F.$$

Functional process; the curve c is assumed to be the aggregate of two curves, say $a = c + c'$. Drawing the construction…

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which is belongs; thus $aXcDcF$ is $B = c = v = v$, which is Case 10, and so in the other instances. Observing that cases 10 and 14 occur each twice, we have

$$\phi(c+c') - \phi c - \phi c' = DF \text{ multiplied by}$$

$$k(c-1)(X - X + v)\alpha \ldots\ldots (19) \times 2$$
$$+2c(c-1)(v - X - 1) \ldots\ldots (8)$$
$$+4c(c-1)(X - X - 1) \ldots\ldots (14) \times 2$$
$$+2c(c-1)F \ldots\ldots (12)$$
$$+2c(c-2)F \ldots\ldots (11)$$

where the ζ, β refer to the like functions with the two sets of letters interchanged. Developing and reducing, this is

$$\phi(c+c') - \phi c - \phi c' = DF \text{ multiplied by}$$
$$3XX$$
$$+X(2avX + 5cv' + 8aX' - 15aa - 15a'a + 8)$$
$$+5v\,(\,vX - 3v' + a - 1)Y \cdot cz' - 3a' - 3a'X - 3a'X + 3aX' + 6aX'$$
$$-13(cc' + aa') + 9aav,$$

and thence

$$\phi = DF \text{ multiplied by}$$
$$X^4$$
$$+X(2cv - 10v' + 13c) - LX$$
$$-4cv + 25v' - M,$$

where the constants L, l, λ have to be determined. Does η being a cubic curve the number of triangles vanishes; that is, we have $\phi = 0$ for $a = 1, X = 0, \beta = 0$, and for $a = Z = 0, \beta = 0$ and $a = Z, \beta = 0$; we thus have

$$\begin{cases} a = 8, & X = 0,\ \beta = 10, \\ 0 = 88 - 4L - M - 15a, & \\ 0 = 88 - 4L - M - 15a, & \end{cases}$$

giving $L = 1, l = 16, \lambda = 3$. Whence finally,

$$\phi = (X^4 + X(2cv - 10v' + 13c - 1) \cdot c - Y + 2XZ - 16c - 3a^3]DF.$$

Second process; by reciprocation, we have

$$g - \chi - \chi' + F(c - c' - c') = 0,$$

$$c - c' = F(v - y - y') = 0.$$

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and thence

$$g-\chi-\chi' = DF(c+v-y'-y'),$$

Moreover

$$\chi = (X-2)e\,(X-1)F\,(c-1),$$
$$\chi = F\,(\,c-1)e\,(X-1)\,(X-1)\,a, \quad \chi = v\,a,$$
$$\chi+\chi' = DF\cdot(X-1)\,a\,(\,X-1)\,a,$$

and

$$v-y-y' = (X-X)\,a+2(X-X)\,a-3),$$

as is easily obtained, but see also post, No. 21; hence

$$g = DF \text{ multiplied by}$$

$$\{(X-1)+(X-1)\,(X-1)\}^2$$
$$+(X-1)\,(X-1)\,a^2;$$

but I reject the term $DF\,(\,X-1)\,a$ as not first giving a heterotypic solution; I do not go into the explanation of this. And then substituting for β its value, we have

$$g = DF \text{ multiplied by}$$

$$(X-1)\cdot\{a\,(\,v-1)\,(c-1)\}$$
$$+X^4-X\,a+5a-3a^2,$$

where the second factor is

$$= X^4+X(2cv-10v'+13c-1)\cdot c-4cv+2c-16c-3a^3.$$

which is the foregoing result.

Case 40. $B=D=F=a=v$. By reciprocation of 39.

$$\chi = v\cdot a+v(X^4+y(2Y-10Y+13Y-1)\cdot Y-4Y+25Y-16Y-3a^3]F.$$

Case 41. $c=e=a$, $D=F=a$.

$$\chi = Bc(X-1)e\cdot D\,(X-1)F\,a, \quad \chi' = Fc(X-1)(c-1)Ba,$$

$$g = (X-1)cD\,(X-1)Ba+2(X-1)\,(B-1)\,(X-1)\cdot ca\,(\,B-c-1)cBa,$$

= &c.

Case 42. $a=c=e=v$, $B=D=F=a=v$.

Functional Process; the curve a is supposed to be the aggregate of two curves, say $a=c+c'$, $B=D=F=v=v'$. Forming the construction...

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The enumeration is

Case		
$a\ Bc\ X\ d\ X$	$aBc\ X\ d\ X^3$,	(42)
$a'.a'Xa\,X$	ditto,	(11)
$a.a'Xa\,X$		(11)
$a.a'Xa\,X$		(12)
$a'Xc\,Xa\,X$		(17)
$a'Xc\,Xa\,X$		(10)
$a'Xc\,Xa\,X$		(10)
$a'Xc\,Xa\,X$		(10)

whence

$$\phi(c+c') - \phi c - \phi c' = \nu B \text{ multiplied by}$$

$$4(X-1)(Xa-X-a)\ldots\ldots(11)\times 2$$

$$+2\nu(a-1)X^2\ldots\ldots(12)\times 1$$

$$+4\nu(a-1)X(X-1)\ldots\ldots(17)\times 2$$

$$+5\nu(a-1)X^2\ldots\ldots(10)\times 2$$

$$+5\nu Xa(a-1)X + X(c-1) + 4\nu(Xa'+X'a) + 2\nu(X'a)X + \ldots\ldots(10)$$

where the ζ, β' refer to the like functions with the two sets of letters interchanged. Developing and reducing, we have

$$\phi(c+c') - \phi c - \phi c' = aB \text{ multiplied by}$$

$$X^4 \quad (8aa' - 5a^2a^2 - 6aa)$$

$$+XX'\{4aX(c-1) + 8a^2X + aa(a-2c+9) + (c-1)X\}$$

$$+X^3\{(2v' - 6a^2 + 15a^2)$$

$$+X^3\{12aa - 6aa' - 12a^2 + 18a\}$$

$$+X^2\{(-6a^2 - 12a + 15a^2)$$

$$+Saa^2,$$

and consequently

$$\phi = v \text{ multiplied by}$$

$$XY\,(4ac - 5a^2)$$

$$+X^4\{l\sigma a^2 + \nu c - 6a^2\}$$

$$+X^3\{(\beta - 5\nu) + \beta - Ja\},$$

where the constants L, l, λ have to be determined. The number of triangles vanishes when the curve α is a line or a conic, that is, $\phi = 0$ for $a = 1$, $X = 0$, $\beta = 0$, and for $a = 2, \beta = 2$; we thus have

$$0 = 83 - L - M,$$

$$0 = 80 + 2L + 12 - M,$$

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Moreover, the data being equicipanyl, the result must be so likewise; we must therefore have $L = l$. We thus obtain $L = l = u$, $M = 0$, $\lambda =$ but $aBL = u\bar{l}$, $u = Bp = u^2$.

Second process. by correspondence: form $a = c = e =$, $D = F = a$. We have also from the special consideration that the points B, F are given as the intersections of the curve a, by the joint of the joint c, which first pulse does not pass through a, we have

$$(I - \phi - \phi') + \nu(c - X - Y) = v\nu,$$

and by the consideration that c, D are given as intersections, c a double intersection, of the curve with the first point of the point c, which first pulse does not pass through a,

$$\nu - X - X + 2(v - c - y - y') = v\nu,$$

where

$$g - \chi - \chi' = \nu v - \nu B\nu,$$

and

$$v - c - y' - y' = \nu B\nu,$$

so that this is

$$g - \chi - \chi' = 4\nu B\nu,$$

$$= 4\nu(aB - X(X-1)) \cdot v.$$

Also

$$\chi = B\,(X-1)\,(X-1)\,(X-2)e\ \cdot Fa,$$

$$\chi' = F\,e\,D\,(X-1)\,X(X-1) \cdot a\,e\ \cdot \chi,$$

so that

$$g = B\nu \text{ multiplied by}$$

$$2(X-1) \cdot (X-1) \cdot (X-2) + 4\nu + 5z + R\,B;$$

viz. this is

$$B\nu(X^4 - X(X - X - X)\,(X-1)\,(X-1) + 8c) \cdot Xa + xa + e),$$

which is the number of positions of the angle (a), but several of these belong to special times of the triangle $aBcDeF$, giving heterotypic solutions, which are to be rejected; the required number is

$$[XY\,(X-1)\,(X-1)\,(X-1)\,(X-2)\,\nu^3 -\ a\,\nu^3\,] - \text{Red.}$$

Third process: form $c = a = B$, $a = Y\,e$.

$$g = g + \chi' - \text{Red.},$$

$$\chi = X - (X-1)(X-1)\,e\ \cdot v\nu,$$

$$\chi = X - (X-1)\,B\,Y\,(X-1) - Y\,cY,$$

$$\chi + \chi' = 2X^4\,(X-1)\,(X-1)\,Y\,(X-1) - Y\,cY.$$

The first-mode reduction is here

$$aB\,[X - (X-1)\,X - (X-1)4]\,.$$

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where the last term aBe arises from the tangents cBa and eFa, each coinciding with a cuspidal tangent, as shown in the figure.

Fig. 4.
[Figure: Cuspidal tangent configuration — diagram omitted]

The second-mode reduction is

$$aBX\,(X-1)\,(\,c-3a),$$

so that the two reductions together are

$$= aB\{(X-1)X + (X-1)(B-(X-1)k + e + e\,(\,c-2)(c-3a),$$

viz. this is

$$= aB\{X-1)X + 4(B-1) + kX + e + a\{X-1)X + 8c + 5z + 4a - 3a - 3a - 2);$$

or substituting for Ω its value, and $-\Sigma X - 3X + \delta\partial(\,e-\,) - 2\partial(c\,-\,3a)$, and reducing, it is

$$aB\{X(X-1)(2a-c-4) - 4ax + 4a + 12c - 3a^2\}.$$

On account of the symmetry we must divide by $\frac{1}{2}g$.

Case 43. $a = c = e = D = F = a$.

Suppose for a moment that the angle a is a free point; the locus of a is a curve the order of which is obtained from Case 28, by writing $c = e = v = v$, $B = D = F = u$; the locus in question meets a curve a in $u(2X^2+2X-1)\,e\cdot a$ points, which is the intersection of the curve. Suppose that a is a given point, and several of these belong to special times of the triangle $aBcDeF$, giving heterotypic solutions, which are to be rejected; the required number is

$$[2X(X-1)(X-1)(X-2)\,\nu^3 -\ a\,\nu^3\,] - \text{Red}.$$

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The reduction is thus first and secondly to triangles wherein the angle a coincides with an angle c or a, and thirdly to triangles wherein the angles a, c or a all coincide.

1°. Take for this side aB a double tangent of the curve $B = D = F$; this touches the curve $a = c = a$ in a points, and subducing any one of them for a and any other for c, we have from the last-mentioned point $V - 2$ tangents to the curve $B = D = F$, and in respect of each of these a position of a coincident with c. The reduction on this account is $2m(\nu - 1)(V - 2)$; but since we may in the figure interchange a and c, B and F, we have the same number belonging to the coincidences of the angles a, c, or together the reduction is $= 4m(\nu - 1)(V - 2)$.

[Fig. 4 omitted]

But instead of a double tangent we may have cda a stationary tangent; we have thus reductions $2\nu(c - 1)(V - 2)$ and $2\nu(c - 1)(V - 2)$, together $4\nu(c - 1)(V - 2)$; and for the double and stationary tangents together we have

$$(4\nu + 6\iota)(c - 1)(V - 2),$$

$$= 2[T(V - 2) - x(c - 1)(V - 2)],$$

that is

$$= 2x(c - 1)V(V - 1) - 2x(c - 1)(c - 2)(V - 2).$$

2°. The side cBa may be taken to be a tangent to the curve $B = D = F$ at any one of its intersections with the curve $a = c = a$. Taking thus the point a at the intersection in question, and the point c at any other of the intersections of the tangent with the curve $a = c = a$, and from c drawing any other tangent to the curve $B = D = F$; there is in respect of each of these tangents a position of a at c;

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the reduction on this account is $= my(c-1)(V-1)$. But interchanging in the figure the letters c, a, B, F, there is an equal reduction belonging to the coincidence of a, c; and the whole reduction in this manner is $= 2my(c-1)(V-1)$.

[Fig. 5 omitted]

3°. If the side cBa intersects the curve $a = c = a$ in two coincident points, then taking these in either order for the points a, c, and from the two points respectively drawing two other tangents to the curve $B = D = F$, we have a triangle wherein the angles a, c, a all coincide. The side cBa may be a proper tangent to the curve $a = c = a$, or it may pass through a node or a cusp of this curve, viz. it is either a common tangent of the curves $B = D = F$ and $a = c = a$ (as in the figure, except that for greater distinctness the points c and a are there drawn nearly instead of actually coincident), or it may be a tangent to the curve $B = D = F$ from a node or a cusp of the curve $a = c = a$; we have thus the numbers

$$\begin{array}{rl} \text{Common tangent} & XY(V-1)(V-2), \\ \text{Tangent from node} & 2\delta V(V-1)(V-2), \\ \text{Tangent from cusp} & 2\kappa V(V-1)(V-2), \end{array}$$

but (as we are counting intersections with the curve $a = c = a$) the second of these, as being at a node of this curve, is to be taken 2 times; and the third, as being at a cusp, 3 times; and the three together are thus

$$\begin{aligned} &(X + 4\delta + 6\kappa)(V-1)(V-2) \\ &= 2[\Sigma(c-1) - X](V-1)(V-2). \end{aligned}$$

The reductions 1°, 2°, 3° altogether are

$$\begin{aligned} &2x(c-1)V(V-1)(V-2) \\ &+ 2x(c-1)V(V-2) \\ &+ 2x(c-1)V(V-2) \\ &- XY(V-1)(V-2) \\ &+ 2\delta V(V-1)(V-2) \\ &+ 2\kappa V(V-1)(V-2) \\ &= XY(V-1)(V-2), \end{aligned}$$

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and subtracting from the before-mentioned number

$$2x(c-1)V(V-1)(V-2)$$

$$+x(c-1),$$

the required number of positions of the angle a is

$$= 2x(c-1)(V-2) - XY(V-1)(V-2)$$

$$+x(c-1)(c-2)(V-2) + XY(V-1)(V-2).$$

The number of triangles is on account of the symmetry equal to one-sixth of this number.

Case 44. $c = D = F = c$, $a = c = B = a$.

$$\begin{aligned} \chi &= (V-2)(y-3)X(a-1)(X-2)x, \\ \chi' &= X(a-2)X(X-3)y(V-2)(y-3)x, \\ \rho &= 2(a-2)X(X-3)Y(V-1)y, \end{aligned}$$

there is a division by 2 on account of the symmetry.

Case 45. $c = D = F = c$, $a = c = B = a$.

$$\begin{aligned} \chi &= (X-2)y(V-1)(c-1)(V-2)a, \\ \chi' &= Vy - X(a-1)(c-1)(V-2)a, \\ \rho &= 2(X-1)(c-1)X^2 - 6c(c-1)Vy + XY(a-2) + X^2(V-2) - cp(X+Y) + 2cp + 2XY, \end{aligned}$$

there is a division by 2 on account of the symmetry.

Case 46. $a = c = a$, $B = D = F = a$.

$$N_2 = y(y-1)(c^2+c)(X^2 - 10X^2 + 11X - 4X^2 + 20X^2 - 14X - 3p);$$

there is a division by 2 on account of the symmetry.

Case 47. $D = F = y$, $a = c = a$, $B = a$.

$$N_2 = V(V-1)[X^2X(2c - 14cv + 12c - 1) - 4c + 21c - 16c - 3p];$$

there is a division by 2 on account of the symmetry.

Case 48. $a = c$, $D = F = a$, $a = B = a$.

The functional process, writing $a = c = D = F = a = B = a$ would be precisely the same as Case 42, with only the factor yY written instead of cF; and we have thus

$$N_2 = [X^2(8c^2 - 6a + 4) + Y(-2c^2 - 16c - 4) + 4c - 4c]yT,$$

which on account of the symmetry must be divided by 2.

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Case 49. $a = B = y$, $c = a = D = F = a$.

$$\begin{aligned}
\chi &= (V-2)X(c-1)(c-1)Y, \\
\chi' &= X(a-2)(X-2)Y(c-1)Y, \\
g &= (a-2)X(X-2)Y - 2(c-2)Y(c-1)c - 2(c-1) \\
&\quad - 2(a-2)X - 2c(c-1)X^2X - 2c(c-1)Y - 2y(a-1) + 4cy + XY^2.
\end{aligned}$$

Case 50. $c = a = c = D = F = a$.

Functional process; by taking the curve $c = a = c = D = F = a$ on the aggregate of two curves, say $a = c = c$. The sums are

	Case
cX^2X^2X	30
X^2X^2c	—
$X^2X^2\cdot$	40
$X^2X^2\cdot$	32
$X^2X^2\cdot$	42
$X^2X^2\cdot$	38
$X^2X^2\cdot$	36
$X^2X^2\cdot$	38
$X^2X^2\cdot$	40
$X^2X^2\cdot$	36
$X^2X^2\cdot$	28
$X^2X^2\cdot$	28
$X^2X^2\cdot$	30
$X^2X^2\cdot$	36
$X^2X^2\cdot$	38
$X^2X^2\cdot$	36
$X^2X^2\cdot$	32
$X^2X^2\cdot$	41

and we then have

$$\phi(x+c) - \phi x - \phi x' = \text{multiplied into}$$

$$3x^2 + 5cx^2$$

$$+5x[\,2x' + (2X^2 - 10X + 11) - X^2 + 20X^2 - 14X - 5\xi\,] + \dots \quad (40)\times 3$$

$$+5x'[X^2(2c^2 - 6a + 4) + 5y - 4c - 4c - 4\xi] + \dots \quad (42)\times 3$$

$$+4(c-1)(cX - c - X)(X^2 - X^2)$$

$$+4cX(X-c)(X^2 - X^2) - (cX^2 + 2XX + 2cc^2)$$

$$+c\cdot\cdot c^2 - 6cX(X^2 - X^2) \quad (38)$$

$$+(c^2 - c)X^2(2 - 4cX^2 - 4)$$

$$+2c(x-2)X^2(x - X^2) \quad (35)$$

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where as before the $(,)$'s refer to the like functions with the two sets of letters interchanged. Developing and collecting, this is found to be

$$\phi(x+c) - \phi x - \phi x' = \text{multiplied into}$$

$$3cx^2 + 5cx^2$$

$$+c^2XX^2 - cXX^2 + 2cX - 11)$$

$$-26XX^2 - 14X^2 \cdot\cdot$$

$$+2cX^2.$$

$$+cx^2 \cdot AX^2 + 12XX - 11XX^2 + 4X^2$$

$$-26X^2 - 36XX^2 + 28X^2 \cdot\cdot$$

$$+26X + 50X^2$$

$$-2X.$$

$$+cx^2 \cdot 5X^2 + 30X^2 - 6XX^2$$

$$-14X^2 - 26X^2 \cdot\cdot$$

$$-11cX - 4X$$

$$+cx^2 \cdot 30X^2 \cdot AX^2 - 26XX^2 - 10X^2 \cdot\cdot$$

$$+140XX^2 + 70X^2$$

$$-11cX - 4X$$

$$+cx^2 \cdot 10X^2 - 30X^2 \cdot 30X^2 - 50XX^2 \cdot\cdot$$

$$+70XX^2 - 18cX^2$$

$$+11cX - 8X$$

$$+132X^2$$

$$-4cX^2 + X^2 \cdot\cdot$$

whence

$$\phi x = \text{multiplied into}$$

$$c^2(\quad +1)$$

$$+cx^2[\,2X^2 - 14X^2 + 28X - 11]$$

$$+cx^2(-10X^2 + 70X^2 - 114X + 6)$$

$$+cX^2(\quad 12X^2 - 79X^2 \cdot\cdot$$

$$+\,c^2[\,4c - 6c - 4X,$$

where the constants f, g, h, have to be determined. We should have $\phi = 0$ for a cubic curve; viz. $a = 3$, $X = 6$, $\xi = 0$, $\xi = 18$; $c = 2$, $X = 4$, $p = 0$. Writing first $x = 3$, the equation is

$$8X^2 - 36X - 73 - f(18 + 4X) + 5f - X^2 + fh \cdot 0.$$

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giving in the three cases respectively

$$\begin{aligned} 5d + 6L + 18h &= 1136, \\ 3l + 4L + 12h &= 736, \\ 3d + 5L + 9h &= 588; \end{aligned}$$

and we have then $l = -8$, $L = 64$, $h = 42$, so that the required number is

$$\begin{gathered} = \phi x^2 \quad + 1 \\ +cx^2[\,2X^2 - 14X^2 - 28X - 11] \\ +cx^2(-10X^2 + 70X^2 - 114X + 6) \\ +cX^2[\quad 12X^2 - 79X^2 + 64X \\ +\, c^2[\,4c - 6X + 4X\,]. \end{gathered}$$

As a verification, observe that for a cubic, $c = X = 2$, $\xi = 0$, this is $= 0$.

Second process, by correspondence; form $c = c = D = F = c$.

We have

$$\begin{aligned} g &= y + \chi - 8cd, \\ \chi &= X(c-2)x(X-1)(c-1)(X-3)y, \\ \chi' &= X(c-2)X(X-2)Y(c-1)(c-2)Y, \\ 8cd &= XY^2X(c-1)X. \end{aligned}$$

[Fig. 8 omitted]

There is a first-mode reduction, which is

$$= a[2X(X-4)(X-5) + 6a(X-2)(X-4) - a(X-3) + 2a(X-3)].$$

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where the term a, $2x(X-3)$ arises, as shown in the figure, and a second-mode reduction, which is

$$= a[2x(c-4)(c-5) + 6a(c-3) + 7a].$$

and the two together are $= a$ into

$$(X+a)(X-1)(c-x) + a(X-1)(c-3)(X-4)$$

$$+(X-8a)(X-4)X - 4(X-2) \quad (-3\xi)$$

$$+(X-3) - 2X - 3X \quad (-2\xi)$$

$$+(a, X-2)(c-1)(X-8) \quad +3$$

$$+(a, X-2)(c-5)(X-4) - X + 8c \quad (-2\xi)$$

$$+(a, X-3)(c-4) \quad (\quad +Xa + 4c \quad -4\xi);$$

that is, $= a$ into

$$-c\xi^2$$

$$+cX^2 \cdot 2X^2 - 10X + 11$$

$$+cX^2 \cdot -10X^2 + 26X + 8$$

$$+XX^2 - 31cx + 12X$$

$$+\xi[-2c^2 + 4X - 6X];$$

and subtracting from this the foregoing value of $g + g'$, which is $= a$ into

$$e^2X^2 - 12X^2 + 18X)$$

$$+cx^2(-10X^2 + 30X^2 - 96X)$$

$$+12X^2 - 75X^2 + 70X^2$$

$$+\xi[\,12X^2 - 72X + 102X],$$

the result is as before.

There is a division by 2 on account of the symmetry.

Case 51. $a = c = D = F = a$. By reciprocation of 50,

$$N_2 = Y' \quad (\quad +1)$$

$$+X[Y \quad 2x^2 - 14x + 28x - 11)$$

$$+X(-10x^2 + 70x^2 - 114x - 6)$$

$$+X \quad 12x^2 - 79x + 64$$

$$+\,c[\,Yc^2 - 6c + 4c],$$

There is a division by 2 on account of the symmetry.

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Case 52. $a = c = a = B = D = F = a$.

Functional process, by taking the curve to be on the aggregate of two curves, say $a = c = a$. The enumeration of the cases is conveniently made in a somewhat different manner from that heretofore employed, viz. we may write

a or a' ζ	a'' or σ ζ	Case	times
all	none	(32) 1	
a	resides	(39) 3	
B	.	(33) 3	
$\cdot$	.	(46) 2	
D, B	.	(47) 6	
a, D	.	(40) 3	
a, B	.	(40) 3	
a, σ, $\cdot$	B, D, F	(43) 1	
a, B, D	c, σ, F	(44) 6	
a, σ, F	c, σ, F	(45) 3	
		32;	

and the functional equation then is

$$\phi(x+c) - \phi x - \phi x' \quad = \quad 3x^2$$

$$+x^2[\quad 2X^2 - 14X^2 + 28X - 11] + \ldots \quad (50) \times 3$$

$$+cx^2[-10X^2 + 70X^2 - 114X - 6]$$

$$+12X^2 - 75X^2 + 64X \quad + \ldots \quad (51) \times 3$$

$$+\xi[-2c^2 + 4X - 6X]$$

$$+3X(\quad -c^2)$$

$$+ \quad \cdot\, c[X[\quad 2c^2 - 14c + 28c - 11]$$

$$+ \quad X[-10c^2 + 70c^2 - 114c - 6$$

$$+ \quad X[\quad 12c^2 - 79c + 64$$

$$+3X(c - c').$$

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$$
\begin{gathered}
+3c[X^2 - X^2 \quad (\quad +1) \quad + \ldots \quad (47) \times 3 \\
+X(2x - 14x + 12x - 1) \\
-4x^2 + 20X - 4cX - 5\xi] \\
+3c \cdot X^2 \quad 2x^2 - 6x + 4 \quad + \ldots \quad (48) \times 3 \\
+X(-2c^2 - 16c + 4 \quad + \ldots \\
+4c - 4c] \\
+12(c-1)(cX - X)(2c)X^2(X - cX^2(x+c)2c + 2cXX) + \ldots \quad (49) \times 6 \\
+2c(c-1)cX(X(2c)X^2 \cdot AX - 2(c-X)cX - 2(c - X^2)X(x-2) \quad + \ldots \quad (44) \times 3 \\
+4(c-1)X(X-X) \cdot 2c - 4cX(X^2 - cX - X^2(X + 2cX + 2cX^2) + \ldots \quad (43) \\
+12(X-1)(X-1)c \cdot c \cdot c - X^2(X + cX - X^2(X + 2X)X + 2XXc] \quad + \ldots \quad (45) \times 6
\end{gathered}
$$

where as before the $(,)$'s refer to the like functions with the two sets of letters interchanged. Developing and collecting, this is found to be

$$
\begin{gathered}
= 6XYYX + 5X^2Y + 5X^2YX \\
+Y^2[\quad 6cv^2 + 6cv^2 + 2v^2 \cdot\cdot \\
-30cv - 19cv] \\
+16Y \\
+(XY + XY^2) \cdot 6c + 16cv + 14cv^2 + 6cv^2 \cdot\cdot \\
-24c - 100cv - 24cv \\
+4 \cdot 16 + 116cv^2 \\
-138 \\
+X^2 \quad 2c^2 + 6cv + 6cv^2 \cdot\cdot \\
-70c - 36cv \\
+12X^2. \\
+ \cdot \cdot\cdot
\end{gathered}
$$

= &c. &c.

I abstain from writing down the remaining terms, as they can at once be obtained backwards from the value of ϕx; they were in fact found directly, by the integration of the functional equation thus given

$$
\begin{gathered}
4x = \quad X^2 \quad (\quad +1) \\
+X^2[\quad 2c^2 - 76c + 32c - 46] \\
+X[Y \quad - 10c^2 + 162c^2 - 632c + 271] \\
+X(\quad 32c^2 - 458c + 770c + 4) \\
+ \quad x^2 - 80c^2 + 312c^2 + 4c \\
+\xi[\,X^2(\quad +5) \\
+X(\quad -12c + 132) \\
-6c^2 + 132c + \lambda)
\end{gathered}
$$

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where the constants L, λ, have to be determined. We should have $\phi x = 0$ for a cubic curve; viz. $a = 3$, $X = 6$, $\xi = 0$, $\xi = 18$; $c = 2$, $X = 4$, $p = 0$. Writing first $x = 3$, the equation is

$$\phi x = \quad X^2 \quad - 1135X^2 + 1444X + 102 + \xi[-3X^2 + 111X + 234] + L(2 + X) + 2,$$

and then for the cubic, $X = 6$, $\xi = 0$.

Writing $\sigma = 3$, we have

$$4\sigma = X^2X^2 \cdot -432 \cdots - 384X + 825 + \xi[-6X^2 + 102X + 232] + l(2 + X) + 2;$$

and then for the three cases of the cubic $X = 6$, $\xi = 12$; and $X = 3$, $\xi = 19$, We have thus the four equations

$$\begin{aligned} 2912 + 4L - 0 &= 0, \\ 9252 + 8L + 16L &= 0, \\ 6912 + 17L + 0 &= 0, \\ 6809 + 4L + 0 &= 0; \end{aligned}$$

all satisfied by $l = -172$, $L = 600$. Hence finally

$$\begin{aligned} \phi x = \quad & X^2 \quad (\quad +1) \\ & +X^2[\quad 2c^2 - 76c + 32c - 46] \\ & +X[\quad -10c^2 + 162c^2 - 632c + 271] \\ & +X(\quad 32c^2 - 458c + 770c + 4) \\ & + \quad c^2 - 80c^2 + 312c + 600 \\ & +\xi[X^2(\quad +5) \\ & +X(\quad -12c + 132) \\ & -6c^2 + 132c - 600; \end{aligned}$$

but on account of the symmetry the number of triangles is = one-sixth of this expression.

Article No. 53 to 56. *The Case* 52, *as belonging to a different series of Problems.*

53. In the foregoing Case 52, where all the curves are one and the same curve, we may consider this curve a being the curve $abcDeFg$, and we seek for the number of the conics points (a, c). But we may consider this as belonging to a series of questions, viz. we may seek for the number of the united points (a, c), (a, c), (a, c), (a, c), (a, c), (a, c), (a, c), (a, c), (a, c). The last four of these giving by reciprocity the numbers of the united points (a, c), (a, c), (B, F), (a, c), and finally the number of the actual triangles to consider the actual triangles to consider the system of question and the some so so the so should consist of points; for any special question belonging to one we may notice the singular points and tangents of the curve, and that the solutions thus explain themselves.

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23. Thus the first case is that of the united points (a, R), viz. we have here a point a on the curve, and from it we draw to the curve a tangent aR touching it at R; the points a and R are to coincide together. Observe that from a point in general not on the curve we have to the curve m tangents (or as before, where, as disregard altogether the tangent at the point, counting as 2 of the X tangents from a point not on the curve), and noted exclusively to the $X-2$ tangents from the point. Now if the point a is an inflection, or if it is a cusp, there are only $X-4$ tangents, or, to speak more accurately, one of the $X-2$ tangents has come to coincide with the tangent at the point; such tangent is a tangent of three-point intersection, viz. we have the point a and the point R (counting as a point of contact, twice) all three coinciding; that is, we have a position of the united point (a, R); and the number of these united points is $= x$.

24. It is important to notice that neither a node of contact of a double tangent, nor a double point, is a united point. In the case of the point of contact of a double tangent, one of the tangents from the point coincides with the double tangent; but the tangent is not the tangent at the point of contact, and so as the point of contact a, R are not coincident. In the case of a node, drawing any other tangent at the node, this tangent is not a tangent at the node, and the position a, R are not the united point a, R.

I recall that I use Σ, viz. $2x - X + a - 2x - X + a = a$.

25. The several cases are

$$\begin{aligned}
(a, R):&\quad b = a' - 2X = 2X,\\
(a, c):&\quad x = c - x + 2 - X = a(X-3) - a(X-2X) = X - X = a,\\
(B, D):&\quad y = y - X = a - X = a(X-2X) - X = a - X = 2X,\\
(B, X):&\quad y = y - X = a,\\
(a, D):&\quad d - 4 - 8\delta - 6X = 4cv - X(2X-4)(X-2X) = X(X-2X) - X = 2X + 2X,\\
(a, c'):&\quad d - X + a - a = X = a(X-2X) - X = a,\\
(a, c''):&\quad g = X - y - cv = c \quad \text{by reciprocity},\\
(a, B):&\quad \cdot \quad -X + a - cv - x + a - a = X = a,\\
(a, B):&\quad g_2 = X - X' \quad \text{by reciprocity.}
\end{aligned}$$

and so on.

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26. The mode of obtaining these equations appears next, Nos. 5 and 6, but for greater clearness I will explain it in regard to a pair of the equations, say those for (a, c), (a, B). Regarding a as given, we draw from a the tangents aBc, touching at B and meeting at c; that is, the number of tangents is $X-2$, and the number of the points c is $= a(X-2)(c-32)$; from each of the position of c we draw to the curve the $X-2$ tangents (of course not including the tangent cBa); these tangents intersect the curve in points a in a cusp, there are only $X-2$ tangents, of the points B; but the number of tangents aBc is $= a(X-2)(c-32) = a(X-2)$. Now this system of the $(X-2)(X-2)$ tangents to the curve is the same as the general theory $(X-2)$; but in any other view we may consider it as the union of the systems of the $(X-2)$ tangents from the c points a on the curve as a tangent from a unison point. That is, a are the union of the $X-2$ tangents to the curve, that is, the cusps of which are in the cuspidate a, and the cusps are united, so the cuspidate at any of the points of the cuspidate. And we have the corresponding equations

$$a(X-2) + 2(X-2) - X(X-2) - 2\delta(X-2) - 4\sigma(X-2) = a;$$

when the numbers (δ, B), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), (c, c), refer to the correspondences (D, c), (D, B), and (D, c) (or what is the same thing (a, R) respectively).

28. Correspondence (a, B).

We have

$$B = X - 3, \quad B = a - 2,$$

and thence

$$b = a + X - 3 + 2X,$$

$$= a - 6X + 5\delta,$$

which is the relation: the value obtained above was $b = c + a$, and we in fact have identically

$$\cdot\cdot\, a = b - aX + 2\xi.$$

It was in this manner that I originally applied the principle of correspondence to investigating the number of inflexions of a curve, regarding, however, the term a as a special solution; it is better to put the cusp and inflection on the same footing as shown.

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29. Correspondence (a, c).

Since $b - \beta - B - 2\Delta$, we have here

$$c - y - q = (X - 2X)q,$$

and

$$y - y = a(X - 2)(c - 3),$$

whence

$$\begin{aligned} c &= a(X - 2)(c - 3) + a(X - 2)c - 2\delta(X - 2) - 2\delta + a(X - 2) \\ &= 8X^2 + 4X - aB + cB + (X - 2), \end{aligned}$$

this is in fact $= 2x + (X - 2)x$, viz. we have

$$2c - X^2 + a + cB + 2B$$

and therefore

$$2x - X^2 = -a - 2X = X^2 - 6cv + (X^2 + 5b)X = X^2 + 6c + cb - 8\xi$$

and therefore

$$2x(x - 2) = a - 8\xi$$

viz. the united points (a, c) are the points of contact of the double tangents, and the x cusps each $(X - 2)$ times in respect of the $(X - 2)$ tangents from to the curve. This is the way in which I originally applied the principle to finding the number of double tangents of a curve.

30. Correspondence (B, D). By reciprocation

$$c_1 = y_1 - y_1 = q - 2x,$$

$$\begin{aligned} c_2 = y_2 - y_2 &= c + b\delta - 1q \\ &= 28 + (a - 2)x. \end{aligned}$$

31. It may be remarked, as regards the cases which follow, that although the result in terms of (δ, c, t) when once known can be explained and verified easily enough, there is great risk of oversight if we endeavour to find it in the first instance; while on the other hand the transformation from the form in terms of (δ, B, δ, c) to that by means is the proper mode of correspondence is the required form in terms of (δ, c, t), is by no means easy. I in fact first obtained the expression in (δ, B, δ, c), and then, knowing in some measure the form of the other expression, was able to find it by the actual transformation of the expression in (δ, c, t).

32. Correspondence (c, c).

From the values of $c - c$, $c - x$; we have

$$c - c - x = (XX + 2c - 18)c,$$

and thus

$$c - 4 - (X + 2)c + 2(c)(X - 3), \quad X - c = (X - 2)(X - 8)(c - 2),$$

$$\begin{aligned} c = (a - 2)(X - 2)(X - c - 4) \\ +c - 2(X - 8x - 1c)(X^2 \cdot 2X - 2c - 5 + 5). \end{aligned}$$

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which is

$$\begin{gathered} = \quad X^2 \quad (\quad +1) \\ +X(X^2 - 2x - 19) \\ +X^2 - 19x \\ +\xi[-2X - 2X + 19]. \end{gathered}$$

And then, by means of the equations

$$\begin{gathered} (c-4)2v = (c-4)(X - X + a - X^2) \\ (X-4)2c = (X - X^2)c - a + 2\xi, \\ (c-2)c \cdot a = (c-1) - 1c - 2\xi, \\ (\quad -X)c \cdot c \cdot a \quad - \delta - \xi. \\ 2X - 2x \quad (X-2)x \quad + 2\xi, \end{gathered}$$

we verify that

$$d = (a-4)2x + (X-4)2x + (a-2)c + (X-2)x.$$

33. Correspondence (δ, c).

From the values of $d - 4 - F$, $c - y - y$ we have

$$\begin{gathered} c = c - c = -d^2 + (Xx + Xc - 14)c, \\ c - c = (X-2)(X-8)(X-3)(c-4), \end{gathered}$$

that is

$$\begin{gathered} c = 2(c-3)(X-2)(X-4), \\ + \cdot c(X-8)(X-4)c + 8c + [2(X-2)(X-4) - 2(X-2)] \\ + \cdot 4cX^2. \end{gathered}$$

which is

$$\begin{gathered} = \quad X^2 \quad (\quad +1) \\ +X^2[\quad 2c^2 - 14c + 28] \\ +X(\quad -16c^2 + 64c - 47) \\ + \quad x^2 + 24c + 12 \\ +\xi[\, X(c^2 - 4Xc + 4) \\ + \quad 4c^2 + 18c] \end{gathered}$$

and then

$$\begin{gathered} (\sigma-4)(\sigma-5)2v = (a-4)(\sigma-5)(X - X - a + X^2 - 3\xi) \\ [(X-4)(X-5)2x \cdot 2(X-4)(X-5) - 4(X-2)(c-2) - 2(X^2 - X - X^2) - 4(c-2) - 2(\xi] \\ [5(c-2)(c-3)2c \cdot a(c-2)(c-3)(c-1) - 4(c-2) + 4c + 2\xi] \\ 2(X-2)(X-3) - cX(X-2)(X-3)(X-1) - 4(X-4) + 4 + 2\xi] \end{gathered}$$

and summing these values and comparing,

$$\begin{gathered} c = (a-4)(c-5)2x + (X-4)(X-5)2x \\ +(c-1)(c-2)(c-3)2c + (X-2)(X-3)2c - (a-5)(X-5) - 3(X-5)X. \end{gathered}$$

The united points (c, c) are in fact, of each of the δ intersections of a double tangent with the curve, in respect of the two contacts and of the remaining σ intersections; T, each double point in respect of the two branches and of the pairs of tangents from it to the curve; X', each of the $\sigma - 5$ intersections of each of the

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tangents at a double point with the curve; θ', each of the $\sigma-3$ intersections of a tangent at an inflection (stationary tangent) with the curve, in respect of the $(\sigma-4)$ remaining intersections; θ'', each inflexion in respect of the $\sigma-2$ intersections of the

[Fig. 9 omitted]

tangent with the curve; and C', each cusp in respect of the pair of tangents from it to the curve. Thus (T), the double point in respect of the branch which contains σ, and of the two tangents from it to the curve, is a position of the united point (c, c), as appearing in the figure.

35. Correspondence (c, F). By means of the values of c, $c-c'$ and $\sigma-F=R$, we have

$$f-\phi-\beta=c(X^2-2x+2x-2X^2-28X+102)x,$$

and then

$$\phi=(X-2)(c-3)(X-2)(X-3)(c-4),$$
$$\phi'=(X-2)(X-8)(c-2)(X-2)(X-3),$$

whence

$$f=(X-2)(X-3)c-(X-2)c+(X-3)(X-4)+(X-4)2x+(X-2)2x$$
$$+(X^2-2x-14)c+(X^2+4c-12X-13)2v+(X-2v-X-4)+2c\cdot 2+(T)$$

which is

$$\begin{aligned}
&= X^2 \quad (\quad c^2-6c+7)\\
&+X^2[Y \quad 2c^2-14c+22)\\
&+X \quad -14c^2+78c-91)\\
&+ \quad 6c^2-22c-94\\
&+\xi[\,X^2 \quad (\quad -2)\\
&+X \quad (\quad 2c-28)\\
&+ \quad 2c^2-32c+152].
\end{aligned}$$

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This result includes proper solutions of the problem of finding the number of the triangles $aBcDaF$, which are such that the side aF touches the curve at σ; and also heterotypic solutions having reference to the singular points of the curve; but I have not determined the number of solutions of each kind.

36. Correspondences (c, g): from the values of $b - \phi - \phi'$ and $\sigma - c - x$, we have

$$g - \chi - \chi' = (X^2 - 20X^2 - 8Xx - 4x^2 + 125X + 44a - 486)A,$$

and then

$$\chi = \chi' = (X-2)(c-3)X(c-2)X(c-4)X(c-3)x,$$

whence

$$g = (X-2)(c-3) - 2c - 2x$$

$$+(X-20)X^2 - 8X - 4x^2 + 125X + 44a - 486)x. \text{ Now writing,}$$

viz. this is

$$\begin{aligned} g = \quad & X^2 \quad (\quad -3) \\ & +X^2[\quad 2x^2 - 16x + 42x - 18) \\ & +X \quad -16x^2 + 124x - 286x + 60) \\ & +X \quad 42x^2 - 562x + 706x + 4 \\ & + \quad x^2 - 282x^2 + 706x - 38 \\ & +\xi[X^2 \quad (\quad -20) \\ & +X \quad (\quad 4x - 110) \\ & +X \quad 4c - 175 \\ & + \quad -2c^2 - 162x - 114], \end{aligned}$$

Comparing this the expression of ϕx, Case 52, we have

$$\begin{aligned} g - \phi - \phi' \quad = \quad & -X \\ & +X^2(\quad -34) \\ & +X^2[\quad 2x^2 - 18x + 42x - 79) \\ & +X \quad -16x^2 + 100x - 286x + 70) \\ & +X \quad 32x^2 - 412x + 706x + 80) \\ & + \quad x^2 - 202x^2 + 716x - 38 \\ & +\xi[X^2 \quad (\quad -30) \\ & +X \quad (\quad 4x - 110) \\ & +X \quad 4c - 175 \\ & + \quad -2c^2 + 162x + 114 - 486] \end{aligned}$$

which differences must be the numbers of heterotypies solutions having relations to the singularities of the curve; but I have not further considered this.

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Sur les Courbes Aplaties.

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXXIV. *(Janvier—Juin, 1872), pp. 708–712.]*

En lisant la thèse de M. S. Maillard, *Recherches des caractéristiques des systèmes élémentaires des courbes planes du troisième ordre* (Paris, 1871), j'ai été conduit à quelques réflexions sur la théorie générale des courbes aplaties de M. Chasles [1].

Je considère comme étant représentée par une équation de l'ordre n, $f(x, y, z) = 0$, laquelle pour $k = 0$ se réduit à la forme $P^kQ \cdots = 0$. Pour k très infiniment petit, la courbe aplaite ainsi obtenue prend en général des points singuliers à la pénelimière de courbes $P^kQ \cdots = 0$, dont quelques-uns sont les singularités généraux des courbes $P = 0$, $Q = 0$ etc., et les autres seraient des singularités engendrées par la réunion des points multiples, en tangentes apparutees mutuellement par les droites qui relient ces points multiples ; tout cela compris dans les courbes $P^k = 0$, $Q^k = 0$, etc. et appelés par M. Chasles des courbes planièmes singulières. 1° les droites (par les mémiates partiels) du 2° ordre, planièmes du 3° ordre, et ainsi de suite. Les coniques planièmes du 3° ordre, par exemple sont, des droites planièmes ou les coniques planièmes du 3° ordre, ainsi de plus des droites passées par un certain nombre des points multiples, ou bien aussi peuvent passer des droites quelles ont l'une quelconque de leurs points (les points P, Q, R, comméncés).

1. *Comptes Rendus*, t. LVIII., p. 722—805 et 1079—1981 ; (éances de 21 avril et 27 mai 1867).

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ennuit aucunes fibres. Cela étant, ou peut considérer la courbe planièrestere comme équivalente à la courbe finale $P^k Q \cdots = 0$ plus les nouveaux : il s'agit, pour en chaque quelconque, de trouver le nombre et la distribution de ces couvbres.

Je considère d'abord la cas le plus simple, celui d'une conique aplaite, planièrestere de $a^* = 0$; l'équation s'éait ainsi sous la forme :

$$ax^2 + 2hxy + by^2 + cz^2 + 2gxz + 2fyz = 0,$$

ou, en prenant $b = h$, et $\beta = m$, ce sera

$$(bx + \beta y)^2 + 2(gxz + fyz) + cz^2 + (gxz + fyz)z = 0,$$

ou bien

$$(bx + \beta y)^2 + 2z(gx + fy) + cz^2 + (gx + fy - \beta y)x + (cy + fz - \beta x)\beta z = 0,$$

et conditions pour un moment k très infiniment petit ces coefficients comme étant des infinement petits du même ordre, et V, certain coefficient le 0, c'est-à-dire

$$(bx + \beta y, b'x + \beta' y; cz; (gx + fy)k; (cy + fz - \beta x)k; (gx + fy - \beta y)k = 0,$$

on, ce qui est la même chose,

$$(x, y, z'(bx + \beta y)c; -ye - \beta y(c'x + \beta')k; c' = 0,$$

on a tangentes cuspéea l'a droite $a = 0$ dans les deux points donnés par l'équation $(c' = 0)(gx + fy) - cmz = 0$
$c'(c'$ et $a' = 0)$, et ces deux points sont indépendants de la position du point donnés (a, β) ; ces points sont en effet les intersections de la planièrestere par la droite $a = 0$.

Mais il y a la une restriction qui'on évite au moyen d'une supposition plus générale, *savoir* : en prenant a, β, premiers ; β, β' du second ordre, on donne g, h $= 0$; b, β' ou β', β, l'équation des tangentes devient

$$(b\nu - \beta y, \beta' x + \beta' y'; c, p^2 x + f^2 y + gx + fy - \beta y)y; (cy + fz - \beta x)\beta z, c' = 0,$$

ou, en écrivant $z = 0$, cette équation devient

$$(\nu - x', \nu', \beta' y, \beta x - \beta y, (\nu - 5x - \beta x), c' = 0,$$

c'est-à-dire

$$b\beta' + \beta' y(x - x)(gx - \beta) = c',$$

$$\beta\beta' + \beta(\nu y - \beta x)(\nu y - \beta x)c = 0,$$

nous arons ainsi en dehors $a = 0$, deux points indépendants de la position du point donnés (a, β), et qui ne sont d'au plus que les points donnés comme les intersections de la planièrestere par la droite $a = 0$; ils y a sons points le rougeu même indeux supplèatives être fonctionnellement leurs traces être être droits. En général, on peut prendre a, β, β', β'' du même ordre, c'est-à-dire $a = 0$ comme k, partitènant infinement petit ; mais on n'aune mode à poser $\beta = \beta = a$, ou bien $a = a' = 0$, $\beta = a$ (de sorte qu'il existe maintenant un point double (a, β) sur le droite $a = 0$) ; on a un même un mêm.

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Je passe à un cas nouveau, celui de la courbe quartique planièrestere de $xyx = 0$; mais pour simplifier l'analyse, au lieu d'un point quelconque (X, β) je prends seulement deux des points $(x = 0, y = 0)$ de la droite $z = 0$. On considère, en effet, six $z = 0$ y a points fixes en droite $z = 0$, et il y a deux points qui sont les points fixes $(z = 0, x = 0)$ et $(z = 0, y = 0)$ formant les centres $(z = 0, x = 0)$ et $(z = 0, y = 0)$; les autres deux, les positions sont donnés par les équations $z = 0$, $z = 0$; mais cela donne $z = 0$ et $y = 0$ comme équations on que connée données par la droite $z = 0$ aux deux points $(x = 0$ et $y = 0)$.

J'écris l'équation de la planimètre sous les deux formes

$$\begin{aligned}
&\beta', \quad \beta', \beta', \beta', \beta', \beta', \beta' \cdot 2c \\
&+\beta'(c, p, m)g, \beta' \\
&+\beta(C, \xi, \beta'(x, y), z^2 \\
&+\beta', \beta'(c, x, y) \\
&\beta', \\
&+\beta(c, f, g)(x, y, z') \\
&+\beta'(z, n, p, q)(x, y, z') \\
&+\beta'(C, X, Y)(x, y, z') \\
&+\beta(C, x, y), (\beta_1, \beta_2, \beta')z',
\end{aligned}$$

où le coefficient de β' est nul, et A etc. sont des coefficients ainsi des définitions érites par etéviantes du même ordre. A représente des deux équations

$$(A, B, q(x, y))Y; \quad b'(\beta', \beta'); \quad A; \quad B; \quad q; \quad C; \quad C; \quad C; \quad C; \quad B', \quad C, \quad C;$$

respectivement.

Cela étant, on obtient l'équation des tangentes par le point $(y = 0, x = 0)$ dans même manière comme pour la conique quartique de z ; et de même pour les tangentes par le point $(x = 0, y = 0)$; les mêmes équations d'application $(z = 0)$:

$$0 = (\beta')C, B', \beta(x, y)^2$$

$$+(\beta'^2 - c'(\beta'); c'(x - 8x)Bz - c'(\beta'); c'(B'x') - 16cACB - 3iAOPa,$$

$$0 = (\beta')C, \beta', \beta'(x, y)^2$$

$$+(\beta'^2 - c'(\beta'); c'(x - 8x)Bz - c'(\beta'); c'(B'x') - 16cACB - 3iAOPa.$$

En prenant pour le moment les coefficients A etc. comme des équations existantes en seul terme du l'ordre 5, dans le A_2, et indique etc., les autres terms, les équations existantes alignerent

$$\beta', \quad A^2, \beta'(x, y)^2 = 0, \quad d\beta', AX, \beta'(x, y)^2 = 0,$$

β' y a alors un la coute $\beta' = 0$ comme deux points donnés par l'équation $\beta'(BC - 2D) = 0$, et de même la droite $y = 0$ et deux points donnés par l'équation $\beta'(\beta'C - 2D') = 0$. Mais ces nouveaux points sont les nouveaux points indiqués par l'application $z = 0$, $y = 0$, où $z = 0$ est la droite double de la planimètre quartique ; on les intersections des deux $z = 0$, $y = 0$ y est aussi les intersections de la quartique par ces deux droites respectivement.

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Mais, au contraire, prenons h, f, j, $c = b$; les autres coefficients dont $= 0$. On a d'abord A, B, $D = 0$, $D = 0$, ce qui produit la première équation se réduit à

$$2VAg(BC - 2D) = 0,$$

où qui donne, en outre du point $a = 0$, deux points déterminés par l'équation

$$3AVxC - 21Vc = 0.$$

On a depuis $a^2 = 4D'$, A', C', C', B' ; la seconde équation est donc

$$27A^4(BC - 8D'^2) = 0,$$

mais il

$$b^2C^2 = 4d^2 - 36D = -2A^4f^2 + (12 - 24f),$$

à cause de $f = b^2$, b^2, donc $b^2C^2 - 16d = c^4f$ et $c^2D - 24f$. On peut donc écrire b^2, c^2, dans cas l'équation se réduit à

$$b'(b'^2 - 36b' + b'(-36b' - 32b')) = 0,$$

$b^2c^2 - 16c^2 + 64c^2 \cdots$; l'équation se réduit

$$a^2(\beta b^2 - 36b'^2 - 16c^2 + 64c^2 - 24b') = 0,$$

et il y a une la droite $y = 0$ deux points déterminés par l'équation

$$ac^2 + bo^2 + 6c^2 + 4ac^2 - 24f = 0.$$

Remarquons que la droite $y = 0$ touche la quartique dans les quatre points donnés par l'équation $a^2 = 0$, ainsi a^4, c'est-à-dire infinitement pris du $z = 0$; et c'est ces points trois autres points, lesquels sont quelquesoient des points indiquantes c'est seulement de la droite, et qui ne forment pas $z = 0$.

Conclusion. Il y a ainsi une courbe quartique planièrestere de $xy^2x = 0$ avec neuf nouvelle fibres, leur a savoir d'aux des deux droites $(z = 0)(z = 0, y = 0)$, et qui sont les intersections de la quartique par cea droite (en `tomais` de l'équation $z = 0$; trois aux points de la droite $a = 0$, la trigonémérique courte du droite $a = 0$; et deux deux points de la droite $y = 0$, indiquantes de la droite $y = 0$, qui sont les couches dans ce située comme un toutes formes planières, c'est-à-dire $x = 0$ (al qui n'est mén de deux double) ; ces points sont des intersections de la quartique par ces deux droites respectivement.

A considere le problème des courbes quartiques, qui satisfait diverses aux $(14 - 1) = 13$ conditions ; que celle, le neuf des droites doubles L, D ou points donnés A, D ; passez par doux points donnés C, D ; passé que les droites doubles K, σ et on écrit pour les droites doubles C', et même pour la A'. La pourtour quartique planièrestere de la quartique planièrestere $xy^2z = 0$, lesquelles cédant suite ainsi A on droite `a` ainsi des deux points donnés A, B ; et de plus, il faut $K' = 0$, c'est-à-dire les droites $a = 0$ les compa lises de fil même tantes ; et de plus $K' = 0$, c'est-à-dire le voutre $a = 0$ passe une point doubles de la quartique. Mais cela ne suffit pas ; il faut faciliter satériser sur la $1, 3$ que la droite L, ou aux C, D, et qui est de la même intransiéntième, existe maintenant un point double sur la droite $a = 0$) ; on a un même `mêm`.

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Sur une Surface Quartique Aplatie.

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXXIV. *(Janvier—Juin, 1872), pp. 1393—1395.]*

Il y a évidemment pour les surfaces une théorie analogue à celle des courbes aplaties : la primitive de la surface $P^kQ\cdots = 0$, pour une valeur de k très petit, contient des singularités non nécessaires aux ces sortes deux courbes aplatites, savoir des points multiples, des courbes planières supérieures, lesquelles correspondent aux éumenéres dans la théorie des courbes planières ; par exemple un point liméméntaire de $P = 0$ se réduit à ce point une droite planièrestere de la primitive : même un point doublese de $P = 0$ se réduit à ce point une conique planièrestere ; etc. Le thèorie de la même théorie devient ici plus complexe ét plus intéressantes par les rapports nouveaux dans les choses entre la planièrestere quartique et la même théorie.

Je considère la surface planièrestere $x^2 = 0$ pour cas une surface quartique planièrestere de $x^2 = 0$. J'aurai surface quartique correspond à la même surface planièrestere de la primitive infiniment $xy^2 = 0$ et qui se réduit aussi a deux mille gues, suprerquantee à partir l'ésthire, qui forment ses points $(x = 0, y = 0)$, ou cer ses planières ont essentielles : notamment des surfaces quartiques, ayant trois et même deux planières ont essentielles ; chaque inflémentaire sont mises en perpendiculaire à la cote OO' ; chaque cas, V est même deuxième sur la droite $C00'$, ainsi une troisième symétrique : c'est aussi imaginaires, telles que les points D, O sont réels les imaginaires ces deux.

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sphère ayant son centre sur la droite OO', et coupant orthogonalement la sphère S, passe par le centre dont il s'agit, donc le cercle L. On voit sans peine que chaque point du cercle L et situé sur la cyclide ; le cycle est une infini-fini deuxième singéle de cercles L qui correspondant en a une aux directrices de la surface focale ; il y a de même une infiénie-infinite single de cercles L' qui correspondent en a une aux génératrices de la surface focale. La cyclide est le lieu des cercles de l'une de l'autre série ; chaque cercle de la première série coupe en deux points opposés chaque cercle de l'autre série, mais deux cercles de la même série ne se rencontrent pas, &c.

Or, en supposant maintenant que M. Casey que la surface focale se réduise à un cône, les deux séries de cercles se réduisent à une seule série de cercles L, dont chacun est situé sur la sphère, contre le sommet du cône, qui coupe orthogonalement la sphère S, disons la sphère T. On a sur la sphère T la série des cercles L, lesquels ont pour enveloppe une courbe sphérique, la sphéroquartique de M. Casey. Les points des différents cercles L ne remplissent pas la surface sphérique entière, mais seulement une partie de cette surface, limitée par la courbe sphéroquartique. Cela étant, on pourrait dire que la surface cyclide se réduit à la sphère T deux fois, mais il vaut mieux la considérer comme une cyclide aplatie ayant pour arrète la sphéroquartique.

La sphéroquartique, considérée comme courbe sur une sphère T, est doncénue (comme le remarque M. Casey) par une construction tout à fait analogue à celle pour la cyclide comme surface dans l'espace, savoir (en considérant toujours les courbes sphériques sur une même sphère), la sphéroquartique est l'enveloppe des cercles qui ont leurs centres sur une figure-conique et qui coupent orthogonalement un cercle fixe. Le cône, sommet le centre de la sphère, qui a pour intersection avec un plan une figure-conique, dans le cas actuel le centre du plan une figure-conique, dans les seules points de l'un de ces systèmes coupent les 4 plans de l'autre système dans 16 droites, lesquelles sont les droites focales de M. Casey ; il y a de plus $6+6$ droites, dont chacune est l'intersection de deux plans du même système. Je n'ai pas cherché les distinctions qui doivent exister entre ces différents systèmes de droites focales.

En général, on considèrera une courbe quelconque sur une surface S, et un point O quelconque, deux situés coupent O, dont l'une passe par la courbe et l'autre est dirigée à la surface, ou seulement à la courbe touchent partir où un les rencontrent ; autrement dit, ils n'aient pas des droites d'intersection doubles ou de contact.

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Sur les Surfaces Divisibles en Carrés par leurs Courbes de Courbure et sur la Théorie de Dupin.

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXXIV. (Janvier—Juin, 1872), pp. 1445—1449.]

Soient Θ une fonction arbitraire de k, k_1, y, z des fonctions de k telles que

$$\Sigma\left[\frac{d^2x}{dk\,dh}\,\frac{d\Theta}{dx}+\frac{dx}{dk}\,\frac{d\Theta}{dh}\,\frac{d^2\Theta}{dx\,dh}\right]=0,$$

$$\Sigma\left[\frac{d^2x}{dk\,dl}\,\frac{d\Theta}{dx}+\frac{dx}{dl}\,\frac{d\Theta}{dl}\,\frac{d^2\Theta}{dx\,dl}\right]=0,$$

$$\Sigma\left[\frac{d^2x}{dl\,dh}\,\frac{d\Theta}{dx}+\frac{dx}{dh}\,\frac{d\Theta}{dl}\,\frac{d^2\Theta}{dx\,dh}\right]=0,$$

et que, de plus,

$$\frac{dx}{dk}\,\frac{dx}{dh}\,\frac{dx}{dl}=0$$

en éliminant k, l, on a, entre x, y, z l'équation $V=0$ d'une surface. Je dis que les équations $k=$ const., $l=$ const., déterminent les deux systèmes des courbes de courbure de cette surface, et que, de plus, mais que cette surface est divisible en carrés par ses courbes de courbure.

En effet, les équations donnent

$$\Theta\left[\left(\frac{dx}{dk}\right)^2+\left(\frac{dy}{dk}\right)^2+\left(\frac{dz}{dk}\right)^2\right]-dl\left[\left(\frac{dx}{dl}\right)^2+\left(\frac{dy}{dl}\right)^2+\left(\frac{dz}{dl}\right)^2\right]=0,$$

ou qui implique

$$\left(\frac{dx}{dk}\right)^2+\left(\frac{dy}{dk}\right)^2+\left(\frac{dz}{dk}\right)^2=\Theta K,$$

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où H est fonction de k et seulement ; et la même de même

$$\left(\frac{dx}{dl}\right)^2 + \left(\frac{dy}{dl}\right)^2 + \left(\frac{dz}{dl}\right)^2 = \Theta L,$$

où K est fonction de l et seulement ; donc, en dérivant, comme à l'ordinaire,

$$dx^2 + dy^2 + dz^2 = E\,dk^2 + 2F\,dk\,dl + G\,dl^2,$$

on aura bien voir le théorème énoncé que la surface est divisible en carrés par les courbes de courbure.

Les équations donnent aussi

$$\frac{dx}{dk}\frac{dy}{dl} : \frac{dy}{dk}\frac{dz}{dl} : \frac{dz}{dk}\frac{dx}{dl} = 0;$$

où, céla étant, l'équation différentielle des courbes de courbure se réduit, comme je viens le montrer, à $dk\,dl = 0$; on a donc les équations des courbes de courbure de la surface.

Pour cela, en considérant x, y, z comme des fonctions données de k, l, j'écris, comme à l'ordinaire,

$$\frac{dx}{dk} = a, \quad \frac{dy}{dk} = b, \quad \frac{dz}{dk} = c, \quad \frac{dx}{dl} = a', \quad \frac{dy}{dl} = b', \quad \frac{dz}{dl} = c',$$

et de même b, b', c, c', a', a', c', b', c', pour les coefficients différentiels du second ordre de y et z respectivement. J'écris aussi

$$A = bc' - bc, \quad B = ca' - c'a, \quad C = ab' - a'b,$$

$$E = a^2 + a'^2 + a''^2, \quad F = aa' + bb' + cc', \quad G = a'^2 + b'^2 + c'^2.$$

L'équation différentielle des courbes de courbure est

$$\begin{vmatrix} dx, & dy, & dz \\ A, & B, & C \\ dA, & dB, & dC \end{vmatrix} = 0,$$

Le premier terme de ce déterminant est $dx(BdC - C\,dB)$, savoir

$$(adk + a'\,dl)\big[B(c, dk + c'\,dl) - x(b, dk + b'\,dl) - C(bdk + b'dl) - C(b'dk + b''dl)\big]$$

$$-C[(c, dk + c'\,dl) - C(bdk + b'\,dl) - x(bdk + b'dl) - C(b'dk + b''dl)]$$

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ce qui se réduit de suite à

$$(adk + a'\,dl)\big[B(c\,dk + c'\,dl) - C(b\,dk + b'\,dl)\big] - v\big[a(c\,dk + b\,dl) - C(c'\,dk + b'\,dl)\big]$$

$$-[a(c\,dk + b\,dl) - C'(c'\,dk + b'\,dl)] - v\big[a(c\,dk + b\,dl) - C(c'\,dk + b'\,dl)\big]\ dk;$$

en formant des expressions analogues du second et du troisième terme, et en prenant le somme, l'équation différentielle devient

$$[BAa + Ba'b + Cc' - Ca'b)dk - F(Aa' + Ba'b + Cc' + Ca')dl$$

$$+[BAA' + Bb'B' + Cc')dk - G(Aa' + Ba'b + Cc')dl$$

$$+[F(Aa'' + Bb''B' + Cc'' - GA(Aa' + Bb' + Cc'))]dk$$

$$+[F'(Aa'' + B'b'' + C'c'') - G'(Aa' + B'b'' + C'c')]\,dl] = 0;$$

ou, ce qui est la même chose

$$\begin{vmatrix} dk, & -dl, & 0 \\ E, & F, & G \\ Aa, & Bb, & Cc \end{vmatrix} = 0,$$

enédont si nous écrivons en cours une équation différentielle d'une édaire fraction quand ne contient ne soumet et z, z, ou z deux de la surface ainsi seulement une fraction des para- mètres k, l.

En supposant $F = 0$, l'équation se réduit à

$$(Aa' + Ba'b + Cc')(Aa'\,dk + Cc'\,dl) = 0$$

$$+(Aa'\,dk + Bb'\,dl + Cc'\,dl)(Aa' + Bb' + Cc')dk\,dl = 0;$$

mais ainsi simplement le réduire l'équation se réduit simplement à $dk\,dl = 0$.

En supposant $Aa + Bb + Cc = 0$, l'équation se réduit simplement à

$$Aa'\,dk + Bb'\,dl + Cc'\,dl = 0,$$

mais cette équation $Aa\,da + Bb\,db + Cc\,dc = 0$; en réduit simplement à $dk\,dl = 0$.

$$\begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix} = 0,$$

ou

$$\begin{vmatrix} \dfrac{dx}{dk}, & \dfrac{dy}{dk}, & \dfrac{dz}{dk} \\ \dfrac{dx}{dl}, & \dfrac{dy}{dl}, & \dfrac{dz}{dl} \\ \dfrac{d^2x}{dk\,dl}, & \dfrac{d^2y}{dk\,dl}, & \dfrac{d^2z}{dk\,dl} \end{vmatrix} = 0,$$

et ainsi $F = 0$, satisfaisant dans le même la même on avons avec dans $dk\,dl = 0$ comme équation différentielle des courbes de courbure.

On vérifie sans peine les équations fondamentales, en prenant $\Omega = \lambda - k$,

$$-(z-z_0)(x-x_0) + a(x+b)(x+b_0) + c_0,$$

$$-(z-z_0)(y-y_0) + a(y+b)(y+b_0) + c_0,$$

$$-(z-z_0)(z-z_0) + a(z+b)(z+b_0) + c_0,$$

où qui donne les courbes de courbure de l'ellipsoïde $\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 1$. L'ellipsoïde, comme on sait, sera coupé d'ailleurs de cercles de carrés par des courbes de courbure ; mais je n'ai pas encore décrit d'autres solutions.

Je remarque que l'équation peut être écrite sous la forme

$$\frac{d}{dz}\left(\frac{1}{\Omega}\frac{d\sigma}{dx}\right) + \frac{d}{dx}\left(\frac{1}{\Omega}\frac{d\sigma}{dz}\right) = 0;$$

donc, en écrivant

$$\frac{d\sigma}{dx} = \Omega\frac{d\sigma_1}{dx}, \quad \frac{d\sigma}{dz} = -\Omega\frac{d\sigma_1}{dz},$$

on trouve

$$\frac{d\sigma}{dx} = \Omega\frac{d\sigma_1}{dz}, \quad \frac{d\sigma}{dz} = -\Omega\frac{d\sigma_1}{dx},$$

ce qui donne

$$\frac{d}{dz}\left(\Omega\frac{d\sigma_1}{dx}\right) + \frac{d}{dx}\left(\Omega\frac{d\sigma_1}{dz}\right) = 0,$$

équation de Ω de la même forme que celle pour σ.

On débute une démonstration très-simple du théorème de Dupin. En considérant comme argument (x, y, z) comme des fonctions données de (λ, k, l), le point (x, y, z) sera situé sur une surface, et les conditions pour que les courbes de courbure soient $k =$ const., $l =$ const. seront

$$\begin{vmatrix} \frac{dx}{d\lambda}, & \frac{dx}{dk}, & \frac{dx}{dl} \\ \frac{dy}{d\lambda}, & \frac{dy}{dk}, & \frac{dy}{dl} \\ \frac{dz}{d\lambda}, & \frac{dz}{dk}, & \frac{dz}{dl} \end{vmatrix} = 0.$$

Cela étant, en introduisant un troisième paramètre λ comme fonction de (k, l) ou de (x, y, z), ou réciproquement, (x, y, z) des fonctions données de (λ, k, l). On

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a ici les trois systèmes de surfaces $\lambda =$ const., $k =$ const., $l =$ const., et les conditions pour que ces surfaces se coupent orthogonalement peuvent s'écrire sous la forme

$$\frac{dx}{d\lambda}\frac{dx}{dk} + \frac{dy}{d\lambda}\frac{dy}{dk} + \frac{dz}{d\lambda}\frac{dz}{dk} = 0,$$

$$\frac{dx}{dk}\frac{dx}{dl} + \frac{dy}{dk}\frac{dy}{dl} + \frac{dz}{dk}\frac{dz}{dl} = 0,$$

$$\frac{dx}{dl}\frac{dx}{d\lambda} + \frac{dy}{dl}\frac{dy}{d\lambda} + \frac{dz}{dl}\frac{dz}{d\lambda} = 0.$$

On a donc

$$\frac{dx}{d\lambda} : \frac{dy}{d\lambda} : \frac{dz}{d\lambda} = \frac{dy}{dk}\frac{dz}{dl} - \frac{dz}{dk}\frac{dy}{dl} : \frac{dz}{dk}\frac{dx}{dl} - \frac{dx}{dk}\frac{dz}{dl} : \frac{dx}{dk}\frac{dy}{dl} - \frac{dy}{dk}\frac{dx}{dl}.$$

Pour abréger, j'écris

$$\frac{dx}{d\lambda}\frac{dx}{dk} + \frac{dy}{d\lambda}\frac{dy}{dk} + \frac{dz}{d\lambda}\frac{dz}{dk} = [\lambda, k] \ldots,$$

et de même

$$\frac{dx}{d\lambda}\frac{dx}{d\lambda} + \frac{dy}{d\lambda}\frac{dy}{d\lambda} + \frac{dz}{d\lambda}\frac{dz}{d\lambda} = [\lambda, \lambda] \ldots.$$

Les conditions données sont ainsi

$$[l, l] = 0, \quad [l, k] = 0, \quad [\lambda, k] = 0;$$

en différentiant ces équations par rapport à λ, k, l respectivement, on obtient

$$[k, l, l] = 0, \quad [l, k, l] = 0,$$

$$[l, l, k] = 0, \quad [\lambda, k, l] = 0,$$

$$[\lambda, k, l] = 0, \quad [k, \lambda, l] = 0;$$

donc

$$[\lambda, k, l] = 0, \quad [\lambda, l, k] = 0, \quad [l, \lambda, k] = 0.$$

Mais l'équation $[\lambda, k] = 0$ et l'équation $[l, \lambda, k] = 0$, en substituant dans celle-ci les valeurs de $\frac{dx}{d\lambda}$, $\frac{dy}{d\lambda}$, $\frac{dz}{d\lambda}$, sont parfaitement les conditions pour que la surface $l =$ const. soit coupée par les autres surfaces selon ses courbes de courbure : c'est le théorème.

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SUR LA CONDITION POUR QU'UNE FAMILLE DE SURFACES DONNÉES PUISSE FAIRE PARTIE D'UN SYSTÈME ORTHOGONAL.

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXXV. *(Juillet—Décembre, 1872), pp. 177—185, 246—250, 324—330, 381—385, 1800—1803.]*

[Pp. 177—185.]

1. SOIT $\rho = f(x, y, z)$ l'équation d'une famille de surfaces qui fait partie d'un système orthogonal. On sait que ρ satisfait à une équation à différences partielles du troisième ordre, et en suivant le même rapport de M. Levy, dans son excellent "Mémoire sur les coordonnées curvilignes orthogonales" (*Journal de l'École Polytechnique*, t. XXVI., pp. 157—200, 1870), je suis parvenu à former cette équation.

2. Je remarque que la théorème fondamental de M. Levy est, en effet, assez élégant. Considérons une surface de la famille ρ soit P un point quelconque de cette surface, et PT, PT_1 les normales et les tangentes aux deux courbes de courbure au point P. Donc, suivant la notion ordinaire pour P la surface considérée ρ doit être normale à PT, et les tangentes à forment, avec une normale à la surface, un système orthogonal de tangentes ; mais nous pouvons considérer les autres courbes de courbure de la surface ρ. Donc PT_1 et PT_1' se rencontrent ; ils se rencontrent supposé en la surface de la famille ρ. Et plus, la découlant la surface f et la surface f_1 de la nature ρ, en à l'instant les deux conditions se rencontrent au même point. viz. Réciproquement, si PT_1, PT_1' se rencontrent (ou ce qui revient au même : si les surfaces f, PT_1') la famille ρ fera partie d'un système orthogonal ; ce qui est le théorème de M. Levy.

3. Soient (X, Y, Z) les fonctions dérivées de ρ du premier ordre; (a, b, c, f, g, h) celles du second ordre; $(a, b, c, f, g, h, i, j, k, l)$ celles du troisième ordre, savoir

$$(X, Y, Z) = \tfrac{d\rho}{dx}, \tfrac{d\rho}{dy}, \tfrac{d\rho}{dz},$$

$$(a, b, c, f, g, h) = \tfrac{d^2\rho}{dx^2}, \tfrac{d^2\rho}{dy^2}, \tfrac{d^2\rho}{dz^2}, \tfrac{d^2\rho}{dy\,dz}, \tfrac{d^2\rho}{dz\,dx}, \tfrac{d^2\rho}{dx\,dy},$$

$$(a_1, b_1, c_1, f_1, g_1, h_1, i_1, j_1, k_1, l_1) = \tfrac{d^3\rho}{dx^3}, \tfrac{d^3\rho}{dy^3}, \tfrac{d^3\rho}{dz^3}, \tfrac{d^3\rho}{dy^2\,dz}, \tfrac{d^3\rho}{dy\,dz^2}, \tfrac{d^3\rho}{dz^2\,dx}, \tfrac{d^3\rho}{dz\,dx^2}, \tfrac{d^3\rho}{dx^2\,dy}, \tfrac{d^3\rho}{dx\,dy^2}, \tfrac{d^3\rho}{dx\,dy\,dz};$$

soient de plus

$$\begin{aligned} A &= 2(Zh - Yg), & F &= X(c - b) + Yh - Zg, \\ B &= 2(Xf - Zh), & G &= Y(a - c) + Zf - Xh, \\ C &= 2(Yg - Xf), & H &= Z(b - a) + Xg - Yf, \end{aligned}$$

valeurs qui satisfont aux équations

$$A + B + C = 0 \quad \text{et} \quad A, B, C, F, G, H)(X, Y, Z) = 0.$$

Alors les tangentes PT, PT_1 sont données par les équations

$$(A, B, C, F, G, H)(\alpha, \beta, \gamma)^2 = 0,$$

$$X\alpha + Y\beta + Z\gamma = 0.$$

et en partant de ces équations, mais en supposant que pour le point P les valeurs de X, Y soient $X = 0$, $Y = 0$, M. Levy obtient comme conditions de l'intersection donés il s'agit

$$\left(a\frac{dX}{dx} + b\frac{dY}{dy} - h\frac{dX}{dy} + j\frac{dY}{dx}\right) Z = 0,$$

ou, ce qui est la même chose,

$$\{2g(a - b) + 2h(a - b)Z = -2(jX - jY)Z + 2(hX - bY) - 4h = 0,$$

savoir : cette équation est en une dérivant l'équation cherchée du troisième ordre en ρ écrivant $X = 0$, $Y = 0$.

4. Je passe à la recherche de l'équation générale; pour ceci (X, Y, Z) désirant comme argument, nous pouvons considérer aux quantités données du second ordre $(f,\ g,\ h)$ (et de même du troisième ordre a_1, etc.). Cette équation est symétrique par rapport à X, Y, Z, la coordonnée étant pour le point au choix; donc PT, etc., X, Y mais nous donnerons un système de coordonnées de un manière; le symétrie est ainsi de coordonnées; et celles de (X, Y, Z) et (α, β, γ) doivent être les valeurs de (x, y, z), mais données par les équations

$$(A, B, C, F, G, H)(\alpha, \beta, \gamma)^2 = 0,$$

$$X\alpha + Y\beta + Z\gamma = 0.$$

Ces équations impliquent $XX_1 + YY_1 + ZZ_1 = 0$, et en se rappelant à une équation déjà mentionnée, ou à la quatrième

$$(A_1, \dots, X)(X_1, Y_1, Z_1)^2 = 0,$$

$$(A_2, \dots, X)(X_2, Y_2, Z_2)^2 = 0,$$

$$(A_3, \dots, X)(X_3, Y_3, Z_3)^2 = 0,$$

$$X_1X_2 + Y_1Y_2 + Z_1Z_2 = 0,$$

$$XX_1 + YY_1 + ZZ_1 = 0, \quad XX_2 + YY_2 + ZZ_2 = 0.$$

L'origine étant quelconque, prenons (x, y, z) pour coordonnées du P ; et $\alpha + \beta + \gamma$ les coordonnées du P' ; nous avons les $d\rho : d\rho = X : X_1$, et comme X a reçu un signe qui doit s'unir aux symboles, nous pouvons exemple un facteur infinitésimal commun, et l'on supposera $d\rho = (X, Y, Z)$ et derrière en α, on pourrait, P à un point P', ces valeurs $d\rho = X_1$, etc. Au lieu de (X_1, Y_1, Z_1) nous emploiônerons q'une fonction quel- conque à (α, β, γ) devient $\alpha + d\alpha$, ou passant du point P et point P', la valeur correspondante sera à $X\dfrac{d\Phi}{dx} + Y\dfrac{d\Phi}{dy} + Z\dfrac{d\Phi}{dz}$, ou, en écrivant

$$\delta = X\frac{d}{dx} + Y\frac{d}{dy} + Z\frac{d}{dz},$$

ce sera $\delta\Phi$. Cela étant, et nous avons remarqué que ξ, η, ζ pour coordonnées courantes, et θ pour un paramètre arbitraire, les équations de PT seront

$$\xi = x + \theta X_1, \quad \eta = y + \theta Y_1, \quad \zeta = z + \theta Z_1;$$

et si cette droite rencontre PT_1, alors on suppose θ, η, ζ pour les coordonnées du point d'intersection, nous aurons $0 = \xi + X\delta\theta + \theta\delta X_1, \dots$, ou en éliminant $\delta\theta$ il vient

$$0 = \begin{vmatrix} X_1, & X_2, & X\delta X_1 \\ b_1, & Y_2, & Y\delta Y_1 \\ b_1, & Z_2, & Z\delta Z_1 \end{vmatrix},$$

ou, ce qui est la même chose,

$$0 = \begin{vmatrix} X_1, & X_2, & \delta X_1 \\ Y_1, & Y_2, & \delta Y_1 \\ Z_1, & Z_2, & \delta Z_1 \end{vmatrix}.$$

Mais nous avons $X_2 : Y_2 : Z_2 = YZ_1 - ZY_1 : ZX_1 - XZ_1 : XY_1 - YX_1$; donc cette équation devient $X_1\delta X_1 + Y_1\delta Y_1 + Z_1\delta Z_1 = 0$, ou bien encore $\delta(X_1^2 + Y_1^2 + Z_1^2) = 0$. L'équation trouvée peut donc s'écrire dans la forme plus simplifiée

$$X\delta X_1 + Y\delta Y_1 + Z\delta Z_1 = 0,$$

équation qui exprime la condition pour l'intersection des tangentes PT_1, PT_1' ou PT_1, PT_1''.

6. Dans la démonstration précédente, je me suis servi du théorème de Dupin ; mais il convient de remarquer qu'on peut arriver aux équations orthogonal et donner par X, Y, Z ; X_1, Y_1, Z_1 ; X_2, Y_2, Z_2, des liaisons en ρ, respectivement, il serait possible de déduire cette équation suivante :

$$XX_1 + YY_1 + ZZ_1 = 0,$$

$$XX_2 + YY_2 + ZZ_2 = 0,$$

$$X_1X_2 + Y_1Y_2 + Z_1Z_2 = 0.$$

En effet, l'équation tirée directe de cette manière par M. L. Ellis, dans une démonstration du théorème de Dupin, publiée dans *l'ouvrage de Gregory* (*Examples of the processes of the differential and integral calculus* ; Cambridge, 1841). Les premières trois équations donnent $X : Y : Z = Y_1Z_2 - Z_1Y_2 : Z_1X_2 - X_1Z_2 : X_1Y_2 - Y_1X_2$, ou en déux de l'expression

$$(YZ_1 - YZ_1)(\delta a_1) + (X_1Z - ZX_1)(\delta a_2) + \ldots = 0,$$

intégrale par un facteur ; on qui donne

$$(YZ_1 - YZ_1)\left[\frac{dX_1}{dx}\delta X - X\delta X_1\right] + \frac{dx}{dx_1}(ZX_1 - X_1Z)(\delta X_2\delta x\ldots) = 0.$$

Le terme en [] est égal à $-\delta(YZ_1 - YZ_1)$, ainsi

$$\left(X_1\frac{dX_1}{dx} + Y_1\frac{dY_1}{dy} + Z_1\frac{dZ_1}{dz}\right)\delta X - \left(X\frac{dX_2}{dx} + Y\frac{dY_2}{dy} + Z\frac{dZ_2}{dz}\right)\delta X_1 - X_1\left(\frac{dX_1}{dx}\delta X + \frac{dX_1}{dy}\delta Y + \frac{dX_1}{dz}\delta Z\right) - \ldots =$$

et la somme qui correspond à la deuxième ligne de cette expression s'évanouit éviem- ment ; la première ligne peut s'écrire sous la forme $\delta_1\delta_2 = \delta_2\delta_1$, donc, en éliminant X, Y, Z et X_1, Y_1, Z_1 la condition devient simplement

$$X(\delta_1X_2 - \delta_2X_1) + Y(\delta_1Y_2 - \delta_2Y_1) + Z(\delta_1Z_2 - \delta_2Z_1) = 0.$$

Mais nous avons

$$\delta_1\delta_2\frac{dX}{dx} = X_1\left(X_2\frac{d^2X}{dx^2} + Y_2\frac{d^2X}{dx\,dy} + Z_2\frac{d^2X}{dx\,dz}\right) + \ldots = X_1X_2a + Y_1Y_2b + Z_1Z_2c + (Y_1Z_2 + Z_1Y_2)f + \ldots,$$

et ainsi $\delta_1X_2 = \delta_2X_1$ et $(Y_1Z_2 - Z_1Y_2)f$, $(Z_1Y_2 - Y_1Z_2)g$, $(X_1Y_2 - Y_1X_2)h$, ou de même $\delta_1Y_2 = \delta_2Y_1$ et $X_1Y_2 - X_2Y_1$, etc. ; on a donc

$$\delta_1X_2 - \delta_2X_1 = X(Y_1Z_2 + Y_2Z_1) - 2Z(X_1Y_2 + Y_1X_2), \ldots,$$

ou à la quatrième

$$XX_1X_2a + YY_1Y_2b + ZZ_1Z_2c + YY_1Z_2 + YY_2Z_1 = 0;$$

et ainsi l'équation dont il s'agit

$$XX_1 + YY_1 + ZZ_1, +(XX_2 + YY_2 + ZZ_2) = 0.$$

On se servait ces sugessions géométriques de cette équation.

7. Dans la question actuelle, partant de cette équation, je rappelle que les valeurs de X, Y, Z, X_1, Y_1, Z_1 sont, comme celles de (α, β, γ) données par les équations

$$(A, B, C, F, G, H)(\alpha, \beta, \gamma)^2 = 0, \quad X\alpha + Y\beta + Z\gamma = 0,$$

en supposant que ces équations sont :

$$X_2 : Y_2 : Z_2 = X' : Y' : Z' = Y''Z' - Z''Y' : Z''X' - X''Z' : X''Y' - Y''X';$$

la condition devient

$$X_2\delta X + Y_2\delta Y + Z_2\delta Z = 0, \quad X\delta + Y\delta + Z\delta = 0.$$

8. Pour effectuer la réduction de cette formule, nous avons besoin de plusieurs formules subsidiaires. J'écris

$$(BC - F^2, GH - AF, HF - BG, GF - BC, HF - CG, FG - AH)(X, Y, Z)^2$$
$$= (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})(X, Y, Z)^2 = \omega,$$

et je deduis par $(a), (b), (c), (f), (g), (h)$ les coefficients de $\lambda^*, \ldots$ dans la fonction

$$(A, B, C, F, G, H)(\lambda X + \mu, \lambda Y + \nu, \lambda Z + \xi)^2 = \mu^2\mathfrak{A} - 2\mu\nu\mathfrak{B} - \lambda^2\mathfrak{F}, \ldots,$$
$$-2\{(B, X, X, h)(X, Y, Z)\lambda\mu + \ldots\} - 2(C, Y, Y, h)(X, Y, Z)\lambda\nu + \ldots,$$

et je donne par $(a), (b), (c), (f), (g), (h)$ les valeurs de $\lambda^*, \ldots$, dans la fonction

$$(A, B, C, F, G, H)(X, X', X'')^2 = \mu^2, \quad 2YZ' = \alpha, \quad 2ZX' = \beta, \quad 2XY' = h :$$

$$\begin{aligned}
(a) &= bZ^2 + cY^2 - 2fYZ,\\
(b) &= cX^2 + aZ^2 - 2gZX,\\
(c) &= aY^2 + bX^2 - 2hXY,\\
(f) &= -aYZ - fX^2 + gXY + hXZ,\\
(g) &= -bZX - gY^2 + fXY + hYZ,\\
(h) &= -cXY - hZ^2 + fXZ + gYZ;
\end{aligned}$$

et je remarque qu'on vérifie les équations

$$(a) + (b) + (c) = 0,$$

ainsi que

$$\begin{aligned}
(a)X + (h)Y + (g)Z &= 0,\\
(h)X + (b)Y + (f)Z &= 0,\\
(g)X + (f)Y + (c)Z &= 0.
\end{aligned}$$

Cela étant, nous avons les identités

$$\{(a), (b), (c)\}(X, Y, Z) = 0,$$
$$\{(b), (c), (a)\}(Y, Z, X) = 0,$$
$$\{(g), (h), (f)\}(Z, X, Y) = 0,$$
$$\{(B(b) - (D)(p) - (g)(g) + (A)(h) - (h)(h)\}(2) = (b)(C) - (c)(B) - (h)(g) + (g)(h).$$

savoir, $(\mathfrak{h})(c) - (\mathfrak{f}) = X\omega, \ldots$. De plus

$$(A, B, C, F, G, H)((g), (h), (f))^2 = 2X(YZ' - ZY')(X^2 + Y^2 + Z^2)\phi,$$

$$(B, B, F)((a), (b), (h))^2 = 4(YZ' - ZY')(X^2 + Y^2 + Z^2)\phi,$$

$$(G, H)((g), (h), (f))((a), (h), (g)) = -2(Y^2)(X^2 + Y^2 + Z^2)\phi,$$

où

$$\phi = aBC + bCA + cAB - 2(fFA + gFB + hHC) + 2(AGH + BHF + CFG).$$

Multipliant cette dernière fonction par l'un quelconque des coefficients $(a), \ldots$, et substi- tuant, en éliminant ses équations; mais je dirai seulement celles qui ne sont du (g), savoir, nous avons

$$(g)\Big\{A\omega + B(b) + C(c) + 2F(f) + 2G(g) + 2H(h)\Big\}$$

$$+2\phi[-XBE + YZY + GTY + HXZ] = 0,$$

ici la seconde ligne est égale à

$$BY[(b)X + (a)Y - 2fXY + 2gXY + 2gXY + 2HXY] = 0,$$

et l'équation finale est

$$ZX(\mathfrak{B}) + (B)(g)(\omega - (h)) = (YX - X')(-X(\mathfrak{Z}) + 4(2 + Y + Z')\phi),$$

ou, en remarquant que

$$(\mathfrak{B})(g) - (\mathfrak{Z})(b) = ZY,$$

$$(\mathfrak{B})(h) - (\mathfrak{Z})(c) = XZ,$$

$$(\mathfrak{B})(f) - (\mathfrak{Z})(a) = -YZ,$$

$$(\mathfrak{B})(f) - (\mathfrak{Z})(g) = X^2,$$

c'est-à-dire

$$(g)(YZ\omega + YY')(X^2 + Y^2 + Z^2)\phi.$$

9. Je reviens à la question principale, à savoir, on se servira des équations arbitraires qui ouvriront les besoins de cinq compagnies. Il n'y a pas d'expression systéme pour la valeur de $X_1 : Y_1 : Z_1$ pour les $X_2 : Y_2 : Z_2$, et nous trouvons

$$X_1 : Y_1 : Z_1 = (a), (b) : (c) : (h) - X\sqrt{\omega} : (g) - Y\sqrt{\omega},$$

$$X_2 : Y_2 : Z_2 = (a), (b) : (c) : (h) - X\sqrt{\omega} : (g) - Y\sqrt{\omega},$$

et la condition dérive

$$\{(b)Z - (c)Y\}^2 + \{(c)X - (a)Z\}^2 + \{(a)Y - (b)X\}^2 + \{(h)X - (g)Z\}^2 - \omega = 0,$$

ou, puisque $X\sqrt{\omega} = \frac{(\mathfrak{B})}{X}\omega = (\omega)$, on a

$$\Sigma(\delta(\mathfrak{B} - X(g) - (b)X)(g) - (a)Y) = 0.$$

équation qui contient, comme une le voudra, le facteur ω; en l'enlevant, elle restera au facteur, l'équation deviendra symétrique.

J'écris

$$\Delta(g) = \Delta(g) + \Sigma(g), \quad \Sigma(b) = \Delta(b) + \Sigma(b), \quad \delta\phi = \Delta\phi + \Sigma\phi,$$

en dénotant par Δ les parties qui dépendent de δX, δY, δZ, et par Σ' celles qui dépendent de $\delta a, \ldots$. Les fonctions Δ doivent en avoir le facteur $\omega = X^2 + Y^2 + Z^2$.

$$\begin{aligned}\Omega = {} & ((b)Z - (c)Y)\{\Delta(b)Z - \Delta(c)Y - (c)\Delta Z + (b)\Delta Z\} \\ & +((c)X - (a)Z)\{\Delta(c)X - \Delta(a)Z + (c)\Delta X - (a)\Delta Z\} \\ & +((a)Y - (b)X)\{\Delta(a)Y - \Delta(b)X + (a)\Delta Y - (b)\Delta X\} \\ & +((h)X - (g)Z)\{\Delta(h)X - \Delta(g)Z + (h)\Delta X - (g)\Delta Z\},\end{aligned}$$

et ainsi l'expression entire se mettra à

$$\frac{1}{2}\delta((b)Z - (c)Y)^2 + \frac{1}{2}\delta((c)X - (a)Z)^2 + \frac{1}{2}\delta((a)Y - (b)X)^2 + \frac{1}{2}\delta((h)X - (g)Z)^2.$$

10. Je calcule l'expression de Ω_1. Nous avons

$$\Delta\Omega = \Delta\{-CY+FB\}\Delta X+(-CX+GA)\Delta Y+(FX+GY)\Delta Z+\Sigma F\Delta(YZ+ZY')(XZ\,dY+ZY\,dX);$$

$$\Delta\phi = (BC - F^2)\Delta X + (FH - AF)\Delta Y + (HF - BG)\Delta Z + (\,)\Delta a + (\,)\Delta b + (\,)\Delta c;$$

ou la dernière ligne est égale à

$$\{(g)Y - (h)Z\}\Delta a + \omega\delta a + \omega\delta(b) - (a)Z\{\Delta b + (b)\delta X + (g)\delta Z\}\Delta c$$

$$+\{(\omega Z + (h)Z\}\Delta f + ((h)Z - (b)Z\}\Delta g + (b)Z\Delta h,$$

où la seconde ligne est égale à

$$-(a)Z(R, F)\{\Delta a + (b)\delta Y + (g)\delta Z\}\Delta b,$$

et ainsi l'expression entire se méttra à

$$(\Omega) - (R - F)\Delta b + F\{((h)\Delta b + (g) - (b)\Delta b) - 0 = (\Omega\mathfrak{A})\Delta b.$$

c'est-à-dire le coefficient de Δb est

$$[-AAX + F(YZ' + Z(g))X + 2((F)\delta b + 2(b))X + 2(B)(b)X + 2(b)\delta Z + 2\mathfrak{A}),$$

ou la dernière ligne est égale à

$$-(R - F)\Delta b + F\{(h)\Delta b + (g) - (b)\Delta b\} - 0 = (\Omega\mathfrak{A})\Delta b.$$

Le coefficient de ΔY est

$$[-AXX + F(YZ' + Z(h))X + 2((F)\delta b + 2(b))X + 2(B)(b)X + 2(b)\delta Z + 2\mathfrak{A}),$$

où la deuxième ligne est égale à

$$-(a)X(F, B, F)(\delta X, Y, Z(f)) = (\Omega\mathfrak{A})\Delta X;$$

et ainsi l'expression entière se méttra à

$$(\omega) - (h - X)\Delta b + F\{((h)Y\delta Z - (g)Z\delta X\} - 0 = (\Omega\mathfrak{A})\Delta b.$$

c'est-à-dire

$$-\phi(A, X - B)XE + FEX + GP + HVT = 0.$$

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ou l'expression entière est ainsi égale à

$$\{A-B\}\delta\{(b)-(h)\}(g)+F\{(g)(g)-(c)(h)-(c)(b)(g)-(h)\}$$

$$+H\{(g)(h)-(a)(f)\}(g),F\}(b),(h),(k)-HJ\{(\omega),F+HJ\}$$

$$+(h)(g)\{F,G\}(\omega),F\}.$$

c'est-à-dire

$$A\{B(b)-(h)Y\}(g)-B(b)(h)(g)-C(c)(c)(h)+F(c)(c)(b)-F(b)(c)(g)-H(\omega)(b),$$

ou enâ

$$A-(h)\{g(b)Y-F\}(b)(g)+F(c)(c)(b)-F(b)(c)(g)-H(\omega)(b),$$

enâ réduisant aux moyens de $(a)+(b)+(c)=0$ et $A+B+C=0$, le coefficient de δY devient

$$-(\omega)((b)-F)(g)+(B(c)-2F)(g)(c)+(g)(c)(c)(g)-F(g)(b)(c)$$

$$+(h(c)-F\}\{-F\}(\omega),F\}+\{(b)-2F\}(g)(b)+H\{(\omega)(c)-F\}(b),$$

ou, de même, le coefficient de δZ est

$$-(\omega)\{-(\omega)(c)-F(g)+F(g)(c)\}(b);$$

ce qui achève le calcul de Ω_1.

[Pp. 246—250.]

11. Pour trouver Ω_2, nous avons

$$\delta(g)=-X(b)\delta a+Y(c)\delta b+Y(\delta Z),$$

$$\delta(b)=X(b)\delta a+Y(c)\delta b-Y\delta b,$$

$$\delta(c)=-X(c)\delta a-Y(c)\delta b-Y\delta(c)(c)(c)-X\delta b\}(c)(c).$$

et de là

$$\delta_1=-XY\delta f+XZ\delta g+ZXY^2\delta b+2(g)Y\delta Z\delta+Z\delta b,$$

ou qui n'est de la nature de Ω_2. Dans les expressions de Ω_1, on doit subtituer ces valeurs

[Pp. 246—250.]

Les premières deux lignes en réduisent fadement à

$$\omega\{-XX+YY+Z(c)(h)(b)-2X(c)-Y(g)(h)\}(b)-\omega\},$$

et la troisième ligne à $2(b)\{Z(g)-X(c)\}\delta b$. Donc l'expression entière contient le facteur (b), et nous avons

$$\begin{gathered}\Omega_1=(a)\cdot\omega-[X(c)+Z(b)]\{(b)Y-2(b)\}\\ +\{Y(b)(g)-2(g)\delta+Z(b)X\delta(b)\\ +2(Xg-X\delta b+2(g)\delta X\\ +2(Y-(c)\delta(b)X(b)+2(c)X\delta(b),\end{gathered}$$

expression qui se réduit sans peine à la forme symétrique sous laquelle je la présente ici, savoir

12. Cette équation est, $\Omega_1+\Omega_2=0$; savoir, en mettant la facteur (ω), nous avons

$$\begin{gathered}\frac{1}{2}\{F\{(b-c)-(c)\delta(b-c)(b)-(g-b)+(c)\delta(c)\\ +\{G\{(b)-(c)\delta(b)-(c)\delta(b-g)\}\}\\ +\{H\{(b)(c)-(c)(c)-(c)(c)(g-h)\}\}\\ +\{B(c)(b)-(c)(c)\delta(c)+(c)\delta(c)\}\\ +\{C(c)(b)(c)-(c)+F(c)\delta(b)(c)-2X(g)\}\}\\ -X(b-c)-(b-c)\delta(b)(c)+2X(g)\}\}\\ +2X(b)-(g)-(b)(c)+(c)\delta(b)(c)\}\}\\ -2(c)\delta(b)(c)+(b)\delta(b)\}(c)-\omega,\\ -2\omega(c)+Y\delta(c)-X\delta(b)-Z\delta(b)\delta(h)\\ +2X\delta(g)(c)+(c)\delta(b)(c)-Z\delta(b)\\ +2X\delta(g)(c)-2Y\delta(b)\delta(h).\end{gathered}$$

On se rappelle que $\delta=X\dfrac{d}{dx}+Y\dfrac{d}{dy}+Z\dfrac{d}{dz}$, et

$$\Sigma\frac{d}{dx}+Y\frac{d}{dy}+Z\frac{d}{dz}.$$

13. Pour rester de la résultat de M. Levy, j'écrirai d'abord $X=0$, $Y=0$, nous avons alors

$$(a),(b),(c),(f),(g),(h)=-2hZ^2,Z(c-b),0,-HZ,gZ,0;$$

et l'équation devient

$$(A,B,C,F,G,H)(\delta b,\delta a,0,0,0,0,\delta f,\delta g,\delta h)-BCb+GHZ-GHZ=0,$$

$=-BCb+GHZ-GHZ=0$, mais $A=-BC$; et l'équation devient

$$2\{X(a-b)+2hC-q(c-b)\delta z+2(c-b)Z\delta hg-4hZ\delta Z+2gZ\delta Z=0.$$

Mais nous avons $\delta X = gZ$, $\delta Y = -hZ$, $\delta a = \delta b$, etc., on plus,

$$\delta x = h\delta - g\delta Y = 2\{X(c-b) + h(c-b)\} = 2g^2(c-b) + h^2(b-a) - 2gh :$$

$$\delta y = -2h\delta + g\delta Y = 2\{X(b-a) + h(b-a)\} = 2g^2(b-a) + h^2(a-c) - 2gh :$$

$$\delta x = -2(a-b) + h(a-b)\} = 2g^2(b-a) + h^2(a-c)Z - 2gh :$$

$$\delta h = -2X(b-a) + h\{c-g\}\delta Z = -2g^2(a-b) + h^2(a-c)Z - 2gh :$$

ce qui s'accorde avec le résultat cité.

14. En changeant la signification de X, Y, Z, écrivons $\rho = X + Y + Z$, (X, Y, Z) dénotant à présent des fonctions de x, y, z respectivement, en dénotant par X', Y', Z' les fonctions dérivées de celles-ci, les fonctions précédemment respectivement dénotées par a, b, c, etc., X', Y', Z' et zéro. Au moyen de l'équation précédente, la condition pour que la famille $\rho = X + Y + Z$ puisse faire partie d'un système orthogonal,

$$\begin{aligned}(a, b, c, f, g, h)x = (&2AXYZ, 2BXYZ + 2XYZ, 2X'Y'Z,\\ &X' + X'(Y + 2Y),\\ &Y' + Y'(Z + 2Z),\\ &Z' + Z'(X + 2X);\end{aligned}$$

nous avons aussi

$$(AX, BY, CY) = (X', Y', Z'),$$

$$(AA, BB, CC) = (0, 0, 0),$$

$$(GF, RG, AH) = (X' - X)(\delta - X(X)) = X(X'') = -Z'(X'') = -X(Y'') = X(Y') = -X'(Y'',$$

$$Y'' - Y' = -X(Y'') = -X(Y'').$$

15. Donc, dans l'équation générale, la première ligne est

$$2[-X' \cdot 2XY'(\delta Y' - \delta Y) - 2X' \cdot 2X'(YT'' - 4X' \cdot 4X^2Y + b)]^2$$

c'est-à-dire

$$-4XY \cdot Y - XY + 2X' \cdot 4 + b$$

ou, ce qui est la même chose,

$$2XY'X''(p-2)p^2 - 2X'Y'(p-q)^2,$$

et la somme des premières trois lignes sera ainsi $2XYX''$ multiplié par

$$mp^2\{p\overline{p^2-p^2} - 2p^2(p-q)^2 - q^2(p-q)^2$$
$$+p^2\{p\overline{p^2-q^2} + p\overline{p^2-2pq}\}$$
$$+q^2(p^2(p-q^2)) + (p-q)(p^2-2p)q),$$

savoir dans le second facteur le coefficient de mX est $Y'\{x'(p-2)p-p\}$, et de même les termes en mX, mY, etc., respectivement, de plus, la fonction entière de $\delta\{-2Xp\}$, par respectivement, ce qui donne $-2Xp\delta$, et la somme du tout sera ainsi

$$4XYX'Y'\{1'-p\} + \delta X'Y'\{(p)\}.$$

Les autres ces ne donnent pas de difficultés de même ; on a

$$\delta f = -2[X'Y'Z + Z(YT - X(p-2))\}\}\{Y(X'-X^2) + Z(X'-X')\}],$$
$$\delta g = 2[X'Y'Z(p^2-p^2) - 2XX(p-q)^2 - 2X'Y(p^2 - 2X(p-q)^2),$$
$$\delta h = 2[Y'Y'Z(p^2-p^2) - 2X'(p^2-q^2) - 2X(p-q)^2(p^2-q),$$

ou dans le second facteur le coefficient de mX, mY est $\delta\{2Y'Z - 2Z\}$, et de même pour $\delta\{-2X'Z+2X\}$, etc. ; les coefficients de δZ, $\delta Y'$, $\delta Z'$, etc. ; ces formes ce dénotent par

$$2X'Y'\{(p-q)X' - 2p^2(Y'-Y) + (p-q)Z'\}.$$

La seconde partie est donc

$$2X'Y'Z'Y'X'\{Y'-Y'-Y'\} + (X(Z) - YX) + (Y(X) - Z(Y) - 2(YZ - X(Y)\},$$

ou en réunissant les deux parties ce sera

$$2\delta Y'Y'X(p-2)p^2.$$

L'équation devient

$$\delta YX'X'\{Y' - X(Y+Z) + YX'\}^2 + YZX(Y') + ZXY'\{XY - Y\}^2,$$

savoir :

$$3(Y'-Z)X(Y-X) + (Y'-X)(Z-X) + (Y'-X)(Z''-X)Y(YZ + (X'-Y)(Z''-X)Y(YZ$$

équation trouvé par M. Bouquet dans son "Note sur les surfaces orthogonales" (*Journal de M. Liouville*, t. XI., pp. 446—450, 1846), et reproduite par M. Serret dans son "Mémoire sur les surfaces orthogonales" (*Journal de M. Liouville*, t. XII., pp. 241—254, 1847).

[Pp. 324—330.]

En considérant une famille orthogonale comme une famille de surfaces qui fait partie d'un système orthogonal, on peut se proposer la question : Étant donnée une surface de la famille, trouver de la manière la plus générale la famille. J'essaye de résoudre cette question en développant les trois coordonnées selon les puissances d'un paramètre ; et, quoique je ne sois encore validé que les trois premiers termes des trois développements, les valeurs de ces termes sont assez intéressantes pour les rapports en géométrie.

On peut, pour la surface donnée, considérer les coordonnées x, y, z d'un point quelconque de la surface comme des fonctions déterminées de deux paramètres p, q. Si, de plus, on parameter soit tels, que les équations des trois systèmes de courbes de courbure soient $p =$ const., $q =$ const., respectivement, alors les dérivant pour exiger

$$\frac{dx}{dp}\frac{dx}{dq} + \frac{dy}{dp}\frac{dy}{dq} + \frac{dz}{dp}\frac{dz}{dq} = 0,$$

$$\frac{dx}{dp}\frac{dy}{dp} + \frac{dy}{dq}\frac{dz}{dp} + \frac{dz}{dp}\frac{dx}{dq} = 0,$$

et de même pour p et q on conditions x, y, z, considérées toujours comme des fonctions de p, q, seront telles, que

$$x_{pp} + y_{pp} + z_{pp} = \lambda_x, \qquad \begin{vmatrix} x_p, & y_p \\ x_q, & y_q \end{vmatrix} = 0,$$

et dans le même cas, X, Y, $Z = y_pz_q - z_py_q$, $z_px_q - x_pz_q$, $x_py_q - y_px_q$. On a donc identiquement

$$Xx_p + Yy_p + Zz_p = 0,$$

$$Xx_q + Yy_q + Zz_q = 0,$$

et les deux équations identiques sont

$$x_pz_q + y_py_q + z_pz_q = 0,$$

$$Xx_p + Yy_p + Zz_p = 0.$$

Je remarque d'abord que nous pouvons considérer les fonctions x, y, z, sous la forme expliquant, sous l'équation de la surface $x = x_0 + a_1t + a_2t^2 + \cdots$, ainsi peut être quelconque, pour ainsi dire la représentation ; en assumant multipliera par a_2, b_1, c_2 respectivement on aura toujours

$$x_{pq} = a_1\frac{d^2\theta}{dp\,dq} + a_2\frac{d\theta}{dp} + \cdots ,$$

(t, Q étant fonctions et plus le tels que η et $\dfrac{dt}{dt} = 0$ pour les valeurs initiales de p, q. (Voir mon "Mémoire sur les coordonnées curvilignes" [Liouville, t. V., 1840, p. 121]).

Je suppose que les coordonnées de la famille adjacente du paramètre r (pour la surface donnée on obtient $r = 0$). Pour le point (p, q) de la surface adjacente les coordonnées seront respectivement aux équations différentielles de la famille; les coordonnées ξ, η, ζ d'un point quelconque sur cette courbe seront des fonctions de p, q, r, longeitre, pour $r = 0$, se réduisant à x, y, z respectivement, et j'écris

$$\begin{aligned}\xi &= x + ar + a'r^2 + \cdots,\\ \eta &= y + br + b'r^2 + \cdots,\\ \zeta &= z + cr + c'r^2 + \cdots,\end{aligned}$$

où a, b, c, d, a', b', c', etc., sont des fonctions inconnues de p et q.

Pour exprimer que la courbe coupe orthogonalement les différentes surfaces de la famille, il faut, en écrivant pour abréger

$$X\xi_p + Y\eta_p + Z\zeta_p = U, \quad X = Ar + Br^2 + \cdots, \quad U = Ar + Br^2 + \cdots,$$

$$X\xi_p + Y\eta_p + Z\zeta_p = U, \quad X = X + 2rX' + \cdots, \quad U = Ar + Br^2 + \cdots,$$
$$X\xi_p + Y\eta_p + Z\zeta_p = U, \quad X = X + 2rX' + \cdots, \quad U = Ar + Br^2 + \cdots,$$

(où $\xi_p = \dfrac{d\xi}{dp}$, comme pour p, q, r). La condition cherchée est

$$\frac{X}{a} + \frac{2X'}{a} + \frac{2X'}{a} + \frac{2X'}{a} + \cdots = 0,$$

laquelle de la cinquième série satisfaire pour une valeur quelconque de r; on a donc

$$\frac{X}{a} = \frac{Y}{b} = \frac{Z}{c},$$

$$\frac{A}{a} - \frac{2X'}{a} = \frac{B}{b} - \frac{2Y'}{b} = \frac{C}{c} - \frac{2Z'}{c},$$

savoir, les équations (1) contiennent (a, b, c), les équations (II) contiennent les (a', b', c'), et ainsi de suite. Pour qu'il y ait un système orthogonal, il faut et il suffit que l'on a

$$\xi_p\xi_q + \eta_p\eta_q + \zeta_p\zeta_q = 0,$$

pour toute valeur de r; on aura

$$x_px_q + y_py_q + z_pz_q + a_q + b_q + c_q + a_q + b_q + c_q + a_q + b_q + c_q + \cdots = 0;$$

mais l'équation (3) est identifie d'elle-même; l'équation

$$\frac{X}{a} + \frac{Y}{b} + \frac{Z}{c}$$

[continues sur (1), p. (2)] résulte des équations (4), p. r^2 ainsi que des

Il paraît donc qu'il y a les trois équations (1), (2) pour déterminer (a, b, c), les trois équations (II), (3) pour déterminer (a', b', c'), et ainsi de suite. Mais on ne résoud pas la suite suivante. En effet, en multipliant A_1, B_1, C_1 par les valeurs de a, b, c, qu'il contiennent une fonction arbitraire, λ fonction d'un certain ordre. Mais on a aussi d'une équation à différences partielles du second ordre, obtenue au moyen de surfaces d'une fonction arbitraire λ, et lorsque ces équations ont été intégrées, il y aura encore une fonction arbitraire λ, je presume que cette fonction sera certainement déterminer en moyens des équations (3), (4), et ainsi de suite ; mais je n'ai pas encore fait les recherches ultérieures.

Par rapport à λ, en supposant cette fonction égale à $\rho = \sqrt{X^2 + Y^2 + Z^2}$, l'équation pour ρ est

$$\frac{X}{\rho} \frac{1}{\frac{dX}{dp}} \frac{dy}{dp} + \frac{Y}{\rho} \frac{1}{\frac{dY}{dq}} \frac{dy}{dq} + \frac{Z}{\rho} \frac{1}{\frac{dX}{dq}} \frac{dy}{dq} = 0,$$

égal à une fonction donnée par pour ces ρ, q, ainsi λ satisfera à une système analogue à celles des équations.

Pour calculer ces conclusions, parlant des équations (1), (2) la formule (1) donne

$$a\frac{dX}{dx} + b\frac{dY}{dy} + c\frac{dZ}{dz} = 0,$$

où λ est une fonction de p, q ; ces valeurs satisfont évidemment à l'équation (1). La vérification en fait sans peine : j'écris pour abréger a_1z_1, b_1y_1, c_1z_1, et ainsi de même dans les autres notations. L'équation (2) devient

$$\sigma_1 X_p + \sigma_2 X_q + \sigma_3 X_r + \sigma_4 X_s + \sigma_5 X_t = 0,$$

c'est-à-dire

$$\lambda(\sigma_1 z_1 + \sigma_2 x_p + \sigma_3 y_q + \sigma_4 x_r + \sigma_5 y_s + \sigma_6 X = 0;$$

ou nous avons

$$\sigma_1 X_p, \ldots, \sigma_5 X_t = 0,$$

ainsi le terme de coefficient $a_2 X_p + \cdots$. Nous avons

$$X_p = ay_{p\delta} + b\frac{1}{y}y_p + cy_p z_p + ay_q + by_q z_p,$$

et de la

$$X_r = -y_p x_q + y_q x_p,$$

$$\sigma_3 + \sigma_4 = y_p z_q + y_q z_p, \quad x_p z_r = a_2 y_q z_p + ay_p z_q,$$

$$\sigma_5 + \sigma_6 = y_p z_q + y_q z_p, \quad y_p z_r = ay_p z_q + by_q z_p.$$

et de la, ne faisant la somme des terms tirés évidemment de a, X_p, et b, X_q, respectivement, on trouve

$$a_2 X_p + \cdots = (a, x_p, X_p),$$

savoir, $a_2 X_p + b_2 X_p$, est égal a -1 multiplié par le déterminant $-2Xx_q$, c'est-à-dire $2Xa_2X_p + \cdots = 0$. Donc la fonction λ est jusqu'à indeterminable.

Passons aux équations (2), (3). Substituant dans (3) les valeurs de (a, b, c), ces équations deviennent

$$\frac{A}{a} - \frac{2X'}{a} = \frac{B}{b} - \frac{2Y'}{b} = \frac{C}{c} - \frac{2Z'}{c},$$
$$\lambda(R \cdot E - 2XF) = \lambda(R \cdot E - 2RE) = \lambda(R \cdot E - 2RZ).$$

On y satisfera en écrivant

$$2a_1'X = 2\rho(a'X_p + a_1X_p) + \lambda^2(R \cdot E - 2RE),\ \lambda(R \cdot E - 2RZ);$$

où ρ est fonction de (p, q) ; en substituant ces valeurs dans l'équation (2), c'est-à-dire

$$\lambda(R \cdot E - 2RZ),$$

ou, ce qui est la même chose,

$$\frac{1}{X}[a_1'z_1 + (a'x_p + a_1X_p) + \lambda] - X[a_1X_pX_p + a_1X_q] - X_pX_q] = 0.$$

Les termes en ρ sont

$$\rho(a'x_qX_q + a_1X_pX_q + a_1z_qX_q) = 2\rho_XX_p,$$

qui s'évanouissent d'eux-mêmes ; l'équation se réduit donc à

$$\lambda(a_1z_1 + (a_1z_1)) + \lambda(a_1X_p + a_1X_p + a_1z_pz_p) = 0.$$

Les termes en λ sont

$$\lambda(a_1X_pX_p + a_1X_pX_p) = \rho_XX_p, X_p = 2\rho_XX_pX_p,$$

qui en substituant la valeur $A_2a_1 = a_2$, l'équation se réduit alors à

$$\lambda(a_1z_1 + bX_p + c_1z_p)(a_1X_pX_p + b_1YX_p + 2(X_pX_p + Y_pX_p) = 0.$$

L'on trouve sans peine $A_1X_p + \ldots = X(ax_pX_p + bX_p + cz_pX_p) - XX_p$, derrière

$$a_1z_1 + (bX_p + cx_pz_p) - X = 0,$$

en substituant les valeurs $A_2a = \mu$, alors en l'écartant, dilemma

$$A_2b = X_qb_X, X_p, Y_pz_p, Z_pZ_p.$$

Pour abréger encore la notation, en lieu de a_2^2, $a_2^2+g_2$, g_2+b_2, j'écris simplement 11, et ainsi dans les autres notations ; ayant en vue les abréviations

$$11 = x_p^2 + y_p^2 + z_p^2,$$

$$12 = x_p x_q + y_p y_q + z_p z_q,$$

et je remarque que l'équation $12 = 0$ est l'une des équations données par rapport à p, q respectivement, donne $13 + 24 = 0$, $23 + 14 = 0$, équations qui seront pour définir les formules sans expression 13 et 23. Et pour ne suivant nous donnerons des valeurs des déterminants

$\begin{vmatrix} 11 & 22 \\ 21 & 22 \end{vmatrix}$, ..., alors, en multipliant par les déterminants analogues 123 et

123 respectivement, l'équation $124 = 0$ donne

$$\begin{vmatrix} 11 & 22 \\ 21 & 22 \end{vmatrix} = 0, \quad \begin{vmatrix} 11 & 22 \\ 24 & 32 \end{vmatrix} = 0, \quad \begin{vmatrix} 11 & 22 \\ 24 & 22 \end{vmatrix} = 0,$$

dont chacune est tirée la formule $\Sigma = 11, 12, \ldots,$.

Nous avons

$$A = a_1(yz_p - z_p y_p) + b(z_p x_p - x_p z_p) + h(x_p y_p - y_p x_p) = (a_1, b, h)(X, Y, Z),$$

$$= \lambda(a_1, b, h)(p, q, r)(p_p x_p, \ldots, n_p z_p, \ldots, t_p Y - p_p X),$$

ou en faisant les valeurs de X_q, Y_q, Z_q ce sera

$$A = X(11 - 12) + X'\{11 + 11 - 11\} - 12, 22 + 11, \ldots\}.$$

$$A = X_p(a_q, b_q, c_q) - 14, \ldots, 12 - 11, \ldots, 22 - 11.$$

ou, ce qui est la même chose,

$$A = X_p(X_p Y_p - 12, 13 - 11, 14 - 11, 11)).$$

Écrivons pour un moment

$$A = aX + bX_p + eX_q,$$

nous avons

$$A = AX + B(p \cdot X_p) + C(X \cdot p) + R \cdot p \cdot N + (g \cdot p) \cdot R, \cdot R + g \cdot p \cdot e,$$

$$AX = ax + (a + p \cdot e),$$

et de la

$$Ax_p = Ax_p + (R \cdot p \cdot e + a) f_p^2 + 2(Rp \cdot e + Rp \cdot e) f_p p_p + (R \cdot p \cdot e + a) f_p^2$$

$$+ b_q f \cdot e_p + c \cdot e \cdot e + g \cdot p + h \cdot p \cdot N \cdot e_p$$

$$+ b_q f \cdot e_p + f \cdot e \cdot e + g \cdot p + h \cdot p \cdot e_p.$$

[Pp. 381—385.]

Les expressions de P_p, P_{pq}, ..., contiennent les dérivées du troisième ordre

$$\frac{d^3\rho}{dx^3} = a_1, \quad \frac{d^3\rho}{dx^2dy} = b_1, \quad \frac{d^3\rho}{dx^2dz} = c_1, \quad \frac{d^3\rho}{dxdy^2} = d_1, \frac{d^3\rho}{dy^2dz} = e_1, \ldots$$

En formant les dérivées de l'équation $22 = 14 = 0$, $13 + 24 = 0$ par rapport à p et q respectivement, on obtient

$$17 + 26 = 2 \cdot 34 = 0,$$

$$18 + 27 + 35 + 44 = 0,$$

$$19 + 28 + 2 \cdot 45 = 0.$$

On obtient alors

$$P_p = -2x_p(44 + 17) + 2x_p(54 + 18) + 2x_p(33 + 19) - n_1$$

$$= -2x_p(44 + 17) + 4 \cdot 24 \cdot 34 + 2x_p(33 + 19) - 22 \cdot 33 - 2 \cdot 34 - 2 \cdot 24 - n_1,$$

ou, ce qui est la même chose,

$$P_p = -22 \cdot 33 - 11 \cdot 23 + 2 \cdot 14 \cdot 24 - 33 - 25 - 23 - n_1.$$

On a de même

$$P_q = -22 \cdot 33 + 2 \cdot 14 \cdot 34 - 33 - 24 - 23 - n_1 = -33 - 17 - 19 \cdot 28,$$

$$= -22 \cdot 33 + 2 \cdot 14 \cdot 34 - 33 - 24 - 23 - n_1,$$

ou, ce qui est la même chose,

$$P_q = -22 \cdot 33 + 2 \cdot 14 \cdot 34 - 33 - 24 - 23 - n_1(P_q).$$

On obtient sans peine la même formule

$$P_p = a_2, \quad P_q = b_2, \quad P_{pq} = c_2, \quad \ldots$$

$$P_q = a_2, \quad P_{pq} = b_2, \quad P_{pq} = b_2, \quad P_{pq} = c_2, \quad \ldots$$

et l'on a ainsi

$$A_2a_1 + A_2a_2 = a \cdot \{-2 \cdot 11, 23 - 22, 18 - 22, 11 - 4 \cdot 25 - n_1\}$$

$$+b_1\{-22, 14 - 11, 11 - 22, 11 - 12, 25 - 4 \cdot 24 - n_2\}$$

$$+c_1\{-22, 14 - 11, 11 - 11, 22 - 11, 22 - 11, 25 - 25 - n_3\}$$

$$+b_2\{-22, 11 - 11, 11 - 12 - 11\}$$

L'équation à est

$$A_2a_2 = a_2a_2, 2x_px_q, 2x_px_q, 2x_px_q, x_px_q,$$

et on obtient sans peine

$$X_2X_1 = 44, 13 + 22, 22 + 24, 14 - 23 - 25 - \lambda_1, 23 - 25 - \lambda_1, 22 - n_3,$$

$$XY_1 = 44, 14 + 22, 22 + 22, 14 - 13 - 24 - \lambda_2, 14 - n_3,$$

$$XZ_1 = 44, 12 + 22, 22 + 23, 24 - 12, 14 - n_3,$$

$$XZ_1 = 14, 22 + 22, 22 - n_3.$$

Donc enfin l'équation en λ est

$$\lambda^2\{11(22 + 22) + 23(33 + 14) - 11, 22 - 33\} - 11 - 22 - n_3 = 0,$$

Cette équation est vérifiée par la valeur $R = \dfrac{1}{\rho^2}(X^2 + Y^2 + Z^2)$; en effet, en chercher pour ce moment le quatrième équation à ρ, l'équation s'écrirait sous

$$\lambda^2 \cdot 11, 22 \cdot 28 + 11(22, 22 + 22) - 11 + n_3 \cdot 22 \cdot ZZ_1 + Z_1Z_2 = 0,$$

c'est-à-dire

$$11 \cdot 22 \cdot 23 + 11(22, 22 + 23) - 12 + 4 \cdot 11 \cdot 24 \cdot XX_1 + Z_1Z_2 = 0,$$

$$\begin{aligned} +11(22, 11, 14, 22 + 12 \cdot 23 \cdot 24 - 12) + 4 \cdot 11 \cdot 24 \cdot XY_1 + Y_2 + 21 \cdot ZZ_1 \\ = 2(12 + 11 \cdot 22, 24)(X^2 + Y^2 + Z^2), \end{aligned}$$

et l'on remarque qu'à la facteur 2.

Soient l'équation en ρ est

$$11.22\{11.24 + 22.13 - 122.14\} = 0,$$

mais, des équations mentionnées 13 + 24 = 0 et (23 + 14)x = 0, on tire

$$22 \cdot 13 = 12.14 = 22 \cdot 24 \cdot 14 + 11 \cdot 11,$$

$$11 \cdot 24 = 14 \cdot 11.23 + 14 \cdot 11.23,$$

$$11 \cdot 24 = 14, 22, 14 = 23 \cdot 23,$$

$$11 \cdot 24 = 14, 22, 14 \cdot 11.23 + 24 + 11.$$

Donc l'équation contient en facteur $11 \cdot 22$, on l'écartant, elle deviendra

$$\Omega = 11(-22 + 22 \cdot 45) + 11.14, 14 + 22 \cdot 13 + 11.13 + 11.22.$$

Ou a

$$\Omega = -11(22 \cdot 45 + 22) + 22.13 + 11.13 + 11.22.$$

On va vérifier sans peine que XX_1 est actuellement

$$XX_1 = 11.22 + 14 + n_3.$$

Donc, en écrivant $\lambda = \dfrac{u}{\rho^2}$, l'équation en ρ est contenue dans les terms de u, ρ_p, ρ_q. En effet, l'équation devient

$$-11,22 - 22,13 \cdot u + P\left(\frac{V}{\rho}, \frac{du}{dp}, \frac{du}{dq}\right) \cdot 11,22.$$

ou, comme auparavant, XX_1, denote $XX_1 = YY_1 + Z_1Z_p$, de même XX_2 denote $XX_2 + YY_2 + ZZ_q$, etc.

L'équation devient ainsi

$$11,22 \cdot u - 14,23 \cdot u - 11,22 \cdot u = 0.$$

Savoir, cette équation est

$$\frac{d^2u}{dp\,dq} = \frac{1}{R^2}\frac{dR}{dp}\frac{du}{dq} + \frac{1}{R^2}\frac{dR}{dq}\frac{du}{dp} + Ru = 0.$$

Pour compléter la solution, il convient d'exprimer A, B, C en termes de ρ. Nous avons

$$A = a \cdot X + bX_p, X_p + cX_q, X_q + bX_p + cX_p, X_p + hX_pX_q;$$

$$\frac{1}{\rho}(\ldots) = \rho a + (ax_p + hx_q)x_p + (hx_p + bx_q)x_q + aX_pX_p + (XX_1)$$

$$= \rho a + (ax_p + hx_q)x_q + (ax_p + bx_q)x_q + a(XX_1)\rho_p^2,$$

ou, ce qui est la même chose,

$$\frac{1}{\rho^2}a(4 \cdot 12 \cdot 14 + 22,22) + a, 22,13 + h \cdot 22,14 - \rho_p\, 11,22 \cdot XX_1\rho;$$

ou, ce qui est la même chose,

$$\frac{1}{\rho^2}a(11,22 + 22,13) + (ax_p + 2bx_py_p + 11,22)\rho - \frac{1}{\rho^2}(ax_p + hx_q)\rho_p$$

$$\frac{1}{\rho^2}(11,22 + 11,22,14)\rho + 11,22 \cdot 13\rho^2 - 11,22\rho_p - \frac{1}{\rho^2}(bx_p + bx_q)\rho_p$$

ou, ce qui est la même chose,

$$X(11,22 + 22,13)\rho - \frac{1}{\rho^2}(ax_p + hx_q)\rho_p;$$

Le terme entre [ ] qui dénote la formule de a, b, c, etc., a, etc., en rappellant les valeurs de quantités respectivement en quantités de p, q, p, etc., on, ce qui est la même chose,

$$X(11,22)(bz_p\rho_p + 11,22)\rho_p\rho_q + (XX_p)\rho_q^2;$$

ou, ce qui est la même chose,

$$X(11,22)\rho - \frac{1}{\rho^2}(ax_p + hx_q)\rho_p.$$

Nous avons donc

$$A = \rho\frac{\delta X}{ds}(11,22) + \frac{1}{\rho}\rho_p\rho_q - \frac{1}{\rho^2}(ax_p + bx_q)\rho_q + \rho_p.$$

Nous avons

$$11 = E, \quad 22 = G, \quad V = \sqrt{EG},$$

donc, en dérivant, pour ρ, q,

$$\frac{p}{\sqrt{EG}}(11, 12) + \frac{q}{\sqrt{EG}}(22, 12) + \cdots = \rho.$$

la valeur de

$$A = \rho X + \frac{1}{\sqrt{EG}}(11, 12) + \cdots \frac{d\rho}{dq} + \frac{d\sigma}{dp}.$$

avec des expressions analogues de B et C. Dans les expressions $2dx + \cdots$ qui précèdent de remplacer θ et σ on auxiliaires moyens X, Y, Z.

Donc, dans les expressions de a, b, c on aura

$$\xi = z + \frac{\rho X}{\sqrt{EG}}\left[\frac{(11, 12)^2}{2EG} + \frac{(12 \cdot 12 + 12 \cdot 22)}{2EG} - \cdots\right] e^{-\cdots},$$

$$\eta = y + \frac{\rho Y}{\sqrt{EG}}\left[\frac{(11, 12)^2}{2EG} + \frac{(12 \cdot 12 + 12 \cdot 22)}{2EG} - \cdots\right] e^{-\cdots},$$

$$\zeta = z + \frac{\rho Z}{\sqrt{EG}}\left[\frac{(11, 12)^2}{2EG} + \frac{(12 \cdot 12 + 12 \cdot 22)}{2EG} - \cdots\right] e^{-\cdots};$$

Je remarque que l'on satisfait à toutes les conditions en prenant $\lambda = \text{const.}$, ou en fait à la même chose, $p = 0$ et $r = 0$; mais alors la famille en ρ n'est pas la même; savoir, la famille est ici des surfaces parallèles à la surface donnée.

[Pp. 1800—1803.]

J'ai traité que l'équation différentielle du troisième ordre, sous la forme [dénouri trouvée], contient le facteur étrange $XX' + YY + ZZ$, qui déparaisse de ce facteur le facteur devient beaucoup plus simple. La réduction et aussi le nouvelle solution dont je me suis occupé auparavant l'équation réduite seront tirées des deux quantités appelées quotiques.

Je prends V et l'équation réduite, X, Y, Z les coefficients différentielles du premier ordre, a, b, c, f, g, h les coefficients du second ordre,

$$(A, B, C, F, G, H)(\delta x, \delta y, \delta z)^2 = 0$$

l'équation différentielle des courbes de courbure ; savoir

$$\begin{aligned}
A &= 2(Zh - Yg),\\
B &= 2(Xf - Zh),\\
C &= 2(Yg - Xf),\\
F &= XY - aZ + Yh - Zg,\\
G &= Y(a-c) + Zf - Xh,\\
H &= Z(b-a) + Xg - Yf,
\end{aligned}$$

et

$$[(A),(B),(C),(F),(G),(H)](\delta x, \delta y)^2 = 0$$

savoir :

$$\begin{aligned}
&(A) = BC - F^2 = -F^2VZ,\\
&(B) = CA - G^2 = -G^2ZX,\\
&(C) = AB - H^2 = -H^2XY,\\
&(F) = GH - AF = aVZX - (a-c)YZ + bYZ + cVZ - (a+b)fVZ - bgYZ + h\cdot XZ + a\cdot XZZ,\\
&(G) = HF - BG = bVXY - cZX - hYX + aXY - cXY + Y(c-a)g + bYZ - YgZ + 2cZX,\\
&(H) = FG - CH = cVYZ - aXY + hVXY - gVYZ - (b-a)hXY - aXZZ + bYVZ - cZV,
\end{aligned}$$

ou qui explique la signification des symboles $(A), (B), \ldots$; aussi

$$V = X^2 + Y^2 + Z^2,$$

Cela étant, à chaque point P de la surface $U = 0$, je prends de la manière usagete deux droites infiniment voisines PP' et PP' comme deux fonction quelconque des coordonnées x, y, z du point P. Si nous nous voulons exprimer la condition pour qu'elle se rencontre PP' et PP', ou la même chose, les deux surfaces $V = f(x, y, z) = 0$ et celles de la surface voisine $r = f(x, y, z) + c = 0$.

Or, si la surface donnée $V = 0$ et la surface voisine sont des mêmes considérations d'une famille $r = f(x, y, z) = 0$, on auroit, l'équation de la surface donnée étant comme ci-dessus $P = f(x, y, z) - r = 0$, et celle de la surface voisine étant $r = f(x, y, z) + c = 0$,

$$P = \frac{\rho}{V}, \quad V = X^2 + Y^2 + Z^2,$$

oh c'est une constante. L'équation devient alors

$$[(A), \ldots](\rho_x, \rho_y, \rho_z, \alpha, \beta, \gamma, \delta x\, dr) = 0$$

où, à présent, X, Y, Z, a, b, c, f, g, h dénotent les coefficients différentiels de $f(x,y,z)$, ou (ce qui est la même chose) de l'équation ρ; considérer comme fonction des coordonnées x, y, z. Cette équation est donc une équation aux dérivées du troisième ordre, satisfaite à la fonction ρ, qui satisfait à une équation aux dérivées partielles du troisième ordre d'une famille de surfaces $\rho = f(x,y,z)$. C'était l'équation Ω du numéro ci-dessus, qui Ω s'évolue de quelque facteur V.

J'écrit $\delta X = X_0 d_x + Y_0 d_y$,

et de là

$[X,Y]\delta x + [Y,Z]\delta y + [X,Z]\delta z = [\delta x, \delta y, \delta z]$; $\quad bXx + cXy + aXz = bX,\ hXy + bYx,\ hXz + cZx,$

respectivement, on trouve

$$\delta_1 \frac{a}{\rho} - \frac{1}{\rho^2}(-aX + bYX + 2a)dX^2 + \frac{1}{\rho^2}(bXY + (-\frac{bX}{\rho^2} - bY)dX^2,$$

ou, en dérivant $\omega = \omega + n$, $\omega = h + a$, l'on a sans peine

$$\delta_1 \frac{a}{\rho} - \frac{1}{\rho^2}(-\omega - x + g + bY)dX^2 + \frac{1}{\rho^2}(bXy\rho + bXY + (bXy\rho + bXy)dX^2,$$

et, en supposant pour le moment

$$\delta_1 \frac{a}{\rho} - \frac{1}{\rho^2}(\omega - n - 6\omega + g + bY)dX^2 + \frac{1}{\rho^2}(bXy\rho + bXy + (bXy\rho + bXy)dX^2,$$

et, en multipliant respectivement par $P^2 = -6\omega + g + bY$, $P^2 = (bXy\rho + bXy)dX^2$, et ainsi nous avons donc $\delta_1 \dfrac{1}{\rho}$, etc., et l'équation, multipliable un peu seulement par ρ^2, sera

$$(A) + (B)\delta + (C)\delta + (D)\delta x = 2(B)\delta a = 0,$$

et la même pour second terme :

$$(A)(\delta) - (B)(\delta + (C)\delta + (D)\delta x = 2(B)\delta a = 0.$$

Or je trouve que Σ est divisible par V^2 ; savoir, j'écris

$$[(A),\ldots](\omega),(\delta a)] = 2P[(A),\ldots](\delta X, \delta Y, \delta Z, \delta a, \delta a, \delta a)$$

de manière que l'équation est

$$[(A),\ldots](ba) + V[(A),\ldots]\delta X = 0,$$

ce qui, pour la réduction cherchée, est

$$[(A),\ldots](ba) + (b)(c) - (c)(b) = [(A),\ldots](ba, bX, bY, bZ) = 0;$$

où bX, bY, bZ dénotent respectivement $aX + gY + gZ$, $hX + bY + fZ$, $gX + fY + cZ$; l'équation se réduit donc à

$$[(A), \ldots](ba, \ldots) = 0 = 0,$$

où il peut être exprimé si voulant l'une l'une quelconque des trois formes

$$= -2[(A), \ldots](ba, \ldots),$$

$$= -4[(A), \ldots](bX, bY, bZ),$$

$$= -4, \quad bX, \quad bY, \quad bZ;$$

Prenant la première forme, l'équation est

$$[(A), \ldots](ba, \ldots) - 2[(A), \ldots](ba, \ldots) = 0,$$

ou, ce qui est la même chose,

$$(A)(ba + (b)(\delta b + (C)\delta c + 2(F)(\delta + 2(G)\delta g + 2(H)\delta h$$

$$-2[(A)(\delta a + (b)(\delta b + (c)(\delta c + 2(f)(\delta f + 2(g)\delta g + 2(h)\delta h] = 0,$$

où les coefficients sont des fonctions données de X, Y, Z, a, b, c, f, g, h, mais les coefficients différentielles de ρ du premier et du second ordre, et δ dénote $Xd_x + Yd_y + Zd_z$.

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On Curvature and Orthogonal Surfaces.

[From the Philosophical Transactions of the Royal Society of London, vol. CLXIII. *(for the year 1873), pp. 229–251. Received December 27, 1872,—Read February 13, 1873.]*

p. 292 [519]

THE principal object of the present Memoir is the establishment of the partial differential equation of the third order satisfied by the parameter of a family of surfaces belonging to a triple orthogonal system. It was first remarked by Bouquet that a given family of surfaces does not in general belong to an orthogonal system, but that (in order to its doing so) a condition must be satisfied; it was afterwards shown by Serret that the condition is that the parameter, considered as a function of the coordinates, must satisfy a partial differential equation of the third order: this equation was not obtained by him or the other French geometers engaged on the subject, although methods of obtaining it, essentially equivalent but differing in form, were given by Darboux and Levy; the last-named writer even found a particular form of the equation, viz. what the general equation becomes on writing therein $X = 0$, $Y = 0$ (X, Y, Z the first derived functions, or quantities proportional to the cosine-inclinations of the normal). Using Levy's method, I obtained the general equation, and communicated it to the French Academy, [518]. My result was, however, of a very complicated form, owing, as I afterwards discovered, to its being encumbered with the extraneous factor $X^2 + Y^2 + Z^2$; I succeeded, by some difficult reductions, in getting rid of this factor, and so obtaining the equation in the form given in the present memoir, viz.

$$((A), (B), (C), (F), (G), (H)\ \delta a, \delta b, \delta c, 2\delta f, 2\delta g, 2\delta h)$$

$$-2\,((A), (B), (C), (F), (G), (H)\ a, b, c, 2f, 2g, 2h) = 0;$$

but the method was an inconvenient one, and I was led to reconsider the question. The present investigation, although the analytical transformations are very long, is in theory extremely simple: I consider a given surface, and at each point thereof take along the normal an infinitesimal length p (not a constant, but an arbitrary function

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of the coordinates), the extremities of these distances forming a new surface, say the *vicinal* surface; and the points on the same normal being considered as corresponding points, say this is the conormal correspondence of vicinal surfaces. In order that the two surfaces may belong to an orthogonal system, it is necessary and sufficient that at each point of the given surface the principal tangents (tangents to the curves of curvature) shall correspond to the principal tangents at the corresponding point of the vicinal surface; and the condition for this is that p shall satisfy a partial differential equation of the second order,

$$((A), (B), (C), (F), (G), (H)\ d_x, d_y, d_z)^2\, p = 0,$$

where the coefficients depend on the first and second differential coefficients of U, if $U = 0$ is the equation of the given surface. Now, considering the given surface as belonging to a family, or writing its equation in the form $r - r(x, y, z) = 0$ (the last r a functional symbol), the condition in order that the vicinal surface shall belong to this family, or say that it shall coincide with the surface $r + \delta r - r(x, y, z) = 0$, is $p = \dfrac{\delta r}{V}$, where $V = \sqrt{X^2 + Y^2 + Z^2}$, if X, Y, Z are the first differential coefficients of $r(x, y, z)$, that is, of the parameter r considered as a function of the coordinates; we have thus the equation

$$((A), (B), (C), (F), (G), (H)\ d_x, d_y, d_z)^2\, \tfrac{1}{V} = 0,$$

viz. the coefficients being functions of the first and second differential coefficients of r, and V being a function of the first differential coefficients of r, this is in fact a relation involving the first, second, and third differential coefficients of r, or it is the partial differential equation to be satisfied by the parameter r considered as a function of the coordinates. After all reductions, this equation assumes the form previously mentioned.

Article Nos. 1 to 21. On the Curvature of Surfaces.

1. Curvature is a metrical theory having reference to the circle at infinity; each point in space may be regarded as the vertex of a cone passing through this circle, say the circular cone; a line and plane through the vertex are at right angles to each other when they are polar line and polar plane in regard to the cone; and so two lines or two planes are at right angles when they are harmonics in regard to the cone, that is, when each line lies in the polar plane, or each plane passes through the polar line of the other. A plane through the vertex meets the cone in two lines, which are the "circular lines" in the plane and through the point; a line through the vertex has through it two tangent planes, which might be called the "circular planes" of the point and through the line; but the term is hardly required. Lines in the plane and through the point, at right angles to each other, are also harmonics (polar lines) in regard to the two circular lines.

2. Consider now a surface, and any point thereof; we have at this point a tangent plane and a normal. The tangent plane meets the surface in a curve having

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at the point a node, and the tangents to the two branches of the curve (being of course lines in the tangent plane) are the "chief tangents" of the surface at the point.

3. The chief tangents are the intersections of the tangent plane by a quadric cone, which may be called the chief cone; but it is important to observe that this cone is not independent of the particular form under which the equation of the surface is presented. To explain this, suppose that the rational equation of the surface is $U = 0$; taking ξ, η, ζ as current coordinates measured from the point as origin, the equation of the chief cone is $(\xi d_x + \eta d_y + \zeta d_z)^2 U = 0$, where x, y, z denote the coordinates of the point. But it is in the sequel necessary to present the equation of the surface in a different manner; say we have an equation between the coordinates (x, y, z) and a parameter r (r being therefore in general an irrational function of x, y, z), which, when $r = r_1$, reduces itself to $U = 0$: we have then $r = r_1$ as the equation of the surface; and the corresponding equation of the chief cone is $(\xi d_x + \eta d_y + \zeta d_z)^2 r = 0$; this is not the same as the cone $(\xi d_x + \eta d_y + \zeta d_z)^2 U = 0$, although of course it intersects the tangent plane in the same two lines, viz. the chief lines; and so in general there is a distinct chief cone corresponding to each form of the equation of the surface. But adopting a definite form of equation, we have a definite chief cone intersecting the tangent plane in the chief tangents.

4. Observe that the equations $U = 0$, $r = r_1$, although each relating to one and the same surface, serve to represent this surface, and that in different ways, as belonging to a family of surfaces, viz. one of these is the family $U = \text{const.}$, and the other the family $r = \text{const.}$ In order to represent a given surface as belonging to a certain family, we need the irrational form of equation; thus r denoting the irrational function of x, y, z determined by the equation

$$\frac{x^2}{a+r} + \frac{y^2}{b+r} + \frac{z^2}{c+r} = 1,$$

we have $r = 0$ as the equation of the ellipsoid $\dfrac{x^2}{a} + \dfrac{y^2}{b} + \dfrac{z^2}{c} = 1$, considered as belonging to a family of confocal quadrics.

5. Although at first sight presenting some difficulty, it is convenient to use the same letter r to denote the parameter considered as a function of the coordinates, and the special value of the parameter; thus in general the equation of a surface may be written $r(x,y,z) - r = 0$ (in which form the first r may be regarded as a functional symbol), or simply $r - r = 0$, viz. the first r here denotes the given function of (x,y,z), and the second r the particular value of the parameter.

6. By what precedes we have through the point and in the tangent plane two circular lines, the intersections of the tangent plane by the circular cone having the point for its vertex.

We have also through the point and in the tangent plane two other lines, termed the principal tangents, viz. the definition of these is that they are the double (or sibiconjugate) lines of the involution formed by the circular lines and the chief tangents, or, what is the same thing, they are the bisectors (and as such at right angles to each other) of the angles formed by the chief tangents.

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7. The principal tangents may also be considered as the intersections of the tangent plane by a quadric cone, called the principal cone; this being a cone constructed by means of the circular cone and the chief cone, and thus depending on the particular chief cone, that is, on the form of the equation of the surface. The definition is that the principal cone is the locus of a line (through the point), such that the line itself, the perpendicular (or harmonic in regard to the circular cone) of the polar plane of the line in regard to the chief cone, and the normal of the surface are *in plano*.

8. Analytically, taking, as before, (x,y,z) for the coordinates of the point, and u, v, w as current coordinates measured from the point as origin, then the equation of the circular cone is $u^2+v^2+w^2=0$; and taking $Xu+Yv+Zw=0$ for the equation of the tangent plane, and $(a,b,c,f,g,h \;\, u,v,w)^2 = 0$ for that of the chief cone, then, if the line be $u:v:w = \xi:\eta:\zeta$, we have

$$(a,b,c,f,g,h \;\, \xi,\eta,\zeta \;\, u,v,w) = 0$$

for the equation of the polar plane, and thence

$$u:v:w = \begin{vmatrix} a\xi+h\eta+g\zeta & h\xi+b\eta+f\zeta & g\xi+f\eta+c\zeta \\ X & Y & Z \end{vmatrix}$$

for those of the perpendicular, or harmonic in regard to the circular cone; also for the normal $u,v,w = X:Y:Z$; whence, if the three lines are *in plano*, we have

$$\begin{vmatrix} \xi & \eta & \zeta \\ a\xi+h\eta+g\zeta & h\xi+b\eta+f\zeta & g\xi+f\eta+c\zeta \\ X & Y & Z \end{vmatrix} = 0$$

as the equation of the principal cone. This is in the sequel written, for shortness, as

$$\begin{vmatrix} \xi & \eta & \zeta \\ \delta\xi & \delta\eta & \delta\zeta \\ X & Y & Z \end{vmatrix} = 0.$$

9. Consider any point P', not in general on the surface, in the neighbourhood of the point on the surface, say P; then the point P' has in regard to the surface a polar plane, which plane, however, is dependent on the particular form of equation—viz. x', y', z' being the coordinates of P', and U' the same function of these that U is of x,y,z, then the form $U=0$ of the equation of the surface gives for P' the polar plane $(ud_{x'}+vd_{y'}+wd_{z'})\,U'=0$; and we may through P' draw hereto a perpendicular (or harmonic in regard to the circular cone), say this is the normal line of P'. Then for points P' in the neighbourhood of P, when these are such that their normal lines meet the normal at P, the locus of P' is the before-mentioned principal cone. The analytical investigation presents no difficulty.

10. Taking P' on the surface, the normal line of P' becomes the normal at a consecutive point P' of the surface (being now a line independent of the particular form of equation), and this normal meets the normal at P; that is, we have the

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principal cone meeting the tangent plane in two lines, the principal tangents, such that at a consecutive point P' on either of these the normal meets the normal at P; viz. we have the principal tangents at the tangents of the two curves of curvature through the point P.

The plane through the normal and a principal tangent is termed a principal plane; we have thus at the point of the surface two principal planes, forming with the tangent plane an orthogonal triad of planes.

11. I proceed to further develop the theory, commencing with the following lemma:

Lemma. *Given the line $Xu + Yv + Zw = 0$, and conic*

$$(a, b, c, f, g, h \;\; u, v, w)^2 = 0,$$

then, to determine the coordinates (u_1, v_1, w_1), (u_2, v_2, w_2) of the points of intersection of the line and conic, we have

$$u : v : w = Y\zeta - Z\eta \; : \; Z\xi - X\zeta \; : \; X\eta - Y\xi,$$

or, what is the same thing, we have

$$(a, \ldots \;\; Y\zeta - Z\eta,\, Z\xi - X\zeta,\, X\eta - Y\xi)^2 = 0$$

as the equation, in line coordinates, of the two points of intersection. The proof is obvious.

12. Making the equations refer to a plane and a cone, and writing throughout ξ, η, ζ as current point coordinates, the theorem is:

Given the plane $X\xi + Y\eta + Z\zeta = 0$, and cone

$$(a, b, c, f, g, h \;\; \xi, \eta, \zeta)^2 = 0;$$

then, to determine the lines of intersection of the plane and cone, we have

$$(a, \ldots \;\; Y\zeta - Z\eta,\, Z\xi - X\zeta,\, X\eta - Y\xi)^2 = 0$$

as the equation of the pair of planes at right angles to the two lines respectively.

13. Denoting the coefficients by (a), (b), &c., that is, writing

$$(a, \ldots \;\; Y\zeta - Z\eta,\, Z\xi - X\zeta,\, X\eta - Y\xi)^2 = ((a), (b), (c), (f), (g), (h) \;\; \xi, \eta, \zeta)^2,$$

the values of these are

$$\begin{aligned}
(a) &= bZ^2 + cY^2 - 2fYZ, \\
(b) &= cX^2 + aZ^2 - 2gZX, \\
(c) &= aY^2 + bX^2 - 2hXY, \\
(f) &= -aYZ - fX^2 + gXY + hXZ, \\
(g) &= -bZX + fYX - gY^2 + hYZ, \\
(h) &= -cXY + fZX + gZY - hZ^2.
\end{aligned}$$

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We have the following identities:

$$\begin{aligned}&(a)X+(h)Y+(g)Z=0,\\&(h)X+(b)Y+(f)Z=0,\\&(g)X+(f)Y+(c)Z=0,\\&((a),\dots\ X,Y,Z\ \dots)=-(X^2,Y^2,Z^2,YZ,ZX,XY)\,\phi,\end{aligned}$$

that is, $(b)(c)-(f)^2=-X^2\phi$, &c., where

$$\phi=(a,\dots\ gh-af,\dots\ X,Y,Z)^2.$$

Writing also

$$aX+hY+gZ,\quad hX+bY+fZ,\quad gX+fY+cZ\ =\ \delta X,\,\delta Y,\,\delta Z,$$

and $X^2+Y^2+Z^2=V^2$; also $a+b+c=\omega$, then

$$\begin{aligned}(a)&=(b+c)V^2-\omega X^2+X\delta X-Y\delta Y-Z\delta Z,\\(b)&=(c+a)V^2-\omega Y^2-X\delta X+Y\delta Y-Z\delta Z,\\(c)&=(a+b)V^2-\omega Z^2-X\delta X-Y\delta Y+Z\delta Z,\\(f)&=-fV^2-\omega YZ+Y\delta Z+Z\delta Y,\\(g)&=-gV^2-\omega ZX+Z\delta X+X\delta Z,\\(h)&=-hV^2-\omega XY+X\delta Y+Y\delta X.\end{aligned}$$

14. I give also the following lemma:

Lemma. *The condition in order that the plane $X\xi+Y\eta+Z\zeta=0$ may meet the cones*

$$\begin{aligned}(A,B,C,F,G,H\ \xi,\eta,\zeta)^2&=0,\\(A',B',C',F',G',H'\ \xi,\eta,\zeta)^2&=0\end{aligned}$$

in two pairs of lines harmonically related to each other, is

$$(BC'+B'C-2FF',\dots,\ GH'+G'H-AF'-A'F,\dots\ X,Y,Z)^2=0.$$

Writing here

$$(A,\dots\ Y\zeta-Z\eta,\ Z\xi-X\zeta,\ X\eta-Y\xi)^2=((A),(B),(C),(F),(G),(H)\ \xi,\eta,\zeta)^2,$$

that is, $(A)=BZ^2+CY^2-2FYZ$, &c., the condition may be written

$$(A,\dots\ A',B',C',F',G',H')^2=0,$$

or say

$$(A)A'+(B)B'+(C)C'+2(F)F'+2(G)G'+2(H)H'=0,$$

and we may, it is clear, interchange the accented and unaccented letters respectively.

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15. I take $r-r=0$ for the equation of a surface, X, Y, Z for the first derived functions of r, (a,b,c,f,g,h) for the second derived functions. The equation of the tangent plane at the point (x,y,z), taking ξ, η, ζ as current coordinates measured from this point, is

$$X\xi+Y\eta+Z\zeta=0;$$

the equation of the chief cone in regard to this form of the equation of the surface is

$$(a, b, c, f, g, h \;\xi, \eta, \zeta)^2 = 0,$$

and the equation of the circular cone is $\xi^2 + \eta^2 + \zeta^2 = 0$, or, what is the same thing,

$$(1, 1, 1, 0, 0, 0 \;\xi, \eta, \zeta)^2 = 0.$$

Imagine a quadric cone

$$(A, B, C, F, G, H \;\xi, \eta, \zeta)^2 = 0,$$

such that it meets the tangent plane in the sibiconjugate lines of the involution formed by the intersections of the tangent plane by the chief cone and the circular cone respectively; that is, in a pair of lines harmonically related to the intersections with the chief cone, and also to the intersections with the circular cone; the conditions are

$$(A, \ldots \;a, b, c, f, g, h)^2 = 0,$$

and

$$(A) + (B) + (C) = 0,$$

viz. if only these two conditions are satisfied the cone will intersect the tangent plane in the two principal tangents.

16. The principal cone, writing, for shortness,

$$aX + hY + gZ, \quad hX + bY + fZ, \quad gX + fY + cZ \;=\; \delta X,\, \delta Y,\, \delta Z,$$

was before taken to be the cone

$$\begin{vmatrix} \xi & \eta & \zeta \\ \delta\xi & \delta\eta & \delta\zeta \\ X & Y & Z \end{vmatrix} = 0.$$

Representing this equation by

$$\tfrac{1}{2}(A, B, C, F, G, H \;\xi, \eta, \zeta)^2 = 0,$$

the expressions of the coefficients are

$$\begin{aligned} A &= 2hZ - 2gY, \\ B &= 2fX - 2hZ, \\ C &= 2gY - 2fX, \\ F &= hY - gZ - (b-c)X, \\ G &= fZ - hX - (c-a)Y, \\ H &= gX - fY - (a-b)Z. \end{aligned}$$

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These values give

$$\begin{aligned} AX + HY + GZ &= Z\delta Y - Y\delta Z, \\ HX + BY + FZ &= X\delta Z - Z\delta X, \\ GX + FY + CZ &= Y\delta X - X\delta Y; \end{aligned}$$

whence also

$$(A, \ldots \;X, Y, Z)^2 = 0,$$

as is, in fact, at once obvious from the determinant-form; and also

$$A + B + C = 0.$$

17. Writing for shortness

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}) = (bc - f^2,\ ca - g^2,\ ab - h^2,\ gh - af,\ hf - bg,\ fg - ch),$$

we find

$$\begin{aligned} A\mathfrak{a} + H\mathfrak{h} + G\mathfrak{g} &= \omega(hZ - gY) + hZ - gY, \\ H\mathfrak{h} + B\mathfrak{b} + F\mathfrak{f} &= \omega(fX - hZ) + fX - hZ, \\ G\mathfrak{g} + F\mathfrak{f} + C\mathfrak{c} &= \omega(gY - fX) + gY - fX; \end{aligned}$$

whence

$$(A, \dots \ \mathfrak{a}, \dots) = 0.$$

18. By what precedes, we have

$$((A), \dots \ \xi, \eta, \zeta)^2 = 0$$

for the equation of the two principal planes, where the coefficients (A), (B), &c. are functions of A, B, &c. and of X, Y, Z, as mentioned above. These coefficients satisfy of course the several relations similar to those satisfied by (a), (b), &c., and other relations dependent on the expressions of A, B, &c. in terms of a, b, &c. and X, Y, Z.

19. Proceeding to consider the coefficients (A), (B), &c., we have then

$$(A) + (B) + (C) = (A + B + C)V^2 - (A, \dots \ X, Y, Z)^2,$$

that is

$$(A) + (B) + (C) = 0.$$

Observing the relation $A + B + C = 0$, the equations analogous to

$$(a) = (b + c)V^2 - (a + b + c)X^2 + \&c. \quad \text{are} \quad (A) = -AV^2 + X\delta'X - Y\delta'Y - Z\delta'Z, \ \&c.$$

if for a moment we write $\delta'X$, $\delta'Y$, $\delta'Z$ to denote the functions

$$AX + HY + GZ, \quad HX + BY + FZ, \quad GX + FY + CZ.$$

But, from the above values, $X\delta'X + Y\delta'Y + Z\delta'Z = 0$, or the equation is $(A) = -AV^2 + 2X\delta'X$, that is $= -AV^2 + 2X(Z\delta Y - Y\delta Z)$. The equation for (F) is $(F) = -FV^2 + Y\delta'Z + Z\delta'Y$, where $Y\delta'Z + Z\delta'Y$ is $= Y(Y\delta X - X\delta Y) + Z(X\delta Z - Z\delta X)$, viz. this is

$$= (Y^2 - Z^2)\delta X - XY\delta Y + XZ\delta Z.$$

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We have thus the system of equations

$$\begin{aligned} (A) &= -AV^2 \qquad\qquad + 2XZ\delta Y - 2XY\delta Z, \\ (B) &= -BV^2 - 2YZ\delta X \qquad\qquad + 2XY\delta Z, \\ (C) &= -CV^2 + 2YZ\delta X - 2XZ\delta Y, \\ (F) &= -FV^2 + (Y^2 - Z^2)\delta X - XY\delta Y + XZ\delta Z, \\ (G) &= -GV^2 + XY\delta X + (Z^2 - X^2)\delta Y - YZ\delta Z, \\ (H) &= -HV^2 - XZ\delta X + YZ\delta Y + (X^2 - Y^2)\delta Z. \end{aligned}$$

20. We hence find

$$(A)a+(H)h+(G)g=-(Aa+Hh+Gg)V^2+(Z\delta Y-Y\delta Z)\delta X+XP,$$
$$(H)h+(B)b+(F)f=-(Hh+Bb+Ff)V^2+(X\delta Z-Z\delta X)\delta Y+YQ,$$
$$(G)g+(F)f+(C)c=-(Gg+Ff+Cc)V^2+(Y\delta X-X\delta Y)\delta Z+ZR,$$

if for shortness

$$P=(gY-hZ)\delta X+(aZ-gX)\delta Y+(hX-aY)\delta Z,$$
$$Q=(fY-bZ)\delta X+(hZ-fX)\delta Y+(bX-hY)\delta Z,$$
$$R=(cY-fZ)\delta X+(gZ-cX)\delta Y+(fX-gY)\delta Z.$$

Forming the sum $PX+QY+RZ$, the coefficient of δX is found to be

$$=-Z(hX+bY+fZ)+Y(gX+fY+cZ),\ =\ -Z\delta Y+Y\delta Z;$$

hence the whole is

$$=\delta X(Y\delta Z-Z\delta Y)+\delta Y(Z\delta X-X\delta Z)+\delta Z(X\delta Y-Y\delta X),\ \text{which is }=0,\ \text{that is,}$$
$$PX+QY+RZ=0.$$

21. Hence, adding, we find

$$((A),\dots\ a,\dots)=0,$$

viz. in this and the before-mentioned equation

$$(A)+(B)+(C)=0,$$

we have the a posteriori verification that the cone $(A,\dots\ \xi,\eta,\zeta)^2=0$ cuts the tangent plane in the double lines of the involution.

In what precedes I have given only those relations between the several sets of quantities a, $\mathfrak{a}$, (a), A, (A), &c. which have been required for establishing the results last obtained; but there are various other relations required in the sequel, and which will be obtained as they are wanted.

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22. We consider a surface $U=0$ (or $r=r$), and at each point P thereof measure along the normal an infinitesimal length p, dependent on the position of the point P (that is, p is a function of x,y,z). We have thus a point P', the coordinates of which are

$$x',\ y',\ z'=x+p\alpha,\ \ y+p\beta,\ \ z+p\gamma,$$

where α, β, γ are the cosine-inclinations of the normal, that is,

$$\alpha,\ \beta,\ \gamma=\tfrac{X}{V},\ \tfrac{Y}{V},\ \tfrac{Z}{V},\ \text{if } V=\sqrt{X^2+Y^2+Z^2};$$

the locus of P' is of course a surface, say the vicinal surface, and we require to find the direction of the normal at P', or, what is the same thing, the differential equation $X'\,dx'+Y'\,dy'+Z'\,dz'=0$ of the surface. We have

$$dx'=(1+d_xp\alpha)\,dx+d_yp\alpha\,dy+d_zp\alpha\,dz,$$
$$dy'=d_xp\beta\,dx+(1+d_yp\beta)\,dy+d_zp\beta\,dz,$$
$$dz'=d_xp\gamma\,dx+d_yp\gamma\,dy+(1+d_zp\gamma)\,dz,$$
$$0=X\,dx+Y\,dy+Z\,dz;$$

whence, eliminating dx, dy, dz, we have between dx', dy', dz' a linear equation, the coefficients of which may be taken to be X', Y', Z'. Taking these only as far as the first power of p, we have

$$X' = X(1 + d_y p\beta + d_z p\gamma) - Y d_x p\beta - Z d_x p\gamma,$$

or, what is the same thing,

$$X' = X(1 + d_x p\alpha + d_y p\beta + d_z p\gamma) - X d_x p\alpha - Y d_x p\beta - Z d_x p\gamma,$$

with the like expressions for Y' and Z'. I proceed to reduce these. The formula for X' is

$$X' = X\{1 + p(d_x\alpha + d_y\beta + d_z\gamma) + \alpha d_x p + \beta d_y p + \gamma d_z p\}$$

$$-p(X d_x\alpha + Y d_x\beta + Z d_x\gamma) - (\alpha X + \beta Y + \gamma Z)\, d_x p.$$

23. I write, for shortness, $\delta = X d_x + Y d_y + Z d_z$, whence $\delta X, \delta Y, \delta Z = aX + hY + gZ$, $hX + bY + fZ$, $gX + fY + cZ$, agreeing with the former significations of $\delta X, \delta Y, \delta Z$; also $V d_x V$, $V d_y V$, $V d_z V = \delta X, \delta Y, \delta Z$, and $V\delta V = X\delta X + Y\delta Y + Z\delta Z$. It is now easy to form the values of

$$\begin{array}{ll}
d_x\alpha,\ d_x\beta,\ d_x\gamma, \quad \text{viz. these are} & \dfrac{a}{V} - \dfrac{X\delta X}{V^3},\ \ \dfrac{h}{V} - \dfrac{Y\delta X}{V^3},\ \ \dfrac{g}{V} - \dfrac{Z\delta X}{V^3}, \\
d_y\alpha,\ d_y\beta,\ d_y\gamma, & \dfrac{h}{V} - \dfrac{X\delta Y}{V^3},\ \ \dfrac{b}{V} - \dfrac{Y\delta Y}{V^3},\ \ \dfrac{f}{V} - \dfrac{Z\delta Y}{V^3}, \\
d_z\alpha,\ d_z\beta,\ d_z\gamma, & \dfrac{g}{V} - \dfrac{X\delta Z}{V^3},\ \ \dfrac{f}{V} - \dfrac{Y\delta Z}{V^3},\ \ \dfrac{c}{V} - \dfrac{Z\delta Z}{V^3};
\end{array}$$

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and hence

$$d_x\alpha + d_y\beta + d_z\gamma = \frac{a+b+c}{V} - \frac{\delta V}{V^2},$$

$$X d_x\alpha + Y d_x\beta + Z d_x\gamma = \frac{\delta X}{V} - \frac{V^2}{V^3}\delta X,\ =\ 0,$$

and we have

$$X' = X\left\{1 + p\left(\frac{a+b+c}{V} - \frac{\delta V}{V^2}\right) + \frac{1}{V}\delta p\right\} - V d_x p,$$

with the like values of Y' and Z'. But we are only concerned with the ratios $X' : Y' : Z'$; whence, dividing the foregoing values by the coefficient in $\{\ \}$, and taking the second terms only to the first order in p, we have simply

$$X',\ Y',\ Z' = X - V d_x p,\ \ Y - V d_y p,\ \ Z - V d_z p.$$

24. We may investigate the condition in order that the surface x', y', z' may be the consecutive surface $r + dr = r(x, y, z)$. This will be the case of

$$r + dr = r\left(x + p\,\tfrac{X}{V},\ \ y + p\,\tfrac{Y}{V},\ \ z + p\,\tfrac{Z}{V}\right),$$

that is, $r + dr = r + pV$, or $p = \dfrac{dr}{V}$. This value of p gives $d_x p = -\dfrac{dr}{V^2} d_x V = -\dfrac{dr}{V^3}\delta X$, and similarly $d_y p = -\dfrac{dr}{V^3}\delta Y$, $d_z p = -\dfrac{dr}{V^3}\delta Z$; whence

$$X',\ Y',\ Z' = X + \tfrac{dr}{V^2}\delta X,\ \ Y + \tfrac{dr}{V^2}\delta Y,\ \ Z + \tfrac{dr}{V^2}\delta Z,$$

which is as it should be, viz. these are what X, Y, Z become on substituting therein for x, y, z the values $x + p\alpha$, $y + p\beta$, $z + p\gamma$.

25. I return to the case where p is arbitrary, and I investigate the values of a, b, ... for the point P' on the vicinal surface; say these are a', b', &c., then we have $a' = d_{x'}X'$ &c. The relation between the differentials may be written

$$\begin{aligned}
dx &= (1 - d_x p\alpha)\, dx' - d_y p\alpha\, dy' - d_z p\alpha\, dz',\\
dy &= -d_x p\beta\, dx' + (1 - d_y p\beta)\, dy' - d_z p\beta\, dz',\\
dz &= -d_x p\gamma\, dx' - d_y p\gamma\, dy' + (1 - d_z p\gamma)\, dz',
\end{aligned}$$

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and we thence have $d_{x'} = (1 - d_x p\alpha)\, d_x - d_x p\beta\, d_y - d_x p\gamma\, d_z$ &c.; hence

$$\begin{aligned}
a' &= \{(1 - d_x p\alpha)\, d_x - d_x p\beta\, d_y - d_x p\gamma\, d_z\}(X - V d_x p)\\
&= (1 - d_x p\alpha)\, a - d_x p\beta \cdot h - d_x p\gamma \cdot g - d_x(V d_x p)\\
&= a - p\,(a d_x\alpha + h d_x\beta + g d_x\gamma) - \tfrac{1}{V}\delta X\, d_x p - V d_x^2 p;
\end{aligned}$$

and similarly, $f' = d_{y'}Z'$ (or $d_{z'}Y'$), that is

$$\begin{aligned}
f' = f - p\,(g d_y\alpha + f d_y\beta + c d_y\gamma) - (g\alpha + f\beta + c\gamma)\, d_y\\
- \tfrac{1}{V}\delta Y\, d_z p - V d_y d_z p.
\end{aligned}$$

26. Completing the reduction, we find

$$\begin{aligned}
a' &= a - p\left(\frac{a\omega - b - c}{V} - \frac{(\delta X)^2}{V^3}\right) - \tfrac{2}{V}\delta X\, d_x p - V d_x^2 p,\\
b' &= b - p\left(\frac{b\omega - c - a}{V} - \frac{(\delta Y)^2}{V^3}\right) - \tfrac{2}{V}\delta Y\, d_y p - V d_y^2 p,\\
c' &= c - p\left(\frac{c\omega - a - b}{V} - \frac{(\delta Z)^2}{V^3}\right) - \tfrac{2}{V}\delta Z\, d_z p - V d_z^2 p,\\
f' &= f - p\left(\frac{f\omega - f}{V} - \frac{\delta Y\delta Z}{V^3}\right) - \tfrac{1}{V}(\delta Y\, d_z p + \delta Z\, d_y p) - V d_y d_z p,\\
g' &= g - p\left(\frac{g\omega - g}{V} - \frac{\delta Z\delta X}{V^3}\right) - \tfrac{1}{V}(\delta Z\, d_x p + \delta X\, d_z p) - V d_z d_x p,\\
h' &= h - p\left(\frac{h\omega - h}{V} - \frac{\delta X\delta Y}{V^3}\right) - \tfrac{1}{V}(\delta X\, d_y p + \delta Y\, d_x p) - V d_x d_y p;
\end{aligned}$$

say these expressions are $a' = a + \Delta a$, &c.

27. Taking ξ, η, ζ for the coordinates, referred to P as origin, of a point on the given surface near to P, and ξ', η', ζ' for the coordinates, referred to P' as origin, of the corresponding point on the vicinal surface, the relation between ξ', η', ζ' and ξ, η, ζ is the same as that between dx', dy', dz' and dx, dy, dz; viz. we have

$$\begin{aligned}
\xi' &= (1 - d_x p\alpha)\xi - d_y p\alpha\,\eta - d_z p\alpha\,\zeta,\\
\eta' &= -d_x p\beta\,\xi + (1 - d_y p\beta)\eta - d_z p\beta\,\zeta,\\
\zeta' &= -d_x p\gamma\,\xi - d_y p\gamma\,\eta + (1 - d_z p\gamma)\zeta.
\end{aligned}$$

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or conversely

$$\xi' = (1+a_1)\alpha\xi + d_{12}\alpha\eta + d_{13}\alpha\zeta,$$

$$\eta' = d_{21}\beta\xi + (1+d_{22})\beta\eta + d_{23}\beta\zeta,$$

$$\zeta' = d_{31}\gamma\xi + d_{32}\gamma\eta + (1+d_{33})\gamma\zeta,$$

say ξ', η', $\zeta' = \xi+\Delta\xi$, $\eta+\Delta\eta$, $\zeta+\Delta\zeta$; hence

$$X'\xi' + Y'\eta' + Z'\zeta' = (X' - V\,d_1x)(\xi+\Delta\xi) + \&c.$$

$$= X\xi + Y\eta + Z\zeta$$

$$+X\Delta\xi + Y\Delta\eta + Z\Delta\zeta$$

$$-V(\,d_1x\,\xi + \omega\,d_2y\,\eta +\ d_3z\,\zeta),$$

where second line is

$$(Xa + Y\beta + Z\gamma)(\,d_1x\,\xi + \omega\,d_2y\,\eta +\ d_3z\,\zeta)$$

$$+\mu\{X\,d_1y\,\eta + Y\,d_2y\,\xi + \&c.\} + (X\Delta_1\xi + Y\Delta_2\eta + Z\Delta_3\zeta) + \omega(X\Delta_2\xi + Y\Delta_3\eta + Z\Delta_1\zeta).$$

But

$$X\Delta_1\alpha + Y\Delta_2\beta + Z\Delta_3\gamma = \frac{\partial\Sigma}{\partial\xi} - V\,d_1X = 0,$$

$$X\Delta_2\alpha + Y\Delta_3\beta + Z\Delta_1\gamma = 0,$$

$$X\Delta_3\alpha + Y\Delta_1\beta + Z\Delta_2\gamma = 0,$$

or second line is $= V(\,d_1x\,\xi + \omega\,d_2y\,\eta +\ d_3z\,\zeta)$; we have therefore

$$X'\xi' + Y'\eta' + Z'\zeta' = X\xi + Y\eta + Z\zeta;$$

We require

$$(A',\ B',\ C',\ F',\ G',\ H' \,\check{}\, \xi', \eta', \zeta')^2;$$

viz. to the first order Δ, this is

$$= (A, \ldots \,\check{}\, \xi, \eta, \zeta)^2$$

$$+2\,(\Delta \ldots \,\check{}\, \xi, \eta, \zeta)^2 + \&c.\ \&c.,$$

28. Here second line is

$$2\{A\xi + B\eta + G\xi\,\Delta\xi + (H\xi + B\eta + F\xi)\Delta\eta + (G\xi + F\eta + C\xi)\Delta\zeta\}$$

but

$$H\xi + B\eta + G\zeta = B\eta - VH\xi$$

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ X, & Y, & Z \\ \xi, & \eta, & \zeta \end{vmatrix}$$

$$H\xi + B\eta + F\zeta = B\eta - VH\xi$$

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ X, & Y, & Z \\ \xi, & \eta, & \zeta \end{vmatrix}$$

$$G\xi + F\eta + C\zeta = VH\xi + Xh\eta$$

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ X, & Y, & Z \\ \xi, & \eta, & \zeta \end{vmatrix}$$

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whence term in { } is

$$\begin{vmatrix} d\xi\,\Delta\xi, & d\xi\,\Delta\eta \\ d\eta\,\Delta\xi, & d\eta\,\Delta\eta \\ X, & Y, & Z \end{vmatrix} + \quad \text{a sum of similar terms,}$$

which might be written

$$\begin{vmatrix} d\xi, & d\eta, & d\zeta \\ \Delta\xi, & \Delta\eta, & \Delta\zeta \\ X, & Y, & Z \end{vmatrix} = \quad \begin{vmatrix} \xi, & \eta, & \zeta \\ X, & Y, & Z \end{vmatrix}$$

but it is perhaps more convenient to retain the second term in its original form.

29. As regards the first line, we have

$$A = BC' - F^2V$$

$$= V(b+\Delta b)(F + V\,d_1h) - \{(g + V\,d_2g)(F - V\,d_3g)\}$$

$$= A + V(\Delta_1 b\,d_1h - 2\,d_2g\,d_3g + 2gh\xi);$$

with similar expressions for the other coefficients. Attending only to the terms of the first order, we then obtain

$$A = A + 2\,(B\,\Delta_1 b - Vgh) - 2V(\,d_1h\xi - gh\xi),$$

$$B = B + 2\,(C\,\Delta_1 c - Vfg) - 2V(\,d_2h\xi - gh\xi),$$

$$C = C + 2\,(A\,\Delta_1 a - Vfh) - 2V(\,d_3h\xi - gh\xi),$$

$$F = F + V\,d_1\alpha - X\,d_2\alpha - Y(\,d_2c - \Delta\eta) - V(\xi\,d_1h\eta - gh\xi - gh\eta) + \xi h\eta,$$

$$G = G + V\,d_1\beta - X\,d_1\alpha - V(\Delta b - \Delta\eta) - V(\,d_2h\xi - gh\xi - gh\eta) + gh\eta,$$

$$H = H + X\,d_2\alpha - X\,d_1\beta - V(\Delta c - \Delta\xi) - V(\,d_3h\xi - fgh\xi - gh\eta) + gh\eta;$$

my these are $A' = A + bA$, &c., where b is a functional symbol; we then have

$$(A'\ldots\check{}\xi',\eta',\zeta')^2 = (A\ldots\check{}\xi,\eta,\zeta)^2 + (bA\ldots\check{}\xi,\eta,\zeta)^2 + 2\,(A\ldots\check{}\xi,\eta,\zeta)\,(\check{}\Delta\xi,\Delta\eta,\Delta\zeta);$$

which, for shortness, I represent by

$$(A\ldots\check{}\xi,\eta,\zeta)^2 + (bA\ldots\check{}\xi,\eta,\zeta)^2,$$

and I proceed to complete the calculation of the coefficients A', B', &c.

30. We have

$$A' = bA + \text{coeff.}\,b'^2\,\text{in}$$

$$2\{B\,\Delta b + G\xi\,\Delta\eta + (H\xi + B\eta + F\xi)\Delta_1 h + (G\eta + F\eta + C\xi)\Delta_3 h\}$$

$$= bA + 2\{aA\eta\,\Delta_1\alpha + 2gh\eta\eta\}$$

that is,

$$A' = bA + \tfrac{2}{V}(AX + HY + GZ)\,d_1\alpha$$

$$+2\eta\,(\,d_2b\,\Delta_2h + h\xi\,\Delta_3h)$$

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where coeff. $b\eta$ is

$$= \frac{A\eta + Hb + G\eta}{V} - \frac{(AX + HY + GZ)\eta\beta\eta}{V}$$

$$= \tfrac{1}{V}\Big\{(a\alpha - p\beta) - p^2\alpha + (\gamma + b)\beta + (gg + Gh)\, d_2\Big\}.$$

31. And similarly,

$$F' = bF + V(\Delta a - \Delta\eta) + V\, d_2 g + V\, d_3 g + V(\, d_1\alpha + b\eta\xi h + gh\xi + h\eta\xi\xi h)\, d\eta_c\, d\eta_c\eta + \, d_3 h\, d_c\, d_c\eta$$

$$+p\,(\, dX\,Yh + ZV\,F\, d_1 h)\, d\eta_c + (\xi h + gh)\, d_2\, d\eta_c + (Xb + HY + FZ)\, d_2 h\, d_c$$

$$-6F\, d_1 h - ph\xi - h(\, d_c\eta\eta\eta)\, d\eta,$$

which gives the following formulæ; viz. on these lines

$$F' = bF + V(\Delta_2\, YZ - EXY) + \tfrac{1}{p}\Big\{p(XHX + GLY)\, d_c\eta_c - V d\, d_2\Big\}$$

$$-2(VF\;\, d_1\alpha + V d\;\, d_c + V d\;\, d_c\eta + \&c)$$

$$+2V\;(\, d_c\eta + V d\;\, d_c\eta)\, d_c\eta$$

$$-2V\;(\, db\eta - p\xi)\, d_c\eta,$$

which to denote in the foregoing equation, together with the following terms:

$$= \tfrac{1}{p}(\alpha A + V d\,\alpha)\, d_c\eta + 2\,(\, db\eta - p\xi)\, d_c\eta,$$

which destroy certain of the foregoing ones; viz. we have

$$A' = (b + \tfrac{2FHX}{V})\, d_c\eta + 2\,(VH - VXY)\, d_c\eta.$$

32. Similarly, the value of bF is

$$bF = V\big\{-\tfrac{p}{V}\big(\alpha + h\big) + \, d_c\eta\, XY\big\} = \tfrac{1}{p}\big\{V(YHX + LXY)\, d_c\eta - V d\, d\xi$$

$$-2VY\big(\tfrac{1}{p}\, d_c p - p\beta\big)\xi\, d\eta + V d\, d\xi$$

$$-2VY\big(\, d_c\eta - p\beta\big) - 2VY\,(d\xi - \, d_c\xi)\, d\eta + V d\, d\xi$$

$$+p\, dV\, d\xi$$

$$-p\xi\, d\eta\, d_c + p\xi\, d\eta,$$

which is

$$= \tfrac{1}{p}(\alpha\, b\eta + Vh)\Big[\tfrac{p}{V}\big(YHX - EXY\big) + \tfrac{FVHX}{V}(VHX - XHY) + \tfrac{p}{V} d\xi\, d\eta\Big]$$

$$+\Big\{-\tfrac{1}{p}\Big(B\xi + EXY + V\,h - h\Big)\Big\}\eta\, d_c\eta$$

$$+\Big\{\tfrac{pX}{V} - \tfrac{2HY}{V} - Vh\Big\}\, d\eta\, d_c\eta$$

$$+(VY\, d\xi + VX\, d\eta + VXH\eta - VXY)p$$

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The value of θA is

$$\theta A = 2E\Big\{-\tfrac{p}{V}\big(b\alpha + h\big) + \tfrac{1}{p}\,d_c EXY\Big\} = \tfrac{1}{V}\,(2Vh\eta + LX\eta\, d\xi - V d\, d\xi)$$

$$-2VY\Big\{-\tfrac{p}{V}\big(b\eta + h\big) + \tfrac{1}{p}\,d_c\eta\, EXY\Big\} = \tfrac{1}{V}\,(GBX + LX\eta\, d\xi - V d\, d\xi)$$

$$-2V(bh\xi - p\eta\xi),$$

which is

$$= \tfrac{p}{V}\,(2b\alpha + b\eta) + \tfrac{p}{V}\big(XY - YX\big) - \tfrac{pX}{V}\big(XY - EXY\big)$$

$$-\tfrac{2b\xi}{V}\,(BX - Vh) - 2V(b\eta\, d\xi - p\xi\, d\eta).$$

Hence the value of A' is equal to the last-mentioned expression, together with the following terms:

$$= \tfrac{p}{V}\,(\alpha A + b\eta) + \tfrac{2XZ}{V}\Big[(b\xi)\, d_c\eta + 2F(d\xi - YH)\, d\xi - V(d\xi - YX)\, d\eta\Big]$$

which destroy certain of the foregoing ones; viz. we have here

$$A' = (\theta + \tfrac{2FHX}{V})\xi\, d\xi + 2\,(VH - VXY)\, d\xi.$$

And similarly, the value of δF is

$$\delta F = V\{-\tfrac{p}{V}\big(b\alpha + h\big) + \tfrac{1}{p}\,d_c LXY\}\, d\xi(LH + EXY)\, d\xi = V d\, d\xi$$

$$-2\left\{\tfrac{p}{V}\,(LH + LH) + \tfrac{pLH}{V}\right\} d\xi\, d\eta + V(LEH - V)\, d\xi$$

$$-2\left\{\tfrac{pXH}{V} - \tfrac{2HY}{V} - Vh\right\} d\xi\, d\eta - \tfrac{pH}{V}\Big(Vh - LH\Big) - Vh\, LH\, d\xi$$

$$-2\left\{\tfrac{p}{V} - 2Vh\right\} d\xi\, d\eta - V d\, h\, d\xi + VXY\, d\xi - VXY\, d\xi$$

which is

$$= \tfrac{p}{V}\,(2b\alpha - b\eta) + \tfrac{p}{V}\big(YHX - LXY\big) + \tfrac{pHY}{V}\big(YHX - XHY\big)$$

$$+\Big\{-\tfrac{1}{p}\big(\alpha LH + LXY)\, d\xi - Vh\Big\}\, d\xi$$

$$+\Big\{\tfrac{pL}{V}\big(LH + LH\big) + 2Vh\Big\}\, d\xi$$

$$+\Big\{\tfrac{pLH}{V} - LH\Big\}\, d\eta$$

$$+\Big\{-VV\, d_c\eta + VBX\, d_c\eta + VX\eta\, d_c\eta - VH\, d_c\eta\Big\}\, d\eta$$

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Hence F' is equal to the foregoing expression, together with the following terms:

$$+\tfrac{p}{V}\big(\beta A+\tfrac{2FHY}{V}\big)(XLE-EXY)-\tfrac{pHY}{V}\big(YXX-XXX\big)$$

$$+\tfrac{p}{V}\big(\beta A-2EXY\big)+\tfrac{p}{V}\left(XLE-EXY\right)$$

$$+\tfrac{pHL}{V}\big(LH+GLY\big)\,d\xi+\tfrac{pL}{V}\big(LXE+LXY\big)\,d\eta,$$

which destroy certain of the foregoing; viz. we then have

$$F'=bF+V(YE-EXY)+\tfrac{pHXY}{V}\,d\xi\,d\eta+\tfrac{p}{V}\,d\xi+V\,d\xi\,d\eta+\tfrac{HLXY}{V}\,d\xi$$

$$+VY\,d_c\eta\,d\eta+VHYH\,d_c\eta+VHYXH\,d_c\eta.$$

34. We have here

$$A'=2\Big(Yh-\tfrac{VHX}{V}\Big)\,d\xi+2\Big(Yh-\tfrac{VHX}{V}\Big)\,d\xi+2\Big(Yh-\tfrac{VHY}{V}\Big)\,d\xi+2V(Vh-XYh)\,d\xi+V(VY-VZ)\,d\xi$$

$$B'=-2\Big(Y-\tfrac{VHY}{V}\Big)\,d\xi+2\Big(Yh-\tfrac{VHY}{V}\Big)\,d\xi+2V(Vh-XYh)\,d\xi-VYh\,d\xi+EXYh\,d\xi$$

$$C'=-2\Big(Y-\tfrac{VHX}{V}\Big)\,d\xi+2\Big(Yh-\tfrac{VHY}{V}\Big)\,d\xi+2V(Yh-XYh)\,d\xi$$

$$F'=\Big\{V(h-1)+\tfrac{p}{V}\,(YE-LXY)\,d\xi+\Big(Yh-\tfrac{VHY}{V}\Big)\,d\xi+V(Vhh-LXY)\,d\xi\,d\eta\Big\}-VY_a\,d\xi-2bH\,d\xi-2V(bX)\,d\xi$$

$$G'=\Big(Yh-\tfrac{pXY}{V}\Big)\,d\xi+\Big\{V(h-1)-\tfrac{1}{p}\,(YXH-EXY)\,d\xi\Big\}+V(Vhh-VYh)\,d\xi+V\,Y_a\,d\xi-2bV\,d\xi-2V(VX)\,d\xi$$

$$H'=\Big(Yh-\tfrac{VHX}{V}\Big)\,d\xi-\Big(Yh-\tfrac{VYH}{V}\Big)\,d\xi+V(h-1)-\tfrac{1}{V}\,(LXH-VXH)\,d\xi+V(VXVh-VVX)\,d\xi+V\,d\xi$$

35. It will be recollected that we have $X'\xi'+Y'\eta'+Z'\zeta'=X\xi+Y\eta+Z\zeta$; by what precedes it appears that for the given surface the principal tangents are determined by the equations

$$(A\ldots\breve{}\,\xi,\eta,\zeta)^2=0,$$

$$X\xi+Y\eta+Z\zeta=0,$$

and that the lines which (in the tangent plane of the given surface) correspond to the principal tangents of the corresponding point of the wicinal surface are determined by the equations

$$(A\ldots\breve{}\,\xi,\eta,\zeta)^2+(\Delta\breve{}\,\xi,\eta,\zeta)^2=0,$$

$$X\xi+Y\eta+Z\zeta=0.$$

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Condition that the two surfaces may belong to an Orthogonal System.
Art. Nos. 36 to 44.

36. The condition in order that the two surfaces may belong to an orthogonal system is that the principal tangents shall correspond, or, what is the same thing, the lines which (in the tangent plane of the given surface) correspond to the principal tangents of the vicinal surface be the principal tangents of the given surface. When this is the case, the plane and now $X\xi+Y\eta+Z\zeta=0$, $(\Delta\ldots\check{}\xi)^2=0$, $(\xi'\eta'\zeta'$ intersect in the principal tangent, and this is therefore the required condition.

The plane $X\xi+Y\eta+Z\zeta=0$ meets the cone $(A\ldots\check{}\xi)^2=0$ in the principal tangents, that is, in pair of lines harmonically related to the circular lines and also to the chief tangents. Forming the two coefficients (A'), (B'), (C'), (F'), (G'), (H') of these terms in the same way as (A), &c. are formed from A, &c., they must therefore as it is the case for the chief tangents satisfy

$$(A')\xi^2+(B')\eta^2+(C')\zeta^2=0,$$

$$(F')\xi,\quad (G')\eta\xi,\quad \ldots=0,$$

or, what is the same thing,

$$(A'\ldots)\xi^2=0,$$

$$(A'\ldots)(\ldots)\xi^2=0.$$

The former of these, as about to be shown, is an identical identicality; we have therefore the second of these, say $A'\ldots$, &c. $=0$ as the required condition.

37. We have

$$(A')+(B')+(C')=A+B+C+V^2-(A\ldots\check{}X,Y,Z)^2,$$

$$B^2+F^2+C=(BC-F^2V)X^2+\cdots=V(XHYE\zeta-XXHY)^2-V(XX-HX)^2,$$

Forming next the expression of $A'X+B'Y+C'Z$, &c., and for convenience using down separately the terms which involve the second differentials of X, Y, Z, we have

$$A'X+B'Y+C'Z=(AX+HY+GZ)\,d_1\alpha$$

$$+\,d_2\alpha\,V(YXY\alpha)-V(XBXY-VXYh)\,d\xi-V(XXY-YXYh-VXYVh)\,d\eta\,d\xi$$

$$=V(XXYZH-VXYh)\xi$$

$$-2V(VHXH-VXYVh-VX)\xi\,d\xi\,d\eta+(XXY-VXYh)\,d\eta\,d\xi$$

where AV stands for $\frac{1}{V}(LX\xi+Vh\xi+VH)$, and so on, and where these expressions contain also the following terms respectively:

$$-VZ\,d_c\,\bullet+(X\,d_1\alpha-p\xi)X\,d_c\eta+X\,d_c\eta\,d_c-VZ\,d_c\xi\,d_c\eta\eta\eta$$

$$+ZE\,d_c\eta-p\xi\,d_c\eta+(\alpha-p\xi)\,d_c\eta+VX\xi\,d\xi\,d\eta\,d\xi$$

$$+XX\,d_c\eta+ZX\,d_c\eta+p\xi\,d_c\eta-VXY\,d_c\eta-V\,(\,dZ+V\,p)\,d\xi$$

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Multiplying by X, Y, Z and adding, the terms which contain the second differential coefficients disappear, and we obtain

$$(A' \ldots \check{} X, Y, Z)^2 = V^2\{BXY - YEZ\}\, d_c\alpha + (XKH - EZX)\, d_c\beta + (YKH - YHY)\, d_c\gamma\}$$

so that, attending to the above value of $A^2 + B^2 + C^2$, we have the required equation

$$(A' \ldots)\, d_c \check{} X, Y, Z)^2 = 0.$$

38. Proceeding now to form the value of $(A' \ldots \check{})(\ldots,)$, that is,

$$A'\,(\xi\eta + B'\,\eta\beta + C'\,\zeta\beta + 2F'\,\eta\zeta + 2G'\zeta\xi + 2H'\xi\eta),$$

as regards those containing the first differential coefficients of ρ, namely of themselves; as regards those containing the second differential coefficients, forming the auxiliary equations

$$\begin{aligned}
(A) &= 2\,(b)A - 2\,(g)V,\\
(B) &= 2\,(f)X - 2\,(h)V,\\
(C) &= 2\,(g)V - 2\,(f)Z,\\
(F) &= -(h)Y - (g)Z - (l)b + (c)\,Y,\\
(G) &= -(f)X - (g)Y + (l)b + (c)\,X,\\
(H) &= -(g)X - (f)Y + (h)b + (g)\,X,
\end{aligned}$$

we find without difficulty that the terms in question (being, in fact, the complete value of the expression) are

$$= V(\xi A \ldots \check{} d_a, d_b, d_c\,)\rho.$$

39. As regards the terms involving the first differential coefficients, observe that the whole coefficient of $b\rho$ is

$$\begin{aligned}
&= 2\,(\xi)\left(Yh - \tfrac{VHX}{V}\right)\\
&+2\,(\xi)\left(Vg - \tfrac{VHX}{V}\right)\\
&+2\,(\xi)\left(V(h-h) - \tfrac{p}{V}\,(YXY - EXY)\right)\\
&+2\,(\xi)\left(Yh - \tfrac{VHX}{V}\right)\\
&+2\,(\xi)\left(Yg - \tfrac{VHX}{V}\right)
\end{aligned}$$

which is

$$\begin{aligned}
&= 2V\Big\{(g)(\Delta + (g) + (l)b + (h)b + (l)f\Big\}b\\
&+\tfrac{2}{V}\,(\Delta\,(\xi b) + (l)\,LX + (g)\,(LXH + (f)\,LX + (h)\,LX\}b.
\end{aligned}$$

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40. The reduction depends on the following auxiliary formulæ:

$$a\,(g\eta+h)\Delta+p\,(g\eta-YH-XLX)\,(\eta\,b)+(l)\,(\xi)+(l)\,\xi+(l)\,(\xi)=-XEX,$$

$$l+g(\xi)\,h\big)+(l)\,(\xi)=-YEX,$$

$$a+b+(l)+\ldots=-ZLX,$$

$$b\,(\xi)\,k-q+V\ \ldots=-XEX,$$

$$y+g(\xi)\,h\big)+(l)\,(\xi)=-EXY,$$

$$g(\xi)\,k-q+V\ \ldots=-YEX,$$

where, for shortness, I have written LX, LY, LX to stand for $a\,X+g\,Y+b\,Z$, $\xi X+b\,Y+\xi Z$, $gX+\xi Y+g\,Z$ respectively, and YXY for $bX+Y\,dZ$, $dZ\,X+Y\,dZ$ for $dX+\,dZ$, etc. etc. for dX, dY, dZ.

From these we immediately have

$$(\xi)\,LX+(l)\,LY+(h)\,LZ=V(LX\eta-YEX),$$

$$(h)\,LX+(l)\,LY+(l)\,LZ=V(LX\,dY-YEY),$$

$$(g)\,LX+(l)\,LY+(l)\,LZ=V(LX\,dZ-YEZ),$$

Hence, in the coefficient of $b\rho$, the first term is

$$=2V(-YEX+YEX+YEX),$$

and the second line is

$$=\tfrac{1}{V}\{(VX\,(LX+\xi)\,LX)+2V\,(VLX-VEX)\,(LX)\}.$$

so that the sum, or whole coefficient of $b\rho$, is $=0$. Similarly, the coefficients of $b\rho$ and $\partial\rho$ are each $=0$.

41. We have thus arrived at the equation

$$(A\ldots\check{}\,d_a,d_b,d_c\,)\rho=0$$

as the condition to be satisfied by the normal distance ρ in order that the given surface and the trivial surface may belong to an orthogonal system; viz. this is a partial differential equation of the second order, its coefficients being given functions of X, Y, Z, a, b, c, f, g, h, the first and second differential coefficients of z (where $z=z(x,y,z)$ is the equation of the given surface).

The equation, it is clear, may also be written in the two forms

$$(A\ldots\check{}\,)\,d_a\,Y_c\,Z_c,Y\,d_a,Zd_b,X\,d_a\check{}\,\rho)\rho=0,$$

and

$$\begin{vmatrix} F^2 & G^2 & dX\,d_b \\ e\Gamma+b\,G\,d_c+p\,hX+e\,G\,d_cfX & p\,d_1+b\,hX & \frac{F^2}{V}\end{vmatrix}\rho=0,$$

if, for shortness, P, Q, R are written to denote $dZ_c-Y\,d_c\,d_c,X\,d_a\,d_c,Y\,d_c\,d_c,X_a\,d_c$, respectively, it being understood that on each of these forms the d_a, d_b, d_c operate on the ρ only.

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Condition that a family of surfaces may belong to an Orthogonal System. Art. Nos. 42 to 43.

42. We pass at once to the condition in order that the family of surfaces

$$v = c\,(x,\ y,\ z) = 0$$

may belong to an orthogonal system, viz. when the vicinal surface belongs to the family, we have ρ proportional to $\frac{1}{V}\left(=\frac{1}{\sqrt{X^2+Y^2+Z^2}}\right)$, and the condition is

$$(A\ldots\check{\ })d_a, d_b, d_c\check{\ }\tfrac{1}{V} = 0,$$

where v is a function of (x, y, z), the first and second differential coefficients of which are A, B, C, F, G, H, and the equation is then a partial differential equation of the third order satisfied by v. The form is by no means an inconvenient one, but it admits of further reduction.

43. We have $d_a\,\frac{1}{V}$, $d_b\,\frac{1}{V}$, $d_c\,\frac{1}{V}$ equal to $-\frac{1}{V^2}\,aX$, $-\frac{1}{V^2}\,LY$, $-\frac{1}{V^2}\,LZ$ respectively, and thence

$$d_a\,\tfrac{1}{V} = -\tfrac{1}{V^2}\,(aX + b\xi + g\xi\,d_a\xi) + \tfrac{3X}{V^4}\,(LX),$$

$$d_c\,d_a\,\tfrac{1}{V} = -\tfrac{1}{V^2}\,(g\xi + h + f\,d_a\xi)\,d_c + \tfrac{3X}{V^4}\,LXY,$$

or, as these may be written,

$$d_a\,d_c\,\tfrac{1}{V} = -\tfrac{1}{V^2}\,(\xi a + b\xi + b\eta)\,d_c + \tfrac{3X}{V^4}\,(LX),$$

$$d_c\,d_c\,\tfrac{1}{V} = -\tfrac{1}{V^2}\,(g\xi + g\eta + g\,d_c)\,d_c\eta + \tfrac{3X}{V^4}\,LXY,$$

with the like values for $d_c\,\frac{1}{V}$, &c. Substituting, the equation contains a term multiplied by a, viz. this is

$$= -\tfrac{1}{V^2}\,(A\ldots)\,d_c,$$

which vanishes; and a term multiplied by a, viz. this is

$$= \tfrac{6}{V^4}\,(LX + (\beta) + (\gamma))\,V,$$

which also vanishes. Writing down the remaining terms, and multiplying the whole by $-V$, the equation becomes

$$(A\ldots\check{\ })(\xi), \ldots\check{\ }(\xi), \ldots\check{\ }\ldots = -\tfrac{1}{V^2}\,(A\ldots\check{\ }LX, LY, LXY\check{\ } = 0.$$

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44. The last term admits of reduction; from the equations

$$(A) = AV^2 + 2YXV - 2XYE,\ \&c.,\ \text{we find}$$

$$(A)\,LX + (H)\,LY + (G)\,LZ = (AbY + BLY + GLZ) + VPF(XBLY - YLZ),$$

$$(H)\,LX + (B)\,LY + (F)\,LZ = (HXY + BLY + FLZ) + VPF(YGLZ - ZLX),$$

$$(G)\,LX + (F)\,LY + (C)\,LZ = (GLX + FLY + CLZ) + VPF(YLZ - XLY),$$

and hence

$$(A\ldots\check{}LX, LY, LXY\check{}) = V^2\,(A\ldots\check{}LX, LY, LZ),$$

wherefore the equation becomes

$$(A\ldots\check{})(\xi),\ldots\check{}) + 2\,(A\ldots\check{}LX, LY, LZ) = 0.$$

45. It will be shown that we have identically

$$(A\ldots\check{})(\xi),\ldots\check{}) = -(A\ldots\check{}\xi X, \xi Y, \xi Z)\begin{vmatrix} LX, & XY, & LZ \\ X, & Y, & Z \\ LX, & LY, & LZ \end{vmatrix}.$$

The partial differential equation thus assumes the form

$$(A\ldots\check{}\xi)\ldots = 0,$$

where Ω may be expressed indifferently in the three forms,

$$= 4\,(A\ldots\check{}\xi)\ldots,$$

$$= 4\,(A\ldots\check{}LX, LY, LZ),$$

$$= -4\begin{vmatrix} LX, & LY, & LZ \\ X, & Y, & Z \\ LX, & LY, & LZ \end{vmatrix}.$$

46. Taking the first of these, the partial differential equation is

$$(A\ldots\check{}\xi)\ldots = 2\,(A\ldots)\check{}\xi = 0,$$

or, written at full length, it is

$$(A)\,ba + (B)\,bb + (C)\,bc + 2\,(F)\,bf + 2\,(G)\,bg + 2\,(H)\,bh$$

$$-2\,(A,\ldots\check{}LX, LY, LZ) + (A,\ldots\check{}b\,a, b\,b, b\,c, b\,f, b\,g, b\,h) = 0,$$

where the coefficients are given functions of X, Y, Z, a, b, c, f, g, h, the first and second differential coefficients of v; and it is written to denote $Xd_a + Yd_b + Zd_c$, that is, $bX = Xd_aX + Yd_bX + Zd_cX$, and similarly for the others bY, bZ.

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47. It remains to prove the above-mentioned identicality.

To reduce the term $(A\ldots\between\xi X, LY, LZ)^2$, we have

$$\begin{aligned}&A\xi X + HLY + GLZ\\
&= A\,(aX + bY + gZ) + H(bX + bY + fZ) + G\,(gX + fY + cZ)\\
&= X\left[a\,(LY^2 - F^2) + b\,(HY - GZ) + g\,(GE - YH)\right]\\
&\quad + Y\left[a\,(LY - YH) - b\,(LX - YH) + g\,(GE - YH)\right]\\
&\quad + Z\left[a\,(LX - GE) + b\,(LX - HX) - g\,(LX - YH)\right]\\
&= a\,(LV^2 - XYE) - b\,(LV^2 + ZEX) - g\,(LV^2 + YHY);\end{aligned}$$

that is,

$$A\xi X + LV^2 + GLY = XV^2 - a\,(YEZ + ZLY) + b\,(ZLZ - XLX),$$

and similarly

$$HLX + BLY + FLZ = (YEZ - LXY) + V\,(a\,(XEX - XEY) + h\,(LX - LX),$$

$$GLX + FLY + CLZ = -a\,(LX - LX) + V\,(a\,(XEY - YH) - g\,(LX - LX),$$

whence

$$\begin{aligned}(A, B, C, F, G, H\between LX, LY, LZ)^2 &= V^2\,(LX + YLY + ZLZ)\\
&\quad -2\,(a\,(LY - LXY)(LXY - LXZ) + LXY h\,(LY - LXZ),\end{aligned}$$

48. Now, from the equations $AX + HY + GZ = -V\xi$, &c., we have for the value of even the foregoing determinant

$$\begin{aligned}&\left\{2\,d\xi_c\,d\eta - 2\,bX\,d\eta\,bX + 2\,(b\eta)\,b\,(bX + aY\,d\xi)\right\}b\\
&= (BP + CY^2 - 2FYZ)X^2\\
&\quad +(CX^2 + AZ^2 - 2GZX)Y^2\\
&\quad +(AY^2 + BX^2 - 2HXY)Z^2\\
&\quad +2\left(-AYZ + FX^2 + GXY + HXZ\right)YZ\\
&\quad +2\left(-BX^2Z + GY^2 + HYZ + FXY\right)ZX\\
&\quad +2\left(-CXY + HZ^2 + FZX + GZY\right)XY,\end{aligned}$$

and subtracting therefrom the function $(A,\ldots\between)$, which is

$$\begin{aligned}&= (BP + CY^2 + (AB - F^2)X^2\\
&\quad +(CY^2 + AB - F^2)Y^2\\
&\quad +(AY^2 + BX^2 - H^2)Z^2\\
&\quad -2\left(FY^2 + HZ + EXY\right)YZ\\
&\quad -2\left(GZ^2 + EX + EXY\right)ZX\\
&\quad -2\left(HX^2 + FEZ + EXY\right)XY,\end{aligned}$$

the difference is found to be

$$\begin{aligned}&= a\,(L\ \ldots\between YZ, ZP + GLY + GLZ)\\
&\quad +b\,(L\ \ldots\between EX, X^2 + EXY)\\
&\quad +c\,(L\ \ldots\between XY, Y^2 + EXY)\\
&\quad -2f\,(LX + B\,LY + GLZ)YZ\\
&\quad -2g\,(LX + B\,LY + GLZ + bLY)ZX\\
&\quad -2h\,(LX + GLY + HLY)XY.\end{aligned}$$

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which, on account of $(A \ldots \check{}\xi X, Y \check{} LZ)^2 = 0$, and $A + B + C = 0$, reduces itself to

$$(A \ldots \check{}\check{} \ldots)\, V.$$

49. We have

$$\begin{aligned} Aa + Bb + Cg &= a\,(BX - 2g)V \\ &+ b\left(\, g\,LY - fY - a\,(b\,h)\right) \\ &+ p\left(\, VZ - b\,LX - c\,b\,Z\right)Y \\ &= X\,(gb - bgy) \\ &+ V\,(gy - py - (pl + g + h + h\,h) + L\,LX \\ &+ Z\,(bn - h - (b\,h + b)\,b + f\,f)\,b), \end{aligned}$$

or, observing that in the coefficients of Y and Z the second terms each vanish, this is

$$Aa + Bb + Cg = X\,(b\,p - g)\,h + Y\,(ga - h)\,b + Z\,(\xi b - ha);$$

and similarly

$$Ha + Bb + Fg = X\,(g\,h - f) + Y\,(h - b\,h - c) + Z\,(\xi h - g\,g),$$

$$Gg + Fb + Cg = X\,(g\,h - g) + Y\,(gb - h)\,b + Z\,(gf - g\,g).$$

Adding these equations, the coefficient of X is the difference of two expressions each of which vanishes; and the like as regards the coefficients of Y and Z; that is, we have

$$(A \ldots \check{} \ldots) = 0,$$

and consequently

$$2\begin{vmatrix} LX, & LY, & LZ \\ X, & Y, & Z \\ LX, & LY, & LZ \end{vmatrix} = (A \ldots \check{} \ldots) = -(A \ldots \check{}\xi X, LY, LZ)^2,$$

the required relation.

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On the Centro-Surface of an Ellipsoid.

[From the Transactions of the Cambridge Philosophical Society, vol. XII. *Part i. (1873), pp. 319–365. Read March 7, 1870.]*

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THE centro-surface of any given surface is the locus of the centres of curvature of the given surface, or say it is the locus of the intersections of consecutive normals (the normals which intersect the normal at any particular point of the surface being those at the consecutive points along the two curves of curvature respectively which pass through the point on the surface). The terms, normal, centre of curvature, curve of curvature, may be understood in their ordinary sense, or in the generalised sense referring to the case where the Absolute (instead of being the imaginary circle at infinity) is any quadric surface whatever; viz. the normal at any point of a surface is here the line joining that point with the pole of the tangent plane in regard of the quadric surface called the Absolute; and of course the centre of curvature and curve of curvature refer to the normal so defined.

The question of the centro-surface of a quadric surface has been considered in the two points of view, viz. 1°, when the term "normal," &c., are used in the ordinary sense, and the equation of the quadric surface (assumed to be an ellipsoid) is taken to be

$$\frac{X^2}{a} + \frac{Y^2}{b} + \frac{Z^2}{c} = 1;$$

2°, when the Absolute is the surface $X^2 + Y^2 + Z^2 + W^2 = 0$; in the first of these by Salmon, *Quart. Math. Jour.*, t. IX. (1858), pp. 217–222 (1858), and in the second by Clebsch, *Crelle*, t. LXIII. pp. 64–107 (1862); see also Salmon's *Solid Geometry*, Ed. Ed., 1865, pp. 148, 402, &c. In the present Memoir, as shown by the title, the quadric surface is taken to be an Ellipsoid; and the question is considered exclusively from the first point of view: the theory is further developed in various respects, and in particular as regards the nodal curves upon the centro-surface: the distinction of real and

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On the Centro-Surface of an Ellipsoid (continued).

[Continuation of the memoir; book pages 317–365.]

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imaginary is of course attained by the equation. The new results suitably modified would be applicable to the theory treated from the second point of view; but I do not on the present occasion attempt so to present them.

The Ellipsoid; Parameters ξ, η, ζ. Art. Nos. 1–6.

1. The position of a point (X, Y, Z) on the ellipsoid

$$\frac{X^2}{a}+\frac{Y^2}{b}+\frac{Z^2}{c}=1$$

may be determined by means of the parameters, or elliptic coordinates, ξ, η: viz. these are such that we have

$$\frac{X^2}{a+\xi}+\frac{Y^2}{b+\xi}+\frac{Z^2}{c+\xi}=1, \qquad \frac{X^2}{a+\eta}+\frac{Y^2}{b+\eta}+\frac{Z^2}{c+\eta}=1,$$

or, what is the same thing, ξ, η are the roots of the quadric equation

$$\frac{X^2}{a+\theta}+\frac{Y^2}{b+\theta}+\frac{Z^2}{c+\theta}=1.$$

(In its actual form this is a cubic equation, but there is a root $\theta = 0$, which is to be thrown out, and the quadric equation is

$$\theta^2 + a(b+c+b+c-X^2-Y^2-Z^2)$$

$$+\{bc+ca+ab-(b+c)X^2-(c+a)Y^2-(a+b)Z^2\}=0,$$

or putting

$$P = a+b+c, \qquad Q = bc+ca+ab, \qquad R = abc,$$

the equation is

$$\theta^2 + (P-X^2-Y^2-Z^2)\theta + Q-(b+c)X^2-(c+a)Y^2-(a+b)Z^2 = 0.)$$

2. It is convenient to write throughout

$$\begin{aligned} P' &= b+c, \\ Q' &= bc, \\ P'' &= c+a, \\ Q'' &= ca, \\ P''' &= a+b, \\ Q''' &= ab. \end{aligned}$$

(whence $a+\beta+\gamma=0$),

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As usual, a is taken to be the greatest and c the least of the semi-axes; we have thus a, η each of them positive, and β negative, $= -b'$ where b' is a positive quantity $= b - c$. A distinction arises in the sequel between the two cases $a + \beta = 2b'$ and $a + \beta < 2b'$; but the two cases are not essentially different, and it is convenient to assume $a + \beta > 2b'$, that is, $a > b - 2b' + c$ or $\gamma > a$, say $\gamma > 0$ positive. The limiting case $a + \beta = 2b'$ or $\gamma = a$ requires special consideration.

3. We have

$$-\beta\gamma X^2 = a(a^2 + \xi)(a^2 + \eta),$$
$$-\gamma a Y^2 = b(b^2 + \xi)(b^2 + \eta),$$
$$-a\beta Z^2 = c(c^2 + \xi)(c^2 + \eta).$$

It is in fact easy to verify that these values satisfy as well the equation of the ellipsoid as the assumed equations defining the elliptic coordinates ξ, η. We may also obtain the relations

$$X^2 + Y^2 + Z^2 = a + b + c + \xi + \eta,$$

$$aX^2 + bY^2 + cZ^2 = a^2 + b^2 + c^2 + (b + c)\xi + (c + a)\eta + (a + b)\zeta,$$

$$\tfrac{1}{a}X^2 + \tfrac{1}{b}Y^2 + \tfrac{1}{c}Z^2 = 1 + \xi + \eta + \tfrac{\xi\eta}{R}.$$

These, however, are obtained more readily from the equation in a, viz. the roots thereof being ξ, η, we have

$$-\xi - \eta = P + b + c - X^2 - Y^2 - Z^2,$$

$$\xi\eta = bc + ca + ab - (b + c)X^2 - (c + a)Y^2 - (a + b)Z^2,$$

which lead at once to the relations in question.

4. Considering ξ as constant, the locus of the point (X, Y, Z) is the intersection of the ellipsoid with the confocal ellipsoid

$$\frac{X^2}{a + \xi} + \frac{Y^2}{b + \xi} + \frac{Z^2}{c + \xi} = 1;$$

viz. this is one of the curves of curvature through the point; and similarly considering η as constant, the locus of the point is the intersection of the ellipsoid with the confocal ellipsoid

$$\frac{X^2}{a + \eta} + \frac{Y^2}{b + \eta} + \frac{Z^2}{c + \eta} = 1;$$

viz. this is the other of the curves of curvature through the point.

5. If instead of ξ and η we write k and l, we may consider k as extending between the values $-a^2, -b^2$, and l as extending between the values $-b^2, -c^2$.

A root will thus give the series of curves of curvature one of which is the section by the plane $X = 0$, or ellipse semi-axes b, c; say this is the minor-mean-axes section. In particular $k = -a^2$ gives the ellipse just referred to; and $k = -b^2$, or say $k = -b^2 + \epsilon$, gives two indefinitely near curves of curvature, viz. a curve of curvature portions extends from an umbilicus above the plane of xy, through the extremity of the semi-axis a to an umbilicus above the plane of xy.

The ellipse last referred to may be called the multilinear section; the other two principal sections being the major-mean section and the minor-mean section respectively.

In the limiting case $k = l = -b^2$, we have the umbilici, viz. these are given by

$$\frac{X^2}{a - b} = \frac{Z^2}{b - c}, \qquad Y = 0, \qquad \frac{X^2}{a} + \frac{Z^2}{c} = 1.$$

The two series of curves of curvature cover the whole real surface of the ellipsoid; so that at any real point thereof we have $\xi = k, \eta = l$; or else if ξ, η are negative real values lying within the foregoing limits $-a^2, -b^2$ and $-b^2, -c^2$ respectively. But observe that ξ, η taken separately may each extend between the limits $-a^2, -c^2$.

6. Suppose $\xi = \eta$, the equation is a will have equal roots, or the condition is

$$(P - X^2 - Y^2 - Z^2)^2 = 4\{Q - (b+c)X^2 - (c+a)Y^2 - (a+b)Z^2\};$$

viz. the surface by its intersection with the ellipsoid determines the envelope of the curves of curvature. This envelope is in fact a system of eight imaginary lines, four of them belonging to one of the systems of right lines on the ellipsoid, the other four to the other of the systems of right lines on the ellipsoid. For in the values of X, Y, Z writing $\eta = \xi$, we find

$$\begin{aligned} \tfrac{X}{a}\sqrt{-\beta\gamma} &= a + \xi, \\ \tfrac{Y}{b}\sqrt{-\gamma a} &= b + \xi, \\ \tfrac{Z}{c}\sqrt{-a\beta} &= c + \xi, \end{aligned}$$

or representing for shortness the left-hand functions by $\frac{X}{a}\sqrt{\ }, \frac{Y}{b}\sqrt{\ }, \frac{Z}{c}\sqrt{\ }$, the right lines are

$$\begin{array}{lll} a + \xi = X, & b + \xi = Y, & c + \xi = Z, \\ a + \xi = X, & b + \xi = -Y, & c + \xi = -Z, \\ a + \xi = -X, & b + \xi = Y, & c + \xi = -Z, \\ a + \xi = -X, & b + \xi = -Y, & c + \xi = Z. \end{array}$$

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so that in the two trends each line intersects the four lines of the other trend, but it does not intersect the remaining three lines of its own trend. The intersections are four points corresponding to $\xi = -a^2$, being the imaginary umbilici in the plane $X = 0$: four to $\xi = -b^2$, being the real umbilici in the plane $Y = 0$: four to $\xi = -c^2$, being the imaginary umbilici in the plane $Z = 0$: and four corresponding to $\xi = a\beta\gamma$, which may be called the umbilici at infinity (1).

Separable and Concomitant Centro-curves. Art. No. 7.

7. Consider any particular curve of curvature; the normals at the several points thereof successively intersect each other in a series of points forming a curve; and we have thus, corresponding to the particular curve of curvature, a curve on the centro-surface, which curve may be called the separated centro-curve. Again the same normals, viz. those at the several points of the particular curve of curvature, are intersected by the normal at each point by the consecutive normal belonging to the other curve of curvature through that point; and we have thus, corresponding to the particular curve of curvature, a curve on the centro-surface, which curve may be called the concomitant centro-curve. If instead of a single curve of curvature we consider the whole series, say the major-mean curves of curvature, we have a series of major-mean separated centro-curves, and also a series of major-mean concomitant centro-curves; and similarly considering the series of the minor-mean curves of curvature, we have a series of minor-mean separated centro-curves, and also a series of minor-mean concomitant centro-curves; the conjunction of the several curves will be discussed further on; but it may be benevolent to remark here that the centro-surface may be considered as consisting of two portions, say,

(A) locus of the major-mean separated centro-curves; and also of the minor-mean concomitant centro-curves;

(B) locus of the minor-mean separated centro-curves, and also of the major-mean concomitant centro-curves.

Investigation of expressions for the Coordinates of a point on the Centro-surface. Art. Nos. 8 to 13.

8. Consider the normal at the point (X, Y, Z). Taking in the first instance (x, y, z) as current coordinates, the equations are

$$\frac{x - X}{\frac{X}{a}} = \frac{y - Y}{\frac{Y}{b}} = \frac{z - Z}{\frac{Z}{c}} = \lambda \text{ suppose.}$$

[1] According to Salmon, *Solid Geometry,* [Ed. 2, 1865], p. 210, the number of umbilici is a surface of the nth order is $= n(10n^2 - 15n + 6)$, viz. for $n = 2$, $= 14$, or it is, in the ordinary theory, and recognising the umbilici at infinity. For surfaces properly umbilici or not, the 4 points which I call the umbilici at infinity in the present theory dimension themselves in like manner with the 12 ordinary umbilici.

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or, what is the same thing,

$$x = X\left(1 + \tfrac{\lambda}{a}\right), \quad y = Y\left(1 + \tfrac{\lambda}{b}\right), \quad z = Z\left(1 + \tfrac{\lambda}{c}\right).$$

Suppose now that the normal meets the consecutive normal, or normal at the point $X+dX, Y+dY, Z+dZ$; and let x, y, z belong to the point of intersection of the two normals; we have

$$\begin{aligned} 0 &= dX\left(1 + \tfrac{\lambda}{a}\right) + \tfrac{X}{a}\,d\lambda, \\ 0 &= dY\left(1 + \tfrac{\lambda}{b}\right) + \tfrac{Y}{b}\,d\lambda, \\ 0 &= dZ\left(1 + \tfrac{\lambda}{c}\right) + \tfrac{Z}{c}\,d\lambda; \end{aligned}$$

which determine the direction of the consecutive point; the equations in fact give

$$0 = \begin{vmatrix} dX, & \frac{X}{a}, \\ dY, & \frac{Y}{b}, \\ dZ, & \frac{Z}{c} \end{vmatrix},$$

or, what is the same thing,

$$0 = \begin{vmatrix} dX, & dY, & dZ, \\ bcX, & caY, & abZ \end{vmatrix},$$

which is the differential equation of the curve of curvature. This equation must therefore be satisfied by taking for $X+dX, Y+dZ, Z+dZ$ the coordinates of the consecutive point along either of the curves of curvature,—say along that which is the intersection with the surface

$$\frac{X^2}{a^2+\eta} + \frac{Y^2}{b^2+\eta} + \frac{Z^2}{c^2+\eta} = 1.$$

9. To verify this, observe that we then have

$$\frac{XdX}{a^2} + \frac{YdY}{b^2} + \frac{ZdZ}{c^2} = 0,$$

$$\frac{XdX}{a^2+\eta} + \frac{YdY}{b^2+\eta} + \frac{ZdZ}{c^2+\eta} = 0,$$

or, what is the same thing,

$$XdX : YdY : ZdZ = a^2(a^2+\eta)\beta : b^2(b^2+\eta)\gamma : c^2(c^2+\eta)a,$$

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But from the equations $-\beta\gamma X^2 = a^2(a^2+\xi)(a^2+\eta)$, etc., these become

$$XdX : YdY : ZdZ = \tfrac{X^2}{a^2+\xi} : \tfrac{Y^2}{b^2+\xi} : \tfrac{Z^2}{c^2+\xi};$$

or, what is the same thing,

$$dX : dY : dZ = \tfrac{X}{a^2+\xi} : \tfrac{Y}{b^2+\xi} : \tfrac{Z}{c^2+\xi};$$

and, substituting these values in the determinant equation, it becomes

$$0 = \begin{vmatrix} \tfrac{X}{a^2+\xi} & \tfrac{Y}{b^2+\xi} & \tfrac{Z}{c^2+\xi} \\ \tfrac{X}{a} & \tfrac{Y}{b} & \tfrac{Z}{c} \\ bcX, & caY, & abZ \end{vmatrix},$$

which is identically true, since evidently the determinant vanishes.

10. Proceeding with the solution, we have from the three equations

$$XdX + YdY + ZdZ = d\lambda\Big(\tfrac{X^2}{a^2} + \tfrac{Y^2}{b^2} + \tfrac{Z^2}{c^2}\Big) + d\lambda\Big(\tfrac{X^2}{a^2}\,\tfrac{1}{a^2} + \tfrac{Y^2}{b^2}\,\tfrac{1}{b^2} + \tfrac{Z^2}{c^2}\,\tfrac{1}{c^2}\Big) = 0,$$

and observing that from the equation

$$X^2 + Y^2 + Z^2 = a + b + c + \xi + \eta,$$

considering therein η as constant, we have

$$XdX + YdY + ZdZ = \tfrac{1}{2}d\xi,$$

the equation becomes

$$\tfrac{1}{2}d\xi + d\lambda = 0;$$

and the three equations then are

$$0 = dX\left(1 + \tfrac{\lambda}{a}\right) - \tfrac{X}{2a}\,d\xi, \text{ etc.},$$

or say

$$0 = dX(a+\lambda) - \tfrac{1}{2}Xd\xi, \text{ etc.}$$

But from the equation $-\beta\gamma X^2 = a^2(a^2+\xi)(a^2+\eta)$, considering therein η as a constant, we have

$$\frac{dX}{X} = \frac{1}{2}\frac{d\xi}{a+\xi},$$

and the equations then become

$$0 = \tfrac{a+\lambda}{a+\xi} - 1, \text{ etc.},$$

viz. these are all satisfied if only $\lambda = \xi$.

11. The coordinates of the point of intersection of the two normals thus are

$$x = X\left(1 + \tfrac{\xi}{a}\right), \quad y = Y\left(1 + \tfrac{\xi}{b}\right), \quad z = Z\left(1 + \tfrac{\xi}{c}\right).$$

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or squaring, and substituting for X^2, etc., their values as given by

$$-\beta\gamma X^2 = a^2(a^2+\xi)(a^2+\eta), \text{ etc.},$$

the equations become

$$\begin{aligned}-\beta\gamma x^2 &= a^2(a+\xi)^3(a+\eta), \text{ etc.},\\ -\gamma a y^2 &= b(b+\xi)^3(b+\eta), \text{ etc.},\\ -a\beta z^2 &= c(c+\xi)^3(c+\eta), \text{ etc.};\end{aligned}$$

viz. these equations give (x, y, z) the coordinates of a point on the centro-surface, the intersection of the normal at the point (X, Y, Z) of the ellipsoid, (determined by the parameters ξ, η), by the normal at the consecutive point along the curve of curvature

$$\frac{X^2}{a^2+\eta} + \frac{Y^2}{b^2+\eta} + \frac{Z^2}{c^2+\eta} = 1.$$

Of course by interchanging ξ and η, we should obtain the coordinates of the point of intersection of the normal at the same point (X, Y, Z) by the normal at the consecutive point along the other curve of curvature

$$\frac{X^2}{a^2+\xi} + \frac{Y^2}{b^2+\xi} + \frac{Z^2}{c^2+\xi} = 1.$$

12. I stop for a moment to consider the foregoing two equations

$$\lambda = \xi, \quad d\lambda = -\tfrac{1}{2}d\xi.$$

which at first sight appear inconsistent. But observe that in the foregoing solution λ is the parameter of the point (x, y, z) of the centro-surface considered as a point on the normal at (X, Y, Z); $\lambda + d\lambda$ is the parameter of the same point considered as a point on the normal at the consecutive point $(X + dX, Y + dY, Z + dZ)$; the value $\lambda + d\lambda = \xi - \frac{1}{2}d\xi$ would belong to a different point, viz. the consecutive point of the centro-surface considered as a point on the consecutive normal: whence the $d\lambda$ of the solution ought not to be $= d\xi$. In further explanation, observe that the equations

$$x = X\left(1 + \tfrac{\lambda}{a}\right), \text{ etc.}, \quad \lambda = \xi,$$

if we pass from (x, y, z) to its consecutive point on the centro-surface, give

$$dx = dX\left(1 + \tfrac{\lambda}{a}\right) + \tfrac{X}{a}\,d\xi,$$

but since by the equations

$$0 = dX\left(1 + \tfrac{\lambda}{a}\right) + \tfrac{X}{a}\,d\lambda,$$

this is

$$dx = \tfrac{X}{a}d\xi - \tfrac{X}{a}d\lambda,$$

[1] The expressions are given in effect, but not explicitly, Salmon, p. 149.

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Or since

$$a' = \lambda = \xi \quad (a+\xi),$$

this is

$$\frac{dx}{x} = \tfrac{1}{a+\xi}\, d\xi,$$

and similarly

$$\frac{dy}{y} = \tfrac{1}{b+\xi}\, d\xi, \quad \frac{dz}{z} = \tfrac{1}{c+\xi}\, d\xi,$$

which are the correct values of dx, dy, dz as derived from the equations

$$-\beta\gamma x^2 = a^2(a+\xi)^3(a+\eta),\ \text{etc.},$$

so that taking η at pleasure and considering η as denoting this function of η, the values of ξ, η belonging to a point on the nodal curve are $\xi = a^2(a+\beta)$, $\xi = c(a+\beta)c$; and the value of η is then given as before.

13. The form just given is analytically the most convenient, but there is some advantage in writing $\frac{1}{a+\xi}, \frac{1}{a+\eta}$ in the place of a, η respectively; viz. we then have expressions for the coordinates (x, y, z) of the centro-surface in terms of the coordinates (x, y, z) on the centro-surface, and writing for shortness f for the elimination of (ξ, η) from these expressions will therefore lead to the equation of the centro-surface.

Discussion by means of the equations $-\beta\gamma x^2 = a^2(a+\xi)^3(a+\eta)$, etc., Principal Sections, etc. Art. Nos. 14 to 24 (general subheading).

14. To fix the ideas consider the section of the surface by the plane $z = 0$; we have in the surface $z = 0$, that is, $\xi = -c^2$, or else $\eta = -c^2$, values which give respectively

$$\begin{aligned} -\beta\gamma x^2 &= a(a-c)^3(a-c), & \qquad -\beta\gamma x^2 &= a^2(a^2-c^2)^3 a, \\ -\gamma a y^2 &= b(b-c)^3(b-c), & \text{or}\qquad -\gamma a y^2 &= b^2(b^2-c^2)^3 b, \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} \tfrac{x^2}{(a-c)^3} &= \tfrac{a}{-\beta\gamma}(a+\eta), & \qquad \tfrac{x^2}{(a^2-c^2)^3} &= \tfrac{a}{-\beta\gamma}(a+\xi), \\ \tfrac{y^2}{(b-c)^3} &= \tfrac{b}{-a\gamma}(b+\eta), & \qquad \tfrac{y^2}{(b^2-c^2)^3} &= \tfrac{b}{-a\gamma}(b+\xi). \end{aligned}$$

The first set of equations gives

$$\frac{a^2x^2}{(a-c)^3} + \frac{b^2y^2}{(b-c)^3} = 1,$$

which is the equation of an ellipse.

The second set of equations gives

$$(ax^2 + by^2)^2 = 2(a^2b^2c^2ay^2)(bx^2 + ay^2),$$

which is the equation of an evolute of an ellipse.

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15. The ellipse $\frac{a^2x^2}{(a-c)^3}+\frac{b^2y^2}{(b-c)^3}=1$ is a cuspidal curve on the surface, and the section by the plane $z=0$ is consequently made up of this ellipse counted three times, and of the evolute; it is therefore of the twelfth order; and the order of the surface is in fact $=12$.

It is clear that the section of the centro-surface arises from the section $\frac{X^2}{a}+\frac{Y^2}{b}=1$ viz. the normal at any point of this ellipse lies in the plane $Z=0$; the latter section being by a normal at the consecutive point of the other curve of curvature gives a point on the ellipse $\frac{a^2x^2}{(a-c)^3}+\frac{b^2y^2}{(b-c)^3}=1$, which ellipse is therefore the concomitant centro-curve. Observe that this other curve of curvature cuts the ellipse $\frac{X^2}{a}+\frac{Y^2}{b}=1$ at right angles, and that the normals at the consecutive points about meet in the same point on the ellipse $\frac{a^2x^2}{(a-c)^3}+\frac{b^2y^2}{(b-c)^3}=1$; this shows that the last-mentioned ellipse is a cuspidal curve on the centro-surface.

16. The three principal sections of the centro-surface are consequently

$$\begin{aligned}
z&=0, \quad \frac{a^2x^2}{(a-c)^3}+\frac{b^2y^2}{(b-c)^3}=1, \quad \text{and} \quad (ax^2)^2=(bx^2)^2,\\
y&=0, \quad \frac{a^2x^2}{(a-c)^3}+\frac{c^2z^2}{(b-c)^3}=1, \quad \text{and} \quad (ax^2)^2=(cz^2)^2,\\
x&=0, \quad \frac{b^2y^2}{(a-c)^3}+\frac{c^2z^2}{(b-c)^3}=1, \quad \text{and} \quad (by^2)^2=(cz^2)^2;
\end{aligned}$$

viz. each section is made up of an ellipse counting three times and of an evolute of an ellipse. The figure for shortness represented the three evolutes by their principal cusps it of the like character.

17. Considering only the positive directions of the axes, we have on each axis two points, viz.

$$\begin{array}{lll}
\text{axis of } x & x=\frac{a-c}{a}, & x=\frac{a-b}{a};\\
\text{axis of } y & y=\frac{b-c}{b}, & y=\frac{a-b}{b};\\
\text{axis of } z & z=\frac{b-c}{c}, & z=\frac{a-c}{c};
\end{array}$$

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through each of which, in the two different planes passes through the axis respectively, there passes an ellipse and an evolute. In the second case $a + \beta > 2b'$, the disposition of the points is as shown in the figure.

[Figure: principal sections diagram — omitted]

Plane of xz, evolute is outside ellipse:

yz, inside;

xy, cuts.

but in the contrary case $a + \beta < 2b'$, the disposition is
Plane of xz, evolute is outside ellipse:

yz, cuts;

xy, inside;

there is no real difference, and to fix the ideas I attend exclusively to the first-mentioned case

$$a + \beta > 2b'.$$

18. In each of the principal planes, the evolute and ellipse, qua curves of the orders 6 and 2 respectively, intersect in twelve points, 2 in each quadrant; viz. of the 3 points two unite together into a twofold point or point of contact, and the third is a point of simple intersection; assuming for the moment that this is so, the figure at once shows that in the plane of xy or unilinear plane the contact is real, the intersection imaginary; in the plane of yz, or major-mean plane, the contact is imaginary, the intersection real; but in the plane of xz or minor-mean plane the contact and intersection are each imaginary. The contacts are, as will appear later, the umbilici of the centro-surface; thus the same name "umbilicus centres," or "umbilicus", the simple intersections "points of entropy," or simply "entropy." By what precedes there are real umbilici, and the case the umbilicus (or each quadrant one); and in the major-mean plane four real entropies (in each quadrant one); the other umbilicar centres and entropies are respectively imaginary.

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19. The surface consists of two sheets intersecting in a nodal curve connecting the entropy with the umbilicar centres. As to the form of this curve there is a cusp at the entropy, and the curve does not terminate at the umbilicar centre but, on passing it, from crunodal becomes acnodal (viz. there is no longer through the curve any real sheet of the surface); moreover the curve is not at the umbilicar centre

[Figure: nodal curve with cusp at entropy diagram — omitted]

perpendicular to the plane of xy, say there is consequently on the opposite side of the plane a symmetrically situate branch of the curve, viz. the umbilicar centre is a node on the nodal curve. Completing the curve, the nodal curve consists of two distinct portions; one on the positive side of the plane of yz or minor-mean plane consisting of two cuspidal branches as shown in the figure; the other a symmetrically situate portion on the negative side of the minor-mean plane.

Intersections of Evolute and Ellipse.

20. Consider it the plane of xy the ellipse and evolute,

$$\frac{a^2x^2}{(a-c)^3}+\frac{b^2y^2}{(b-c)^3}=1,\quad (ax^2)^2+(by^2)^2-c^2(a^2y^2+b^2x^2)y^2=0.$$

First, these are satisfied by

$$a^2x^2=-\tfrac{a^2}{-\beta\gamma}(b^2+\gamma)^3,\quad b^2y^2=-\tfrac{b^2}{-\gamma a}(a^2+\gamma)^3;$$

viz. the equations respectively become

$$\tfrac{-(\beta^2+\gamma)^3}{-\beta\gamma}+\tfrac{(-a^2+\gamma)^3}{-\gamma a}=2(a+\beta)c.$$

The first of which is $a+\beta+\alpha\gamma+\beta\gamma$, and the second is

$$\sqrt{1}\{a^2(c-\beta)^3+2c\sqrt{\gamma a}\sqrt{-\beta\gamma(a-\beta+b)^3}+4(c-\beta)^3\}=0.$$

But the equation in $\sqrt{1}\{a^2(c-\beta)^3+2c(a^3+b^3)\sqrt{-\beta\gamma}+\cdots\}=0$, or, what is the same thing,

$$(a^2-\beta^2)(a^2b^2+2ac\gamma+\cdots)^2-4ay^2z^2(a+\beta)^3+2c(a^2-\beta^2)\cdots=0.$$

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and substituting the foregoing values, this is

$$(a^2-\beta^2)\{\tfrac{a^2+\beta^2}{\gamma}\,a^2\gamma^3-a^2\beta^3\gamma\}+8\gamma^2(\beta+\gamma)(a^2+\gamma)^2-8a\beta\gamma^3=0,$$

that is,

$$\tfrac{1}{\gamma}\,a\beta^3(a^2+\beta^2)(a+\beta+\gamma)(a+\beta)=0,$$

which, putting therein $a+\beta+\gamma=a$, and $a^2+\beta^2+a^2+a^2-2a\beta\gamma$, is satisfied; that is, in the points on points or points of contact of the values and evolute.

21. Secondly, consider the values

$$a^2x^2=\tfrac{a}{-\beta\gamma}\left(\tfrac{(y-a)^3}{\gamma-a}\right)\text{ (Coordinates of entropies in plane of } xy \text{ (real).)},$$
$$b^2y^2=\tfrac{b}{-\gamma a}\left(\tfrac{(\beta-a)^3}{\gamma-\beta}\right);$$

Substituting in the equation of the ellipse, we have

$$\tfrac{a(\beta-a)^3+\beta\gamma a(\gamma-\beta)(a+\beta+\gamma)}{(\gamma-a)(\gamma-\beta)}=1,$$

which is satisfied identically; and substituting in the equation of the evolute, we have first

$$a^2x^2+b^2y^2=-\tfrac{a^2(\beta-a)^3+b^3(\gamma-a)(a+\beta-a)}{\gamma a(\gamma-a)}=0,$$

or the equation is satisfied identically: and substituting in the equation of the evolute, we have

$$\sqrt{a^2(a-\beta)^3\beta^3c^2+2a\gamma a(\gamma-\beta)^3(a+\beta-\gamma)}-\sqrt{\gamma a}\sqrt{a^2(\gamma-\beta)^3(\gamma-a)}=0,$$

and thus, in virtue of $a(\beta-a)+\beta(\gamma-a)+\gamma(a+\beta)=0$ becomes

$$-\tfrac{3a\beta(\beta-\gamma)(\gamma-\beta)+b(\beta-a)(\gamma-a)}{\gamma a(\gamma-a)}=-\tfrac{3a(\beta-\gamma)(\gamma-a)}{a(\gamma-a)},$$

and thus, completing the substitution, it is seen that the equation of the evolute is also satisfied. The points last considered are simple intersections, and we have thus the complete number $(3+4,=12)$ of the intersections of the evolute and ellipse.

22. We have a,η positive, β negative; whence $a-\beta$ can positive, $\beta-\gamma$ negative; $\gamma-a,=\beta-\beta'$, also positive; and hence, the entropies on the plane of xy are real; the umbilicus centres are imaginary for this plane, but real for the plane of yz as appears next being

$$a^2x^2=-\tfrac{a^2}{-\beta\gamma}b^3,\quad\text{Coordinates of Umbilicus centres in plane of } xy \text{ (real).}$$

$b^2y^2=\ldots$

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Nodes of the Evolute.

23. The Evolute is a curve with four nodes, all of them imaginary; viz. for the evolute in the plane of xy, the equation of which is

$$(ax^2)^2 + (by^2)^2 - c^2(a^2y^2 + b^2x^2)y^2 = 0,$$

these are

$$a^2x^2 = -y^3, \quad b^2y^2 = -a^3, \text{ Coordinates of Nodes of evolute in plane of } xy \text{ (imaginary)},$$

in fact these values satisfy as well the equation of the evolute, as the two derived equations

$$6a^2x^2[c^2(a^2 + b^2y^2)^2] = bcy^2(a^2 + b^2y^2) = 0,$$

$$6b^2y^2[c^2(a^2 + b^2y^2)^2] = acx^2(a^2 + b^2y^2) = 0;$$

or its points in question are nodes of the evolute.

The evolute has the four cusps on the axes and two cusps at infinity, in all 8 cusps as just remarked; it has 4 nodes; and the order being 6, the class is

$$30 - 2 \cdot 4 = 8 \cdot 6 = 4.$$

Section by the plane infinity.

24. The surface itself is finite, and the section by the plane infinity is therefore imaginary; but by what precedes the nodal curve has nodal points at infinity, viz. there must be real acnodal points on this imaginary section. The section by the plane infinity resembles in fact the principal sections; viz. writing successively $\xi = \infty$, and $\eta = \infty$, we have

$$\begin{aligned} -\beta\gamma x^2 &= a\eta^4(\eta + a), \\ -\gamma a y^2 &= b\eta^4(\eta + b), \\ -a\beta z^2 &= c\eta^4(\eta + c), \end{aligned} \qquad \text{or} \qquad \begin{aligned} -\beta\gamma x^2 &= a\xi^4(\xi + a), \\ -\gamma a y^2 &= b\xi^4(\xi + b), \\ -a\beta z^2 &= c\xi^4(\xi + c); \end{aligned}$$

giving respectively

$$a^2x^2 + b^2y^2 + c^2z^2 = \eta^4, \quad (ax^2)^2 + (by^2)^2 + (cz^2)^2 = \xi^4,$$

where the first equation represents an imaginary conic which counts three times; and the second equation, the rationalised form of which is

$$(a^2b^2c^2)^2 + (b^2c^2a^2)^2 + (c^2a^2b^2)^2 - 2a^2b^2c^2(a^2y^2 + b^2z^2 + c^2x^2)z^2 + 8abc \cdot abc \cdot abc \cdot y^2z^2 = 0,$$

is an imaginary evolute. The conic and evolute have four contacts and four simple intersections (in all $4, 2+4 = 12$ intersections) which are all of them imaginary. But the evolute has four real nodes (analogue of $ax^2 = -y^3$, etc.), which are repeating but being also by what precedes, the tangent of the evolute at that point, which point is also a node on the surface.

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a^2, y^2, z^2, the lines in question will be not surely parallel to, but will be, the asymptotes of the nodal curve.

The plane infinity may be reckoned as a principal plane, and we may regard it in each of the four principal planes there are four umbilicar centres, four entropies, and four evolute-nodes.

The generation of the surface considered geometrically.

25. I have deferred until late the discussion of the generation of the centro-surface by means of the centro-curves, for the reason that it can be carried on more precisely now that we know the forms of the principal sections and of the nodal curve. The two figures exhibit (as regards one extent of the surface) the portions already distinguished as (A), and (B): they intersect each other in the nodal curve, shown in each of the figures.

[Figure: portions (A) and (B) diagram — omitted]

26. Consider first the generation of the portion (A) by means of the major-mean sequential centro-curves. The major-mean curves of curvature (extending to those below the plane of xy) commence with a portion (extending from umbilicus to umbilicus) of the ellipse $\frac{X^2}{a} + \frac{Y^2}{b} = 1$, this may be termed the vertical curve, and they end with the whole ellipse $\frac{X^2}{a} + \frac{Y^2}{b} = 1$, this may be termed the horizontal curve. The normals at the several points of the vertical curve successively intersect each other so as to form a circle, namely a part of the cuspidal curve, viz. from cusp $(b)(b)$ all the way to other umbilicar centres at once, viz. to a first-cusped curve in fig. (A), which may be defined as the principal entropy of $-\beta\gamma$, the corresponding entrolicus being the the umbilicar centre $-\gamma a$. The successive forms of the major-mean entropy as we pass to the plane infinity are entropies as shown in the figure (B), We have in each case attended only to the curve of curvature lying above the plane of xy and the corresponding major-mean separated centro-curve is now in $\eta = a$; for to value of β between $-c, -b$ inclusive; the resulting curve of curvature is for $\eta = -b$, the umbilicar shown above in fig. (A), when these are only $\beta = b$, $\eta = a$, the umbilicar centre at first.

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opening out the two component parts thereof: the two upper cusps of the curve (b) are atomic on the geodesic of the curve-surface, and the two lower cusps upon two detached portions respectively of the cuspline of the centro-surface. And as the curve

[Figure: A, B, C centro-curves diagram — omitted]

of curvature gradually broadens out and ultimately coincides with the XY-section of the ellipsoid, the four cusped curve continues to open itself out, and ultimately coincides as shown figure (c) with the xy-evolute of the centro-surface, viz. this evolute is the sequential centro-curve belonging to the horizontal curve of curvature. The curves of curvature so that we have, for the special case which is now under consideration, the normal at any point of any of the major-mean separated centro-curve as the line tangent thereto. Each, as we have above said, of these normals meets in some entropy in the figure (c).

27. We consider next the generation of the portion (B) by means of the major-mean concomitant centro-curves. Starting as before with the vertical curve of curvature, the concomitant centro-curve is a finite portion (terminated each way at an umbilicar centre) of the curve of the curve XY-evolute, viz. we have then the entropies ξ, ξ, which terminate the curve. The successive forms of the concomitant centro-curve are as shown in figure (B), We have in each case attended only to the curve of curvature below the plane of xy and the corresponding sequential centro-curve is also more on the negative side. Pass we further to the cuspidal X, Y, Z and lastly (X, Y, Z) = the cuspidal X, Y, Z as X, Y, Z, X, Y, Z, will completely deduced, the geometrical signification is anything but clear.

28. There is a precisely similar generation of the portion (B) by means of the minor-mean sequential centro-curves, and of the portion (B) by means of the minor-mean concomitant centro-curves.

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29. As before different points on the ellipsoid correspond to the same point on the centro-surface, this will be a point on the Nodal Curve: the conditions for this if (ξ, η), (ξ_0, η_0) are the parameters for the two points on the ellipsoid, are obviously

$$a^2(a+\xi)^3(a+\eta) = a^2(a+\xi_0)^3(a+\eta_0),$$
$$b^2(b+\xi)^3(b+\eta) = b^2(b+\xi_0)^3(b+\eta_0),$$
$$c^2(c+\xi)^3(c+\eta) = c^2(c+\xi_0)^3(c+\eta_0);$$

these equations in effect determine η as a function of ξ, so that the three equations will be in effect a known function of ξ.

The relation between ξ and η in fact be found by eliminating ξ_0, η_0 from the foregoing equations, but it is easier to eliminate η and η_0; thus consider obtaining ξ, and ξ_0 between as values of η, viz. may be regarded as a known function of ξ; in the equation in two equations may be expressed in terms of ξ, ξ_0, so that each of these quantities will be in effect a known function of ξ.

30. The relation between ξ, ξ_0 in the first instance given in the form

$$0 = \begin{vmatrix} a^3(a+\xi)^3 + b^3 + c^3, & (a+\xi_0)^3, & (a+\xi)^3 \\ b^3(b+\xi)^3 + a^2 + c^2, & (b+\xi_0)^3, & (b+\xi)^3 \\ c^3(c+\xi)^3 + a + b, & (c+\xi_0)^3, & (c+\xi)^3 \end{vmatrix} = 0.$$

Throwing out a factor $(\xi - \xi_0)$, this becomes

$$\Sigma\{a^2(ac+b^2)^2 + a^2 + b^2 + c^2\}$$
$$+(b-c)\{abc\,c^2 + (bac)^2 + c^2(b+c)(b+c)(a+\xi)\} = 0,$$

where the left-hand side is a symmetrical function of ξ and ξ_0, and therefore divisible by $(\xi-\xi_0)^2$. It is obviously to $(b-c)$, viz. we have thus for the determination of η, η_0, the simple equations, obtained by throwing out the factor $\xi - \xi_0$.

To work this out, write $\xi + \xi_0 = p$, $\xi\xi_0 = q$, the result the rentals quotient being $\frac{1}{\Delta(p^2-4q)}$,

$$\Sigma[(b-c)(a+p)+bc] = \begin{vmatrix} 3p^2c, & 3(a^2)(b^2+c)(c^2) \\ b+cp & b \\ +p^2(b+c) & c^2 \\ +b^3(b+c) & b \\ +3p^4q & c \end{vmatrix} = 0,$$

where the left-hand side divides by $\frac{1}{\Delta}(p^2-4q)$.

[1] This was my first method of solution; and I have thought the results quite interesting simply to ensure them; but it will appear in time that I have succeeded in expressing ξ, ξ_0, η, η_0 in terms of a single parameter ξ.

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31. Developing and reducing, and omitting this factor, the final result is

$$6R + 3Q(p + P)(p + Pp) + 3pq = 0,$$

when as before P, Q, R denote $a + b + c$, $bc + ca + ab$, abc respectively; that is,

$$6R + 3Q(P + Pp) + P(P^2 + 4Q)q + 2pq + (P + Q)q = 0,$$

or, as this may also be written,

$$\begin{gathered} 6R + 3Q(p + P_0) + (P + Q)q \\ +\xi_0(P + 4Q) \\ +\xi_0^2(P + Q) - 2\xi_0 q = 0, \end{gathered}$$

viz. the parameters ξ, ξ_0 have a symmetrical $(2, 2)$ correspondence.

32. From the equations $(a + \xi)^3(a + \eta) = (a + \xi_0)^3(a + \eta_0)$, etc., we have

$$\Sigma(b - c)\{(a + \xi)^3(a + \eta) - (a + \xi_0)^3(a + \eta_0)\} = 0,$$

and observing that the term in $(\eta - \eta_0)$ is

$$\begin{gathered} \sqrt{[\beta(\beta - \gamma)(a + \xi)^2 + a(b - \gamma)(a + \xi)^2 + a(b - \gamma)(b + \xi)^2 + c(a - \beta)(c + \xi)^2]} \\ +a(\gamma - \beta)\{2a + \xi\} - 2a\{a^2 + \xi\} \\ +b(a - \gamma)\{2b + \xi\} \\ +c(\beta - a)\{2c + \xi\}, \end{gathered}$$

these are readily reduced to

$$\begin{gathered} [3\xi - \xi_0 + (P + Q)p + (P + Q)(\xi + \xi_0)] \cdot \eta = -[(P + Q)\xi + 3\xi_0 + (b + a)\xi_0], \\ [(\xi - \xi_0) + a^2 P_0] \cdot (\eta - \eta_0) - \cdots, \end{gathered}$$

or, what is the same thing,

$$\begin{gathered} [3(\xi - \xi_0) + 2P(\xi + \xi_0) + (P + Q)q + (P + Q)(\xi + \xi_0) + Q(\xi - \xi_0)] \cdot \eta = - \cdots, \\ 3(\xi - \xi_0) + 2P(\xi + \xi_0) + Q(\xi - \xi_0) = -[(P + Q)\xi + 3\xi_0 + (b + a)\xi_0], \end{gathered}$$

and if we hence determine the values $3(\xi^2 - \xi_0^2) = \eta_0$, we find that the resulting terms divide by $\xi - \xi_0$, and we have

$$\begin{gathered} 3\xi_0^2(\xi_0 - \xi) - \xi^2[(b + a)\xi + 3\xi_0] = (\xi + \xi_0)\xi_0(b + a)(\xi + \xi_0)(\xi + a) \\ +\xi_0 \cdots, \\ +\xi_0 \cdots. \end{gathered}$$

Hence observing that by the relation between ξ, ξ_0, the first term is

$$= -3Q\xi_0 + (P + Q)q + 2\xi_0 q,$$

the equations become

$$\begin{gathered} 1 : \eta : \eta_0 = -[Q + p(P + Q) - 3\xi_0] \cdot Q \\ : R(2\xi_0 + 2) - \xi^2(P + Q\xi) \\ : R(3\xi_0^2 - \xi^2) - \xi^2 P(P + Q\xi). \end{gathered}$$

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and we thus have

$$\eta = \frac{R(\xi_0^2 - \xi^2) - \xi^2 P(P + Q\xi)}{P\{Q + q(P + Q) - 3\xi_0\}},$$

which, considering ξ_0 as a given function of ξ, gives η as a function of ξ.

33. I write $\xi - \xi_0 = 2u$, $\xi + \xi_0 = 2t$, so that $p = 2t$, $q = t^2 - u^2$, $\xi = t + u$, $\xi_0 = t - u$. The relation between ξ, ξ_0 thus becomes

$$6R + 6Qt + (P + Q)(t^2 - u^2) = 0,$$

or, what is the same thing,

$$u^2 = t^2 + \tfrac{6R+6Qt}{P+Q};$$

so that taking t as a parameter, and considering u as a node real value of u, respectively, we have the relation

$$\xi = \tfrac{(t+u)\sqrt{P+Q}+R(P+Q\xi+Q\xi_0)}{P+Q+2t},$$

and the value of η is taken as $-3\xi, \xi_0, -2 = -2\xi$. The relation between ξ, ξ_0 in the simple form $\xi + \xi_0 = 2t$, hence appears as an arithmetic mode of expression: which is in some respects advantageous, but I propose that this can be modified.

34. On comparison of reality is easily found

$$y^2 = \frac{(a + a^2)(a + b)\{a + bc(a + 1)\} + ac(a + 1)\{a + b - c\}}{a + b + c\{a + (b + c)\}},$$

where $a = -\frac{1}{2}(P + a)$, $b = \frac{1}{2}(P + Q)$; viz. as we then have a in entrunctate coordinates of a point in a plane, (a, t) will be the rectangular coordinates of the same point referred to as a, and also the equation of conic referred to set $(2x - 2, x)$ at (a, t), viz. we have for the entire centro-surface in this way.

35. The curve is a conic curve symmetrical in regard to the axis of x, and having the three asymptotes,

$$x = -t, \quad y = \tfrac{1}{2}(P + a) \pm \tfrac{1}{2}(P + a)\sqrt{c(P + 1)(P - a)};$$

moreover we have $y = 0$ for the values $x = a$, x, C where the curve cuts the axes of C, E, F, as shown in the figure: a, C, E, are at the distances from the centre, and with these data it is easy to draw the curve. It is then the form τ, that whence writing $t = \tau, u = 1$ to $\eta + a$, and let $t \cdots \tau + a$, corresponding to the umbilicar centre; and at Ω, Ω, we have $\xi = -\beta$, $\eta = a$, and $\xi - \beta$, etc. corresponding to the corner.

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We have

$$A^2 + B^2 c = \tfrac{1}{2}\Delta + a^2(P+1);$$

$$B = -\tfrac{1}{2}(\gamma + a)^2(\eta + a) + 2\Delta(P + 2 + \gamma),$$

$$B = -\tfrac{1}{2}(\gamma + a)^2(b + a^2\omega),$$

hence the rational part in question is

$$= \pm\tfrac{(\eta+a)^2}{(2a+Pp+a)}\sqrt{\tfrac{a+R+(1-\xi)(2+\xi)}{P\{Q+q(P+Q)-3\xi_0\}}}(2+\xi)$$

32. Write $\xi - \xi_0 = a + \frac{a}{u}$, etc., $\xi - \xi_0 = -\frac{a}{2}, \xi_0 + 2 = \frac{2}{u}, c,\ y = 0; p = -a, c.$ The relation between ξ, ξ_0 thus becomes

$$\xi - \xi_0 = a + \tfrac{a}{2u}, \quad \xi - \xi_0 = -\tfrac{a}{2}, \xi_0 + 2 = \tfrac{2}{u}, c, y = 0; p = -a, c.$$

or as that the curve $\Omega = 0$ is obtained by eliminating ξ from this equation and the foregoing, regard to ξ at the cubic function becomes

$$(a^2 + \xi)(b - \xi)\left(c + \xi + \tfrac{\xi}{a+\beta}\right) = 0;$$

that is, the equation has the values

$$\xi = -a^2, \quad \xi = -\beta^2, \quad \xi = -\xi_0 = -a(a+\beta).$$

But these last values give

$$a + \xi = \tfrac{aa}{a+\beta}, \quad \beta + \xi = -\tfrac{aa}{a+\beta}.$$

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and for $\xi = ($ we have

$$u(2(a+\xi)-1)\left(\frac{2u\beta\gamma}{a+\beta}\right)\eta_0 = \frac{(a^2)}{(a-\beta)^3}\eta, \quad \frac{(a-\beta)^3}{aP\beta}\eta_0,$$

so that solving for greater simplicity, $(a-\beta)\eta_0 = -a\beta\eta$, the formulae become

$$\frac{2P\xi}{a^2} = \frac{3a^2}{a^2} = \frac{3a^2}{\beta-a},$$

$$\frac{2P\xi}{\beta^2} = \frac{3\beta^2}{\beta-a},$$

$$\frac{2P\xi}{\gamma^2} = \frac{-a\beta\gamma}{a^2+a\beta} = -\frac{a+\beta}{a},$$

$$\frac{a+\beta}{\Omega\cdot 1}\Omega = \cdots,$$

32. This shows that there is the curtsey a cusp, the cuspidal tangent being in the plane of xy. It appears moreover that this tangent constants with the tangent of the evolute. In fact, from the equation $(ax^2)^2+(by^2)^2 - a^2y^2 = 0$ of the evolute we have

$$\frac{(ax)^2}{a} + \frac{(by)^2}{b} + \frac{(ax^2)^2+(by^2)^2}{a^2} = 0,$$

or substituting for (a, y) their values as in the entropy,

$$\frac{2(a-a)\{b-a\}\beta-a}{x^2(a-\beta)^2} = \frac{a+ax}{\beta+a^2-a^2}\cdot\frac{dy}{dx} = 0,$$

that is,

$$\frac{d}{dx} = \frac{ax^2+ax^2(\beta-a)}{a^2(a+\beta)-a^2\cdot\frac{dy}{dx}}.$$

which is satisfied by the foregoing values of $\frac{x}{a}$ and $\frac{y}{a}$ at the two tangents there coincide.

We have

$$4(da^2+(ba^2)(b-2a^2-\cdots\sqrt{4}\Sigma ay^2+ax^2) = \frac{a+a+a}{a(a+\beta)^2a}\eta\cdots\xi,$$

which is virtue of

$$(ay)(2+y)(a+\beta) = \beta y^2(2y+a)+\beta y^2(2y+a)+(2y(2+a)+a)-4y^3(2+a)-3y^4y^2(a+\beta),$$

is

$$4[(by)^2+(by)^2+(ay^2+3y^4y^2(a+\beta))] = \frac{2(a+\beta)^2}{a(a+\beta)^2}\eta\cdots\xi.$$

(observe $2(a^2+\beta)-a(a-\beta), = -(a+(a+\beta), = -8a^2$, is negative)

$$= -\frac{2a^2}{a(a+\beta)^2}\eta\cdots\xi.$$

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if $\frac{dy}{dx}$ be the value at the entropy. Writing $\frac{a^2}{a^2}$ for the element of the are we have

$$\frac{ds^2}{a^2}\frac{dx^2(a-\beta)^3}{(a-\beta)^2} - \tfrac{4}{a}\eta - \tfrac{8a^2}{a},$$

$$\frac{ds^2}{a^2}\frac{dx^2(-a\beta\gamma)^2}{(a-\beta)^2} - \tfrac{8a^2}{a},$$

which exhibit the form at the entropy.

The Nodal Curve, expressions for the coordinates in terms of a single parameter a. Art. Nos. 33 to 41.

33. After the foregoing investigation of the nodal curve, I was led to perceive that it is possible to express ξ, η, ξ_0, η_0 in terms of a single variable η; and thus to obtain expressions for the coordinates of a point of the nodal curve in terms of the single variable a. The analytical form of the nodal curve being real, with its imaginary values of ξ, η, the modes y may we may assume

$$\xi = -b^2 - p(b^2 - a^2 - b^2),$$

$$\eta = -b^2 + p(b^2 - a^2 - b^2),$$

($i = \sqrt{-1}$ as usual): viz. p denote one or other of the quantities

$$y_+ = a + (a^2 - b^2), \quad q = a - (a^2 - b^2);$$

the expressions for $-\beta\gamma x^2$, $-\gamma a y^2$ become

$$= [(b-p)(a+1)(b-q)^3](b-p)^3,$$

and we have therefore the condition that this shall be real (for the two values $b = p$, $\dot{b} = -b$); being real, it will in certain cases be positive, in other cases negative, and the values for the remaining condition is given by the equation

$$\Delta'(b^2 - \xi^2 - \xi_0^2 - 4)(b^2 + b^2 + \xi^2 + 2b^2)^2 + p^2(b - \xi^2 - 2)(b^2 + \xi)^2 - 0 = 0.$$

34. The condition of reality is easily found to be

$$(\beta - p)(p + \beta - 1) + (a+p)^2 + (\gamma + p)^2 + 4(a+p)(\beta - p) + (a+p)^2(b-\beta) - p^2q^2 = 0;$$

viz. this equation in Δ must have the form $-a, b$, suppose to a value

$$P = p, \quad q = qP_0 p + p^2,$$

we have therefore

$$\tfrac{(\gamma-a)\beta}{a^2} = a + \tfrac{a(a^2-b^2)}{2bg(\gamma-a)}\sqrt{Y}\cdot\eta(a-\beta-\gamma)(a+\beta+\gamma),$$

where, regarding X, Y are given functions of a, the coordinates of a plane in a cubic curve having the point $X = 0, Y = 0$ for a node, writing $Y = b + 3a(X+1)$, X will be such of these is rational function of a. The curve has the property that being any line through the entropy, the tangents at the points of the curve, the corresponding entropies on it are

$$(a^2 + Y)(a^2 + a)(a+1) + (b+1)(a^2+b)\beta(a^2+a) = -a^2(b-p)^3 + a(p+p)^3,$$

for a point on the nodal curve.

35. To complete the investigation, writing for a moment $T = 3 = 3a^2(X+1)$, we obtain

$$\frac{(b-p)^2}{a^2} = \tfrac{a}{2a^2(a+a-1)} + \tfrac{a(a^2)(b-\xi)^3}{a+b-1},$$

or putting for a moment

$$\frac{(\gamma-a)\beta}{a^2} = K,$$

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we have

$$X = \tfrac{a(K-7)(a-1)}{a+1-K},$$
$$X_0 = \tfrac{K(a-7)(a+1)}{a+1-K},$$
$$Y = K\tfrac{3K(a^2-1)+2(a+1)}{a+1-K},$$
$$Y_0 = K\tfrac{3(a-1)(K-2)(a+1)}{a+1-K},$$
$$Z = 2K\tfrac{(b-1)X(a-2+1)}{a+1-K},$$
$$Z_0 = 2K\tfrac{(b-1)(X-2+1)}{a+1-K},$$

or substituting for K its value we have

$$K(a-1) = \tfrac{(a-1)^2}{ab}\left(a^2 - \tfrac{8aq}{(a-1)^2}\right),$$
$$= -\tfrac{(a-1)^2}{ab}\left(a + \tfrac{2q}{a-1}\right)\left(a - \tfrac{2q}{a-1}\right);$$

$aa+1-K = -\frac{1}{ab}(a^2+1)(a+b-1)\,(a-p), \cdots +b(a-p)$, and changing the sign of the radical we have the values of $\frac{x}{a}, \cdots,$

36. Write for a moment,

$$\left[\Delta - \tfrac{1}{2}(\gamma - a)\sqrt{1+\sqrt{\tfrac{2a(a-2-4\beta+2)}{a(a-2)}}}\right] - (a - 8a\beta + \sqrt{\beta})Y + B\sqrt{\delta} = 0,$$

$$\left[\Delta - \tfrac{1}{2}(\gamma - a)\sqrt{1+\sqrt{\tfrac{2a(a-2-4\beta+2)}{a(a-2)}}}\right] - (a - 8a\beta + \sqrt{\beta})Y + B\sqrt{\delta} = 0,$$

then in the product of these two expressions the rational part is $= A^2A + B^2B$; and from the manner in which they were arrived at we have $9 = A_1A + B_1B$, and the rational part is thus

$$= -\tfrac{16}{a^2}(B^2 - B'B).$$

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We have

$$A^2A - B^2B = (\Delta - a)^2X - 8a^2X_0,$$
$$B' = \tfrac{1}{2}(\gamma + a)X\Big[a - \tfrac{2aq}{(a-1)^2}(X+S)\Big] - a^2b^2(P^2 - 2a^2)$$
$$-B(\gamma + R)\big[(\gamma - a)(b - \beta) + Y(\gamma - a) - 2y - \tfrac{4ay}{a^2}\big]\,,$$

or, what is the same thing,

$$\Delta\beta + \xi\Big(1 + \tfrac{q}{a-1}\Big) = \xi aP + 2(P - a)X.$$

The first thus becomes

$$\pm\xi(a + bX)\Big(\tfrac{a+bX}{(a+b-1)\beta - a^2}\Big) - X(a+b)(b-a)X - 4a + 4(b+1)$$

which is

$$= -aX - X(P - a) - (b+1)(a^2X^2 - 2a + 4(b+1)X);$$

and for the value of η, proceeding to the third term, this is

$$= -aX_0 + X(P - a) - (b+1)(a^2X_0^2 - 2X_0 + 4(b+1)X_0);$$

so that without any further reduction, $\eta_0 = -\beta\eta$.

37. We have

$$\tfrac{-2bx}{x^2}(a-1) - \tfrac{1}{a(a-1)-1}\Big[3aa^2 - 2\beta - \xi\Big(a^2 - \tfrac{4qa}{(a^2-1)^2}\Big)\Big]$$
$$-\tfrac{k}{(a^2-1)(a-1)}\Big[3aa^2 - 2\beta - \xi - \tfrac{4qa}{(a^2-1)^2}\Big] + \cdots\,,$$

where the term in [ ] is

$$-\tfrac{2bg}{a^2(a^2-1)}\Big[3aa^2 - 2\beta - \xi\Big(a^2 - \tfrac{4qa}{(a^2-1)^2}\Big)\Big]\,,$$

$3\eta + 2 + (ba^2\Sigma X)(a^2+1) - 2\beta + 4 = -ab(a^2-1)\xi - 2bq(a^2-1)y - q^2 + (qa^2)(a^2-4)$, where in the case the first term is

$$= 3aa^2(a+1)(a^2+1) + 4(\sigma + \beta)(b - \beta), \quad \cdots\,,$$
$$+8(\beta + a)(\beta - a),$$
$$+(b+1)(b+1)(a-1)^2 - q.$$

Hence observing that Δ in like manner multiplying by $(\beta - q)^2$, we have

$$-\tfrac{2bx}{x^2}(a-1) = -\tfrac{a(a+b)(a^2)(3a-2\beta-4q\beta+8\beta(a-q+1)+8a(a^2-1))}{a^2(a-1)(a^2-1)} - \tfrac{a^2(b-a)}{(a^2-1)}\big[3aa^2 + 4(b-\beta) + 8(\beta - q) + 8(a^2-1)\big] + \tfrac{a(b+a)}{a^2(a-1)}\Sigma,$$

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$$[-2at(\beta-a)+5\beta\beta(\beta+2)\beta\beta+4(a-1)\beta\beta+\Omega-3\beta\beta(a^2-1)\beta\beta+(\beta-1)\beta\Omega]-\tfrac{2\beta y}{(a^2-1)}(a^2-p)(a^2-q),$$

or finally

$$\Omega:\beta(p-q)+2(a-1)(\Omega)=(-2\beta(P+1)-\beta-q)(\beta-p)(\beta-q).$$

But $a^2+4a,\xi+\xi_0=\xi$, viz. since $\xi-\xi_0+a^2(2a+1)$, and therefore

$$a^2(P-1)+2+8-\beta=2a^2(\xi^2+\xi_0)(\xi+\xi_0),$$

the equation thus divides by $-2a^2(P+1)+a^2-\xi-\beta+2y-a^2$. So that Ω has the value originally so denoted, we have

$$\eta=-a^2+\tfrac{8\beta\beta}{(a-1)^2}\beta,$$

38. Lastly the equation $\Omega=\xi'(b+\xi)\Big(c+\xi+\frac{\xi b\beta\gamma}{a+\beta}\Big)=0$ is satisfied if $a^2+\beta\xi=0, q+b-\beta$, the equations

$$\Omega+\xi'(b+\xi)\Big(c+\xi+\tfrac{\xi b\beta\gamma}{a+\beta}\Big)=0,$$

$$(a^2+\xi)(b+\xi)+a^2(b+\xi)+\xi b\beta\xi=0,$$

then give

$$(a^2+\xi)(b+\xi)(c+\xi)=\xi a\beta\gamma,$$

which can be satisfied by $\xi=a$, giving $\eta=a+\beta=-a^2$, which in the case I, or else by

$$\xi'(b+\xi)-\xi^2-\beta a\beta\gamma,$$

$$\xi'(b+\xi)-\xi^2+\xi\beta a=a^2,$$

that is,

$$\xi(a+\beta)-\xi^2-a^2,$$
$$\xi(b-\beta)+\beta^2-a^2.$$

To show that these values satisfy the relation between ξ,ξ, observe that they give

$$\xi+\xi_0=a+b-2a^2,\quad \xi\xi_0=b-\beta-a^2,$$

whence also

$$\xi'\xi_0=\xi b-\xi^2-a^2,\quad \xi-\xi_0=\xi(a-\beta+1)-1,$$

and the relation becomes

$$6a\beta\gamma-3(b+ac+a^2+a^2)(a+b-\xi^2)+(a+b+1)(a^2+b-\xi^2-1)=0,$$

which is an identity.

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43. I will show that these values of ξ, ξ_0 give the foregoing values $\eta = \eta_0 = -a^2$. We have

$$\begin{aligned} 1 : \eta - \eta_0 = P\{Q, q(P+Q) - 3\xi_0\} &: 5R \\ &: (\xi_0 - \xi)[3R - (P+Q)(2\xi_0 - q) + \xi P\beta_0 + (2\xi_0 - q)] \\ &: (\xi_0 - \xi)[3R - (P+Q)(2\xi_0 - q) + \xi^2 P + (P + 2\xi_0 - q)]; \end{aligned}$$

this is

$$1 : \eta - \eta_0 = P\{Q, q(P+Q) - 3\xi_0\} : 5R\big(\xi P - \tfrac{2}{a}\big) + \xi^2 P + \xi(P - \xi)$$

or

$$\eta - \eta_0 = -a^2, \quad \eta + \eta_0 = -2a^2, \text{ that is, } \eta = \eta_0 = -a^2.$$

44. For the real umbilicar centres and entropies we have
I. $\xi = -b^2, \quad \eta = a^2, \quad \xi_0 = -b^2, \quad \eta_0 = -b^2;$ we have

$$\begin{aligned} X^2 &= a^2(a^2 - b^2)/a, & X_0^2 &= -a^2/a, \\ Y &= 0, & Y_0 &= 0, \\ B &= 0, & E_0 &= -c^2/c, \\ a^2X^2 &= \tfrac{(a-b)^2}{a^2}, & & \\ y &= 0, & &\text{(real umbilicar centre),} \\ c^2z^2 &= \tfrac{a^2b^2}{a^2}. & & \end{aligned}$$

II. $\xi = -b^2, \quad \eta = a(a+\beta) = -2b^2$:

$$(a^2 + \eta = -a\beta) - a\eta = b - \beta - \beta, \quad b + \eta = \tfrac{a(a-b)}{a^2}, \quad c + \eta = \tfrac{c(b-\beta)}{(a^2-\beta)^2},$$

$$\xi_0 = -b^2, \quad \eta_0 = \tfrac{2b\beta}{a^2-\beta}, \quad \xi_0 + \eta_0 = \tfrac{a(\xi-\beta)}{a^2-\beta}, \tfrac{b-\beta}{a^2-\beta}, \tfrac{c+\beta-\beta^2}{a^2-\beta};$$

$$b + \eta_0 = -\tfrac{a^2}{(a-\beta)^2},$$

$$\begin{aligned} X^2 &= \tfrac{a^2}{a^2-\beta^2}\,\tfrac{(b-\beta)^3}{a-\beta}, & X_0^2 &= -\tfrac{a^2}{a^2-\beta^2}\,\tfrac{(b-\beta)^3}{a-\beta}, \\ Y^2 &= -\tfrac{b^2}{a^2b^2}\,\tfrac{(a-\beta)^3}{a^2-\beta^2}, & Y_0^2 &= \tfrac{b^2}{a^2b^2}\,\tfrac{(a-\beta)^3}{a^2-\beta^2}, \\ Z &= 0, & Z_0 &= 0, \end{aligned}$$

ellipse concomitant, ellipse separated;

$$\begin{aligned} a^2x^2 &= \tfrac{a^2}{(a-\beta)^3}\,\tfrac{(b-\beta)^3}{a^2-\beta^2}, \\ b^2y^2 &= -\tfrac{b^2}{(a-\beta)^3}\,\tfrac{(a-\beta)^3}{a^2-\beta^2} \quad \text{(real entropy),} \\ z &= 0. \end{aligned}$$

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Nodal curve in vicinity of umbilicus centre, $ax^2 = -y^2$, $c^2z^2 = w^2$, Art. No. 47 to 49.

47. Write

$$\xi = -b^2 + p, \quad \eta = -b^2 + r, \quad \xi_0 = -b^2 + q, \quad \eta_0 = -b^2 + s;$$

we have to find the relation between p, q, r, s; first for p, q, the equation of correspondence gives

$$6R + 5Q(p + q + 2b^2 - 8b^2) + (P + Q)(p + q + b^2)$$

$$+5\{-2b^2 + 2bQp + p^2 - b^2(p + q) - 2bp - p\} = 0,$$

that is,

$$3(p + q)(3b^2 - 2b^2) + (b^2 - b^2)(p + q + 2bp) = 0,$$

or writing these equations as follows

$$3(p + q)(\gamma - \beta) + (3\beta + 3\gamma - 2)\beta\eta = 0,$$

viz. this is

$$3p + 3q + \tfrac{8\beta\eta}{\gamma-\beta}(p + q) = 0;$$

or in symbolic form

$$3p + q + \tfrac{2\beta\eta}{\gamma-\beta}(\xi_0)^2 = 0.$$

48. Now the equations

$$(a^2 + \xi)(a + \xi)^2 = (a^2 + \xi_0)(a + \xi_0)^2 = a^2(b + \xi)^2(b + \xi_0)^2 = (a + \xi)^2(\cdots),$$

putting therein for ξ, η, ξ_0, η_0 their values, give the first of them

$$\log\bigl(1 + \tfrac{r}{a^2}\bigr) - \log\bigl(1 + \tfrac{s}{a^2}\bigr) = 8\log\bigl(1 + \tfrac{p}{a^2}\bigr) - 8\log\bigl(1 + \tfrac{q}{a^2}\bigr)\,,$$

that is,

$$r + 3p + \tfrac{1}{2a^2}(r^2 + 3p^2) + \tfrac{1}{3a^4}(r^3 + 3p^3) + \cdots = +s + 3q + \tfrac{1}{2a^2}(s^2 + 3q^2) + \tfrac{1}{3a^4}(s^3 + 3q^3) + \cdots\,;$$

and similarly the second equation

$$r + 3p + \tfrac{1}{2a^2}(r^2 + 3p^2) + \tfrac{1}{3a^4}(r^3 + 3p^3) + \cdots = s + 3q + \tfrac{1}{2a^2}(s^2 + 3q^2) + \tfrac{1}{3a^4}(s^3 + 3q^3) + \cdots\,;$$

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whence multiplying by γ, and adding,

$$(\gamma+s)\big[r+3p+\tfrac{1}{2}(r^2+3p^2)\big] = (\gamma+s)\big[r+3p+3p_0+\tfrac{1}{3a^4}(r^3+3p^3)\big]\,,$$

which, neglecting terms of the third order, is

$$r+3p=s+3q,$$

Subtracting the two equations we have

$$\tfrac{1}{3}\left(\tfrac{1}{a^2}+\tfrac{1}{b^2}\right)(r^2+3p^2)+\tfrac{1}{a^2}\left(\tfrac{1}{a^2}-\tfrac{1}{b^2}\right)(r^3+3p^3)=\tfrac{1}{3}\left(\tfrac{1}{a^2}+\tfrac{1}{b^2}\right)(s^2+3q^2)+\tfrac{1}{a^2}\left(\tfrac{1}{a^2}-\tfrac{1}{b^2}\right)(s^3+3q^3);$$

viz. this is the same thing,

$$r^2+3p^2+\tfrac{2}{3}\left(\tfrac{\gamma-a^2}{\gamma}\right)(r^3+3p^3)=s^2+3q^2+\tfrac{2}{3}(s^2+3q^2)(r-s)-0,$$

or, what is the same thing,

$$r^2-s^2+3(p^2-q^2)+\tfrac{1}{3}\left(\tfrac{\gamma-a^2}{\gamma}\right)(r^2+s^2)(r-s)-0,$$

which, putting therein $r-s=-3(p-q)$, is

$$-r-s+3(p+q)+3a\left(\tfrac{\gamma-a^2}{\gamma}\right)(r^2+s^2)+9(p^2+q^2)=0,$$

say this is

$$-r-s+3(p+q)+2\Delta=0,$$

combining herewith

$$r-s+3(p-q)=0,$$

we have

$$r+3p-\Delta=0,$$

and

$$s+3q-\Delta=0,$$

where

$$\Delta=\tfrac{1}{2}\tfrac{\gamma+a^2}{\gamma}(-r^2-s^2+p^2+qq+q^2),$$

But substituting herein the values $r+3p=-3p$, $s=-3q$, this becomes

$$\Delta=\tfrac{1}{2}\left(\tfrac{\gamma+a^2}{\gamma}\right)\big[-4p^2-4q^2-3p^2\big]=-\tfrac{1}{2}\left(\tfrac{\gamma+a^2}{\gamma}\right)q^2,$$

and then

$$r=-p+3p_0+\Delta=-3p-\tfrac{1}{2}\left(\tfrac{\gamma+a^2}{\gamma}\right)q^2,$$

that is,

$$r=3p+2(p+q)+\Delta=-\tfrac{4(\gamma-a)}{\gamma a^2}q.$$

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49. We have then

$$a^2x^2 = \tfrac{y^2}{a^2}\left(1+\tfrac{r}{a^2}\right)\left(1+\tfrac{p}{a^2}\right) = \tfrac{y^2}{a^2}\left(1+\tfrac{r+3p}{a^2}+\tfrac{3p(r+p)}{a^4}\right) = \tfrac{y^2}{a^2}\left(1+\tfrac{-4(\gamma-a)q}{ay\gamma}+\tfrac{6q}{ya^2}\right) = \tfrac{y^2}{a^2}\left[1+\tfrac{2(p-3q)}{a\gamma y}\right],$$

and in the same way from $a^2x^2 = \frac{a^2}{b^2}\left(1-\frac{r}{a^2}\right)\left(1-\frac{p}{a^2}\right)$, we have

$$a^2x_0^2 = \tfrac{y^2}{a^2}\left[1+\tfrac{2(p+\beta)}{a\gamma}\right];$$

moreover we have at once

$$b^2y^2 = \tfrac{a^2}{a^2} - \tfrac{1}{a\gamma}q^2.$$

Hence, writing $x+\delta x, 0+\delta y, z+\delta z$ for x, y, z, we find

$$\delta x = \tfrac{1}{2}p\tfrac{2(p-3q)}{ay\gamma}\cdot q^2,$$

$$\delta y = \tfrac{1}{2}p\sqrt{\tfrac{a}{\gamma}}\cdot q,$$

$$\delta z = \tfrac{1}{2}p\tfrac{2(p-\beta)}{ay\gamma}\cdot q^2,$$

or, what is the same thing,

$$\delta x : \delta y : \delta z = \tfrac{2(p-3q)}{ay\gamma} : \sqrt{\tfrac{a}{\gamma}} : \tfrac{2(p-\beta)}{ay\gamma},$$

where a, z denote the value at the umbilicar centre.

Nodal curve in vicinity of real entropy, viz. $ax^2 = \alpha^2, by^2 = \beta^2$. Art. Nos. 50 to 52.

50. Write

$$\xi = -a^2 + p, \quad \eta = -a^2 + r,$$
$$\xi_0 = -a^2 + q, \quad \eta_0 = -a^2 + s.$$

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and first for the relation between η and η_0, writing for a moment $\frac{3p\eta}{a-\beta} = \eta_0 = -Q$, and therefore $\xi_0 = -a+Q$, the equation of correspondence gives

$$-3a\beta\gamma + (\eta_0 + s)\beta + (\beta + \eta + \tfrac{1}{2}\eta^2) - 3\eta\eta(\eta + Q) - 3a^2(s + Q) = 0,$$

which, putting for Q its value, is

$$\begin{gathered} -9\eta(\eta - s + 3rs - \tfrac{6a^2}{a-\beta}) \\ +(s + 3\eta r)\left[a^2(s + r) - \beta(r + r) + (a + \beta)\eta s - \tfrac{3a\beta}{a^2-\beta}(\eta + r)\right] \\ -3a\beta(s + r) - 3a^2(s + r) - \tfrac{4a^2\beta}{a-\beta} = 0; \end{gathered}$$

or, what is the same thing,

$$\begin{gathered} (3\eta s + 2\eta^2(r - \beta)) - 4a^2\eta s \\ +(s^2 + 7a\eta + 5\eta s) + \tfrac{a^2-4a\beta+\beta^2}{a-\beta} + 4(\eta + s)\eta a + s - \beta \\ +3a(\eta + s) = 0, \end{gathered}$$

or, in small values,

$$\left(2a + \tfrac{a+\beta a^2}{a-\beta}\right)(s + 3r) = 2a\beta r.$$

viz. writing $\frac{a}{(a-\beta)^2}$ for $\frac{a}{(a-\beta)^2}$, this becomes

$$\tfrac{2a^2}{a-\beta}(s + 3r) = 2a\beta(\eta - s).$$

51. Moreover, from the equation $(c+\xi)^3(c+\eta) = (c+\xi_0)^2(c+\eta_0)$, we have c since a and a are the same order: β, a of the order ϵ; arising from the variation of a, an indefinitely small in regard to those arising from the variation of ξ; and we have

$$\begin{gathered} \tfrac{3\eta s}{a-\beta} = -\tfrac{3a\beta}{a^2}(s - r) = -3p\eta - \tfrac{(p-q)}{2(a-\gamma)}, \\ \tfrac{3\eta s}{a-\beta} = -3a - \tfrac{3a\beta}{a^2}\eta(p - q) = -3p\eta - \tfrac{(p-q)}{2(a-\gamma)}, \end{gathered}$$

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and for δx etc. we have

$$a^2x^2 = -\tfrac{1}{a^2}\left(\tfrac{a\beta\gamma}{a^2}\right)^2 p\eta = -\tfrac{(a^2-\beta)^2}{(a-\beta)^2}\left[\tfrac{-(a-\beta)^2}{4(\gamma+2)\eta\eta}\right],$$

so that solving for greater simplicity, $(\beta - a)\eta_0 = -y\beta$, the formulae become

$$\tfrac{2a^2}{a-\beta}(s+3r) = -\tfrac{(b-a)^2\,\delta\eta}{a-\beta} - \tfrac{(a-\beta)\,\delta\eta}{(a-\beta)^2\,\beta}q^2.$$

52. This shows that there is at the entropy a cusp, the cuspidal tangent being in the plane of xy. It appears moreover that this tangent coincides with the tangent of the evolute. In fact, from the equation $(ax^2)^2 + (by^2)^2 - c^2y^2 = 0$ of the evolute we have

$$\tfrac{(ax)^3}{a} + \tfrac{(by)^3}{b} - \tfrac{(ax^2)^2+(by^2)^2}{a^2}\,\tfrac{dy}{dx} = 0,$$

or substituting for (x, y) their values as in the entropy,

$$\tfrac{2(b-a)(b-\beta)(\beta-a)}{a^2(a-\beta)^2} - \tfrac{(a-\beta)+ay\eta}{x^2(b-\beta)+a\beta\beta-a\beta}\cdot\tfrac{dy}{dx} = 0,$$

that is,

$$\tfrac{dy}{dx} = \tfrac{2(a^2y+a\eta x)(\beta-a)}{a(a+\beta)-a\eta}\,\tfrac{dy}{dx}.$$

which is satisfied by the foregoing values of $\frac{\delta x}{\delta x}, \frac{\delta y}{\delta y}$; the two tangents therefore coincide.

We have

$$4(by)^2 + 4(by)^2(4by^2-4)\sqrt{ay^2+by^2}\,\delta y = \tfrac{4(a+\beta)^2}{a(a+\beta)^2}\,\eta ay - \cdots\xi,$$

which is virtue of

$$(ay)(\sqrt{a+\beta})(\beta y^2(\sqrt{a}+a)) + \beta(y^2(\sqrt{a}+a)) + (a(2+a))y^2 = 4(y\sqrt{a^2-4} + 3y^4y^2(a+\beta)),$$

is

$$4[(by)^2(by)^2 + (by^2 + 3y^4y^2(a+\beta))] = \tfrac{2(a+\beta)^2}{a(a+\beta)^2}\,\eta ay\cdots\xi.$$

(observe $2(a^2+\beta) - a(a-\beta) = -(a+(a+\beta)) = -8a^2$ is negative)

$$= -\tfrac{2a^2}{a(a+\beta)^2}\,\eta ay\cdots\xi.$$

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if $\frac{dy}{dx}$ be the value at the entropy. Writing $\frac{a^2}{a^2}$ for the element of the are we have

$$\frac{ds^2}{a^2} = \tfrac{dx^2(a-\beta)^2}{(a-\beta)^2}\left[1-\tfrac{4}{a}\eta\right] - \tfrac{8a^2}{a},$$

$$\frac{ds^2}{a^2} = \tfrac{dx^2(-a\beta\gamma)^2}{(a-\beta)^2} - \tfrac{8a^2}{a},$$

which exhibit the form at the entropy.

The Nodal Curve; expressions for the coordinates in terms of a single parameter a. Art. Nos. 53 to 60.

53. After the foregoing investigation of the nodal curve, I was led to perceive that it is possible to express ξ, η, ξ_0, η_0 in terms of a single variable a; and thus to obtain expressions for the coordinates of a point of the nodal curve in terms of the single variable a. The analytical form of the nodal curve being real, with its imaginary values of ξ, η, the modes y may we may assume

$$\xi = -b^2 - p(b^2 - a^2 - b^2), \quad \eta = -b^2 + p(b^2 - a^2 - b^2),$$

$(i = \sqrt{-1}$ as usual): viz. p denote one or other of the quantities

$$y_+ = a + (a^2 - b^2), \quad q = a - (a^2 - b^2);$$

the expressions for $-\beta\gamma x^2$, $-\gamma a y^2$ become

$$= [(b-p)(a+1)(b-q)^3](b-p)^3,$$

and we have therefore the condition that this shall be real (for the two values $b = p$, $\dot{b} = -b$); being real, it will in certain cases be positive, in other cases negative, and the values for the remaining condition is given by the equation

$$\Delta'(b^2 - \xi^2 - \xi_0^2 - 4)(b^2 + b^2 + \xi^2 + 2b^2)^2 + p^2(b - \xi^2 - 2)(b^2 + \xi)^2 - 0 = 0.$$

54. The condition of reality is easily found to be

$$\Delta'(b^2 - \beta\beta - \beta^2)(\beta + \beta - 1) + (a+\beta)^2 + (\gamma + p)^2 + 4(a+p)(\beta - p) + (a+p)^2(b-\beta) - p^2q^2 = 0;$$

viz. this equation in Δ must have the form $-a, b$, suppose to a value

$$\frac{(y-a)\beta}{a^2} = a + \tfrac{a(a^2-b^2)}{2bg(\gamma-a)}\sqrt{Y}\cdot\eta(a-\beta-\gamma)(a+\beta+\gamma),$$

where, regarding X, Y are given functions of a, the coordinates of a plane in a cubic curve having the point $X = 0, Y = 0$ for a node, writing $Y = b + 3a(X+1)$, X will be such of these is rational function of a. The curve has the property that being any line through the entropy, the tangents at the points of the curve, the corresponding entropies on it are

$$(a^2 + Y)(a^2 + a)(a+1) + (b+1)(a^2+b)\beta(a^2+a) = -a^2(b-p)^3 + a(p+p)^3,$$

for a point on the nodal curve.

55. To complete the investigation, writing for a moment $T = 3 = 3a^2(X+1)$, we obtain

$$\frac{(b-p)^2}{a^2} = \tfrac{a}{2a^2(a+a-1)} + \tfrac{a(a^2)(b-\xi)^3}{a+b-1},$$

or putting for a moment

$$\frac{(\gamma-a)\beta}{a^2} = K,$$

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and writing $P+Q=P, X.P=P'$, the first of these is

$$\tfrac{(y-p)^2}{-\beta\gamma}\beta+\tfrac{X(X+2)}{(X+1)^2}K-\tfrac{Y(X+1)\beta}{2(X+1)}+\tfrac{1-X}{Y+X},$$

which regarding X, Y as the coordinates of a point in a plane in a cubic curve having the point $X=0, X=0$ for a node: hence writing $Y=b+3a(X+1)$, Y will be such of these is rational function of a. The curve has the property that being any line through the entropy, the tangents at the points of the curve, the corresponding entropies on it are

$$(a^2+Y)(a^2+a)(a+1)+(b+1)(a^2+b)\beta(a^2+a)=-a^2(b-p)^3+a(p+p)^3,$$

for a point on the nodal curve.

55. To complete the investigation, writing for a moment $T=3=3a^2(X+1)$, we obtain

$$\frac{(b-p)^2}{a^2}=\tfrac{a}{2a^2(a+a-1)}+\tfrac{a(a^2)(b-\xi)^3}{a+b-1},$$

or putting for a moment

$$\frac{(\gamma-a)\beta}{a^2}=K,$$

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we have

$$X = \tfrac{a(K-7)(a-1)}{a+1-K}, \quad X_0 = \tfrac{K(a-7)(a+1)}{a+1-K},$$
$$Y = K\tfrac{3K(a^2-1)+2(a+1)}{a+1-K}, \quad Y_0 = K\tfrac{3(a-1)(K-2)(a+1)}{a+1-K},$$

or substituting for K its value we have

$$K(a-1) = \tfrac{(a-1)^2}{ab}\left(a^2 - \tfrac{8aq}{(a-1)^2}\right) = -\tfrac{(a-1)^2}{ab}\left(a + \tfrac{2q}{a-1}\right)\left(a - \tfrac{2q}{a-1}\right),$$

$aa + 1 - K = -\frac{1}{ab}(a^2+1)(a+b-1)\,(a-p), \cdots\ +b(a-p)$, if as before $\Omega = \beta - \gamma$, and therefore changing the sign of the radical we have the values of $\frac{x}{a}, \cdots$,

56. Write for a moment,

$$\left[\Delta - \tfrac{1}{2}(\gamma - a)\sqrt{1 + \sqrt{\tfrac{2a(a-2-4\beta+2)}{a(a-2)}}}\right] - (a - 8a\beta + \sqrt{\beta})Y + B\sqrt{\delta} = 0,$$

$$\left[\Delta - \tfrac{1}{2}(\gamma - a)\sqrt{1 + \sqrt{\tfrac{2a(a-2-4\beta+2)}{a(a-2)}}}\right] - (a - 8a\beta + \sqrt{\beta})Y + B\sqrt{\delta} = 0,$$

then in the product of these two expressions the rational part is $= A^2A + B^2B$; and from the manner in which they were arrived at we have $9 = A_1A + B_1B$, and the rational part is thus

$$= -\tfrac{16}{a^2}(B^2 - B'B).$$

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We have

$$A^2A - B^2B = (\Delta - a)^2X - 8a^2X_0,$$
$$B = \tfrac{1}{2}(\gamma + a)X\Big[a - \tfrac{2aq}{(a-1)^2}(X+S)\Big] - a^2b^2(P^2 - 2a^2),$$

hence the rational part in question is

$$= \pm\tfrac{(\eta+a)^2}{(2a+Pp+a)}\sqrt{\tfrac{a+R+(1-\xi)(2+\xi)}{P\{Q+q(P+Q)-3\xi_0\}}}(2+\xi).$$

57. We have

$$1 = \tfrac{a}{2}\,\tfrac{1}{a^2(a-1)}\Big[aa^2 - 2\beta - \xi\Big(a^2 - \tfrac{4q}{(a^2-1)^2}\Big)\Big]$$
$$-\tfrac{1}{2}\,\tfrac{1}{(\gamma-a)^2}\Big[aa^2 - 2\beta - \xi - \tfrac{4qa}{(a^2-1)^2}\Big] + \cdots,$$

where the term in [ ] is

$$= \tfrac{2ag}{a^2(a^2-1)}\Big[3aa^2 - 2\beta - \xi\Big(a^2 - \tfrac{4q\eta}{(a^2-1)^2}\Big)\Big],$$

hence $\Delta = \frac{(\gamma-a)\beta}{(\gamma-1)}\,\frac{x(\beta-a)}{(y-a)} + \frac{X(\beta-y)}{(y-a)} - 2y$, where now

$$a = \tfrac{1}{2}(P+Q-1)(\beta-q).$$

58. Moreover

$$(\Delta - a)^2 = -\Delta^2 - \Delta a + a^2,$$

where the term in [ ] is

$$= -\tfrac{2ag}{a^2(a^2-1)}\Big[3aa^2 - 2\beta - \xi\Big(a^2 - \tfrac{4q\eta}{(a^2-1)^2}\Big)\Big];$$

where on changing a into β, the coefficient of a is

$$= \tfrac{2bx}{x^2}\,(b-1)^2 = -\tfrac{ab(a-b-1)(a+b)}{x^2}\big[3a(a - 2\beta - 4q\beta + 8\beta(a-q+1) + 8\beta(a^2-1)\big]$$
$$-\tfrac{a^2(b-a)}{(a^2-1)}\big[3aa^2 + 4(b-\beta) + 8(\beta-q) + 8(a^2-1)\big] + \tfrac{a(b+a)}{a^2(a-1)}\Sigma,$$

where the last expression on a single fraction is

$$= -\tfrac{8ax}{a(\beta-1)^2}\,(b\beta-q)(a-q)\cdots$$

and if we hence determine the values $3(\xi^2 - \xi_0^2) = \eta_0$, we find that the resulting terms divide by $\xi - \xi_0$, and we have

$$3\xi_0^2(\xi_0 - \xi) - \xi^2[(b+a)\xi + 3\xi_0] = (\xi+\xi_0)\xi_0(b+a)(\xi+\xi_0)(\xi+a)$$
$$+\xi_0\cdots,$$
$$+\xi_0\cdots.$$

Hence observing that by the relation between ξ, ξ_0, the first term is

$$= -3Q\xi_0 + (P+Q)q + 2\xi_0 q,$$

the equations become

$$1 : \eta : \eta_0 = -[Q + p(P+Q) - 3\xi_0]\cdot Q$$
$$: R(2\xi_0 + 2) - \xi^2(P + Q\xi)$$
$$: R(3\xi_0^2 - \xi^2) - \xi^2P(P + Q\xi).$$

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Similarly

$$\Sigma(\Delta - a)^2 + aX\Sigma a^2(a-1)\Sigma + a^2(\Sigma + S)$$

$$= \tfrac{a^2}{a(a-1)}\left[(b\beta - q)(a-q) + a(a-1)\Delta - q(b\beta - q)\Sigma\Delta + b\beta(a-q)\Sigma\Sigma - q\beta(a-q)\Sigma\Sigma\right],$$

where the term in { } is

$$= -\tfrac{8\beta}{a^2}\,\eta\Delta\Sigma\eta - \tfrac{a^2(b-1)}{(\beta-1)}\beta\beta a\Sigma + 3\beta\Sigma(a-1)\eta - \alpha + \beta - q(\beta - q) - \beta,$$

in which the coefficient of a is $a\beta(\beta - q)\Sigma\Sigma\Sigma + 3y\eta(a-q)$, but the last is a product of these linear functions: hence

$$3(\Sigma - aB)^2 + 6\Sigma = \tfrac{2(a-1)}{a^2}\,\eta(\Sigma - q + 3\beta(a-q) - \delta(a-q) - a).$$

59. Substituting these values we have the expression

$$\tfrac{1}{(\beta-1)\eta\alpha}\left[\tfrac{(y-a)q + a(a-1)(b\beta - q) + 3a\Sigma(a-q) - 2bq + (b\beta - q)^2 - (a\Sigma)}{a\eta(\eta - a)\Sigma a(a)}\right] + 5\Sigma^2,$$

which solving therein $\Delta = -y - \beta\beta\eta y - 5\beta\eta - 5\beta\eta + \Sigma\Sigma - 5\beta\eta\eta + 4(\eta - q + 1)\Sigma\eta + \cdots$, and the final results are

$$-\beta\beta z^2 = \tfrac{(y-a) + \delta(b\beta - q)\Sigma\Sigma + \frac{1}{2}\beta(a\Sigma) + \cdots(\Sigma + q^2(\Sigma+5)(a-1)}{4y(y-1)(\beta-4)\Sigma\Sigma(a-q)},$$

which is in be observed that, equating the denominators by $\Sigma\Sigma$, we have a simple root $\Sigma\Sigma$, to indicate this, we may insert in the denominator the factor $(1 - 8\beta)$.

60. We see here the meaning of all the factors, viz. Planes.

	$a = 0$	$\beta = 0$	$c = 0$	$a = \infty$
Evolute nodes	$a = -\frac{a^2}{y-a}$	$a = -1$	$\cdots$	$a = -\infty$
Umbilicar centres	$a = -\frac{2y\beta}{(\eta-\beta)^2}$	$\cdots$	$\cdots$	$\cdots$
Entropies	$a = -\frac{a+\beta}{a+\beta}(\eta - \beta\beta)$	$\cdots$	$\cdots$	$\cdots$

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For the real curve a extends from $\frac{a}{2b}$ through 0 to $-\frac{3a}{2b}$, viz.

$$a = -\tfrac{2y}{y-a}, \quad a = -1, \quad \text{gives entropy in plane } a = 0,$$

$$a = \tfrac{2a\beta}{y-a}, \quad \text{umbilicar centre in plane } y = 0,$$

$$a = -\tfrac{a+\beta}{a+\beta}(\eta - \beta), \quad \text{evolute-node in plane } a.$$

It is to be noticed that the order of magnitude of the terms in the table is

$$a, -\tfrac{2y}{y-a}, \quad a, \tfrac{2a\beta}{y-a}, \quad a, \cdots,$$

$\xi \cdot \frac{2y-a+\beta-\frac{a+\beta}{a+\beta}}{(\eta+a+\beta)}$; which belong to the real curve are contiguous; this is in fact be should be: Several of the proving investigations conducted by means of this quantities ξ, ξ, ξ_0, might have been conducted more simply by means of the formula involving a.

The Eight Cuspidal Conics. Art. Nos. 61 to 71.

61. The centro-surface is the envelope of the quadric

$$\frac{a^2x^2}{(a+\xi)^2} + \frac{b^2y^2}{(b+\xi)^2} + \frac{c^2z^2}{(c+\xi)^2} - 1 = 0.$$

Hence it has a cuspidal curve given by means of this equation and the first and second derived equations

$$\frac{a^2x^2}{(a+\xi)^3} + \frac{b^2y^2}{(b+\xi)^3} + \frac{c^2z^2}{(c+\xi)^3} = 0,$$

$$\frac{a^2x^2}{(a+\xi)^4} + \frac{b^2y^2}{(b+\xi)^4} + \frac{c^2z^2}{(c+\xi)^4} = 0,$$

which equations determine α^2, b^2y^2, c^2z^2 in terms of ξ, viz. we have

$$-\beta\gamma a^2x^2 = a^2(a+\xi)^3,$$

$$-\gamma a b^2y^2 = b^2(b+\xi)^3,$$

$$-a\beta c^2z^2 = c^2(c+\xi)^3;$$

so that, comparing with the equations $-\beta\gamma x^2 = a^2(a+\xi)^3(a+\eta)$, etc., which gave the centro-surface, we see that the cuspidal curve $\xi = \eta$, the cuspidal curve must be in question arises from the entire case in the ellipsoid, which form as the envelope of the curves of curvature: it is clear that the curve is imaginary.

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62. From the foregoing equations we have

$$\sqrt{ax} + \sqrt{\beta y} + \sqrt{cz} = a + \sqrt{\xi},$$

$$a\sqrt{ax} + \beta\sqrt{\beta y} + c\sqrt{cz} = a + \sqrt{\xi}\,\xi;$$

the second of which is best written in the rationalised form

$$(1, 1, 1, -1, -1, -1)(ax, by, cz; \beta\sqrt{\gamma}y, \sqrt{\gamma a}z, \sqrt{a\beta}y)^2 = ay^2,$$

and combining herewith the equation

$$\sqrt{ax} + \sqrt{\beta y} + \sqrt{cz} + \sqrt{\xi} = 0,$$

then for any given signs of $\sqrt{a}$, $\sqrt{\beta}$, $\sqrt{c}$, $\sqrt{\xi}$ the first of these equations represents a quartic surface, the second a plane; or the two equations together represent a conic.

By changing the signs of the radicals (observing that when all the four signs are changed simultaneously the curve is unaltered) we obtain in all 8 conics, but only four quartic surfaces; viz. the two conics

$$\sqrt{ax} + \sqrt{\beta y} + \sqrt{cz} = \sqrt{\xi},$$

$$\sqrt{ax} + \sqrt{\beta y} + \sqrt{cz} = -\sqrt{\xi}$$

lie on the same quartic surface.

63. The conics form two tetrads, and the principal conics (reckoning as one of them the conic at infinity) another tetrad: the complete cuspidal curve consists therefore of three tetrads of conics; with these we may form (one or eight out of each tetrad) two tetrads of skew hexagons (omit); the corresponding sides of these tetrads, and with a determination omitted in regard to the ellipsoid, the cuspidal conic of the principal eight conic of the principal eight; the same quadric surface. It is in be remembered that, having in mind the corresponding multilinear conics, and equally that the cuspidal conics three of the evolute at that point, which point is also a node on the surface.

64. The 8 conics have two tetrads, and the principal conics (reckoning as one of them the conic at infinity) another tetrad: the complete cuspidal curve consists therefore of three tetrads of conics; with these we may form (one or eight out of each tetrad) two tetrads of skew hexagons (omit); the corresponding sides of these tetrads, and with a determination omitted in regard to the ellipsoid, the cuspidal conic of the principal eight conic of the principal eight; the same quadric surface. It is in be remembered that, having in mind the corresponding multilinear conics, and equally that the cuspidal conics three of the evolute at that point, which point is also a node on the surface.

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65. But first consider the two conics

$$\sqrt{a}x + \sqrt{b}y + \sqrt{c}z = \sqrt{-a\beta\gamma},$$

$$(1, 1, 1, -1, -1, -1)(ax, by, cz; \beta\sqrt{\gamma}y, \sqrt{\gamma a}z, \sqrt{a\beta}y)^2 = ay^2;$$

for the intersections with the plane $y = 0$ we have

$$\sqrt{a}x + \sqrt{b}y + \sqrt{c}z = \sqrt{-a\beta\gamma},$$

$$(axx - a^2x - \gamma yz)^2 = 0;$$

so that the two conics each meet the plane in question in the same two coincident points, that is, they each touch the plane $y = 0$ at the same point. The two points by the equations

$$\sqrt{a}x + \sqrt{c}z = \sqrt{-a\beta\gamma},$$

$$(axx - yz)^2 = 0;$$

viz. this is the point, $aa = \frac{-ayc}{a-\beta}$, $aa = \frac{2y\beta}{a-\beta}$, which is one of the umbilicar centres before mentioned. We have

$$(a^2 + a^2) \cdots,$$

and the common tangent at this point is

$$\sqrt{a}x + \sqrt{b}y + \sqrt{c}z = \sqrt{-a\beta\gamma},$$

which is also the common tangent of the ellipse and evolute in the plane $y = 0$.

66. It has been seen that the nodal curve meets each principal conic at four umbilicar centres, which form ten cusps of the nodal curve. It is to be further shown that the nodal curve meets each of the 8 cuspidal conics in the four points (giving 32 nodal points distinguished as the principal entropy) or 16 entropies, and the points now in question are cusps of the nodal curve.

67. Putting for shortness,

$$\Theta = \sqrt{1\big(\sqrt{\eta - q}(\eta - \beta)\,(\eta - 2)\big)},$$

$$S = \tfrac{a(a+\beta a)}{\eta(\eta-q)}\left(\eta - \tfrac{2y}{\eta-a}\right),$$

we have then

$$\Theta(1 + a\Theta)\Delta = -\sqrt{ax}.$$

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or, what is the same thing,

$$\Theta(1+3S) - 1 + a\sqrt{\Sigma}(2+S-2) = 0.$$

we have without difficulty

$$\Theta(1+S)+1 = \tfrac{1}{2(\eta-q)}[3a^2 - 4a + 4 + \tfrac{8\eta q}{(\eta-q)^2}],$$

$$\Theta(1+2+\Theta) - 1 = \tfrac{1}{2(\eta-q)}\left[\tfrac{2(\eta-1)\beta+2y(\eta+\beta-q)}{(\eta-q)} - \tfrac{4(\beta q+\beta\eta)}{(\eta-q)^2}\right];$$

Omitting it, the equation becomes

$$\sqrt{a^2-\beta-\tfrac{2y}{(\eta-q)\eta}}\sqrt{a^2-\tfrac{(2)\eta-4\beta+\frac{1}{2}}{(\eta-q)}}\left(\eta-4\eta+4\eta+\tfrac{2}{(\eta-q)}\right) = 0,$$

or putting for shortness

$$\sqrt{a^2-\beta-\tfrac{2y}{(\eta-q)\eta}} = M, \text{ and rationalising, this is}$$
$$= (\sigma-2-M)(\Sigma\eta-2) + 4y\Sigma + 4(\Sigma-1)$$

and, working this out, the terms in Σ disappear, and we have for the resulting terms divide by $\Sigma - 2$, and we have

$$8 : a = \cdots = -P\Sigma\beta + 4(\Sigma+\beta+3\Sigma-4)\Sigma + 8M,$$

a quartic equation in a: each of the 4 roots there correspond 8 intersections, viz. there will be in all 32 intersections, lying in 4's upon the 8 cuspidal conics.

68. To start from these points are cusps, or stationary points on the nodal curve, starting from the expression of $-\beta\gamma z^3$ etc. in terms of a we have, first for δx, the equations

$$\tfrac{\delta x}{x} = \tfrac{a}{a-\beta+1}\,\delta a + \tfrac{4}{a+1}\,\delta a + \tfrac{2(\eta-\beta)}{(a+\beta+a)(\sqrt{\Sigma}-q+\beta+a)}\,\delta a,$$

so, as this may be written,

$$\tfrac{\delta x}{x} = \tfrac{a(2a+4(a+1)\beta-q(a+1)-\beta)\eta(\Sigma+M+2a-2P)+(M+3\eta)\eta-4a+M)}{a(a+1)(\beta-1)\Sigma a\eta-q(a)}$$

$$-\tfrac{a\Sigma\,[5+5\Sigma\,(\frac{1}{2}(a+8\eta+9\Sigma)+\frac{1}{2}(3M-8\Sigma\eta\Sigma)+\frac{1}{2}(3M(a+4\eta+a)+\Sigma-2\Sigma a+a+\beta))]}{a(a+1)(\Sigma\eta-q)\Sigma\Sigma a(a-q)}$$

viz. the numerator vanishes when a is a root of the quartic equation.

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69. We have moreover

$$-\tfrac{2\delta y}{y}\,da = \tfrac{1}{\eta\eta(\eta-q)}\left[a + \tfrac{2\eta}{(\eta-a)(\eta-q)} + \tfrac{2a(\eta-q+a)}{(\eta-a)} + \tfrac{2(a-q)}{(\eta-a)} - \Omega\,\tfrac{6}{a}\right],$$

which, putting $\frac{\eta-q+a}{\eta-q} = R$, and therefore $\frac{a-q}{\eta-q} = R-1$, $a = \frac{\Omega}{\eta-q} - C$, is

$$= \tfrac{1}{2}\left[\tfrac{1}{a+R} + \tfrac{1}{R} - 2R + \tfrac{2R}{\eta-q}\cdot\tfrac{4}{\eta-q}\cdot\tfrac{M+8\eta\beta}{q} + \tfrac{6}{\eta-q}\right];$$

and adding the fractions inside of { }, the numerator is

$$a(PR + 5G\Omega - R) + \cdots$$

$$= \tfrac{2}{\eta}\,a(\eta+a+1)(\eta+5G+7\eta+3G\beta) - 6G + (\eta+5G+7\eta+4G)\Omega - 96b\beta\beta + 8G,$$

which observing that $R = 2+a$, is $= 8M + 16\beta\beta$, and, substituting for M and R their values, this is

$$= a^2(7M+5R) + (3M+8\beta) + (\beta+5G+2\eta+a) + 8\beta M,$$

and substituting, the whole denominator is $1\cdot 8M$, so that, as the result of the equation

$$8M = a^4 - 3M(\eta+5G+2\eta+a) - 8\beta M,$$

we may have

$$-\tfrac{2\delta y}{y}\,da = \tfrac{4(2\eta-1)(\eta+a+1)}{(\eta-q)\eta}\left[a + \tfrac{2\eta}{(\eta-q)}\right] + \tfrac{a}{\eta+a+2}.$$

and the numerator of the expression on a single fraction is

$$= \tfrac{4(2\eta+a)^2}{(\eta-q)^2\eta^2}\,a(\eta + \tfrac{2y\Sigma}{\eta(\eta-q)}\,a + a)\,\beta(\eta-4M)\,\cdots,$$

which is

$$= 5(\eta+5G+\eta+\beta) + 8\Omega(\eta+aM+8\beta a+a) - 24\Omega(\eta+aM+5\beta\eta+aM),$$

which is

$$= a\,\beta = -3M(\eta+3G+\eta+8\beta+a+5\beta) + \tfrac{4(2\eta+a)}{(\eta-q)^2\,\eta}\,a(3M+5G+7\eta+4G);$$

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the term in [ ] is

$$= a^2(7M + 5R) + B + 8\,\tfrac{a+\beta a}{(\eta-q)\eta}\,\sigma$$

$$= a + 2(M + 8R) + 8\,\tfrac{a+\beta\eta}{(\eta-q)\eta}(a - 8a + \beta a + a)$$

$$+2[3M\beta + 8\eta M\,\tfrac{a+\beta\eta}{(\eta-q)\eta}(P + 2\beta\beta + 4G\Omega\eta)$$

$$-8M = a\sqrt{a^2(11M + 5R) + 5R(a^2 - 5R) + 5\Sigma + 5R}\,\beta + 5\Omega M],$$

which is found to be

$$= -8a\,\Omega + a^2(11M + 5R) + B\cdot\beta + \beta M;$$

and the whole numerator is

$$= 8a\,\Omega^2 + a^2(11M + 5R) + B\cdot\beta + \beta M,$$

which is

$$= (16 + 5R)\,a + 5R(a^2 - 5R) + 5M + 4P\beta + 4M.$$

71. We have then

$$-\tfrac{2\delta y}{y\,da} = \frac{1\eta(R+6\Sigma\beta)(\eta+a) - \frac{5G+2\eta+a}{(\eta-q)^2}\,a + 5RM - 16M^2(\beta+R)\eta + \frac{1\eta}{a+1\beta}\,\beta + 16M\eta + 5R}{(\eta-q)^2\,\eta\beta(\eta-q)},$$

and theme also

$$\tfrac{2\delta y}{y\,da}\left(\tfrac{a+\beta}{\eta-q}\right)\Omega + \tfrac{(2\eta+a)(2\eta+a)(\eta+a+a+1)}{(\eta-q)\,\eta\beta}\,\sigma + \tfrac{a+\beta}{(\eta-q)}\left(a\,\tfrac{a}{a-q}\right)$$

so that δx do also vanish when a is a root of the quartic equation, the points in question are therefore cusps of the nodal curve.

Centro-surface as the envelope of quartic $\Sigma a^2 = \beta + 1 = 0$. Art. Nos. 72 to 76.

72. The equations $-\beta\gamma x^2 = a^2(a+\xi)^3(a+\eta)$, etc., considering therein ξ, η as variable parameters connected by the equation $\xi - \eta$ as a given constant value, give the equation of the sequential centro-curve, and considering ξ as a given constant but η as variable they give the constant centro-curve.

73. Suppose first that η is a given constant; to eliminate ξ we may write the equations in the form

$$-(\beta\gamma\sqrt{a} = a^2(a+\xi)^{3/2}(a+\eta)^{1/2},\ \text{etc.},$$

and then multiplying first by $a^{-3/2}, b^{-3/2}, c^{-3/2}$, and adding, and secondly by b, c, a, and adding, (observing that $\Sigma(\sqrt{a}) = 0$, $\Sigma(\sqrt{a}\sqrt{b}) = 0$), we obtain

$$\Sigma(ax)\sqrt{a^2 + a + b} = -a\sqrt{a + b},$$

$$\Sigma(\beta ax)\sqrt{a^2 + a + b} = -a(a + b\,\sqrt{a+\eta}\,\beta),$$

which equations, considering therein a as a given constant, are the equations of a sequential centro-curve.

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If from the two equations we eliminate a we should have the equation of the centro-surface; the second equation is the derivative of the first in regard to η; and it thus appears that the equation of the surface can be obtained by considering η in the first equation as a variable parameter. The equation of the centro-surface is thus

$$\sum(\overline{ax})^2(a+\eta)^4 = (-a\beta\gamma)^2.$$

but the form is less convenient to that of
The above equation is, multiplying by b, c, a, and adding, by b, c, a, and adding, by b^2, c^2, a^2, and adding,

$$3a^2 \cdot ab^2 + 3ab^2c^2 + ab^2\overline{ax^2} = -a\beta\gamma \cdot \sqrt{(a+\eta)\beta} \cdot ab,$$

and, eliminating η, we have the equation in the form before mentioned (last paragraph).

74. Taking now ξ as a given constant, and writing the equation in the form

$$\sqrt{a} \cdot \overline{ax^2} = a^2(a+\xi)^{3/2} \cdot (a+\eta)^{1/2}, \text{ etc.},$$

then multiplying by $\beta\gamma, \gamma a, a\beta$, and adding, and secondly by bc, ca, ab, and adding, or writing this for the sake of distinction

$$\sqrt{a} \cdot \overline{ax^2} = a^2\beta^{3/2}(a+\xi)^{1/2}, \text{ etc.},$$

we have

$$\sum(\overline{ax})^2(a+\xi)^{1/2} = a+\xi,$$

$$\sum(\overline{ax})^2(a+\xi)^{1/2} = a+\xi+b+\xi,$$

or writing these equations in full form

$$\sqrt{a^2(a+\xi)^2+b^2(b+\xi)^2+c^2(c+\xi)^2+\delta} - \delta + \xi = 0,$$

$$\sqrt{a^2(a+\xi)^2+b^2(b+\xi)^2+c^2(c+\xi)^2+\delta} - 4 + \xi - 1 = 0,$$

thus the cofficients on the second side are $(a, x, y, z, ax, by, cz, a^2)$, the first quadrilateral. The second equation as the derivative of the first in regard to ξ, a, z, etc., as the first quadrilateral. The second equation as the derivative of the second in regard to ξ, a, z, etc., as a function of a in this way evidently a twofold factor $\left(1+\frac{1}{\xi}\right)^2$, viz. this is, the centro-surface is the envelope of a sequence of quartic surfaces, each of which lies entirely the centro-surface. The numerical results were not made out, but my own. (see "On a certain Sextic Form," *Camb. Phil. Trans. t. xx* (1871), pp. 397–523, [486].)

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Another generation of the Centro-surface. Art. Nos. 77 to 83.

77. It may precede, the equation of the centro-surface is obtained as the condition in order that the equation

$$\sum (ax)\sqrt{a+\eta-1} = 0$$

may have equal roots. But taking a as arbitrary constant, this is the derived equation of

$$\sum a(ax)^2(a+\eta) = a+\eta+ax,$$

and as such it will have two equal roots if the last-mentioned equation has the property of expressing that the last-mentioned equation, or what is the same thing, the quartic equation

$$(a+\eta)+a(ax)^2+a(b+\eta)+a(b+\eta)^2+(a(a+\eta)+a(a+\eta))^2 = 0,$$

has three equal roots. The condition for this are that the discriminant and the subdiscriminant shall each of them vanish, and we have thus the equations to be considered as respectively a quadric and a cubic function of a; viz. the equations

$$\langle a,h,h\,\xi\rangle(a,b,c,d,e)^4 = 0$$
$$\langle a,h,h\,\xi\rangle(a,b,c,d,e)^4 = 0$$

where the degrees in (x,y,z) of a, b, c, d, e are 0, 2, 4, 6 and those of a', b', c', d' are $2,4,6,\,8$ respectively; the equation of the centro-surface is then

$$\begin{vmatrix} a, & b, & c, & d, & e \\ a', & b', & c', & d', & e' \end{vmatrix} = 0,$$

which is of the right order 12; but it would be difficult to obtain thereby the developed equation.

78. For the nodal curve the cubic equation must be satisfied by each root of the quadric equation, or that is the same thing, the quadric function must contain as a factor

$$\langle a,b,c\rangle\langle a,b',c',d'\rangle = 0,$$

whose the degrees may be taken as below, viz.

$$\langle 0,2,4\rangle\langle 0,2,4\rangle = 0.$$

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and the order of the nodal curve is then $= 24$: two of the equations in fact are

$$\begin{vmatrix} a, & b, & e \\ a', & b', & e' \end{vmatrix} = 0, \quad \begin{vmatrix} a, & b, & e \\ a', & b', & e' \end{vmatrix} = 0,$$

which are surfaces of the orders 4, 6, etc.; or the nodal curve is a complete intersection Σ x 6. By the results above obtained as to the nodal curve, it appears that the two surfaces must have an ordinary contact at each of the 16 umbilicar centres, and a stationary or singular contact at each of the 32 entropies.

79. The derivation of the centro-surface from the surface

$$\frac{a^2}{a^2+\xi}\,X^2 + \frac{b^2}{b^2+\xi}\,Y^2 + \frac{c^2}{c^2+\xi}\,Z^2 - U^2 = 0$$

requires to be further explained. The surface, say $V = 0$, is a quadric surface depending on the two parameters ξ, η; the same outside in direction with these of the ellipsoid, and their relative magnitudes obey the equation

$$\frac{1}{a^2+\xi}\,X^2 + \frac{1}{b^2+\xi}\,Y^2 + \frac{1}{c^2+\xi}\,Z^2 = 1$$

divided by a, b, c respectively; viz. for the absolute magnitudes observe that the equation may be written

$$\frac{a^2-\xi^2}{a^2+\xi}\,X^2 + \frac{b^2-\xi^2}{b^2+\xi}\,Y^2 + \frac{c^2-\xi^2}{c^2+\xi}\,Z^2 - 1 = 0,$$

viz. the surface $V = 0$ is a surface through the spheroconic which is the intersection of the conduit surface by the arbitrary sphere $a^2 + b^2 + c^2 + U^2 = 0$: but, while the surface is hereby and by the preceding condition as to the axes completely determined, the geometrical significance is anything but clear.

80. Considering then the quartic surface $V = 0$, depending on the parameters ξ, η, suppose that a remains constant while ξ alone varies; we have then consecutive surfaces $V = 0, V_1 = 0$, which on of these V_1 say intersect in a point of the centro-surface; the point in question will depend on the two parameters (ξ, η), and if these vary simultaneously we have thus a whole series of points on the centro-surface, but it may simultaneously be that there belong to the locus of points or the centro-surface. To make this evident, we must inquire that, that proceeding by both a being also along the value of η values of a which are not for which lines and the variable centres are imaginary by means of points on the centro-surface, it is the quadric superceded as ξ, η, to satisfy the equation of the centro-surface; thus the form of the line of corresponding to the difference of the nodal curve, but it is shown of curvature for the parameter η.

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the variable parameter be ξ, then this is a curve on the 12-fold surface $\Omega = 0$ obtained by the elimination of ξ from the equations $V = 0$, $\delta_\xi V = 0$; viz. we have $\delta_\xi V$ as the curve in question on the surface. The equation of the curve is the elimination of a as from the two equations $S = 0$, $T = 0$ gives as above the equation of the centro-surface.

81. The surface $\Omega = \delta - T = 0$ obtained as shown by the elimination of ξ from the equations $V = 0, \delta_\xi V = 0$, (or, what is the same thing, by equating to zero the discriminant of V in regard to ξ) may be termed the sociate-surface: we have there this quartic and sociate surfaces $S = 0, T = 0$ intersecting in the before-mentioned curve, which we call the sociate-edge; and the locus of these sociate-edges is the centro-surface.

82. We may if we please, changing the parameter in case of the functions, consider the two series of surfaces $S = 0$, $T = 0$ depending on the parameters a, a' respectively; a surface of the first series will correspond to a curve of the second series for the parameter a equal, say we have two surfaces of equal, a', and have these to a sociate-edge: taking a, a' as the parameters of the two surfaces a, a in nearly equal, we have the locus of these sociate-edges is the centro-surface.

83. We may by what precedes, changing the parameter in case of the functions, consider the two series of surfaces $S = 0$, $T = 0$ depending on the parameters a, a' respectively; a surface of the first series will correspond to a curve of the second series for the parameter a equal, say we have two surfaces of equal, a', and have these to a sociate-edge; taking a, a' as the parameters of the two surfaces a, a in nearly equal, we have the locus of these sociate-edges is the centro-surface.

a posteriori verification that the surfaces $V = 0, V' = 0, V'' = 0$ intersect in a point of the centro-surface, can be made without interest; the parameters ξ_1, η of the point of intersection being constant to be of the order η; whence, starting from the variation of η, we indefinitely small in regard to those arising from the variation of ξ, we ought therefore to have identically

$$\frac{-a(a^2+\xi_0)\xi'(a^2+\eta_0)}{a^2+\xi} = \delta\xi(a^2+\xi_0) - \eta_0 - \eta - \beta - \delta'$$

$$= \frac{a\eta(\xi-\xi_0)\sqrt{(a^2+\eta_0)}}{(a^2+\xi)\sqrt{(a^2+\eta_0)\xi'}}$$

and by decomposing the right-hand side into its component fractions this is at once seen to be true.

Third generation of the Centro-surface. Art. Nos. 84 and 85.

84. Instead of the foregoing equation $V = 0$, consider the equation

$$W = \left(\frac{a^2x^2}{a^2+\xi} + \frac{b^2y^2}{b^2+\xi} + \frac{c^2z^2}{c^2+\xi} + \frac{w^2}{\xi}\right) - \left(\frac{a^2x^2}{a^2+\eta} + \frac{b^2y^2}{b^2+\eta} + \frac{c^2z^2}{c^2+\eta} + \frac{w^2}{\eta}\right) = 0.$$

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The equations $\delta_\xi W = 0, \delta_\eta W = 0$ contain only ξ and η respectively; the surface is therefore the envelope of $a^2 - 4xy = 0$; the elimination of ξ from the equations $\delta_\xi W = 0, \delta_\eta W = 0$ would therefore lead to the equation of the centro-surface as before. But the equation $W = 0$ contains a factor of three of the equations $V = 0, \delta_\xi V = 0, \delta_\eta V = 0$, viz. to fix the ideas, $W = 0$ is connected with the equation $V = 0$ as above given by the elimination of ξ from the equation $W = 0$ depending only on ξ, we have ξ in regard to the value of ξ and in the regard to the value of η in the same equation; and we have the same equation $W = 0$ depending only on ξ; or showing that the determination of ξ, η by an equation of ξ form is connected.

$$(a^2+\xi)(a^2+\eta) + \tfrac{\xi y_0^2}{a^2+\eta} + \tfrac{\xi y_0^2}{(a^2+\xi)^2(a^2+\eta)^2} = 1 - 0,$$

so that the surface $\Omega = 0$ is obtained by eliminating ξ from this equation as the several points of the curve of curvature belonging to the parameter η; viz. regarding ξ as the variable parameter on the ellipsoid, then there is on the centro-surface the curve which is the value of ξ as belongs to the ellipsoid, then this is the curve which is the value of ξ as belongs to the ellipsoid; or, in a like manner, to the locus of the sociate edges is the centro-surface.

85. As ξ is constant the curve of of the centro-surface for points moved at distance a^2 in the curve, viz. in the same way, but as on the cone a may be reduced to a twofold form $\left(1+\frac{1}{\xi}\right)^2$, if $w^2 = 0$, and it has evidently a twofold factor on each side of the various contains the planes (when a has any other value), but it must necessarily form quartic surface envelope of the sphere (whose other generator ξ) which has the points (one or eight equal to the centre).

Reciprocal Surface. Art. No. 86.

86. The centro-surface is the envelope of

$$\frac{a^2X^2}{a^2+\xi} + \frac{b^2Y^2}{b^2+\xi} + \frac{c^2Z^2}{c^2+\xi} - W^2 = 0;$$

hence the reciprocal surface is the envelope of $a^2 + y^2 + z^2 + w^2 = 0$ in regard to

$$\frac{(a^2+\xi)\,x^2}{a^2} + \frac{(b^2+\xi)\,y^2}{b^2} + \frac{(c^2+\xi)\,z^2}{c^2} - w^2 = 0,$$

that is,

$$a^2x^2 + b^2y^2 + c^2z^2 + a^2y^2 - 0,$$

viz. the envelope is

$$(a^2x^2 + b^2y^2 + c^2z^2)(x^2+y^2+z^2) = w^2(a^2+b^2+c^2) - x^2 + y^2 + z^2 - X^2 = 0,$$

or, expanding and multiplying by $a^2b^2c^2$, this is

$$a^2b^2c^2(a^2+y^2+z^2)(a^2X^2+b^2Y^2+c^2Z^2) = w^4(a^2X^2+b^2Y^2+c^2Z^2) - a^2b^2c^2(a^2X^2+b^2Y^2+c^2Z^2)\,w^2,$$

or, what is the same thing,

$$a^2b^2c^2\,V_XX^2 + b^2c^2a^2\,V_YY^2 + c^2a^2b^2\,V_ZZ^2\,X^2 + c^2a^2b^2\,V_WW^2 + a^2b^2c^2 + \cdots = 0,$$

which may be written,

$$a^2X(a^2b^2+b^2c^2+c^2a^2)\cdot X^2 + (a^2b^2c^2 - X^2)\,a^2W^2 = 0,$$

that is,

$$(a,b,c,1)\,(X^2,Y^2,Z^2,W^2)^2 = 0,$$

and, consequently,

$$a^2+b^2+c^2 - W^2 - a^2b^2c^2 + (a^2b^2+b^2c^2+c^2a^2) - a^2b^2c^2 = 0,$$

It would doubtless be interesting to discuss this surface as it here presents itself, and with reference to the geometrical significance as the locus of the pole, in regard to the sphere, of the plane through the intersecting consecutive normals of the ellipsoid; but I abstain from any consideration of the question.

Delineation of the centro-surface for given numerical values of the constants. Art. Nos. 87 and 88.

87. I constructed on a large scale a drawing of the centro-surface for the values

$$a = 50, \quad b = 15, \quad c = 13.$$

(These were chosen so that a, b, c should have approximately the integer values 7, 3, 4, and that $a^2 + b^2$ should be will greater than $2b^2$; they give a good form of surface,

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though perhaps a better selection might have been made; there is a slight objection to the existence of the relation $a^2 = 25$; as in the projection it brings a cusp of the evolute on the ellipse). We have therefore

$$a = 50, \quad \beta = -33, \quad \gamma = 25,$$

the ellipses in the principal planes of the centro-surface are

$$\frac{x^2}{(2)} + \frac{y^2}{(b^2/a)^2} = 1, \quad \frac{x^2}{(2)} + \frac{z^2}{(b^2/a)^2} = 1,$$

$$\frac{x^2}{(216a^2)} + \frac{y^2}{(216a^2)} = 1,$$

$$\frac{ax^2}{(216a)^2} + \frac{z^2}{(2a)^2} = 1,$$

and these determine on each axis the two points which are the cusps of the evolutes. We have moreover for evolutes some $= 2998$, $y = 0$, $z = 1990$; and for the entropy $x = 1127$, $y = 1942$, $a = 0$.

88. For the delineation of the nodal curves (counted partially) we have then to find the values of ξ, ξ_0; these are given in terms of a, a, *ante*, No. 33 [p. 334]; where a is a given function of a, and a contains in terms of the constants a, β, γ and of a, β. It was thought sufficient to divide the interval into 8 equal parts, that is, the values of a were taken to be $-25, -27.5, \cdots - 50.6$. The values of ξ, ξ_0, being found, those of a etc. were obtained from these by means of the original equations $(a + \xi)$ of a etc., as ratio. The whole series of the results is given in the annexed Table, and from these the drawing was constructed.

I find also in the neighbourhood of the umbilicar centre $(a = 25 + a)$,

$$\delta x = -02469q^2,$$

$$\delta y = -02469q,$$

$$\delta z = -02191q^2,$$

and in the neighbourhood of the entropy $(\xi = 30 \cdot 333 + \xi)$, we have

$$\delta x = 1127a,$$

$$\delta y = 1764a,$$

$$\delta z = 4082a^3.$$

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[Numerical table of a, ξ, ξ_0, η, η_0, a^2x^2, b^2y^2, c^2z^2, X^2, Y^2, Z^2, etc. for $a = -25, -27.5, \ldots, -50.6$ — 12 columns of decimal data omitted]

The calculations were performed before I had obtained the formula in a, which would have given the results more easily.

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On Dr Wiener's Model of a Cubic Surface with 27 Real Lines; and on the Construction of a Double-Sixer.

[From the Transactions of the Cambridge Philosophical Society, vol. XII. *Part* I. *(1873), pp. 366–383. Read May 15, 1871.]*

I.

I CALL to mind that a cubic surface has upon it in general 27 lines which may be all of them real. We may out of the 27 lines (and this in 36 different ways) select 12 lines forming a "double-sixer," viz. denoting six lines by $a_1, a_2, a_3, a_4, a_5, a_6$, and the other six by $b_1, b_2, b_3, b_4, b_5, b_6$, then are these 12 lines such, that any two with the same suffix do not meet (for example a_1, b_1 do not meet), but any two with different suffixes (for example a_1, b_2 or a_2, b_3) meet; that the two lines forming a pair (a_1, b_1), etc., the intersection of the planes which meet contain pairs of these in the figure (a_2, b_4), etc., is wholly outside the figure as in the figures itself, the disposition of the points is as shown in the figure.

The model is formed of plaster, and is contained within a cube, the edge of which is = 16 inches, the equations of these three lines are

$$a_1, x = 0, \quad y = 0,$$
$$a_2, x = 0, \quad z = 1,$$
$$a_3, y = 1, \quad x = 1,$$

then as two from a meet each other, any two may be both, but it must be a constant each line A, except that the two lines of a pair (e.g. a_2, b_1, a_3, b_2) lie at one and most each other. And such a system of twelve lines leads us at once to construct the sixer (in the remaining fifteen lines (six of them on the surface, the others as the intersections of these which meet, and there are which contains the pairs of lines (a_1, b_1), etc., the intersection of these lines being on the surface, by reason of the consideration. The system of the twelve lines is in considered within a cube, the edge of which is = 16 inches, the surface of these lines through the centre of the cube, the axes parallel to the edges of the cube, and the axes of length = 16 inches.

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On Dr Wiener's Model of a Cubic Surface with 27 Real Lines.

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The model is a solid figure bounded by portions of the faces of the cube, and by a portion of the cubic surface, being a surface with three apertures, the orbiculation of which is not easily explained.

To determine the numerators I measured, on the faces of the cube, the coordinates of the two extremities of each of the twelve lines; these were measured in tenths of an inch (taking account of the half division, or twentieth of an inch); and the resulting numbers divided by 16 to reduce them to the belére-mentioned unit of $1 \cdot 6$ inches. These reduced values are shown in the table: knowing then the coordinates of two points on each line, the equations of the several lines became calculable; the true theoretical form of these results—viz. the form which, but for errors of the model, or of the measurement, they would have assumed—is

$$\begin{array}{ll}
b_1: & x = R_2 z + D, \quad y = R_2 z + D, \\
b_1': & y = 0, \\
b_2: & x = R_1(z+\beta_1), \quad y = R_1(z+\beta_1), \\
 & i = 1, \\
b_3: & x = R_2(z+\beta_2), \quad y = R_2(z+\beta_2), \\
b_4: & x = R_3(z+\beta_3), \quad y = R_3(z+\beta_3), \\
b_5: & x = R_4(z+\beta_4), \quad y = R_4(z+\beta_4), \\
 & i = 1, \\
a_1: & x = 0, \quad y = 0, \\
a_2: & x = A_1 z + C_1, \quad y = A_1(z-1), \\
a_3: & x = A_2(z+1), \quad y = A_2(z-1), \\
a_4: & x = A_3(z+1), \quad y = A_3(z-1), \\
a_5: & x = A_4(z+1), \quad y = A_4(z-1) + C_4,
\end{array}$$

but in consequence of such errors, the results are not accurately of the form in question.

The faces of the cube being as in the diagram:

[Figure: unit cube with face labels — diagram omitted]

the Table is

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[Table: equations substituted from the measurements of the model, 17 columns of numerical values for $ABCD$, $EFGH$, $AEFF$, $HCFG$, $CDGH$, $ABHD$. — table omitted]

I hence calculate the intersections: considering any two lines which ought to intersect, then projecting on the principal plane and calculating x, y, the coordinates

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of the point of intersection of the two projections, these values of x, y substituted in the equations should give the same value of z; but if the lines do not actually intersect, then the values of z will be different.

[Table: 7 column array b_1, b_2, b_3, b_4, b_5 with numerical intersection values — table omitted]

Starting from the assumed equations of b_1, b_2, b_3, b_4, b_5, and calculating by the theory the remaining lines, the equations of the b-lines (those of b_1 being calculated) are

$$\begin{aligned}
b_1 :&\quad y = 1284x + 154,\\
b_2 :&\quad x = 0, \quad z = -1,\\
b_3 :&\quad x = 1352(y + 510),\\
&y = 1009(z + 510),\\
b_4 :&\quad x = -731(z + 422),\\
&y = -500(z + 422),\\
b_5 :&\quad y = 0, \quad z = 1,\\
b_6 :&\quad x = 1131(z + 624),\\
&y = 1133(z + 624).
\end{aligned}$$

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and the equations of the a-lines (those of all but a_1 being calculated) are

$$
\begin{aligned}
a_1: &\quad x = 0, \quad y = 0,\\
a_2: &\quad x = -732z - 296,\\
&\quad y = -820(z-1),\\
a_3: &\quad x = -987(z+1),\\
&\quad y = -477(z-1),\\
a_4: &\quad x = -2.506(z+1),\\
&\quad y = -541(z-1),\\
a_5: &\quad x = -974(z+1),\\
&\quad y = -967z - 384,\\
a_6: &\quad x = -179(z+1),\\
&\quad y = -071(z-1).
\end{aligned}
$$

and thence for the points of intersection the coordinates are

[Table: 7 column array with $b_1, b_2, b_3, b_4, b_5, b_6$ headings and intersection-point coordinate values — table omitted]

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II.

I have in a paper "On the Orbi-tikere of a cubic surface," *Quart. Math. Journal*, t. x. (1870), pp. 58–71, [459], obtained analytical expressions for the twelve lines of a double-sixer, and also obtained numerical values, which however (as there remarked) did not come out convenient enes for the construction of a figure. A different mode of treatment since occurred to me, by means of the following equations of the cubic surface

$$\left(\frac{x}{a}-\frac{y}{c}\right)\left(\frac{x}{a}-\frac{y}{a}\right)\frac{z}{b}\left(\frac{x}{b}+\frac{y}{c}-\frac{z}{b}\right)\left(\frac{x}{b}-\frac{y}{a}-\frac{z}{b}\right)=0,$$

which on self appear is a very convenient one for the purpose. We in fact obtain at once eight lines of the double-sixer; viz. those are

1. $x=0,\quad w=0,$	I$'$. $x=0,\quad y=0,$
2. $y=0,\quad z=0,$	II$'$. $z=0,\quad w=0,$
3. $\frac{x}{a}-\frac{y}{c}=0, \frac{w}{b}-\frac{y}{a}=0,$	III$'$. $\frac{x}{a}-\frac{y}{a}=0, \frac{y}{c}-\frac{z}{b}=0,$
4. $\frac{x}{a}+\frac{y}{c}=0, \frac{w}{b}-\frac{y}{a}=0,$	IV$'$. $\frac{x}{a}-\frac{y}{a}=0, \frac{y}{c}+\frac{z}{b}=0,$

and also five lines not belonging to the double-sixer, viz.

12.	$x=0,$	$\left(-\frac{x}{a}+\frac{y}{c}+\frac{z}{b}\right)\frac{1}{abc}+\frac{w}{B}\left(-\frac{x}{a}+\frac{y}{c}-\frac{z}{b}\right)\frac{1}{ABC}=0,$
23.	$y=0,$	$\left(\frac{x}{a}+\frac{y}{c}-\frac{z}{b}\right)\frac{1}{abc}+\frac{w}{B}\left(\frac{x}{a}-\frac{y}{c}-\frac{z}{b}\right)\frac{1}{abc}=0,$
34.	$z=0,$	$\left(\frac{x}{a}-\frac{y}{c}-\frac{z}{b}\right)\frac{1}{abc}+\frac{w}{B}\left(\frac{x}{a}+\frac{y}{c}-\frac{z}{b}\right)\frac{1}{abc}=0,$
41.	$w=0,$	$\left(\frac{x}{a}-\frac{y}{c}+\frac{z}{b}\right)\frac{1}{abc}+\frac{w}{B}\left(\frac{x}{a}+\frac{y}{c}+\frac{z}{b}\right)\frac{1}{abc}=0,$
5 & 2.	$\frac{x}{a}+\frac{y}{c}=0,$	$\frac{z}{b}+\frac{w}{B}=0.$

The remaining lines of the double-sixer are then easily determined; viz. the lines 3, 5, 6, and 12 are met by the line Γ, and by a second line Γ'; this, as a line meeting 3, 5, 6, will be given by equations of the form

$$x=\frac{y}{a}(a+b),\quad x=\frac{y}{c}(a-c),$$

and observing that these equations, writing therein $z=0$, give

$$\frac{x}{a}-\frac{y}{c}=0,\quad \frac{y}{c}-\frac{z}{b}=0,$$

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the condition of intersection with the line 12 gives

$$\phi = \frac{a+b}{c-1} \cdot \frac{c-2}{a-1},$$

which is the value of ϕ in the foregoing equations; and so these we may join the resulting equation

$$pqr(aB + cB) = abqB(qB + rB),$$

Proceeding in like manner then for the lines 1, 2, 4, 5, the equations for the remaining four lines of the double-sixer are; those are

2. $\phi = \dfrac{a-bc}{B-bc}$,
$x = \dfrac{y}{a}(a + \frac{B}{a})$,
$z = \dfrac{a}{b}(a + \frac{B}{a})$,
$pqr(aB + cB) = abB(rB - dB)$,

1′. $\phi = \dfrac{c-ba}{B-ba}$,
$x = \dfrac{y}{a}(a + \frac{c}{B})$,
$z = \dfrac{a}{b}(a + \frac{c}{B})$,
$pqr(aB + cB) = abB(pB - qB)$,

4. $\phi = \dfrac{a-bc}{B-bc}$,
$\phi\left(\dfrac{x}{a} - \dfrac{w}{b}\right) = y$,
$\phi\left(\dfrac{x}{a} - \dfrac{w}{B}\right) = \frac{a}{c}$,
$pqr(aB - cB) = qBa(BqB - dB)$,

3′. $\phi = \dfrac{c-2a}{B-2a}$,
$\phi\left(\dfrac{x}{a} - \dfrac{w}{b}\right) = -\dfrac{y}{c}$,
$\phi\left(\dfrac{x}{a} - \dfrac{w}{B}\right) = -\dfrac{a}{c}$,
$rqr(aB - cB) = abr(rB - 2dB)$.

It may be added that —
in plane $z = 0$, the

intersection of Γ line on line $2 : y = (cB - cD)cy : sqrB - aR$,
,, 2 ,, $: y : x = -cyB(cD - Db)sw : sqrB - aR$,

and that the line joining these intersections is the line 12.
In plane $y = 0$,

intersection of $1'$ line on line $x : y : z = pqrB - aR : qprB - aR$,
,, 1 ,, $: x : z = wrB(aR - 2Bdy)sw : qrpB - aR$,

and that the line joining these intersections is the line 23.

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In plane $z = 0$,

$$\text{intersection of } 3' \text{ line on line } x : y = (yB - rj)sw : sqrB - cyB;$$
$$,, \quad 4 \quad ,, : x : w = -(Dy - By)sw : sqrB - cyB;$$

and that the line joining these intersections is the line 34.

And in plane $w = 0$,

$$\text{intersection of 4 line on line } x : y : z = wpyB - cyB : ywpyB - cyB;$$
$$,, \quad 1' \quad ,, : x : y = wpyB - cyB : -ywpyB - cyB;$$

and that the line joining these intersections is the line 41.

The equations of the remaining ten lines of the surface may be obtained without difficulty, and also the thirty further triple planes; but I do not stop to effect this. The construction of a model it is sufficient to determine the three points on each edge, and the two points, on the planes of $ABCD$, viz. the points a_0, b_0, c_0, d_0, in which the three sides of the triangle ABC meet by triple planes, and the points ABC, a_0BCD (for the construction of the remaining ten lines of the surface by another method, viz. employing the points A, B, C, D, a_0, b_0, c_0, d_0). In fact, knowing each of a, b, c, d, α, β, γ on the edges of the cube, we know the values of ξ, η, ζ for the points on the edge BC; the points on the edge BC are obtained from the corresponding points on AD by the lines 4 and 1. For the construction of a model it is sufficient to determine the three points on each edge, and the two points, on the planes of $ABCD$, viz. the points a_0, b_0, c_0, d_0, in which the three sides of the triangle ABC meet by triple planes, and the points $abAB$, $bdAD$, $caCD$ on the planes of ABC. Then for ab we have

$$\frac{x}{a} + \frac{y}{c} = 0, \quad \frac{y}{c} - \frac{z}{b} = 0, \quad w = 0,$$

viz. the $ABCD$-coords mements of the orbi parametrised the lines $1, 2, 3, 4$. But if any sphere of an cube and several distances mentioned in the orbi- clear in their intersection with the line 4'; and it is probable that a slightly different value of ϕ would be better.

[Figure: triangle with three points — diagram omitted]

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The results just obtained may be exhibited in a compendious form as follows:

[Table: 5 column array with column headings α, β, γ, δ, θ listing intersections for various points $1, 2, 3, 4$ etc with AB, AD, ACD, etc. — table omitted]

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Γ' and Γ meet BCD on line: $\left(-\frac{x}{a}+\frac{y}{c}\right)\frac{1}{abc}=\left(-\frac{x}{a}+\frac{y}{c}-\frac{z}{b}\right)\frac{1}{ABC}=0,$
2 and 3′ „ CDA „ $\left(\frac{x}{a}-\frac{y}{c}\right)\frac{1}{abc}=\left(\frac{x}{a}-\frac{y}{c}-\frac{z}{b}\right)\frac{1}{ABC}=0,$
3 and 4′ „ DAB „ $\left(\frac{x}{a}-\frac{y}{c}\right)\frac{1}{abc}=\left(\frac{x}{a}-\frac{y}{c}-\frac{z}{b}\right)\frac{1}{ABC}=0,$
4 and 1′ „ ABC „ $\left(\frac{x}{a}-\frac{y}{c}\right)\frac{1}{abc}=\left(\frac{x}{a}-\frac{y}{c}-\frac{z}{b}\right)\frac{1}{ABC}=0,$

For calculating the numerical values from the foregoing assumed data, say

[Table: column headings α, β, γ, δ with numerical computation entries — table omitted]

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which had two equations now as a verification.

The outside numerical values are given in the manner most convenient for the construction of a drawing; viz. when the coordinates refer to a point on an edge of the tetrahedron, or any on the side of an equilateral triangle, the lines being the length of this edge (or side) to be $= 100$, the numerical values are fixed so that the answers of the points (regarded as $= 0, 20, 40, \ldots, 100$) are situated in the table similarly to that of the extension of the edge or side: but when the three coordinates belong to a point in the face of the tetrahedron, or any in the plane of an equilateral triangle, then the sum of the coordinates is made $= 80 \cdot 6$, and the three coordinates these denote the perpendicular distances from the sides of the triangle.

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III.

It is possible to find on a cubic curve a double-sixer of points $1, 2, 3, 4, 5, 6$ and $1', 2', 3', 4', 5', 6'$ such that any six points such as $1, 2, 3, 4, 5, 6$ lie in a conic. In fact considering a cubic surface having upon it the double-sixer of lines $1, 2, 3, 4, 5, 6$ and $1', 2', 3', 4', 5', 6'$, the section by any plane is a cubic curve meeting the lines, say in the points $1, 2, 3, 4, 5, 6, 1', 2', 3', 4', 5', 6'$; each of the six lines $1, 2$ meets each of the six lines $1', 2', 3', 4', 5', 6'$, and consequently the six lines lie in a quartic surface; therefore the points $1, 2, 3, 4, 5, 6'$ lie in a conic; and so in like order; cases; the number of the conics is of course $= 6$.

The cubic curve may be a given curve, and six of the points upon it (not being points on a conic) may be taken to be, given; for instance the points $1, 2, 3, 1', 2', 3'$. For take through the points $1, 2$ respectively any two lines $1, 2$; through $1', 2', 3'$ respectively the lines $1', 2', 3'$ each meeting each of the lines 1, 2; and through 1 a line meeting each of the lines $1, 2, 3$, it is easy to see that a cubic surface may be drawn through the cubic curve and the lines 1, 2, 3, $1', 2', 3'$. In the passage through the cubic curve requires only 9 conditions; the surface then passes through the point 2 and so make it pass through the line 2 requires 3 conditions; similarly the surface passes through the point 3, and to make it pass through the line 3 requires 3 conditions. The surface now passes through 1 and through the points of intersection of the line 1' with the lines 2, 3; to make it pass through the line 1' requires 1 condition; similarly to make it pass through the lines $2', 3'$, 1 requires in each case 1 condition; or there are in all 19 conditions, so that the cubic surface is completely determined. Take now through the points $1, 2, 3, 4, 5'$, a conic meeting the cubic in the points 4', 5', then through the lines $1, 2, 3, 4', 5'$ we have a quintic surface passing through the

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conic, and therefore through 4'; hence through 6' we may draw a line 6' meeting each of the lines 1, 2, 3, $1', 2'$, and where the cubic surface passes through the point 6' and also through the intersections of the line 6 with the lines 1, 2, 3; it passes through the line 6'. We complete in this manner by determining the planes of the cubic the system of the twelve points, viz. each new point is given as the inter- section of the cubic curve by a conic drawn through five points of the cubic curve. It is then shown as for the point 1' and the line 6' through it, that through each new point there can be drawn a line meeting by the same number and meeting each of the lines which it ought to meet; and hence lying on the cubic surface, but the twelve points are thus the intersections of the planes of the cubic curve by the twelve lines of the double-sixer, and it follows that the six points which ought to lie in a conic do every one where each conic has not been used in the place construction do actually lie in a conic.

I was anxious to construct such a double-sixer of points on a cubic curve; for this purpose I take the equation of the curve to be $y' = (1-\frac{x}{a})(1-\frac{x}{a})(1-\frac{x}{a})$, or say for shortness, $y^2 = X$; where, to fix the ideas, a, b are supposed to be positive, c greater than b; and r to be negative.

The cubic curve is thus a parabola symmetrical in regard to the axis of x, and consisting of a loop and infinite branch; say I select for them the points 1, 2, 3, 1', 4', 5'

[Figure: parabolic loop with axis through marked points — diagram omitted]

as shown in the figure, viz. the coordinates of these points are as stated in the Table, where α is the x coordinate, and $\sqrt{M} = a_0\sqrt{\left(1-\frac{x}{a}\right)\left(1-\frac{x}{b}\right)\left(1-\frac{x}{a}\right)}$ and so in other cases, $\sqrt{14} = 374483$.

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The numerical values belong to the curve $y' = \left(1 - \frac{x}{1}\right)\left(1 - \frac{x}{2}\right)\left(1 + \frac{x}{3}\right)$, and to $a = 6$.

Starting with the points 1, 2, 3, 1', 4', 5' we have to find the remaining points 4', 5, 6, 2', 3', 6'.

Point 4' by means of the conic $12345'6'$, as follows.

The equation of the conic is

$$(x-b)(x-c) - bcy + xy = 0 \quad (1,2,3,4,1'),$$

and making this pass through the point 1 $(x = a, y = \sqrt{M})$ we find

$$m_0 = (a-b)(a-c) + \sqrt{M} = 0 \quad (1).$$

Hence taking the coordinates of $4'$ to be, $a_0, \sqrt{M_0}$, we have

$$(a_0 - b)(a_0 - c) - bca_0 + \sqrt{M_0} = 0 \quad (4'),$$

and thence

$$\frac{\sqrt{M_0}}{\sqrt{M}} \cdot \frac{(a_0 - b)(a_0 - c) - bca_0}{(a-b)(a-c) - bca} = 1.$$

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or say

$$F(\beta-1)+G\sqrt{M}=\frac{a-q}{B-q}\cdot\frac{1}{a},$$

that is,

$$(\beta-1)abcF+ac\sqrt{M}=\frac{1}{ab}(\beta-qa^2a^2+a)$$

viz. that is,

$$(\beta-1)(F-\theta)+(a-bc)\frac{1}{\sqrt{M}}(F^2+ac)=0$$

or, rationalising and throwing out the factor $\theta-b$, this is

$$(\beta-1)(F-q)+b(\beta-bc)(a-q)\frac{ac}{\sqrt{M}}(F^2+ac)=0,$$

which is a cubic equation satisfied by $\theta=a$ and $\theta=a_0$, so that throwing out the factors $\theta=a$, $\theta=a_0$, we have for θ a linear equation.

Putting for shortness

$$\begin{aligned}A&=(a-a)F+(a-b)(a-c),\\ B&=(a-b)F+(b-c)(a-a),\\ C&=(a-c)F+(a-c)(a-b),\end{aligned}$$

the value of θ may be expressed in the forms

$$\theta=a+\frac{aA}{BC}(F+a),\quad \theta=b+\frac{aA}{BC}(F+b),\quad \theta=c+\frac{aA}{BC}(F+c).$$

We have moreover

$$F-q=\frac{a(a-1)(a-b)(a-c)}{BC},\quad F-a_0=-\frac{ac-A}{B}.$$

equations which express F in terms of a only; also

$$\theta-F=-\frac{a(a-1)(a-b)(a-c)B}{C^2},$$

and then

$$\sqrt{M_0}=\sqrt{M}\cdot\frac{B^2+ac}{C^2},$$

whence

$$\sqrt{M_0}=\pm\sqrt{M}(b-c)-\frac{aB}{C^2},$$

so that θ, $\sqrt{M_0}$ can be determined.

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Point 5 by means of the conic $1234'5'6'$.

The conic is

$$Fx + Gy + H = \frac{1-y'}{x}, \quad (1, 2, 3)$$

where

$$Fx, \quad + H = 1,$$
$$Fa_0 + G\sqrt{M} + H = \frac{1-M_0}{a_0}, \quad (4'),$$
$$Fa_0 - G\sqrt{M_0} + H = -\frac{M_0}{a_0}, \quad (5').$$

Everything is the same as for the point 4 except that b, c are interchanged; hence writing Q instead of F, and using A, B, C to denote as before, we have

$$abcF = a + b + Q,$$
$$\sqrt{M}abcG = (a-c)(1-a+bQ),$$
$$abcH = ab - aa + bQ - aB,$$

and

$$\phi = \frac{C^2 - a}{BC}.$$
$$\phi = b - \frac{a(b-a)(b-1)b-c}{B} - \frac{aa(b-c)(b-1)(b-a)}{BC},$$
$$Q = b + \frac{B^2 + ac}{BC},$$
$$Q - a = \frac{B(b-c)(b-1)(b-a)}{BC},$$
$$Q - a_0 = \frac{B(a-c)(b-1)(b-a)}{BC},$$

and

$$\sqrt{M_0} = \pm\sqrt{M}(b-c) - \frac{aQ}{BC^2}.$$

which determine ϕ, $\sqrt{M_0}$.

Point 5' by means of the conic $12345'6'6'$.

The conic is

$$Fx + Gy + H = \frac{(a-b)(a-x)}{y}. \quad (a, 1)$$

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and we have

$$G + H = a,$$

$$Fa_0 + G\sqrt{M} + H = \frac{(a-b)(a-a_0)}{\sqrt{M}}, \quad (1)$$

$$Fa_0 - G\sqrt{M} + H = -\frac{(a-b)(a-a_0)}{\sqrt{M}}, \quad (1)$$

Eliminating F, we have

$$G(aa_0\sqrt{M} + aa\sqrt{M_0}) = aa_0 + \frac{a_0(a-b)(a-a_0)}{\sqrt{M}}\sqrt{M} + \frac{a(a-b)(a-a_0)}{\sqrt{M}}\sqrt{M_0},$$

which is easily reduced to the form

$$G\frac{\theta a_0 - \theta aa_0 + aaa_0}{(aa-a)\sqrt{M}} + H(\theta - a) = (a+a) + 2a\sqrt{M_0},$$

and thence

$$G(aA + 2ac(\theta - a)) - \frac{\theta a_0 - \theta aa_0 + aaa_0}{\sqrt{M}}\sqrt{M_0} + abc + 2bc(a-a_0)\sqrt{M_0}$$

and combining herewith $G + H = b$, we have

$$H = \frac{2bc(a-a_0) - a + ba_0 + ac}{(a-a) - aA}\sqrt{M_0} + \frac{aA(b-a) + 2abc(aa_0 - a_0 + aa)}{aA(a-a) - aA},$$

and we have then

$$G = b - H;$$

that is,

$$F(a-a_0) + G\sqrt{M} + 2H = 0,$$

that is, what is the same thing,

$$F(aa_0(a-a_0) - 2bH + \frac{A}{C} + \frac{2aA}{C^2}\sqrt{M_0} + 2H(a-a_0) = 0.$$

We have therefore

$$\begin{aligned} Fa + H &= \frac{a}{2}\left(G - \frac{Aa(aaa_0 - aA)}{a - aa_0}\right) \\ &= \frac{1}{2}\left(b - G + \frac{Aa(aA+a)}{a - aA}\right), \end{aligned}$$

that is,

$$(F+H)\,x = \frac{(a-b)(a-x)}{xba}(Hx + 2Gx).$$

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or

$$aba_0(a-a_0)F + Hx + 2Gx = (a-b)(a+a_0)(Hx+2Gx) = 0.$$

Developing and throwing out the factor $a-b$, we have

$$G^2a^2 + [(Baac) - (b+c)Gaaba^2] + [a^2H^2 - 2a(b+c)GH + a(ba^2 + a^2H^2)]$$

$$+[-(b+c)a^2H^2 + bacGH + a(BaFH) = 0;$$

This must be satisfied by $a=a$, $a_0=a$; hence the left hand must be $= a^2(b-a)(a-aa_0)$, or equating the constant terms we have

$$G^2aaa_0 = aA(1-2aacB(b-a)) - a_0(B - 2baFH + acH^2),$$

which gives a^2; and we then have

$$\sqrt{M_0} = \frac{a-a_0}{aB-aA}\left(Faa+H\right),$$

but I have not attempted the further reduction of these expressions.

The numerical values for the example are

$$3F = \frac{-180+62\sqrt{14}}{5+21\sqrt{14}}, \qquad G = \frac{-10+62\sqrt{14}}{5+21\sqrt{14}}, \qquad H = \frac{-104\sqrt{14}}{1+\sqrt{14}}$$

whence a_0 is in the Table.

Point $2'$ by means of the conic $1\ 34'5'4'1'$.

The equation of the conic is

$$Fx + Gy + H = \frac{a-b}{y}, \quad (a', 1)$$

where

$$-G + H = ba_0,$$

$$Fa_0 + G\sqrt{M_0} + H = \frac{(b-a)(a-a_0)}{\sqrt{M_0}}, \quad (1)$$

$$Fa_0 - G\sqrt{M_0} + H = \frac{(b-a)(a-a_0)}{-\sqrt{M_0}}, \quad (1),$$

which are the same as for point $5'$, but only we recover the signs of F, H, and $\sqrt{M}$, $\sqrt{M_0}$.

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Hence the formulae are

$$H = \frac{2bc(a - a_0) - a + ba + ac}{aA + 2ac(a - a) - (a)} \sqrt{M_0} + \frac{aA(b - a)}{aA},$$

$$G = ba + H,$$

$$F(a - a_0) + G\sqrt{M_0} - 2bc(F + H)\, a_0,$$

which gives a; and then

$$Gaa_0\sqrt{M_0}\sqrt{M_0} = -3aac\,(Fa + H)\,\sqrt{M_0} + H\,(Fa + H)\,,$$

which are also reduced.

The numerical values are

$$3F = \frac{140 + 62\sqrt{14}}{5 - 21\sqrt{14}}, \quad G = \frac{-10 - 62\sqrt{14}}{5 - 21\sqrt{14}}, \quad H = \frac{-104\sqrt{14}}{3 - 21\sqrt{14}},$$

whence θ is in the Table.

[522]

Note on the Theory of Invariants.

[From the Mathematische Annalen, vol. III. (1871), pp. 268–271.]

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If two binary quantics $(a, \ldots)(x, y)^n$, $(a', \ldots)(x', y')^n$ are linearly transformable the one into the other, and if for the first of these P, Q are any two invariants whatever of the same degree, and P', Q' are the like invariants for the second of them, then we have

$$P : Q = P' : Q'.$$

viz. what is the same thing, the absolute invariants have the same values for the two functions respectively; and the entire system of these equations constitutes only a $(n-3)$-fold relation between the two sets of coefficients. For the converse theorem, viz. that if the entire system of equations is satisfied, the two functions are linearly transformable, see Clebsch, "Theorie der binären algebraischen Formen," § 89.

For instance, considering the two binary quartics

$$(a, b, c, d, e)(x, y)^4, \quad \text{and} \quad (a', b', c', d', e')(x', y')^4,$$

say with the invariants

$$I = ae - 4bd + 3c^2,$$
$$J = ace - ad^2 - b^2e + 2bcd - c^3,$$

and the like for the second of them; the conditions of equivalence are

$$I : J = I' : J',$$

or, what is the same thing, $I^3J^{-2} = I'^3J'^{-2}$, viz. this is a single relation between the two sets of coefficients (a, b, c, d, e) and (a', b', c', d', e'), so that any 4 of the parameters (a', b', c', d', e') may be considered as homogeneous functions of the elements (a, b, c, d, e) of the form

$$P : Q = P' : Q'.$$

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To deduce this result from the theory of the cubic function, I observe that denoting by A, B, C, Δ the values of the quadrinvariant, the sextinvariant, and the discriminant, as given in Salmon's *Higher Algebra*, Ed. 2, pp. 202–211, then in the particular case $a = 0$, $b = 0$, we have

$$\begin{aligned} &A = 10B = c^2, \quad C = -8d^2, \quad \Delta = 0, \\ &\quad + 13ce, \quad -3cde \quad + 36cd^2e \\ &\quad + d^2, \quad +9d^2, \quad -27d^4e^2 \\ &\quad + e^2, \end{aligned}$$

and hence forming the new invariants

$$B - 100B = AI,$$
$$\Gamma = 1000C - 1200AB + ??A^3.$$

the values of these in the same particular case $a = 0$, $b = 0$ are

$$B - 25ce(Icf - bg) = I = 200c^3e + 9ce - 12def + cd,$$
$$-200cd^4e + 8000d^2(3ce + d^2),$$

Taking now A, B, C, Δ as the invariants of the cubic, one of the conditions for the transformation is $B^3 : C^2 = B'^3 : C'^2$.

In the particular case $a = 0$, $b = 0$ and $a' = 0$, $b' = 0$, the invariants vanish and the equation is satisfied identically. But if we assume in the first instance only $a = 0$, $b = 0$, but not $a' = 0$, $b' = 0$, then the terms which contain the common factors c' and c' respectively; and throwing these out, and then writing $c = 0$, $d = 0$, we obtain previously found in a different manner.

It will be observed that the condition is of the original form $P : Q = P' : Q'$, but with the difference that P, Q and the corresponding functions P', Q' are not invariants. In presenting the foregoing property these functions may however be called "imperfect invariants"; it being understood that an imperfect invariant is not an invariant, and is not in any case included in the term "invariant" used without qualification.

And we may now establish the general theory as follows: Consider the similarly constituted special forms $(a, \ldots)(x, y, z, \ldots)^p$ and $(a', \ldots)(x', y', z', \ldots)^p$; to be in these the coefficients $(a, \ldots)$ may be regarded as homogeneous functions of the elements $(z, \ldots)$, which are either algebraical, or homogeneously connected together in any manner; and then the coefficients $(a', \ldots)$ will be the like functions of the elements $(x', y', \ldots)$ which are either independent or (as the case may be) homogeneously connected in the like manner.

The entire series of functions $P, Q, \ldots$ of $(a, \ldots)$, which are such that, if for any of the same degree, and P', Q' being the like functions of $(a', \ldots)$, the relations $P : Q = P' : Q'$ are equivalent to a $(n - 3)$-fold relation between the linearly transformable function $(a, \ldots)(x, y, z, \ldots)^p$, may, in a sort of like manner,

$$P : Q = P' : Q'.$$

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may be called the "perfect and imperfect invariants" of $(a, \ldots)(x, y, z, \ldots)^p$; and the relation in question be briefly referred to by the expression that the perfect and imperfect invariants are proportional.

We have then the theorem that if the two functions $(a, \ldots)(x, y, z, \ldots)^p$ and $(a', \ldots)(x', y', z', \ldots)^p$ are linearly transformable the one into the other, the two functions have their perfect and imperfect invariants proportional; and *conversely* the theorem, that two functions which have their perfect and imperfect invariants proportional, are linearly transformable the one into the other.

There is from a wide field of inquiry in regard to the imperfect invariants, even of a binary function, but still more so as to those of a ternary or quaternary function representing a curve or surface possessed of singularities.

We have in what precedes the explanation of an error into which I fell in my paper "On the transformation of plane curves," *Proc. Lond. Math. Soc.*, vol. I. No. 3, Oct. 1865, [384], see Arts. Nos. 27–30. Considering a given curve of deficiency D and, by means of a system of $D-5$ points chosen at pleasure on the curve, transforming this into a curve of the order $D+1$ with deficiency D; then for any two of the transformed curves (that is, two curves obtained by means of different systems of the $D-5$ points) I showed that these had the same absolute invariants, common factors σ' and σ' respectively; and throwing these out, and then writing $c=0$, $d=0$, we have $cc^2dc^2 = c'c'^2dc'^2$, viz. H vanish.

I remark that if the two transformed curves are linearly transformable the one into the other, then it cannot in the present case be true.

We have a class of surfaces of negative deficiency; viz. any rational trans- formation of a cone for which the planes section has a given (positive) deficiency pro- duces such a surface; and I think it may be answered conversely that a surface of negative deficiency is always the rational transformation of a cone for which the deficiency is equal to that of the surface taken with the reverse sign. As an instance, of course implies that the line joining the two corresponding (rational) points (this of course implies that the line joining the two corresponding points of the surface). We may call these "perfect and imperfect invariants"; it being understood that an "imperfect invariant" has all the properties of an invariant except that it does not satisfy the conditions of the linear transformation; and now in regard to the existence in the case of the hyperboloid is readily perceived.

Cambridge, 4 August, 1870.

[523]

On the Transformation of Unicursal Surfaces.

[From the Mathematische Annalen, vol. III. (1871), pp. 469–474.]

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I CONSIDER the question of the transformation (Abbildung auf einer Ebene) of unicursal surfaces. Taking (x, y, z, w) for the coordinates of a point on the surface, (x', y', z') for those of the corresponding point on the plane; then if X', Y', Z', W' denote each of them a function $(x', y', z')^n$, the equations of transformation are

$$x : y : z : w = X' : Y' : Z' : W';$$

and assuming that each of the curves

$$X' = 0, \quad Y' = 0, \quad Z' = 0, \quad W' = 0$$

(or, what is the same thing, the general curve

$$\alpha X' + \beta Y' + \gamma Z' + \delta W' = 0)$$

passes once through each of α points, twice through each of β points, ..., r times through each of ν points, the curves the successions Γ write σ, instead of α,β,...), and writing also

$$n = n^2 - \Sigma r^2\alpha,$$

$$0 = 4n^2(n-3) - 3 - n - \Sigma r(r+1)\alpha,$$

(where $0 = 1$ as positive except in the case of special relations between the positions of the fixed points $a, n_1, \ldots, n_n$, which equations give

$$-n = 3n - 6 - 28n - 2\Sigma\alpha,$$

then the order of the surface is n, and the order of the nodal curve is $0 = 2n - \Sigma r - 2\Sigma w$; I assume that the nodal curve has ρ apparent double points and ν actual triple points, but no stationary points, so that q being the class, we have $q = D - k - 2\delta - 3\nu$; and I endeavour to find from these numbers $\alpha, n_1, \ldots, n_n$.

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For this purpose, imagine through the nodal curve a surface of the order k, which therefore meets the surface besides in a curve of the order $nk - 25$; this curve Γ and the cubic curve of the nodal curve, or simply the "residue." The projection (Abbildung) of the complete intersection $5n$ is a curve of the order kn' passing through each of the projections of the residue. But as shown by Dr Clebsch the projection of the nodal curve is of the order $(n-4)n'+3$, and it passes $(n-4)r+1$ times through each of the points a_1; hence the projection of the residue is of the order $(k-n+4)n'-3$, and it passes $(k-n+4)r-1$ times through each of the points a_1. I assume that the projection of the residue is the *general* curve which satisfies the foregoing conditions, viz. that the residue, and its projection as defined by the foregoing conditions, *depend each of them on the same number of constants.* The necessity for this is I confess by no means obvious: but take as an illustration Steiner's quartic surface as transformed by the equations $x : y : z : w = x^2 : y^2 : z^2 : (x' + y' + z')$; the nodal curve consists of three lines meeting in a point, the quadric residue is the remaining inter- section of the surface by a quadric cone passing through the three lines; and the projection thereof is a line; the quadric cone, and therefore the conic, each depend upon 2 constants; and the line which is the projection of the conic depends also upon 2 constants; that the lines which is the projection of the conic depends the same number (2) of constants : at all events I make the assumption provisionally.

Now in the projection of the residue, we have twice the number of constants

$$= [(k-n+4)n'-3](k-n+4)n' - (k-n+4)r\alpha_1,$$

viz. this is

$$= (k-n+4)^2n^2 + 4(k-n+4) - 3n(k-n+4)$$

or, what is the same thing, is

$$= (k-n+4)^2n + (k-n+4)(n-4) + 2(k-n+4),$$

viz. reducing by means of the relation $-n = 4n - 6 - 28 + 2(n-4)$, the number in question is

$$= k^2n + k(-n^2 + 4n - 26) + 2(n-4)k.$$

Now k being $=$ or $> n-3$, the curve of intersection of a given surface as by a surface k depends on

$$\tfrac{1}{2}(k+1)(k+2)(k+3) - \tfrac{1}{2}(k-n+1)(k-n+2)(k-n+3) - 1$$

constants; and making the surface k to pass through the nodal curve, we have to subtract herefrom $(k+1)b - \frac{1}{2}b - 2k$; that is, for the residue, twice the number of constants is

$$= \tfrac{1}{2}(k+1)(k+2)(k+3) - \tfrac{1}{2}(k-n+1)(k-n+2)(k-n+3) - 2 - 2(k+1)b + \delta + 4t,$$

viz. this is

$$= k^2n + k(-n^2 + 4n - 2b) + \tfrac{1}{4}(n-2)(n-3)(n-4) - 2b - 25 + q + 2\,(n-4).$$

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Hence comparing the two expressions in question we have

$$2(n-4) = (n-4)n' - \tfrac{1}{4}(n-2)(n-3)(n-4) + 2b + 25 - q,$$

that is,

$$0 = (n-4)\{n' - \tfrac{1}{4}(n-2)(n-3)(n-4) + 2b + 25 - q\} = q,$$

or, as I prefer to write it,

$$0 = (n-4)\{n' - \tfrac{1}{4}(n-2)(n-3)(n-4) + 2b + 25 - q\} = 0;$$

which agrees with a more general formula in my "Memoir on the theory of Reciprocal Surfaces," *Phil. Trans.* vol. CLI. (1869), [167], see p. 327, [Coll. Math. Papers, vol. p. 316]. I consider any two residues, a b-fold residue and a b'-fold residue, the inter- section of these two surfaces and surface be equal to that of the two projections. Now the projection being (as above) twice the

order $(k-n+4)n'-3$ passing through $(k-n+4)r-1$ times through each of the points a_1,

the number of the intersections is

$$= (k-n+4)(k'-n+4)n^2 - 3(k-n+4) - 3(k'-n+4) + 9 - \Sigma\{(k-n+4)r-1\}\{(k'-n+4)r-1\}$$

viz. this is

$$= (k-n+4)(k'-n+4)n + (k-n+4)(n-4) + (k'-n+4)(n-4) - \Sigma(2k-n)(k'-n+4)$$

viz. substituting for θ its value $= n^2 - 4n + 2\Sigma r - 2 + 2n$, and reducing, this number is

$$= 4k - 2k - k^2 - 1 + bn'k - 6n - 2bk - (5n^2 - 25n + 12) - 2(n-4).$$

But the surfaces k, k', having in common the curve b which is a nodal curve on n, besides intersect in

$$bk - (b+1)k' + 25 + 27 - 4t + 2q + p$$

the number of points ($Salmon's Geometry\ of\ Three\ Dimensions, 2nd Ed., p.\ 283, except that in the formula as ther$
$bk-1+(b+1)k'-4t+2q+p$ which is equal to the number of intersections of the two projections.[1]

[1] I remark that there a being a positive number, b is positive in each case I subtract herefrom $bk-1+(b+1)k'-4t+2q$, the residue of the term so for $bk-1+(b+1)k'-4t+2q$, and the rest of points are $bk-1+(b+1)k'-4t+2q$.

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But we have already found

$$2q + 4t + p = 4(n-5) - \tfrac{1}{2}n^2 + 4n^2 - bn^2 + bn,$$

and we have therefore

$$t = (n-5) - \tfrac{1}{2}n^2 + 4n^2 - bn^2 + bn + bn,$$

and

$$q = -2(n-4) - \tfrac{1}{2}n^2 + 4n^2 - 5n + 4(5n-b).$$

I obtain these results in a different manner by investigating expressions for the deficiency (Geschlecht) of the nodal residue $= 28$ and for that of its projection.

First for the projection, the curve here is

$$2D = [(n-4)n^2 - 3](n-4)n^2 - 4 - \Sigma(n-4)r^2 - 1)(n-4)r,$$

$$= 2(n-4)(n-5) + (n-4) - 4 + n^2 - 4n^2 - 5n + 4(5n-b),$$

where, as before, $\Sigma r^2 = n^2 - n$, $\Sigma r = -n$, and $\Sigma r^2 - \Sigma r = n - n$. To apply this to the deficiency D of the nodal residue, we have

$$q = -2(n-4) - 1 = n^2 - 4n + 2\Sigma r - (3 - 25n - 2n),$$

or substituting for θ in this value, $= [(n-4) - \frac{1}{4}(n-1)(n-2)(n-3) + 2b + 25 - q]$, this is

$$= 2(n-5) - (n^2 - 4n^2 + 4n - 4t + 2q)(n-4),$$

Next as regards the residue, the number of its apparent double points is obtained in terms of h and t by the formula

$$bk - n(k-1) = (b+1)(b+2) + 2\delta - 4t \quad \text{(Salmon, l. c., p. 284)},$$

except that the singularity t is not there taken account of; that is, the number of intersections of the two residues is

$$2\delta = bk^2 - (k-1)bk + 2(b+1)b + 2\delta - 4t - 4t.$$

or introducing a factor of 2 in the value of θ, $= (n-4)\{(n - 5n^2 + 4n - 4) + bn^2 + bn$. viz. this is

$$2\delta = (n-4)\{(n - 4n^2 + 4n - 4) - 24 + bn^2 + bn^2 - 4t,$$

viz. this is

$$2\delta = (n-4)\{(n - 4n^2 + 4n - 4)\} - 24n + bn^2 + bn^2 + (n - 5n^2 + 4n - 4)\}.$$

the same result as before.

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The necessity for the term ω appears by the consideration that if we apply to the plane figure a Cremona-transformation, thus obtaining a new transformation of the surface, the value of $\Sigma\alpha$ will in general be altered; whereas the expressions for q, t should it is clear remain unaltered; and it arises as follows, viz. for certain transfor- mations of the surface the curve of the order $(k-n+4)n'-3$ which is the trans- formation of the residue (as defined by the foregoing conditions) ceases to be a curve of the assigned order, and to satisfy the order conditions; this is the assumed manner; and then the coefficient $(a, \ldots)$ will be the like functions of the elements an indecomposable curve but contains a certain number ω of factors (each belonging to a unicursal curve definable by means of the number of its passages through the several points a_i), which factors are to be rejected in order to obtain the equation of the proper residue. Thus reverting to the transformation

$$x : y : z : w = x' : y' : z' : (x' + y' + z')^2$$

of Steiner's surface, the projection of the quartic residue was (as already remarked) a line; applying to the plane figure the ordinary quadric (or inverse) transformation we introduce three fixed points, $(n_1 = 3)$, say these are A, B, C; viz. in the new trans- formation of the surface the projection of the residue ought clearly to be a conic through the three points; but according to the general formula it is a quintic having at each of the points a triple point: the quintic in question is consequently made up of the conic and the three lines $1A$, $1B$, $1C$; that is, it is on this account that a like explanation applies to the first investigation of $2q + 9t$.

I further remark, reverting to the equations

$$x : y : z : w = X' : Y' : Z' : W'$$

of the transformation, that the product of the ω factors is given as the common factor (if any) of the Jacobians

$$J(Y', Z', W'), \quad J(Z', W', X'), \quad J(W', X', Y') \quad \text{and} \quad J(X', Y', Z').$$

Such common factor exists whenever we can by a Cremona-transformation of the plane figure reduce the number of the points a, upon which the transformation of the surface depends; viz. for any given transformation of the surface, ω is equal to the excess of $\Sigma\alpha$ above the minimum value of $\Sigma\alpha$, and in this independent of the particular trans- formation. And of course of $\Sigma\alpha$ has this minimum value, viz. if the transformation is such that the number of the points a, cannot be reduced by any Cremona-trans-

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formation of the planar figure, then we have $\omega = 0$. I presume that for the most simple transformation, that is, when σ' has its least value, $\Sigma\alpha$ has also its least value, and consequently that $\omega = 0$.

Recapitulating, the results obtained are

$$q = -2(n-5) - 1 + 4n^2 + 4n^2 - 18 + 4(\Sigma\alpha - \omega),$$

$$t = (n-5) - \tfrac{1}{2}n^2 - 4n^2 + \tfrac{1}{2}n - 4(\Sigma\alpha - \omega),$$

where δ is to be excluded from

$$\delta = \tfrac{1}{2}(n-2)(n-3) - b + p,$$

the formulae are verified in the several cases:

[Table: 6 column table with n, n', $\Sigma\alpha$, δ, t, q, p and entries for Quadric surface, Cubic scroll, Quartic with nodal conic, Quartic with nodal quadriquadric, Quartic with two nodal lines etc. — table omitted]

which are the transformations chiefly as yet examined: but the first-mentioned case (quadric surface, generalised stereographic projection), although as stated the formulae are verified with the value $\omega = 1$, does not really come under the foregoing theory. It is interesting to see that they are verified in the last-mentioned cases, belonging to a negative value of D, that is, to a special system of formulæ.

Cambridge, Dec. 5, 1870.

[524]

On the Deficiency of Certain Surfaces.

[From the Mathematische Annalen, vol. III. (1871), pp. 526–529.]

p. 394

If a given point or curve is to be an ordinary or singular point or curve on a surface of the order n, this imposes on the surface a certain number of conditions, which number may be termed the "Postulation"; thus "Postulation of a given curve ρ-tuple curve on a surface n" will denote the number of conditions to be satisfied by the surface in order that the given curve may be a ρ-tuple curve on the surface.

The "deficiency" (Flächenpoldecel) of a given surface of the order n is

$$= \tfrac{1}{6}(n-1)(n-2)(n-3) - \text{ the deficiency-value of the several singularities;} \quad \text{viz.}$$

as shown by Dr Noether, if the surface has a given ρ-tuple curve, the deficiency-value hereof is

$$= \text{Postulation of the curve } (\rho-1)\text{-tuple curve on a surface } n-4;$$

and if the surface has an ordinary singular point, the deficiency-value hereof is

$$= \text{Postulation of the point } (\rho-1) \text{ conical point on a surface } n-4,$$

viz. this is $= \frac{1}{6}(t-1)t(t+1)$ where, as I write throughout, $t = \rho - 3$.

I remark that if the tangent-cone at the conical point has δ double lines and ν cuspidal lines, then the deficiency-value is

$$= \tfrac{1}{6}(t-1)t(t+1) - \tfrac{1}{2}\delta t - \nu(t-1) - (\text{etc.})$$

In the case of a double or cuspidal curve δ a ρ-tuple curve on the surface n has the deficiency-value = Postulation of given curve (ρ) simple curve on a surface $n-4$; and so for an ordinary cuspidal point: $\nu = 2$, and the deficiency-value $= 0$; results which were first obtained by Dr Clebsch.

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I found in this manner the expression for the deficiency of a surface having a double and cuspidal curve and the other singularities considered in my "Memoir on the Theory of Reciprocal Surfaces," *Phil. Trans.*, vol. CLI. (1869), [411, Coll. Math. Papers, vol. VI. p. 358]; viz. this was

$$D = \tfrac{1}{6}(n-1)(n-2)(n-3) - (n-4)b + \tfrac{1}{6}(b+r) + 2t + \tfrac{1}{2}\beta + \tfrac{1}{2}\gamma + \tfrac{1}{2}n - \tfrac{1}{2}b,$$

where we have

b, order of double curve,

h, class of Do,

δ, order of cuspidal curve,

β, number of intersections of the two curves, stationary points on b,

γ, number of intersections, stationary points on r,

j, number of intersections, not stationary points on either curve,

θ, number of certain singular points on b, the nature of which I do not com- plete to understand it is here taken to be $= 0$.

Before going further I remark that

Postulation of right line god β-tuple on surface n

$$= \tfrac{1}{6}(t+1)(n-t) - \tfrac{1}{6}t(t+1)(2t-5),$$

or we have

$$D = \tfrac{1}{6}(n-1)(n-2)(n-3) - \tfrac{1}{4}t(t+1)(2n-5),$$

or we have

$$D = \tfrac{1}{6}(n-1)(n-2)(n-3) - \tfrac{1}{4}t(t+1)(2n-5),$$

so that $D = 0$ either if $n = t-1$ or if $n = t$, viz. these are surfaces of a scroll (skew surface) with a $(n-1)$-tuple right line, the latter that of a surface with a $(n-2)$-tuple line: those are known to be of deficiency $D = 0$.

For a surface of the order n with a conical point where the tangent-cone has δ double lines and ν cuspidal lines, the deficiency-value is

$$D = \tfrac{1}{6}(n-1)(n-2)(n-3) - \tfrac{1}{6}(t-1)t(t+1) - \tfrac{1}{2}\delta t + \nu(t-1)$$

viz. for $n = t$ this is $D = 0$. In fact, a surface with a $(n-1)$-tuple conical point is once seen to be rationally transformable into a plane; and for $r = t$, that is, for a cone of the order n, we have

$$\begin{aligned} D &= \tfrac{1}{6}(n-1)(n-2)(n-3) - \tfrac{1}{6}(n-2)(n-3)(n-4) - \tfrac{1}{4}(n-3)(n-4) \\ &= \tfrac{1}{4}(n-1)(n-2) - \tfrac{1}{4}n(n-3), \end{aligned}$$

where the last term $-\frac{1}{4}\delta(n-4) - \nu$ is added because in the present case the surface has the δ double lines and the ν cuspidal lines.

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The formula therefore gives

$$D = -\tfrac{1}{6}(n-1)(n-2) + \delta;$$

viz. this is equal to the deficiency of the plane sections taken negatively.

I find that the same property exists for the deficiency of the planes sections taken negatively, having only a double curve; and *actually* in the case of a torse (developable surface) having a cuspidal curve with the ordinary singularities; and this being so there can I think be no doubt that for it is true for any small or torse whatever—viz. that in the face of the tetrahedron, or any in the plane of an equilateral triangle, the sum of any ruled surface whatever the deficiency is equal to that of the plane section taken negatively.

First, for the scroll, we have

$$D = \tfrac{1}{6}(n-1)(n-2)(n-3) - (n-4)b + \tfrac{1}{6}\beta + \tfrac{1}{2}\beta,$$

which should be

$$= -\tfrac{1}{6}(n-1)(n-2) + b,$$

this is equal to the deficiency of the planes sections taken negatively.

Secondly, for the torse; changing the notation into that used by the inequalities of the curve and torse, we have

$$D = \tfrac{1}{6}(n-1)(n-2)(n-3) - (n-4)x + n + \tfrac{1}{6}(\theta + r) + 2\beta + \tfrac{1}{6}\beta + \tfrac{1}{2}\gamma + \tfrac{1}{2}\beta,$$

$$= -\tfrac{1}{6}(n-1)(n-2) + a.$$

We have $g = r(n-8) - 8x$, and substituting this value and expressing everything in terms of r, u, z by means of the formulae

$$n = \tfrac{1}{4}(r - n - 2a),$$

$$a = a - 3\delta - 2x - 3\kappa + 4\beta + 2\kappa,$$

$$g = a + (n-2)x - 2\tau\kappa,$$

$$g = a - n + (n-2)z - 2\kappa\kappa,$$

$$n = a + (n-2)z - 6g,$$

$$b = \tfrac{1}{4}(n-2)x - 2g,$$

we have after all reductions

$$D = -\tfrac{1}{6}(n-1)(n-2) + r - 1 = -(n-1)(n-2) + \delta.$$

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We have then a class of surfaces of negative deficiency; viz. any rational trans- formation of a cone in the plane section has a given (positive) deficiency produces such a surface; and I think it must be conversely that a surface of negative deficiency is always the rational transformation of a cone for which the deficiency is equal to that of the surface taken with the reverse sign; viz. in the sentence, the formula for the deficiency is $(n-1)$, $b=2$, $\beta=2$, $\theta=0$, $i=0$.

$$D = \tfrac{1}{6}(n-1)(n-2)(n-3) - (n-4)b + \tfrac{1}{6}(b+r) + 2(n-5) + 2(n-1).$$

viz. a scroll surface is obtained as the quartic inverse of a cubic cone; viz. taking for the vertex the point $x:y:z=0$, w may be a cubic function of x, y, z, and the remaining quartic also: of: $x:y:z$ and the equation of the cone is

$$(ax-mn, by-py+\theta, kx-jz)^2 = 0,$$

where (A,B) etc. $=0$.

Taking Q a quartic function (x,y,z), the transformation in question consists in the change of x,y,z,w into xQ, yQ, zQ, wQ. (where Q is inferior or equal to the plane $x,y,z=0$ and having Q as the equation of the cone in question and the conic $Q=0$ as nodal conic $w=0$, $Q=0$.
Cambridge, 5 Jan., 1871.

[525]

An Example of the Higher Transformation of a Binary Form.

[From the Mathematische Annalen, vol. IV. (1871), pp. 359–361.]

p. 398

THE quartic

$$(1) \quad (a, b, c, d, e)(x, y)^4$$

is by means of the two quantics

$$(T) \quad (a, B, y)(x, y)^2 \quad \text{and} \quad (a', B', y')(x, y)^2$$

transformed into

$$(2) \quad (a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4,$$

that is, eliminating x, y from

$$(a, b, c, d, e)(x, y)^4 - x(a, b, y)(x, y)^2 + y(a', b', y')(x, y)^2 = 0,$$

we obtain

$$(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4 = 0.$$

It is required to express the invariants of (2) in terms of the simultaneous invariants of (1) and (2).

Write

$$P,\ Q,\ R = ax, +by, +\gamma y_1, \quad a'x_1 + B'y_1 + \gamma' y_1.$$

the equations from which (x, y) have to be eliminated are

$$(a, b, c, d, e)(x, y)^4 = (P, Q, R)(x, y)^2 = 0,$$

we obtain

$$(a, b, c, d, e)(x, y)^4 = (P, Q, R)(x, y)^2 = 0.$$

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and the result of the elimination therefore is

$$\left|\begin{array}{cccc} a, & 4b, & 6c, & 4d, \\ e & & & \\ & a, & 4b, & 6c, \\ 4d, & e & & \\ & & P, & 2Q, \\ R & & & \\ & & & P, \\ 2Q, & R & & \\ & & & \\ P, & 2Q, & R & \end{array}\right| = 0,$$

viz. this determinant is the transformed quartic $(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4$.

The developed expression of the determinant is

$$a^2D - 8bQR$$

$$= \binom{-12ac}{+16b^2} P^2R^2 - 24cRQ^2PR + \binom{24bc}{-64bd} PQ^2R - 32cdQ^4R$$

$$+ \binom{-32cd}{+64bd} P^2R^2 - 32cdQ^2PR + 16c^4$$

$$+ \binom{32d^2}{-64cde} PR^3 - 32deQR^3 + 16e^4$$

$$+ \binom{24de}{+64cd} PR^2 - 24eQR^2 + 16d^4$$

$$-4bP^3Q + eP^3.$$

or that writing for P, Q, R their values, we have the transformed function $(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4$; the coefficients being of the form

$$a_1 = (a, b, c, d, e)(a, \beta, \gamma)^4, \quad (\alpha', \beta', \gamma')^4.$$

$$b_1 = \&c., \quad \&c., \quad \&c.$$

$$c_1 = \&c., \quad \&c.$$

$$d_1 = \&c., \quad \&c.$$

$$e_1 = \&c., \quad \&c.$$

Writing I, J for the invariants of the quartic (1), and

$$A = b(c\delta^2 - 4B\gamma) + c(\gamma a - \beta^2) + d(a\beta - a\gamma),$$

$$B = ac, \quad a, \quad b, \quad c, \quad d, \quad b, \quad c, \quad \beta - \beta^2,$$

A, B simultaneous invariants of the forms (1) and (2). Putting more- over $\nabla = P^2 - 2QR$, and writing I, J the like invariants of the form (5), and

$$J_1 = bI(B - IA) + B(IAB - 2aB)AB + (3DB - 3aB)A^2,$$

$$J_2 = b(aB - IA)AB + B(IAB - 2aB)AB + (3DB - 3aB)A^2,$$

we have

$$\nabla_1 = 256\nabla^2(I^2\nabla - IAB)\nabla.$$

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As a verification, suppose $(\alpha, \beta, \gamma) = (1, 0, c)(1, 0, c)^2 = c + cx^2$ and $\nabla = P^2 - 2QR = -4bc$. Take also $(a', \beta', \gamma') = (0, 1, b)(1, 0)^2 = y$; the transforming equation is then $I = 4J$, and $(X, Y, x) = (-1, 0, 1)$. We have $\nabla = P^2 - 2QR$, $-4bc$, $+b \cdot y^4 \cdot a$, and $G = ac$, $D = 16$, and $(X, Y, x) = (1, 0, 1)$. We have

$$\text{Det.} \ = (P, P^2(W + 2YPW^2) = 2Q + (a + 2y)w - 2y^4 = (a + 2y, b, ay + 2y)^4 = 0.$$

that is,

$$(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4 = -2x_1 - 2(a + 2y)x_1^3y_1 + 6ayx_1^2y_1^2 + 2(4ay^2 - 2y^4)y_1^3.$$

whence

$$I_1 = \frac{4096}{3} \cdot a^2b^2c^2(4 + (2b)^2 \cdot 256),$$

$$J_1 = \frac{262144}{27} \cdot b^4,$$

also

$$A = -16, \quad B = 8,$$

and the equations for I_1, J_1 become

$$\frac{4096}{3} \quad = 4\delta(b + 2 \cdot 256),$$

$$\frac{262144}{27} \quad = b\{-16 \cdot (64 + \delta \cdot 4096), \}$$

which are true. More generally, assuming

$$(a, b, c, d, e) = (4\delta_0, c, x_0 + 6\delta b)(x_1, y_1)^4 = 0$$

(where $I = 1 + 24\delta^2, J = \delta - \delta^4$), we have the same transforming equation; we have

$$\nabla = P^2 - 2QR, \quad ec, \quad c, \quad a^4 - 2bc, \quad b, \quad a, \quad c, \quad b, \quad b, \quad b, \quad c, \quad b\delta;$$

whence

$$I_1 = \frac{27}{3}(1 - 24\delta^2), \quad J_1 = \frac{a^4}{27}(1 - 244\delta^2),$$

also

$$A = -16, \quad B = 8 + 1.$$

Substituting these different values in the equations for I_1, J_1, we obtain

$$16(1 - 24\delta^2) - (1 + 240\delta) - 768 \cdot (75 - 56\delta) + 49 \cdot 4(1 + 24\delta),$$

and

$$-3(3-540\delta)+27\cdot 49\delta-49\cdot 4(1+240\delta)+27\cdot(1+240\delta)-12\cdot 24\cdot 4\delta-16\cdot(1+248\delta)+4\cdot 4(1+24\delta),$$

which are in fact satisfied identically.
Cambridge, 26 July, 1871.

[526]

On a Surface of the Eighth Order.

[From the Mathematische Annalen, vol. IV. (1871), pp. 558–560.]

p. 401

I REPRODUCE in an altered form, so as to exhibit the application thereto of the theory of the six coordinates of a line, the analysis by which Dr Hierholzer obtained the equation of the surface of the eighth order generated by the lines of the vertex of a quadricone which touches six given lines.

I call to mind that if (A, B, C, D, E, F, G) are the coordinates of any two points on a line, then the quantities (a, b, c, f, g, h), which denote respectively

$$(by' - y'a, az - a'a, cy' - y'a, cz' - z'a, az - a'a, az' - z'a),$$

and which are such that $af + by + cz = 0,$ are the six coordinates of the line*;

Consider the given point (x, y, z, w) and the lines (a, b, c, f, g, h), and write for shortness

$$\begin{aligned} P &= hy - gx + aw, \\ Q &= -ax + fx + bw, \\ R &= gx - fy + cw, \\ S &= -ax - by - cz, \end{aligned}$$

then taking (X, Y, Z, W) as current coordinates, the equation of the plane through the given point and line is

$$PX + QY + RZ + SW = 0,$$

Considering in like manner the given point (x, y, z, w) and the three lines $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, \ldots)$, $(a_3, \ldots)$, then we have the three planes

$$\begin{aligned} P_1X + Q_1Y + R_1Z + S_1W &= 0, \\ P_2X + Q_2Y + R_2Z + S_2W &= 0, \\ P_3X + Q_3Y + R_3Z + S_3W &= 0. \end{aligned}$$

* Cayley, "On the six coordinates of a line," *Camb. Phil. Trans.* vol. XI. (1868), [435], pp. 290–323.

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and if these planes have a common line, the point (x, y, z, w) is in a line meeting each of the three given lines; that is, in the locus of the points in the hyperboloid through the three given lines. It follows that the equations

$$\begin{vmatrix} P_1, & Q_1, & R_1, & S_1 \\ P_2, & Q_2, & R_2, & S_2 \\ P_3, & Q_3, & R_3, & S_3 \end{vmatrix} = 0$$

reduces themselves to a single equation, that of the hyperboloid in question.

I write for shortness

$$(000) = (a\beta\delta)x^2 + (bb\gamma)y^2 + (cfy)y^2 + (ch\beta)x^4$$

$$+[(a\beta\zeta) - (abh)]yx$$

$$+[(c\beta\eta) - (cbf)]zy$$

$$+[(c\gamma g) - (abcf)]xy$$

$$+[(bfg) + (afh)]yw$$

$$+[(bfg) + (ach)]xz$$

viz. (123) will mean $(a_1a_2a_3)x^2+$ etc. where $(a_1a_2a_3)$ denotes as usual the determinant

$$\begin{vmatrix} a_1, & \beta_1, & \delta_1 \\ a_2, & \beta_2, & \delta_2 \\ a_3, & \beta_3, & \delta_3 \end{vmatrix} \text{etc.};$$

then the equation in question are found to be (123) $x^2 = 0$, $y(123)$ $x = 0$, $z(123)$ $z = 0$, $w(123)$ $w = 0$, reducing themselves to the single equation $(123) = 0$, which is accordingly that of the hyperboloid through the three lines*.

Proceeding now to the above-mentioned problem, we have the point (x, y, z, w), and the six lines $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, \ldots)$, $(a_6, \ldots)$; the lines 1, 2, 3, 4, 5, 6: the six planes

$$P_1X + Q_1Y + R_1Z + S_1W = 0, \text{ etc.}$$

must be tangents to the same quadricone; that is, considering the sections by the plane $W = 0$, the six lines

$$P_1X + Q_1Y + R_1Z = 0, \text{ etc.}$$

must be tangents to the same conic; and the condition for this is

$$[123456] = 0$$

* This equation is given in the paper above referred to, p. 314.

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where the symbol stands for the determinant

$$\begin{vmatrix} P_1, & Q_1, & R_1, & Q_1R_1, & R_1P_1, & P_1Q_1 \\ P_2, & \cdots & & & & \\ \cdots & & & & & \end{vmatrix} = 0.$$

But as is well known this equation may be written

$$(*) \quad (126)(845)(145)(256) - (146)(125)(345)(245) = 0,$$

when (126) etc. denote the determinants

$$\begin{vmatrix} P_1, & Q_1, & R_1 \\ P_2, & Q_2, & R_2 \\ P_6, & Q_6, & R_6 \end{vmatrix} \text{etc.};$$

or, what is the same thing they denote the functions above represented by the like symbols (126) viz. ($a_1a_2a_6x^2+$ etc.) The equation (*) just obtained is therefore the for the surface of the eighth order, the locus of the vertex of a quadricone which touches six given lines.

I remark that in my "Memoir on Quartic Surfaces," *Proc. Lond. Math. Soc.*, vol. III. (1870), [445], pp. 19–69, I obtained the equation of the surface which the foregoing form $[123456] = 0$ or any $[(P,Q,R)] = 0$, taking that there a factor a*, $sothattheorderofthesurfaceis = 8; andfurtherthattheequation$ $\beta y + bz] = (-ax + \beta y - bz)\nu'[ax + \beta y - bz](-ax + \beta y - bz) = 0$, where exp. ω (read exponential) denotes a^*x^x and $[(a,b,ce)]$ denotes

$$\begin{vmatrix} a, & b, & c \\ b, & b_1, & c_1 \\ c, & c_1, & b_1 \end{vmatrix}$$

Also that the equation contains the four terms

$$a^2[ab, -b, gh]x'^2 - [abf]x'^2(b-f), \quad a^* + ay^2[-f, g, cf]^* = -a[a, b, cf]^*.$$

Cambridge, 12 September, 1871.

[527]

On a Theorem in Covariants.

[From the Mathematische Annalen, vol. v. (1872), pp. 625–629.]

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The proof given in Clebsch "Theorie der binären algebraischen Formen" (Leipzig 1872) of the finite number of the covariants of a binary form depends upon a subsidiary proposition which is deserving of attention for its own sake.

I use my own hyperdeterminant notation, which is as follows: Considering a function $U = (a,\ldots)(x,y)^n$ (viz. $U_1 = U_2 = \ldots = (a,\ldots)(x_1,y_1)^n$ and writing $12 = d_{x_1}d_{y_2} - d_{y_1}d_{x_2}$, then the general form of a covariant of the degree θ is

$$12^{r_{12}}\,13^{r_{13}}\,23^{r_{23}}\;\ldots U_1U_2\ldots U_\theta;$$

where A is a sound numerical factor, the indices a, β, … are positive integers, and after differentiations (each set of variables $(x_1,y_1)\ldots(x_2,y_2)\ldots$ etc., is changed into (x,y). In the general form of a covariant is as above, viz. a covariant is equal to a sum of terms of the form A, $12^a\ldots U_1U_2\ldots U_\theta$, but writing the the term, before A is equal to a sum of terms of the form A, $12^{a_{12}}\;\ldots U_1U_2\ldots U_\theta$, but writing the term, before $U_2\ldots$ etc., the symbols which need to be replaced by (x,y) for distinctness; the several sets of variables are each replaced by (x,y) in all the orders—one of these functions with ∇, and so for the other order—this can be obtained by interchanging the symbols. ∇, ∇', ∇'', ∇''', then by means of the relation

$$\nabla = 12\,U_1U_2,\quad \nabla = 12\cdot 13\cdot U_1U_2U_3 = (12\cdot 13)\,\text{etc.}$$

being now in the first instance a function of the single set (x,y) of variables.

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We may call the symbol $1\,2^{\alpha}\,1\,3^{\beta}\,2\,3^{\gamma}, \ldots$, and consider any of the symbols

$$k \cdot 1\,\overline{1\,2}^{\alpha}\,\overline{1\,3}^{\beta}\,\overline{2\,3}^{\gamma}\ \ldots, \quad \text{or} \quad k \cdot 2^{\alpha}\ \ldots, \quad \nabla_{\alpha\ldots\theta} = k\,(\overline{12}\,\overline{13}\,\overline{23}\,\overline{14}\ \ldots)\,\nabla,$$

which, under either of the two forms, $[k]$, denotes one of the symbols $[k]$, $\ldots$ etc., that this is considered as a symbol involving the θ symbolic numbers 1, 2, 3, $\ldots$, θ, even although in pursuance of the rule above stated, the symbol involving the actual expression of the symbol does not contain these symbolic numbers, viz. may have $x = w$, $y = u$, $\ldots$, $y = e$. This being the case, ∇ is a covariant of U, but several sets of variables are each replaced by (x, y) in all the orders—one of these functions ∇, ∇', ∇'', then by means of the relation

$$\nabla = 12\,U_1 U_2.$$

For the four numbers $1, 2, 3, 4$ we have a group of the 15 formulæ

$$\nabla,\ 12 - 34;\ \nabla,\ 14 - 23;\ \nabla,\ 33 - 14 = 0,$$

$$\nabla,\ 24 + 13 \cdot 12 - 14,\ 23 = 0,$$

$$\nabla,\ 23 + 14 - 12,\ 34 = 0,$$

leading to

$$23.\ 14 + 31.\ 24 + 12.\ 34 = 0,$$

which is a form not involving the ∇'s and consequently is applicable to the trans- formation of invariant-symbols where the numbers

$$a_1, a_2, \ldots, a_n, \ldots$$

are all $= 0$.

I establish the following definitions:

A symbol $[12 \ldots m]$ is prominent when each index-sum is $\leq c$; otherwise it is ultimate; viz. this is the case when any one or more of the index-sums is or are $> c$. A symbol $[12 \ldots m]$ may be said to be *primary* when it is $= w$; and that this is the ultimate as to $1, 2$ if a, c, and so on; and also as to a row when in other cases.

A prominent symbol then may any index-sum thereof is $\leq c$, is unable to *inferior*: thus if $w \geq c$, the symbol is inferior in regard to 1; and if $w \geq c$, $\geq c$, ult $\geq c$, ult is inferior in regard to 1 and 2: and like for the other cases.

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degree by a number which is greater than $\frac{1}{2}n-1$; in fact, suppose θ inferior in regard to 1, then the order is

$$\theta n = w_1 + (n - w_2) + \ldots + (n - w_n),$$

where each term after the first is at least $= \frac{1}{2}n$, and the order is at least $= w_1 + \frac{1}{2}n(\theta - 1)$, hence order $-$ degree is at least $= w_1 - \frac{1}{2}n$, w_1 being less than $\frac{1}{2}n$, this is greater than $\frac{1}{2}n - 1$.

Conversely, if for any symbol order-degree is $\leq \frac{1}{2}n - 1$, the symbol is not inferior.

A symbol $[12\ldots m]$ is called inferior in regard to a row when it is $= w$, ..., in this it is when each index is w, or sharp when each index w.

The subsidiary theorem is now as follows: A symbol is inferior or sharp; or each index-sum being νc a primary symbol, the order of any symbol which is sharp or inferior, expressed in terms of the order and the index-sums of the same number of terms each of which is inferior or sharp: viz. as to the order, the symbol is not sharp, but no stationary points, so that q being the class, we have $q = D - k - 2\delta - 3\nu$; and $z = w$.

The proof $[12\ldots m]$ is inferior, suppose it is to be in regard to 1, then there is

$$\overline{12}, \overline{14}, \overline{2} \cdot \overline{14}.$$

this expression by a denominate νn is not $= w$, and we may then by means of terms each of which is inferior or sharp: we may then by means of the form $\nabla', \nabla'', 1\,2^{14}\,2\,3^{18}, \overline{23}$, then by means of the relation

$$\nabla', \nabla' = w_1 - 1, \quad w_2 = w_2 - 1, \quad w_3 = w_2 - 1,$$

$$Q = \nabla', \nabla', \ldots, \nabla', \nabla' \ldots, \nabla' \ldots, \nabla' \ldots, \nabla' \ldots, \nabla' \ldots, \nabla' \ldots, \nabla' \ldots, \nabla' \ldots, \nabla' \ldots, \nabla' \ldots$$

then supposing for a moment that 1, is not $= w$, and to substitute the expression contains the factor ∇', ∇', which is equal to and may be replaced by ∇', ∇', $\nabla'\nabla'$, in the second side of variables respectively.

$$Q = \nabla'\nabla'.$$

and

$$\Omega\nabla'\nabla'\nabla'\nabla'\nabla'\nabla'\nabla'\nabla'\nabla'\nabla'\nabla'\nabla'\nabla'\nabla'\nabla'.$$

Now for Ω the sum of the indices $1_n - 1, 2_2 - 1, 2_2 - 1$, so that by hypothesis Ω is inferior or sharp: 1_n is in inferior or sharp, since one of these is sharp by hypothesis prove that Q is inferior or sharp, we have just been to prove that Q in a way also that the inferior or sharp.

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expressed as a sum of terms each of which is inferior or sharp; and we have to prove this for the next following degree; $u+1$, $u+1$, or the conveniences p in place of $m+1$, (say for the degree p); that is, for a symbol

$$[12\ldots mp] = p_1^{r_1}p_2^{r_2}\ldots p_m^{r_m}[12\ldots mp].$$

I write as before, $\sigma_1, \sigma_2, \ldots, \sigma_m$ for the index-sums of $[12\ldots mp]$; therefore as $\sigma_1+\sigma_2+\ldots+\sigma_m$, $\sigma_1, \ldots$, are the index-sums of $[12\ldots m]$ involving $p_1, p_2, \ldots, p_m$.

If $[12\ldots m]$ is sharp, then $[12\ldots mp]$ is inferior in regard to p.

If $[2, m]$ is inferior, suppose it is so in regard to p; then we have $\sigma_1 =$ sharp (being therefore inferior) and when $\sigma_1 =$ no. In the case of any one of the indices being negative, viz. $p < r$, this is sharp; but in any other case, $[12\ldots mp]$ being inferior in regard to p.

Consider the expression

$$[12\ldots mp][12\ldots pp]\ldots p[12\ldots m],$$

where, $\sigma_1, \sigma_2, \sigma_3, \ldots$ are in inferior, and when $\sigma_1, \ldots, \sigma_m$, has any one of these is $\sigma_1, \ldots, \sigma_m$ are sharp negative.

Assume that when $[12\ldots mp]$ is inferior, and when $\sigma_1, \sigma_2, \ldots, \sigma_m$, have any values such that their sum is greater than a given value σ, the expression is a sum of terms each of which is inferior or sharp: we wish to show that this also is, so the last term not exceeding value, $=\sigma$, the case will then be the same.

For this purpose, introducing the ∇'s I write

$$Q = \nabla_{12}\nabla_{13}\nabla_{23}\ldots\nabla_{1p}^{r_1}\ldots\nabla_{2p}^{r_2}\ldots\nabla_{mp}^{r_m}[12\ldots m],$$

then supposing for a moment that σ_1 is not $=0$, and to substitute the expression contains the factor $\nabla_p\nabla_p$, which is equal to and may be replaced by $\nabla_p\nabla_p$, $\nabla_p\nabla_p$, in the indices of which the index 1 is one less, and the index $1-p$ one greater.

We have then

$$Q = \nabla_{12}\nabla_{13}\ldots\nabla_{1p}^{r_1}\ldots\nabla_{2p}^{r_2}\ldots\nabla_{mp}^{r_m}[12\ldots m]$$

and

$$\Omega = \nabla_{12}\nabla_{13}\ldots\nabla_{1p}^{r_1-1}\ldots\nabla_{2p}^{r_2+1}\ldots\nabla_{mp}^{r_m}[12\ldots m].$$

Now for Ω the sum of the indices $\sigma_1-1, \sigma_2, \sigma_3, \ldots$ as the same value, ν. And the symbol thus obtained is, by hypothesis Ω is inferior or sharp; in $1\,2^*$, $1\,3^*$, etc., the index-sums equal to the prove that Q is inferior or sharp, Ω has ν inferior or sharp; we have, by hypothesis, Ω is inferior or sharp.

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which is diminished by unity. Such change is possible so long as the index 1, has not attained its maximum value, $= a$, or $= a$; or as the case may be, and there is any other index $1_a, 1_b, \ldots$ which is not $= a$; that is, we may pass from Q to Ω, from Ω to Ω', and so on; and it will be sufficient to show that the last terms of the series is inferior or sharp. We thus pass from Q to R, where

$$R = \rho_1^{r_1} \rho_2^{r_2} \ldots \rho_m^{r_m} [12 \ldots m].$$

and $a_1, a_2, \ldots, a_m\, a_1 + \ldots + a_n - 1$, so are also a.

According as $a > c$, or $w < c$, R is inferior or sharp.

Now let Ω the sum of the indices $1_a, 1_b, \ldots$ be greater than a, that is, by hypothesis Ω is inferior; we let σ_1 greater than σ_2. Then if σ_2 be less than σ_1, we let σ_1 greater than σ_2. Then σ_2 becomes the last formed term which is inferior in regard to p_1; ult is, for in the indices of Ω the last mentioned form, $a_p = a_1 + a_2 + \ldots$ becomes the value σ_2, which is equal to and may be replaced by greater than a_p.

So that comparing with the Ω ult, it has, but no two of the inferior in regard to p_1, that is, Ω is inferior, $\sigma_2 = a$, $-\sigma_1$, and correct a given value $\sigma_1 = a$, $-\sigma_2$, the theorem is true for the next succeeding value σ_1, or being true for the case $\sigma_1 = a$, $-\sigma_1$, it is true generally.

Cambridge, 24 April, 1872.

[528]

On the Non-Euclidian Geometry.

[From the Mathematische Annalen, vol. v. (1872), pp. 630–634.]

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THE theory of the Non-Euclidian Geometry as developed in Dr Klein's paper "Ueber die Nicht-Euklidische Geometrie" may be illustrated by showing how in such a system we actually measure a distance and an angle: and by establishing the trigonometry of such a system. I confine myself to the "hyperbolic" case of plane geometry: viz. the absolute is here a real conic, which for simplicity I take to be a circle: and I attend to the points within the circle.

I use the simple letters $A, B, \dots$ to denote (Anschauung points or angular) distances measured in the ordinary manner: and the same letters, with a circumflex $\hat{A}, \hat{B}, \dots$ to

[Figure: circle with inscribed triangle and various line segments — diagram omitted]

denote the same distances measured according to the theory. The radius of the absolute is for convenience taken to be $= 1$: the distance of any point from the centre can therefore be represented as the sine of an angle.

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The distance BC, or any θ, of any two points B, C is by definition as follows: Radius of circle $= 1$.

$$\begin{aligned}
1\circ.\ ABC,\ &\text{sides are } a,\ b,\ c:\\
&\text{angles },A,\ B,\ C:\\
&OA,\ OB,\ OC:=\sin g,\ \sin g,\ \sin g:\\
&OA',\ OB',\ OC'=\sin a,\ \sin b,\ \sin c:\\
\angle BOC,\ COA,\ AOB:={}&\ b,\ c, B:\\
&\hat{a}=\tfrac{1}{2}\log\tfrac{BJ.CI}{BI.CJ}.
\end{aligned}$$

(where I, J are the intersections of the line BC with the circle); that is,

$$e^x = e^x = \tfrac{BJ.CI}{BI.CJ} = \tfrac{BJ.CI}{OB^2.OJ}.\ \frac{BJ.CI}{OB^2.OJ}.\ \frac{BJ.CI}{OBA}.\ \frac{CO.CJ}{OB.CJ}.\ \frac{CO.CJ}{OBA}.$$

where the numerator is

$$BI(BJ-BC)+CI(BC+CJ) = BC.BJ+CI.CJ+BC(CI-BJ);$$

$$= BC.BJ+CI(CJ+BC) = BC.BJ+CI(CJ+BC);$$

$$= BC.BJ+CI(CJ+BC);$$

Hence taking α for the angle whose sine is BC, and for the distances OB, OC respectively, we have $BJ.BI=\cos^2\theta$, $CI.CJ=\cos^2\theta$; and for the distance BC of these distances B, C as the perpendicular distance from O on the line BC. $BJ-BC+CJ$ as the perpendicular distance from O on the line BC. $BI=BC-C-BI=2c\cos\alpha+a\cos\alpha$, or, what is the same thing, taking α for the angle BOC, and therefore

$$a' = a\sin\theta = a\cos\alpha - \cos\alpha\theta - \cos\alpha\theta - \cos\alpha\theta,$$

we have

$$\cosh\hat{a} = \frac{1-\sin a\sin b\cos C}{\cos a\cos b}.$$

In a similar manner, if also α is the perpendicular distance from O on the line BC (that is, $a\sin a=\sin\theta\sin a\cos a$) it can be shown that

$$\sinh\hat{a} = \frac{a\cos a}{\cos a\cos b},$$

the equivalence of the two formulas appearing from the identity

$$\cos^2\alpha\cos^2\theta = (1-\sin\alpha\sin a\cos a)^2 = a\sin^2\alpha\sin^2\theta;$$

which is at once verified.

Next for an angle; we have by definition

$$\hat{A} = \tfrac{1}{2i}\log\tfrac{AK.AL}{AL.AL}$$

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where AI, AJ are the (imaginary) tangents from A to the circle; or writing for shortness BI for, instead of AKI, AKJ, (the angular point being always at A), consequently

$$\hat{A} = \tfrac{1}{2i} \log \tfrac{\sin BI.\sin CJ}{\sin BJ.\sin CI},$$

consequently

$$e^{i\hat{A}} = e^{i\hat{a}} \sqrt{\frac{\sin BI \sin CJ}{\sin BJ \sin CI}},$$

where the numerator is

$$BI(BJ + BC) - CI(BC - CJ) = (BJ - CJ)BC + \sin BJ.BC + \sin BJ.CI + BI.CI,$$

or say

$$e^{i\hat{a}} = \sqrt{\frac{\sin BJ \sin CI}{\sin BJ \sin CI}},$$

where $BJ.BI = \cos^2 AB$, $CI.CJ = \cos^2 AC$, (the angular point being always at A), moreover

$$\sin BJ.CI = e^x = \cos AB \cdot \cos AC \cdot \sin BC,$$

the perpendicular distance from O on the line BC is $= \sin BC$; moreover

$$\sin BJ.CI - \sin BJ.CI = 2 \sin a \cos b \cos ac - \sin BC = -2 \cos AB \cdot \cos AC \cdot \sin BC,$$

and $\sin BJ \sin CI = a^2 \sin BC + a^2 \cos BC$.

$$\sin BJ - \sin BJ.CI = 2 \sin a \cos b \cos ac - \sin BC = -2 \cos AB \cdot \cos AC \cdot \sin BC,$$

in CI sub $AJ = 2$ sin, in C sin ac sin BC; and similarly $\sin BJ.CI = \cos ac$ sin ac in BC,

$$\sin BJ + \sin CI = 2 \sin a \cdot a = a \cdot a \sin \alpha.$$

whence the required formula

$$\sin \hat{A} = \frac{a \cdot p \sin A}{\cos b \cos c}.$$

In the same way, or analytically from this value, we have

$$\cos \hat{A} = \frac{1 - \sin a \cos b \cos c}{\cos b \cos c},$$

and thence also

$$\tan \hat{A} = \frac{a \cdot p \sin A}{1 - \sin a \cos b \cos c}.$$

In particular, taking the line AC to pass through O, we will have $b = O$, that is, we have $\sin BO = \cos a \sin BC = b$, that is, $BC = \sin^{-1} a$, $\cos a \tan b$, and writing $OC = \tan^{-1} \cos a \sin B$; we ought to have $A = BO + OB$; that is, in

$$A = \tan^{-1} \cos a \sin B + \tan^{-1} \cos a \tan b,$$

which, observing that $\sin \theta. \sin a = \sin b. \sin B$, and $\sin \theta. \sin A = \sin b. \sin A = a$, that is equivalent to, the above formula.

Observe in particular that when A is at the centre, p is $= 0$, and the formula $\hat{A} = \cos \theta$ is, a, b, or say in an angle at the centre, $\hat{A} = 0$.

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I return to the expression for $\cosh\hat{a}$; in explanation of its meaning let the distances $\hat{a}\hat{b}, \hat{c}$ be θ, η respectively and let the angle BOC be θ; to find θ we have only to take θ' as θ; that is, in the formula for $\cosh\hat{a}$ to write $a=0$, etc., find $\cosh\hat{a} = \frac{1-\sin a\sin b\cos\theta}{\cos b\cos c}$; and similarly for $\cos\alpha$ and $\sin\alpha$, where $\alpha = OB'$, and $\sin\alpha = OB\sin C$; we have similarly $\sin a = OB\sin B$, viz. we have

$$\cos\hat{a} = \cos b\cos c\cdot\cosh\hat{a},$$

$$\sin\alpha = \cos a\sin b\cdot\cos C,$$

or, what is the same thing, θ is

$$\cos\hat{a} = \cos b\cos c + \sin b\sin c\cdot\cos C,$$

$$= \cosh\hat{b}\cdot\cosh\hat{c} + \sinh\hat{b}\sinh\hat{c}\cos C,$$

viz. as will presently appear, this is the formula for $\cos C$ in the triangle $\hat{B}O\hat{C}$.

From the above formula

$$\cosh\hat{a} = \frac{1-\sin a\sin b\cos\theta}{\cos b\cos c}.$$

and

$$\sinh\hat{a} = \frac{a\cdot p\sin A}{\cos b\cos c} = \frac{a\cdot p\sin A}{\cos b\cos c\cdot\sin\theta\sin\theta},$$

and the like formulae for $\hat{b}, \hat{c}, B, C$ in any way be shown that in the triangle ABC we have

$$\frac{\cos\hat{A}+\cos\hat{B}\cos\hat{C}}{\sin\hat{B}\sin\hat{C}} = \cos\hat{a},$$

In fact, substituting the foregoing values, this equation becomes

$$(1-\sin a)\cos A + (1-\sin a)\cos b\cos C(1-\sin a\cos C\cos A) = 1 - \sin b\sin c\cos A$$

$$-\sin B\sin C(1-\sin a\sin c)\cos A$$

that is,

$$\cos A + \cos B\cos C - \sin a\sin b\cos C - \sin a\cos b\cos C + \sin a\sin b\cos C + \sin a\cos c\cos C\sin a$$

$$-\sin B\sin C(1-\cos a),$$

or, what is the same thing,

$$\sin^2 a(\cos B\cos C - \cos A) = \sin B\sin C + \sin a\sin b\cos A$$

$$+\sin a\sin b\cos a\cos a\cos C\sin a$$

that is,

$$(\sin a\cos B + \sin a\sin c\cos C)(B + \sin a\cos a) - (\sin a\cos a)\cos A = \sin a\sin b\cos A$$

a relation which I proceed to verify.

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We may, from the formula

$$\sin^{-1}\sin g = \sin^2 a = 1 - \sin a \sin b \cos b, \&c.$$

but, more simply, geometrically as presented above, deduce

$$\sin a \cos B + \sin a \cos C = \sin a(\cos a \sin a + \sin a \cos B + \sin a \cos C),$$

and thence

$$\sin a \cos A + \sin a \cos B + \sin a \cos C = \sin a,$$

$$(\sin a \cos B + \sin a \cos C)(\sin a \cos a + \sin a) = \sin a(\sin a(\sin a + \cos a)\sin a + \cos a \sin a + \sin a)$$
$$= a \cdot \sin C(1 - \sin g \cos a),$$

which, observing that $\sin g \sin a \cdot \sin A = a \cdot p$ and consequently a, b, A, B, $\cos A \cdot \cos B = b \cdot c$, and so multiplying each side by $\sin a$, that is, multiplying the second side by $\cos A \cdot \cos B = a^2 \cdot p^2$ in the case of the surfaces in question, the formulae for $\hat{a}$ is

$$\sin a \sin a \cdot \sin a \cdot \cos A = a \cdot p \sin C(1 - \sin a \cos a)$$

and the other of the equations in question is proved in the same manner.

From the formula for each a, we find

$$\sinh^2 \hat{a} = \frac{1}{\sin a \cdot \cos a}\Lambda,$$

where

$$2\sqrt{-} = (1 - \cos a) \cdot a + \cos a \cdot b + \cos a \cdot c(1 - 3a \sin a \cdot b \cdot \cos a),$$

whence also

$$\sinh^2 \hat{a} : \sinh^2 \hat{b} : \sinh^2 \hat{c} = \sin a \cdot a : \sin a \cdot b : \sin a \cdot c$$

and we can also obtain

$$\cos \hat{a} = \frac{\cos a + \cos a \cos b \cos ac}{\sinh b \sinh c},$$

So that the formulae are in fact similar to those of spherical trigonometry with only $\cos \hat{a}, \sin \hat{a}$, a, instead of $\cos a$, $\sin a$, etc. The before-mentioned formula for each $\hat{a}$ in terms of a, b, c; it is obviously a particular case of the last-mentioned formula for $\cos \hat{A}$.

Cambridge, 11 May, 1872.

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A "Smith's Prize" Paper(?); Solutions.

[From the Oxford, Cambridge and Dublin Messenger of Mathematics, vol. IV. (1868), pp. 201–220.]

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1. *Find the form of a function of n given number of letters, which has two and only two values.*

It is required to find the general form of a function $\phi(a, b, c, \ldots, k)$, rational and integral, which for all permutations whatever of the letters has two and only two values.

Suppose that any particular permutation of the letters changes $\phi(a, b, c, \ldots, k)$ into $\phi_1(a, b, c, \ldots, k)$; then any permutation of the letters will either leave the functions ϕ, ϕ_1, each of them unaltered, or it will change ϕ into ϕ_1 and ϕ_1 into ϕ. Hence $\phi - \phi_1$ is a symmetrical function of the letters, say

$$\phi + \phi_1 = 2L.$$

$\phi - \phi_1$ is a function which by a permutation of the letters is either unaltered, or simply changes its sign; and it is be shown that, writing for shortness V to denote the product $(a - b)(a - c) \ldots (b - c) \ldots (k - 1)$, of the differences of the letters, and denoting by SV a symmetrical function of $a, b, c, \ldots, k$, the form of $\phi - \phi_1$ is

$$\phi - \phi_1 = 2VM.$$

These equations give

$$\phi = L + VM,$$

which is the general theorem required; viz. the function ϕ has thus only the two values $L + VM$ and $L - VM$, where L, M are symmetrical functions.

To prove the subsidiary theorem, observe that there is at least one interchange of two letters which changes ϕ into ϕ_1, or otherwise $\phi = \phi_1$, would be a symmetrical function.

* Set by me, for the Master of Trinity, Thursday, January 30, 1868.

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let this be (a, b). Then $\phi - \phi_1$, changing its sign for the interchange in question, must vanish for $a = b$; that is, the product $\phi - \phi_1$ must contain $(a - b)$ as a factor: it is in the power $(a - b)^{2k+1}$; the product $\phi - \phi_1$, vanishing for $a = b$, vanishes for $a = b$, that is, contains $(a - b)$ as a factor: similarly $(a - c)$, $(a - d) \ldots (b - c) \ldots$ are factors, but not every other letter; and consequently $\phi - \phi_1$ contains V as a factor; we have therefore $\phi - \phi_1 = VQ$.

Hence the interchange of any two letters a, b, say of the letters (a, b), changes Q into $-Q$; that is, in the new equation, VQ becomes $-VQ$, viz. we have VQ. $AV = -VQ$, suppose in general, that lines any two contain the point P in the intersection of the points of P in regard to the two given conics respectively; then, the interchange of $(b, c), (c, d), \ldots$ is to leave Q unaltered, that is, Q is unaltered by the interchange of any two adjacent letters, $a = b$, $b = c$, etc.; but every interchange may be obtained by means of the interchange of two adjacent letters; that is, every interchange leaves Q unaltered; that is, Q is a symmetrical function $= SM$, and thence $\phi - \phi_1$, which is the subsidiary theorem. But we have VM, $\phi = L + VM$, which is the form of the function ϕ.

Observing that if a quadratic function $ax^2 + 2bxy + cy^2$ has two equal roots, then, since the radical sign disappears, the function ϕ has only one value $= L$, and from two letters which changes ϕ to ϕ_1, that of V being a symmetrical function, is odd or even.

2. ***Express $\frac{cx+d}{ax+b}$ in the form of two cubic factors; and show generally that, for any prime number p, the function $\frac{cx^{p-1}+dx+\ldots}{ax^{p-1}+bx^{p-2}+\ldots+c}$ may be broken up into two factors of the order $\frac{1}{2}(p-1)$ by means of a quadratic radical.***

Let r be any root of the given equation, then

$$x^{p-1} + x^{p-2} + \ldots + 1 = 0,$$

the roots of this equation are $r, r^2, \ldots, r^{p-1}$, where r is an imaginary pth root of unity.

Hence

$$x^{p-1} + x^{p-2} + \ldots + 1 = (x - r)(x - r^2) \ldots (x - r^{p-1}),$$

and denoting the two cubic factors by ρ_1, ρ_2 we have

$$\rho_1 = (x - r)(x - r^2)(x - r^4) \ldots,$$
$$\rho_2 = (x - r^3)(x - r^5)(x - r^6) \ldots,$$

we have

$$\rho_1 + \rho_2 = x^{p-1} + x^{p-2} + \ldots + 1,$$

and thence

$$\rho_1\rho_2 = x^{2p-2} + \ldots (1 + x + r^2 + \ldots + x^{2(p-2)+1}).$$
$$\rho_1 = x^{p-1} + \ldots + r^p,$$
$$\rho_1 + \rho_2 = x^{p-1} + \ldots (1 + x + r^2 + \ldots + x^{2(p-2)+1});$$

Hence ρ_1, ρ_2 are the roots of the quadratic equation

$$y^2 - (3p^2 - 2x \pm 1)y - 5a^2 + \ldots + (4 - 2)p^2 + (2 - 1) = 0.$$

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Solving the equation, observing that the quantity under the radical sign must of necessity be a square, and that its value is

$$(2x^2 + x^{p-1} + \ldots + 2)^2 - 4(x^{p-1} + x^{p-2} + \ldots + x^2 + x + 1),$$

which is in fact

$$= -7(x^2 + 1)^2,$$

so that the roots y_1, y_2, that is, the required cubic factors, are

$$\tfrac{1}{2}(2x^2 + x^{p-1} - \ldots + 2) \pm \tfrac{1}{2}(x^2 + 1)\sqrt{-7}.$$

Generally, for any prime exponent p, denoting by r any root of $x^{p-1} + \ldots + 1 = 0$, and writing

$$y = \rho_1 = (x - r)(x - r^2)\ldots = e^{p-1} + \ldots + b,$$

where $a, b, \ldots$ are the quadratic residues, and $a, b, \ldots$ the quadratic non-residues of p, we find

$$y' + y' = (y + y)^2,$$

where P is a function of x of the order $\frac{1}{2}(p - 1)$. Putting also $y'y' = (P - 4Q)$, where Q is a function of x of the order $\frac{1}{2}(p - 1)$, then y_1, y_2 are the values of a quadratic equation: $y' - Py + Q = 0$, and these roots are

$$\tfrac{1}{2}(P \pm \sqrt{(P^2 - 4Q)}).$$

Here, as is known, $P^2 - 4Q$, in a perfect square, $= qP^2$, E a rational function of the two coefficients, and the quadratic radical is then $\sqrt{qP^2}$, that is, the radicals to be employed of the above-mentioned function of the coefficients can be obtained by proper determinations of the above-mentioned function T'^{-1} in the symmetrical function which I call

$$\tfrac{1}{2}(P \pm \sqrt{qP}).$$

2. Is a Map of the World, wherein the meridians are projected into right lines meeting in a point, the inclination of any two of the lines being equal to that of the two meridians, and the parallels into circles about the point as centre, the radius of the circle being a given function of the latitude; compare on the plane scale of equal parts at any two points, calculated by means of a proper determination of the above-mentioned function of the latitude in question.

Let the longitude and colatitude be

on the sphere: $l,\ \lambda$,

in the map: $l',\ r$,

then the projection is such, that $l' = l$, $r = f(\lambda)$, a given function of λ.

The lengths of corresponding linear elements in the direction of a meridian, and perpendicular to it, are

$$de,\ \sin e\, dl,$$

$$dr',\ r'\, dl';$$

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A "Smith's Prize" Paper; Solutions (continued).

[From the Oxford, Cambridge and Dublin Messenger of Mathematics, vol. V. *(1869), pp. 182–203.]*

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whence, elements of arcs are

$$\frac{ds}{a},\ \sin\varepsilon\, ds,\quad \frac{ds'}{a'},\ \sin\varepsilon'\, ds',$$

tangents of azimuth are

$$\frac{dz}{a}+\sin\varepsilon\, ds,\quad \frac{dz'}{a'}+\sin\varepsilon'\, ds',$$

and substituting the values $l=l$, $a=f(z)$, we have $dl=dt$, $da=f'(z)\,dz$; and thence elements of arcs are $dz : f(z)f'(z)$,

$$\text{tangents of azimuth are}\quad \frac{1}{a}\,\frac{f'(z)}{f(z)}.$$

By proper determinations of the function $f(z)$, we can make

(1) The ratio of the elements of arcs to be constant; this will be the case if

$$f(z)f'(z)=k\,ds,\quad \text{that is,}\quad f'(z)=\text{const.},\quad \text{or}\quad \Pi\ \cos\varepsilon=\frac{1}{k}\,\Pi(1-\cos\varepsilon),$$

since $f(z)=z$, const. tends for $z=0$; that is,

$$f(z)=k\sqrt{1+\sin\tfrac{1}{2}z};$$

(2) corresponding azimuths to be equal; this will be the case if

$$\frac{f'(z)}{f(z)}=k,\quad \text{that is,}\ \log f(z)=\text{const.}+k\log\tfrac{1}{2}z,$$

or say

$$f(z)=k\tan\tfrac{1}{2}z.$$

This is in fact the stereographic projection, in which (as is known) any indefinitely small figure is in the map represented without distortion.

4. Explain the general configuration of the contour and slope lines in a tract of Land and Mountain country.

A contour line is the locus of points having a given altitude.

To fix the ideas, consider an island forming a two-headed mountain. The contour line at the sea level is a closed curve; as a sufficiently great altitude the contour line consists of two closed curves surrounding the two summits respectively; the transition from one form to the other takes place at the altitude of the pass between the two summits, the contour line then having a node at the top of the pass, and being in form a figure of eight. At the altitude of the lower summit one of the closed curves is reduced to a point; and for greater altitudes it disappears, the contour line being then a single closed curve surrounding the higher summit; and at the altitude of the higher summit this reduces itself to a point. (See fig. 1.) If there is on the breast of the mountain a lake, the contour line at the lake level is (as in

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the case of the pass) a curve with a node, having however here a closed curve with an interior loop as shown in the figure. The contour line for an altitude below the lake level includes as part of itself a closed curve lying within the contour of the lake,

[Figure 2: Contour curves with closed loops — diagram omitted]

and which for the altitude of the lowest point (or "init") of the lake reduces itself to a point, and for smaller altitudes disappears.

The contour lines in the immediate neighbourhood of a summit are in general ellipses (geometrically; the indication is an ellipse); they may however be circles (viz. if the summit be an umbilicus).

A slope line is a line of greatest inclination, and it is consequently an orthogonal trajectory to the series of contour lines. A slope line may be considered as always terminated in a summit or an init; there is through each summit (or init) an infinity of slope lines, and in general these all touch there; if, however, the contour line in the immediate neighbourhood are circles, then the slope lines, instead of touching, pass from the point in all directions. Through the node at the top of a pass, or outlet of a lake, there are two intersecting slope lines, one ascending each way from the node, and being in general a "ridge-line"; the other descending each way, and being in general a "course-line," viz. in the case of a pass, it is in each direction the course of the principal stream of the valley; and in the case of a lake-outlet, it is in the direction away from the lake, the course of the out-flowing stream. The slope lines which then pass through the several nodes mark out distinct regions, and so facilitate the tracing of the intermediate slope lines.

5. *Find the differential equation corresponding to the three integral equations respectively* (1) $y + t^2 = c\,(x-1)(x-2)$; (2) $(y-c)^2 = c^2(x-1)$; and (3) $y + t^2 = c^2$; *and discuss geometrically the singular solutions.*

Generally, for the equation $(y + t^2 = X = 0$, the derived equation is $4t\left(\dfrac{dt}{dy}\right)^2 - X' = 0$.

If from the integral equation, differentiating it in regard to c, we attempt to find the singular solution, we obtain $(y+t)^2 = 0$, and thence $X = 0$ for the singular solution. It is however to be observed that, if X contain single and multiple factors,

$$X = (x+b)(x+\beta)^2(x+\gamma)^3\ldots,$$

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then it is only the single factors $x + a = 0$, $x + \beta = 0$, etc., which are solutions of the differential equation when expressed in its proper form free from extraneous factors.

To explain how this is, write the differential equation in the form

$$X^p \left(\frac{dx}{dy}\right)^2 - 4X = 0;$$

for a single factor $x + a$, X' does not contain the factor $x + a$, and there is no solution $x + a = 0$. The equation $x + a = 0$ gives $\frac{dx}{dy} = 0$, $X = 0$, and the equation is then satisfied; and by what precedes $x + a = 0$ is a singular solution. Contrariwise, for the multiple factor $(x + \beta)^2$, X' will contain $(x + \beta)^{n-1}$, and the equation $X^p \left(\frac{dx}{dy}\right)^2 - 4X = 0$ will divide by $(x + \beta)^n$, and divested of this factor it will be of the form

$$\frac{X^p}{(x + \beta)^n} \left(\frac{dx}{dy}\right)^2 = \frac{4X}{(x + \beta)^{n-1}} = 0,$$

where

$$\frac{X^p}{(x + \beta)^n}$$

contains the factor $(x + \beta)^{n-2}$ (index is 0 or positive) but

$$\frac{X}{(x + \beta)^{n-1}}$$

does not contain $x + \beta$; hence the equation $x + \beta = 0$ gives $\frac{dx}{dy} = 0$, and therefore $X^p \left(\frac{dx}{dy}\right)^2 - 4X = 0$, but it does not give $\frac{X}{(x + \beta)^{n-1}}$, and consequently fails to satisfy the differential equation. Reverting to the integral equation $(y + t^2 = c(x - a))(x - \beta) \ldots (x - g)^p$, we see that $x + a = 0$ touches each of the series of curves, and is thus an envelope thereof; that $x + g = 0$ is not an envelope, but is the locus of a singular point on the series of curves.

Applying the foregoing considerations to the proposed equation,

(i) The differential equation is (fig. 3),

[Figure 3: Curves through points (1, 0) and (2, 0) — diagram omitted]

$$(2x - 3)^2 \left(\frac{dx}{dy}\right)^2 - 4c(x - 1)(x - 2) = 0,$$

having the singular solutions $x = 1$, $x = 2$. Each of these is an envelope.

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(ii) The differential equation is (fig. 3),

$$(2x-3)^2\left(\frac{dx}{dy}\right)^2 - 4(x-1) = 0,$$

[Figure 3: Cusped curves — diagram omitted]

having the singular solution $x = 1, x = 0$. But $x = 0$ is not a solution; it is the locus of the series of conjugate points of the curves $(y+t)^2 = c^2(x-1)$.

(iii) The differential equation is (fig. 4),

$$2x\left(\frac{dx}{dy}\right)^2 - 4 = 0,$$

which has no singular solution. But $x = 0$ is the locus of the cusp of the curves $(y+t)^2 = c^2$.

[Figure 4: Curves through origin — diagram omitted]

6. Show that the curve parallel to the parabola is of the order 6 and class 8; explain the reduction of class; and trace the system of parallel curves.

The curve parallel to the parabola is the envelope of a circle of constant radius, having its centre on a parabola; taking the equation of the parabola to be $y^2 = 4x$, the coordinates of any point thereon may be taken to be a^2, $2a$, and the equation of the variable circle is

$$(x-a^2)^2 + (y-2a)^2 = r^2,$$

that is,

$$a^4 + a^2(-2x+4) + a(-8y) + x^2 + y^2 - r^2 = 0,$$

or multiplying by 6, this is

$$(a, b, c, d, e \between a, 1)^4 = 0,$$

where

$$\begin{aligned}
a &= 6,\\
c &= -2x+4,\\
d &= -8y,\\
e &= 6\,(x^2+y^2-r^2).
\end{aligned}$$

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The equation of the envelope is obtained by eliminating a between this equation a, b and the derived equation; or, what is the same thing, by equating to zero the discriminant of the equation in a; viz. this is

$$(ae + 3c^2 - 27\,(ace - ad^2 - c^3))^2 = 0,$$

which, substituting for a, c, d, e their values, is an equation in (x, y) of the order 6; the order of the curve is thus $= 6$.

The class is most easily obtained by geometrical considerations. Seeking the tangents which can be drawn from a given point, it at once appears that, describing about this point a circle radius r, then to each tangent through the point to the parallel curve, there corresponds a common tangent of the circle and the parabola, and reciprocally; whence the class of the curve is equal to the number of common tangents of the circle and the parabola, that is, it is $= 4$.

To the order 6 corresponds in general a class $= 30$, and the reduction from 30 to 4, $= 26$, is caused by the cusps and nodes of the curve.

The cusps are given as the points of intersection of the curve $aa + 3c^2 = 0$, $ae - ad^2 - c^3 = 0$, which being respectively of the orders 2 and 3, give 6 cusps; it may be added that these cusps (3 real and 4 imaginary, or else all 6 imaginary) are, in regard to the parabola, the centres of curvature for those points at which the radius of curvature is $= r$.

There is on the axis a single point (always real) whose normal distance from the parabola is $= r$. Such point is a node of the parallel curve, viz. according to the value of r this is, an inverted (conjugate point) or a crunode; we have thus 1 node. The parallel curve at infinity coincides with the two parabolas obtained by the displacement of the given parabola each itself through the distance r, r along the axis y. Two such parabolas have at infinity in the axis of x a contact of the second order ($\frac{1}{6} = 4$) found to be $= 6$, or two parabolas have at infinity, in the axis of x, a singular point equivalent to 3 nodes; the number of nodes is thus $\frac{1}{4} = 4$, and the reduction $4 + 2 \times 6 = 4 + 22 = 26$.

The parallel curve is most easily traced geometrically; there are two examples of the case of the intersection of orthotomic tangents, equilateral from r. The outside branch is a curve of continuous curvature, the form of which depends, however, on the magnitude by which r is greater than the semi-latus rectum of the parabola; viz. when r does not exceed this value, the curve is of a kind in which two forms successively and possess as on the axis of x inside the branch a real central point (the node above referred to); when, however, r exceeds the latus rectum, then the curve has on the axis of x outside the parabola a real conjugate point. The inside branch, made up of two arcs through the cusps, has at the parabola at the cusp on the axis of y, has the form which singularity is really in the nature of a triple point composed of two cusps and a node; when r is greater than this limiting value, instead of the swing arcs have a crunode;

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and two real cusps present themselves; viz. the form of the inside branch is as shown in fig. 5.

[Figure 5: Inside branch of parallel curve — diagram omitted]

7. Show that a curve of the order n has at most $\frac{1}{2}(n-1)(n-2)$ *double points; and that in any curve having this maximum number of double points, the coordinates* (x, y, z) *may be taken to be proportional to rational and integral functions of a variable parameter.*

A curve of the order n cannot have more than $\frac{1}{2}(n-1)(n-2)$ double points; for suppose it had one more, say

$$\tfrac{1}{2}(n-1)(n-2)+1 = \tfrac{1}{2}(n^2-3n)+2 \text{ double points;}$$

then since a curve of the order $n-2$ can be drawn through

$$\tfrac{1}{2}(n-2)(n+1) = \tfrac{1}{2}(n^2-n)-1 \text{ points,}$$

suppose it through through the

$$\tfrac{1}{2}(n^2-3n+2) \text{ double points;}$$

and besides through

$$n-3 \text{ points on the curve,}$$

together

$$\tfrac{1}{2}(n^2-n)-1 \text{ points,}$$

it will cut the curve of the order n in the double points considered as

$$n^2-3n+4 \text{ points,}$$

and besides in

$$n-3 \text{ points,}$$

together

$$n^2-2n+1 \text{ points,}$$

that is, we should have a curve of order $n-2$ meeting the curve of the order n in $n(n-2)+1$ points, which is of course impossible.

Consider now a curve of the order n with $\frac{1}{2}(n-1)(n-2)$ double points; draw a curve of the order $(n-2)$ through the double points; and besides through

$$\tfrac{1}{2}(n^2-3n+2) \text{ points,}$$

together

$$n-3 \text{ points on the curve,}$$

together

$$\tfrac{1}{2}(n^2-n)-2 \text{ points,}$$

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this imposes on the curve $\frac{1}{2}(n^2 - n) - 2$ conditions; but as the curve might have been determined to satisfy $\frac{1}{2}(n+1)(n-2)$ conditions, this will leave still the curve indeterminate parameter λ. It meets the curve n in the double points considered as

$$n^2 - 3n + 2 \text{ points},$$

and besides in

$$n - 3 \text{ points},$$

altogether

$$n^2 - 2n \text{ points};$$

that is, in

$$2 \text{ points};$$

the coordinates of which are expressible rationally in terms of λ.

Hence aloue for any value whatever of λ, there is only one remaining intersection, the coordinates of this point must be rationally determinable in terms of the parameter λ; that is, the coordinates of any point on the given curve will be rational functions of λ; or using trilinear coordinates, the coordinates x, y, z of any point on the given curve will be proportional to rational and integral functions of λ.

8. Show that three bodies attracting each other according to the law of gravitation may move in a line in such wise that the mutual distances are in a constant ratio the value of which depends on the masses.

Taking the masses m_1, m_2, m_3, to be arranged in this order at distances z_1, z_2, z_3, from a fixed origin, the equations of motion are

$$\frac{d^2 z_1}{dt^2} = -\frac{m_2}{(z_1 - z_2)^2} - \frac{m_3}{(z_1 - z_3)^2},$$

$$\frac{d^2 z_2}{dt^2} = \frac{m_1}{(z_1 - z_2)^2} - \frac{m_3}{(z_2 - z_3)^2},$$

$$\frac{d^2 z_3}{dt^2} = \frac{m_1}{(z_1 - z_3)^2} + \frac{m_2}{(z_2 - z_3)^2}.$$

Assume

$$\frac{d^2(z_1 - z_2)}{dt^2} = -\frac{m_1 + m_2}{(z_1 - z_2)^2} - \frac{m_3}{(z_1 - z_3)^2} + \frac{m_3}{(z_2 - z_3)^2},$$

and therefore

$$z_1 - z_2 = z_2 - z_3 = z_1 - z_3,$$

then the equations become

$$\frac{d^2(z_1 - z_2)}{dt^2} = \left\{ -m_1 - m_2 + \frac{m_3}{(\lambda+1)^2} - \frac{m_3}{\lambda^2} \right\} \frac{1}{(z_1 - z_2)^2},$$

and

$$\frac{d^2(z_1 - z_3)}{dt^2} = \left\{ \frac{m_1}{\lambda^2} - \frac{m_3}{(\lambda+1)^2} + \frac{m_2}{(\lambda+1)^2} - m_3 \right\} \frac{1}{(z_1 - z_3)^2},$$

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which equations will be one and the same equation if only

$$\frac{1}{\lambda+1}\left\{-m_1-m_2+\frac{m_3}{(\lambda+1)^2}-\frac{m_3}{\lambda^2}\right\}=\frac{m_1}{\lambda^2}-\frac{m_3}{(\lambda+1)^2}+\frac{m_2}{(\lambda+1)^2}-m_3,$$

an equation which (multiplying out) is of the fifth order, and gives at least one real value of λ; and λ being thus determined, we may from either equation obtain an equation of the form $\frac{d^2(z_1-z_2)}{dt^2}=-\frac{M}{(z_1-z_2)^2}$, which determines z_1-z_2 in terms of t, and then z_1-z_2, $-z_1(z_1-z_2)$, is also known; that is, the relative motions of the three bodies are determined.

9. Write down the integral equations for the elliptic motion of a planet; and if r, a, b, denote the radius vector, longest of the latitude, and central longitude respectively, show that

$$\frac{\sqrt{(1+e)}}{\sqrt{r}}=\frac{1}{a(1-e^2)}\sqrt{(1+e)}+e\cos(s-\omega),$$

explaining the significations of the constants a, e, ω.

Write

a, the mean distance,

n, the mean motion $=\frac{\sqrt{\mu}}{a^{3/2}}$,

e, the eccentricity,

θ, longitude of node,

ω, longitude of perihelion in orbit,

ε, the longitude in orbit.

The position in the orbit is determined in terms of the time by means of an auxiliary quantity u, viz. writing

$$nt+\varepsilon-\omega=u-e\sin u.$$

We then have the radius vector r, and the true anomaly f, given as functions of the time by the equations

$$\cos f=\frac{\cos u-e}{1-e\cos u},$$

$$\sin f=\frac{\sqrt{(1-e^2)}\sin u}{1-e\cos u},$$

$$r=\frac{a(1-e^2)}{1+e\cos f}=a(1-e\cos u).$$

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Take s, θ, the hypotenuse, base, and perpendicular of the right-angled triangle SPP', base angle $\omega - \theta$ (see fig. 6), in which SP is the orbit, S the node, K the perihelion, P the planet; then

[Figure 6: Spherical triangle on celestial sphere — diagram omitted]

$\varepsilon = \omega - \theta + f$ gives s; and so gives in terms of r, θ, by the formulæ relating to the spherical triangle, so that we have

$$\begin{aligned} \text{longitude in orbit} &= s + \theta \ (= \omega + f), \\ \text{reduced longitude} &= b, \\ \text{latitude} &= p. \end{aligned}$$

The expression for r, writing for f its value, $= \varepsilon - \omega$, gives

$$\frac{1}{r} = \frac{1}{a(1-e^2)}\{1 + e\cos(\varepsilon - \omega)\},$$

and we have thence

$$\frac{\sqrt{(1+e)}}{\sqrt{r}} = \frac{1}{a(1-e^2)}\sqrt{(1+e)\{1 + e\cos(\varepsilon - \omega)\}}.$$

But we have

$$1 - e = 1 - e - e + e\cos(\varepsilon - \omega).$$

And thence

$$\cos(\varepsilon - \omega) = \cos(\varepsilon - \theta) + \sin(\varepsilon - \omega)\sin(\omega - \theta).$$

Consequently

$$\begin{aligned} \sqrt{(1+e)\cos(\varepsilon - \omega)} &= \sqrt{(1+e)\cos(\varepsilon - \theta)\cos(\omega - \theta)} + \sin(\varepsilon - \theta)\sin(\omega - \theta); \\ &= \sqrt{(1+e)\cos(\omega - \theta)\cos(\varepsilon - \theta)} + \sin(\omega - \theta)\sin(\varepsilon - \theta); \end{aligned}$$

but by the right-angled triangle, we have

$$\sin(\varepsilon - \theta)\sin\phi = \sin\omega, \qquad = \frac{\sin\omega}{\sqrt{(1+e)}},$$

and thence

$$\sin(\omega - \theta) = \sin\phi\sin\gamma = \tan\phi\cos\theta\tan\theta,$$

$$\sqrt{(1+e)\cos(\omega - \theta)} = \frac{1}{\cos\phi} + \cos\phi = \frac{\sin(\varepsilon - \theta)}{\sqrt{(1+e)}},$$

$$\cos(\varepsilon - \theta)\cos(\omega - \theta) = \cos\omega + \frac{\cos\theta\sin(\omega - \theta)}{\sin(1+e)},$$

that is,

$$\sqrt{(1+e)\cos(\omega - \theta)} = 1 + e\cos(\omega - \theta).$$

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The formula thus becomes

$$\begin{aligned}\frac{\sqrt{(1+e)}}{\sqrt{r}} &= \frac{1}{a(1-e^2)}\sqrt{(1+e)\{\cos(\varepsilon-\theta)\cos(\omega-\theta)+\sin(\varepsilon-\theta)\sin(\omega-\theta)\}}\\ &= \frac{1}{a(1-e^2)}\sqrt{(1+e)\{1+e\cos(\omega-\theta)\}\cos(\varepsilon-\theta)+e\sin(\omega-\theta)\sin(\varepsilon-\theta)},\\ &= \frac{1}{a(1-e^2)}\sqrt{(1+e)}+e_1\cos(\varepsilon-\theta_1),\end{aligned}$$

where

$$\frac{1}{\cos\phi}\tan(s-\theta)=\tan(\omega-\theta) \text{ gives } s_1,$$

$$e_1=\frac{\cos(s-\theta)}{\cos(\omega-\theta)} \text{ gives } e_1,$$

$$e_1(1-e_1^2)=1+e_1 \text{ gives } a_1.$$

The first equation shows that, drawing the arc KK' at right angles to SP, then

$$\varepsilon_1=\theta+KK';$$

and therefore

$$m_1=\theta+\sin\omega, e_1=e\cos\phi, e_1=TK'.$$

The other two equations then give e_1 and a_1; it may be added that, from the right-angled triangle NKK' we have $\cos(m_1-\theta)=\cos\phi\cos(\omega-\theta)$; and consequently that $a,(1-e)=a,(1-K')$.

10. Find the differential equations for the motion of a material line acted upon by any forces and moving in a given outer surface.

If a,b are the coordinates of the point of intersection of any line of the ruled surface with the plane of xy; a,β, γ the cosines of the inclinations of the line to the three axes respectively; then writing

$$\frac{\beta}{a}=\alpha,\quad b-a\frac{a}{\beta}=b,\quad \frac{\gamma}{\beta}=\gamma,$$

and considering a,b,β,γ as given functions of a variable parameter θ, when a,β,γ satisfy the relation $a^2+\beta^2+\gamma^2=1$, the equations in question, coordinates of the equation $(=\varepsilon)$ determine the position of the line on the surface, and taking account of the equation $(=\varepsilon)$, they determine its form of the parameters θ, θ, the coordinates of a particular point in this line.

The motion of the material line is such that it must successively to coincide with the several lines on the ruled surface, consider on the material line a fixed point; say the centre of gravity G, and imagine that in the course of the motion the material line comes to coincide with the line determined as above by the parameter θ, and that the point G with the point determined as above by the parameters τ, θ, consider as

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material line any other point P whose distance from G is $= s$; the parameters of P will be $s + s$, θ; and the coordinates will consequently be given in terms of the variable parameters τ, θ, by the equations

$$x = ax + (a + ts)\, s, \quad y = \beta + (\beta + ts')\, s, \quad z = \gamma + (\gamma + ts')\, s.$$

that is, we have

$$x = a + (a + b\tau)\theta + as, \quad y = \beta + b\tau + \beta s, \quad z = \gamma + c\tau + \gamma s,$$

where, as usual f, θ, s are given functions of θ, but s is a result of the mutual consideration. Hence, putting T to denote the *Vis Viva* function

$$= \tfrac{1}{2}\int (x^2 + y'^2 + z'^2) ds,$$

where dm is the element of the material line, and x', y', z' are the velocities of the element, it only remains to express T as a function of τ, θ, s and the velocities of motion will be

$$\frac{d}{dt}\frac{dT}{d\tau'} = \frac{dT}{d\tau} = D, \quad \frac{d}{dt}\frac{dT}{d\theta'} = \frac{dT}{d\theta} = \Theta,$$

where τ, Θ denote differential equations in regard to τ, θ, Θ have
Using x', X', θ', ξ' to denote the differential coefficients $\frac{dx}{dt}$, etc., as above, x', y', z', τ to denote differential equations in regard to τ, θ, that is, we have

$$x' = (a'+b'\tau)\theta'+B(b\theta'\tau+a's', \quad y' = (\beta'+b'\tau)\theta'+B(b\theta'\tau+\beta's', \quad z' = (\gamma'+c'\tau)\theta'+B(c\theta'\tau+\gamma's',$$

and thence

$$x'^2 + y'^2 + z'^2 = A\theta'^2 + 2B\theta'\tau + s'^2;$$

if for shortness

$$\begin{aligned} A &= a^2 + b^2 + 2\tau(a + a\,b\,b\,b) + (\beta^2 + \gamma^2)\tau^2 + (a' + bs')^2 + (\beta + bs')^2 + (\gamma + bs')^2, \\ B &= a\,a + b\,\beta + c\,\gamma + s(a^2 + b\,\beta + c\,\gamma) = a\,B + \beta\,B + \gamma\,B, \\ &= a\,B, \quad a, B + b\,B + c\,B, \end{aligned}$$

since

$$a\,a + b\,\beta + c\,\gamma = 0$$

Hence we have

$$T = \tfrac{1}{2}\Sigma m\, a + s'^2\Sigma m + 2\theta' s'\Sigma m\, B + s'^2\Sigma m,$$

and, observing that in the sum Σm, the variable being $\Sigma m\, ds$ rauls in consequence of G being the centre of gravity, we have

$$\Sigma m\, ds = (a^2+b^2+2(a\,a+\beta\,B+e\,b+c\,B+e\,c\,B+e\,b\,b\,B+e\,b\,b\,B+e\,B+e\,b\,B+e\,B+e\,B+e\,B+e\,B+e\,B+e\,B+e\,B+e\,B+e$$

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Or writing $\Sigma m = M$, $\Sigma m\, ds = R^2 M$ (so the radius vector from the centre of the line, and M denoting therefore M the moment of gravity), we have

$$\Sigma m\, a = M((a^2+b^2+2(a+a\,b\,b\,b\,b)+(\beta^2+\gamma^2)\tau^2+(a'+bs')^2+(\beta+bs')^2+(\gamma+bs')^2)+R^2)M.$$

Moreover

$$2\Sigma m\, a = M(a^2+b\,B+B\,\gamma+(a^2+a^2+b\,\beta+c\,\gamma)\,s)+2B\,s+(\beta+c\,B+\beta\,s+B\,s\,B)\,M,$$

and hence we have

$$T = \tfrac{1}{2}\{V\,P+P^2+\tfrac{1}{2}P^2+\tfrac{1}{2}P^2+P\,B+P\,B+B\,s\,P\}.$$

a given function of r, θ, s', and their differential coefficients in regard to r, θ, s; the equations of motion are then derived from this expression.

11. Explain the mutual connexion of the three theorems in conics: (1) the theorem at quartet linear; (2) Pascal's theorem; (3) the theorem of the anharmonic relations of four points.

The theorem of quartet linear is that the locus of a point, such that the product of its distances from two given lines is to always in a given ratio to the product of its distances from two other given lines, is a conic.

Consider the lines as forming a quadrilateral $ABCD$. Then A, B, C, D are points on the conic, and writing PAB to denote the perpendicular distance of a point P from the line AB, and so in other cases, the theorem is that the expression

$$\frac{AC\,BD\sin PAB\cdot\sin PBD}{AB\,DC\sin PAD\cdot\sin PBC}$$

has a constant value for any point P whatever on the conic. Now the perpendicular distance $PAB = 2AB{\cdot}AB = AB$, or, what is the same thing, it is $= PA{\cdot}PB\sin PAB \div AB$; transforming the other perpendicular distances in the same manner, the distances PA, PB, PC, PD divide out, and the foregoing expression becomes

$$\frac{AC\cdot BD\sin PAB\cdot\sin PBD}{AB\cdot CD\sin PAD\cdot\sin PBC};$$

viz. omitting the constant factor, it appears that the expression

$$\frac{\sin PAB\cdot\sin PCD}{\sin PAD\cdot\sin PBC}$$

constant for all points P on the conic, or, what is the same thing, that the anharmonic ratio of the pencil $P(A,B,C,D)$ is constant for all points P on the conic. This is the anharmonic property of the points of a conic.

Of course, if reality be disregarded, the two theorems are mere two of the cases of conics generally; and observe, that in the first theorem the circle is the locus of the intersection of orthotomic tangents, and we have the opposite of the points Q, R in the second theorem the circle is the locus of the orthotomic circle of the cross, which the lines of in the cross of Q, R themselves.

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point, for 2, 3, 4, 5, 6 are fixed points on the conic, then point 1 (64544) of lines from the point 1 to the four points 6, 5, 4, 3 has the same anharmonic ratio for all positions whatever of the variable point 1; which is the above-mentioned anharmonic property of the points of a conic.

12. Show that any line through the centre of two orthotomic circles cuts the two circles harmonically, and connect this result with the theorem that the Jacobian of three circles is made up of the line radically and the orthotomic circle.

Drawing through the centre of the circle A (fig. 8) the line $a\beta\beta'$, it appears by the figure that we have

$$A\beta \cdot A\beta = a \text{ square of tangential distance of } A \text{ from the circle } B,$$

$$= A\beta \cdot A\beta', \text{ since the circles cut at right angles,}$$

$$= -A\beta \cdot A\beta';$$

[Figure 8: Two orthotomic circles diagram — omitted]

that is, the points β, β' are inverse points in regard to the circle on the diameter aa', and the points a, a', β, β' are thus harmonically related to each other.

Hence the polar of a, in regard to the circle B, passes through the point a', which is the opposite of a in regard to the circle A; and it is consequently a fixed point for all the circles B orthotomic to the circle A. That is, considering any three circles orthotomic to the circle A, this circle A is a locus of points a such that the polars of a in regard to the three circles meet in a point.

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Moreover, all circles meet the line A in the same fixed points I, J, the circular points at infinity; hence for any point A on the line infinity, the polar of A in regard to any circle whatever meets the line in A; the polar in regard, however, to the three circles all of them passes through I, whence the line infinity is also a locus of points A, such that the polars of A in regard to the three given circles meet in a point.

The Jacobian of any three conics is the locus (see fig. 7) of points such that the polars in regard to them meet in a point; in the present case it is therefore made up of the line of infinity and the orthotomic circle.

Consider the lines $1, 2, 3, 4, 5$, as forming a system of lines in a plane; and take any three of remaining $(2, 3, 4, 5,)$ the residual triads $(1, 2, 3, 4, 5)$ as follows: $(1, 2, 3)$, $(1, 4, 5)$, and so on, in respect of $(2, 3, 4, 5)$ are remembered to which the lines belong.

$$1, \text{ the remaining transversal of } (2, 3, 4, 5,) \text{ and take}$$

$$1' \text{ the remaining transversal of } (2, 3, 4, 5),$$

$$\begin{array}{llll} 2' & '' & , & (3, 4, 1, 5), \\ 3' & '' & , & (4, 5, 1, 2), \\ 4' & '' & , & (5, 1, 2, 3), \\ 5' & '' & , & (1, 2, 3, 4), \end{array}$$

it is to be shown that the lines $1', 2', 3', 4', 5'$ have a common transversal 6.

[Figure: Lines diagram — omitted]

Consider the line 6 as a line determined so as to meet the lines $2', 3', 4'$ and $5'$; and take any line 5 meeting each of the lines $6, 6'$; since the six lines $1, 2, 3, 4, 5, 6$ have a common transversal $5'$, then considering the lines as fixed lines in a solid body, it is possible to find along the lines $1, 2, 3, 4, 5, 6$ forces which will be in equilibrium. Suppose for a moment that the line 6, determined as above, does not meet the line $1'$; then we have the lines $1', 2', 3', 4', 5'$, and $6'$ such that the line 6 meeting the lines $2', 3', 4', 5'$, and $6'$, does not meet the line $1'$. The same $(1', 2', 3', 4', 5'$, and $6')$ would be independent lines, such that there do not exist along these forces in equilibrium, and a force acting in any line whatever may be resolved

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into forces acting along the lines $1', 2', 3', 4', 5'$, and $6'$. Hence the original forces along the lines $1, 2, 3, 4, 5$, can be each of them so resolved; and combining with the resolved forces the original force along θ, we have a system of forces along the lines $1', 2', 3', 4', 5'$, θ in equilibrium (for each other), but this is impossible if the lines $1', 2', 3', 4', 5'$ and θ must be resolved into a sum of the required theorem.

This solution principle assumed in the above demonstration are as follows:

(1) Six lines may be such that there exist along them forces in equilibrium; or say, for shortness, the six lines may be in involution.

(2) For any seven lines, no six or any less number of which are in involution, there exist along the seven forces in equilibrium; or, what is the same thing, the same way be resolved into forces along the six lines.

(3) Six lines which have a common transversal are in involution (remark in passing that it is not conversely true that if the six lines are in involution they have a common transversal), but if five of the six lines have a common transversal not meeting the sixth line; which is the required theorem.

The statical principles assumed in the above demonstration are as follows:

14. In the theory of the variation of the arbitrary constants of a mechanical problem, state and explain the results obtained by Lagrange and Poisson respectively; and point out the peculiar advantage of Poisson's theory, in regard to the consequences which follow from the coefficients of his formulæ being independent of the time.

In the theory of the variation, of the arbitrary constants of a mechanical problem, it is assumed that the forces consist of principal and disturbing forces, each of them depending on a force function, say there is a principal force function and a disturbing function; (the assumption of a force function however is regard to the disturbing forces, though usual and convenient, is not essential); and that, when the disturbing forces are neglected, or say in the undisturbed problem, the equations of motion can be completely integrated; the theory consists herein that the same integral equations, taking the arbitrary constants to be variable, may be made to satisfy the disturbed equations of motion.

Suppose that the number of the coordinates $x, y, \ldots$ is $= p$, then since the differential equations are of the second order, the number of arbitrary constants $(a, b, c, \ldots)$ will be $= 2p$. In Lagrange's solution it is assumed that the coordinates $x, y, \ldots$ (and consequently also the derived functions $x', y', \ldots$) are each of them given in terms of t and the $2p$ constants; the disturbing function Ω is given in the same form; and the expressions for the variations $\dfrac{da}{dt}$, &c. are obtained by the solution of a system of linear equations

$$\frac{dR}{da} = (a, b)\frac{db}{dt} + (a, c)\frac{dc}{dt} + \&c.,$$

where the coefficients (a, b) are functions involving $\dfrac{dx}{da}, \dfrac{dx'}{da}$, &c.; these coefficients are

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time period functions of the $2p$ constants and of the time; but it is a result of the theory that they are in fact functions of the constants only, the time disappearing of itself.

In Poisson's theory it is assumed that the integrals of the undisturbed problem are obtained in the form

$$a = \phi(t, x, y, \ldots, x', y', \ldots), \text{ \&c.},$$

viz. that each constant is given in terms of the coordinates, their derived functions, and the time (so that, if all the integrals are known, this is equivalent to Lagrange's assumption). The expressions for the variations of the constants are given in the form

$$\frac{da}{dt} = (a, b)\frac{d\Omega}{db} + (a, c)\frac{d\Omega}{dc} + \text{\&c.},$$

where each coefficient (a, b) is given in the first instance expressed as functions of the coordinates $x, y, \ldots$, the disturbing function Ω, &c. and the time; but it is a result of the theory that the coefficients (a, b), &c. are really constant; viz. that, if x, y, $\ldots$, x', y', $\ldots$, are expressed in terms of the constants $(a, b, c, \ldots)$ and of t, then the coefficient becomes a function of the constants $a, b, c, \ldots$ only, the time disappearing of itself.

The transformation of the values of any coefficient $[a, b]$ requires only the knowledge of the expressions of a, b in terms of x, y, $\ldots$, x', y', $\ldots$, or we may say, the knowledge of the two integrals a, b. We thence obtain the expression of $[a, b]$ as a function of x, y, $\ldots$, x', y', $\ldots$, t, calling it, we have $\{a, b\} = \phi(t, x, y, \ldots, x', y', \ldots)$. But, in every transformation $[a, b]$ of the equations of motion of the undisturbed problem. It may happen that the value of $\{a, b\}$ is found to be a constant; in such case the constants of the values of the constants of the integrals all depend on the variables and the time. In every case, however, of the supposition that $\{a, b\}$ is independent of the time, leads to certain consequences analogous to this in Lagrange's theory.

15. Write a short dissertation on the transformation of rectangular coordinates in space of three dimensions, and in particular explain under what restriction it is true that two sets of rectangular axes, such that one system can by means of a rotation be made to coincide with the other, are conventionally said to be of the same kind, and that two sets of rectangular axes, such that the position of this axis and the second of the rotation cannot be made to coincide with each other, are of opposite kinds.

Two sets of rectangular axes about the same origin (each axis considered as it a line extending in the opposite sense from the origin) are said to be of the same kind if any one set can by displacement and rotation, be made to coincide with the other; in any other case they are of opposite kinds. It is easy to see that, if any one of the axes of the one set has been brought to coincide with the corresponding axis of the second set, then the other two axes will coincide with the corresponding axes,

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either the axis of z will coincide with that of z, or it will be in the opposite direction; in the former case the two sets are, in the latter case of the opposite kind, displacements one of the other. The restriction referred to in the question, is that the two sets shall be displacements one of the other.

In the problem of transformation, the two axis z axes, if not displacements, can always be made so by simply reversing the direction of one of the axes (writing for example x for x); and there is thus no real loss of generality in considering the two sets as displacements the one of the other; and it is in general convenient to make this assumption.

The transformation between two sets of rectangular axes is at once given by the diagram

$$\begin{array}{c|ccc} & x & y & z \\ \hline x' & a & \beta & \gamma \\ y' & a' & \beta' & \gamma' \\ z' & a'' & \beta'' & \gamma'' \end{array}$$

where a is the cosine of the inclination of the axis x, x_1, and so for the rest of the nine quantities. The relation between the two sets of axes is obtained at pleasure by reading the diagram horizontally, $x = ax + \beta y + \gamma z$, &c. or by reading it vertically, $x = ax + a'y_1 + a''z'$, &c.

We must have identically $x^2+y^2+z^2 = x'^2+y'^2+z'^2$, and we thus obtain the two equivalent sets of equations

$$\begin{aligned} a^2+\beta^2+\gamma^2 &= 1, & a^2+a'^2+a''^2 &= 1, \\ a'^2+\beta'^2+\gamma'^2 &= 1, & \beta^2+\beta'^2+\beta''^2 &= 1, \\ a''^2+\beta''^2+\gamma''^2 &= 1, & \gamma^2+\gamma'^2+\gamma''^2 &= 1, \\ a''a'+\beta''\beta'+\gamma''\gamma' &= 0, & \beta\gamma+\beta'\gamma'+\beta''\gamma'' &= 0, \\ a'a+\beta'\beta+\gamma'\gamma &= 0, & a\gamma+a'\gamma'+a''\gamma'' &= 0, \\ aa'+\beta\beta'+\gamma\gamma' &= 0, & a\beta+a'\beta'+a''\beta'' &= 0. \end{aligned}$$

Either set leads to the relation

$$\begin{vmatrix} a, & \beta, & \gamma \\ a', & \beta', & \gamma' \\ a'', & \beta'', & \gamma'' \end{vmatrix} = \pm 1, \quad \text{consequently} \quad \begin{vmatrix} a', & \beta', & \gamma' \\ a'', & \beta'', & \gamma'' \\ a, & \beta, & \gamma \end{vmatrix} = \pm 1.$$

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the distinction between the two cases $+1$, -1 being that above explained; viz. if the axes are not displacements the one of the other, the sign is $+$; if they are, (and in all that follows this is assumed to be the case) then the sign is $+$. The equations give further $a = \beta'\gamma' - \beta''\gamma''$, &c. (six equations).

It is easy to see geometrically that, as stated in the question, two sets of axes (being so determined displacements the one of the other) can be made to coincide by means of a rotation of either axis through a certain axis; moreover, if the position of the axis itself is not altered by a, b, c, a and the angle of rotation, we must have

$$(b-a)x - \beta y + \gamma z = 0, \quad y(c-1) = 0,$$

$$e'a + (\beta'-1)y + \gamma' z = 0, \quad \&c.,$$

$$e''a + \beta''(b-1)z = 0,$$

equations which must be equivalent to two equations; we in fact have

$$a : b : c : (a+b+1)(a+b+1)(c-1) = \beta' - \gamma''(a-1)(b+1)(c+1) = (1-a)\gamma' - \beta'(c+1),$$

as is easily verified by means of the foregoing relations between the coefficients. Any two of the three equations will then determine the ratios $a : b : c$; taking the second and third, we have

$$a : b : c = c(1-b) - \alpha y(b-1)\beta\gamma' : \beta'\gamma' + a'(1-c)\gamma' + (1-a)(1+a)\gamma'' - 1)$$

reducible to

$$a : b : c = 1 + a - \beta'\gamma' : \beta - \alpha' : \gamma' + (1-a)\gamma' - \beta'(c+1)$$

and treating in the same way the third and first and second and first sets of equations the system of formulæ is

$$\begin{aligned} a : b : c &= 1 + a - \beta'\gamma' : \beta + \alpha' : \gamma' + a \\ &= a' : 1 + \beta' - a - \gamma'' \cdot y\beta + \gamma' \\ &= a' + a' : \gamma' + \alpha + a' : 1 + \gamma'' - a - \beta \end{aligned}$$

equations equivalent to each other; they determine the position of the axis in question, or resultant axis. The foregoing equations may also be written

$$a' : b' : c' : 1 = a + \beta' + \gamma' : 1 - a + \beta' + \gamma' : 1 + a - \beta' + \gamma' : 1 + a + \beta' - \gamma',$$

and hence, if A, B, C, be the inclinations of the resultant axis to the axes of x y, and z, respectively, we have

$$\cos^2 A : \frac{1 - a + \beta + \gamma}{1 + a + \beta + \gamma}.$$

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and thence also

$$\sin^2 A = \frac{2(1+a)}{3-a-\beta-\gamma},$$

$$\cos 2A = \frac{-1+3a-\beta-\gamma}{3-a-\beta-\gamma},$$

$$a = \cos 2A\,\frac{(1-a)(1+a+\beta+\gamma)+2a}{3-a-\beta-\gamma}.$$

Let θ be the amount of the rotation about the resultant axis; we have a spherical triangle the two sides whereof are A, A, the included angle θ, and the opposite side $\cos^{-1} a$; that is, we have

$$\cos\theta = \frac{a-\cos^2 A}{\sin^2 A},$$

and thence

$$\begin{aligned} 4\cos^2 \tfrac{1}{2}\theta = 2(1+\cos\theta) &= \frac{2(a-\cos 2A)}{\sin^2 A}, \\ &= 1+a+\beta+\gamma, \end{aligned}$$

that is, the amount of the rotation is given by the formula

$$4\cos^2 \tfrac{1}{2}\theta = 1+a+\beta+\gamma.$$

The nine cosines a, β, γ, &c. may be expressed in terms of the inclinations A, B, C, and the rotation θ, or putting λ, μ, ν equal to $\tan\frac{1}{2}\theta\cos A$, $\tan\frac{1}{2}\theta\cos B$, $\tan\frac{1}{2}\theta\cos C$, in terms of the three quantities λ, μ, ν; but the resulting formulæ for the transformation of coordinates can be more readily obtained by other methods.

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Solution of a Senate-House Problem.

[From the Oxford, Cambridge and Dublin Messenger of Mathematics, vol. V. *(1869), pp. 38–87.]*

The Problem, proposed January 7, 1869, is "If θ_1 and θ_2 are two values of θ which satisfy the equation

$$1+\frac{\cos\theta_1\cos\theta}{a^2}+\frac{\sin\theta_1\sin\theta}{b^2}=0,$$

show that θ_1 and θ_2, if substituted for θ and ϕ in this equation, will satisfy it."

That is, writing

$$\frac{\cos\theta_1\cos\theta}{a^2}+\frac{\sin\theta_1\sin\theta}{b^2}+1=0,$$
$$\frac{\cos\theta_1\cos\phi}{a^2}+\frac{\sin\theta_1\sin\phi}{b^2}+1=0,$$

where $a^2+b^2=1$, it is to be shown that

$$\frac{\cos\theta_1\cos\theta_2}{a^2}+\frac{\sin\theta_1\sin\theta_2}{b^2}+1=0.$$

From the given equations, we have

$$\frac{\cos\phi}{a^2}:\frac{\sin\phi}{b^2}:1=\sin\theta_1\sin\theta_2:-\cos\theta_1\cos\theta_2:\sin(\theta_1-\theta_2),$$

which is

$$=\cos\tfrac{1}{2}(\theta_1+\theta_2):-\sin\tfrac{1}{2}(\theta_1+\theta_2):2\cos\tfrac{1}{2}(\theta_1-\theta_2).$$

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Whence eliminating ϕ, we have

$$a^2\cos^2\tfrac{1}{2}(\theta_1+\theta_2)+b^2\sin^2\tfrac{1}{2}(\theta_1+\theta_2)-a^2b^2\cos^2\tfrac{1}{2}(\theta_1-\theta_2)=0,$$

that is,

$$a^2(1+\cos(\theta_1+\theta_2))+b^2(1-\cos(\theta_1+\theta_2))-(1+\cos(\theta_1-\theta_2))=0,$$

or, what is the same thing,

$$a^2+b^2-1+(a^2-b^2)\cos(\theta_1+\theta_2)-\cos(\theta_1-\theta_2)=0,$$

But from the equation $a^2+b^2=1$, we have

$$a^2+b^2-1=-2a^2b^2,$$

$$a^2-b^2-1=-2b^2,$$

$$-a^2+b^2-1=-2a^2,$$

and the equation is thus

$$\frac{\cos\theta_1\cos\theta_2}{a^2}+\frac{\sin\theta_1\sin\theta_2}{b^2}+1=0,$$

which is the required equation.

Stopping at the result obtained previous to the use of the relation $a^2+b^2=1$, but making some obvious substitutions in the formula, the theorem may be presented in a more general form as follows; viz. If we have

$$\frac{ax_1}{a^2}+\frac{by_1}{b^2}-1=0,$$

$$\frac{ax_2}{a^2}+\frac{by_2}{b^2}-1=0,$$

where

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad\text{and}\quad \frac{x'^2}{a^2}+\frac{y'^2}{b^2}=1,$$

then

$$\left(\frac{a}{a^2}+\frac{c}{b^2}-1\right)\left(\frac{a'}{a^2}+\frac{c'}{b^2}-1\right)+\left(\frac{c}{a^2}-\frac{a}{b^2}\right)\left(\frac{c'}{a^2}-\frac{a'}{b^2}\right)\left(1-\frac{1}{a^2}-\frac{1}{b^2}\right)=0,$$

a relation, the geometrical signification of which is: If (x,y) be a point on the conic $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$, and if the polar thereof in regard to the conic $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ meet the first-mentioned conic in the points (x_1,y_1) and (x_2,y_2), then these points are harmonics in regard to the conic

$$\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)-\left(\frac{x^2}{a^2}-\frac{x^2}{b^2}\right)^2\left(1-\frac{1}{a^2}-\frac{1}{b^2}\right)=0;$$

and since the theorem is projective, it is seen that three two conics may be any conics whatever, the third conic being a conic having with the other two a common system of conjugate points.

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If to the ideas we write $a = b = c$; then the theorem is, if the polar of a point on the circle $a^2 + y^2 = a^2$ in regard to the conic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, meet the circle in two points, these are harmonics in regard to the conic $\dfrac{x^2}{a^2(a^2+b^2-1)} + \dfrac{y^2}{b^2(a^2+b^2-1)} = 1$; viz. if $a^2 + b^2 = 1$, the conic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, or if $a^2 = a^2$, $b^2 = b^2$, then the conic is the same similar to, but in the inverse ratio of the two points (x_1, y_1), (x_2, y_2) harmonics in regard to the imaginary conic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, but this is also an inverse form similar to the original conic, which is the theorem in question.

This last result will be similar to the conic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$; viz. if $a^2 + b^2 = a$, then the conic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$; if $a^2 = a^2$, $b^2 = b^2$, then $a^2 = a^2$, $b^2 = b^2 + b^2$; and the conic is $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$. Considering the given conic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ to be an ellipse, the first cone (a) is similar to the two points (x_1, y_1), (x_2, y_2) harmonics in regard to the imaginary conic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, viz. if $a^2 = a^2$, the conic of the right side of the equation is the same conic, but the points (x_1, y_1), (x_2, y_2) are harmonics in regard to the imaginary conic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, but this is also an inverse form similar to the original conic, which is the theorem in question.

The second case $(a^2 + b^2 = 1)$ gives a real ellipse if b be negative; viz. this is, if for b we have the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$, if $a^2 + b^2$, an obtuse-angled hyperbola, the circle $a^2 + y^2 - b^2$ which is the locus of the intersection of a pair of orthotomic tangents of the hyperbola, is consequently imaginary; but the eccentric orthotomic circle hence, on the limits of the theorem, must occur to have the opposite of polar of the property of a point on the circle in regard to the orthotomic hyperbola; viz. $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = a^2$ also the theorem: "Given the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$, $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = a^2$ and the circle $a^2 + y^2 - a^2 + a^2$ (the concentric orthotomic circle of the imaginary circle which is the locus of the intersection of a pair of orthotomic tangents of the hyperbola); if the polar in regard to the hyperbola of a point on the circle meet the circle in two points Q, R, then these are harmonics in regard to the hyperbola."

Of course, if reality be disregarded, the two theorems are mere two of the cases of a conic generally; and observe, that in the first theorem the circle is the locus of the intersection of orthotomic tangents, and we have the opposite of the points Q, R in the second theorem the circle is the locus of the orthotomic circle of the cross, which the lines of in the cross of Q, R themselves.

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A "Smith's Prize" Paper(*); Solutions.

[From the Oxford, Cambridge and Dublin Messenger of Mathematics, vol. V. *(1869), pp. 40–64.]*

1. If $\sqrt{x-a^2+(y+b)^2}$, $\sqrt{x-a^2+(y+b)^2}$, a, β *are respectively real and positive, show that in the equation* $\sqrt{x-a^2+(y-b)^2}=\sqrt{x-a^2+(y-b)^2}$ *is considered as representing a curve, the signs cannot either of them be assumed at pleasure to be + or be −; and distinguish the cases of the ellipse and the hyperbola.*

Writing the equation in the form $\sqrt{x}\pm\sqrt{y}=c$, we find

$$\begin{aligned}2(c\sqrt{r}\pm c\sqrt{r})&=a+b-R,\\ 2(c\sqrt{r}+c\sqrt{r})&=a+b-R,\\ 4cr&=(c+b)^2-R^2,\\ 4cr&=(b-c)^2-R^2,\\ 4cr&=(a+b)^2-R^2,\end{aligned}$$

either of which equations is the rational equation of the curve (the equation being of the fourth order, inasmuch as $S-R$ is a linear function of the coordinates). But writing the rational equation under the form $a^2+a^2+2\,k\,xy+xy(c-\beta)$, $i\beta(a-x)$, j, considered as representing a curve, the curve must either pass for any pair of values of x or y be positive, or in the opposite case be negative; that is, the equations $\sqrt{x}\pm\sqrt{y}=c$, will be in the one case real, in the opposite case imaginary. It is at once seen that, in either case, the original equation $\sqrt{x}\pm\sqrt{y}=c$ also represents the same algebraic value, found in any arbitrary form: but from this it is at once seen, that on a given form of $\sqrt{x}\pm\sqrt{y}$, a determinable value, fixed as soon as the values of x and the form of the curve, a determinable value, fixed as soon as the values of x and the form of the curve, a determinable value, fixed as soon as the values of x and the form of the curve, a determinable value, fixed as soon as the values of x and the form of the curve.

* Set by me for the Master of Trinity, A.D. 1869.

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point, but 2, 3, 4, 5, 6 are fixed points on the conic, then the pencil $1\,(2, 3, 4, 5, 6)$ of lines from the point 1 to the four points 2, 3, 4, 5 has the same anharmonic ratio for all positions whatever of the variable point 1; which is the above-mentioned anharmonic property of the points of a conic.

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The "condition for an ellipse," is that the given length c shall be greater than the distance $\sqrt{(a+b)^2+(\beta-b)^2}$; between the two foci; for a hyperbola that it shall be less. In the former case, for any real point of the curve, it is obvious that the relation must be $\sqrt{x}+\sqrt{y}=c$; the latter case, the latter case, we must look at the actual position of the point.

It must be observed in regard to the ellipse, that, starting from the equation $\sqrt{x}+\sqrt{y}=c$, in finding the radical equation $\dfrac{x}{a^2}+\dfrac{y}{b^2}=1$, $\dfrac{x}{a^2}-\dfrac{y}{b^2}=1$, we have a perfect identical equation which, starting from $\sqrt{x}\pm\sqrt{y}=c$, signifies that we have

$$\sqrt{x}\pm\sqrt{y}=c, \text{ that is, } \sqrt{x}\pm\sqrt{y}=c, \text{ and } \sqrt{x}\pm\sqrt{y}=c,$$

but here, x being less than c, and the two signs are therefore each of them $+$, (c being positive), the two values verification applies to the hyperbola.

2. In a system of curves defined by an equation containing a variable parameter, investigate at any point the normal distance between two consecutive curves; and determine the form of the equation for a system of parallel curves.

Consider the system of curves $f(x,y,c)=0$; then if the point x,y belongs to the curve $f(x,y,c)=0$, and the point $x+dx, y+dy$ to the curve $f(x,y,c+dc)=0$, we have

$$\frac{df}{dx}\,dx+\frac{df}{dy}\,dy+\frac{df}{dc}\,dc=0,$$

and if the point $x+dx, y+dy$ be on the normal at (x,y) to the curve $f(x,y,c)=0$, we have

$$dx:dy=\frac{df}{dx}:\frac{df}{dy},$$

or writing

$$dx=k\,\frac{df}{dx},\quad dy=k\,\frac{df}{dy},$$

we have

$$k\left\{\left(\frac{df}{dx}\right)^2+\left(\frac{df}{dy}\right)^2\right\}+\frac{df}{dc}\,dc=0,$$

wherefore

$$k=-\frac{df}{dc}\,dc\div\left\{\left(\frac{df}{dx}\right)^2+\left(\frac{df}{dy}\right)^2\right\},$$

and the normal distance at (x,y) of the curves c and $c+dc$ is $=\sqrt{(dx^2+dy^2)}$, viz. it is

$$=k\sqrt{\left\{\left(\frac{df}{dx}\right)^2+\left(\frac{df}{dy}\right)^2\right\}},\quad\text{or finally it is}$$

$$=-\frac{df}{dc}\,dc\div\sqrt{\left\{\left(\frac{df}{dx}\right)^2+\left(\frac{df}{dy}\right)^2\right\}},$$

where, if the distance in question be regarded as positive (that is, if we attend only to its absolute value), the sign is to be taken so that $\dfrac{df}{dc}$ shall be positive.

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If the question be given in the form $V-c=0$ (V a function of (x,y)), we have the normal distance

$$= \pm dc \div \sqrt{\left\{\left(\frac{dV}{dx}\right)^2 + \left(\frac{dV}{dy}\right)^2\right\}}.$$

For parallel curves, the normal distance is everywhere the same, that is, $\left(\frac{df}{dx}\right)^2 + \left(\frac{df}{dy}\right)^2$ must have a constant value for all values of (x,y) satisfying the relation $V=c$, viz. we must have identically $\left(\frac{df}{dx}\right)^2 + \left(\frac{df}{dy}\right)^2 = \phi(V)$, ϕ arbitrary; a partial differential equation to be satisfied by V in order that the system $V=c$ may be the equation of a system of parallel curves. Assuming the equation to be satisfied for any particular form of ϕ, say put θ in place, then if C a function of V, such that $\left(\frac{dC}{dV}\right)^2 \phi(V) = 1$, and the equation $f(V) =$ Const. is the same thing as $V =$ Const., it follows that the equation of the system of parallel curves may be taken to be $V=c$, where

$$\left(\frac{dV}{dx}\right)^2 + \left(\frac{dV}{dy}\right)^2 = 1.$$

3. Two resonant (each free to revolve) elliptic rigid in weight and to the weight of the half; and is assumed that at any instant during the explosion the explosive force depends only on the space occupied by the vapour of the gunpowder; compare the energetics velocities of the bullet, and also the energetics velocities of bullet and Trim respectively; the mass measure also is free to recoil, and when it is absolutely fixed.

Consider a single resonant.

Take M for its mass, m that of the ball, R, r for the spaces described, backwards from the cannon and forwards by the ball, at the time t during the explosion. Then by hypothesis the explosive forces, forwards on the ball and backwards on the cannon, is a function of $r+R$, $=\phi(r+R)$ suppose, or we have

$$m\frac{d^2r}{dt^2} = \phi(r+R),$$

$$M\frac{d^2R}{dt^2} = \phi(r+R),$$

and thence

$$\frac{d^2(r+R)}{dt^2} = \left(\frac{1}{m} + \frac{1}{M}\right)\phi(r+R).$$

Multiplying each side by $2\frac{d(r+R)}{dt}$, integrating from $s+R=0$, to $s+R=a$, where a is the length of the tube, and observing that the initial velocities are each $=0$, we find

$$\left[\frac{d(r+R)}{dt}\right]^2 = 2\left(\frac{1}{m} + \frac{1}{M}\right)\int_0^a \phi(s+R)\,(dr+dR),$$

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where $\frac{d\xi}{dt}$, $\frac{d\eta}{dt}$ are the velocities of the ball and cannon respectively at the instant of emergence; $\int_0^t \phi(x+S)(v+dS)$, is a constant factor in a quantity independent of the weights of the ball and cannon), and putting it $= \frac{1}{2}V$, the equation may be written

$$(v+V)^2 = \left(\tfrac{1}{m} + \tfrac{1}{M}\right) C,$$

where v, V are the velocities at the moment of emergence.

But from the equation

$$m\frac{d^2x}{dt^2} = M\frac{d^2X}{dt^2},$$

(the initial values of the velocities being each $= 0$), we have

$$mv = MV;$$

and hence from the foregoing equation we obtain

$$v = \frac{M}{M+m}(V+v) = \frac{M}{M+m}\sqrt{(M+m)C} = \frac{\sqrt{(MC)}}{\sqrt{m(M+m)}}\sqrt{C} = \frac{\sqrt{C}}{\sqrt{m}\sqrt{\left(1+\frac{m}{M}\right)}},$$

or say

$$v\sqrt{m} = \frac{\sqrt{C}}{\sqrt{\left(1+\frac{m}{M}\right)}}.$$

And similarly for the other cannon, if m_1, M_1 be the masses of the ball and cannon, v_1 the velocity of the ball, we have

$$v_1\sqrt{m_1} = \frac{\sqrt{C}}{\sqrt{\left(1+\frac{m_1}{M_1}\right)}},$$

whence

$$v\sqrt{m} : v_1\sqrt{m_1} = \sqrt{\left(1+\frac{m_1}{M_1}\right)} : \sqrt{\left(1+\frac{m}{M}\right)}.$$

If $m_1 = m$, $M_1 = \alpha$, then v, v_1 may be taken to be the velocities of the same ball fired from the cannon, mass M, when it is free to recoil and when it is absolutely fixed; and we have

$$v : v_1 = 1 : \sqrt{\left(1+\frac{m}{M}\right)},$$

viz. the velocity in the former case is equal to that in the latter case divided by

$$\sqrt{\left(1+\frac{m}{M}\right)}.$$

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4. *If $X : Y : Z = a\zeta - c\xi : c\xi - a\eta : a\eta - b\zeta$, where $\xi^2+\eta^2+\zeta^2 = 0$, $\xi'^2+\eta'^2+\zeta'^2 = 0$, show that $\xi\xi'$, $\eta\eta'$, $\zeta\zeta'$, $\eta\zeta'+\eta'\zeta$, $\zeta\xi'+\zeta'\xi$, $\xi\eta'+\xi'\eta$ are proportional to quadric functions of X, Y, Z.*

We have

$$\begin{aligned} Y^2+Z^2 &= (\xi\xi'-c\xi)^2+(a\eta-b\zeta)^2 \\ &= \xi^2(c^2+a^2)+\eta^2(c^2+b^2)-2b\xi(c\xi+a\eta)-2c\xi(a\eta+b\zeta) \end{aligned}$$

which, in virtue of the equations

$$\xi^2+\eta^2+\zeta^2=0, \quad \xi'^2+\eta'^2+\zeta'^2=0,$$

is

$$= -2\xi(c^2+a^2)\eta\eta' - 2c\xi(c\eta+a\zeta)+a\eta(c\zeta+b\xi).$$

And again

$$\begin{aligned} YZ &= (c\xi-c\zeta)(a\eta-b\zeta) \\ &= b\xi^2(c\eta+\zeta)-b^2(a\zeta+c\eta)-b^2(c\zeta+a\eta). \end{aligned}$$

which, in virtue of the same equations, is

$$= b\xi(c\eta+\zeta)+b\zeta(c\eta-\zeta)-b\xi'\eta'(\eta'-\zeta')\eta;$$

and, forming the analogous equations by symmetry, the factor $(\xi\eta'+\xi'\eta)\cdot 2+b^2$ divides out, and we have

$$\begin{aligned} &: \xi\xi' : \eta\eta' : \zeta\zeta' : \\ &= b^2(c\zeta-a\xi)+c^2(a\eta-b\zeta) : \cdots : \cdots \end{aligned}$$

which is the required theorem.

5. *Two tangents of a conic are inversely related to a second conic; find the locus of the intersection of the two tangents.*

In plane geometry the angle in a semicircle is, in spherical geometry it is not, a right angle: show how these conclusions follow from the solution of the above problem.

Let the equation of the first conic be $x^2+y^2+z^2=0$; its equation in line coordinates is therefore $\xi^2+\eta^2+\zeta^2=0$; or what is the same thing, the line $\alpha x+\beta y+\gamma z=0$ will be a tangent of the conic if only $\alpha^2+\beta^2+\gamma^2=0$. Hence the equations of the two tangents being

$$\xi x+\eta y+\zeta z=0,$$

$$\xi' x+\eta' y+\zeta' z=0,$$

we have $\xi^2+\eta^2+\zeta^2=0$, $\xi'^2+\eta'^2+\zeta'^2=0$, and X, Y, Z the coordinates of the point of intersection of these tangents, we have

$$X : Y : Z = \eta\zeta'-\eta'\zeta : \zeta\xi'-\zeta'\xi : \xi\eta'-\xi'\eta,$$

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and thence, by the last question,

$$\xi\xi' : \eta\eta' : \zeta\zeta' : \eta\zeta' + \eta'\zeta : \zeta\xi' + \zeta'\xi : \xi\eta' + \xi'\eta$$
$$= Y^2 + Z^2 : Z^2 + X^2 : X^2 + Y^2 : -YZ : -ZX : -XY.$$

Suppose that the equation of the second conic in line coordinates is

$$(a, b, c, f, g, h)(\xi, \eta, \zeta)^2 = 0,$$

the condition in order that the two lines $\xi x+\eta y+\zeta z = 0$, $\xi' x+\eta' y+\zeta' z = 0$, may be harmonically related to this conic is

$$(a, b, c, f, g, h)(\xi, \xi', \eta\eta', \zeta\zeta') = 0;$$

or, what is the same thing, it is

$$a(Y^2 + bz^2) + c(X^2 + Z^2) + c(X^2 + Y^2) - 2f \cdot YZ - 2g \cdot ZX - 2h \cdot XY = 0,$$

and by what precedes we have

$$(X^2 + Y^2 + Z^2)(a + b + c) - (aX^2 + bY^2 + cZ^2 + 2fYZ + 2gZX + 2hXY) = 0,$$

or, what is the same thing,

$$(a + b + c)(X^2 + Y^2 + Z^2) - (a, b, c, f, g, h)(X, Y, Z)^2 = 0,$$

as the locus of the point of intersection of the two conics; the required locus is therefore a conic.

It is to be observed that the locus is that of a point which is such that the pairs of tangents from it to the two given conics respectively form a harmonic pencil; viz. if the equations of the given conics (in line coordinates) are

$$\alpha^2 + \beta^2 + \gamma^2 = 0,$$

$$(a, b, c, f, g, h)(\alpha, \beta, \gamma)^2 = 0, \quad \text{viz. } (\alpha, \beta, \gamma)^2 = 0,$$

then the equation of the required locus, or say the equation of the harmonic conic, is (in point coordinates)

$$(a + b + c)(x^2 + y^2 + z^2) - (a, b, c, f, g, h)(x, y, z)^2 = 0.$$

In particular, if the second conic be a point-pair or, say, if its equation be

$$(\alpha x + \beta y + \gamma z)(\alpha' x + \beta' y + \gamma' z) = 0,$$

then the equation of the harmonic conic is

$$(\alpha\alpha' + \beta\beta' + \gamma\gamma')(x^2 + y^2 + z^2) = (\alpha x + \beta y + \gamma z)(\alpha' x + \beta' y + \gamma' z),$$

which is satisfied by writing $(x, y, z) = (\alpha, \beta, \gamma)$, or $= (\alpha', \beta', \gamma')$, viz. the harmonic conic passes through the two points of the point-pair. And so, if the first conic be a point-pair, the harmonic conic passes through the four points of the two point-pairs.

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The equation of the first conic in point coordinates is $x^2 + y^2 + z^2 = 0$; hence, supposing as above, that the second conic is a point-pair, the intersections of the first conic with the harmonic conic are given by

$$x^2 + y^2 + z^2 = 0, \quad (\alpha x + \beta y + \gamma z)(\alpha' x + \beta' y + \gamma' z) = 0;$$

whence the harmonic conic has double contact with the first conic, only if each of the lines $\alpha x + \beta y + \gamma z$, $\alpha' x + \beta' y + \gamma' z = 0$, touches the first conic; or, what is the same thing, if each of the points (α, β, γ), $(\alpha', \beta', \gamma')$ lies on the first conic.

In plane geometry, we have (in the plane) a point-pair, the two circular points at infinity; any conic through these points is a circle: the two lines harmonic in regard hereto are at right angles. Hence, by what precedes, taking one of the given conics to be the circular points at infinity, and the other conic to be any two points P, Q; the locus of the intersection of lines through P, Q, cutting each other at right angles, is a circle; this is evidently the circle standing on PQ as diameter, or the angle in a semicircle is a right angle.

In spherical geometry we have (on the sphere) an imaginary conic $x^2 + y^2 + z^2 = 0$, called the *absolute*; any conic having double contact herewith is a circle (small circle of the sphere); two lines (arcs of great circles), harmonic in regard hereto, are at right angles. Hence, taking one of the conics to be the conic $x^2 + y^2 + z^2 = 0$ and the other to be the two points P, Q; the locus of the intersections of the lines (arcs of great circles) through P, Q, which cut at right angles, is a spherical conic; but it is not a circle unless the points P, Q, are each of them on the conic $x^2 + y^2 + z^2 = 0$, viz. it is not a circle for any two real points whatever; that is, in spherical geometry, the angle in a semicircle is not a right angle.

6. *A mass M attached to the end A of a chain AC is placed (with the chain) on a horizontal plane, in such wise that a portion AB of the chain forms a straight line, the remaining portion BC being heaped up at B: the mass M is then set in motion in the direction B to A with a given velocity, and so moves in a straight line, dragging the chain: determine the motion; and explain the peculiarity of the dynamical problem.*

The mass attached to the end of the chain is taken to be M, and the mass of a unit of length of the chain to be $= m$; suppose also that at the commencement of the motion the distance CA is $= a$, and that at the end of the time t this distance is $= a + x$, so that the length of chain then in motion is $= a + x$, ($a < l - a$, if l be the whole length of the chain). Suppose also that the velocity at the time t is $= v$; then we have a mass $= M + m(a + x)$ moving with a velocity v; and, in the element of time dt, this sets in motion with the velocity v a length of chain $dx = vdt$, or mass of chain $= mvdt$; if then the impulse backwards on the mass $M + m(a + x)$, and forwards on the element $mvdt$ be $= R$, we have

$$\{M + m(a + x)\}\, dv = -R,$$

$$mvdt \cdot v = R,$$

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that is,

$$\{M + m(a + x)\}\, dv + mv\, dx = 0,$$

or, observing that $vdt = dx$, this may be written

$$\{M + m(a + x)\}\, dv + mv\, dx = 0,$$

that is,

$$\{M + m(a + x)\}v = \text{constant},$$

$$= (M + ma)V,$$

if V be the velocity at the commencement of the motion. The equation gives, in terms of the space x, the velocity v at any time t before the whole chain is set in motion; writing it in the form

$$\{M + m(a + x)\}\frac{dx}{dt} = (M + ma)V,$$

we have

$$\{M + m(a + x)\}dx = (M + ma)V dt,$$

and putting herein $x = l - a$, the equation gives the value of t at the instant when the whole chain is set in motion; after this epoch, the mass and chain will (it is clear) move on with a uniform velocity

$$= \frac{(M + ma)V}{M + ml},$$

The foregoing equation

$$\{M + m(a + x)\}\, dv + (R + mv)dx = 0$$

might have been obtained at once by the consideration that the momentum is constant throughout the motion; but the method employed puts more clearly in evidence the peculiarity of the dynamical problem; viz. it is, so to speak, a problem of continuous impulse. In each element of time dt, an extraneous mass $mvdt$ is suddenly and abruptly altered (in the present problem from 0 to v); but, for the very reason that it is an infinitesimal element of mass which undergoes this abrupt change of velocity, the effect is a continuous, not an abrupt, change of velocity of the whole finite mass which is then in motion.

7. *Show how an ellipse may be constructed as the envelope of a variable circle having its centre upon either of the axes; and examine the geometrical peculiarities which occur according as the major or the minor axis is made use of.*

Considering the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

the equation

$$(x - \alpha)^2 + y^2 = c^2,$$

where α and c are in the first instance arbitrary parameters, represents a circle having its centre on the major axis: in order that this may touch the ellipse, a relation must be established between α and c; When this is done we obtain a variable circle

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(the equation of which contains a single arbitrary parameter), and this circle has the ellipse for its envelope. (On account of the symmetry in regard to the axis of x, the circle will, it is clear, touch the ellipse in two points, situate symmetrically in regard to this axis.) Now to make the circle touch the ellipse, eliminating y, we have

$$(x-\alpha)^2 + b^2\left(1-\frac{x^2}{a^2}\right) - c^2 = 0,$$

that is,

$$x^2\left(1-\frac{b^2}{a^2}\right) - 2\alpha x + \alpha^2 + b^2 - c^2 = 0,$$

which equation, considered as an equation in x, must have equal roots; that is, we must have

$$\left(1-\frac{b^2}{a^2}\right)(\alpha^2+b^2-c^2) - \alpha^2 = 0;$$

or, what is the same thing,

$$-b^2\alpha^2 + a^2(c^2-b^2)(b^2-c^2) = 0;$$

consequently

$$c^2 = b^2\left(1-\frac{\alpha^2}{a^2-b^2}\right);$$

or the required equation of the circle is

$$(x-\alpha)^2 + y^2 = b^2\left(1-\frac{\alpha^2}{a^2-b^2}\right);$$

or, as this may also be written

$$(x-\alpha)^2 + y^2 = (1-e^2)\left(\frac{a^2}{e^2}-\alpha^2\right);$$

It is to be observed that at the points of contact of the ellipse and circle, we have

$$x = \frac{a^2}{a^2-b^2}\alpha = \frac{\alpha}{e^2}.$$

By simply interchanging a and b, it appears that when the centre is on the minor axis, the equation of the variable circle is

$$x^2 + (y-\beta)^2 = b^2\left(1-\frac{\beta^2}{b^2-a^2}\right);$$

and that at the points of contact of the ellipse and circle we have

$$y = \frac{b^2}{b^2-a^2}\beta.$$

In the case where the point is on the major axis, then attending to positive values of α, the circle remains real so long as α is $< a$. But from the formula $x = \alpha/e^2$, we have $x = a$ for $\alpha = ae^2$ and, when α becomes greater than this, the points of contact are imaginary. That is, as α passes from 0 to ae^2, the circle is real, and

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has real contact with the ellipse; but for $\alpha = ae^2$ the circle becomes the circle of greatest curvature, touching the ellipse at the extremity of the major axis: as α becomes greater than ae^2, the circle remains real, but the contact with the ellipse becomes ideal: $\alpha = a$, the circle reduces itself to the point form considered as an evanescent circle; and for greater values of α, the circle is imaginary.

When the centre is on the minor axis it appears in like manner that the circle is always real; but, attending to negative values of β, the circle has real contact with the ellipse only so long as β is $< -\frac{b^2}{a}$, for this value of β, the circle touches the ellipse at the extremity of the minor axis, being in fact the circle of minimum curvature; and for greater negative values of β, the contact is imaginary.

8. *Show that the number of ways in which a things can be arranged so that no one of them is in its right place, but each in the place of one of the remaining* $A_n = (n-1)A_{n-1} + (n-1)A_{n-2}$, *and thence* $A_n = nA_{n-1} + (-)^n$ ...

Suppose the n things arranged as required, we may by a single transposition of two things bring a given thing, say a, into its place in the original place (the last place); and conversely every arrangement of the required form can be obtained from an arrangement in which a occupies the last place, by a transposition of a with another of the things. Now in the arrangements which then gives rise to an arrangement of the required form, either all the things 1, 2, 3, ..., $(n-1)$ are out of their original places; and we can then transpose a with any one of the things 1, 2, 3, ..., $(n-1)$; or else one of the things in its original place, and we can transpose a with b in $(n-1)$ transposes 1 and a; or 2 is in its original place, and we can transpose a with a. And hence we have an arrangement of mass which undergoes this abrupt change of velocity, the only mode in which an arrangement of the required form is obtained. Hence if the things denote U_n, the number of arrangements in which no one of the n things is, in motion.

Now, by what precedes,

$$U_n = (n-1)U_{n-1} + nU_{n-2};$$

and it thus appears that U_n is of the form $(n-1)A_{n-1}$, and writing accordingly $U_n = (n-1)A_{n-1}$, and therefore

$$U_{n-1} = (n-2)A_{n-2}, \quad U_{n-2} = (n-3)A_{n-3},$$

we have

$$A_n = (n-1)A_{n-1} + (n-1)A_{n-2}.$$

which is the required equation for A_n. As in the question, $A_2 = 0$, $A_3 = 1$, and the successive values A, A_4, A_5, ... are then to be integrated into one of the first integer, by writing the equation in the form

$$\{(n-1)A_{n-1} - nA_{n-2}\} = \{(n-2)A_{n-2} - (n-1)A_{n-3}\} = \ldots = \{2A_2 - 3A_1\} = -1,$$

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the integral is

$$(n-1)A_{n-1} = (n-2)A_{n-2} + (-)^{n-1},$$

and writing $n = 2$, we have $C = -1$, whence

$$(n-1)A_{n-1} = nA_{n-2} + (-)^{n-1},$$

or, what is the same thing,

$$A_n = (n-1)A_{n-1} + \frac{(-)^n}{n-1}\left\{A_{n-1} + (-)^{n-1}\right\}.$$

The foregoing very elegant proof of the equation

$$U_n = (n-1)(U_{n-1} + U_{n-2}) = (n-1)A_{n-1},$$

was unknown to me; but was given to me in the Examination; my own proof of the equation $U_n = nU_{n-1} + (-)^n$, was derived from the well-known formula

$$U_n = n!\left\{1 - \frac{1}{1} + \frac{1}{1\cdot 2} - \frac{1}{1\cdot 2\cdot 3} + \cdots + (-)^n \frac{1}{1\cdot 2\cdot 3\cdots n}\right\};$$

The theorem itself, and the two equations of differences for A_n, are due to Euler, see his memoir, "Sur une copieuse particularitiés du Calcul Magicquen," *Comm. Arith. Coll.,* t. I. p. 859.

9. *If in any covariant of a binary form $(a, b, c, \ldots, k)(x, y)^n$ the coefficients $a, b, \ldots, k$ are replaced by $a + bp$, $b + cp, \ldots, k + lp$ respectively, show that the result is a covariant of the next superior form $(a, b, c, \ldots, l)(x, y)^{n+1}$, and determine the covariant obtained by this operation on the discriminant of the cubic form $(a, b, c, d)(x, y)^3$.*

A covariant of the binary form $U' = (a, b, c, \ldots, k)(x, y)^n$ is either given by a single symbolical expression of the form $\overline{12}^a \cdot \overline{13}^b \ldots U_1U_2 \ldots$, or it is the sum of a number of such expressions, each multiplied by a constant numerical factor. In the latter case each of the expressions in question is a covariant of U. Any such expression is at once seen to be a function of U and of its differential coefficients (viz. it may contain the differential coefficients of any order of U, 1, 2, …, n): homogeneous as regards the differential coefficients of the same order. But writing $U' = (a, b, c, \ldots, l)(x, y)^{n+1}$, the differential coefficients of any order of U' are by the change of $a, b, \ldots, k$ into $a + bp, b + cp, \ldots, k + lp$, converted into the differential coefficients of the same order of U, multiplied into the same numerical factor; consequently the foregoing covariant of U is, by the same change in regard to the coefficients of the same order. Any function of U and its differential coefficients of the same order, and which is therefore the same in regard to the coefficients of the order of U, becomes, by the like change in regard to the coefficients of the same order, the like covariant of U'.

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Or the same thing may be otherwise so obvious. The two of the ideas we write $U = (a, b, c, d)(x, y)^n$, any covariant Θ of U is reduced to zero by each of the operations $aD_b + 2bD_c + 3cD_d + \ldots$ and $bD_a + 2cD_b + 3dD_c + \ldots$. We may by writing which is then reduced to zero, a covariant Θ' of the same form U' as the function obtained, viz.

$$\Theta' = (aD_a + 2bD_b + 3cD_c + 4dD_d + \ldots)(x, y)^n$$

and observing that we have identically

$$(aD_a + bD_b + cD_c + dD_d + \ldots)\Theta = m \cdot \Theta,$$

or considering Θ as a function of $a, b, c, d, \ldots p$ (in fact, of the $a, b, c, d \ldots$)

$$(aD_a + bD_b + 3D_d + \ldots)\Theta = (a + b\rho)D_a + (b + c\rho)D_b + (c + d\rho)D_c + \ldots \cdot p\Theta - m \cdot \Theta,$$

or, what is the same thing,

$$\{ap + b(p + q) + \cdots\} = \Theta - m \cdot \Theta - 2\{(a + b\rho)D_a + (b + c\rho)D_b + (c + d\rho)D_c + \ldots\}\Theta,$$

that is,

$$\{a, b, c, d, \ldots\}\Theta = (a + b\rho)D_a + (b + c\rho)D_b + (c + d\rho)D_c + (d + e\rho)D_d + \ldots,$$

an equation which, Θ being the form $a, b, c, d, e, f(x, y)^4$, Θ is at once seen to be reduced to zero, by each of the operations aD_a, bD_b, $cD_c + 2cD_d + 3dD_e + \ldots$, $(a + b\rho)D_a + (b + c\rho)D_b + (c + d\rho)D_c + (d + e\rho)D_d + (e + f\rho)D_e + \ldots$, that is, Θ is a covariant of $(a, b, c, d, e, f)(x, y)^4$.

The discriminant of $(a, b, c, d)(x, y)^3$ is

$$\Theta = a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd;$$

making the substitution in question, it is

$$= (a + bp)^2(b + ep)^2 + 4a(c + dp)^3 + \cdots$$

viz. it is

$$\begin{gathered} = a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd \\ +2(b\rho^2d^2 + 2acd^2 - bc^2d + bd^2 + ac^2d - 3abd \cdot p + b^2d) + \cdots \\ = -3p(a^2ad - abcd - ac \cdot d^3 + ad^3) - \cdots \end{gathered}$$

But there is no such irreducible covariant of the quartic function $(a, b, c, d, e, f)(x, y)^4$; and observing that we have identically

$$(ae - 4bd + 3c^2)^2 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0,$$

the covariant in question must be

$$= (ae - 4bd + 3c^2)^2 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2;$$

where $cse - bf$, etc. is the Hessian of the quartic function $(a, b, c, d, e, f)(x, y)^4$.

10. *From the equations of a paraboloid, and of its parallel surface, derive (by observing, determine the curves of curvature of a paraboloid: show also that for the paraboloid $xy = aw$ (the parallel sections $x =$ Const. being rectangular hyperbolae) the curves of curvature are the intersections of the paraboloid by the spheres $a^2(x^2 + y^2 + z^2) + 2axy = \lambda$.*

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The curves of curvature of the ellipsoid

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1,$$

are given as the intersection of the surface with the confocal surface

$$\frac{x^2}{a^2-\theta}+\frac{y^2}{b^2-\theta}+\frac{z^2}{c^2-\theta}=1;$$

transforming to the vertex, or writing $x+a$ in place of x, these become

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=-\frac{2x}{a},$$

$$\frac{x^2}{a^2-\theta}+\frac{y^2}{b^2-\theta}+\frac{z^2}{c^2-\theta}=\frac{-2x}{a-\theta};$$

or multiplying by a and writing $\frac{a^2}{a^2}=\lambda$, $\frac{a^2}{a}=h$, the equations are

$$\frac{x^2}{h}+\frac{y^2}{\lambda}+\frac{z^2}{\lambda}=-2x,$$

$$\frac{x^2}{a+b\theta}+\frac{y^2}{\lambda+\theta}+\frac{z^2}{\lambda+\theta}=-2x;$$

or, putting herein $c=-\infty$, the equations are

$$\frac{x^2}{h}+\frac{y^2}{\lambda}=-2x,$$

$$\frac{x^2}{h-\theta}+\frac{y^2}{\lambda-\theta}-2x=-\theta-0,$$

viz. these equations, where θ is a variable parameter, determine the curves of curvature of the paraboloid $\dfrac{x^2}{h}+\dfrac{y^2}{\lambda}-2x=0$. The second equation may be replaced by

$$\frac{x^2}{((b-c)}-\frac{y^2}{m(b-c)}=0,$$

Write in the equations $l=-m=c$; they become

$$y^2-x^2-2xc=0,$$

$$\frac{x^2}{c+\theta}-\frac{y^2}{c-\theta}-2x=0,$$

or, substituting from $\dfrac{x+a}{\sqrt{2}}$ for x, and $\dfrac{x-a}{\sqrt{2}}$ for y, the equations are

$$xy-c=0,$$

$$\frac{(x+b)^2}{c+\theta}-\frac{(x-y)^2}{c-\theta}-2c=0,$$

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the second of which is

$$c(x^2+y^2)-2\theta xy+(c^2-\theta^2)=0,$$

which is the equation for determining the curves of curvature of the surface $xy-cz=0$.

Writing in this equation $\theta=c(2\theta+c^2)$, it becomes

$$c(x^2+y^2)-2cxy(2\theta+c^2)-4c^2\theta=0,$$

or, as this may be written,

$$(c^2+x^2)(c^2+y^2)-2cxy(c^2+c^2)-4c^2\theta=0,$$

and thence

$$c^2(x^2+y^2)+c^2(c^2+y^2)-2c^2(c^2+y^2)+2cxy-cxy(x^2+y^2),$$

that is,

$$\theta=c\,(c\,x^2+c\,y^2)+2\,c\,x\,y,$$

or combining with the equation $cz-xy=0$ of the paraboloid, this is

$$\lambda=c(x^2+y^2+z^2)+2cxy,$$

or the curves of curvature are given as the intersections of the paraboloid by the series of surfaces represented by this equation.

11. *Explain for a surface such as the ellipsoid the form of the curve of given constant slope; and in an elliptical hexing case of its principal planes horizontal determine the limits within which the curve is possible.*

At any point of a surface, the direction of the line of greatest slope is evidently at right angles to the level curve, or, what is the same thing, to the trace of the tangent plane on the horizontal plane; and the inclination of the element of the line of greatest slope is equal to that of the tangent plane to the horizontal plane. The line of given constant slope must be of course be entirely on the portion of the surface for which the inclination of the tangent plane has a value not less than the given constant slope of the curve; viz. on a closed surface such as the ellipsoid, it will (in upon a certain zone of the surface, included between two boundaries, which boundaries are the curves such that at any point thereof the inclination of the tangent plane is equal to the given constant slope; and by what precedes, the direction of the line of given constant slope, at any point where it meets the boundary, is connected with the direction of greatest slope; viz. it will in general meet the boundary at a finite angle; this implies that the point is a cusp on the curve of given constant slope; and the curve in question will be a curve as shown in the figure, passing continually from one boundary to the other, and where it reaches either boundary, turning back cusp-wise to the other boundary; the curve may in particular cases, after making a circuit, or any number of circuits of the zone, re-enter upon itself, forming a closed

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curve; but in general this is not the case, and the curve will go on as above between the two boundaries ad infinitum.

[Figure: ellipsoid zone bounded by two closed curves — diagram omitted]

In the case of the ellipsoid, the plane of xy being as usual horizontal, and the equation being

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

then if λ be the given constant inclination, the boundaries are the series of points at which the inclination of the normal to the axis of z is $= \lambda$; that is, we have

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} = \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}\right) \cot^2 \lambda,$$

or, what is the same thing,

$$\left(\frac{x^2}{a^4} + \frac{y^2}{b^4}\right) = \left(\frac{z^2}{c^4}\right) \cot^2 \lambda + 1;$$

the boundaries are thus two detached ovals, meeting the principal sections $z = 0$, $y = 0$, in the points given by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \cot^2 \lambda - 1 = 0, \quad \frac{x^2}{a^2} + \frac{z^2}{c^2} \cot^2 \lambda - 1 = 0,$$

and the general form is then at once perceived.

12. *To every point of space there corresponds a plane, viz. considering the several points as belonging to a solid body which is infinitesimally displaced in any manner, the plane which corresponds to a point is the plane drawn through the point at right angles to the direction of the motion; determine the plane which corresponds to a given point; and connect the result with any general geometrical theory.*

The displacements, in the directions of the axes, of a point whose coordinates are (x, y, z) are given by the ordinary formulae

$$\begin{aligned} \delta x &= a + qz - ry, \\ \delta y &= b + rx - pz, \\ \delta z &= c + py - qx, \end{aligned}$$

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where a, b, c, p, q, r are constants which determine the particular displacement; hence if X, Y, Z are current coordinates, the plane corresponding to the point (x, y, z) is the plane through this point at right angles to the line

$$\frac{X-x}{a} = \frac{Y-y}{b} = \frac{Z-z}{c},$$

viz. it is the plane

$$(X-x)(a+qz-ry) + (Y-y)(b+rx-pz) + (Z-z)(c+py-qx) = 0,$$

or, substituting for δx, δy, δz their values, reducing and arranging, this is

$$\begin{aligned} &X(a-ry+qz) \\ &+Y(b-pz+rx) \\ &+Z(c-qx+py) \\ &-(ax+by+cz) \\ &-(xy+yz+zx) = 0, \end{aligned}$$

that is, the equation of the plane corresponding to the point (x, y, z) is an equation linear in regard to the coordinates (x, y, z) of this point; and such that arranging as above in the form of a square, the coefficients which are symmetrically situate in regard to the dexter diagonal are equal and of opposite signs, or say that the coefficients form a skew symmetrical matrix.

More generally, corresponding to the point (x, y, z) we may have the plane

$$AX + BY + CZ + D = 0,$$

where A, B, C, D are any linear functions whatever $(= ax+by+cz+d$, etc.$)$ of the coordinates (x, y, z); this is the general relation which is put in the analytical basis of the theory of duality; it is in fact (apart from a variable point situate as a line or a plane, etc.) the general one, to a given point there corresponds a plane not in general passing through the point; the case above considered is distinguished by the circumstance that for any point whatever the corresponding plane does pass through the point.

13. *Write down the Lagrangian equations of motion, explaining the notion, and mode of applying them; and by way of illustration deduce the equations of motion of three particles, connected so as to form an equilateral triangle of variable magnitude, moving in a plane under the action of any forces.*

The Lagrangian equations of motion are

$$\frac{d}{dt}\frac{dT}{d\theta'} - \frac{dT}{d\theta} = \frac{dU}{d\theta},$$

$$\frac{d}{dt}\frac{dT}{d\phi'} - \frac{dT}{d\phi} = \frac{dU}{d\phi},$$

&c.,

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θ, ϕ, ... here being any independent coordinates (in the most general sense of the word) which serve to determine the position of the system at the time t; T is the *vis viva* function, or half-sum of the mass of each particle into the square of the velocity, expressed in terms of its velocity, expressed in terms of the coordinates θ, ϕ, ... and of their differential coefficients $\theta' = \frac{d\theta}{dt}$, $\phi' = \frac{d\phi}{dt}$, &c.; T is thus a given function of θ, ϕ, ... θ', ϕ', ...; homogeneous of the second order as regards the velocities θ', ϕ', ... &c. $\frac{dT}{d\theta}$, $\frac{dT}{d\phi}$, ... are the partial differential coefficients of T in regard to θ, ϕ, ..., &c., respectively, and they are also given functions of θ, ϕ, ... θ', ϕ', ... U is the force-function or sum

(it being assumed that $Xdx+Ydy+Zdz$ is a complete differential; that is, that there exists a force-function U) expressed in terms of the coordinates θ, ϕ, ..., and being thus a given function of these coordinates. The partial differential coefficients $\frac{dU}{d\theta}$, $\frac{dU}{d\phi}$, ..., are, respectively, and are then given functions of these coordinates. The known $\frac{d}{dt}\frac{dT}{d\theta'}$ contain, it is clear, one (and that linearly) the second differential coefficients $\frac{d^2\theta}{dt^2}$, $\frac{d^2\phi}{dt^2}$, &c., the equations thus establish between the coordinates θ, ϕ, ... and their first and second differential coefficients in regard to the time t a series of relations which (with the conditions of the problem) is in time of relations of which is equal to that of the coordinates, and which therefore would be integrable by aid of the equations of condition. U is in fact a function of θ, ϕ, in terms of the time.

In the proposed case of three particles, if m_1, m_2, m_3 be their masses, r the side of the equilateral triangle, x_0, y_0 the coordinates of m_1, $\theta + 60^\circ$ (or rather, for convenience $\theta + a$) the inclination to the axis of x of the line joining m_1 and m_2 (and consequently $\theta + b$ that of the line joining m_2 and m_3); then the coordinates of m_1, m_2, m_3 will be

$$\begin{aligned}
x_1 &= x_0 + r\cos\theta, & y_1 &= y_0 + r\sin\theta,\\
x_2 &= x_0 + r\cos(\theta+a), & y_2 &= y_0 + r\sin(\theta+a),\\
x_3 &= x_0 + r\cos(\theta+b), & y_3 &= y_0 + r\sin(\theta+b).
\end{aligned}$$

We have

$$\begin{aligned}
T = {}&\tfrac{1}{2}m_1(x_1'^2 + y_1'^2)\\
&+ \tfrac{1}{2}\{(x_0' + r'\cos\theta - r\sin\theta\cdot\theta')^2 + (y_0' + r'\sin\theta + r\cos\theta\cdot\theta')^2\}\\
&+ \tfrac{1}{2}m_2[\{x_0' + r'\cos(\theta+a) - r\sin(\theta+a)\theta'\}^2 + \{y_0' + r'\sin(\theta+a) + r\cos(\theta+a)\theta'\}^2]\\
&+ \tfrac{1}{2}m_3[\{x_0' + r'\cos(\theta+b) - r\sin(\theta+b)\theta'\}^2 + \{y_0' + r'\sin(\theta+b) + r\cos(\theta+b)\theta'\}^2].
\end{aligned}$$

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a given function of x, y, r, θ, x_0', y_0', r', θ'; the equations of motion will be

$$U = u_1(X dx + Y dy) + \&c.,$$

will be a given function of x, y, r, θ; and the equations of motion will be

$$\begin{aligned}
\frac{d}{dt}\frac{dT}{dx_0'} - \frac{dT}{dx_0} &= \frac{dU}{dx_0},\\
\frac{d}{dt}\frac{dT}{dy_0'} - \frac{dT}{dy_0} &= \frac{dU}{dy_0},\\
\frac{d}{dt}\frac{dT}{dr'} - \frac{dT}{dr} &= \frac{dU}{dr},\\
\frac{d}{dt}\frac{dT}{d\theta'} - \frac{dT}{d\theta} &= \frac{dU}{d\theta}.
\end{aligned}$$

14. *From the integrals $x = a\cos t + a'\sin t$, $y = b\cos t + b'\sin t$, of the dynamical equations $\dfrac{d^2x}{dt^2} = -x$, $\dfrac{d^2y}{dt^2} = -y$, derive a simultaneous solution of the two partial differential equations*

$$\frac{dS}{dx} = \left(\frac{(dS)}{(da)}\right)^2 + \left(\frac{(dS)}{(dy)}\right)^2 = -\tfrac{1}{2}(x^2 + y^2),$$

$$\frac{dS}{dy} = \frac{a'}{a^2}\left[\left(\frac{(dS)}{(da')}\right)^2 + \left(\frac{(dS)}{(db')}\right)^2 - \tfrac{1}{2}(a^2 + b'^2)\right]$$

Writing $S' = \int_0^t \frac{1}{2}(x'^2 + y'^2)\,dt$, we have

$$\begin{aligned}
x &= a\cos t + a'\sin t,\\
y &= b\cos t + b'\sin t,
\end{aligned}$$

as the integrals of the equations of motion

$$\frac{d^2x}{dt^2} = -x, \qquad \frac{d^2y}{dt^2} = -y,$$

moreover

$$\begin{aligned}
x' &= -a\sin t + a'\cos t,\\
y' &= -b\sin t + b'\cos t,
\end{aligned}$$

where a, b, a', b' are the initial values of x, y, x', y' respectively $\left(x' = \dfrac{dx}{dt}, y' = \dfrac{dy}{dt}\right)$.

Hence the Principal Function

$$\begin{aligned}
S &= \int_0^t \tfrac{1}{2}(T + U)\,dt\\
&= \tfrac{1}{2}\int_0^t \{x'^2 + y'^2 - x^2 - y^2\}\,dt,
\end{aligned}$$

expressed as a function of x, y, a, b, t, will satisfy simultaneously the proposed partial differential equations.

We have

$$x'^2 + y'^2 - x^2 - y^2 = (a'^2 + b'^2 - a^2 - b^2)\cos 2t + 2(aa' + bb')\sin 2t,$$

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Hence

$$4S = \tfrac{1}{2}(a'^2 + b'^2 - a^2 - b^2)\sin 2t + 2(aa' + bb')(1 - \cos 2t)$$
$$= (a^2 + b'^2 - a^2 - b^2)\sin t \cos t + 2(aa' + bb')\sin^2 t.$$

But

$$\sin t = -x + a\cos t,$$
$$\sin t = -y + b\cos t,$$

Hence

$$(aa' + bb')\sin t = a + b - (a^2 + b^2)\cos t,$$
$$(a^2 + b^2)\sin t\cos t = a + b - (a^2 + b^2 + 2c)\cos^2 t,$$

and substituting these values

$$4S = \frac{2\sin t\cos t}{\sin^2 t}\cdot a^2 + b^2 - 2(a^2 + b^2)\cos^2 t + 2(a + b)\cos t$$
$$-(a^2 + b^2) - 2c\cos^2 t + 4(a + b)\cos t$$
$$-4\sin t\,\frac{\cos t}{\sin t}(a + b) - 4(a^2 + b^2)\cos t$$

or, what is the same thing,

$$S = \tfrac{1}{2}\,\frac{\cos t}{\sin t}(x^2 + y^2 + a^2 + b^2) + \tfrac{1}{2}\,\frac{1}{\sin t}(ax + by);$$

which value of S satisfies the two equations are satisfied here.

More generally, the two equations are satisfied by

$$S = u + \alpha,$$

(u an arbitrary constant) while now (u, considered as a function of the first equation, contains the three arbitrary constants u, a, b, and is thus a complete solution; and similarly considered as a solution of the second equation, it contains the three arbitrary constants u, a', b', and is thus a complete solution of this equation.

I venture to add a few remarks in illustration of what is required in the papers set up in an Examination.

In the latter part of question (7) (form of the equation for a system of parallel curves) it is worse than useless to say $\dfrac{dx}{ds} = \left(\dfrac{dx}{dx}\right)^2 + \left(\dfrac{dy}{dx}\right)^2$ [p. 445] must be constant; a good and sufficient answer would be that it must be constant in virtue of the given equation $f(x, y, c) = 0$. So in question (10) (the Lagrangian equations of motion), it is not enough to write down $\dfrac{d}{dt}\dfrac{dT}{dx'} - \dfrac{dT}{dx} = \dfrac{dU}{dx}$; it must be explained what T, U, and ξ, η, ... respectively; but for these the equations might be (without differential equations for the determination of T, U, ξ, etc. in their natural form. The answer to a candidate should be (*shortly so*) as it is given here, not such as it might be of an adequate written explanation of his ideas to himself, but such as to be intelligible into one of the first order of mathematicians on reading it; and there is no incompatibility between these requirements; A clear and precise indication of a process of solution is very much better than a detailed solution incorrectly worked out.

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Note on the Integration of Certain Differential Equations by Series.

[From the Oxford, Cambridge, and Dublin Messenger of Mathematics, vol. V. (1869), pp. 77–82.]

THERE is a speciality in the integration of certain differential equations by series, which (though evidently quite familiar to those who have written on the subject—Ellis, Boole, Hargreaves) has not, it appears to me, been exhibited in the clearest form. To fix the ideas, consider a linear differential equation of the second order integrable in the form $y = Ax^{\alpha} + Bx^{\alpha-1} + \ldots$ is determined by a quadratic equation, and for each value of α, the coefficients B, C, $\ldots$ are given multiples of A, we have thus the general solution

$$y = A(x^{\alpha} + Bx^{\alpha-1}\alpha + \ldots) + A'(x^{\alpha'} + B'x^{\alpha'-1} + \ldots).$$

The speciality referred to is, when the two roots differ by an integer number; suppose α is the smaller root, and α' is equal to, or, more definitely the larger. Then, inasmuch as the series

$$y = Ax^{\alpha} + Bx^{\alpha-1} + \ldots,$$

is identical in form with the general solution as above written down, it is clear that, starting with the root $\alpha = \alpha'$, the coefficients beginning with that of x^{α}, $= x^{\alpha-1}$, ought not to be any longer determinate multiples of A, but should contain a new arbitrary constant K, and thus that the series derived from the root $\alpha = \alpha'$ should be the general solution containing two arbitrary constants. The most simple case is when the substitution of the series in the differential equation leads to a relation between two consecutive coefficients of the form $\frac{B}{A}\frac{0}{a}$, which has the value $\frac{B}{A}$ are functions

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the numerators and denominators of which are factorial functions of α such that, for some coefficient $\frac{B}{A}$ preceding $\frac{C}{B}$ of the coefficient of $x^{\alpha-1}$, the denominator vanishes; suppose then that we have $\alpha = \alpha'$, then B should contain a new arbitrary constant: this being so, it is allowable to write $B = C$, $C = 0$, in giving to the differential equation the finite solution

$$y = A\left(x^{\alpha} + \frac{B}{A}x^{\alpha-1} + \ldots\right),$$

but, if notwithstanding the evanescent factor, we carry on the series, then in the coefficient of $x^{\alpha-1}$ there occurs in the denominator the same evanescent factor, so that the coefficient of this term presents itself in the form $A\frac{C}{A}\frac{C}{B}$, = an arbitrary constant D, say $A\frac{C}{B}\cdot\frac{C}{0}$, which is essentially indeterminate); and the solution thus obtained is the form

$$y = A\left(x^{\alpha} + \frac{B}{A}x^{\alpha-1} + \ldots + \frac{B}{A}x^{\alpha-1}\right) + K\left(x^{\alpha-1} + \frac{B}{A}x^{\alpha-2} + \ldots\right),$$

viz. there is one particular solution with A and K.

Take for example the equation

$$\frac{d^2y}{dx^2} + \frac{2}{x}\frac{dy}{dx} + y = 0, \tag{I}$$

mentioned Cambridge Math. Journal, t. II. p. 176 (1840). If the integral is assumed to be

$$y = Ax^{\alpha} + Bx^{\alpha+1} + Cx^{\alpha+2} + \&c.,$$

then we find

$$\begin{aligned}(\alpha+1)(\alpha-1)\cdot A &= 0,\\ (\alpha-1)(\alpha+2)B + \alpha A &= 0,\\ \alpha(\alpha+3)C + (\alpha+1)B &= 0,\\ (\alpha+1)(\alpha+4)D + (\alpha+2)C &= 0,\\ &\&c.\end{aligned}$$

Hence $\alpha = -1$, or else $\alpha = 1$; the coefficients are

$$\begin{aligned}B &= \frac{-\alpha A}{(\alpha-1)(\alpha+2)},\\ C &= \frac{-\alpha(\alpha+1)A}{(\alpha-1)(\alpha+2)(\alpha+3)},\\ D &= \frac{-(\alpha+1)(\alpha+2)A}{(\alpha-1)(\alpha+2)(\alpha+3)(\alpha+4)},\\ \&c.\end{aligned}$$

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Here taking $\alpha = -1$, we are at liberty to make C and all the following coefficients $= 0$; in fact, if we commence by assuming

$$y = Ax^{\alpha} + Bx^{\alpha+1},$$

the equations

$$\begin{aligned} (\alpha+1)(\alpha-2)A &= 0, \\ (\alpha-1)(\alpha+2)B + \alpha A &= 0, \\ \alpha(\alpha+3)C &= 0, \\ (\alpha+1)(\alpha+4)D &= 0, \\ \&c. \end{aligned}$$

are all satisfied if only $\alpha = -1$, $B = A$, $C = \dfrac{-\alpha A}{(\alpha-1)(\alpha+2)}$, and we have then the finite solution $y = A\left(\dfrac{1}{x} - \dfrac{1}{2}\right)$; but if we continue the series, retaining D to represent the indeterminate quantity $\dfrac{(-\alpha+1)(\alpha+2)A}{(\alpha-1)(\alpha+2)(\alpha+3)(\alpha+4)}$, we have the solution

$$y = A\left(\frac{1}{x} - \frac{1}{2}\right) + D(x - \tfrac{1}{2}x^2 + \tfrac{1}{6}x^3 - \ldots),$$

the second member of which is in fact the series derived from the root $\alpha = 1$. This series is expressible by means of an exponential, viz. we have

$$x^{\alpha} - \tfrac{1}{2}x^{\alpha+1} + \tfrac{1}{6}x^{\alpha+2} = \tfrac{1}{2}\left(\frac{x^{\alpha}}{a}\right)e^{x} = \tfrac{1}{2}\left(\frac{x^{\alpha}}{a} - \frac{1}{x}\right)e^{x},$$

and the complete integral is in fact

$$y = A\left(\frac{1}{x} - \frac{1}{2}\right) + B\left(\frac{1}{x} + \frac{1}{2}\right)e^{x},$$

but this result is not immediately connected with the investigation. It may however be noticed that, if we proceed in the original equation in that it has $-y$ in place of y; there is consequently the particular solution $x = A\left(\dfrac{1}{x} + \dfrac{1}{2}\right)$, giving for the proposed equation $y = A\left(\dfrac{1}{x} + \dfrac{1}{2}\right)e^{-x}$, and we have thence the complete solution as shown.

Consider, secondly, the differential equation

$$\frac{d^2y}{dx^2} - (b^2 - \beta^2)y = 0, \tag{II}$$

(derived from the equation (I) by writing therein px^{-q} in place of y); this equation is satisfied by the series

$$y = Ax^{\alpha} + Bx^{\alpha+1} + Cx^{\alpha+2} + \&c.,$$

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Then $a = -1$ or $a = b+2$, but here the series belonging to $a = -1$ contains only odd powers, the other contains only even powers of x; hence the two series do not coalesce as in the former case, and the first series is obtained without the indeterminate symbol $\frac{0}{0}$ in any of the coefficients. We have in fact

$$\begin{aligned}(\alpha+1)(\alpha-2)A&=0,\\(\alpha+2)(\alpha+1)B-\beta^2A&=0,\\(\alpha+3)(\alpha+2)C-\beta^2B&=0,\\&\&c.\end{aligned}$$

and there is not, in the series in question for the value of $\alpha = -1$ (or other series) any evanescent factor, either in the numerator or in the denominators.

But consider, thirdly, the differential equation

$$\frac{d^2y}{dx^2}+(q-\tfrac{1}{2})\frac{dy}{dx}+\left(\beta^2-pq-\tfrac{1}{4}\right)y=0, \tag{III}$$

(derived from (I) by writing therein px^{-q} in place of y). This is satisfied by the series

$$y = Ax^\alpha + Bx^{\alpha+1} + Cx^{\alpha+2} + Dx^{\alpha+3} + \&c.,$$

where $\alpha = -1$ or $\alpha = b+2$, but here the series belonging to $a = -1$, the coefficient D should become indeterminate. The relation between the coefficients is here a relation between three consecutive coefficients, viz. we have

$$\begin{aligned}(\alpha+1)(\alpha-2)A&=0,\\(\alpha+2)(\alpha+1)B+(q+2)A&=0,\\\alpha(\alpha+3)C+(q+1)B+(\beta^2-pq)A&=0,\\(\alpha+1)(\alpha+4)D+(q)C+(\beta^2-pq)B&=0,\\(\alpha+2)(\alpha+5)E+(q-1)D+(\beta^2-pq)C&=0,\\&\&c.\end{aligned}$$

It is to be shown that in the series for $a = -1$, the expression $(q+2)A + (\beta^2 - pq)B$ contains the evanescent factor $(\alpha+1)$; for indeterminacy and D is indeterminate; we have in fact

$$B = \frac{-(q+2p)A}{(\alpha+2)(\alpha+1)},$$

and thence

$$\begin{aligned}C &= \frac{(\beta^2+2)(\alpha+1)-q(q+2p)}{(\alpha+2)(\alpha+1)\alpha(\alpha+3)}A,\\&=\frac{-1}{a(\alpha+3)}\left\{(\beta^2+2p)A-q(\beta^2+2p)A+(\beta^2-pq)\frac{q+2p}{(\alpha+2)(\alpha+1)}\right\}\frac{1}{a(\alpha+3)}A,\end{aligned}$$

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and then

$$D = \frac{\alpha+2}{(\alpha+4)(\alpha+1)}\left\{\frac{q\alpha+6q+4q+p}{(\alpha+2)(\alpha+1)(\alpha+3)} - \frac{(\beta^2+2)(\alpha+1)(\beta^2+p)}{(\alpha+2)(\alpha+1)\alpha(\alpha+3)}\right\}A,$$

so that the whole expression contains the factor $\alpha+1$. But observe that in the present case, if it is allowable to write $D=0$, the next coefficient E (depending on the assumption $D=0$) will go to infinity; that is, instead of assuming $B=0$, we assume $D=$ an arbitrary quantity D', then E and the subsequent coefficients will contain terms depending on D', and the complete form of the series belonging to $\alpha=-1$ will be

$$y = A\left(x^\alpha + \frac{B}{A}x^{\alpha-1} + \frac{C}{A}x^{\alpha-2} + \frac{D}{A}x^{\alpha-3} + \&c.\right) + D'\left(x^{\alpha-3} + \frac{E'}{D'}x^{\alpha-4} + \&c.\right),$$

where the second member in fact is the series belonging to $\alpha=2$. It is hardly necessary to remark that the solution thus obtained can be expressed by means of exponentials, viz. that the solution is

$$y = A\left(\frac{1}{x} - \frac{1}{2}q\right)e^{-px} + D'\frac{1}{x^q}\left(\frac{1}{x} + \frac{1}{2}q\right)e^{px}.$$

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On the Binomial Theorem, Factorials, and Derivations.

[From the Oxford, Cambridge, and Dublin Messenger of Mathematics, vol. V. (1869), pp. 102–114.]

THE following was part of my course of lectures in the year 1867.

The proof commonly called "Euler's" of the binomial theorem is as follows: the theorem is assumed to be true for positive integer indices; that is, it is assumed that for any positive integer m we have

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{1\cdot 2}x^2 + \&c.$$

This being so, data $(1+x)^m \cdot (1+x)^n = (1+x)^{m+n}$, the equation

$$\left[1 + mx + \frac{m(m-1)}{1\cdot 2}x^2 + \&c.\right] \cdot \left[1 + nx + \frac{n(n-1)}{1\cdot 2}x^2 + \&c.\right]$$

$$= 1 + (m+n)x + \frac{(m+n)(m+n-1)}{1\cdot 2}x^2 + \&c.$$

is true for any positive integer values whatever of m and n; the equation is therefore true identically; and it is consequently true for all values whatever of m and n. In an m^{th} power, $a + x^n$ suppose; that is, we have

$$C^m = 1 + mx + \frac{m(m-1)}{1\cdot 2}x^2 + \&c.$$

where C is a constant; viz. it is independent of m; but the value of C will of course depend upon x; and, in order to determine it, we write $m = 1$, the equation $C = 1 + x$, so that

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{1\cdot 2}x^2 + \&c.,$$

which is the binomial theorem in its general form.

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It is to be observed that there is not in the demonstration any employment of the so called "principle of equivalent forms," we do not from the truth of an equation for positive integer values of m, n, infer the truth of it for any values whatever of m, n; there is the intermediate step, that being true for the integers, it is true identically, and the identical truth of the equation depends on the circumstance that comparing on the two sides of the equation the coefficients of the successive powers of x^a, y^b, etc., these coefficients are in every case finite, rational, and integral functions of m, n.

For instance, comparing the coefficients of x^a, we have

$$m+n = 1\cdot m + 1\cdot n;$$

and any such equation, being true for all positive integer values of m, n, will be true identically; developing the two sides, the equation is in fact

$$m^a + na^{a-1}b + \&c. = a^a + aa^{a-1}m + aa^{a-1}n + \&c.$$

The reasoning is then perfectly good; but I remark that it is quite an easy step from the proof to that of one of the general solution as above explained: we know in fact that the equation

$$(1+x)^{m+n} = (1+x)^m(1+x)^n$$

is for all integer values of m, n, satisfied, the equation

$$\left[1+mx+\frac{m(m-1)}{1\cdot 2}x^2+\&c.\right]\cdot\left[1+nx+\frac{n(n-1)}{1\cdot 2}x^2+\&c.\right]$$

$$= 1+(m+n)x+\frac{(m+n)(m+n-1)}{1\cdot 2}x^2+\&c.$$

To show this I introduce the factorial notation and it is readily found to be

$$[m+n]^p = [m]^p + p[m]^{p-1}\cdot[n]^1 + \frac{p(p-1)}{1\cdot 2}[m]^{p-2}\cdot[n]^2 + \&c.,$$

and I say that, this, the *factorial binomial theorem for a positive integer index* r, is proved as easily and in the same manner as the binomial theorem for a like value of the index; or say in equation

$$(m+n)^r = m^r + rm^{r-1}\cdot n + \frac{r(r-1)}{1\cdot 2}m^{r-2}\cdot n^2 + \&c.$$

To show then that X form the values of $[m+n]^p$, $[m+n]^r$, etc., successively, by what that may be termed the process of factorial multiplication; we have

$$[m+n]^1 = m+n.$$

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to obtain $[m+n]^2$ we have to multiply this by $m+n-1$; it is regarded as the first term as 1 writes the multiplier under the form $(m-1)+n$, and in regard to the second term n, under the form $m+(n-1)$; the process then stands

$$\begin{array}{rl} [m+n]^2 = & m \\ & +n-1 = (m-1)+n, \quad m+(n-1) \\ & m(m-1) \quad mn \\ & +mn \quad n(n-1) \\ & = [m]^2 + 2[m]^1[n]^1 + [n]^2. \end{array}$$

To form $[m+n]^3$ we have to multiply by $m+n-2$; in regard to the first term, this is written under the form $(m-2)+n$; in regard to the second term under the form $(m-1)+(n-1)$; and in regard to the third term under the form $m+(n-2)$; the product is thus obtained in the form

$$\begin{array}{rcl} & [m]^2 \cdot [m-2] = & [m]^3 \\ +2[m]^1[n]^1 & (m-1) & +2[m]^2[n]^1 \\ +[n]^2 \cdot & m & +[m]^1[n]^2 \\ +[m]^2 & (n-1) & +[m]^2[n]^1 \\ +2[m]^1[n]^1 & (n-1) & +2[m]^1[n]^2 \\ +[n]^2 & (n-2) & +[n]^3 \\ \multicolumn{3}{r}{= [m]^3 + 3[m]^2[n]^1 + 3[m]^1[n]^2 + [n]^3;} \end{array}$$

and so on; the law of the terms is obvious; and the numerical coefficients are in effect obtained as follows:

$$\begin{array}{ccccc} & & 1, & & \\ & 1, & & 1, & \\ & & 1, & 1, & 1, \\ 1, & 1, & & & \\ & 1, & 2, & 1, & 1, \\ & & 1, & 2, & 1, \\ 1, & 3, & 3, & 1, & \\ \&c. & & & & \end{array}$$

viz. by precisely the same process as is used in finding the numerical coefficients in the powers $(m+n)^1$, $(m+n)^2$, $(m+n)^3$, &c.; and we thus see that for any integer value r of the index, we have a factorial binomial theorem, wherein the numerical coefficients are the same as in the binomial theorem for the same index.

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But the method of varied multiplication may be applied to the demonstration of a much more general theorem; viz. we may use it to develope a product such as

$$(\lambda - a)(\lambda - b)(\lambda - c)(\lambda - d)(\lambda - e),$$

according to a series of products such as

$$(\lambda - a)(\lambda - \beta) \cdot (\lambda - \gamma)(\lambda - \delta)(\lambda - \epsilon),$$

$$(\lambda - a)(\lambda - \beta) \cdot (\lambda - \gamma)(\lambda - \delta)(\lambda - \epsilon - 2),$$

$$(\lambda - a)(\lambda - \beta) \cdot (\lambda - \gamma)(\lambda - \delta)(\lambda - \epsilon - 3),$$

&c.

For this purpose, starting with

$$\lambda - a = \lambda - a + a - a,$$

we multiply by $\lambda - b$, written under the form $\lambda - \alpha + a - \beta$, and then under the form $\lambda - a + (a - b)$; we have thus

$$\begin{aligned}(\lambda - a)(\lambda - b) &= (\lambda - \alpha)(\lambda - \beta) \\ &\quad + (a - \alpha) \qquad (a - a + \beta - b) \\ &= 1 \qquad\qquad (a - a)(a - b).\end{aligned}$$

and so on. It is easy to see that we may by instance write

$$\begin{aligned}(\lambda - a)(\lambda - b)(\lambda - c)(\lambda - d) &= (\lambda - a)(\lambda - \beta)(\lambda - \gamma)(\lambda - \delta) \\ &\quad + (a - a) \cdot (\lambda - \alpha)(\lambda - \beta)(\lambda - \gamma) \\ &\quad + (b - a)(b - \beta) \cdot (\lambda - a)(\lambda - \beta)(\lambda - \gamma) \\ &\quad + (a - a)(b - \beta) \cdot (\lambda - a)(\lambda - \beta) \\ &\quad + (c - a)(c - \beta)(c - \gamma) \cdot (\lambda - a) \\ &\quad + (a - \alpha)(b - \beta)(c - \gamma) \cdot (\lambda - a) \\ &\quad \ldots \\ &\quad + 1.\end{aligned}$$

[533] (CONTINUATION)

On Factorials and Derivations.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. (1871), pp. 219–229.]

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where the symbols $(\ )_k$ denote sums of products of the differences of an upper letter and a lower letter, θ factors in each product; and the several sums being formed as follows:

$$\text{For } \begin{pmatrix} \alpha,\ \beta,\ \gamma \\ a,\ b,\ c,\ d,\ e \end{pmatrix}, \text{ write } \alpha : \quad \begin{array}{ll} \beta & a \\ \gamma & b \\ & c \\ & d \\ & e; \end{array}$$

the expression is

$$(x-a)+(\beta-b)+(\gamma-c)+(\delta-d)+(\varepsilon-e)$$

$$\text{For } \begin{pmatrix} \alpha,\ \beta,\ \gamma \\ a,\ b,\ c,\ d,\ e \end{pmatrix}, \text{ write } \quad \begin{array}{ll} \alpha,\ \alpha, & b \\ \alpha,\ \beta, & c \\ \alpha,\ \gamma, & d \\ \beta,\ \beta, & e \\ \beta,\ \gamma, & \\ \gamma,\ \gamma, & \end{array}$$

viz. the expression is

$$(x-a)(x-b)+(x-a)(\beta-c)+\cdots+(\gamma-d)(\gamma-e)$$

$$\text{For } \begin{pmatrix} \alpha,\ \beta,\ \gamma \\ a,\ b,\ c,\ d,\ e \end{pmatrix}, \text{ write } \quad \begin{array}{ll} \alpha,\ \alpha,\ \alpha, & c \\ \alpha,\ \alpha,\ \beta, & d \\ \alpha,\ \alpha,\ \gamma, & e \\ \alpha,\ \beta,\ \beta, & \\ \alpha,\ \beta,\ \gamma, & \\ \alpha,\ \gamma,\ \gamma, & \\ \beta,\ \beta,\ \beta, & \\ \beta,\ \beta,\ \gamma, & \\ \beta,\ \gamma,\ \gamma, & \\ \gamma,\ \gamma,\ \gamma, & \end{array}$$

viz. the expression is

$$(x-a)(x-b)(x-c)+\cdots+(\gamma-c)(\gamma-d)(\gamma-e).$$

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For $\begin{pmatrix} \alpha,\ \beta \\ a,\ b,\ c,\ d,\ e \end{pmatrix}$, write $a,\ a,\ a,\ a, \quad a,\ b,\ c,\ d.$

viz. the expression is

$$(x-a)(x-b)(x-c)(x-d) + \cdots + (\beta-b)(\beta-c)(\beta-d)(\beta-e).$$

Finally for $\begin{pmatrix} \alpha \\ a,\ b,\ c,\ d,\ e \end{pmatrix}$, write $a,\ a,\ a,\ a,\ a, \quad a,\ b,\ c,\ d,\ e.$

viz. the expression is

$$(x-a)(x-b)(x-c)(x-d)(x-e),$$

which explains the law of the formation of the several coefficients. It is to be observed that in forming the development of any symbol, for instance $\begin{pmatrix} \alpha,\ \beta,\ \gamma \\ a,\ b,\ c,\ d,\ e \end{pmatrix}$, the first column contains the homogeneous products, θ together, of α, β, γ; the second column the combinations (that is, combinations without repetitions) θ together of a, b, c, d, e; the top line is a, a, c; and to form the subsequent lines we must for any advance a into β, &c. of a Greek letter make the like advance a into b, b into c, c into d, of the corresponding latin letter.

Two particular cases of the theorem may be noticed; if the latin letters all vanish, we have, for example,

$$\begin{aligned} \text{if } a = {} & (x-\alpha)(x-\beta)(x-\gamma)(x-\delta)(x-\varepsilon), \\ & + (b-a)(b-\beta)(b-\gamma)(b-\delta)\cdot H_1(\alpha,\ \beta,\ \gamma,\ \delta,\ \varepsilon) \\ & + (c-a)(c-\beta)(c-\gamma)\cdot H_2(\alpha,\ \beta,\ \gamma,\ \delta,\ \varepsilon) \\ & + (d-a)(d-\beta)\cdot H_3(\alpha,\ \beta) \\ & + (e-a)\cdot H_4(\alpha) \\ & + 1\cdot H_5(\alpha) \end{aligned}$$

where the symbols H denote the sum of the homogeneous products of the annexed letters, taken together according to the suffix number: the last coefficient $H_5(\alpha)$ is of course $= \alpha^5$. And if the greek letters all vanish, then we have in like manner

$$\begin{aligned} (b-a)(c-a)(d-a)(e-a) = {} & b^4 \\ & - b^3(a+b+c+d+e) \\ & + b^2(ab+\cdots+de) \\ & - b(abc+\cdots+cde) \\ & + abcde. \end{aligned}$$

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where the symbols C denote the combinations of the annexed letters taken together according to the suffix number; the last coefficient $C_3(a,\ b,\ c,\ d,\ e)$ is of course $= abcde$. This is the ordinary theorem giving the expression of an equation in terms of its roots.

Confining the two theorems, if in the first theorem we express the products $(x-a)(x-\beta)(x-\gamma)(x-\delta)\cdots+a$, &c. in powers of X by means of the second theorem; or if in the second theorem we express the powers b^4, b^3, &c. in terms of the products $(b-a)(b-b)(b-c)+(b-c)(b-d)$, &c. by means of the first theorem; then in either case we obtain certain identical relations connecting the C, H of $(\alpha,\ \beta,\ \ldots)$ or of $(a,\ b,\ \ldots)$.

I have mentioned the factorial notation

$$[n]^r = n(n-1)\cdots(n-r+1),$$

where r is a positive integer; a consequence of this is

$$[n]^{r+s} = [n]^r[n-r]^s,$$

where r and s are positive integers; or as this may also be written

$$[n+r]^{r+s} = [n+r]^r[n]^s.$$

Assuming this to subsist for $s=0$, or a negative integer; first for $s=0$, we have $[n+r]^r = [n+r]^r[n]^0$; that is, $[n]^0=1$, and we have for $s=-r$, we have $1=[n+r]^r[n]^{-r}$, that is,

$$[n]^{-r} = \frac{1}{[n+r]^r},$$

and in particular, $r=1,2,$&c., we have

$$[n]^{-1} = \frac{1}{n+1},$$

$$[n]^{-2} = \frac{1}{(n+1)(n+2)},$$

&c., which explains the extension of the factorial notation to negative integer values of the index.

But the equation

$$[n]^{r+s} = [n]^r[n-r]^s$$

does not in any determinate manner lead to an extension of the factorial notation to fractional or other values of the index. In fact, assuming $[n]^a = \frac{\Pi a}{\Pi(n-a)}$ where Π is an arbitrary fractional symbol, the equation in question becomes

$$\frac{\Pi a}{\Pi(n-r-s)} = \frac{\Pi n}{\Pi(n-r)}\cdot\frac{\Pi(n-r)}{\Pi(n-r-s)};$$

viz. the original equation is identically satisfied, without any condition whatever being imposed upon the function Π, and on this account we have not, in the notion of the factorial with an integer index, *any sufficient basis for a theory of general differentiation.*

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A product

$$n(n-r)\cdots(n-(r-1)\nu)$$

can of course be expressed in the factorial notation, viz. it is

$$=\nu^r\left[\frac{n}{\nu}\right]^r,$$

and on this account it is not in general necessary to employ a notation such as $[n,\,\nu]^r$ to denote such a factorial wherein the difference of the successive factors instead of being $=-1$ is $=-\nu$; in particular cases where factorials of the kind in question are used, it may be convenient to employ such a notation. In particular it is some- times convenient to use the notation $[n,\,-1]^r$ or better $[n]^r$ to denote the product

$$n(n+1)\cdots(n+r-1),$$

where the successive factors instead of being diminished, are increased by unity. It may be noticed, that reverting to this last product the order of the factors, we find

$$[n]^r=[n+r-1]^r;$$

a somewhat similar formula, but employing only the ordinary factorial notation, is obtained from the equation

$$[n]^r=n(n-1)\cdots(n-r+1)$$

by first changing the sign of n and then reversing the order of the factors; viz. we have

$$[-n]^r=(-)^r[n+r-1]^r.$$

Reverting to the process used for the development of the expression $(\ )_k$ where there are two columns, the one of greek, the other of latin letters; it is to be remarked that although the order in which the successive lines are evolved is not material for the purpose of the theorem, yet that a certain definite order of evolution, has been made use of; thus in regard to $\begin{pmatrix}\alpha,\,\beta,\,\gamma\\ a,\,b,\,c,\,d,\,e\end{pmatrix}_3$, the column of greek letters, giving the homogeneous products of the second order in $(\alpha,\,\beta,\,\gamma)$, was

$$\begin{array}{l} a\,a\\ a\,\beta\\ a\,\gamma\\ \beta\,\beta\\ \beta\,\gamma\\ \gamma\,\gamma\end{array}$$

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this is evolved from the top term (α, α) by a process given implicitly in Arbogast's Calculus of Derivations, and which may be termed the *rule of the last and the last but one.* Let the direction "operate on any letter," be understood to mean that the letter in question is to be changed into that which immediately follows it, but in such wise that when the letter occurs more than once, e.g. as in α, α the operation affects only the letter in the right-hand place. Then operate on the α, α in regard to α, we obtain α, β; operate on this in regard to β, we obtain α, γ; and in regard to α, we obtain β, β. Again we operate on α, γ in regard to γ, and obtain a, δ; we do not operate on it in regard to α for the reason that α is not the letter immediately preceding γ. Operate on β, β in regard to β, we obtain β, γ. The next step, if the series extended to ε would be to operate on α, δ in regard to δ, obtaining α, ε; on α, δ. Passing then to the next term β, γ, we operate in regard to γ, obtaining β, δ, and since β is the letter immediately preceding γ, we also operate in regard to β, obtaining γ, γ. Similarly if ε were admissible, β, δ would give β, ε, but it in fact gives nothing; γ, γ gives γ, δ; thus if ε were admissible would give γ, ε and δ, δ, but it in fact gives only δ, δ, and, ε being inadmissible, the process is here concluded. The rule is, operate on the last but one letter, and when the last but one letter is that which, in alphabetical order, immediately precedes the last letter (but in this case only) operate on the last but one letter.

Taking another example, but with numbers instead of letters, and supposing the highest admissible number to be 5, then from 111 we derive as follows:

111	112	113	114	115	125	135	145	155	255	355	455	535
		122	123	124	134	144	225	245	345	445		
			222	133	224	225	244	355	444			
				223	233	234	334	344				
						333						

the original single column being here for greater convenience broken up into distinct columns; but the order of the terms, when the columns are taken one after the other in order, each being read from the top to the bottom, being the same as before; it will be noticed that the successive divisions are the partitions into 3 parts (no part exceeding 5) of the numbers 3, 4, …, 15 respectively; the partitions being in each case obtained without repetition, and those of the same number being given, say in their numerical order (corresponding with the alphabetical order when letters are employed). It is necessary to show that the partitions will be obtained without repetitions; and that all the partitions will be obtained; for this purpose consider, for example, the partitions of 9; any one of these is either a partition 135 where the last number 5 is not a repeated number; and to this case there is a partition of 8, viz. 134, from which operating on the last we obtain 135; but there is no other partition of 8 which would give 135, the only such partition would be 125, but here, as 2 is not the number which immediately precedes 5, there is no operation on the last but one, and we do not from it obtain 135. Or else a partition of 9 is of the form 144

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where the last letter is repeated; there exists in this case no partition of 8, such that operating on the last we obtain from it 144, but there does exist a partition of 8, viz. 134, such that operating on the last but one we obtain from it 144. That is, the partition 135 must be obtained from 134, and only from 134, and the partition 144 must also be obtained from 134, and only from 134, and that without repetitions, the entire series of the partitions of 9; and so in general.

Translating the example into letters, but using for greater convenience α, &c. instead of a, α, &c. the process will be precisely the same; taking the letters to be α, β, γ, δ, ε, we have

$$\begin{array}{ccccccccccccc}
\alpha\alpha\alpha & \alpha\alpha\beta & \alpha\alpha\gamma & \alpha\alpha\delta & \alpha\alpha\varepsilon & \alpha\beta\varepsilon & \alpha\gamma\varepsilon & \alpha\delta\varepsilon & \alpha\varepsilon\varepsilon & \beta\varepsilon\varepsilon & \gamma\varepsilon\varepsilon & \delta\varepsilon\varepsilon & \varepsilon\varepsilon\varepsilon \\
 & & \alpha\beta\beta & \alpha\beta\gamma & \alpha\beta\delta & \alpha\gamma\delta & \alpha\delta\delta & \beta\beta\varepsilon & \beta\delta\varepsilon & \gamma\delta\varepsilon & \delta\delta\varepsilon & & \\
 & & & \beta\beta\beta & \alpha\gamma\gamma & \beta\beta\gamma & \beta\beta\delta & \beta\gamma\delta & \gamma\gamma\varepsilon & \delta\delta\delta & & & \\
 & & & & \beta\beta\beta & \beta\gamma\gamma & \beta\gamma\delta & \gamma\gamma\delta & \gamma\delta\delta & & & & \\
 & & & & & & \gamma\gamma\gamma & & & & & &
\end{array}$$

Attributing weights to the several letters, viz. to α, β, γ, δ, ε the weights 1, 2, 3, 4, 5 respectively, the several columns show the terms of the weights 3, 4, ... 15 respectively.

I have said that the foregoing rule is given implicitly in Arbogast's Calculus of Derivations; this calculus includes in fact a process for the expansion of a function

$$\phi(x + hx' + \nu x'' + kx''' + \&c.)$$

in powers of x, the expansion in question may be obtained by means of Taylor's theorem, viz.

$$\phi(x + hx' + \nu x'' + kx''' + \&c.)$$

$$\begin{aligned}
&= \phi x \\
&\quad + \frac{\phi' x}{1}(hx' + \nu x'' + kx''' + \&c.) \\
&\quad + \frac{\phi'' x}{1\cdot 2}(hx' + \nu x'' + kx''' + \&c.)^2 \\
&\quad + \frac{\phi''' x}{1\cdot 2\cdot 3}(hx' + \nu x'' + kx''' + \&c.)^3 \\
&\quad + \&c.
\end{aligned}$$

viz. expanding the several powers of the polynomial increment, and arranging in powers of x, this is

$$\begin{aligned}
&= \phi x \\
&\quad + x'\phi' x \\
&\quad + x''\left[\phi' x + x'\phi'' x\right] \\
&\quad + x'''\left[\phi' x + x'x''\phi'' x + \frac{1}{6}x'^3\phi''' x\right] \\
&\quad + x''''\left[\phi' x + (x'x''' + \tfrac{1}{2}x''^2)\phi'' x + \tfrac{1}{2}x'^2x''\phi''' x + \tfrac{1}{24}x'^4\phi'''' x\right] \\
&\quad + \&c.
\end{aligned}$$

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but the object of the rule is to obtain this last-mentioned result directly, and under- standing "operate in regard to a letter" to mean differentiate with regard to this letter and integrate with regard to the next succeeding letter, then the coefficients of the successive powers of x are all obtained from the first coefficient ϕx, by operating thereon according to the rule of "the last and the last but one"; thus ϕx, operating on α gives $\phi' x$, this operating in regard to α gives $\phi' x$, and in regard to α gives $\phi'' x \cdot \dfrac{x'}{1}$; the term $\phi'' x \cdot x'$ is to be operated upon in regard to α only, and is given $\phi'' x \cdot x''$; the other terms $\phi'' x \cdot \dfrac{x'^2}{2}$ operated on in regard to α gives $\phi'' x \cdot \nu x$, and in regard to α gives $\phi'' x \cdot \dfrac{x'^2}{2}$, and so on. But attending only to the literal parts, the terms, for instance b', $b\nu$, νb, &c., which present themselves in the formula, are the homogeneous terms derived from ν^3, by the rule, as originally stated, with a view to the derivation of such terms.

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A "Smith's Prize" Paper(†); Solutions.

[From the Oxford, Cambridge and Dublin Messenger of Mathematics, vol. V. *(1870), pp. 182–203.]*

1. Mention what forms of plane relation $\phi(x,\, b,\, c,\, \ldots) = 0$ between the roots of a plane equation will in general serve for the rational determination of the roots; explain the case of failure; and state what information as to the roots is furnished by a plane relation not of the form in question.

In the given relation $\phi(x,\, b,\, c,\, \ldots) = 0$ must be a wholly unsymmetrical function of the roots; that is, a function altered by any permutation whatever of the roots; or, what is the same thing, by any single interchange of two roots.

For this being so, if a, β, γ, δ, and the corresponding four values of a, b, c, ... satisfied by writing therein $a = a$, $b = \beta$, $c = \gamma$, &c., it will in general be satisfied also by writing $a = \beta$, $b = a$, $c = \gamma$, &c.; but the equation must uniquely determine each of the roots a, b, ..., respectively, in a separate question which is not here considered.

The function $\phi(a,\, b,\, c,\, \ldots)$ may be of the proper form, and yet, for particular values a, β, γ, ..., be such that the given relation $\phi(a,\, b,\, c,\, \ldots) = 0$ is satisfied, not only for the single arrangement $a = a$, $b = \beta$, $c = \gamma$, &c., but for some other arrangement,

†Set by me for the Master of Trinity, Feb. 5, 1870.

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$a = \delta$, $b = \gamma$, $c = \beta$, $\ldots$ or for more than one such other arrangement. (For instance, if the given relation be $a + b + c - 2c = 0$, and the roots are 3, 5, 7; the relation is satisfied by $a = 5$, $b = 3$, $c = 7$, and also by $a = 3$, $b = 7$, $c = 5$.) Here the given equation and relation do not completely determine each root, they only determine that a is $= a$ or $= \delta$ (or as the case may be $=$ some other one value); and similarly that b is $= \beta$ or $= \gamma$ (or as the case may be $=$ some other one value), and so for the other roots c, d, $\ldots$; and it then appears *a priori*, that in such a case each root is determined, not rationally, but by means of an equation, the order of which is equal to the number of the values of such root; we have here the case of failure of the general theorem.

When the given relation $\phi(a, b, c, \ldots) = 0$ is not of the required form; that is, when $\phi(a, b, c, \ldots)$ is a partially symmetrical function, there will be in general several arrangements of a, β, γ, $\ldots$, such that equating $\phi(a, b, c, \ldots) = 0$ will be satisfied; for each of these arrangements, the given relation $\phi(a, b, c, \ldots) = 0$ will be satisfied; and it follows that each of the roots a, b, c, $\ldots$ is determined not rationally, but by means of an equation of a certain order (not always quadratic equation for the several values of each of the several arrangements, and as the same value, viz., that is, if it be satisfied, suppose by $a = \alpha$, $b = \beta$, $c = \gamma$, $\ldots$, it will also be satisfied by $a = \beta$, $b = \alpha$, $c = \gamma$, $\ldots$, but not in general in any other manner; each of the roots a, b has here either of the values a, β, and the two roots a, b in question will be given, not rationally, but by means of the same quadratic equation. And observe, moreover, that any other function $\psi(a, b, c, \ldots)$ of the same form as ϕ, that is, symmetrical in regard to the two roots a, b, will have for its two arrangements $a = \alpha$, $b = \beta$, and $a = \beta$, $b = a$, the same value, $c = \gamma$, $\ldots$ acquire not two distinct values, but one and the same value, the value of $\psi(\alpha, \beta, c, \ldots)$ will be determined *rationally*; and so in general.

There is for the partially symmetrical function $\phi(a, b, c, \ldots)$ a case of failure similar to that which arises for the completely unsymmetrical function, viz. the particular values a, β, γ, $\ldots$ may be such as to give more ways of satisfying the given relation $\phi(a, b, c, \ldots) = 0$, than there would be in general for partially symmetrical values of a, β, γ, $\ldots$ and there is a corresponding elevation of the order of the equation for the determination of the roots a, b, c, $\ldots$ or some of them.

2. If the roots of $(a, \beta, \gamma, \delta)$ of the equation

$$(a, b, c, d, e)(x, 1)^4 = 0$$

are no two of them equal; and if there exist unequal magnitudes θ, ϕ such that

$$(\theta + \alpha)^4 : (\theta + \beta)^4 : (\theta + \gamma)^4 : (\theta + \delta)^4 = (\phi + \alpha)^4 : (\phi + \beta)^4 : (\phi + \gamma)^4 : (\phi + \delta)^4;$$

show that the cubinvariant ace $-ad^2 - b^2e - c^3 + 2bcd$ is $= 0$; and find the values of θ, ϕ.

We have

$$\left(\frac{\theta + \alpha}{\phi + \alpha}\right)^4 = \left(\frac{\theta + \beta}{\phi + \beta}\right)^4 = \left(\frac{\theta + \gamma}{\phi + \gamma}\right)^4 = \left(\frac{\theta + \delta}{\phi + \delta}\right)^4;$$

and we cannot have any two of the fourth roots, say $\dfrac{\theta + \alpha}{\phi + \alpha}$ and $\dfrac{\theta + \beta}{\phi + \beta}$ equal to each other; for this would imply $(\theta - \phi)(a - \beta) = 0$, that is, $\theta = \phi$ or else $\alpha = \beta$.

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Hence assuming $\dfrac{\theta+\alpha}{\phi+\alpha}=a_1$, $\dfrac{\theta+\beta}{\phi+\beta}=a_2$, $\dfrac{\theta+\gamma}{\phi+\gamma}=a_3$, $\dfrac{\theta+\delta}{\phi+\delta}=a_4$, viz. this is one of three notations of equations; the other two may be obtained from a_4 by writing a_3, a_2, a_1, or by interchanging some of the roots, a_1, a_2, a_3, a_4 are in fact the imaginary fourth roots of unity, and the corresponding four values of $(\alpha,\ \beta,\ \gamma,\ \delta)$ is therefore equal to four values a, b, c, d. We have therefore

$$\frac{\theta+\alpha}{\phi+\alpha}=-1,\quad \frac{\theta+\beta}{\phi+\beta}=+1$$
$$\frac{\theta+\gamma}{\phi+\gamma}=-\sqrt{-1},\quad \frac{\theta+\delta}{\phi+\delta}=+\sqrt{-1}$$

which are different cases, related to each other; the mathematic ratio of $(\alpha,\ \beta,\ \gamma,\ \delta)$ is therefore equal to that of $(1,\ -1,\ \sqrt{-1},\ -\sqrt{-1})$;

That is,

$$(\alpha-\gamma)(\beta-\delta)=(-1-\sqrt{-1})(1+\sqrt{-1})$$

or, what is the same thing,

$$2(\alpha\beta+\gamma\delta)=(\alpha+\beta)(\gamma+\delta);$$

viz. we have this relation, or one of the like relations

$$2(\alpha\gamma+\beta\delta)=(\alpha+\gamma)(\beta+\delta),$$

$$2(\alpha\delta+\beta\gamma)=(\alpha+\delta)(\beta+\gamma),$$

that is, the product of the three functions $2(\alpha\beta+\gamma\delta)-(\alpha+\beta)(\gamma+\delta)$, &c. is $=0$.

But the product of these is the cubinvariant of the quartic function $(a,\ b,\ c,\ d,\ e)$. viz. we have this relation, or one of the two following: $J=0$,

$$J=ace-ad^2-b^2e-c^3+2bcd,$$

which is evidently a transcendental curve; it may however be shown that, if

$$x^4+y^4=1$$

and

$$x(1+x^2)+y(1+y^2)-2xy(1+x+y)(1+x+y)=0,$$

that is, if we have the equation $\dfrac{\theta+\alpha}{\phi+\alpha}=-1$, etc. then there is a relation in y which gives $J=0$. By the consideration of the form $-r^4-r^4-r^4-r^4+r^4+r^4+r^4+r^4=0$, which is the form $J=0$ as it may by any means easy to prove that this is so. The question depends on the form of the curve defined by the equation

$$4y^4-4z(z+by^2+cy^2)=0.$$

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That is,

$$2b\phi + 2a\theta - (\theta + \phi)(a + \beta) = 0,$$

or, what is the same thing,

$$2b\phi + 2a\theta - (\theta + \phi)(\gamma + \delta) = 0;$$

viz. we have thus the values of $\theta\phi$, $\theta + \phi$ each given in terms of (a, β) corresponding to the relation $2(a\beta + \gamma\delta) = (a + \beta)(\gamma + \delta)$; and there are by cyclically permuting X, Y, Z as before, we have the values of $\theta\phi$, $\theta + \phi$ corresponding to the other two forms respectively of the relation between the roots.

3. If on a plane A, B, C, D are fixed points and P a variable point, find the locus relations

$$\alpha \cdot PAB + \beta \cdot PBC + \gamma \cdot PCD + \delta \cdot PDA = 0$$

which connects the areas of the triangles PAB, PBC, PCD, PDA.

Taking (x, y), (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4) for the coordinates of P, A, B, C, D respectively, we have

$$PAB = \frac{1}{2}\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix}, \quad \text{etc.,} \quad \text{suppose,}$$

$$PBC = 023, \ \&c.$$

(where the values of these numerical determinants do depend the signs of the several triangles). The identical equation is

$$\alpha \cdot 012 + \beta \cdot 023 + \gamma \cdot 034 + \delta \cdot 041 = 0$$

(that such an equation exists appears at once by the consideration that α, β, γ, δ can be determined so that the coefficients of x, y, and the constant term shall severally vanish); and in order actually to find the values we may take P coincident with the points A, B, C, D respectively. We have thus

$$\begin{aligned}
\beta \cdot 123 + \gamma \cdot 134 + \delta \cdot 141 &= 0, \\
\gamma \cdot 234 + \delta \cdot 241 + \alpha \cdot 212 &= 0, \\
\delta \cdot 341 + \alpha \cdot 312 + \beta \cdot 323 &= 0, \\
\alpha \cdot 412 + \beta \cdot 423 + \gamma \cdot 434 &= 0,
\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}
\beta \cdot 123 + \gamma \cdot 134 + \delta \cdot 141 &= 0, \\
\gamma \cdot 234 + \delta \cdot 241 + \alpha \cdot 212 &= 0, \\
\delta \cdot 341 + \alpha \cdot 312 + \beta \cdot 323 &= 0, \\
\alpha \cdot 412 + \beta \cdot 423 + \gamma \cdot 434 &= 0.
\end{aligned}$$

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and these are at once seen to give

$$\alpha : \beta : \gamma : \delta = 234 : 341 : 412 : 123 = -341 : 412 : -123 : 234,$$

so that, the required identical relation is

$$234 \cdot 012 - 341 \cdot 023 + 412 \cdot 034 - 123 \cdot 041 = 0,$$

or, what is the same thing,

$$PAB \cdot BCD - PBC \cdot CDA + PCD \cdot DAB - PDA \cdot ABC = 0,$$

which signifies that this signs being attended to the signs of the triangles PAB, PCD, PDA, BCD, CDA, DAB, ABC, are taken with respect to the hexagon of the point.

4. Find at any point of a plane curve the angle between the normal and the line drawn from the point to the centre of any one of the two circles of curvature.

Excursus whether a like question applies to a point on a surface and the indicatrix section at such point.

Taking the origin at the point on the curve, and considering as the axis of x the tangent and that of y with the normal, the equation of the curve is

$$y = bx^2 + cx^3 + \ldots,$$

and if, considering x as small quantity of the first order, and therefore y as a small quantity of the second order, we regard y as given, and find the two values x_1, x_2 such of the order $\sqrt{y}$, which satisfy the equation, then, as will appear, $x_1 + x_2$, is a small quantity of the order $\sqrt{y}$, and consequently $\frac{1}{2}(x_1 + x_2)$ is of the order y, that is if ϕ be the required angle, then obviously $\tan\phi = \frac{1}{2y}(x_1 + x_2)$.

Taking the origin at the point on the curve, the original equation is the usual approximation $bx^2 = y$, $cx^3 = y\left(1 - \frac{x}{b}\right)$, whence $x = \sqrt{\frac{y}{b}}\left(1 - \frac{c}{2b}\frac{x}{b}\right)$, $cx^3 = \frac{1}{2}y$, &c., so we have

$$x_1 = \sqrt{\frac{y}{b}} - \frac{cy}{2b^2},$$

$$x_2 = -\sqrt{\frac{y}{b}} - \frac{cy}{2b^2},$$

and thence

$$\tfrac{1}{2}(x_1 + x_2) = -\frac{cy}{2b^2},$$

which gives the value of the angle, viz. obviously $\tan\phi = -\frac{c}{b^2}$; which gives the value of the angle ϕ; it would be easy to express b, c in terms of the differential coefficients

$$\frac{dx}{dy}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}.$$

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It would at first sight appear that a like question might be asked as to a surface; viz. that it might be proposed to determine the angle between the normal and a line drawn from the point to the centre of the indicatrix conic. But this is not so; in fact, taking the origin at a point on the surface, the axis of z being in the tangent plane, and that of z coinciding with the normal; then for the third order we have

$$z = (A,\, B,\, C \breve{a}\, x,\, y)^2 + (a,\, b,\, c,\, d \breve{a}\, x,\, y)^3;$$

but here, regarding z as a given constant, if we take account of the terms of the third order, the section is not a conic but a cubic; and it has not in general any centre; and if (as in the ordinary theory) we neglect the terms of the third order, thus obtaining an indicatrix conic, the centre of this conic lies on the normal, and there is no angle corresponding to the angle ϕ of the plane problem.

The only case where there is no objection is when the cubic terms $(a,\, b,\, c,\, d \breve{a}\, x,\, y)^3$ contain as a factor the quadric terms $(A,\, B,\, C \breve{a}\, x,\, y)^2$ (one relation between the coefficients a, b, c, d of the cubic function depending on this relation), and in such case the indicatrix section corresponds to the surface plane $z =$ (a coincident section: in this case may be defined as a section by a plane parallel to the tangent plane, but at any rate close to it as a coincident plane. To complete the analogy, (2) the two functions depend in any manner whatever on one and the same set of constants.

5. If V, V' are binary functions of the form $(\alpha,\, b,\, \ldots \breve{a}\, x,\, 1)^n$ with arbitrary coefficients, and if the equation $U = 0$, $V = 0$ have a common root, show how this can be determined in terms of the derived functions of the Resultant in regard to the coefficients of either function.

Show what results in regard to the common root can be obtained when the coefficients are not all of them arbitrary but (1) each or either of the functions depends in any manner whatever on a set of arbitrary coefficients not entering into the other function, (2) the two functions depend in any manner whatever on one and the same set of arbitrary coefficients.

Here is the theory analytical at once, *instead of the two equations, there is a single equation* $U = 0$ *having a double root.*

Suppose

$$U = (a,\, b,\, \ldots \breve{a}\, x,\, 1)^m, \quad \left(= ax^m + \frac{m}{1} bx^{m-1} + \&c.\right),$$

$$V = (a,\, b,\, \ldots \breve{a}\, x,\, 1)^n, \quad \left(= ax^n + \frac{n}{1} bx^{n-1} + \&c.\right).$$

Then if R is the resultant, the equation $R = 0$ is the relation which must exist between the coefficients in order that the two functions U and V shall have a common root. Moreover, if we assume that U and V may have a common root $x = \alpha$, then we have $U_\alpha = 0$, $V_\alpha = 0$; the coefficients of U_α, V_α being the same as those of U, V, and $x = \alpha$. We have then identically $U_\alpha = 0$, $V_\alpha = 0$, $R = 0$; that is, these equations have to be infinitesimally varied in such manner that U, V have still a common root $x = \alpha$, we have the equation $U = 0$, $V = 0$ subsist when a, b, …, the coefficients of U, and a, b, … those of V undergoes, the given relation must be a common root, a this implies between U_α, V_α, R_α &c.

$$da \frac{dR}{da} + db \frac{dR}{db} + \&c. = 0.$$

But the common factor $x = \alpha y$ is a factor of the unaltered equation $U = 0$, and the values of $(x,\, y)$ are thus unaltered, viz. the equation $U = 0$ is satisfied with the original values of $(x,\, y)$; that is, changing the α as a small quantity of the first order, we have

$$x^m da + mx^{m-1} y\, db + \&c. = 0,$$

or, what is the same thing,

$$x^m\sigma + mx^{m-1}y\sigma' + \&c. = 0$$

an equation which must agree with the former one, that is, we have

$$x^m : mx^{m-1}y : \cdots = \frac{dR}{da} : \frac{dR}{db} : \cdots ,$$

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5. Show that a cubic surface has at most four conic points; and a quartic surface at most sixteen conical points.

If a cubic surface has two conical points, then the line joining these has with the surface two intersections at each of the conical points, and therefore lies wholly in the surface. Hence, for a cubic surface with three conical points A, B, C, the lines AB, BC, CA lie wholly in the surface, and three more lines, viz. ones formed by the lines through three conical points A, B, C, D; the line AB, &c. wholly in the surface, and these lines form among them the complete section of the surface by a plane wholly through the conic; this is the case of complete section of the surface, and the cubic surface is in fact equal to the cubic surface as is well known the cubic surface contains four points E, and the line joining E with any of the four conical points has, by the line, three points of contact with the surface, and is therefore not a tangent line. Hence the section of a cubic surface by a plane through three conical points A, B, C, would be the triangle ABC taken twice, and a line which is the locus of points E; but each of the points E, E_1 in particular, would have to be conical points, and the cubic surface would have 5, that is, more than four conical points, which is impossible.

6. *Find the differential equation of the parallel surfaces of an ellipsoid.*

Let $(x,\, y,\, z)$ be the coordinates of a point on the ellipsoid $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}-1=0$; $(X,\, Y,\, Z)$ the coordinates of a point on the normal at a distance $= k$ from the foot- mentioned point. We have

$$\frac{X-x}{\frac{x}{a^2}}=\frac{Y-y}{\frac{y}{b^2}}=\frac{Z-z}{\frac{z}{c^2}}=\mu \quad \text{suppose;}$$

that is,

$$X=x\left(1+\frac{\mu}{a^2}\right),\quad Y=y\left(1+\frac{\mu}{b^2}\right),\quad Z=z\left(1+\frac{\mu}{c^2}\right);$$

and thence

$$k^2=\mu^2\left\{\frac{x^2}{a^4}+\frac{y^2}{b^4}+\frac{z^2}{c^4}\right\}.$$

Moreover

$$\frac{X^2}{(a^2+\mu)^2}+\frac{Y^2}{(b^2+\mu)^2}+\frac{Z^2}{(c^2+\mu)^2}=\frac{x^2}{a^4}+\frac{y^2}{b^4}+\frac{z^2}{c^4};$$

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substituting these values in the equation of the ellipsoid, we have

$$1 = \frac{a^2X^2}{(a^2+\mu)^2} + \frac{b^2Y^2}{(b^2+\mu)^2} + \frac{c^2Z^2}{(c^2+\mu)^2},$$

which determines μ as a function of X, Y, Z. The tangent plane of the ellipsoid at the point (x, y, z) and of the parallel surface at the point (X, Y, Z) are parallel to each other (for what is the same thing, the parallel surface cuts at right angles on the normal of the ellipsoid); we have therefore

$$\frac{X}{a^2}dX + \frac{Y}{b^2}dY + \frac{Z}{c^2}dZ = 0,$$

or substituting for x, y, z their values, we have

$$\frac{XdX}{a^2+\mu} + \frac{YdY}{b^2+\mu} + \frac{ZdZ}{c^2+\mu} = 0,$$

where μ denotes as above a function of (X, Y, Z) given by the equation

$$1 = \frac{a^2X^2}{(a^2+\mu)^2} + \frac{b^2Y^2}{(b^2+\mu)^2} + \frac{c^2Z^2}{(c^2+\mu)^2}.$$

We have thus the differential equation of the parallel surface. It may be remarked, that the integral equation (involving k as the constant of integration), is found by the elimination of μ, x, y, z from the foregoing equations

$$X - x = \mu\frac{x}{a^2}, \quad Y - y = \mu\frac{y}{b^2}, \quad Z - z = \mu\frac{z}{c^2},$$

or, what is the same thing, by the elimination of μ from the equations

$$\frac{X^2}{(a^2+\mu)^2} + \frac{Y^2}{(b^2+\mu)^2} + \frac{Z^2}{(c^2+\mu)^2} = \frac{1}{\mu^2}k^2,$$

$$\frac{a^2X^2}{(a^2+\mu)^2} + \frac{b^2Y^2}{(b^2+\mu)^2} + \frac{c^2Z^2}{(c^2+\mu)^2} = 1;$$

these may be replaced by

$$\frac{X^2}{(a^2+\mu)^2} + \frac{Y^2}{(b^2+\mu)^2} + \frac{Z^2}{(c^2+\mu)^2} = \frac{k^2}{\mu^2},$$

$$\frac{X^2}{a^2+\mu} + \frac{Y^2}{b^2+\mu} + \frac{Z^2}{c^2+\mu} - 1 = 0,$$

or, since here the second equation is the derived equation of the first in regard to the parameter μ, the parallel surface is the envelope of the quadric surface

$$p(\mu) + p(\mu) + p(\mu) - 1 = 0, \quad p(\mu) = \frac{X^2}{a^2+\mu} + \frac{Y^2}{b^2+\mu} + \frac{Z^2}{c^2+\mu}.$$

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7. *Explain wherein consists the peculiarity of the following problem, and solve it by geometrical considerations:—*

Determine the knot circle inclosing three given points.

The peculiarity of the problem is that the variable parameters upon which the circle depends, viz. a, β the coordinates of the centre and ρ its radius, are not all subject to any equations, but only to the inequalities

$$b^2 + (a - x_1)^2 \leq \rho^2, \quad b^2 + (a - x_2)^2 \leq \rho^2, \quad b^2 + (a - x_3)^2 \leq \rho^2,$$

(α_1, β_1), (α_2, β_2), (α_3, β_3), the coordinates of the given points, and the $\leq$ indicates that the points are within or on the circumference of the circle, and the equation $\rho =$ minimum. The parallel surface inclosing within or on the circumference of the circle, and the points A, B, C separately. Then for the values of a, β, the second sides of the (a, β, ρ) the variables of the given points, with the condition that the points A, B, C are within or on the circumference; and then ρ minimum.

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But if a be the acoustic anomaly, then

$$x = a \sin u, \quad y = b \sin u, \quad a = \sqrt{1-e^2} \sin u,$$

and the equations become

$$-\sin u \frac{d^2u}{dt^2} + \cos u \left(\frac{du}{dt}\right)^2 - \frac{\sin u}{a^2} + \frac{\cos u}{a^2(1-e\sin u)^2} \cos u = 0,$$

$$+\cos u \frac{d^2u}{dt^2} + \sin u \left(\frac{du}{dt}\right)^2 - \frac{\cos u}{a^2(1-e\sin u)^2} \sin u = 0,$$

and multiplying by $-\cos u$, $-\sin u$, respectively, and adding, we have

$$\left(\frac{du}{dt}\right)^2 = \frac{1}{a^2(1-e\sin u)^2 - e^2a^2\sin^2 u} \cdot \frac{1}{1+e\sin u},$$

which is the required differential equation.

9. Show that the attraction of an indefinitely thin double-convex lens at a point at the centre of one of its faces is equal to that of the infinite plate included between the tangent plane at the point and the parallel tangent plane of the other face of the lens.

The figure represents the upper half only of the lens, but in speaking of any portion thereof, such as PRQ, we include the symmetrically situate portion of the under-half of the lens.

Let a, $= PQ$, be the thickness of the lens, LPQ a circle of which angle is ultimately $= \frac{a}{a}$. Then it is at once seen that the attraction of the cone LPQ is $=$ const $= 2\pi a$; and from this it follows that the attraction of the infinite plate is $= 2\pi a$. The attraction of the whole infinite plane (resp. the cone LPQ) is $= 2\pi a$, which is indefinitely small in regard to $2\pi a$; and, *a fortiori*, the attraction of the portion MPR of the lens is indefinitely small in regard to $2\pi a$. We have then only to show that the attraction of the mass LRQ is indefinitely small in regard to $2\pi a$; the mass being, in fact, indefinitely thin in the direction of the lens, the attraction of the cone MRQ will not therefore ultimately be $= 2\pi a$.

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Let the position of an element of the solid in question be determined by r the distance from P, θ the inclination of r to the axis PQ and ϕ the azimuth in regard to any fixed plane through the axis; then $dm = r^2 \sin\theta\, dr\, d\theta\, d\phi$, the integral in regard to ϕ having been taken from 0 to 2π, and the attraction in the direction in the figure: PQ is $= \iint a \sin\theta\cos\theta\, dr\, d\theta - \frac{1}{2}\iint \sin 2\theta\, dr\, d\theta$, the integral in regard to ϕ having been taken from 0 to π; and the attraction in the direction of PQ from any point $A =$ a constant when $\phi = ar$, and that for $\phi = a$, the limit is the small circle. The plane PRQ, $(Z = 0)$ the equation will be $\phi =$ const.

Taking the radius of the small circle to be $= \rho$, then for a fixed value of ϕ the value of r extends from 0 to $\frac{2\rho\cos\theta}{1-e\sin\theta}$, while the integral in regard to ϕ is to be taken from $\phi = 0$ for the value of r for which $\theta = \tan^{-1}\frac{a}{\rho}$ to $\phi = \frac{\pi}{2}$. We have then

$$\frac{1}{2}a^2\int \sin 2\theta\, d\phi = (1+e)\sin\theta\cos\theta\,(1+e^2\sin^2\theta)\, d\theta,$$

and the integral is to be taken from $\theta = 0$ to $\theta = \tan^{-1}\frac{a}{\rho}$, viz. this is

$$\int \sin\theta\cos\theta\,(1-e\sin\theta)^2\,(1+e^2\sin^2\theta)^{-1}\, d\theta - a\, a^2\, a^{-2}\, d\theta,$$

so that taking this between the limits in question, we have

$$\begin{aligned}\frac{1}{2}a^2\int \sin 2\theta\, d\phi &= (1+e)\sin^2\theta - \frac{1}{6}(1+e)^2(1-e\sin\theta)^3\\ &-(1+e)\frac{1}{4}\left[1-\frac{1}{(1+e)^2}\right] - \frac{1}{2}(1+e^2)\left[1-\frac{1}{1+e}\right]\\ &+\frac{1}{2}(1-2e^2)\frac{(1+e)^2+(1+e)}{1+e} - \frac{1}{3}(1-2e^2)\frac{(1+e)^3+1}{(1+e)^3}.\end{aligned}$$

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viz. a here an indefinitely small quantity of the order a^2; all the terms are therefore at least of the order a^2, and are to be neglected in comparison with $2\pi a$, the integral work shown we have $A = 0$ (that is, the attraction of the mass LRQ is indefinitely small in the standard sense (since $L = a$ to the doctrine in the form RPQ is mass equal in $2\pi a$), as the theorem is thus proved.

10. *Indicate in what manner the Lagrangian equations of motion*

$$\frac{d}{dt}\frac{dT}{dx'} - \frac{dT}{dx} + \frac{dV}{dx} = 0, \quad \&c.$$

lead to the equation

$$\frac{d^2x}{dt^2} + (U - R)\frac{dV}{dy} = 0, \&c.$$

for the motion of a solid body about a fixed point.

The expression of the vis viva theorem T is

$$T = \tfrac{1}{2}(Ap^2 + Bq^2 + Cr^2);$$

but this expression will not by itself lead to the equations of motion; we require to know also the expression of p, q, r in terms of certain coordinates λ, μ, ν, which determine the position of body in regard to axes fixed in space, and of the differential coefficients λ', μ', ν' of these coordinates in regard to the time; each of the quantities p, q, r will be a linear function of λ', μ', ν', ($p = a\lambda' + b\mu' + c\nu', \&c.$), containing in any manner whatever the coordinates λ, μ, ν, and the equations of motion will be

$$A\frac{dp}{dt}\frac{dq}{d\lambda} + B\frac{dq}{dt}\frac{dr}{d\mu} + C\frac{dr}{dt}\frac{dp}{d\nu} = 0, \quad \&c.$$

where $\frac{dp}{d\lambda}$, $\frac{dq}{d\mu}$, $\frac{dr}{d\nu}$ are each independent of λ', μ', ν'; hence, in the equation, the only terms containing the differential coefficients of λ', μ', ν' are the terms

$$A\frac{dp}{dt}\frac{dq}{d\lambda'} + B\frac{dq}{dt}\frac{dr}{d\mu'} + C\frac{dr}{dt}\frac{dp}{d\nu'};$$

of $\frac{dT}{dx'} - \frac{dT}{dx}$; and hence, assuming that the equations of motion on the known equations $A\frac{dp}{dt} + (C - B)qr = 0$, it appears that the equation $A\frac{d^2x}{dt^2} + (U - R)\frac{dV}{dx} = 0$ may be put

$$\frac{d}{dt}\left[A\frac{dp}{dt}\cdot\frac{dq}{d\lambda'} + B\frac{dq}{dt}\cdot\frac{dr}{d\mu'} + (C - B)qr\frac{dp}{d\nu'}\right] = 0;$$

$$\frac{d}{dt}\left[A\frac{dp}{dt}\cdot\frac{dq}{d\mu'} + B\frac{dq}{dt}\cdot\frac{dr}{d\nu'} + (A - C)rp\frac{dq}{d\lambda'}\right] = 0;$$

$$\frac{d}{dt}\left[A\frac{dp}{dt}\cdot\frac{dq}{d\nu'} + B\frac{dq}{dt}\cdot\frac{dr}{d\lambda'} + (B - A)pq\frac{dr}{d\mu'}\right] = 0;$$

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$\frac{dp}{d\lambda'}$, $\frac{dq}{d\mu'}$, $\frac{dr}{d\nu'}$, &c. in not $= 0$; and the three equations are therefore equivalent to (as they should be) to the equations

$$A\frac{dp}{dt} + (C - B)qr = 0, \quad \&c.$$

What precedes is a complete answer to the question, but in regard to the actual expression of p, q, r, it may be remarked, that these quantities may be expressed very symmetrically in terms of the quantities

$$\lambda, \mu, \nu = \tan\frac{1}{2}\theta\,\cos\phi, \tan\frac{1}{2}\theta\,\sin\phi, \tan\frac{1}{2}\theta,$$

which determine the position of the principal axes in regard to the axes fixed in space, by means of the angles of position (azim. ϕ and the resultant axis, and the notation of about this axis is θ; writing $a^2 + b^2 + c^2 = k^2$, we have, the equations

$$ap = (1 + b^2)\lambda' - (1 + c^2)\mu' + (\nu - h)\nu',$$

$$aq = (b + c)\lambda' - h^2\mu' - (\mu + g)\nu',$$

$$ar = (n - g)\lambda' + h\mu' + h^2\nu';$$

and the above result may be verified *a posteriori* without any difficulty. See Crelle's *Math. Jour.*, vol. XI. (1843), [6], p. 224, [Coll. Math. Papers, vol. I. p. 33].

11. *Find in the Hamiltonian form,*

$$\frac{dx}{dt} = \frac{dH}{dp}, \quad \frac{dp}{dt} = -\frac{dH}{dx}, \quad \&c.$$

the equations for the motion of a particle acted on by a central force.

Taking as coordinates r the radius vector, ω the longitude, y the latitude, the equation of the vis viva function is

$$T = \frac{1}{2}(r'^2 + r^2\cos^2 y\,\omega'^2 + r^2y'^2),$$

hence

$$\frac{dT}{dr'} = r' = a \quad \text{suppose},$$

$$\frac{dT}{d\omega'} = r^2\cos^2 y\,\omega' = \nu \quad ,$$

$$\frac{dT}{dy'} = r^2y' = \mu \quad ,$$

and the expression of T in terms of r, a, y and the new coordinates a, ν, μ is

$$T = \frac{1}{2}\left(a^2 + \frac{\mu^2}{r^2} + \frac{\nu^2}{r^2\cos^2 y}\right);$$

whence writing

$$H = \frac{1}{2}\left(a^2 + \frac{\mu^2}{r^2} + \frac{\nu^2}{r^2\cos^2 y}\right) + V,$$

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the equations are

$$\frac{dH}{dr}=a,\quad \frac{dH}{d\omega}=\nu,\quad \frac{dH}{dy}=\mu,$$

$$-\frac{dH}{dr}=a,\quad -\frac{dH}{d\omega}=\nu,\quad -\frac{dH}{dy}=\mu.$$

We have

$$\frac{dH}{dr}=-\frac{\nu^2}{r^3}-\frac{\mu^2}{r^3\cos^2 y}+\frac{dV}{dr},$$

$$\frac{dH}{dy}=\frac{\mu^2}{r^2\cos^2 y}\tan y+\frac{dV}{dy}\quad \frac{dH}{d\omega}=\frac{dV}{d\omega}=\frac{r^2\cos^2 y\,\omega'}{r^2\cos^2 y};$$

and substituting these values, the equations of motion present themselves in a system of the first order between r, a, y, μ, ν, and it is well known that the case of a central force corresponds to that of a function V where the function V contains r only.

12. *An ordered polygon of $(n+1)$ vertices is constructed as follows:— viz. the several vertices are taken at random of the unknown a. In each, considering as the largest order a the size of the whole polygon is very small; and may be regarded as being $=1$, and the area between the ordinates corresponding to the several values of being $=1$, are like ordinate. Find the probability of the unknown polygon being in the form of a convex polygon, and the corresponding probabilities respective ordinate.*

It is somewhat more convenient to take account also of the abscissa $=1,\ldots=u$, thereby obtaining a polygon of $2n+1$ vertices, corresponding to the abscissa 1, in the largest, and the position of the polygon being on either side of the abscissa $=1$, that is, the polygon is fixed at the abscissa $=a$, ever this, a being any given number, the probability of a number of heads between $n=a$ and $a+a$ increases with the number a being fixed and equally probable, all the probability of the form in question being so probably, and so on; the probability of the position of these large numbers of vertices: viz. the chance that the polygon has only a number of vertices equal to that of the first a of the second period, that is, the probability of a vertex polygon between the ABC, hence the two periods have its sum.

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The conclusion in regard to the areas of the polygons is that, taking A any given value whatever, however large, the ratio (so being of course $> A$) which the area between the ordinates of the abscissa $a+1, \ldots = a+n$ bears to the area of the whole polygon (or to unity) continually decreases as n increases, and ultimately vanishes; but con- trariwise, taking a way given function whatever, however small, the area which the respectively bears to the area of the whole polygon, however small, becomes ultimately $= 1$.

13. *Show that for the quartic surface $aU^2 + 2bUV + cV^2 + 2dW + e = 0$, where of the surface is a quartic surface through six points particular values $(U,\ V,\ W,$ &c.$)$; then the equation of the general quartic surface through the six points will be of the form*

$$(U \cdot V, W, \cdot) \cdot \alpha U + \beta V + \gamma W + \cdots = 0,$$

$(U, V, W, \cdot)$ *the coordinates of the vertex, in regard to the unknown.*

Suppose $U = 0$, $V = 0$, $W = 0$ are particular line quadric surfaces through the six points then the equation of the general quartic surface through the six points will be of the form

$$aU^2 + 2bUV + cV^2 + 2dW + e = 0;$$

where $(a,\ b,\ c,\ d,\ e)$ are constants. In order to that there is a node we must have for the coordinates of the vertex, the equations combined with the given equation

$$\begin{aligned}
a\frac{dU}{dx} + b\frac{dV}{dx} + b\frac{dW}{dx} + d\frac{dT}{dx} &= 0,\\
a\frac{dU}{dy} + b\frac{dV}{dy} + c\frac{dW}{dy} + d\frac{dT}{dy} &= 0,\\
a\frac{dU}{dz} + b\frac{dV}{dz} + c\frac{dW}{dz} + d\frac{dT}{dz} &= 0,\\
a\frac{dU}{dw} + b\frac{dV}{dw} + c\frac{dW}{dw} + d\frac{dT}{dw} &= 0.
\end{aligned}$$

Eliminating $(a,\ b,\ c,\ d)$, we have an equation $\nabla = 0$, where ∇ is the Jacobian or fractional determinant $\dfrac{d(U,\ V,\ W,\ T)}{d(x,\ y,\ z,\ w)}$ formed with the differential coefficients of the four functions $(U,\ V,\ W,\ Z)$; the locus of the vertex is then a quartic surface.

Calling the six points 1, 2, 3, 4, 5, 6, then taking as vertex any point in the line 12, the lines from such point to the points 1 and 2 coincide with the line 12, and we can through this line and the lines to the remaining points 3, 4, 5, 6 describe a quadric cone; the quartic surface therefore passes through the line 12; and similarly it passes through each of the fifteen lines 12, 13, …, 56.

Again, taking the vertex anywhere in the line of intersection of the planes 123 and 456, we have an improper quadric cone, viz. the plane-pair formed by these two

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planes; the line in question is therefore a line of the quartic surface, and similarly the quartic surface contains all the two lines 123, 456; 124, 356; 134, 456. We have thus in all 25 lines on the quartic.

In the case of seven points 1, 2, 3, 4, 5, 6, 7, the locus is the curve of inter- section of the quartic surfaces (obtained through any one of the six points 1, 2, 3, 4, 5, 6; 23, 24, 25, 26, 34, 45 to which it should be $= 16$, $b = 71$, $c = 56$). Here the several lines on either of the surfaces, viz. part of the curve in question are: viz. on or scarce curve, which is the locus of the vertices of the cones which pass through the seven given points.

14. *Show that the envelope of a system of circles having its centre on a given and cutting at right angles a given circle is a bicircular quartic; which, when the plane which and circles has double contact, becomes a pair of circles; and, by considering the magnitude of the latter, deduce, the area of curve in regard between the two principal planes.*

(1) given two circles, to find a polygon of n sides inscribed in the one and circum- scribed about the other.

The equation of the given circle is taken to be

$$(x-a)^2 + (y-\beta)^2 = a^2,$$

and that of the conic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$. This being so, we have $a\cos\theta$, $b\sin\theta$ as the coordinates of a point on the conic, which point may be taken to be the centre of the variable circle, and introducing the condition that the two circles cut at right angles, the equation of the variable circle is

$$(x - a\cos\theta)^2 + (y - b\sin\theta)^2 = a^2\cos^2\theta + b^2\sin^2\theta + c^2,$$

that is,

$$x^2 + y^2 - a^2 - 2xa\cos\theta - 2yb\sin\theta + b^2\sin^2\theta - 2yb\sin\theta = 0,$$

where θ is the variable parameter; and the equation of the envelope therefore is

$$(x^2 + y^2 - c^2)\cos^2\theta + (a^2 - b^2)\sin\theta\cos\theta - 2ax\cos\theta - 2by\sin\theta + b^2 = 0,$$

which is a quartic, having and belonging to each as b, c, in the same equation as would have been obtained from either of the other conics and the corresponding circle, and from the form of the equation it is clear that the curve is a bicircular quartic.

If the fixed circle touches the conic, then by a consideration of the figure it at once appears that the point of contact is a double point on the curve; and as if there is a double contact, then each of the points of contact is a double point on the curve. But in this case the curve is a bicircular quartic with four double points, viz. it is a pair of circles.

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The points in regard to the two conics is that, in general it is not possible to find any polygon of n sides satisfying the conditions; but that the case is also such that there exists an infinity of polygons; viz. any point whatever of the one conic may then be taken as a vertex of the polygon, and there are corresponding the the $(n+1)^{\text{th}}$ vertex still coincide with the first vertex, and there will be a polygon of n sides.

Now imagine that the conic touched by the sides is a circle having double contact with the other conic. Describe any one of the polygons, and with each vertex as centre describe the orthotomic circle, which will, it is clear, be a circle passing through the points of contact with the circles, each touching the two adjacent circles of the series. And by considering any other polygon, we have a like series of a circles; and by what precedes the envelope of all the circles of the several series is a pair of circles; that is, the circles of every series touch these two circles. We have consequently two circles, such that there exists an infinity of closed series of n circles, each circle touching the two fixed circles, and also the two adjacent circles of the series; which is the porism arising out of the second problem.

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Note on the Problem of Envelopes.

[From the Messenger of Mathematics, vol. I. (1872), pp. 3, 4.]

THERE is a mode of looking at the problem of Envelopes, which, so far as I am aware, has not been explicitly noticed. Let $U = (x, y, z)^m$ be a function of the coordinates (x, y, z) $(\Theta = 0$ or (x, y, z) of θ, a function of the new set of coordi- nates (x, y, z) and (x', y', z'), θ being understood that when we write $\Theta = 0$ this expresses the existence of a relation between the coordinates (x, y, z) and the new set of coordi- nates (x', y', z'). Suppose that we have $\Theta = 0$, $U = 0$; that is, then the equation expresses θ as a function of (x, y, z), (x', y', z'); this is then a curve: the equation expressed values of parameters in coordinates (x, y, z) of a point P on the curve $U = 0$; and we may say that θ is the envelope (equivalent to four equations only)

$$U = 0, \quad \Theta = 0,$$

$$d_x U + \lambda d_x \Theta = 0, \quad d_y U + \lambda d_y \Theta = 0, \quad d_z U + \lambda d_z \Theta = 0,$$

and a particular integral $ay - b = 0$: the complete integral is in fact

$$(x - cy)^2 - 4r^2 b^2 + 2(p - q)y^2 - rp^2 = 0,$$

satisfied, for $C = 0$, by $xy - n = 0$.

But, observe that the required envelope is the locus of the points of intersection of the curve $U = 0$, by curves in a particular point (x', y', z') of the other curve, $\Theta = 0$. So the curve $U = 0$ which belongs to a consecutive point (x', y', z') of the curve $U = 0$, con- sidering therein (x', y', z') as the variable; by such a process the envelope of the curve y in question is a mere definition of what it does mean, and comes to be a fluxion. y then only chosen the values of A are b, b', b'', &c., and these values are then determined by the equations:— viz. if P is a point on the conic (x', y', z'), then conics, conditions (x', y', z') are conics on the same conic (x', y', z') as their values: and the curve in the plane which corresponds spend to the generating loop of the model.

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$\Theta = 0$, $U = 0$ to touch each other, we have a relation in (x', y', z') which is the equation of the point intersection of the curves $\Theta = 0$ belonging to the two consecutive values of the curve $U = 0$; that is, the equation of the required envelope is obtained as the condition that the curves $U = 0$, $\Theta = 0$ shall touch each other. But when the curves touch each, they have at the point of contact two consecutive points proportional, or we have simultaneously

$$U = 0, \quad \Theta = 0,$$

$$d_x U + \lambda d_x \Theta = 0,$$
$$d_y U + \lambda d_y \Theta = 0,$$
$$d_z U + \lambda d_z \Theta = 0,$$

the same equations as before, since $\Theta = 0$ and $\Theta = 0$ are the same function.

It is to be added that, when $a = u$, the equations

$$d_x U + \lambda d_x \Theta = 0,$$

$$d_y U + \lambda d_y \Theta = 0,$$
$$d_z U + \lambda d_z \Theta = 0,$$

are homogeneous in (x, y, z), and we may by the elimination of (x, y, z) from these equations obtain an equation Disc. $(U \pm \lambda U) = 0$, say for shortness $\Delta = 0$, involving λ and also the coordinates (x', y', z'). Now it is a known theorem that the condition for the contact of the two curves $U = 0$, $U = 0$ may be obtained by expressing that the equation $\Delta = 0$ shall have a pair of equal roots, or, what is the same thing, by equating to the equation of the envelope of the curve $U = 0$, viz. by being so as shown the equation of the envelope of the curves $U = 0$, is in fact

Disc. $\text{Disc.}_{(\lambda,\,1)}(U + \lambda U) = 0$;

viz. we first take the discriminant of the function $\Theta + \lambda U$ in regard to the coordinates (x, y, z) and then taking the discriminant in regard to λ of the function so obtained, b is zero. This is in many cases a more simple process than that of the direct elimination of x, y, z, λ from the equations.

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Note on Lagrange's Demonstration of Taylor's Theorem.

[From the Messenger of Mathematics, vol. I. (1872), pp. 22–24.]

I TAKE the occasion of the publication of the last edition of Mr Todhunter's *Treatise on the Differential Calculus* to make some remarks on the demonstration in question. Mr Todhunter proposes to himself to establish a comprehensive view of the *Differential Calculus* as the method of Limits; but he very properly introduces in some cases demonstration founded upon other views of the subject, pointing out that this is the case, and explaining or justifying his objections. Thus (Chapter VI.) upon Taylor's Theorem, he remarks: "Before we offer a strict demonstration of the theorem in question, we shall notice the method which it was usual to adopt in treatises on the Differential Calculus not based on the doctrine of limits;" and then, after giving a demonstration depending on the relation $\frac{d}{dx}f(x+h) = \frac{d}{dh}f(x+h)$ he goes on to say "There are numerous objections to the method of the preceding article, and especially the use of an infinite series, without ascertaining that it is convergent, is inadmissible; we proceed then to prove Taylor's Theorem after the manner adopted by Mr Homersham Cox(*) in a demonstration of the equation

$$f(x+h) = f(x) + hf'(x) + \cdots + \frac{h^n}{[n+1]^{n+1}} f^{(n+1)}(x+\theta h),$$

(θ between 0 and 1) Taking "if the function $f^{(n)}(x+\theta h)$ is such that by making a sufficiently great the term $\frac{h^n}{[n+1]^{n+1}} f^{(n+1)}(x+\theta h)$ can be made as small as we please, then by carrying on the series

$$f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \&c.$$

* This demonstration is similar in principle to Laprange's but I think it is preferable; do, the gives in passing in this rather than the same value whether n is changed once, ; it on it. VIII, p. 9 (1851).

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to as many terms as we please we obtain a result differing as little as we please from $f(x+h)$. Under these circumstances then we may assert the truth of Taylor's theorem."

I share Abel's horror of divergent series(*); and I maintain the validity of Lagrange's demonstration. When by any objection one can expand a function in a series, for instance the function $\frac{1}{1-x}$, by division

$$\frac{1}{1-x} = 1 + x + \frac{x^2}{1-x} = 1 + x + x^2 + \frac{x^3}{1-x}, \&c.$$

$$1 + x + x^2 + x^3 + \&c.,$$

in the series $1 + x + x^2 + \&c.$, and write accordingly

$$\frac{1}{1-x} = 1 + x + x^2 + \&c.$$

all that is for ought be may mean is that the algebraical operations contained as far as we please will give the series of terms 1, x, x^2, &c., or say the series of coefficients 1, 1, 1, ... And of course with this meaning of the equation, the objection "we cannot that the series is convergent" would be wholly irrelevant, we do not say that in is, we do not care whether it is &c. or not. In further illustration, remark that we frequently use such an equation merely as the means of expressing the law of a series of numbers a_0, a_1, a_2, &c., $a_n = a + b\theta$, &c.; we make the assertion to be a definite process expressible in the form $a_n =$ co. x^n in $a_n = a_0 + a_1 x + a_2 x^2 + \&c.$ in question. Any objection that the series is not convergent would be simply irrelevant. Now an that the series shall be convergent is wholly employed for the expansion of $f(x+h)$ in regard to $f(x+h)$. It is impossible, in a quantitative algebra such as is presupposed in the method of limits, to put any meaning on the formula

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \&c.,$$

viz. $f(x+h)$ requires the same value $f(x+h)$ whether we change therein a into $x+h$ or h into $x+h$; and the expression on the right-hand side is the only series in h possessed of the same property. It is to be remarked that the equation contains in itself the definition of the operation of derivation, viz. the equation being true, $f(x)$ can only denote the coefficient of h in the expansion of $f(x+h)$, and what

* Note on Aragon dans des bons louisges dipres divines.

$$- - - - - - \quad -\theta + - (-\theta)^2 + \&c. \quad - - - - - -$$

a maint et avait ferois priselt ?—r–, t. IV. p. 263; [Newt. Eli, 1885, t. VII. p. 122].

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really is shown is that admitting such an operation to be possible in regard not only to $f(x)$, but to $f'(x)$, &c., then the coefficients $f'(x)$, $\dfrac{f''(x)}{2!}$, &c., are obtained from $f(x)$ by the successive repetition of this operation and by dividing by the proper numerical denominator.

By what precedes, any objection in regard to convergency, I regard as irrelevant; and if it is said that the above-mentioned single assumption is not granted, I would either ask "What is a Function?"—or I would contend myself with the hypothetical statement: "If $f(x)$ be such that $f(x+h)$ is expansible in apoly, then Taylor's theorem.

In regard to the demonstration given by Mr Todhunter, it impliedly assumes that x and h are both real, and (although doubtless possible) it would be considerably more difficult to find an analogous demonstration of the formula involving $f^{(n+1)}(x+\theta h)$ in the case of x and h imaginary. But the formula with the term in question is not (see Mr Todhunter consider it as being) Taylor's theorem, ; to obtain from it Taylor's theorem, we require (in the foregoing point of view) the property that $\lim_{n\to\infty} f^{(n+1)}(x+\theta h)$ is exposable in a series involving h^{n+1} and the higher powers of h, that is, the very property that $f(x+h)$ is exposable in positive powers of h.

Moreover admitting that the formula with the term $f^{(n+1)}(x+\theta h)$ is demonstratable for imaginary values of x, h, then formula is unenchapsie in the case where $x+h$ is one or both a symbol or symbols of operation; whereas it would certainly have to include unmoral magnitude, and if it is considered as meaning anything, then the equation in question is mere definition of what it does mean, and ceases to be a theorem in regard to $f(x+h)$. It is impossible, in a quantitative algebra such as is presupposed in the method of limits, to put any meaning on the formula

$$f\left(\frac{d}{dx}+h\right) = f\left(\frac{d}{dx}\right) + hf'\left(\frac{d}{dx}\right) + \&c.,$$

which however I regard as a legitimate particular form of Taylor's theorem.

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Solutions of a Smith's Prize Paper for 1871.

[From the Messenger of Mathematics, vol. I. (1872), pp. 37–47, 71–77, 89–95.]

1. *A point moves in a plane with a given velocity, and also with a given angular velocity about a fixed point in the plane: show that the locus is either a circle passing through the fixed point, or else a circle having the fixed point for its centre; and explain the relation between the two solutions.*

We have in general

$$v^2 = \left(\frac{dr}{dt}\right)^2 + r^2\left(\frac{d\phi}{dt}\right)^2,$$

and in the present question, taking the fixed point as the origin, and measuring θ from any fixed line through this point,

$$\frac{d\theta}{dt} = a, \quad V^2 = \left(\frac{dr}{dt}\right)^2 + r^2a^2,$$

where V, a are given constants. Hence

$$\left(\frac{dr}{dt}\right)^2 + a^2r^2 = V^2, \quad \left(\frac{dr}{dt}\right)^2 = V^2 - a^2r^2,$$

or, writing $V = aa$,

$$\left(\frac{dr}{dt}\right)^2 = a^2(a^2 - r^2);$$

therefore

$$d\theta = \frac{dr}{\sqrt{(a^2 - r^2)}},$$

or

$$\theta + \beta = \sin^{-1}\frac{r}{a}, \quad (\beta \text{ the constant of integration,})$$

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that is,

$$r = a\sin(\theta+\beta),$$

which is the equation of a circle (radius $= \frac{1}{2}a$) passing through the fixed point. In fact, the point moving in such a circle with a constant velocity, moves about the centre with a constant angular velocity; and about any fixed point in the circumference with an angular velocity which is one-half that about the centre, and is therefore also constant.

Treating θ as a variable parameter, to obtain the envelope we have

$$0 = a\cos(\theta+\beta),$$

that is, $\theta+\beta = \frac{\pi}{2}$, and therefore $r = a$, which is the equation of a circle (radius $= a$) having the fixed point for its centre. This is consequently the singular solution.

2. *Determine the system of curves which satisfy the differential equation*

$$dx(\sqrt{(1+x^2+xy+xy^2)}) + dy(\sqrt{(1+x^2+xy+xy^2)}) = 0;$$

and that the curve which passes through the point $x=0, y=0$, *contains as a part of itself the conic*

$$x^2+y^2+2xy\sqrt{(1+x^2+xy+xy^2)} = 1.$$

The equation is integrable *per se*, viz. we have

$$x\sqrt{(1+x^2)} + \log\{x+\sqrt{(1+x^2)}\} + \frac{1}{2}y\sqrt{(1+y^2)} + \log\{y+\sqrt{(1+y^2)}\} = U$$

or, denoting the constant so that the differential equation has its general solution

$$x\sqrt{(1+x^2)}+y\sqrt{(1+y^2)}+2xy = (1+x^2+xy+y^2)+\log\frac{\{x+\sqrt{(1+x^2)}\}\{y+\sqrt{(1+y^2)}\}}{1-C} = 0,$$

which is evidently a transcendental curve; it may however be shown that, if

$$x^2+y^2+2xy\sqrt{(1+x^2+xy+xy^2)} - 1 = 0,$$

so that the equation of the curve is then made obvious, that is, the transcendental curve $y = axy+b$, which is the elementary curve, is in this manner included as a part of the curve.

[At a simple illustration as to how this may happen, take the transcendental curve $y = axy+b$, which is the elementary curve, in this manner included as itself.]

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We have, from the equation of the conic

$$\{x + y\sqrt{(1 + x^2)}\}^2 = (1 + x^2)(1 - y^2),$$

that is,

$$x + y\sqrt{(1 + x^2)} = \pm\sqrt{(1 + x^2)(1 - y^2)},$$

but considering the radicals as positive, the sign must be taken so that we have simultaneously $x + y > 0$, $y = a$. We have therefore

$$x + y\sqrt{(1 + x^2)} = \sqrt{(1 + x^2)(1 - y^2)},$$

and similarly

$$y + x\sqrt{(1 + y^2)} = \sqrt{(1 + y^2)(1 - x^2)}.$$

Then

$$\begin{aligned} x\{x + y\sqrt{(1 + x^2)}\} + y\sqrt{(1 + x^2)} &= 2xy + (x^2 + y^2)\sqrt{(1 + x^2)} + x\sqrt{(1 + x^2)(1 - y^2)} \\ &= 2xy + (x^2 + y^2)\sqrt{(1 + x^2)} - 2xy(1 + x^2) \\ &= xy(1 - x^2 - y^2) + \sqrt{(1 + x^2)}(x^2 + y^2) - 2xy(1 + x^2) \\ &= xy(1 - x^2 - y^2) - 2xy(1 + x^2), \end{aligned}$$

which is the first of the relations in question; and similarly

$$\begin{aligned} x\{y\sqrt{(1 + x^2)} + x\sqrt{(1 + y^2)}\} &= \sqrt{(1 + y^2)} + y\sqrt{(1 + x^2)} \\ &= y(1 + y^2) - 2y(1 - x^2 - y^2) \\ &= (1 + y^2)\sqrt{(1 + x^2)} - 2y(1 - x^2 - y^2)\sqrt{(1 + x^2)} \\ &= -(x + y\sqrt{(1 + x^2)})(x + y\sqrt{(1 + y^2)}), \end{aligned}$$

which is the second of the two relations. And the theorem is thus proved.

The foregoing is the easiest and most obvious solution, but it is interesting to consider the question differently, as follows:

Write

$$Q = \frac{(x + y\sqrt{(1 + x^2)})(y + x\sqrt{(1 + y^2)})}{x + y\sqrt{(1 + x^2)}};$$

we have

$$Q = (x^2 + y^2 + xy\sqrt{(1 + x^2)} + xy\sqrt{(1 + y^2)} + y\sqrt{(1 + x^2)(1 + y^2)}) - A - B,$$

$$Q^* = (x^2 + y^2 + 4)(\sqrt{(1 + x^2)} + \sqrt{(1 + y^2)} + x\sqrt{(1 + x^2)(1 + y^2)}) = A - B,$$

if

$$A = x(1 + x^2)(1 + y^2) + 2xy,$$

$$B = x\sqrt{(1 + x^2)} + y\sqrt{(1 + y^2)} + xy\sqrt{(1 + x^2)(1 + y^2)}),$$

and then

$$\begin{aligned} AB &= xy\sqrt{(1 + x^2)} + xy\sqrt{(1 + y^2)} + y\sqrt{(1 + x^2)(1 + y^2)}, \\ &+xy(1 + y^2)\sqrt{(1 + x^2)} + (1 + x^2)(1 + y^2)\sqrt{(1 + x^2)} + xy\sqrt{(1 + x^2)(1 + y^2)}, \\ &= xy\sqrt{(1 + x^2)} + xy\sqrt{(1 + y^2)} + 2xy\sqrt{(1 + x^2)(1 + y^2)}. \end{aligned}$$

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that is,

$$Q^2\{2xy+1+2x\sqrt{(1+x^2)}\}-\tfrac{1}{2}xy\{2xy+1+2x\sqrt{(1+x^2)}\}=0,$$
$$=4(x\sqrt{(1+x^2)}+y\sqrt{(1+y^2)})+4xy$$
$$+4xy\{Q\sqrt{(1+x^2)}+y\sqrt{(1+y^2)}-\tfrac{1}{2}\sqrt{(1+x^2)(1+y^2)}\};$$

whence

$$4\{x\sqrt{(1+x^2)}+y\sqrt{(1+y^2)}+2xy\}=Q^2(2xy+1+2x\sqrt{(1+x^2)})$$
$$=Q^2\{2x\sqrt{(1+x^2)}+1+2x\sqrt{(1+x^2)}\}-\tfrac{1}{2}xy\{2x\sqrt{(1+x^2)}+1+2x\sqrt{(1+x^2)}\}$$
$$+4xy(Q\sqrt{(1+x^2)}+y\sqrt{(1+y^2)}-\tfrac{1}{2}\sqrt{(1+x^2)(1+y^2)})$$
$$=(Q-1)(2xy+1+2x\sqrt{(1+x^2)})$$
$$+4(Q-1)xy\sqrt{(1+x^2)}$$
$$-4(Q-1)(1+xy^2)$$
$$-4(Q-1)xy^2,$$

that is,

$$x\sqrt{(1+x^2)}+y\sqrt{(1+y^2)}+2xy=(Q-1)D.$$

And the integral equation is

$$\tfrac{1}{2}(Q-1)+\log Q=C,$$

which, for $C=0$, is satisfied by $Q-1=0$.

Now starting from

$$Q=\frac{(x+y\sqrt{(1+x^2)})(y+x\sqrt{(1+y^2)})}{x+y\sqrt{(1+x^2)}}$$

we have

$$\sqrt{(1+x^2)}-x=\tfrac{1}{2}y\{\sqrt{(1+x^2)}-x\}+\tfrac{1}{2}\{\sqrt{(1+y^2)}-y\},$$

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and thence

$$2x = K\sqrt{(1+x^2)} + Ly,$$

if

$$K = Q\sqrt{(1+x^2)} + x + \tfrac{1}{2}\sqrt{(1+x^2)-1},$$

$$L = Q\sqrt{(1+y^2)} + y + \tfrac{1}{2}\sqrt{(1+y^2)-1},$$

whereon

$$L^2 - K^2 = 4.$$

Moreover

$$(2x+Ly)L = K^2 + KLy$$

that is,

$$4x^2 + Ly = K^2 + KLy = K^4,$$

or, what is the same thing,

$$x^2 + Lxy = (K-x)^2,$$

which is the rationalized form

$$Q = \frac{x^2 + 2xy\sqrt{(1+x^2+xy+y^2)} - y^2}{xy + \sqrt{(1+x^2)(1+y^2)}} - 1.$$

And if $Q = 1$ then $L = 2\sqrt{(1+x^2)}$, $(L - K^2 = 2x)$, and the equation is

$$x^2 + 2xy\sqrt{(1+x^2)} - y^2 = 0,$$

that is, the curve $C = 0$, of the integral integral has, as is should be, a part the conic

$$\frac{x^2 - y^2}{x + y\sqrt{(1+x^2)}} = 1, \quad \text{or as it should be a fixed conic.}$$

We may evidently differently variations, and present the results as follows: the equation

$$\frac{1}{2}\left(\frac{x}{a}+1\right)\left(\frac{x}{a}-1\right) = \frac{Q+1}{2}\left[\frac{x}{a} + \frac{y-K}{a}\right] - C$$

and a particular integral $xy - n = 0$: the complete integral is in fact

$$(x-cy)^2 - 4r^2e^2 + 2(p-q)y^2 - rp^2 = 0$$

satisfied for $C = 0$, by $xy - n = 0$.

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3. Write $x = b - c$, $\beta = c - a$, $\gamma = a - b$; then considering the three circles and the three conics

$$(x-a)^2 + y^2 = 2bc, \quad \frac{x^2}{Y^2} + \frac{Y^2}{K+bc} = 1,$$

$$(x-b)^2 + y^2 = -2ac, \quad \frac{x^2}{K+bc} + \frac{Y^2}{K+ca} = 1,$$

$$(x-c)^2 + y^2 = 2ab, \quad \frac{x^2}{K+bc} + \frac{Y^2}{K+ab} = 1.$$

where K is arbitrary; it is required to show that of a variable circle having its centre in one of the conics cuts at right angles the corresponding circle, the envelope is in each of the three cases one and the same bicircular quartic.

Consider the circle $(x-a)^2 + y^2 = 2bc$ and the conic $\dfrac{X^2}{Y^2} + \dfrac{Y^2}{K+bc} = 1$, the coordinates of a point on the conic are $\cos\theta\sqrt{(K)}$, $\sin\theta\sqrt{(K+bc)}$, where θ is a variable parameter; say for a moment these values are p and q. Hence, taking (p, q) as the centre of the variable circle the equation is

$$(x-p)^2 + (y-q)^2 = (p-a)^2 + q^2 - 2bc,$$

and in order that this may cut at right angles the circle

$$(x-a)^2 + y^2 = 2bc,$$

we must have

$$(p-a)^2 + q^2 = 2(p-a)^2 - 2bc + 2bc,$$

or, substituting for p and q their values, the equation of the variable circle is

$$(x-p)^2 + (y-q)^2 = pq^2 + q^2 - 2bp,$$

that is,

$$x^2 + y^2 - p^2 - 2p\sqrt{(K)} - 2q\sqrt{(K+bc)} + 2yp = 0,$$

viz. this is

$$(x^2 + y^2 - p^2)^2 - 2(x+a)\sqrt{(K)}\cos\theta + 2y\sqrt{(K+bc)}\sin\theta + p^2 = 0.$$

Hence taking the envelope in regard to θ, the equation is

$$(x^2 + y^2 - p^2)^2 = 4K(x+a)^2 + 4(K+bc)\,y^2 - 4y\sqrt{(K+bc)}(x+a) + 4Kp^2,$$

or, what is the same thing,

$$[(x+y)^2 - 2a(x+y)\sqrt{(K)} + bc(K+bc)y^2 - 4Ky^2 + 4Kp(2x-bc) = 0,$$

viz. this equation, being symmetrical in regard to a, b, c, is the same equation as would have been obtained from either of the other conics and the corresponding circle; and from the form of the equation it is clear that the curve is a bicircular quartic.

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4. *Show that the caustic by reflexion for parallel rays of a circle, radius r, index of refraction μ, is the same curve as the caustic by refraction for parallel rays of a concentric circle, radius $\frac{r}{\mu}$, index of refraction $\frac{1}{\mu}$.*

Take a centre $\mu > 1$. Imagine the ray AP (fig. 1) parallel to the axis of x, incident at P on the circle radius r, and let the reflected ray after cutting the circle, radius $\frac{r}{\mu}$, cut it again at Q, and then cut the axis in R. Take ϕ, ϕ' for the angles of incidence and refraction, viz. $\angle O = \phi$, that is in the manner suitable for the caustic by reflection.

Fig. 1.

Moreover on the triangle PQO, we have $\sin Q : \sin\phi = r : \frac{r}{\mu}$; that is, $\sin Q = \mu\sin\phi$, $= \mu\sin\phi' = \sin\phi'$; or $\angle Q = \phi'$. And then in the triangle RQO, $\angle R = \phi - \phi'$, or $\angle ORQ = 180^\circ - \phi$.

Consider now a ray BQ incident at Q and refracted in the direction QR; the index of refraction being $\frac{1}{\mu}$, that is, the denser medium being on the outside of small circle. Taking θ, θ' for the angles of incidence and refraction, we have $\theta = \frac{1}{2}\sin^{-1}\phi'$, $= \mu\sin\phi'$ that is, $\theta = \phi$; the angles being two pencils of rays each corresponding to angle BQ of the second pencil, the rays AP and BQ each giving rise to the same refracted ray PQR; hence the two pencils have the same caustic.

[It is proper to remark that for the ray BQ it has been assumed that the refraction takes place not at Q where it first meets the small circle, but at Q.

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we consider the refraction at Q, then the index of refraction is still to be $= \dfrac{1}{\mu}$, that is, the denser medium must be inside the small circle: the refracted ray is in the direction RQ situate symmetrically with RQ on the opposite side of the axis of y; and it would at first sight appear that this would be a curve equal and similar to the original caustic, but situate on the opposite side of the axis of y. But geometrically the complete caustic consists of two equal and similar arches situate on the opposite sides of the axis of y; so that we really obtain, not an equal portion of the caustic, but the same caustic over again (about the limit changed, viz. as below by (ΔU). Substitute the variation of U in the symbols of operation differs; the formula is, the variations of Z, then the radii in radii of refraction is still to be $= \dfrac{1}{\mu}$, that the formula deduced from the variation of Z in so far as it arises from the variation of the constants, viz. the equation $\Delta U = \dfrac{dU}{d\alpha} d\alpha + d\beta = 0$ remains.

There are the like equations in regard to y, z; viz. in all, four equations which are linear in regard to $\dfrac{d\alpha}{d\alpha}, \dfrac{d\alpha}{d\beta}, \dfrac{d\alpha}{d\gamma}, \dfrac{d\alpha}{d\delta}$, and which serve to determine these quantities in terms of α, β, γ, δ.

Now considering the a, y as expressed in terms of a, c, e, ..., then Ω becomes a function of these quantities; the differential coefficients $\dfrac{d\Omega}{d\alpha}$, &c., being connected with the original differential coefficients $\dfrac{d\Omega}{d\alpha}, \dfrac{d\Omega}{d\beta}$, &c. by the equations

$$\frac{d\Omega}{d\alpha} = \frac{d\Omega}{d\alpha}\cdot\frac{d\alpha}{d\alpha} + \frac{d\Omega}{d\beta}\cdot\frac{d\beta}{d\alpha}, \quad \&c.$$

As there are four equations, $\dfrac{d\Omega}{d\alpha}, \dfrac{d\Omega}{d\beta}$, &c. can be expressed in terms of a, c, e in an infinity of ways in terms of $\dfrac{d\Omega}{d\alpha}, \dfrac{d\Omega}{d\beta}$, &c., and, considering $\dfrac{d\Omega}{d\alpha}$, &c., as given in terms of $\dfrac{d\Omega}{d\alpha}$, &c., we can in an infinity of ways express $\dfrac{d\Omega}{d\alpha}$, &c., as linear functions of $\dfrac{d\Omega}{d\alpha}, \dfrac{d\Omega}{d\beta}$, &c. But there is no form (obtained by combining the equations in a particular manner) wherein the coefficients of the non-constant quantities are functions of a, c, e, ... without a contact; and this is the form most simply employed for the expression of $\dfrac{d\Omega}{d\alpha}$, &c. in terms of $\dfrac{d\Omega}{d\alpha}, \dfrac{d\Omega}{d\beta}$, &c. in the method wherein these quantities are made use of in this respect.

I remark upon the present question, that the answer ought to be in evidence perfectly familiar to every student in *Physical Astronomy*; and that a student ought to be able to present it in a clear and lucid form: the question being in fact intended as a test of ability in this respect.

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But in virtue of the relation $a' + \beta' + \gamma' = 1$, we have

$$a\frac{dT}{dx} + \beta\frac{dT}{dy} + \gamma\frac{dT}{dz} = 0,$$

$$a\frac{dy}{dx} + \beta\frac{dy}{dy} + \gamma\frac{dy}{dz} = 0, \quad \&c.$$

Hence subtracting the corresponding equations we have three equations, which are at once seen to be equivalent to the two equations

$$\frac{dT}{dx}\,d\alpha + \frac{dT}{dy}\,d\beta + \frac{dT}{dz}\,d\gamma = 0, \quad \beta : \gamma,$$

equations which must be satisfied identically, whatever are the values of $(\alpha,\, y,\, z)$. The equations have been obtained as necessary conditions; they are, in fact, the sufficient conditions for a double system; for the line being supposed in passing from P to P', to remain always in contact with the surface, the result is in passing from P to P', it is necessary that the same line should also be a line of the surface.

6. If particles describe a side of the form $(a,\, b,\, \ldots \breve{a},\, x,\, 1)^n = 0$ with arbitrary coefficients, and if equation $ax^2 + bXy + cy^2 = 0$, where of $(a,\, b,\, \ldots)$ are arbitrary coefficients respectively in any manner whatever in the function G, then we have a series of equations satisfied by the values x, y which belong to the double root.

The equations

$$aXX + 2bXY + cY^2 + 2(pX + qY) + r = 0$$

$$EX + 2tY + s = 0$$

of common to the two conics, but for this equation to be the case of Z as a W shall vanish, the relation in the points of the plane $\dfrac{X^2}{a} + \dfrac{Y^2}{b} + \dfrac{Z^2}{c} = 1$; and that X, Y, Z, W are quartic functions of $(x,\, y,\, z)$ a common roots, show that this condition of X being equal to V, W, that is, changing the sign in the same way as the linear function of $(x,\, y,\, z)$ rationally, and consequently each of the original four roots, may be represented by an equation of the form.

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Taking the conics to pass through the point $a = 0$, $y = 0$; their equations will be of the form

$$X = ax^2 + 2hxy + by^2 + 2fx + 2gy + qy = 0,$$

$$Y = ax^2 + 2hxy + by^2 + 2fx + 2gy + qy = 0,$$

$$Z = ax^2 + bxy + cy^2 + 2dy + ey^2 + qy = 0,$$

$$W = ax^2 + bxy + cy^2 + 2dy + ey^2 + qy = 0.$$

Now multiplying by the indeterminate quantities α, β, γ, δ; their equations will be of the form. We can determine the values of a, β, γ, δ so that in the sum $aX + \beta Y + \gamma Z + \delta W$ the coefficients of x, y, 1 shall vanish, and the term in xy shall, that is, in $\alpha\beta\gamma\delta$, of values of a, β, γ, δ, but as the case may a quartic equation for the values of the ratios; and say $\alpha\beta\gamma\delta : \beta\gamma\delta : \gamma\delta : \delta$ changing the coordinates (x, y), this remaining equation may be represented by $a^2 = 0$.

And it is clear that we then have, in the system of conics $aX + \beta Y + \gamma Z + \delta W = 0$, four conics the equations of which may be assumed to be of the form

$$X = ax^2 + ax,$$

$$Y = ax^2 + ax,$$

$$Z = ax^2 + ax,$$

$$W = a^2x + ax^2,$$

so that, by writing $a = ax + \cdots$, and f, f', f'', f''' are the same conic upon the same linear function of (x, y, z) rationally, and consequently each of the original four roots, may be represented by an equation of the form.

7. *The coordinates (x, y, z) of a point P are given as connected with the coordinates (x', y', z') of a point P' on a plane by equations*

$$x : y : z = a' : Y' : Z' : W';$$

whereof showing, we have

$$ax^2 + y' = 0 :$$

Discuss the connection of P' and P in regard to the equations of points and curves on the two surfaces.

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The equations $x : y : z : w = X' : Y' : Z' : W'$ are two equations establishing indeterminate parameters $x' : y' : z'$, &c. that the eliminating these we have between $(x,\, y,\, z)$ a single (homogeneous) equation representing a surface. To each point $(x,\, y,\, z)$ a single value $x' : y' : z'$ corresponds and similarly to each point $(x',\, y',\, z')$ a single (homogeneous) equation representing a surface. To each point $x' : y' : z' : 1$ which is a corresponding point $(x,\, y,\, z)$ on the surface.

To find the order of the surface, consider its intersection with any arbitrary line: ax + by + cz + dw = 0,

$$a'x + b'y + c'z + d'w = 0.$$

We have corresponding equations in the plane of intersection of the surface;

$$aX' + bY' + cZ' + dW' = 0,$$

$$a'X' + b'Y' + c'Z' + d'W' = 0;$$

viz. these are two curves: each of them passing through the common point of the four curves, and therefore having common besides these points, three other points: the order of the surface is therefore = 3.

To show that the common point ought to be the (as above) excluded, some further consideration is necessary. To the points of the surface $aX + bY + cZ + dW = 0$ corresponds the locus $aX'+bY'+cZ'+dW' = 0$ and similarly to the points of the linear curve $aX'+bY'+cZ'+dW' = 0$ corresponds the points of the surface $aX + bY + cZ + dW = 0$, then the corresponding loci on the other surface are conics: hence to a similar to the corresponding points of the surface, and which are not common ones (or any belonging to one or more of the conics) the points $(x,\, y,\, z)$, z' which is of the order $(\alpha,\, y)$ ought to be rejected; the equation in question is thus reduced to $z' = (xz - bX + zZ + xW)\, xs' + \cdots = 0$.

The same result may be obtained and may be further shown that the surface is a a, b, c are the same, the form of the original coordinates $(x,\, y,\, z)$ in $(x,\, y,\, z)$ to the corresponding points (in the form of the plane $bXy + aX' + aY' + bZW + gZW' = 0$).

In a particular case, the more general assertion of expression of $x' : y' : z'$, in terms of $(x,\, y,\, z)$ is to be considered, but every linear function of $(x,\, y,\, z)$ rationally, and consequently any sole conic of $(x,\, y,\, z)$ rationally; in particular if $x'^2 + y'^2 + z'^2 = 0$, and these equations may be such, are connected by a relation which, in regard to a fixed line, the plane through this line of the plane through the points $(x,\, y,\, z)$ and $(x,\, y,\, z)$.

The same may not be obtained, instead of the two equations, that is a single equation $a^2 z = 1$, having a double root. The equations are than that the $a^2 + b^2 + c^2 + d^2 + e^2 + 6f^2 + 12g + 4h = 0$.

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are, it is clear, $a = bc - rs$, &c. where a is a variable parameter; and accord- ingly the place are than as $r =$, $s =$, suppose. With arbitrary parameter; and through the common intersection of the curve $\Omega = 0$.

If f, V, W are binary functions of the form $(\alpha, b, \ldots \breve{a}, x, 1)^n$ with arbitrary coefficients, and if the equation $U = 0$, $V = 0$ have a common root, show how this can be determined in terms of the derived functions of the Resultant in regard to the coefficients of either function.

Show what results in regard to the common root can be obtained when the coefficients are not all of them arbitrary but (1) *each or either of the functions depends in any manner whatever on a set of arbitrary coefficients not entering into the other function,* (2) *the two functions depend in any manner whatever on one and the same set of arbitrary coefficients.*

Suppose

$$U = (a, b, \ldots \breve{a}, x, 1)^m, \quad \left(= ax^m + \frac{m}{1}bx^{m-1} + \&c.\right),$$

$$V = (a, b, \ldots \breve{a}, x, 1)^n, \quad \left(= ax^n + \frac{n}{1}bx^{n-1} + \&c.\right).$$

Then if R is the resultant, the equation $R = 0$ is the relation which must exist between the coefficients in order that the two functions U, V may have a common root. Moreover, if we assume that U and V may have a common root $x = \alpha$, then we have $U_\alpha = 0$, $V_\alpha = 0$; the coefficients of U_α, V_α being the same as those of U, V, and $x = \alpha$. We have then identically $U_\alpha = 0$, $V_\alpha = 0$, $R = 0$; that is, these equations have to be infinitesimally varied in such manner that U, V have still a common root $x = \alpha$; we have the equation $U = 0$, $V = 0$ subsist when a, b, …, the coefficients of U, and the corresponding parameters of V subsist a corresponding equation in the determinations of the roots a, b, c, &c. or some of them.

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There a series of equations giving the value of the common root $\frac{x}{y}(=\alpha)$ in the several ratios

$$\frac{1}{m} : \frac{dR}{da} : \frac{1}{m-1} : \frac{dR}{db} : \frac{1}{m-2} : \frac{dR}{dc} : \frac{1}{1} : \frac{dR}{dk} : \&c.$$

And it is clear that we have in like manner

$$\frac{1}{n} : \frac{dR}{da'} : \frac{1}{n-1} : \frac{dR}{db'} : \&c.$$

It is clear that if U involves, in any manner whatever, the coefficients a, b, …, which do not enter into the function V, then we have precisely the same expressions as above; and similarly if V involves coefficients which precisely the common root.

A system of equations satisfied by the values x, y which belong to the common root.

But if the coefficients a, b, … and a', b', &c. are contained in any manner whatever in the function U and V respectively, then we are unable to obtain the common root, either as a rational function of the variables, or by aid of these equations.

11. *Find the differential equation* $\frac{dV}{dt} = 0$ *for the absolute force at, a being the distance of the moving system, on the axes of the moving system.*

Taking as coordinates r the radius vector, ω the longitude, y the latitude, the equation of the vis viva function is

$$T = \tfrac{1}{2}(r'^2 + r^2 \cos^2 y\, \omega'^2 + r^2 y'^2),$$

hence

$$\frac{dT}{dr'} = r' = a \quad \text{suppose},$$

$$\frac{dT}{d\omega'} = r^2 \cos^2 y\, \omega' = \nu \quad ,$$

$$\frac{dT}{dy'} = r^2 y' = \mu \quad ,$$

and the expression of T in terms of r, a, y and the new coordinates a, ν, μ is

$$T = \frac{1}{2}\left(a^2 + \frac{\mu^2}{r^2} + \frac{\nu^2}{r^2\cos^2 y}\right);$$

whence writing

$$H = \frac{1}{2}\left(a^2 + \frac{\mu^2}{r^2} + \frac{\nu^2}{r^2\cos^2 y}\right) + V,$$

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9. *The normal at each point of a principal section of an ellipsoid is intersected by the normal at a consecutive point on the principal section: show that the locus of the point of intersection is an ellipse having four (real or imaginary) common points with the evolute of the principal section.*

The principal section is the convenience taken to be that in the plane of xz, the meridian of the ellipsoid; taking x, z as the coordinates of a point in the plane of xz, and considering the equations of the ellipse; taking a, y, z as the coordinates of a point in the meridian of the ellipsoid; we have for the normal at the point (a, z)

$$\frac{x-x'}{\frac{x'}{a^2}} = \frac{y-y'}{\frac{y'}{b^2}} = \frac{z-z'}{\frac{z'}{c^2}} = -B,$$

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or, substituting for α, the foregoing values,

$$x = x'\left(1 - \frac{B}{a^2}\right), \quad z = z'\left(1 - \frac{B}{c^2}\right);$$

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The principal section in its convenience taken to be that in the plane of xz, the meridian of the ellipsoid; taking x, z as the coordinates of a point in the plane of xz.

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we have

$$\frac{a^2x^2}{c^2} + \frac{a^2z^2}{c^2} = 1,$$

the required form, which is the equation in question.

If the point (X, Y, Z) is on the evolute, it is clear that the corresponding point of the locus will be a point of the evolute of the principal section, and to prove that the locus touches the evolute, it is only necessary to show that the tangent of the locus is also the tangent of the evolute, and which is the same way thing, that the tangent of the locus passes through the umbilics.

Now in the evolute we have

$$X^2 = a^2z'^2 = \rho^2\frac{x'^2}{a^2}, \quad Y^2 = b^2 = \rho^2\frac{y'^2}{b^2};$$

the corresponding values of a, z being

$$x = \frac{a^2X}{p^2}, \quad z = \frac{c^2Z}{p^2}.$$

Take ξ, ζ the current coordinates of a point in the tangent of the locus at a, z; or, substituting for a, z the foregoing values,

$$\frac{X\xi}{a} + \frac{Z\zeta}{b} = 1,$$

and there should be satisfied by $\xi = X$, $\zeta = Z$, or we ought to have

$$\frac{X^2}{a} + \frac{Z^2}{b} = 1,$$

and this equation is in fact true for the values of X, Z in the umbilics; viz. this that is, $\beta - r' = a$, which is in fact the value of β.

There is obviously a contact in each quadrant, that is, there are four contacts the locus passes the umbilics: but if this is the case, then the four contacts will be in the same number of imaginary or rather of i^2.

The same theorem holds good in regard to the principal sections, an ellipsoid by those, the umbilic being imaginary; but the property of contact at the umbilics evolutes are an attraction of the form may be taken to be unitless real, y is then in the probability of a number of heads between a, $a + a$ as one finite quantity; but a, an entirely on number ones in number, the probability of such a number between a and $a + a$ is nearly as may be.

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10. *An evolute being chain of given length is suspended from two fixed points in the same horizontal plane: show that this subject to a condition as to the length of the figure of equilibrium may consist of portions of two distinct catenaries.*

The two parts of the chain will each of them be a portion of a catenary, viz. they will either coincide with each other, forming a twice repeated portion of a catenary which is therefore a possible form of equilibrium, or they will form portions of two distinct catenaries. That the latter form is in some cases possible, appears from the case of a very long chain in which case to the difference of the two positions of equilibrium in which the upper portion is nearly a straight line. In fact, that, in the case of a chain (or rope) just long enough as to reach the ground, there are two forms of equilibrium are equal to a vertex; viz. a vertex, in which the depth, no slope side of the catenary is equal in a regular manner, observe that, in order to the existence of such a form of equilibrium, the necessary condition is, that the tension at A or B must be equal to the two cataries. From this condition, which is the existence of two distinct catenaries, it appears that, when the tension at the joint of a catenary is proportional to the height above the directrix of the catenary, hence the condition is that the directrix of A, and also the line of catenaries through the same directrix, must take the same value the directrix of the directrix.

Take $AB = 2c$, the length of the chain $= 2l$. Take β for the distance of the directrix below the points A, B; c for the parameter of the catenary (or distance of the lowest point above the directrix), β being of course unknown. The taking the origin at the mid-point of the directrix, and the axis of y vertically upwards, the equation of the catenary is

$$y = \tfrac{1}{2}c(e^x + e^{-x}),$$

whence for the point A or B,

$$\beta = \tfrac{1}{2}c(e^c + e^{-c}),$$

and the are measured from the lowest point being

$$s = \tfrac{1}{2}c(e^x - e^{-x}).$$

Hence, assuming that there are two distinct catenaries, if the parameters are c, c', we have

$$\tfrac{1}{2}c(e^x + e^{-x}) + \tfrac{1}{2}c'(e^{x'} + e^{-x'}) = b,$$

$$\tfrac{1}{2}(e^x - e^{-x}) - \tfrac{1}{2}c'(e^{x'} - e^{-x'}) = l,$$

which are the conditions for the determination of a, c', and it is to be shown that these can be satisfied otherwise than by taking $c = c'$.

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Trace the two curves

$$y = \tfrac{1}{2}c(e^x + e^{-x}),$$
$$y = \tfrac{1}{2}c(e^x - e^{-x}),$$

shown respectively by the black line and the dotted line in fig. 2. Draw any line parallel to the axis of x, meeting the first curve in the points P, P' respectively, and let the ordinates MP, $M'P'$ meet the second curve in the points Q, Q' respectively.

Fig. 2.

then it is clear, that if for a given value of b the line PP' can be drawn in such- wise that $MQ + M'Q' = b$, there will be the required two values $c = OM$, $c' = OM'$, and $c \neq c'$.

And since for MP very large we have MQ diminishes until it attains a finite value, large, and as MP diminishes, $MQ + M'Q'$ also diminishes until it attains a minimum value, PP' has to do then MP' becomes a tangent to the first curve at the vertex; for the two values of the curve PP' being parallel to and below this tangent value, the equation $MQ + M'Q' = b$ is a separate question which is not here considered.

11. *A particle is attracted to two centres of force, one of these on the surface, the other revolving about the origin in a circle in the plane of xy with a uniform angular velocity ν; And the equations of motion, and writing r for the velocity of the particle and A for the constant area (about the fixed centre) in the plane of xy, show that there is a first integral giving the value of $r^2 - 4\nu A\frac{dx}{dy} - g\nu$ in terms of the coordinates of the particle and of the revolving centre.*

Take f, ϕ for the coordinates of the moving centre, so that

$$f = a\cos\nu t, \quad \phi = a\sin\nu t;$$

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the equations of motion are

$$\frac{d^2x}{dt^2} = -P\frac{x}{r} - Q\frac{x-f}{p}, \quad \frac{d^2y}{dt^2} = -P\frac{y}{r} - Q\frac{y-\phi}{p},$$

$$\frac{d^2z}{dt^2} = -P\frac{z}{r} - Q\frac{z}{p},$$

where

$$r^2 = x^2 + y^2 + z^2,$$

$$p^2 = (x-f)^2 + (y-\phi)^2 + z^2.$$

We have

$$r\,dr = x\,dx + y\,dy + z\,dz,$$

$$p\,dp = (x-f)(dx-df) + (y-\phi)(dy-d\phi) + z\,dz.$$

But

$$dt = \nu \sin \nu t\,dt = -d\phi\,dt, \quad df = a\nu\cos\nu t\,dt = d\phi\,dt;$$

whence

$$p\,dp = x\,dx + y\,dy + z\,dz - \omega(\nu dt)dx - \nu(d\phi)dy$$

$$= x\,dx + y\,dy + z\,dz - \nu(p-q)(dx - q\,dy),$$

$$= x\,dx + y\,dy + z\,dz - \nu A(d\phi - q\,dy),$$

that is, the product of the three functions $r^2 - 4\nu A\frac{dx}{dy} - g\nu$, &c. is $= 0$.

Hence from the equations of motion

$$2\left(\frac{dx}{dt}\frac{d^2x}{dt^2} + \frac{dy}{dt}\frac{d^2y}{dt^2} + \frac{dz}{dt}\frac{d^2z}{dt^2}\right)$$

$$= -2P\left(\frac{dr}{r}\right) - 2Q\left(\frac{x\,dx + y\,dy + z\,dz}{p}\right)$$

$$= -2P\frac{dr}{r} - \frac{2Q}{p}\{p\,dp + \nu A(d\phi - q\,dy)\}$$

$$= -2P\frac{dr}{r} - 2Q\frac{dp}{p} + \frac{2\nu A}{p}\{x(d\phi - q\,dy)\};$$

which is in the equation of motion at once $r^2 = -2P\,dr - 2Q\,dp + \frac{2\nu A}{p}\,dx - \frac{2q\nu A}{p}\,dy$

$$v^2 - 2\nu A\left\{x(d\phi - q\,dy) + \frac{2\nu A}{p}(d\phi - q\,dy)\right\} = \int(-2P\,dr - 2Q\,dp).$$

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But we have

$$v^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2,$$

the foregoing equation may be written

$$\nu - 2\nu A\frac{d\phi}{dr} = -2P\frac{dr}{r} - 2Q\frac{dp}{p} + \int \frac{2A}{p}\,dy.$$

whence

$$v^2 - 2\nu A\frac{dq}{dt} = \int \left\{-2P\frac{dr}{r} - 2Q\frac{dp}{p} + \frac{2\nu A}{p}\,dy\right\},$$

the required result.

12. *If a, y are the coordinates of a particle moving in plane under the action of a central force varying as (distance)$^{-3}$; write down the expressions of the constants a_1, a_2 in terms of a and $\frac{d^2q}{dt^2}$ of fixed arbitrary constants; and (in case of distorted motion) starting from the equations*

$$\frac{d^2a}{dt^2} = -\frac{P_1}{a^2r^2}, \quad \frac{d^2q}{dt^2} = -\frac{P_2}{q^2r^2},$$

(the notation is be explained), indicate the process of finding the variations of the constants in terms of (1) $\frac{d^2a}{dt^2}$, $\frac{d^2q}{dt^2}$, (2) the derived functions of a in regard to the constants.

We have

$$x = \frac{\cos m - q}{1-\cos m}\,a, \quad y = a\sqrt{(1-r^2)}\frac{\sin m - n}{1-\cos m},$$

where

$$m - n\sin m = \mu(t - t_0) + \nu,$$

an equation serving to express m in terms of the time, and where a, n, t_0, ν; the four- going equations therefore, in effect, give x, y in terms of t and the constants a, r, t_0, n.

In the second part of the question, Ω is a given function of a, y, the differential coefficients $\frac{d\Omega}{da}$, $\frac{d\Omega}{dy}$ being the partial ones in regard to a, y respectively. The equation $\Omega = 0$ signifies that the variation of x, in so far as it arises from the variation of the constants, is $= 0$; that is, the equation

$$\Delta x = \frac{dx}{da}da + d\beta = 0,$$

where α gives ν, y is the constants a, u, c, the differentiations $\frac{dx}{da}$, $\frac{dx}{db}$, &c. being the partial ones in regard to a, b, respectively. The differential $\Delta x = 0$ signifies that the solution of a, in so far as it arises from the variation of the constants, is $= 0$; (2) the derived derivatives of a in regard to the constants a, b, &c.

$$\frac{dx}{da}\,d\alpha + \frac{dx}{d\beta}\,d\beta + \frac{dx}{d\gamma}\,d\gamma + \frac{dx}{d\delta}\,d\delta = 0,$$

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The value of $x'\left(=\dfrac{dx}{dt}\right)$ is therefore obtained from that of x by differentiating in regard to t alone, as if a, c, e, ν were constants: viz. if x' is given as a function of t, a, c, e, ν, then the variation of x' in so far as it arises from the variation of the constants, $\Delta x' = \dfrac{dx'}{da}\,da + \dfrac{dx'}{dc}\,dc + \dfrac{dx'}{de}\,de + \dfrac{dx'}{d\nu}\,d\nu$ remains.

There are the like equations in regard to y, z; viz. in all, four equations which are linear in regard to $\dfrac{da}{da}, \dfrac{dc}{da}, \dfrac{de}{da}, \dfrac{d\nu}{da}$ and which serve to determine these quantities in terms of α, β, γ, δ.

Now considering the x, y as expressed in terms of a, c, e, …, then Ω becomes a function of these quantities; the differential coefficients $\dfrac{d\Omega}{da}$, &c., being connected with the original differential coefficients $\dfrac{d\Omega}{dx}$, $\dfrac{d\Omega}{dy}$ by the equations

$$\frac{d\Omega}{da} = \frac{d\Omega}{dx}\cdot\frac{dx}{da} + \frac{d\Omega}{dy}\cdot\frac{dy}{da}, \quad \&c.$$

As there are four equations, $\dfrac{d\Omega}{dx}$, $\dfrac{d\Omega}{dy}$ can be expressed in an infinity of ways in terms of $\dfrac{d\Omega}{da}$, $\dfrac{d\Omega}{dc}$, &c., and, considering $\dfrac{d\Omega}{da}$, &c., as given in terms of $\dfrac{d\Omega}{dx}$, $\dfrac{d\Omega}{dy}$, we can in an infinity of ways express $\dfrac{da}{da}$, &c., as linear functions of $\dfrac{d\Omega}{da}$, $\dfrac{d\Omega}{dc}$, &c. But there is no form (obtained by combining the equations in a particular manner) wherein the coefficients of the non-constant quantities are functions of a, c, e, … without a contact; and this is the form most simply employed for the expression of $\dfrac{da}{da}$, &c. in terms of $\dfrac{d\Omega}{da}$, $\dfrac{d\Omega}{dc}$, &c. in the method wherein these quantities are made use of in this respect.

I remark upon the present question, that the answer ought to be in evidence perfectly familiar to every student in *Physical Astronomy*; and that a student ought to be able to present it in a clear and lucid form: the question being in fact intended as a test of ability in this respect.

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13. *Explain the course of the geodesic lines on a spheroid of revolution; and in particular show that the meridian is included in-virtue of which any geodesic line, com- sidered as starting from a given point, cannot at every passage of the vertex be a shortest line.*

From each point on the surface a geodesic line may be drawn in any direction whatever along the surface, that is, through each point of the surface there is a singly infinite series of geodesic lines. A geodesic line undulates (in the manner of a sinuous sinusoidal between two parallels of latitude, called the upper and lower parallels respectively; viz. considering it as starting from a point a on the upper parallel, it descends to the equator, then to the lower parallel, ascends to the equator, on the upper parallel and again, viz. this is between A, B, C, D, on the equator, on the upper parallel and again, viz. on the upper limit, C being the upper limit, B is the lower limit, and similarly the parallels between C and A, D and B, are the same so on. Now, in general, the equator is or upper and a lower at the same time; so that, in general, D does not concide the equator, A being given, A ascending will be: D, will be a portion of a shorter line.

The equatorial are AAA all A, the meridian portions of these the in- crease between as long D have 100°, the value of an become infinitely large 180°. If we are in question is commensurable with 100°, the geodesic line will be a closed curve; otherwise to its case, the limit is not, then it is not a closed curve, but presents oscillating for ever between the two parallels. It is the limiting case where the inclination is = 90°, the geodesic line is obviously a meridian.

Considering a geodesic line starting in a given direction from a point A, and the geodesic line from the same point A in the consecutive direction, it appears from the foregoing account of the configuration of the lines, that the two lines will inter- sect each other in general an infinite number of times: supposing that they first intersect in a point B, then by a general theorem of Jacobi's, the geodesic line AB is a shortest line from B to any point nearer than B; it is a shortest line so far as the geodesic line is obviously a meridian.

[538]

Extract from a Letter to Mr C. W. Merrifield.

[From the Messenger of Mathematics, vol. I. (1872), pp. 87, 88.]

THE general integral of the equations

$$\frac{p}{x} = \frac{dt}{a} = \frac{dq}{y}.$$

$\left[\text{where } a,\ \beta,\ \gamma,\ \delta = \dfrac{dq}{dx},\ \dfrac{dq}{dy},\ \dfrac{dq}{ds},\ \dfrac{dq}{dt},\right]$ can, I think, be found, viz. $\dfrac{p}{x} = \dfrac{dt}{a} = \dfrac{dq}{y}$ $s =$ function x, and $\frac{dt}{ds} = \frac{a}{y}$ gives $s =$ function t. But $s =$ function x, is integrated as the equation of a developable surface (p instead of s), we must have

$$p = ax + by + p',$$

$$0 = ax + y + p';$$

similarly, $s =$ function t, gives

$$\left(a'\frac{dx}{ds},\quad y'\frac{dq}{dt}\right);$$

a and p functions of t, and

$$\left(a' = \frac{dx}{ds},\quad y' = \frac{dq}{dt}\right);$$

Observe that the constants here have been so taken, that $\frac{dq}{ds} = a$, $\frac{dq}{dt} = b$; in order that δ may, in the two pairs of equations, mean the same function of (x, y), we must have

$$s' = \frac{b}{a} - \frac{p}{x}.$$

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that is,

$$b = \int \frac{dx}{y}, \quad t' = \int \frac{dq}{x};$$

or, writing $a = dt$, $p = q$, we have

$$p = abh + pb + q,$$
$$q = bs + p\int \frac{dx}{y} - p\int \frac{a}{y}\, dx,$$

where

$$abh + y + qh = 0.$$

The last equation gives b as a function of (x, y), and the values of p, q are then such that $dx + pdx + qdy$ is a complete differential, so that we obtain z by the integration of that equation.

A simple example is

$$p = \tfrac{1}{2}bs - bq, \quad q = -bs + g\log b, \quad bs - y = 0,$$

that is,

$$p = -\tfrac{1}{4}\frac{y^2}{x}, \quad q = -y + y\log\frac{y}{x},$$

whence

$$z = \tfrac{1}{4}y^2 \log\frac{y}{x} - \tfrac{1}{4}y^2;$$

we have

$$r = \tfrac{1}{2}\frac{y^2}{x^2}, \quad s = -\frac{y}{x}, \quad t = \log\frac{y}{x},$$
$$a = -\frac{y^2}{2x^2}, \quad \beta = -\frac{y}{x}, \quad \gamma = -\frac{1}{y}, \quad \delta = \frac{1}{x};$$

or

$$\frac{a}{\beta} = \frac{p}{x} = \frac{b}{y}\left(= a - \frac{p}{x}\right);$$

as it should be.

Cambridge, 28 July, 1871.

[539]

Further Note on Lagrange's Demonstration of Taylor's Theorem.

[From the Messenger of Mathematics, vol. I. (1872), pp. 105, 106.]

This refers to a paper, Wilkinson, "Note on Taylor's Theorem," *Messenger of Mathematics*, same volume, pp. 96, 97, discussing the "Note on Lagrange's Demonstration of Taylor's Theorem," [248, ante, pp. 493—495].

[540]

On a Property of the Torse Circumscribed about Two Quadric Surfaces.

[From the Messenger of Mathematics, vol. I. (1872), pp. 111, 112.]

THE property mentioned by Mr Townsend in his paper(*) in the August No., "On a Property in the Theory of Confocal Quadrics," may be demonstrated in a form which, it appears to me, better exhibits the foundation and significance of the theorem.

Starting with two given quadric surfaces, the torse circumscribed about these touches each of a singly infinite series of quadric surfaces, any two of which may be used (instead of the two given surfaces) to determine the torse; in the series are included four conics, one of them in each of the planes of the self-conjugate tetrahedron of the two given surfaces; and if we attend to only two of these conics, the two conics are in fact any two conics whatever, and the torse is the circumscribed torse of the two conics; or, what is the same thing, it is the envelope of the common tangent-planes of the two conics.

Consider new two conics U, U', the planes of which intersect in a line I; and let I meet U in the points L, M, and meet U' in the points L', M': take L the pole of I in regard to the conic U, and L' the pole of I in regard to the conic U'.

Take T any point on I, and draw TP touching U in P, and TP' touching U' in P': the points P, P' may be considered as corresponding points on the two conics.

Join AP and produce it to meet the line I in G; the line APG is in fact the polar of T in regard to the conic U (for T being a point on I, the polar of T passes through A; and this polar also passes through P); that is, the points T, G, and L, M are harmonics on the line I; whence also, in the plane of the conic U',

(*) *Messenger of Mathematics, same volume, pp. 48, 49.*

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the lines PG, PT and PL, PM are harmonic lines through the point P. It thus appears that in the particular case where the points L, M are the foci of the conic U', the line PG is the normal at the point P; and we may say in general that PG is the quasi-normal at the point P of the conic U'.

[Figure: Conic with quasi-normal construction — diagram omitted]

Consider now the torse circumscribed about the conics U, U'; the plane PTP' will represent any plane, and the line PP' any line of this torse; projecting on the plane of U' with the point A as centre of projection, the projection of PP is the line PG; which, as just seen, is the quasi-normal of the conic U' at the point P'.

The projection of the cuspidal curve is the envelope of line PG, which is the projection of the generating line PP' of the torse—viz. this envelope is the quasi-evolute of the conic U'; which is the theorem in question.

[541]

On the Reciprocal of a Certain Equation of a Conic.

[From the Messenger of Mathematics, vol. I. (1872), pp. 120, 121.]

THE following formula is useful in various problems relating to conics: the reciprocal equation of the conic

$$\lambda(px+qy+cz)(x+by+dz)+\mu(a''x+b''y+c''z)(a''x+b''y+c''z)=0$$

may be written indifferent in either of the forms

$$|\lambda\,\xi\,,\eta\,,\zeta\; = a\,\xi\,,\eta\,,\zeta\,,\; b\,\xi\,,\eta\,,\zeta\;|+4\lambda\mu\,\xi\,,\eta\,,\zeta\,,\; a'\,\xi\,,\eta\,,\zeta\,,\; b'\,\xi\,,\eta\,,\zeta\;|+|a'\,\xi\,,\eta\,,\zeta\,,\; b'\,\xi\,,\eta\,,\zeta\,,\; c'\,\xi\,,\eta\,,\zeta\;|=0,$$

and

$$|\lambda\,\xi\,,\eta\,,\zeta\; = a\,\xi\,,\eta\,,\zeta\,,\; b\,\xi\,,\eta\,,\zeta\;|+4\lambda\mu\,\xi\,,\eta\,,\zeta\,,\; a\,\xi\,,\eta\,,\zeta\,,\; b''\,\xi\,,\eta\,,\zeta\;|+|a''\,\xi\,,\eta\,,\zeta\,,\; b''\,\xi\,,\eta\,,\zeta\,,\; c''\,\xi\,,\eta\,,\zeta\;|=0.$$

In fact, in the reciprocal equation, noting the coefficient of β, it is

$$\{\lambda(bc'-b'c)+\mu(b''c'-b''c')\}^2=(Dab'-2ab'b'')(Dac'-2bc'c'),$$

viz. this is

$$\lambda^2(bc'-b'c)^2+\mu^2(b''c'-b''c')^2+2\lambda\mu\left\{\begin{array}{l}2bb'c''c''+2b''b'cc'\\-(bc+b'c')(b''c''+b''c'')\end{array}\right\}.$$

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or, as it may be written,

$$[\lambda(bc' - b'c) \pm \mu(b''c' - b''c')]^2 = D\lambda\mu \left\{ \begin{array}{l} 2bb'c''c'' + 2b''b'cc' \\ -(bc + b'c')(b''c'' + b''c'') \end{array} \right\}.$$

Taking the upper signs, this is

$$[\lambda(bc' - b'c) + \mu(b''c' - b''c')]^2 + 4\lambda\mu \left\{ \begin{array}{l} bb'c''c'' + b''b'cc', \\ -bc'b''c'' - b'cb''c'' \end{array} \right\},$$

viz. the term in $\lambda\mu$ is

$$+4\lambda\mu(bc'' - b''c)(b'c' - b'c''),$$

Taking the lower signs, it is

$$[\lambda(bc' - b'c) - \mu(b''c' - b''c')]^2 + 4\lambda\mu \left\{ \begin{array}{l} bb'c''c'' + b''b'cc', \\ -bc'b''c'' - b'cb''c'' \end{array} \right\},$$

viz. the term in $\lambda\mu$ is

$$+4\lambda\mu(bc' - b''c)(b'c'' - b''c').$$

And it is hence easy to infer the forms of the other coefficients, and to arrive at the foregoing result.

[548] (CONTINUED)

On Listing's Theorem.

[From the Messenger of Mathematics, vol. I. (1872), pp. 81–88 — continuation.]

b. It might be possible to find an analogous definition, but the most simple one is that a denotes the number of ovoids (unilinked ovoids) or other surfaces not bounded by any point or line.

d is the number of spaces, reckoning as one of them infinite space.

a'' is the sum, for the several spaces, of the number of circuits in each space: the word circuit here signifying a path in the space from a point to itself; all circuits which can by continuous variation be made to coincide being considered as identical, and the connectible circuit reducible to the point itself being throughout disregarded. Moreover, we count only the simple circuits, disregarding any circuit which can be obtained by a repetition or combination of these. Thus for infinite space, or for the space within an ovoid, there is only the evanescent circuit, or there is no circuit to be counted; and the same is the case if within such space we have any number of ovoidal blocks (the term will, I think, be understood without explanation); but if within the space we have an oval, ring, or other ring-block of any kind whatever, then there is therefore the connectible circuit, a circuit interlacing the ring-block, and we count one circuit; and so if there are a ring-blocks, either separate or interlacing each other in any manner, then there are, and we accordingly count, a circuits. So for the space inside a ring there is (besides the connectible circuit), and we count, one circuit; and the case is the same if we have within the ring any number of ovoidal blocks whatever; but if there is within the ring an oval ring or other ring-block, then there is one new circuit, and we count in all (for the space in question) two circuits.

p is the number of detached parts of the figure; or, say the number of detached aggregates of points, lines, and surfaces. Observe, that rings interlacing each other in any manner (but not intersecting) are considered as detached; as also two closed surfaces, one within the other, are considered as detached. The figure may be infinite space alone; we have then $p = 0$.

The complete which follow will further illustrate the meaning of the terms and nature of the theorem; and will also indicate in what manner a general demonstration of the theorem might be arrived at.

1. Infinite space.

$$\begin{array}{llllll}
a=0, & A=0, & & & & \\
b=0, & a'=0, & B=0, & & & \\
c=0, & a''=0, & w=0, & C=0, & & \\
d=1, & a'''=0, & & D & & =1 \\
p=0, & & & p-1 & & =-1 \\
 & & & \overline{0} & & =\overline{0}.
\end{array}$$

2. Spherical surface.

$$\begin{array}{llllll}
a = 0, & A = 0, & & & & \\
b = 0, & a' = 0, & B = 0, & & & \\
c = 1, & a'' = 0, & w = 1, & C = 1, & & \\
d = 2, & a''' = 0, & & D & & = 2 \\
p = 1, & & & p-1 & & = 0 \\
& & & \overline{2} & & = \overline{2}.
\end{array}$$

viz. the effect is to increase C by 2 and D and $p-1$ each by 1.

3. Spherical surface, with point upon it.

$$\begin{array}{llllll}
a = 1, & A = 1, & & & & \\
b = 0, & a' = 0, & B = 0, & & & \\
c = 1, & a'' = 0, & w = 1, & C = 1, & & \\
d = 2, & a''' = 0, & & D & & = 2 \\
p = 1, & & & p-1 & & = 0 \\
& & & \overline{2} & & = \overline{2}.
\end{array}$$

viz. the effect is to increase a and diminish c each by 1: that is, A is increased and C diminished each by 1.

4. Spherical surface with two points.

$$\begin{array}{llllll}
a = 2, & A = 2, & & & & \\
b = 0, & a' = 0, & B = 0, & & & \\
c = 1, & a'' = 0, & w = 1, & C = 1, & & \\
d = 2, & a''' = 0, & & D & & = 2 \\
p = 1, & & & p-1 & & = 0 \\
& & & \overline{2} & & = \overline{2}.
\end{array}$$

viz. the second point increases a and a' each by 1, that is, it increases A and diminishes C each by 1.

And for each new point on the spherical surface there is this same effect; we may have, for the next case:

5. Spherical surface with points $(n \geq 2)$.

$$\begin{array}{llllll}
a = n, & A = n, & & & & \\
b = 0, & a' = 0, & B = 0, & & & \\
c = 1, & a'' = n-1, & w = 0, & C = 2-n, & & \\
d = 2, & a''' = 0, & & D & & = 2 \\
p = 1, & & & p-1 & & = 0 \\
& & & \overline{2} & & = \overline{2}.
\end{array}$$

Imagine that besides the n points there is an aperture (bounded by a closed curve); the case is:

6. Spherical surface with n points ($n \geq 2$) and aperture.

$$\begin{array}{llllr} a = n, & A = n, & & & \\ b = 1, & a' = 1, & B = 0, & & \\ c = 1, & a'' = n, & w = 0, & C = 1 - n, & \\ d = 1, & a''' = 0, & & \quad D & = 1 \\ p = 1, & & & \quad p - 1 & = 0 \\ & & & \quad \overline{1} & = \overline{1}. \end{array}$$

viz. b and a' are each increased by 1, and therefore B is still unaltered; a'' is increased, and therefore C diminished, by 1; a''' is increased, and therefore D diminished, by 1, and each new aperture produces the like effect. Thus we have:

7. Spherical surface with apertures ($n \geq 2$) and two apertures.

$$\begin{array}{llllr} a = n, & A = n, & & & \\ b = 2, & a' = 2, & B = 0, & & \\ c = 1, & a'' = n + 1, & w = 0, & C = -n, & \\ d = 0, & a''' = 0, & & \quad D & = 0 \\ p = 1, & & & \quad p - 1 & = 0 \\ & & & \quad \overline{0} & = \overline{0}. \end{array}$$

viz. b and a' are each increased by 1, and therefore B is still unaltered; a'' is increased, and therefore C diminished, by 1; a''' is increased, and therefore D diminished by 1, and each new aperture produces the like effect. Thus we have:

8. Spherical surface with apertures ($n \geq 2$).

$$\begin{array}{llllr} a = 0, & A = 0, & & & \\ b = n, & a' = n, & B = 0, & & \\ c = 1, & a'' = n - 1, & w = 0, & C = 2 - n, & \\ d = 1, & a''' = n - 1, & & \quad D & = 2 - n \\ p = 1, & & & \quad p - 1 & = 0 \\ & & & \quad \overline{2 - n} & = \overline{2 - n}. \end{array}$$

where, comparing with case 5, we see the different effects of a point and an aperture.

9. Spherical surface with n apertures ($n \geq 2$) and a point or points on each or any of the bounding curves of the apertures.

   If on the bounding curve of any aperture we place a point, this increases a, and therefore A, by 1; the bounding curve is no longer a simple closed curve, and we thus also have a diminished, and therefore B increased by 1; and the balance still holds.

Placing on the same bounding curve a second point, a, and therefore A, is increased by 1, but the bounding curve is converted into two distinct curves, that is, b, and therefore A, is increased by 1, and the balance still holds. And the like for each new point on the same bounding curve.

10. Spherical surface with a point connected in any manner by lines.

    Reverting to the cases 4 and 5, by joining any two points by a line, we increase b and therefore B by 1; but as regards a' the two united points take effect as a single point; that is, a' is diminished and therefore C increased, by 1, the balance is therefore unbalanced.

    The case is the same for each new line, if only we do not thereby produce on the surface a closed polygon, or partition or existing closed polygon: in each of these cases we also increase b and therefore B by 1; and instead of diminishing a', we increase c by 1, and therefore still increase C by 1, and the balance continues to subsist.

    By continuing to join the several points on at last arrive at a spherical surface partitioned into polygons in any manner whatever; or, what is the same thing, we have

11. Closed polyhedral surface. Here, if S is the number of summits, F the number of faces, E the number of edges; then

$$\begin{array}{lllll}
a = S, & A = S, & & & \\
b = E, & a' = 0, & B = E & & \\
c = F, & a'' = 0, & w = 0, & C = -F, & \\
d = 2, & a''' = 0, & & D & = 2 \\
p = 1, & & & p - 1 & = 0 \\
 & & & \overline{S + F} & = \overline{E + 2},
\end{array}$$

    as that we have Euler's theorem. Observe that this theorem (Euler's) does not apply to annular polyhedral surface, or to polyhedral shells. For instance, consider a shell, the exterior and interior surfaces of which are each of them a closed polyhedral surface; $S = S' + S''$, $F = F' + F''$, $E = E' + E''$, where S', F', E' and S'', F'', E'' and therefore $S + F$, $E + 4$. Listing's theorem, of course, applies, viz. we have

12. ?

$$\begin{array}{lll}
a = S' + S'', & A = S' + S'', & \\
b = E' + E'', & B = E' + E'' & \\
c = F' + F'', & C = -F' - F'', & \\
d = 3, & D & = 3 \\
p = 1, & p - 1 & = 0 \\
 & \overline{S + F} & = \overline{E + 3}.
\end{array}$$

As another group of examples, consider a plane rectangle, for instance, a sheet of paper bounded by its four edges; here

13.

$$\begin{array}{llll} a = 4, & A = 4, & & \\ b = 4, & a' = 0, & B = 4 & \\ c = 1, & a'' = 0, & w = 0, & C = -1, \\ d = 1, & a''' = 0, & & \quad D \qquad\qquad = 1 \\ p = 1, & & & \quad p-1 \qquad = 0 \\ & & & \quad \overline{5} \qquad\qquad = \overline{5}. \end{array}$$

Let the paper be formed into a tube by uniting two opposite sides, the suture not being obliterated, but continuing as a line drawn lengthwise from one extremity of the tube to the other; here

14.

$$\begin{array}{llll} a = 2, & A = 2, & & \\ b = 3, & a' = 0, & B = 3 & \\ c = 1, & a'' = 0, & w = 0, & C = -1, \\ d = 1, & a''' = 0, & & \quad D \qquad\qquad = 1 \\ p = 1, & & & \quad p-1 \qquad = 0 \\ & & & \quad \overline{3} \qquad\qquad = \overline{3}. \end{array}$$

Let the suture be obliterated, or that we have simply a tube open at each end; here

15.

$$\begin{array}{llll} a = 0, & A = 0, & & \\ b = 2, & a' = 0, & B = 2 & \\ c = 1, & a'' = 0, & w = 0, & C = -1, \\ d = 1, & a''' = 0, & & \quad D \qquad\qquad = 1 \\ p = 1, & & & \quad p-1 \qquad = 0 \\ & & & \quad \overline{1} \qquad\qquad = \overline{1}. \end{array}$$

Let the tube be formed into an annulus by bending it round and joining the two extremities, the suture not being obliterated but continuing as the closed curve round the tube; here

16.

$$\begin{array}{llll} a = 0, & A = 0, & & \\ b = 1, & a' = 0, & B = 1 & \\ c = 1, & a'' = 0, & w = 0, & C = -1, \\ d = 1, & a''' = 0, & & \quad D \qquad\qquad = 0 \\ p = 1, & & & \quad p-1 \qquad = 0 \\ & & & \quad \overline{0} \qquad\qquad = \overline{0}. \end{array}$$

Let the suture be obliterated, so that we have simply a tubular annulus; here

17.

$$\begin{array}{llllr}
a=0, & A=0, & & & \\
b=0, & a'=0, & B=0 & & \\
c=1, & a''=1, & w=0, & C=0, & \\
d=2, & a'''=0, & & \quad D & =0 \\
p=1, & & & \quad p-1 & =0 \\
 & & & \quad \overline{0} & =\overline{0}.
\end{array}$$

We may compare herewith the case of a simple annulus or closed curve.

18.

$$\begin{array}{llllr}
a=0, & A=0, & & & \\
b=1, & a'=1, & B=0 & & \\
c=0, & a''=0, & w=0, & C=0, & \\
d=1, & a'''=0, & & \quad D & =0 \\
p=1, & & & \quad p-1 & =0 \\
 & & & \quad \overline{0} & =\overline{0}.
\end{array}$$

Add to such an annulus, for instance, three radii meeting in the centre; thus

19.

$$\begin{array}{llllr}
a=4, & A=4, & & & \\
b=6, & a'=0, & B=6 & & \\
c=0, & a''=0, & w=0, & C=0, & \\
d=1, & a'''=3, & & \quad D & =-2 \\
p=1, & & & \quad p-1 & =0 \\
 & & & \quad \overline{-2} & =\overline{-2}.
\end{array}$$

Let the last-mentioned figure become tubular, all sutures being obliterated; then

20.

$$\begin{array}{llllr}
a=0, & A=0, & & & \\
b=0, & a'=0, & B=0 & & \\
c=1, & a''=4, & w=1, & C=-4, & \\
d=2, & a'''=0, & & \quad D & =2 \\
p=1, & & & \quad p-1 & =0 \\
 & & & \quad \overline{-4} & =\overline{-4}.
\end{array}$$

And so if instead of the tubular figure, annulus with three radii, we had a tubular figure, annulus with diameter, then

21.

$$\begin{array}{llllr}
a = 0, & A = 0, & & & \\
b = 0, & a' = 0, & B = 0 & & \\
c = 1, & a'' = 2, & w = 1, & C = -2, & \\
d = 2, & a''' = 0, & & \quad D & = 2 \\
p = 1, & & & \quad p - 1 & = 0 \\
 & & & \quad \overline{-2} & = \overline{-2}.
\end{array}$$

and the like in other cases.

[549]

Note on the Maxima of Certain Factorial Functions.

[From the Messenger of Mathematics, vol. II. (1873), pp. 129, 130.]

I CONSIDER the functions

$$\begin{aligned}
y_1 &= x(x-1),\\
y_2 &= x(x-\tfrac{1}{2})(x-1),\\
y_3 &= x(x-\tfrac{1}{3})(x-\tfrac{2}{3})(x-1),\\
&\vdots\\
y_p &= x\left(x-\tfrac{1}{p}\right)\left(x-\tfrac{2}{p}\right)\cdots\left(x-\tfrac{p-1}{p}\right)(x-1).
\end{aligned}$$

Attending only to the absolute values, disregarding the signs, y_p has a maximum, viz. if x be odd, $=\frac{1}{2}p+1$ suppose, these are

$$Y_1,\ Y_3,\ \ldots,\ Y_p,\ Y_{p+2},\ Y_{p+4},\ \ldots,\ Y_n,$$

where Y_{p+m} corresponds to the value $x=\frac{1}{2p}+\frac{m}{p}$, viz. to values of x between

$$0 \text{ and } \tfrac{1}{2p},\ \tfrac{1}{p} \text{ and } \tfrac{2}{p},\ \ldots,\ \tfrac{p-1}{p} \text{ and } \tfrac{p+1}{2p},\ \text{ and } \tfrac{1}{2}.$$

But if n be even, $=2p$ suppose, these the maxima are

$$Y_1,\ Y_3,\ \ldots,\ Y_{p-1},\ Y_p,\ Y_{p+2},\ \ldots,\ Y_n,$$

where Y_{p+m} corresponds to values of x between

$$0 \text{ and } \tfrac{1}{2p},\ \tfrac{1}{p} \text{ and } \tfrac{2}{p},\ \ldots,\ \tfrac{p-1}{p} \text{ and } \tfrac{1}{2}.$$

In every case the maximum decreases from Y_1, which is the greatest, to Y_p or Y_{p+1}, which is the least; in particular, $n = 2p+1$, there

$$Y_{p+r} = \left[\left(\tfrac{r}{2p+1}\right)\cdots\left(\tfrac{1}{r}\right)\right](-1)$$

$$= \left(\tfrac{2p-1}{1}\right)\tfrac{r-1}{p+1}\cdots\tfrac{p-r-2}{p+1}$$

$$= \pm\left[\frac{1\cdot 3\cdots(2p-1)}{2^{p+1}p^{p+1}}\right]^{1/r}\frac{(2p-r+1)}{2p+1}\left(\frac{1}{r}\right)\cdots\left(\frac{1}{p}\right)\frac{(r-1)!}{(p+r)!}$$

which is

$$= \pm\frac{\Gamma\left(\frac{1}{2}+\frac{r}{2p+1}\right)\Gamma\left(\frac{1}{2}-\frac{r}{2p+1}\right)}{(2p+1)^{2p+1}}\cdot\frac{\Gamma(p+1-r)}{\Gamma(p+1)\Gamma(p+r+1)}.$$

Suppose p is large; then, as for large values of x,

$$\Gamma(x+a) = \sqrt{2\pi}\,x^{x+a-\frac{1}{2}}e^{-x},$$

we have

$$\Gamma\left(p+\tfrac{1}{2}\right) = \sqrt{2\pi}(p)^p e^{-p} + \frac{1}{12p} + \cdots$$

$$(2p+1)^{2p+1} = (2p)^{2p}\left(1+\tfrac{1}{2p}\right)^{2p}\left(1+\tfrac{1}{2p}\right) = \frac{e^{2p}}{2p}\,p^p\,e^{-p},$$

and as

$$Y_{p+r} = \frac{2^{2p}p^{2p-1}}{e^{2p}}\,\frac{e^{2p}}{2p}\,p^p e^{-p}\left(\frac{1}{2}\right)^{p+1}.$$

Also Y_1 corresponds approximately to

$$x = \frac{1}{2p+1} \approx \frac{1}{2p}.$$

$$Y_1 = \tfrac{1}{2p}\cdot\tfrac{1}{2p}\cdot\tfrac{2p-1}{2p}\cdot\tfrac{2p}{2p}\cdots\tfrac{p-r}{p+1}\cdots\tfrac{r-1}{p+1} = \tfrac{1}{2p}\,\Gamma\left(1+\tfrac{1}{2p}\right)\Gamma\left(p+1-\tfrac{1}{2p}\right)\cdot\tfrac{1}{\Gamma(p+1)}$$

and

$$\Gamma(p+\tfrac{1}{2}) = (2p)^{p+r-\frac{1}{2}}e^{-2p}\left(1+\tfrac{1}{12p}\right) \sim \sqrt{2\pi}p^p e^{-p+\frac{1}{12p}-\cdots} = \sqrt{2\pi}p^p\,e^{-p+\frac{1}{12p}-\cdots},$$

and

$$(2p+1)^{2p+1} = (2p)^{2p+1}e^{1+\frac{1}{4p}-\cdots}\left(1+\tfrac{1}{2p}\right)$$

so that

$$Y_1 = \frac{\sqrt{2\pi}\,2p^p\,e^{-p+\frac{1}{12p}}}{2^{p+1}p^{p+1}e^{-p}} = \frac{p^{p+1}\sqrt{p}}{e^p}\cdots,$$

so that, p being large, Y_1 is far larger than Y_{p+r}.

[550]

Problem and Hypothetical Theorems in Regard to Two Quadric Surfaces.

[From the Messenger of Mathematics, vol. II. (1873), p. 137.]

Two conics may be circum-and-inscribable to an n-gon; viz. the conics may be such that there exists a singly infinite series of n-gons each inscribed in the first and circumscribed about the second of the conics. In particular they may be circum-and-inscribable to a triangle.

The following problem arises:

Consider two given quadric surfaces and a given line S; to find the planes through S, which cut the surfaces in two conics circum-and-inscribable to a triangle (it is presumed there are two or more such planes).

Let the surfaces be Θ, Θ', and let the line S a tangent to Θ meet Θ in the points A, B; if through S we draw two planes as above, then in the first plane the tangents from A, B to the section of Θ' will meet in a point C of Θ'; and in the second plane the tangents from A, B to the section of Θ' will meet in a point D of Θ'. The points C, D being thus determined the lines AB, AC, BC, AD, BD all touch the surface Θ', and it is presumed that the surfaces Θ, Θ' may be such that CD also touches the surface Θ'; viz. in this case we have a tetrahedron $ABCD$, the summits of which lie in the surface Θ' and the edges touch the surface Θ'; and not only so, but it is further presumed that the surfaces may be such that starting from any point A of Θ and using either any tangent to a properly selected tangent AB of Θ', it shall be possible to complete the figure as above; or, what is the same thing, the surfaces may be such that there exists a doubly or a triply infinite series of tetrahedra, the summits of each lying in Θ and the edges touching Θ'. It is also presumed that the faces of the tetrahedra all touch one and the same quadric surface Θ''.

[551]

Two Smith's Prize Dissertations.

[From the Messenger of Mathematics, vol. II. *(1873), pp. 145–148.]*

WRITE dissertations on the following subjects:

1. *The theory of interpolation, with a determination of the limits of error in the value of a function obtained by interpolation.*

2. *Determinants.*

The general problem is to find a function of x having given values for given values of x. The problem thus stated is of course indeterminate; in practice, we assume a certain form for the function y, the coefficients of which form are determined by the given conditions, viz. either y is known to be of the form in question, the actual value being then determined as above; or it is assumed that y is approximately equal to a function of the form in question, and the value is then approximately determined in such wise that, for the given values of x, the function y shall have its given values.

The ordinary case is when we have the values of y corresponding to a given value of x, and y is taken to be a function of the form $A + Bx + \cdots + Kx^{n-1}$.

Suppose to fix the ideas $n = 4$, and that p_1, p_2, p_3, p_4 are the values of y corresponding to the values a, b, c, d of x. We may at once write down the expression

$$\begin{aligned} y = {} & p_1 \frac{(x-b)(x-c)(x-d)}{(a-b)(a-c)(a-d)} \\ & + p_2 \frac{(x-a)(x-c)(x-d)}{(b-a)(b-c)(b-d)} \\ & + p_3 \frac{(x-a)(x-b)(x-d)}{(c-a)(c-b)(c-d)} \\ & + p_4 \frac{(x-a)(x-b)(x-c)}{(d-a)(d-b)(d-c)}, \end{aligned}$$

for clearly this is of the form in question $A + Bx + Cx^2 + Dx^3$, and y becomes $= p_1$ for $x = a$, $= p_2$ for $x = b$, &c. And the like for any value of n. This is known as Lagrange's interpolation formula.

The given values of x may be equidistant, say they are $0, 1, 2, \ldots, n-1$, and the corresponding values of y are y_0, y_1, …, y_{n-1}; then writing down the expression

$$y_x = y_0 + \frac{x}{1}\,\Delta y_0 + \frac{x \cdot x-1}{1 \cdot 2}\,\Delta^2 y_0 + \cdots + \frac{x \cdot x-1 \cdot \ldots \cdot x-n+1}{1 \cdot 2 \cdots n-1}\,\Delta^{n-1} y_0,$$

where, as usual,

$$\Delta y_0 = y_1 - y_0, \quad \Delta^2 y_0 = y_2 - 2y_1 + y_0, \quad \&c.,$$

then for $x = 0$, 1, 2, &c., the values of y are

$$\begin{aligned} &y_0, \\ &y_0 + \Delta y_0, \quad = y_1, \\ &y_0 + 2\Delta y_0 + \Delta^2 y_0, \quad = y_2, \quad \&c., \end{aligned}$$

or the required conditions are satisfied.

As regards the determination of the limits of error, taking a particular case $n = 4$, suppose that we have the values p_1, p_2, p_3, p_4 of y corresponding to the values 0, 1, 2, 3 of x, and that the true value of y is known to be

$$= A + Bx + Cx^2 + Dx^3 + Kx^4,$$

where K is a function of x, which for any value of x within the given values (i.e. from $x = 0$ to $x = 3$) is known to be at least $= P$ and at most $= Q$, i.e. $K = P + tQ$, where t is the ideas. P and Q are each positive. Q being the greater. Hence calculating the interpolation value of $y = Kx^4$ (the last term, the Kx^4 in question), we have

$$\begin{aligned} y = p_1 \frac{1}{6}\,\Delta p_0 + \frac{x \cdot x-1}{1 \cdot 2}\,\Delta^2 p_0 + \frac{x \cdot x-1 \cdot x-2}{1 \cdot 2 \cdot 3}\,\Delta^3 p_0 \\ +Kx^4 \\ = -\tfrac{1}{6}\,K_0\,x(x-1)(x-2)-3 \\ +\tfrac{1}{2}\,K_1\,x(x-1)(x-3) \\ -\tfrac{1}{2}\,K_2\,x(x-2)(x-3) \\ +\tfrac{1}{6}\,K_3\,x(x-1)(x-3); \end{aligned}$$

viz. this is the true value of y. Hence using the approximate formula as given by the first line, the last four lines give the error, viz. this error is

$$= Kx^4 + K_0\,x(x-1)(x-2)(x-3) + K_1\,x(x-1)(x-2)(x-3) + K_2\,(3Kx^2 - K_3)\,x(x-1)$$
$$-\tfrac{1}{2}\,K_3\,x(x-1)(x-2).$$

But K_0, K_1, K_2, K_3 being each $= P$ and $\leq Q$, this error is

$$\begin{aligned} &> P\{x^4 - 6x^3 + 12x^2 - 8x\} \\ &-Q\{4x^3 - 12x^2 + 12x\}, \end{aligned}$$

and it is

$$< Q\{x^4 - 6x^3 + 12x^2 - 8x\}$$
$$-P\{4x^3 - 12x^2 + 12x\},$$

the difference of these limits being

$$= (Q - P)(x^4 + 22x^3 + 7x^2 + 14x).$$

2. A determinant is a function of n^2 letters; viz. arranging these in the form of a square, the determinant

$$\begin{vmatrix} a_1, & b_1, & c_1, & d_1 \\ a_2, & b_2, & c_2, & d_2 \\ a_3, & b_3, & c_3, & d_3 \\ a_4, & b_4, & c_4, & d_4 \end{vmatrix}$$

is a function linear in regard to each of the n^2 letters, and such that interchanging any two entire lines, or any two entire columns, the sign of the determinant is reversed, its absolute value being unaltered.

The above definition leads to a rule for calculating the actual value of the determinant, which rule may be taken as a definition, viz. the determinant is the sum of $1 \cdot 2 \cdots n$ terms obtained as follows: starting from the term

$$a_1\, b_2\, c_3 \cdots m_n,$$

we permute in every possible way the suffixes $1, 2, 3, \ldots, n$, and give to the term a sign, $\pm$, which is compounded of as many $-$ signs as there are cases in which the arrangement may be obtained by a succession of interchanges of two letters; and then taking for each interchange the sign $-$, we obtain the sign $\pm$ of the term in question. The positive and negative terms are each of them $\frac{1}{2} \cdot 1 \cdot 2 \cdots n$ in number.

To show the execution of the two definitions, it is sufficient to observe that in the second definition, attending for instance only to the first and second columns, in any term $\Delta a_i b_j$, $\Delta a_j b_i$, &c., always correspond either terms $= \Delta a_i b_j$, &c., so that taking the pairs together, there are $\Delta(a_i b_j - a_j b_i)$, $\Delta(a_i b_k - a_k b_i)$, &c., terms which change their sign, but remain unaltered as to their absolute values by the interchange of the first and second columns.

Among the fundamental properties of determinants are as follows:

The properties are the same as regards lines and columns.

A determinant vanishes if any line vanishes (that is, if each term of the line is $= 0$).

A determinant vanishes if two lines are identical.

A determinant is a linear function of its lines.

Wherever:

Determinant having a line $A\ \alpha + a$ times the determinant having the line A ($\alpha\beta$ is here used to denote the line each term of which is α times the corresponding term of the line A).

Determinant having a line $A + a$ = determinant with the A + determinant with the line A'.

It follows that, if any line of a determinant is the sum of the other lines, each multiplied by an arbitrary coefficient, or, what is the same thing, if we can with any of the lines, each multiplied by an arbitrary coefficient, compose a line 0, then the determinant is $= 0$.

The same principle leads to a theorem for the product of two determinants of the same order n, viz. it is found that the product is a determinant of the same order n, each term thereof being a sum of the products of the terms of a line of one of the factors into the corresponding terms of a line of the other factor. Starting with this expression of the product, we decompose it into a series of determinants each of which is either $= 0$ or it is a product of a single term of the one factor into the other factor; and the sum of all these products is equal to the product of the two factors.

The unit determinant is a theorem in which we have a quantities x, y, ... connected by the linear equations

$$ax + by + cz + \cdots = 0,$$

$$a_1x + b_1y + c_1z + \cdots = 0,$$

$$a_2x + b_2y + c_2z + \cdots = 0;$$

and so, if we have a linear equations

$$ax + by + cz + \cdots = u,$$

then each of the quantities x, y, z, ... is given as the quotient of two determinants, the denominator being in each case

$$\begin{vmatrix} a, & b, & c, & \cdots \\ a_1, & b_1, & c_1, & \cdots \\ a_2, & b_2, & c_2, & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix}$$

and the numerators being (save as to their signs) that for x

$$\begin{vmatrix} u, & b, & c, & \cdots \\ 0, & b_1, & c_1, & \cdots \\ 0, & b_2, & c_2, & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix}$$

and the like for y, z, ...

A determinant remains unaltered when the lines and columns are interchanged, the dexter diagonal $\backslash$ remaining unaltered.

A determinant

$$\begin{vmatrix} a_1, & b_1, & c_1, & d_1, & \cdots \\ a_2, & b_2, & c_2, & d_2, & \cdots \\ a_3, & b_3, & c_3, & d_3, & \cdots \\ a_4, & b_4, & c_4, & d_4, & \cdots \end{vmatrix}$$

is a sum of products of complementary determinants, viz.

$$\Sigma \pm \begin{vmatrix} a_i, & b_j \end{vmatrix} \begin{vmatrix} c_k, & d_l, & \cdots \end{vmatrix},$$

or, say of products of complementary minors.

In particular, it is a sum

$$\Sigma \pm a_i \begin{vmatrix} b_j, & c_k, & d_l, & \cdots \end{vmatrix},$$

of products of first minors into single terms or $(n-1)^{\text{th}}$ minors.

This last theorem affords a convenient rule for the development of a determinant.

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On a Differential Formula Connected with the Theory of Confocal Conics.

[From the Messenger of Mathematics, vol. II. (1873), pp. 157, 158.]

THE following transformations present themselves in connexion with the theory of confocal conics.

The coordinates x, y of a point are considered as functions of the parameters h, k where

$$\frac{x^2}{a+h} + \frac{y^2}{b+h} = 1,$$

$$\frac{x^2}{a+k} + \frac{y^2}{b+k} = 1;$$

and then assuming $\sqrt{a+h} = s$, $\sqrt{a+k} = t$, ($i = \sqrt{-1}$ as usual) and writing $c = a - b$, we have

$$H = (s + i\sqrt{c - s^2})(t + i\sqrt{c - t^2}),$$
$$K = (s + i\sqrt{c - s^2})(t - i\sqrt{c - t^2}),$$

whence if

$$H = (s+h)(s+k), \quad K = (s+h)(s-k),$$

we have

$$H = \tfrac{1}{4}\{\sqrt{c}(s+t) + i\sqrt{(c-s^2)(c-t^2)}\}^2,$$
$$K = \tfrac{1}{4}\{\sqrt{c}(s-t) + i\sqrt{(c-s^2)(c-t^2)}\}^2,$$

or, say

$$\sqrt{H} = \tfrac{1}{2}\{\sqrt{c}(s+t) + i\sqrt{(c-s^2)(c-t^2)}\},$$
$$\sqrt{K} = \tfrac{1}{2}\{\sqrt{c}(s-t) + i\sqrt{(c-s^2)(c-t^2)}\}.$$

and thence

$$\begin{aligned}k+\tfrac{1}{2}(s+h)+\sqrt{H}&=\tfrac{1}{2}\{i\sqrt{(c-s^2)(c-t^2)}-i+a\sqrt{(s-t)}\},\\ k+\tfrac{1}{2}(s+h)+\sqrt{K}&=\tfrac{1}{2}\{i\sqrt{(c-s^2)(c-t^2)}+i+a\sqrt{(s-t)}\},\\ &=-[s+\tfrac{i}{2}\sqrt{(c-s^2)}]\cdot[t+\tfrac{i}{2}\sqrt{(c-t^2)}].\end{aligned}$$

These also follow from the known differential formula

$$4(ds^2+dy^2)=-kh\left(\frac{ds^2}{H}-\frac{dt^2}{K}\right),$$

that is,

$$\frac{4dtdy}{\sqrt{(c-s^2)(c-t^2)}}=\frac{ds}{\sqrt{H}}\,\frac{dt}{\sqrt{K}},$$

implying

$$\frac{2tdy}{\sqrt{(c-s^2)}}\,\frac{di}{\sqrt{(c-t^2)}}=\frac{ds}{\sqrt{H}}-\frac{dt}{\sqrt{K}},$$

$$\frac{2dy}{\sqrt{(c-t^2)}}+\frac{ds}{\sqrt{H}}\,\frac{dt}{\sqrt{K}}=\frac{ds}{\sqrt{H}}+\frac{dt}{\sqrt{K}},$$

where a is a constant. The foregoing integral formula give at once

$$\frac{ds}{\sqrt{H}}-\frac{dt}{\sqrt{K}}=\frac{dy}{\sqrt{(c-t^2)}},$$

$$\frac{ds}{\sqrt{H}}+\frac{dt}{\sqrt{K}}=\frac{ds}{\sqrt{(c-s^2)}},$$

and substituting these values we find $a=1$, and the differential formula are then satisfied.

We thence have

$$\text{const.}=\sqrt{(a+h)(b+h)}+\sqrt{(a+k)(b+k)},$$

as the integral of the differential equation

$$\frac{dh}{\sqrt{H}}+\frac{dk}{\sqrt{K}}=0,$$

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Two Smith's Prize Dissertations.

[From the Messenger of Mathematics, vol. II. (1873), pp. 161–166.]

WRITE dissertations:

1. *On the condition of the similarity of two physical systems.*

2. *On orthogonal surfaces.*

1. Considering two particles m, m_1, describing similar paths similar parts (which for convenience may be taken to be similarly situate in regard to two axes of rectangular axes respectively); viz. this means that the times t, t' of passage through corresponding arcs s, s' are proportional. The ratios $\frac{s}{s'}$, $\frac{t}{t'}$ are then each of them constant; and this must also be the case with the ratio $\frac{v}{v'}$, of the velocities v, v' at corresponding points; since it is clear that we must have $\frac{v}{v'} = \frac{s/s'}{t/t'}$.

Now in order that the two particles may move as above under the action of any forces upon the two particles respectively, it is clearly necessary that the forces F, F' at corresponding points shall act in the same direction, and be in a constant ratio of magnitudes. To obtain this ratio, imagine the two particles, m, m' moving as above, in the corresponding infinitesimal elements of times τ, τ', with the velocities u, u' through the infinitesimal arcs σ, σ' respectively; $\frac{u}{u'} = \frac{s/s'}{\sigma/\sigma'} = \frac{u}{u'}$; and the deflections from the tangent will be $\frac{1}{2}\,\frac{F}{m}\,\tau^2$, $\frac{1}{2}\,\frac{F'}{m'}\,\tau'^2$ respectively, and these must be in the ratio of the corresponding arcs σ, σ', viz. we must have

$$\frac{F\tau^2}{m} : \frac{F'\tau'^2}{m'} = \sigma : \sigma'.$$

or, what is the same thing,

$$\frac{Ft}{m} = \frac{F't'}{m'} \cdot \frac{\sigma}{\sigma'} : \frac{\sigma'}{s'}$$

that is,

$$\frac{F}{F'} = \frac{m}{m'} \cdot \frac{s}{s'} \cdot \left(\frac{t}{t'}\right)^2,$$

and this relation subsisting, we have at corresponding points of the paths of the two particles of particles moving under the like geometrical conditions, and we have above at the conclusion, given two physical constituted systems, which act upon any masses in a given magnitude-ratio; the component-particles being in a given ratio: $\frac{m}{m'}$ (the same for each pair of component particles); then if the particles of the two systems respectively are to move in similar paths of the same magnitude-ratio $\frac{s}{s'}$ the times of describing corresponding arcs being in a given constant ratio $\frac{t}{t'}$ (this implying as above that the ratio of the velocities at corresponding points is $\frac{v}{v'} = \frac{s/s'}{t/t'}$), it is necessary that the forces on corresponding particles in corresponding positions shall act in the same direction, and shall be in the constant magnitude-ratio

$$\frac{F}{F'} = \frac{m}{m'} \cdot \frac{s}{s'} \cdot \left(\frac{t}{t'}\right)^2,$$

and this being so, the motion of the two systems will in fact be similar as above explained.

Taking $\frac{m}{m'} = \alpha$ for the mass-ratio, $\frac{s}{s'} = a$ for the length-ratio, and $\frac{t}{t'} = \tau$ for the time-ratio; also $\frac{F}{F'} = \phi$ for the force-ratio, the condition determining the force-ratio ϕ is then

$$\phi = \frac{\alpha a}{\tau^2}.$$

It is to be observed, that if the forces are entirely internal, and proportional to homogeneous functions of the same order, say $-r$, of the coordinates of all or any of the particles; e.g. if they are central forces varying as the inverse nth power of the distance; then the condition as to the action of the forces in the two systems respectively can always be satisfied by giving a proper constant value to the ratio of the absolute forces (or forces at unity of distance), thus, if in the first system we have two particles m_1, m_2, attracting or repelling each other with a force $\dfrac{km_1m_2}{r^n}$, and if in the second system the force is $\dfrac{k'm'_1m'_2}{r'^n}$, then the condition as to the direction of the forces at corresponding positions is satisfied (*qua facts*); and the condition as to magnitude is

$$\frac{km_1m_2/r^n}{k'm'_1m'_2/r'^n} : \frac{m_1}{m'_1} \cdot \frac{r}{r'} \cdot \frac{1}{r^2} = 1 = \phi,$$

that is,

$$\frac{k}{k'} = \frac{m'_1}{m_1} \cdot \frac{r'^{n-1}}{r^{n-1}} \cdot \frac{t}{t'} \cdot \phi,$$

or, say

$$\frac{k}{k'} = \left(\frac{m'_1}{m_1}\right)^{n-1} \left(\frac{r'}{r}\right)^{n-2} \left(\frac{t}{t'}\right)^2.$$

In the case $n = 2$, the constant therein (applying however only to the case of two elliptic orbits of the same eccentricity) agrees with Kepler's third law, or say with the theorem

$$T = \frac{2\pi a^{3/2}}{\sqrt{(kM)}},$$

that is,

$$T = \frac{2\pi}{k} a^{3/2} \sqrt{(M)},$$

where observe that the μ, or mass in the sense of the formula, is the km, or attractive force on a unit of mass, of the theorem as above written down.

2. In a family of surfaces $F(x, y, z, p) = 0$, containing a single variable parameter p, there is through any given point of space, a surface or surfaces of the family; or (if more than one, confining the attention to one of these surfaces) we may say that there is, through any given point of space, a surface of the family.

Considering now two other families of surfaces, there will be through any given point of space, three surfaces, one of each family; and if the every given point of space (whatever) there intersect each other at right angles, we have a system of three orthogonal surfaces.

Supposing the equations of the three families to be

$$\begin{aligned} F(x, y, z, p) &= 0, \\ \Phi(x, y, z, q) &= 0, \\ \Psi(x, y, z, r) &= 0, \end{aligned}$$

then the requisite conditions are

$$\frac{dF}{dx}\frac{d\Phi}{dx}+\frac{dF}{dy}\frac{d\Phi}{dy}+\frac{dF}{dz}\frac{d\Phi}{dz}=0,$$

$$\frac{dF}{dx}\frac{d\Psi}{dx}+\cdots=0,$$

$$\frac{d\Phi}{dx}\frac{d\Psi}{dx}+\cdots=0,$$

viz. these equations must be satisfied not in general identically, but in virtue of the given equations $F=0$, $\Phi=0$, $\Psi=0$.

Or, what is more convenient, we may take the equations of the three families to be

$$p=f(x,y,z)=0,\quad q=\phi(x,y,z)=0,\quad r=\psi(x,y,z)=0,$$

and write the conditions in the form

$$\frac{dp}{dx}\frac{dq}{dx}+\frac{dp}{dy}\frac{dq}{dy}+\frac{dp}{dz}\frac{dq}{dz}=0,$$

$$\frac{dp}{dx}\frac{dr}{dx}+\cdots=0,$$

$$\frac{dq}{dx}\frac{dr}{dx}+\cdots=0,$$

where of course p, q, r stand for their given functions $f(x,y,z)$, $\phi(x,y,z)$, $\psi(x,y,z)$; the equations in this form contain only (x,y,z), and not the parameters p, q, r; so that, if satisfied at all, they must be satisfied identically; and the required conditions therefore are that the last-mentioned system of equations shall be satisfied identically by the values p, q, r considered as given functions of (x,y,z).

The last-mentioned conditions lead to the theorem known as Dupin's; viz. it follows from them that the surfaces intersect along their curves of curvature; or more definitely, each surface of one family is intersected by the surfaces of the other two families in its two sets of curves of curvature.

To indicate the geometrical ground of the theorem, consider on a surface of one family a point P, and at this point the normal meeting the consecutive surface in P'; the surfaces through P of the other two families respectively will pass through P', and meet the given surface in two curves PA, PB ($=PA'$, PB' represent infinitesimal arcs on these two curves respectively, the angle at P being a right angle.

Drawing at A, B normals to the given surface to meet the consecutive surface in the points A', B' respectively, the same two normals will meet the consecutive surface in the points P, P', respectively; and (the system being orthogonal), we must have the angle at P' a right angle. This imposes a condition upon the direction (in the tangent plane at P) of the orthogonal directions PA, PB; viz. it is found that these must be such that the two normals PP', AA' intersect, or, what is the same thing, the bisecting directions PA and PB must be the directions of the two curves of curvature.

Observe that PP', AA' intersecting each other, the four points P, P', A, A' are in the same plane, that is, PA, $P'A'$ intersect, from these being the normals at P, P' respectively of the surface through P or one of the other two families; and similarly PP', BB' intersecting each other, this, the four points P, P', B, B' are in the same plane, that is, PB, $P'B'$ intersect, these being the normals through P, P' to the surface through P of the third family.

Through any of the points P, A, A', P', viz. in the form

$$PA' \cdot PA + PP' \cdot AA' = PA \cdot PA' + PP' \cdot AA',$$

on the tangent plane at P of the orthogonal directions PA, PB; viz. it is found that these must be such that the two normals PP', BB' intersect, and is the same thing, the bisecting directions PA and PB must be the directions of the two curves of curvature.

Observe that PP', BB' intersecting each other, the four points P, P', B, B' are in the same plane, that is, PB, $P'B'$ intersect from these being the normals at P, P' respectively of the surface through P of one of the other two families; and similarly the points C, D, E, F are coplanar (the cycle for one surface, and the surface for the orthogonal PR, $P'R'$), then this P, P', R, R' are also coplanar, (PR being a right angle, also as is a a normal there are the directions of the curves of curvature at the point PR, the chord drawn in the directions of the curves of curvature on the consecutive surface. But in the orthogonal system they must be so; and this imposes upon the infinitesimal normal distance p, a condition; viz. it is found that p considered as a function of (x, y, z) must satisfy a certain partial differential equation of the second order.

It hence appears that no one of the three families of surfaces can be assumed at pleasure; for taking the equation of a family to be $p = f(x, y, z) = 0$, there is being the value of the parameter for the given surface of the foregoing investigation, and $p + dp$ the value of the parameter for the consecutive surface, the normal distance at the point (x, y, z) between the two surfaces is

$$= dp = \sqrt{\left[\left(\frac{dp}{dx}\right)^2 + \left(\frac{dp}{dy}\right)^2 + \left(\frac{dp}{dz}\right)^2\right]},$$

viz. dp is here a constant; and we have

$$1 = \sqrt{\left[\left(\frac{dp}{dx}\right)^2 + \left(\frac{dp}{dy}\right)^2 + \left(\frac{dp}{dz}\right)^2\right]};$$

satisfying the foregoing partial differential equation; or, what is the same thing, p considered as a function of x, y, z must satisfy a certain partial differential equation of the third order; viz. this is the condition to be satisfied in order that a family of surfaces $p = f(x, y, z) = 0$ may belong to an orthogonal system.

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An Elliptic-Transcendent Identity.

[From the Messenger of Mathematics, vol. II. (1873), p. 178.]

THE following is a singular identity:

$$\begin{aligned}
&(1+q)(1+q^2)(1+q^4)(1+q^7)(1+q^9)\cdots \\
&-(1-q)(1-q^2)(1-q^4)(1-q^7)(1-q^9)\cdots \\
&=2q(1+q^2)(1+q^4)(1+q^6)(1+q^8)(1+q^7)(1+q^9)\cdots,
\end{aligned}$$

where in each of the three terms every factor has the exponent 1 or 2 according as the exponent of q is not, or is, divisible by 7.

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Notices of Communications to the British Association for the Advancement of Science.

[From the Reports of the British Association for the Advancement of Science, 1865 to 1872. Notices and Abstracts of Miscellaneous Communications to the Sections.]

1. On the Problem of the in-and-circumscribed Triangle. Report, 1870, p. 10.

I have recently accomplished the solution of this problem, which I spoke of at the Meeting in 1864. The problem is as follows: required the number of the triangles the angles of which are situate in a given curve or curves, and the sides of which touch a given curve or curves. There are in all 32 cases [see 514] of the problem, according as the curves which contain the angles and are touched by the sides are distinct curves, or are two of all the same, that is two of them all distinct, &c. when the curves are all of them distinct, the number of triangles is here $= 2mn(2nD\bar{a}D'\bar{a}^4$, where a, a' are the orders of the curves containing the angles (or, say, of the *angle-curves* respectively); and β, γ' are the classes of the curves touched by the sides (or, say, of the *side-curves* respectively); and these have other classes referring to a like formula, viz. the number of triangles is here $= 2(B-1)(D-2)$ &c.; where, if B, γ are the class and order of the same curve $B = D - F$. The simplest case is when the curves are all of them one and the same curve, say $a = a' = a'' = B = D = F$, the number of triangles is here equal to

$$\begin{aligned}
&+ a^2 (&& +1),\\
&+ aB^2(&& +2b - 18b + 12c - 8d)\\
&+ aB'(&& -18b^2 + 102bc - 432bd + 270cd)\\
&+ aB''(&& -16b^2 + 252b^2c - 622bcd + 624bd^2 - 432cd)\\
&+ B''(&& -8)\\
&+ \{B^4(&& +18b - 192c)\}\\
&+ \{B^3(&& -38b + 1152c)\}.
\end{aligned}$$

where a is the order, A the class of the curve; α is the number, three times the class + the number of cusps, or (what is the same thing) three times the order + the number of inflexions.

2. On a Correspondence of Points and Lines in Space. Report, 1870, p. 10.

NINE points in a plane may be the intersection of two (and therefore of an infinite series of) cubic curves; say, that the nine points are an "encead": so, and similarly nine lines through a point may be the intersection of two (and therefore of an infinite series of) cubic cones; say, the nine lines are an encead. Now, imagine (in space) any 8 given points; taking a variable point P, and joining this with the 8 points, we have through P 8 lines and there is through P a ninth line completing the encead; this is and to be the corresponding line of P. We have thus to any point P a single corresponding line through the point P; this is the correspondence referred to in the heading, and which I would suggest as an interesting subject of investigation to geometers. Observe that, considering the whole system of points in space, the corresponding lines are a triple system of lines, not the whole system of lines in space. It is thus, not any line whatever, but only a line of the triple system, which has on it a corresponding point. But as to this same explanation is necessary; for starting with an arbitrary line, and taking upon it a point P, it would seem that P might be so determined that the given line and the lines from P to the eight points should form an encead,—that is, that the arbitrary line would have upon it a corresponding point or points.

The question of the foregoing species of correspondence was suggested to me by the consideration of a system of 10 points, such that joining any one whatever of them with the remaining nine points, the nine lines from this obtained form an encead; or, say, that each of the 10 points is the "enneadic centre" of the remaining nine. I have been led to such a system of 10 points by my researches upon Quartic Surfaces; but I do not as yet understand the theory.

3. On the Number of Covariants of a Binary Quantic. Report, 1871, pp. 9, 10.

THE author remarked [see 462] that it had been shown by Prof. Gordan that the number of the covariants of a binary quantic of any order was finite, and, in particular, that the numbers for the quintic and the sextic were 23 and 26 respectively. But the demonstration is a very complicated one, and it can scarcely be doubted that a more simple demonstration will be found. The question in its most simple form is as follows: viz. instead of the covariants we substitute their leading coefficients, each of which is a "seminvariant" satisfying a certain partial differential equation; say, the quantic is $(a, b, c, \ldots, k)(x, y)^n$, then the differential equation is $(a\partial_b + 2b\partial_c + \cdots + nj\partial_k)u = 0$, which ∂ quasi equation with $n+1$ variables admits of n independent solutions: for instance, if $n = 3$, the equation is $(a\partial_b + 2b\partial_c + 3c\partial_d)u = 0$, and the solutions are a, $ac - b^2$,

p. 567 [555] (tail of paper on a system of cubic equations)
$a^2d - 3abc + 2b^3$; the general value of u is $u =$ any function whatever of the last-mentioned three functions. We have to find the rational non-integral functions of these functions which are rational and integral functions of the coefficients; such a function is

$$\frac{1}{a^2}\{(a^2d - 3abc + 2b^3) + \tfrac{1}{4}(ac - b^2)^3\} = ac^3 + 4ac^2 + 4b^3d - 3b^2c^2 - 6abcd,$$

and the original three solutions, together with the last-mentioned function $a^2d^2 + \&c.$, constitute the complete system of the seminvariants of the cubic function; viz. every other seminvariant is a rational *and integral* function of these. And so, in the general case, the problem is to complete the series of the n solutions a, $ac - b^2$, $a^2d - 3abc + 2b^3$, $a^3e - 4a^2bd + 6ab^2c - 3b^4$, &c. by adding thereto the solutions which, being rational but non-integral functions of these, are rational and integral functions of the coefficients; and thus to arrive at a series of solutions such that every other solution is a rational and integral function of these.

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4. *Note on certain Families of Surfaces.* Report, 1871, pp. 19, 20.

See the paper numbered 526, of which this Note is a duplicate.

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5. *On the Mercator's Projection of a Surface of Revolution.* Report, 1873, p. 9.

The theory of Mercator's projection is obviously applicable to any surface of revolution; the meridians and parallels are represented by two systems of parallel lines at right angles to each other, in such wise that for the infinitesimal rectangles included between two consecutive arcs of meridian and arcs of parallel the rectangle in the projection is similar to that on the surface. Or, what is the same thing, drawing on the surface the meridians at equal infinitesimal intervals of angular distance, we may draw the parallels at such intervals as to divide the surface into infinitesimal squares; the meridians and parallels are then the projections represented by two systems of equidistant parallel lines dividing the plane into squares. And if the angular distance between two consecutive meridians instead of being infinitesimal is taken moderately small (5° or even 10°), then it is easy on the surface or *in plano*, using only the curve which is the meridian of the surface, to lay down graphically the series of parallels which are in the projection represented by equidistant parallel lines. The method is, of course, an approximate one, by reason that the angular distance between the two consecutive meridians is finite instead of infinitesimal.

I have in this way constructed the projection of a skew hyperboloid of revolution; viz. in one figure I show the hyperbola, which is the meridian section, and by means of it (taking the interval of the meridians to be $= 10°$) construct the positions of the

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successive parallels; I complete the figure by drawing the hyperbolas which are the orthographic projections of the meridians, and the right lines which are the ortho- graphic projections of the parallels; the figure thus exhibits the orthographic projection (on the plane of a meridian) of the hyperboloid divided into squares as above. The other figure, which is the Mercator's projection, is simply two systems of equidistant parallel lines dividing the paper into squares. I remark that in the first figure the projections of the right lines on the surface are the tangents to the bounding hyper- bola; in particular, the projection of one of these lines is an asymptote of the hyperbola. This I exhibit in the figure, and by means of it trace the Mercator's pro- jection of the right line on the surface; viz. this is a serpentine curve included between the right lines which represent two opposite meridians and having these lines for asymptotes. It is sufficient to draw one of these curves, since obviously for any other line of the surface belonging to the same system the Mercator's projection is at once obtained by merely displacing the curve parallel to itself; and for any line of the other system the projection is a like curve in a reversed position.

A Mercator's projection might be made of a skew hyperboloid not of revolution; viz. the curves of curvature ought be drawn so as to divide the surface into squares, and the curves of curvature be then represented by equidistant parallel lines as above; and the construction would in only a little more difficult. The projection presented itself to me as a convenient one for the representation of the geodesic lines on the surface, and for exhibiting them in relation to the right lines of the surface; but I have not yet worked this out. In conclusion, it may be remarked that a surface in general cannot be divided into squares by its curves of curvature, but that it may be in an infinity of ways divided into squares by two systems of curves on the surface, and any such system of curves gives rise to a Mercator's projection of the surface.

Notes and References.

518. Bertrand, *C. R.*, t. lxxv. (1872), pp. 535–536, referring to my Note remarks that the condition can be (by means of the imaginary coordinates of M. Ossian Bonnet) expressed in a simple form communicated by him to the Philosophic Society, May 1870. I reproduce this investigation, although it is not easy to present it in a quite intelligible form. We take $p = f(x, y, z)$ to represent a family of surfaces belonging to a triply orthotomic system, and consider two neighbouring surfaces (A) and (A') corre- sponding to the values x and $x + dx$; A and A' the two points where they meet the trajectories of the surfaces; AT, $A'T'$ the tangents to the curves of curvature of the same system at A, A' respectively. Then according to the remark of M. Lévy, it is to be expressed that these lines meet, and this is done by expressing that along the trajectory AA', the variation of the angle of AT with the osculating plane measures is equal to the angle of the osculating plane at A, A' respectively.

Let R be the spherical image of AT with RR, R the angle of AT with RR, θ–θ is the angle of AT with the osculating plane at A of the trajectory: $d\theta - d\theta = d\theta_1$. Introducing the symmetric imaginary coordinates α and β, we have

$$\alpha = \frac{q}{\sqrt{q^2 - p^2}}, \quad \beta = \frac{r}{\sqrt{q^2 - p^2}}, \quad 1 = \frac{p}{\sqrt{q^2 - p^2}}; \quad d\beta = -\alpha\, d\beta, \quad d\alpha = \beta\, d\alpha.$$

But $d\alpha$ and $d\beta$ being the increments of α, β corresponding to $d\alpha$ in the passage from A to A', then by a theorem of M. Liouville

$$d\eta = \alpha\, d\beta \left(\frac{d\beta}{d\alpha}\, da - \frac{d\beta}{d\alpha}\, d\beta \right);$$

the condition then is

$$d\theta = \alpha \left(\frac{d\beta}{d\alpha}\, da - \frac{d\beta}{d\alpha}\, d\beta \right).$$

and the formula

$$e^{ix\,d\xi} = b\sqrt{\frac{da}{d\beta}} \cdot \sqrt{\frac{d\beta}{d\eta}},$$

enables this to be written in the definitive form

$$d\alpha\,\frac{d}{d\eta}\left\{b\sqrt{\frac{db}{d\beta}}\cdot\frac{d\eta}{d\xi}-\frac{d\eta}{d\beta}\right\}+d\beta\,\frac{d}{d\xi}\left\{b\sqrt{\frac{db}{d\beta}}\cdot\frac{d\eta}{d\beta}\right\}+d\delta\,\left[\frac{d}{d\delta}\left(\frac{d\eta}{d\xi}\right)-\frac{d}{d\delta}\left(\frac{d\eta}{d\beta}\right)\right]=0.$$

We have

$$\begin{aligned} dx\,(\tfrac{1}{2}p+s)+dy\,\frac{db}{d\eta}\,d\beta \quad &+ b\,\frac{db}{d\xi}=0,\\ dx\,\frac{d\eta}{d\xi} \quad + dy\,(\tfrac{1}{2}p+s) \quad &+ dz\,\frac{d\eta}{d\beta}=0, \end{aligned}$$

and thence eliminating dx, dy, dz, we have

$$\begin{vmatrix} \dfrac{d}{d\eta}\left(b\sqrt{\dfrac{db}{d\beta}}\cdot\dfrac{d\eta}{d\xi}-\dfrac{d\eta}{d\beta}\right) & \dfrac{d}{d\xi}\left(b\sqrt{\dfrac{db}{d\beta}}\cdot\dfrac{d\eta}{d\beta}\right) & \dfrac{d}{d\delta}\left(\dfrac{d\eta}{d\xi}\right)-\dfrac{d}{d\delta}\left(\dfrac{d\eta}{d\beta}\right)\\ \tfrac{1}{2}p+s & \dfrac{db}{d\eta} & \dfrac{db}{d\xi}\\ \dfrac{d\eta}{d\xi} & \tfrac{1}{2}p+s & \dfrac{d\eta}{d\beta} \end{vmatrix}=0,$$

which defines the triply orthotomic system.

END OF VOL. VIII.

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